Mathematics for Computer Science and Artificial Intelligence

Lecture Notes

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Chapter 1

Introduction

1.1 Overview

In high school, the main aim of doing mathematics is often to score well on exams. In that setting, studying mathematics is useful in the sense that it can help you gain admission to a better university.

At the undergraduate level, studying mathematics is, of course, still associated with practical goals. A solid mathematical foundation may help you pursue graduate studies, enter competitive programs, or access research opportunities.

But it is worth pausing a moment to ask: *what lies at the end of this path?*

Historically, the development of mathematics was rarely driven by examinations. Mathematics has evolved partly from the need to solve practical problems, and partly from deep intellectual curiosity about patterns, structures, and the laws governing the world.

Geometry, for example, originated from the need to measure land and understand spatial relationships. Calculus emerged from attempts to describe motion, change, and accumulation in classical physics. In the early twentieth century, many physicists had little formal training in linear algebra, yet it rapidly became an indispensable language for quantum mechanics.

Some areas of mathematics were once considered purely theoretical. Number theory is a well-known example: for centuries it was admired for its elegance but believed to have little practical value. Today, it forms the backbone of modern cryptography, secure communication, and digital security. Every time you use HTTPS to access a website, you are enjoying the fruits of number theory.

Although learning the mathematics we are going to cover in this course is useful in the sense that it can add value to your résumé, this is not our primary purpose. We aim to present a different notion of usefulness than what you may have experienced in high school: mathematics provides a unique language and a powerful set of tools to model, analyze, and even solve challenging problems that arise in computer science and artificial intelligence.

From this perspective, mathematics is better viewed as a friend rather than an obstacle — not something to "fight against," nor something to be "conquered" merely for a certificate, but something to think with. Throughout this course, we will see how mathematical ideas help us design algorithms and machine learning models that are better, faster, and safer.

You cannot decline this invitation — it is a required course for every Yao Class student. Nevertheless, we will do our best to make challenging concepts accessible and intuitive.

1.2 Mathematics for Computer Science

The first part of the course will cover the following topics with applications in computer science:

- Discrete Probability (Chapter 2): How probability is defined in discrete settings; Some interesting math puzzles and applications.
- Sampling and Concentration (Chapter 3): When random variables concentrate around their means; How to provide guarantees on sample averages.
- Generating Functions (Chapter 4): How to use generating functions to solve counting problems; Some interesting tools from complex analysis.

1.3 Mathematics for Artificial Intelligence

The second part of the course will cover the following topics with applications in artificial intelligence:

- Continuous Probability (Chapter 5): How probability is defined in continuous settings; Gaussian distributions and their applications.
- Central Limit Theorem (Chapter 6): How sums of independent random variables approach Gaussian behavior; How variance-sensitive concentration inequalities refine Hoeffding's inequality.
- Optimization (Chapter 7): How to optimize convex functions with gradient descent; How to solve online problems with online convex optimization.
- Continuous Approximations (Chapter 8): How to approximate discrete problems with continuous ones; Ordinary differential equations and stochastic differential equations.

1.4 Notations and Conventions

We collect a few standard notations that will be used throughout the notes.

▶ Summation

The symbol $\sum$ denotes a *sum*. We write $\sum_{i=a}^{b} x_i$ for the sum of $x_a, x_{a+1}, \ldots, x_b$, and $\sum_{x \in S} f(x)$ for the sum of $f(x)$ over all elements x in a finite set S. For example, $\sum_{i=1}^{n} i = 1 + 2 + \cdots + n = n(n+1)/2$.

▶ Product

Similarly, the symbol $\prod$ denotes a *product*. We write $\prod_{i=a}^{b} x_i$ for the product of $x_a, x_{a+1}, \ldots, x_b$, and $\prod_{x \in S} f(x)$ for the product of $f(x)$ over all elements x in a finite set S. For example, $\prod_{i=1}^{n} i = 1 \times 2 \times \cdots \times n = n!$.

▶ Set Notations

A set can be specified as $\{x : \text{condition on } x\}$, read as "the set of all x such that the condition holds." We use the colon rather than the vertical bar $|$ so as not to confuse it with the bar in conditional probability (e.g., $\Pr[A \mid B]$). For example, $\{2k : k \in [5]\} = \{2, 4, 6, 8, 10\}$. We use $\mathbb{N}$ for the natural numbers, $\mathbb{Z}$ for the integers, $\mathbb{R}$ for the real numbers, and $[n] = \{1, 2, \ldots, n\}$ for the set of positive integers from 1 to n.

▶ Set Operations

For sets A and B: *union* $A \cup B = \{x : x \in A \text{ or } x \in B\}$, *intersection* $A \cap B = \{x : x \in A \text{ and } x \in B\}$, and *set difference* $A \setminus B = \{x : x \in A,\ x \notin B\}$. When working inside a universe Ω, the *complement* of A is $\bar{A} = \Omega \setminus A = \{x \in \Omega : x \notin A\}$. The *Cartesian product* of sets A and B is $A \times B = \{(a, b) : a \in A,\ b \in B\}$, the set of all ordered pairs with first element from A and second from B. We say that A and B are *disjoint* if $A \cap B = \emptyset$. When A and B are disjoint, we write their union as $A \sqcup B$, called the *disjoint union*. We use $|A|$ or $\#A$ to denote the *cardinality* of A, i.e., the number of elements in A.

▶ Functions

We write $f : X \to Y$ to denote a function from X to Y, and we denote by $f(x)$ the value of f at x. We say that f is *injective*, or equivalently, that f is an *injection* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$ for all $x_1, x_2 \in X$. We say that f is *surjective*, or equivalently, that f is a *surjection* if for every $y \in Y$, there exists an $x \in X$ such that $f(x) = y$. We say that f is *bijective*, or equivalently, that f is a *bijection* if it is both injective and surjective. When f is bijective, we use f^{-1} to denote the *inverse function* of f, which is the unique function $f^{-1} : Y \to X$ such that $f(f^{-1}(y)) = y$ for all $y \in Y$.

▶ Vectors.

A (real) vector in $\mathbb{R}^d$ is an ordered d-tuple of real numbers, written as $\boldsymbol{x} = (x_1, \ldots, x_d)$. We write vectors in bold lowercase. We write $\mathbf{0}$ for the zero vector, and $\mathbf{1}$ for the vector with all entries equal to 1. A linear combination of vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$ is a vector of the form $\sum_{i=1}^k c_i \boldsymbol{x}_i$, where $c_1, \ldots, c_k$ are real numbers. The *linear span* of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$ is the set of all linear combinations of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$. A set of vectors $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k\}$ is *linearly independent* if the only solution to the equation $\sum_{i=1}^k c_i \boldsymbol{x}_i = 0$ is $c_1 = c_2 = \cdots = c_k = 0$. The inner product of two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \sum_{i=1}^d x_i y_i$. We may also write the inner product as $\boldsymbol{x}^\top \boldsymbol{y}$. Two vectors are *orthogonal* if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$. The *$L^2$-norm* of a vector $\boldsymbol{x}$ is defined as $\|\boldsymbol{x}\|_2 = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle} = \sqrt{\sum_{i=1}^k x_i^2}$. A vector is a *unit vector* if $\|\boldsymbol{x}\|_2 = 1$.

▶ Matrices

A (real) matrix in $\mathbb{R}^{m \times n}$ is an $m \times n$ rectangular array of real numbers, written as $\boldsymbol{A} = [a_{ij}]_{1 \le i \le m,\, 1 \le j \le n}$. We write matrices in bold uppercase. The matrix-vector product $\boldsymbol{A}\boldsymbol{x} \in \mathbb{R}^m$ and the matrix-matrix product $\boldsymbol{A}\boldsymbol{B} \in \mathbb{R}^{m \times p}$ are defined in the usual linear-algebra sense. The *transpose* of a matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ is the matrix $\boldsymbol{A}^\top \in \mathbb{R}^{n \times m}$ such that the (i, j)-th entry of $\boldsymbol{A}^\top$ is the (j, i)-th entry of $\boldsymbol{A}$. Here are some important categories of matrices:

- The *zero matrix* $\mathbf{0} \in \mathbb{R}^{m \times n}$ is a matrix with all entries equal to 0.
- A matrix $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ is *diagonal* if all off-diagonal entries are zero. That is, the (i, j)-th entry of $\boldsymbol{A}$ is 0 if $i \neq j$.
- The *identity matrix* $\boldsymbol{I} \in \mathbb{R}^{d \times d}$ is a diagonal matrix with all diagonal entries equal to 1. That is, the (i, j)-th entry of $\boldsymbol{I}$ is 1 if $i = j$ and 0 otherwise.
- A matrix $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ is *symmetric* if $\boldsymbol{A} = \boldsymbol{A}^\top$.
- A matrix $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ is *orthogonal* if $\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{I}$. This condition is equivalent to $\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{I}$.
- A matrix $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ is *invertible* if there exists a matrix $\boldsymbol{B} \in \mathbb{R}^{d \times d}$ such that $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{I}$ and $\boldsymbol{B}\boldsymbol{A} = \boldsymbol{I}$. The matrix $\boldsymbol{B}$ is called the *inverse* of $\boldsymbol{A}$ and is denoted by $\boldsymbol{A}^{-1}$.

▶ Limits

For any sequence $(a_n)_{n \ge 1}$, we write

$$\lim_{n \to \infty} a_n = A$$

if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - A| < \epsilon$ holds for all $n \ge N$. In the same spirit, for any function $f : \mathbb{R} \to \mathbb{R}$, we write $\lim_{x \to +\infty} f(x) = A$ if for every $\epsilon > 0$ there exists $M > 0$ such that $|f(x) - A| < \epsilon$ holds for all x with $x > M$, and similarly for $\lim_{x \to -\infty} f(x) = A$. For a multivariate function $f : \mathbb{N}^d \to \mathbb{R}$, we write $\lim_{n_1, \ldots, n_d \to \infty} f(n_1, \ldots, n_d) = A$, or equivalently,

$\lim_{(n_1,\ldots,n_d)\to(\infty,\ldots,\infty)} f(n_1,\ldots,n_d) = A$, if for every $\epsilon > 0$ there exists $N > 0$ such that $|f(n_1,\ldots,n_d) - A| < \epsilon$ holds whenever $n_i \geq N$ for all $i = 1,\ldots,d$. Similar definitions apply for functions $f : \mathbb{R}^d \to \mathbb{R}$ when $x_i \to +\infty$ or $x_i \to -\infty$ for all $i = 1,\ldots,d$. For example, $\lim_{n\to\infty} \frac{1}{n} = 0$ and $\lim_{(x,y)\to(+\infty,-\infty)} \frac{1}{x^2+y^2} = 0$.

▶ Limits to a Finite Point

Limits can also describe convergence to a finite point. We write $\lim_{x\to x_0} f(x) = A$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that $|f(x) - A| < \epsilon$ holds for all x with $0 < |x - x_0| < \delta$. For $f : \mathbb{R}^d \to \mathbb{R}$, we similarly write $\lim_{\boldsymbol{x}\to\boldsymbol{x}_0} f(\boldsymbol{x}) = A$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that $|f(\boldsymbol{x}) - A| < \epsilon$ holds for all $\boldsymbol{x}$ with $0 < \|\boldsymbol{x} - \boldsymbol{x}_0\| < \delta$. Typical examples are $\lim_{x\to 0} \sin x = 0$ and $\lim_{(x,y)\to(0,0)}(x^2 + y^2) = 0$.

▶ Divergence to Infinity

We extend the notion of limits to allow the limit to be infinity. For a sequence $(a_n)_{n\geq 1}$, we write $\lim_{n\to\infty} a_n = +\infty$ if for every $A > 0$ there exists $N \in \mathbb{N}$ such that $a_n > A$ holds for all $n \geq N$, and similarly for $\lim_{n\to\infty} a_n = -\infty$. The definition for functions is analogous. For example, $\lim_{x\to\infty} x^2 = +\infty$, and $\lim_{x\to-\infty} x = -\infty$. With this extension, limits can take values in the extended real line $\mathbb{R} \cup \{\pm\infty\}$. It is worth noting that the limit can still fail to exist. For example, $\lim_{x\to+\infty} \sin x$ does not exist.

▶ Limit Superior and Limit Inferior

Let $(a_n)_{n\geq 1}$ be a sequence and let $u_n := \sup_{m\geq n} a_m$ and $\ell_n := \inf_{m\geq n} a_m$ be the supremum and infimum of the tail sequence starting from the n-th term. We define the *limit superior* and *limit inferior* of $(a_n)_{n\geq 1}$ as $\limsup_{n\to\infty} a_n := \lim_{n\to\infty} u_n$ and $\liminf_{n\to\infty} a_n := \lim_{n\to\infty} \ell_n$, These quantities describe the eventual lower and upper bounds of a sequence, and can be defined analogously for functions. Note that both quantities are always well-defined in the extended real line $\mathbb{R} \cup \{\pm\infty\}$, but the ordinary limit exists if and only if the two quantities are equal. For example, $\limsup_{x\to+\infty} \sin x = 1$ and $\liminf_{x\to+\infty} \sin x = -1$, but $\lim_{x\to+\infty} \sin x$ does not exist. On the other hand, $\limsup_{x\to+\infty} \frac{1}{x} = 0$, $\liminf_{x\to+\infty} \frac{1}{x} = 0$, and $\lim_{x\to+\infty} \frac{1}{x} = 0$.

▶ Series

Given a sequence $(a_n)_{n\geq 0}$, we use $\sum_{n=0}^{\infty} a_n$, or equivalently $\sum_{n\geq 0} a_n$, to denote the corresponding *series*. The n-th partial sum of this series is $s_n = \sum_{i=0}^{n} a_i$. If the sequence of partial sums $(s_n)_{n\geq 0}$ converges to a limit s, then we say that the series *converges* and write $\sum_{n=0}^{\infty} a_n = s$. We say that the series *converges absolutely* if the series of absolute values $\sum_{n=0}^{\infty} |a_n|$ converges.

▶ Derivatives

For a function $f : \mathbb{R} \to \mathbb{R}$, the derivative of f is defined as

$$f'(x) := \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

whenever the limit exists. Intuitively, the derivative of f at x is the quotient $\frac{\Delta f}{\Delta x}$ when Δx is very small. Another way to write this is $f'(x) = \frac{\mathrm{d}f}{\mathrm{d}x}(x)$, where d stands for *differential*. We say that f is *differentiable* at x if $f'(x)$ exists. We say that f is *differentiable* if it is differentiable at every point in its domain. Some useful properties of derivatives are:

- $c' = 0$ for any constant c.
- $(f+g)' = f' + g'$, $(f-g)' = f' - g'$.
- $(cf)' = cf'$ for any constant c.
- Product Rule: $(fg)' = f'g + fg'$.
- Quotient Rule: $(\frac{f}{g})' = \frac{f'g - fg'}{g^2}$.
- Chain Rule: $(f(g(x)))' = f'(g(x))g'(x)$.
- $(x^n)' = nx^{n-1}$ for any integer n.
- $(e^x)' = e^x$, $(\ln x)' = \frac{1}{x}$.

If f' is differentiable, then we can further define *the second-order derivative* $f''(x) := (f'(x))'$. If f'' is differentiable, then we can further define *the third-order derivative* $f'''(x) := (f''(x))'$. In general, we write $f^{(n)}(x)$ for *the n-th order derivative* of f at x, and it is defined recursively as $f^{(n)}(x) := (f^{(n-1)}(x))'$. We say that f is *n-times differentiable* if $f^{(n)}$ exists.

▶ Taylor Expansion

For a function $f : \mathbb{R} \to \mathbb{R}$, the Taylor expansion of f around x_0 is given by

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + R_n(x),$$

where $R_n(x)$ is the *remainder term*. If f is $(n+1)$-times differentiable, then $R_n(x)$ can be written as the Lagrange form of the remainder term:

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1},$$

for some ξ between x_0 and x.

▶ Integrals

For a function $f : \mathbb{R} \to \mathbb{R}$, the integral of f over an interval $[a, b]$ is defined as $\int_a^b f(x)\,\mathrm{d}x$. There are many ways to define the integral, but in general, the integral of f over $[a, b]$ is the area under the curve $y = f(x)$ between $x = a$ and $x = b$. For anyone who wondered what these symbols $\int$ and $\mathrm{d}x$ really mean,

$\int$ is a stylized letter S that stands for *sum*, and $\mathrm{d}x$ indicates an infinitesimally small change in x. Roughly speaking, an integral means we sum over all x in the interval $[a, b]$, each contributes $f(x)$ times an infinitesimal interval length Δx:

$$\int_a^b f(x)\,\mathrm{d}x \approx \sum_{x\in[a,b]} f(x)\Delta x.$$

If $f(x) = F'(x)$ for some differentiable function $F(x)$, then the *fundamental theorem of calculus* states that

$$\int_a^b f(x)\,\mathrm{d}x = F(x)\Big|_a^b, \quad \text{where} \quad F(x)\Big|_a^b := F(b) - F(a).$$

In this case, we say that $F(x)$ is an *antiderivative* of $f(x)$. The indefinite integral of $f(x)$ is the set of all antiderivatives of $f(x)$, denoted as $\int f(x)\,\mathrm{d}x = F(x) + C$, where C is an arbitrary constant. Some useful properties of integrals are:

- $\int_a^b (f+g)\,\mathrm{d}x = \int_a^b f\,\mathrm{d}x + \int_a^b g\,\mathrm{d}x.$
- $\int_a^b (cf)\,\mathrm{d}x = c\int_a^b f\,\mathrm{d}x.$
- Integration by Substitution / Change of Variables:

  $$\int_a^b f(g(x))g'(x)\,\mathrm{d}x \overset{u=g(x)}{=\!=} \int_{g(a)}^{g(b)} f(u)\,\mathrm{d}u.$$

  A heuristic to remember this is to think of $\mathrm{d}u = \mathrm{d}(g(x)) = g'(x)\,\mathrm{d}x$.
- Integration by Parts:

  $$\int_a^b F(x)G'(x)\,\mathrm{d}x = F(x)G(x)\Big|_a^b - \int_a^b F'(x)G(x)\,\mathrm{d}x.$$

  This is derived by integrating the product rule for derivatives. A heuristic to remember this is to think of $G'(x)\,\mathrm{d}x = \mathrm{d}(G(x))$ and $F'(x)\,\mathrm{d}x = \mathrm{d}(F(x))$. Then $F\,\mathrm{d}G = \mathrm{d}(FG) - \mathrm{d}F\,G$, and thus $\int F\,\mathrm{d}G = FG - \int \mathrm{d}F\,G$.
- $\int x^n\,\mathrm{d}x = \begin{cases} \frac{x^{n+1}}{n+1} + C & \text{if } n \neq -1, \\ \ln x + C & \text{if } n = -1. \end{cases}$
- $\int e^{\alpha x}\,\mathrm{d}x = \frac{1}{\alpha}e^{\alpha x} + C,\ \int a^x\,\mathrm{d}x = \ln a \cdot a^x + C.$

▶ Permutations

For a finite set S with n elements, a *permutation* of S is an ordered sequence $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ such that every element of S appears exactly once in the sequence. We denote by $\mathrm{Perm}(S)$ the set of all permutations of S. In particular, we write $\mathrm{Perm}(n) := \mathrm{Perm}([n])$ for convenience. The number of permutations of S is $|\mathrm{Perm}(S)| = n! = n \times (n-1) \times \cdots \times 1$, the *factorial* of n. For a permutation $\sigma \in \mathrm{Perm}(S)$, it is often convenient to view σ as a bijection from $[n]$ to S. Under this view, $\sigma(i) \in S$ is the i-th element of the permutation and $\sigma^{-1}(s) \in [n]$ is the position of the element s in σ.

▶ Binomial Coefficient

The *binomial coefficient* $\binom{n}{k}$ is the number of ways to choose k elements from a set of n elements. It is defined as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. For example, $\binom{5}{2} = 10$.

▶ Modular Arithmetic

We use $\mathbb{Z}_m$ to denote the integers modulo m, i.e., $\mathbb{Z}_m = \{0, 1, \ldots, m-1\}$. For integers x and $m \geq 2$, the expression $x \bmod m$ denotes the remainder when dividing x by m. So $x \bmod m \in \mathbb{Z}_m$. For example, $17 \bmod 5 = 2$, $(-3) \bmod 5 = 2$. We write $x \equiv y \pmod{m}$ if m divides $x - y$. This is equivalent to saying that x and y have the same remainder modulo m, i.e., $x \equiv y \pmod{m}$ if and only if $x \bmod m = y \bmod m$.

When the modulus is a prime p, every non-zero residue has a multiplicative inverse modulo p: for each $a \in \mathbb{Z}_p \setminus \{0\}$, there exists a unique $b \in \mathbb{Z}_p \setminus \{0\}$ such that $ab \equiv 1 \pmod{p}$. We use the notation a^{-1} to denote this unique number b and call it the *multiplicative inverse* of a modulo p.

Chapter 2

Discrete Probability

Probability theory is a branch of mathematics that models and analyzes random events and the likelihood of their occurrence. It provides a quantitative framework for measuring uncertainty and plays a fundamental role in computer science and artificial intelligence.

The first half of this course will focus on *discrete* probability theory, where all events are *discrete* in the sense that outcomes can be enumerated and probabilities can be computed as sums over individual outcomes. This includes examples such as coin flips, dice rolls, and randomized algorithms that rely on discrete random number generators.

Of course, the course will also cover *continuous* probability theory, where outcomes vary over a continuum and probabilities are computed using integrals rather than sums. This part of the study focuses on continuous random variables, such as the height of a person or the prediction of a regression model trained on randomly sampled data.

In this chapter, we begin with discrete probability theory, which is more directly connected to classical algorithms in computer science. Continuous probability theory has broad applications in artificial intelligence and will be covered in the second half of the course.

2.1 Probability Spaces and Events

Our world is full of randomness. For example, when playing Monopoly, a player rolls two dice at each turn, and the token moves forward by the sum of the outcomes. While it is impossible to predict the exact result in advance, experienced players would develop an intuition about the likelihood of different outcomes. For instance, obtaining a sum of 7 is more likely than obtaining a sum of 2.

2.1.1 Definitions

To reason about such likelihoods in a precise way, probability theory introduces the concept of *probability*, which assigns a numerical value to each event. The formal definition of probability is built upon two fundamental notions: the *probability space* and the *event*.

Definition 2.1 (Finite discrete probability space)**.** A finite discrete probability space is a pair $\mathcal{P} = (\Omega, p)$ consisting of:

- A non-empty finite set Ω, called the *sample space*, whose elements represent all possible outcomes of an experiment.
- A function $p : \Omega \to [0, 1]$, called the *probability mass function*, which assigns probabilities to outcomes and satisfies $\sum_{\omega \in \Omega} p(\omega) = 1$. That is, the total probability of all outcomes is 1.

Remark 2.2. For now, we only consider finite discrete probability spaces. In general, the sample space can also have countably infinite number of elements, and it is straightforward to extend the definitions to that setting. See also Section 5.1 for more details.

A probability space provides a precise description of a random experiment. The sample space contains all possible outcomes that can be observed in the experiment, and each outcome ω occurs with probability $p(\omega)$, a non-negative number between 0 and 1.

We next introduce the notion of an *event*, which is simply a set of outcomes. An event is said to occur if the observed outcome belongs to this set. The probability of an event is defined as the sum of the probabilities of all outcomes in the event.

Definition 2.3 (Event)**.** An event A is a subset of Ω ($A \subseteq \Omega$), and its probability is defined as

$$\Pr[A] := \sum_{\omega \in A} p(\omega). \tag{2.1}$$

Remark 2.4. For any outcome $\omega \in \Omega$, we have $\Pr[\{\omega\}] = p(\omega)$. We call $\{\omega\}$ an *elementary event* for every $\omega \in \Omega$. Another trivial fact is that $\Pr[\Omega] = 1$.

2.1.2 Example: Rolling Two Dice

Consider the experiment of rolling two fair dice. The sample space is

$$\Omega := \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}. \tag{2.2}$$

Since the dice are fair, the probability mass function is

$$p(\omega) := \frac{1}{36}, \quad \text{for } \omega \in \Omega. \tag{2.3}$$

We compare the probabilities of two events:

- Event A: the sum of the two dice is 7;
- Event B: the sum of the two dice is 2.

These two events consist of the following elementary events:

- $A = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$;
- $B = \{(1,1)\}$.

Therefore, we have $\Pr[A] = \frac{6}{36} = \frac{1}{6}$ and $\Pr[B] = \frac{1}{36}$. So $\Pr[A] > \Pr[B]$.

2.1.3 Example: The Birthday Paradox

▶ Problem Statement

Suppose you are in a class of $n = 30$ students. *Is it very likely or very unlikely that at least two students in your class share the same birthday?* We ignore Feb 29 and assume that birthdays are independently and uniformly distributed over the 365 days of a year.

▶ The Paradox

At first glance, one might reason naïvely that $30 \ll 365$, and therefore the probability of having a birthday collision should be very small. Indeed, the probability that *your own* birthday matches someone else's seems small, since only 29 other birthdays are sampled from the 365 days.

However, this is not the question we are asking. The real question is whether *any pair* of students shares the same birthday. The answer is surprising: the probability of having a birthday collision is over 70%!

This surprising phenomenon is known as the *birthday paradox*. Here, "paradox" does not mean a logical contradiction, but rather a result that runs contrary to our expectation. In the following, we will show why this paradox occurs.

▶ Solution

First, we set up the probability space. The sample space is the set of all possible assignments of birthdays to students. That is,

$$\Omega := \{(b_1, \ldots, b_n) : b_i \in [365] \text{ for all } i \in [n]\} \tag{2.4}$$

Since birthdays are independently and uniformly distributed, the probability mass function is

$$p(\omega) := \frac{1}{365^n}, \quad \text{for } \omega \in \Omega. \tag{2.5}$$

The event of having a birthday collision is:

$$A := \{(b_1, \ldots, b_n) : b_i = b_j \text{ for some } i \neq j\}. \tag{2.6}$$

A convenient way to compute the desired probability is to consider the complement event ($\bar{A}$ means "the event A does not happen"):

$$\bar{A} = \{(b_1, \ldots, b_n) : b_i \neq b_j \text{ for all } i \neq j\}. \tag{2.7}$$

That is, there is no birthday collision in the assignment. Once we have the probability of this event, we can deduce the probability of A by $\Pr[A] = 1 - \Pr[\bar{A}]$.

To compute $\Pr[\bar{A}]$, we can count the number of birthday assignments in $\bar{A}$ and divide by the total number of assignments in Ω. To avoid birthday collision, the birthday of the first student can take any of the 365 days. The birthday of the second student cannot coincide with the first student's birthday, so it can take any of the 364 days remaining. The birthday of the third student cannot coincide with the first two students' birthdays, so it can take any of the 363 days remaining. And so on. For the i-th student, there are $365 - i + 1$ choices of birthdays. Therefore,

$$\Pr[\bar{A}] = \frac{|\bar{A}|}{|\Omega|} = \frac{365 \times 364 \times 363 \times \cdots \times (365 - n + 1)}{365^n} \tag{2.8}$$

$$= \prod_{k=0}^{n-1} \frac{365 - k}{365} \tag{2.9}$$

$$= \prod_{k=0}^{n-1} \left(1 - \frac{k}{365}\right). \tag{2.10}$$

And the probability of A is

$$\Pr[A] = 1 - \Pr[\bar{A}] = 1 - \prod_{k=0}^{n-1} \left(1 - \frac{k}{365}\right). \tag{2.11}$$

How large is it when $n = 30$? We can compute it using a simple Python script:

```python
def get_collision_p(n):
    p = 1
    for k in range(n):
        p *= 1 - k / 365
    return 1 - p
print(get_collision_p(30))
```

Surprisingly, the probability is $\Pr[A] \approx 70.63\%$, already larger than 70%!

▶ Understanding the Birthday Paradox

We list the probability of birthday collision $\Pr[A]$ for different numbers of students n in Table 2.1 and plot it in Figure 2.1. Two numbers are striking: the curve passes 50% already at $n = 23$, and is very close to 100% around $n \approx 50$. *Why does collision appear so early?*

The key idea is to think in terms of *pairs*. Among n students, there are

$$N := \binom{n}{2} = \frac{n(n-1)}{2} \tag{2.12}$$

Table 2.1: The number of pairs $\binom{n}{2}$ and the probability of birthday collision $\Pr[A]$ for different numbers of students n.

n	Number of pairs $N = \binom{n}{2}$	Pr[collision]	$1 - e^{-N/365}$
5	10	2.71%	2.70%
10	45	11.69%	11.60%
20	190	41.14%	40.58%
30	435	70.63%	69.63%
40	780	89.12%	88.20%
50	1225	97.04%	96.51%
60	1770	99.41%	99.22%
70	2415	99.92%	99.87%

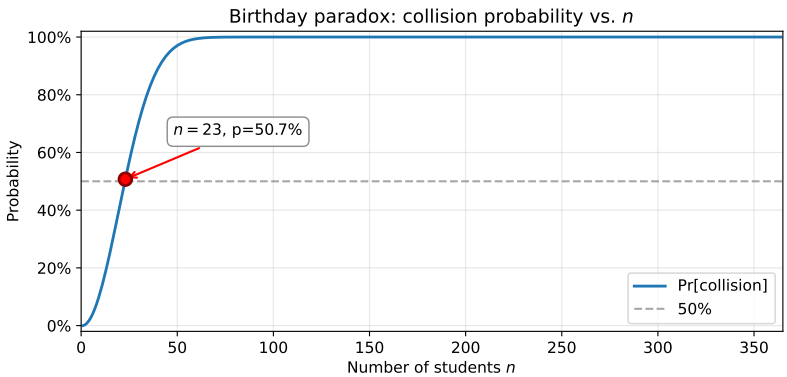


Figure 2.1: The probability of birthday collision $\Pr[A]$ exceeds 50% at $n = 23$ and is very close to 100% around $n \approx 50$.

pairs, and each pair has collision probability 1/365. Of course, these pairs are not independent, since different pairs may share a common student. However, such dependency is not very strong, and we will explain later that it is a good approximation to treat them as independent. Once we have this approximation, we can compute the probability of having a birthday collision as

$$\Pr[A] \approx 1 - \left(1 - \frac{1}{365}\right)^N, \tag{2.13}$$

which is 1 minus the probability of having no birthday collision among N independent pairs.

Using the Taylor approximation $\ln(1 + x) \approx x$, we get

$$\Pr[A] \approx 1 - \exp\left(\ln\left(1 - \frac{1}{365}\right) \cdot N\right) \approx 1 - e^{-N/365}. \tag{2.14}$$

So the collision probability converges to 1 exponentially as N increases. In particular, 365 serves as a half-life of the collision probability. when N becomes larger than 365, the collision probability can be very close to 1.

When $n = 30$, we have $N = 435$, which is larger than 365. We can check that $1 - e^{-N/365} \approx 69.63\%$, which is very close to the exact value $\Pr[A] \approx 70.63\%$. As shown in Table 2.1, the approximation is indeed very accurate across a wide range of n.

Therefore, the key insight is that n students lead to $N \approx n^2$ pairs, which grows quadratically fast as n increases. Although $n = 30$ is much smaller than 365, $N = 435$ is already large enough to induce birthday collision.

Finally, we show why Equation (2.13) is a good approximation in a more mathematical way. This only requires us to apply another Taylor approximation, $(1 + x)^k \approx 1 + kx$ for small x. Applying this approximation to each term in the product in Equation (2.11), we get

$$\Pr[A] = 1 - \prod_{k=0}^{n-1} \left(1 - \frac{k}{365}\right) \tag{2.15}$$

$$\approx 1 - \prod_{k=0}^{n-1} \left(1 - \frac{1}{365}\right)^k \tag{2.16}$$

$$= 1 - \left(1 - \frac{1}{365}\right)^{\sum_{k=0}^{n-1} k} \tag{2.17}$$

$$= 1 - \left(1 - \frac{1}{365}\right)^{N}, \tag{2.18}$$

which is exactly Equation (2.13).

▶ Back-of-the-Envelope Calculation

A *back-of-the-envelope calculation* refers to a quick, rough calculation of complex expressions based on simple approximations, much like one might sketch on the back of an envelope. The term is historically rooted in the estimation style popularized by Enrico Fermi. See the Wikipedia page on back-of-the-envelope calculations[1] for historical background.

In the above example, we used back-of-the-envelope calculations to quickly estimate the probability of birthday collision, based on Taylor approximations $\ln(1+x) \approx x$ and $(1+x)^k \approx 1+kx$ for small x. Although these calculations are not rigorous (the symbol "$\approx$" is used informally rather than as a mathematically defined relation), they are often valuable for developing intuition about the order of magnitude of the result before performing a more careful analysis.

In principle, one could make the argument fully rigorous by bounding the Taylor remainder. However, this may obscure the main idea and distract from the intuition.

Is there a middle ground between informal approximation and full rigor? The answer is yes. We will introduce the asymptotic notation in the next section.

[1] https://en.wikipedia.org/wiki/Back-of-the-envelope_calculation

2.2 A Useful Tool: Asymptotic Notations

Suppose there is a class consisting of n aliens, whose planet has m days in a year. The number m here can be much larger than 365. For example, Martians have 669 Martian days in a year, and Saturnians have 24,225 Saturnian days in a year. *What is the probability of having a birthday collision among n aliens?*

Following the same reasoning as in the Earth case, it is not hard to show that the probability is approximately

$$\Pr[A] \approx 1 - e^{-N/m} \quad \text{where} \quad N = \frac{n(n-1)}{2}, \tag{2.19}$$

Therefore, a birthday collision becomes likely once N becomes larger than m, which occurs when n is larger than $\sqrt{m}$.

However, such approximation is based on back-of-the-envelope calculation. Of course, for $m = 365$ we can list the exact values of $\Pr[A]$ and $1 - e^{-N/m}$ as what is shown in Table 2.1 and verify that the approximation is very accurate. But in general, how do we know the approximation is good?

Now we introduce a useful tool: *asymptotic notation*. This tool allows us to quantify the approximation error with rigorous mathematics, while keeping the calculation concise and intuitive.

Remark 2.5. This alien birthday paradox is not entirely hypothetical. As we will see in Section 2.4.3, the birthday paradox has a deep connection to hashing in computer science. The number of days in a year is analogous to the hash table size, and a birthday collision is analogous to a hashing collision. Studying birthday paradox can help us understand how collisions arise and how to choose an appropriate hash table size to avoid collisions.

2.2.1 Big-$\mathcal{O}$ Notation

Building on the motivation above, we now introduce the *big-$\mathcal{O}$ notation*, which is an asymptotic notation for upper bounding the growth rate of a function. A concrete warm-up is pair counting. With n students, the number of unordered pairs is

$$N = \frac{n(n-1)}{2} = \frac{1}{2}n^2 - \frac{1}{2}n.$$

At small n, both terms matter numerically; at large n, the n^2 term controls the scale, and the linear term only contributes a lower-order correction. So if our goal is to say "roughly speaking, N grows like n^2 as n increases", big-$\mathcal{O}$ notation allows us to write

$$N = \frac{n(n-1)}{2} = \mathcal{O}(n^2). \tag{2.20}$$

Here, the big-$\mathcal{O}$ notation $\mathcal{O}(n^2)$ means that N is eventually upper bounded by a constant multiple of n^2, with constants and lower-order terms absorbed into the notation. Now we make this statement formal.

Definition 2.6 (Big-$\mathcal{O}$ notation)**.** Let $f, g : \mathbb{N} \to \mathbb{R}$ be functions defined on $\mathbb{N}$. We write $f(n) = \mathcal{O}(g(n))$ if there exist constants $C > 0$ and $N \in \mathbb{N}$ such that

$$|f(n)| \leq Cg(n), \quad \forall n \geq N. \tag{2.21}$$

In other words, $f(n) = \mathcal{O}(g(n))$ denotes an eventual upper bound: after some threshold n_0, the magnitude of $f(n)$ is controlled by a constant multiple of $g(n)$.

Example 2.7. The following examples are easy to verify:

$$\log n = \mathcal{O}(n), \quad 3n^3+5n^2+100n = \mathcal{O}(n^3), \quad \frac{1}{n^2+10} = \mathcal{O}\left(\frac{1}{n^2}\right). \tag{2.22}$$

Note that the statement $f(n) = \mathcal{O}(g(n))$ should be understood as a whole, rather than as an ordinary algebraic equality. So does $\mathcal{O}(g(n))$ mean something in this statement?

Strictly speaking, $\mathcal{O}(g(n))$ should be written as $\mathcal{O}(g)$, defined as the following set of functions:

$$\mathcal{O}(g) := \{f : \exists C > 0, \exists N \in \mathbb{N} \text{ s.t. } |f(n)| \leq Cg(n), \text{ for all } n \geq N\}. \tag{2.23}$$

The notation $f(n) = \mathcal{O}(g(n))$ is therefore shorthand for the statement that the function f belongs to the set $\mathcal{O}(g)$.

Remark 2.8. Since $\mathcal{O}(g)$ is a set of functions, the correct notation would be $f \in \mathcal{O}(g)$. In practice, however, nobody cares, and computer scientists just simply write $f(n) = \mathcal{O}(g(n))$ instead of $f \in \mathcal{O}(g)$. Here are four reasons listed in Section 9.2 of Concrete Mathematics [GKP94]:

1. **Tradition.** Number theorists adopted the equals sign with big-$\mathcal{O}$ notation early on, and this usage has since become standard. We cannot hope to change this tradition in the mathematical community.
2. **Tradition.** Computer scientists are quite used to seeing equals signs abused, such as $N = N + 1$ in programming languages.
3. **Tradition.** The symbol "=" is often read as "is." And in English, "is" is one-way. For example, a bird is an animal, but we do not say that an animal is a bird.
4. **It is natural.** When $\mathcal{O}(g(n))$ appears in the middle of a formula, it is natural to interpret it as representing a quantity rather than a set.

Anyway, we will use the equals sign!

2.2.2 Basic Properties of Big-$\mathcal{O}$ Notation

For any functions $\alpha, g : \mathbb{N} \to \mathbb{R}$, we write $\alpha(n) + \mathcal{O}(g(n))$ to denote the set

$$\alpha(n) + \mathcal{O}(g(n)) := \{\alpha + f : f \in \mathcal{O}(g)\}. \tag{2.24}$$

In other words, $\alpha(n) + \mathcal{O}(g(n))$ means all functions of the form $\alpha(n) + f(n)$ where $f \in \mathcal{O}(g)$. Similarly, we can interpret $\alpha(n) - \mathcal{O}(g(n))$, $\alpha(n)\mathcal{O}(g(n))$, and $\mathcal{O}(g(n))/\alpha(n)$ by treating $\mathcal{O}(g(n))$ as an unknown function $f(n)$ with $f \in \mathcal{O}(g)$ and then computing as usual.

More generally, whenever $\mathcal{O}(g(n))$ appears in a formula, the result should be interpreted as the set of all possible outcomes arising from substituting $\mathcal{O}(g(n))$ by all functions $f \in \mathcal{O}(g)$. Formally, for $h : \mathbb{N} \times \mathbb{R} \to \mathbb{R}$, $h(n, \mathcal{O}(g(n)))$ is defined as the following set of functions:

$$h(n, \mathcal{O}(g(n))) := \{n \mapsto h(n, f(n)) : f \in \mathcal{O}(g)\}. \tag{2.25}$$

Further, the following rules can be easily derived from Definition 2.6:

Lemma 2.9. Let $f, g : \mathbb{N} \to \mathbb{R}$ and $c \in \mathbb{R}$. If $f(n), g(n) \geq 0$ for all n, then

$$f(n) = \mathcal{O}(f(n)) \tag{2.26}$$
$$c \cdot f(n) = \mathcal{O}(f(n)) \tag{2.27}$$
$$\mathcal{O}(\mathcal{O}(f(n))) = \mathcal{O}(f(n)) \tag{2.28}$$
$$\mathcal{O}(f(n)) + \mathcal{O}(g(n)) = \mathcal{O}(f(n) + g(n)) \tag{2.29}$$
$$\mathcal{O}(f(n)) + \mathcal{O}(g(n)) = \mathcal{O}(\max\{f(n), g(n)\}) \tag{2.30}$$
$$\mathcal{O}(f(n)) \cdot \mathcal{O}(g(n)) = \mathcal{O}(f(n) \cdot g(n)) \tag{2.31}$$
$$\mathcal{O}(f(n)g(n)) = f(n) \cdot \mathcal{O}(g(n)) \text{ if } f(n) > 0 \text{ for all } n. \tag{2.32}$$

We leave the proof as an exercise.

Remark 2.10. Here we list the formal definitions of some of the expressions above. The key spirit is to replace $\mathcal{O}(\,\cdot\,)$ with any function in the set and then compute as usual.

$$\mathcal{O}(f(n)) + \mathcal{O}(g(n)) := \{f_0 + g_0 : f_0 \in \mathcal{O}(f), g_0 \in \mathcal{O}(g)\} \tag{2.33}$$
$$\mathcal{O}(f(n)) \cdot \mathcal{O}(g(n)) := \{f_0 \cdot g_0 : f_0 \in \mathcal{O}(f), g_0 \in \mathcal{O}(g)\} \tag{2.34}$$
$$\mathcal{O}(\mathcal{O}(f(n))) := \{h : g \in \mathcal{O}(f), h \in \mathcal{O}(g)\} \tag{2.35}$$
$$f(n) \cdot \mathcal{O}(g(n)) := \{f \cdot g_0 : g_0 \in \mathcal{O}(g)\}. \tag{2.36}$$

Remark 2.11. Continued from Remark 2.8, whenever an equation involves big-$\mathcal{O}$ notation, the equals sign "=" should be interpreted as a unidirectional relation (membership or set inclusion), rather than a bidirectional equality. More precisely, if $\mathcal{F}$ is a set of functions and we write $\mathcal{F} = \mathcal{O}(g(n))$, this actually means $\mathcal{F} \subseteq \mathcal{O}(g(n))$ rather than $\mathcal{F} = \mathcal{O}(g(n))$. For example, $\mathcal{O}(\mathcal{O}(f(n))) = \mathcal{O}(f(n))$ should be understood as $\mathcal{O}(\mathcal{O}(f(n))) \subseteq \mathcal{O}(f(n))$. Similarly, $\mathcal{O}(f(n)g(n)) = f(n) \cdot \mathcal{O}(g(n))$ should be understood as $\mathcal{O}(f(n)g(n)) \subseteq f(n) \cdot \mathcal{O}(g(n))$.

Example 2.12. If $f(n) = c + \mathcal{O}(\frac{1}{n})$, then

$$f(n)^2 = (c + \mathcal{O}(\tfrac{1}{n}))^2 = c^2 + 2c \cdot \mathcal{O}(\tfrac{1}{n}) + \mathcal{O}(\tfrac{1}{n^2}) \tag{2.37}$$

$$= c^2 + \mathcal{O}(\tfrac{1}{n}) + \mathcal{O}(\tfrac{1}{n^2}) \tag{2.38}$$

$$= c^2 + \mathcal{O}(\tfrac{1}{n}). \tag{2.39}$$

In other words, if an approximation to a constant value has error $\mathcal{O}(\frac{1}{n})$, then the induced approximation to its square still has error $\mathcal{O}(\frac{1}{n})$.

2.2.3 Extended Definitions of Big-$\mathcal{O}$ Notation

In the above definition of big-$\mathcal{O}$ notation $f(n) = \mathcal{O}(g(n))$, we only require $g(n)$ to dominate $f(n)$ for all sufficiently large n. It is also common to use the big-$\mathcal{O}$ notation to describe that $f(n)$ is dominated by $g(n)$ for all n in a given domain, such as $n \in \mathbb{N}$.

Definition 2.13 (Big-$\mathcal{O}$ notation over a domain)**.** Let $f, g : D \to \mathbb{R}$ be functions defined on a domain D. We say that $f(n) = \mathcal{O}(g(n))$ for all $n \in D$ if there exists a constant $C > 0$ such that

$$|f(n)| \leq Cg(n), \quad \forall n \in D. \tag{2.40}$$

More precisely, $f(n) = \mathcal{O}(g(n))$ for all $n \in D$ means $f \in \mathcal{O}_D(g)$, where $\mathcal{O}_D(g)$ is the following set of functions:

$$\mathcal{O}_D(g) := \{f : \exists C > 0, \forall n \in D, |f(n)| \leq Cg(n)\}. \tag{2.41}$$

To distinguish this extension to finite domains from the original definition of big-$\mathcal{O}$ notation (Definition 2.6), we write $f(n) = \mathcal{O}(g(n))$ as $n \to \infty$ to refer to Definition 2.6 and $f(n) = \mathcal{O}(g(n))$ for all $n \in D$ to refer to Definition 2.13. We may omit mentioning $n \to \infty$ or $n \in D$ when it is clear from the context. By default, the big-$\mathcal{O}$ notation is understood with respect to the limit $n \to \infty$.

Example 2.14. Let $f(n) = n + 1$. Then $f(n) = \mathcal{O}(n)$ as $n \to \infty$, but we cannot say $f(n) = \mathcal{O}(n)$ for all $n \in \mathbb{N}$, since $f(0) = 1 > 0$. Still, we can safely say $f(n) = \mathcal{O}(n)$ for all $n \geq 1$ because $f(n) = n + 1 \leq 2n$ for all $n \geq 1$.

Remark 2.15. In fact, in typical applications, there is not much difference between the two definitions of big-$\mathcal{O}$ notation if the domain is $\mathbb{N}$. If $g(n) > 0$ for all $n \in \mathbb{N}$, then $f(n) = \mathcal{O}(g(n))$ for all $n \in \mathbb{N}$ is equivalent to $f(n) = \mathcal{O}(g(n))$ as $n \to \infty$. We leave the proof as an exercise.

The big-$\mathcal{O}$ notation can also be extended to describe the asymptotic behavior of multivariate functions.

Definition 2.16 (Big-$\mathcal{O}$ notation for multivariate functions)**.** Let $f, g : \mathbb{N}^d \to \mathbb{R}$ be functions defined on $\mathbb{N}^d$. We write

$$f(n_1, \ldots, n_d) = \mathcal{O}(g(n_1, \ldots, n_d)) \tag{2.42}$$

if there exist constants $C > 0$ and $N \in \mathbb{N}$ such that

$$|f(n_1, \ldots, n_d)| \leq Cg(n_1, \ldots, n_d), \quad \forall n_1, \ldots, n_d \geq N. \tag{2.43}$$

Example 2.17. In algorithm analysis, we often implicitly assume that the input size goes to infinity. When you multiply two rectangular matrices $\boldsymbol{A} \in \mathbb{R}^{n \times d}$ and $\boldsymbol{B} \in \mathbb{R}^{d \times n}$ together, a naïve implementation may require you to compute $n^2 d$ multiplications and store n^2 elements in the result matrix. The total time complexity is $\mathcal{O}(n^2 d) + \mathcal{O}(n^2) = \mathcal{O}(n^2 d)$.

The big-$\mathcal{O}$ notation can also be extended to describe the behavior of functions when some input variables go to infinity while others range over some domains that may or may not depend on the input variables that go to infinity.

This situation is far from unusual. Although in computer science, we often analyze problems as the input size tends to infinity, the input itself may depend on several parameters, and we may be particularly interested in the cases where certain relationships between these parameters are satisfied.

For example, for rectangular matrix multiplication (see Example 2.17), the time complexity can be reduced to $\mathcal{O}(n^2)$ when $d \leq n^{0.3}$ [LG12]. For graph algorithms, simple graphs (those with at most one edge between any two vertices) are frequently studied, where the number of edges m is bounded by $\frac{n(n-1)}{2}$. In machine learning theory, the dataset size n and the model size m are both important parameters, and a learning algorithm's behavior can differ significantly depending on whether $m < n$, $m \approx n$, or $m > n$.

We omit the formal definition of big-$\mathcal{O}$ notation in this setting and instead illustrate the idea with the following example.

Example 2.18. A simple undirected graph consists of a set of vertices and a set of undirected edges, where each edge connects two vertices and no two edges connect the same pair of vertices. Let n be the number of vertices, then the number of edges m must satisfy $0 \leq m \leq \frac{n(n-1)}{2}$. A naïve implementation of Dijkstra's algorithm performs $\mathcal{O}(n + m)$ operations on a binary heap data, each requiring $\mathcal{O}(\log n)$ time. Therefore, the total time complexity is $\mathcal{O}((n+m) \log n)$. Here, the big-$\mathcal{O}$ notation means that there exist a constant $C > 0$ and $N \in \mathbb{N}$ such that the running time is bounded by $C \cdot (n + m) \log n$ for all $n \geq N$ and $m \in \{0, 1, \ldots, \frac{n(n-1)}{2}\}$.

The big-$\mathcal{O}$ notation can also be extended to limiting regimes on real-valued variables. For example, when $x \to 0$, we define big-$\mathcal{O}$ notation as follows.

Definition 2.19 (Big-$\mathcal{O}$ notation for $x \to 0$)**.** Let $f, g : \mathbb{R} \to \mathbb{R}$ be functions defined on $\mathbb{R}$. We say that $f(x) = \mathcal{O}(g(x))$ as $x \to 0$ if there exist constants $C > 0$ and $\delta > 0$ such that

$$|f(x)| \leq Cg(x), \quad \forall x \in \mathbb{R} \text{ with } 0 < |x| < \delta. \tag{2.44}$$

The big-$\mathcal{O}$ notation for $x \to +\infty$, $x \to -\infty$, or limits involving multiple variables is defined analogously.

Example 2.20. The following examples illustrate big-$\mathcal{O}$ notation under different limiting regimes:

- $x \in \mathbb{R}$ and $x \to 0$: $\ln(1+x) = x + \mathcal{O}(x^2)$.
- $x \in \mathbb{R}$ and $x \to +\infty$: $\frac{1}{1+x} = \mathcal{O}(\frac{1}{x})$.
- $x \in \mathbb{R}$ and $x \to -\infty$: $e^x = \mathcal{O}(1)$.
- $(x, y) \in \mathbb{R}^2$ and $(x, y) \to (0, 0)$: $\ln(1 + x + y) = x + y + \mathcal{O}(x^2 + y^2)$.
- $(x, y) \in \mathbb{R}^2$ and $(x, y) \to (+\infty, +\infty)$: $x \sin y = \mathcal{O}(x)$.

A typical source of approximation formulas is Taylor expansion. We refer the reader to calculus textbooks for a detailed proof.

Proposition 2.21 (Taylor Expansion)**.** Let $\varphi : \mathbb{R} \to \mathbb{R}$ be $(r+1)$-times continuously differentiable in a neighborhood of 0. Then, as $x \to 0$,

$$\varphi(x) = \sum_{j=0}^{r} \frac{\varphi^{(j)}(0)}{j!} x^j + \mathcal{O}(x^{r+1}), \tag{2.45}$$

where $\varphi^{(j)}(0)$ is the j-th order derivative of φ evaluated at 0.

Example 2.22. Examples of Taylor expansions when $x \to 0$:

- $\ln(1 + x) = x + \mathcal{O}(x^2)$.
- $e^x = 1 + x + \mathcal{O}(x^2)$.
- $(1 + x)^\alpha = 1 + \alpha x + \mathcal{O}(x^2)$.
- $(1 - x)^\alpha = 1 - \alpha x + \mathcal{O}(x^2)$.

Remark 2.23. We can also take D to be a neighborhood of 0 and write Taylor expansion based on the definition of big-$\mathcal{O}$ notation over the domain D:

$$\varphi(x) = \sum_{j=0}^{r} \frac{\varphi^{(j)}(0)}{j!} x^j + \mathcal{O}(x^{r+1}), \quad \forall x \in D. \tag{2.46}$$

2.2.4 Other Asymptotic Notations

Beyond the big-$\mathcal{O}$ notation, a few other asymptotic notations are worth mentioning. Before that, we first introduce an equivalent definition of big-$\mathcal{O}$ notation using lim sup:

Definition 2.24 (Big-$\mathcal{O}$ notation defined using lim sup)**.** Let $f, g : \mathbb{N} \to \mathbb{R}$, where $g(n) > 0$ for all sufficiently large n. We write $f(n) = \mathcal{O}(g(n))$ if

$$\limsup_{n\to\infty} \frac{|f(n)|}{g(n)} < +\infty. \tag{2.47}$$

Here, $\limsup_{n\to\infty} a_n$ is the *limit superior* of a sequence $(a_n)_{n\geq 1}$ as n goes to infinity, defined as

$$\limsup_{n\to\infty} a_n := \lim_{n\to\infty} \left(\sup_{m\geq n} a_m \right). \tag{2.48}$$

We leave it as an exercise to verify that the two definitions are equivalent.

Remark 2.25. Strictly speaking, Definitions 2.6 and 2.24 are *not* exactly equivalent. The first definition allows $0 = \mathcal{O}(0)$ while the second definition does not. However, if we restrict g to be positive for all sufficiently large n, then the two definitions are equivalent.

Note that $f(n) = \mathcal{O}(g(n))$ is an asymptotic upper bound, so it plays the role of a "$\leq$" type relation up to constant factors. We next introduce asymptotic notations that act as "$\geq$" and "$\approx$" type relations up to constant factors.

Definition 2.26 ($\Omega, \Theta, \sim$ Notations)**.** Let $f, g : \mathbb{N} \to \mathbb{R}$, where $g(n) > 0$ for all sufficiently large n.

- We write $f(n) = \Omega(g(n))$ if $\liminf_{n\to\infty} \frac{|f(n)|}{g(n)} > 0$.
- We write $f(n) = \Theta(g(n))$ if $f(n) = \mathcal{O}(g(n))$ and $f(n) = \Omega(g(n))$.
- We write $f(n) \sim g(n)$ if $\lim_{n\to\infty} \frac{|f(n)|}{g(n)} = 1$.

The Ω notation may be viewed as the opposite of the big-$\mathcal{O}$ notation. If both $f(n)$ and $g(n)$ are positive for all sufficiently large n, then $f(n) = \Omega(g(n))$ if and only if $g(n) = \mathcal{O}(f(n))$. The Θ notation specifies a tight order of growth, since $f(n) = \Theta(g(n))$ means that $f(n)$ is both asymptotically upper and lower bounded by $g(n)$. Finally, the $\sim$ notation indicates that $f(n)$ and $g(n)$ have exactly the same order of growth, which is a stronger statement than $f(n) = \Theta(g(n))$.

Example 2.27. For $N = \frac{n(n-1)}{2}$, we have: Examples of $\mathcal{O}, \Omega, \Theta, \sim$ notations:

- $\frac{n(n-1)}{2} = \mathcal{O}(n^2) = \mathcal{O}(n^3)$.
- $\frac{n(n-1)}{2} = \Omega(n^2) = \Omega(n)$.
- $\frac{n(n-1)}{2} = \Theta(n^2)$.
- $\frac{n(n-1)}{2} \sim \frac{1}{2}n^2$.

Remark 2.28. Some people may write $f(n) \ll g(n)$ or $f(n) \lesssim g(n)$ for $f(n) = \mathcal{O}(g(n))$, $f(n) \gg g(n)$ or $f(n) \gtrsim g(n)$ for $f(n) = \Omega(g(n))$, and $f(n) \asymp g(n)$ for $f(n) = \Theta(g(n))$.

All the above notations accept the case of $f = g$. The following two asymptotic notations describe the strict versions: little-o corresponds to a strict "$<$" type comparison, and little-ω corresponds to a strict "$>$" type comparison.

Definition 2.29 (Little-o and Little-ω Notations)**.** Let $f, g : \mathbb{N} \to \mathbb{R}$, where $g(n) > 0$ for all sufficiently large n.

- We write $f(n) = o(g(n))$ if $\lim_{n\to\infty} \frac{|f(n)|}{g(n)} = 0$.
- We write $f(n) = \omega(g(n))$ if $\lim_{n\to\infty} \frac{|f(n)|}{g(n)} = +\infty$.

The little-o notation means that $f(n)$ grows strictly slower than $g(n)$, and the little-ω notation means that $f(n)$ grows strictly faster than $g(n)$.

Example 2.30. Examples of little-o and little-ω notations:

- $\frac{n(n-1)}{2} = \mathcal{O}(n^2) = o(n^3)$.
- $\frac{n(n-1)}{2} = \Omega(n^2) = \omega(n)$.

2.2.5 Revisiting the Birthday Paradox with Big-$\mathcal{O}$ Notation

We now return to the birthday approximation and quantify its error. We consider the asymptotic regime where the number of aliens satisfies $1 \le n \le \mathcal{O}(\sqrt{m})$ as the number of days in a year m goes to infinity. This is the regime where the collision probability transitions from unlikely to likely. We now give a rigorous error analysis of the approximation in Equation (2.19).

Theorem 2.31. For n aliens living on a planet with m days in a year, if $n = \mathcal{O}(\sqrt{m})$ as $m \to \infty$, the collision probability is

$$\Pr[A] = 1 - e^{-N/m} + \mathcal{O}(\tfrac{1}{\sqrt{m}}), \tag{2.49}$$

where $N := n(n-1)/2$.

Proof. Following a similar argument as in Section 2.1.3, we can derive the exact formula for $\Pr[A]$:

$$\Pr[A] = 1 - \Pr[\bar{A}] = 1 - \prod_{k=0}^{n-1} \left(1 - \frac{k}{m}\right). \tag{2.50}$$

Taking the logarithm of $\Pr[\bar{A}]$, we get

$$\ln \Pr[\bar{A}] = \sum_{k=0}^{n-1} \ln \left(1 - \frac{k}{m}\right). \tag{2.51}$$

By Taylor expansion of $\ln(1-x)$ at 0, we have

$$\ln(1+x) = x + \mathcal{O}(x^2), \qquad \forall x \in (-\tfrac{1}{2}, \tfrac{1}{2}). \tag{2.52}$$

Under $n = \mathcal{O}(\sqrt{m})$, for all sufficiently large m we have $0 \le \frac{k}{m} \le \frac{1}{2}$ uniformly for $0 \le k \le n-1$. Hence, for all k in this range,

$$\ln \left(1 - \frac{k}{m}\right) = -\frac{k}{m} + \mathcal{O}\left(\frac{k^2}{m^2}\right). \tag{2.53}$$

Summing over k,

$$\ln \Pr[\bar{A}] = \sum_{k=0}^{n-1} \left(-\frac{k}{m} + \mathcal{O}\left(\frac{k^2}{m^2}\right)\right) = -\frac{1}{m} \sum_{k=0}^{n-1} k + \mathcal{O}\left(\frac{1}{m^2} \sum_{k=0}^{n-1} k^2\right) \tag{2.54}$$

$$= -\frac{N}{m} + \mathcal{O}\left(\frac{1}{\sqrt{m}}\right), \tag{2.55}$$

where we used $\sum_{k=0}^{n-1} k = N$ and $\sum_{k=0}^{n-1} k^2 = \mathcal{O}(n^3) = \mathcal{O}(m^{3/2})$.

Taking the exponential of both sides, we get

$$\Pr[\bar{A}] = \exp\left(-\tfrac{N}{m} + \mathcal{O}(\tfrac{1}{\sqrt{m}})\right) = \exp(-\tfrac{N}{m}) \cdot \exp(\mathcal{O}(\tfrac{1}{\sqrt{m}})) \tag{2.56}$$

$$= e^{-N/m} \left(1 + \mathcal{O}(\tfrac{1}{\sqrt{m}})\right) \tag{2.57}$$

$$= e^{-N/m} + \mathcal{O}(\tfrac{1}{\sqrt{m}}), \tag{2.58}$$

where we used the Taylor expansion $\exp(x) = 1 + x + \mathcal{O}(x^2)$ and $e^{-x} = \mathcal{O}(1)$ when $x \ge 0$. Substituting the result for $\Pr[\bar{A}]$ into the expression for $\Pr[A]$ completes the proof. □

This theorem provides a precise quantification of the error of our back-of-the-envelope approximation in Equation (2.19), suggesting that $\Pr[A] \approx 1 - e^{-N/m}$ is accurate up to an $\mathcal{O}(\frac{1}{\sqrt{m}})$ error term when $n = \mathcal{O}(\sqrt{m})$. Of course, in general,

the number of aliens n can take other values such as $n = \Theta(m^{0.9})$. We leave the analysis for the full range of n as an exercise.

2.2.6 Summation of Big-$\mathcal{O}$ Terms

The previous proof hides one technical point that is easy to misuse: how to sum a sequence of error terms that can be expressed in big-$\mathcal{O}$ notation. In particular, in Equation (2.54), we wrote

$$\sum_{k=0}^{n-1} \ln\left(1 - \frac{k}{m}\right) = \sum_{k=0}^{n-1} \left(-\frac{k}{m} + \mathcal{O}\left(\frac{k^2}{m^2}\right)\right) = -\frac{1}{m}\sum_{k=0}^{n-1} k + \mathcal{O}\left(\frac{1}{m^2}\sum_{k=0}^{n-1} k^2\right) \quad (2.59)$$

Intuitively, this makes sense and is very convenient. However, in what sense is the entire manipulation correct in mathematical rigorous sense?

We have to think carefully about this question because the summation of big-$\mathcal{O}$ terms is subtle and can be wrong if not done correctly. Below, we first discuss separately the univariate case and the multivariate case.

▶ Univariate Case

We start with a simple example that serves as a cautionary tale. Consider the following sum:

$$F(n) = \underbrace{10 + 10 + \cdots + 10}_{n \text{ times}} \quad (2.60)$$

Each term satisfies $10 = \mathcal{O}(1)$. Since the sum of two $\mathcal{O}(1)$ terms is still $\mathcal{O}(1)$, we may naïvely apply this rule $n - 1$ times to merge all the terms into a single $\mathcal{O}(1)$ term. However, this is clearly wrong! In fact, $F(n) = n \cdot 10 = \Theta(n)$.

What went wrong? All the rules we have seen in Lemma 2.9 are valid only when applied a constant number of times. If they were applied n times, the hidden constant factors that appear in the proof of these rules will accumulate and become unbounded. This is similiar to the phenomenon in calculus that $\frac{1}{n} \to 0$ and $\frac{2}{n} \to 0$ as $n \to \infty$, but $\sum_{i=1}^{n} \frac{1}{n} = 1$.

To handle such situations properly, we need to study sums of the form:

$$\sum_{k=1}^{n} f(k) \quad \text{where} \quad f(k) = \mathcal{O}(g(k)). \quad (2.61)$$

Or more conventionally, we may write it as

$$\sum_{k=1}^{n} \mathcal{O}(g(k)). \quad (2.62)$$

We are interested in the asymptotic behavior of $\sum_{k=1}^{n} f(k)$ as $n \to \infty$.

Luckily, the following lemma provides a principled way to handle such sums.

Lemma 2.32. Let $f, g : \mathbb{N} \to \mathbb{R}$ with $g(n) > 0$ for all $n \geq 1$. If

$$f(k) = \mathcal{O}(g(k)) \quad \text{as } k \to \infty, \tag{2.63}$$

then

$$\sum_{k=1}^{n} f(k) = \mathcal{O}\left(\sum_{k=1}^{n} g(k)\right) \quad \text{as } n \to \infty. \tag{2.64}$$

Proof. Let $F(n) := \sum_{k=1}^{n} f(k)$ and $G(n) := \sum_{k=1}^{n} g(k)$. It suffices to show that $F(n) = \mathcal{O}(G(n))$. By definition of big-$\mathcal{O}$ notation, there exist constants $C > 0$ and $N \geq 1$ such that $|f(n)| \leq Cg(n)$ for all $n \geq N$. When $n \geq N$, split the summation into two parts:

$$F(n) = \underbrace{\sum_{k=1}^{N-1} f(k)}_{=: I_1} + \underbrace{\sum_{k=N}^{n} f(k)}_{=: I_2(n)}. \tag{2.65}$$

The first part I_1 is independent of n, so $|I_1| = \mathcal{O}(1) = \mathcal{O}(g(1)) = \mathcal{O}(G(n))$. For the second part, we bound the absolute value of each term $f(k)$ by $Cg(k)$:

$$|I_2(n)| \leq \sum_{k=N}^{n} |f(k)| \leq \sum_{k=N}^{n} Cg(k) \leq C \sum_{k=1}^{n} g(k) = C \cdot G(n) = \mathcal{O}(G(n)). \tag{2.66}$$

Adding the two bounds proves $F(n) = \mathcal{O}(G(n))$. □

Example 2.33. For $f(n) = 10$, we have $f(n) = \mathcal{O}(1)$. Therefore,

$$\sum_{k=1}^{n} f(k) = \mathcal{O}\left(\sum_{k=1}^{n} 1\right) = \mathcal{O}(n). \tag{2.67}$$

Example 2.34. For $f(n) = (1+\frac{1}{n}) \cdot 2^n$, we have $f(n) = \mathcal{O}(2^n)$. Therefore,

$$\sum_{k=1}^{n} f(k) = \mathcal{O}\left(\sum_{k=1}^{n} 2^k\right) = \mathcal{O}(2^n). \tag{2.68}$$

▶ Multivariate Case

We now return to the birthday paradox problem, where each summand depends on two variables:

$$\sum_{k=0}^{n-1} f(m,k) \quad \text{where} \quad f(m,k) = \ln\left(1 - \frac{k}{m}\right) + \frac{k}{m} = \mathcal{O}\left(\frac{k^2}{m^2}\right). \tag{2.69}$$

A natural continuation of the univariate lemma is to ask whether a termwise big-$\mathcal{O}$ bound still passes through summation. Specifically, given two functions

$f, g : \mathbb{N}^2 \to \mathbb{R}$, can we claim the following?

$$f(m,k) = \mathcal{O}(g(m,k)) \quad \Longrightarrow \quad \sum_{k=1}^{n} f(m,k) = \mathcal{O}\left(\sum_{k=1}^{n} g(m,k)\right). \tag{2.70}$$

In general, the answer is:



The intuition is that multivariate big-$\mathcal{O}$ controls behavior only in an eventual region of the joint limit $m, k \to \infty$. It may ignore small-k slices, while those few small-k terms can still carry large dependence on m and dominate the sum. We leave the construction of a counterexample as an exercise. For a more systematic discussion, we refer readers to Howell's note: *On Asymptotic Notation with Multiple Variables* [How08].

So does Equation (2.54) in our proof really work? In fact,



The key insight is that we are not writing $\ln(1-\frac{k}{m}) = -\frac{k}{m} + \mathcal{O}(\frac{k^2}{m^2})$ in the joint limit $m, k \to \infty$, but rather for the regime where $m \to \infty$ but k is constrained in the range $0 \le k \le n-1 = \mathcal{O}(\sqrt{m})$. In this case, the function inside the big-$\mathcal{O}$ notation provides a uniform bound for all k that can appear in the sum.

In general, it is easy to show the following lemma:

Lemma 2.35. Let $f, g : \mathbb{N}^2 \to \mathbb{R}$ with $g(m,n) > 0$ for all $m, n \ge 1$. If for some function $U(m) \ge 1$, we have

$$f(m,k) = \mathcal{O}(g(m,k)), \quad \text{for } 1 \le k \le U(m), \text{ as } m \to \infty, \tag{2.71}$$

Then,

$$\sum_{k=1}^{n} f(m,k) = \mathcal{O}\left(\sum_{k=1}^{n} g(m,k)\right), \quad \text{for } 1 \le n \le U(m), \text{ as } m \to \infty. \tag{2.72}$$

Proof. Let $F(m,n) := \sum_{k=1}^{n} f(m,k)$ and $G(m,n) := \sum_{k=1}^{n} g(m,k)$. It suffices to show that $F(m,n) = \mathcal{O}(G(m,n))$. By definition of big-$\mathcal{O}$ notation, there exist constants $C > 0$ and $M \ge 1$ such that $|f(m,k)| \le C g(m,k)$ for all $m \ge M$ and $1 \le k \le U(m)$. Therefore, for all $m \ge M$ and $1 \le n \le U(m)$,

$$|F(m,n)| \le \sum_{k=1}^{n} |f(m,k)| \le \sum_{k=1}^{n} C g(m,k) = C \cdot G(m,n). \tag{2.73}$$

Hence, $F(m,n) = \mathcal{O}(G(m,n))$ for $1 \le n \le U(m)$ as $m \to \infty$. □

Remark 2.36. The lemma can be extended in several straightforward ways. For example, the index set for k may be shifted from $1 \le k \le U(m)$ to $0 \le k \le U(m) - 1$ and the range for summation can be changed accordingly from $1 \le k \le n$ to $0 \le k \le n - 1$. More generally, the index set of k can be replaced by any domain that may or may not depend on m. The result also extends to settings where multiple variables tend to infinity simultaneously.

Example 2.37. In Equation (2.54), we have assumed $n = \mathcal{O}(\sqrt{m})$, i.e., $n \le U(m)$ for some function $U(m) = \mathcal{O}(\sqrt{m})$ as $m \to \infty$. Let $f(m,k) = \ln(1 - \frac{k}{m}) + \frac{k}{m}$. Then $f(m,k) = \mathcal{O}\left(\frac{k^2}{m^2}\right)$ for $0 \le k \le U(m) - 1$ as $m \to \infty$. Therefore,

$$\sum_{k=0}^{n-1} f(m,k) = \mathcal{O}\left(\sum_{k=0}^{n-1} \frac{k^2}{m^2}\right). \tag{2.74}$$

In summary, we have elaborated on the use of big-$\mathcal{O}$ notation in this section. From now on, we will use it liberally throughout the lectures. However, beyond this course, like when writing a paper whose theoretical analysis relies heavily on big-$\mathcal{O}$ notation, you should:

Always state the definition of big-$\mathcal{O}$ notation before using it!

In the computer science community, several different definitions of big-$\mathcal{O}$ notation coexist. If you do not specify which one you are using, your AI autocompletion may apply a different convention, your readers may be confused, and your reviewers may reject your paper. But if you use big-$\mathcal{O}$ notation only in its standard univariate form, such as the one in Definition 2.6, you do not need to restate the definition explicitly.

2.3 Uniform Probability Means Counting

Now we return to our introduction to probability theory. An important special case is when the probability space is *uniform*, i.e., $p(\omega) = \frac{1}{|\Omega|}$ for every $\omega \in \Omega$. In this case, computing the probability of an event A reduces to counting the number of elements in A:

$$\Pr[A] = \frac{|A|}{|\Omega|}. \tag{2.75}$$

In many examples, such as the birthday paradox example in Section 2.1.3, computing probability is essentially a counting problem.

2.3.1 Addition and Multiplication Principles

We begin with two elementary principles of counting, which you may have already encountered in middle school or even earlier.

Theorem 2.38 (Addition Principle)**.** If a set A is the union of m disjoint sets $S_1, \ldots, S_m$, i.e., $A = S_1 \sqcup \cdots \sqcup S_m$, then

$$|A| = |S_1 \sqcup \cdots \sqcup S_m| = \sum_{i=1}^{m} |S_i|, \tag{2.76}$$

where $\sqcup$ denotes the union of two disjoint sets.

Theorem 2.39 (Multiplication Principle)**.** If a set A is the Cartesian product of m sets $S_1, \ldots, S_m$, i.e., $A = S_1 \times S_2 \times \cdots \times S_m$, then

$$|A| = |S_1 \times S_2 \times \cdots \times S_m| = \prod_{i=1}^{m} |S_i|, \tag{2.77}$$

where $\times$ denotes the Cartesian product of two sets, i.e., $A \times B = \{(a, b) : a \in A, b \in B\}$.

The addition principle states that the size of the union of disjoint sets is the sum of the sizes of the sets. The multiplication principle states that the size of the Cartesian product of sets is the product of the sizes of the sets. In other words, disjoint cases add, and sequential choices multiply.

These simple rules are surprisingly useful in solving many probability problems. For example, in the birthday paradox, we implicitly used the multiplication principle to compute the size of $\bar{A}$. Below we make this usage explicit.

Example 2.40. In the birthday paradox, the event $\bar{A}$ is the set of all assignments of birthdays to n students such that no two students share the same birthday.

$$\bar{A} = \{(b_1, \ldots, b_n) \in [365]^n : \ b_i \neq b_j \text{ for all } i \neq j\}. \tag{2.78}$$

For each assignment $\boldsymbol{b} = (b_1, \ldots, b_n) \in \bar{A}$ and each $i \in [n]$, define $s_i(\boldsymbol{b})$ to be the rank of b_i in the set $[365] \setminus \{b_1, \ldots, b_{i-1}\}$. That is, $s_i(\boldsymbol{b}) = k$ if b_i is the k-th smallest number in the set $[365] \setminus \{b_1, \ldots, b_{i-1}\}$. Then the following map is bijective:

$$\phi : \bar{A} \to [365] \times [364] \times \cdots \times [365 - n + 1], \quad \boldsymbol{b} \mapsto (s_1(\boldsymbol{b}), \ldots, s_n(\boldsymbol{b})). \tag{2.79}$$

Therefore,

$$|\bar{A}| = |[365] \times [364] \times \cdots \times [365 - n + 1]| \tag{2.80}$$

$$= 365 \times 364 \times \cdots \times (365 - n + 1), \tag{2.81}$$

where the last equality follows from the multiplication principle.

In the rest of this section, we will see more involved examples where both addition and multiplication principles are used together.

2.3.2 Example: The Secretary Problem

▶ Background

Suppose you are moving to a new city for an internship or a job, and you need to find an apartment to live in. However, good apartments are scarce and highly competitive. You will inspect exactly n apartments, one by one following a uniformly random order. After each visit, you will assign a quality value to the apartment, and then you must make an immediate decision: sign now, or reject and continue. Once rejected, an apartment cannot be recalled.

This creates a fundamental tradeoff: if you commit too early, you may miss a much better option later; if you wait too long, the best option may already be gone. Your goal is to maximize the probability of selecting the best apartment, namely the one with the highest quality value. So the central question is: *Can we design a strategy that gives a high probability of selecting the best apartment?*

This is the classical *secretary problem*. The name originates from an equivalent formulation: a manager interviews candidates sequentially for a secretary position and must decide immediately after each interview, without the ability to return to earlier candidates. The objective is to maximize the probability of hiring the best candidate overall.

At first glance, it seems impossible to design a strategy with a non-trivial success probability when n is large. If we choose an option completely at random, the success probability is only $1/n$. But how could we do any better? Perhaps surprisingly, we will show below that it is indeed possible to succeed with a probability of over 35%, even when n is large.

Remark 2.41. In computer science, the secretary problem is closely related to *online algorithms*, where the input is revealed sequentially and decisions must be made immediately and irrevocably. From this viewpoint, the secretary problem can be seen as one of the simplest online decision problems: selecting the maximum element from an online sequence of numbers. In Chapter 7, we will see more examples of online algorithms in the context of online learning and online optimization.

▶ Problem Formulation

We now formalize the problem.

- There are exactly n options, each associated with a distinct value $v_i \in \mathbb{R}$. The options are presented in a random order σ, where σ is a permutation of $[n]$ sampled uniformly at random.
- At step t, you observe the value of the current option v_{σ_t} and only know the values of the first t observed options $v_{\sigma_1}, \ldots, v_{\sigma_t}$. You must make an immediate decision: accept or reject the current option.

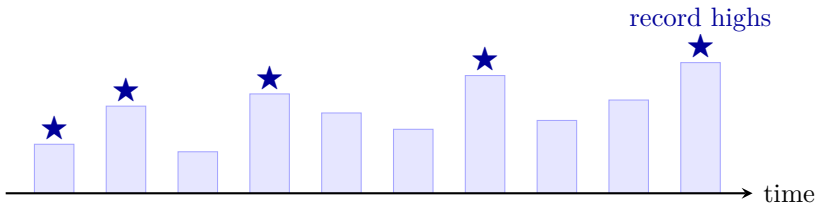


Figure 2.2: Record highs in a sequence: only positions that set a new prefix maximum are candidates worth considering.

- If an option is rejected, it cannot be recalled. If an option is accepted, the process terminates immediately and no subsequent options can be inspected or accepted.

The final goal is to maximize the probability of accepting the global best option, namely the option with the highest value $v_{\max} := \max_{i\in[n]}\{v_i\}$.

Remark 2.42. We do not know the set of values $\{v_i\}_{i=1}^n$ in advance. That is, an adversary can choose the values of the options arbitrarily before randomly shuffling them. In this sense, the only effective information we can leverage at step t is the relative order of the observed values $v_{\sigma_1}, \ldots, v_{\sigma_t}$. The problem would be trivial if we knew the values in advance: just wait until the current value matches the maximum value and accept it.

Remark 2.43. If the values are sampled independently from known distributions (the sampled values are still not known in advance), then the problem falls into the framework of *prophet inequalities*, a topic at the intersection of online algorithms and algorithmic mechanism design. For interested readers, we refer to Hu Fu's notes [Fu21] for a concise introduction.

▶ Threshold Strategy: Intuition and Description

At time t, if the current value is *not* the largest among the first t observations, then it can never be the global best, so we should reject it immediately. Therefore, only the options that set a new prefix maximum are candidates worth considering. We call such options *record highs*.

So the real design question becomes: *when we see a record high, should we stop now or keep waiting?*

First, we should not rely on the absolute values of the observed options. The values $v_1, \ldots, v_n$ may be arbitrary, and without additional assumptions, their magnitudes alone reveal nothing about their ranks among all n options.

Second, we also should not rely on the relative order of the observed options because of a symmetry property induced by the uniformly random arrival order. After observing the first t options, we may imagine that there are $t+1$ possible gaps (before, between, and after the observed options). Each unseen option is equally likely to be inserted into any of these gaps. Consequently, when we

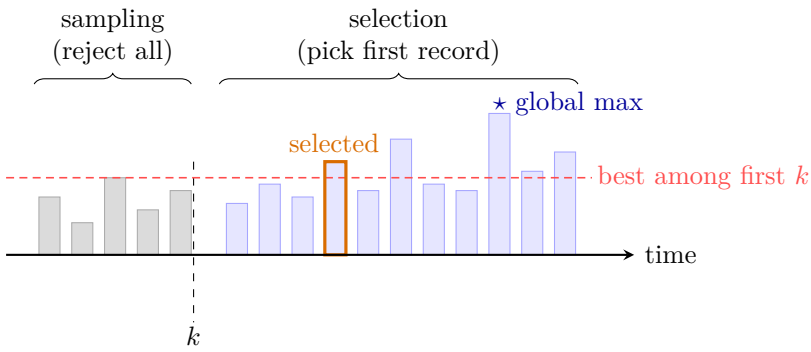


Figure 2.3: An illustration of the threshold strategy: reject the first k options, then select the first option that beats the maximum among the first k options.

observe a record high at time t, the probability that one of the remaining $n - t$ options has a value larger than the current prefix maximum is unaffected by the relative ordering of the first t observed options.

Therefore, for this decision problem, the only critical factor is the prefix length t (equivalently, how long we have observed). Stopping too early is risky because the prefix is short and provides limited information, while waiting too long increases the risk of having already missed the best option. This naturally suggests a simple one-parameter rule: wait for an initial period, then accept the first record high after that.

This approach is equivalent to the following strategy known as a *threshold strategy*, also called the *k-strategy*.

Definition 2.44 (Threshold Strategy)**.** For a fixed integer k satisfying $1 \leq k \leq n - 1$, the *k-strategy* is an online decision strategy defined as follows:

- Reject the first k options and record the maximum value observed;
- Starting from step $k+1$, accept the first option whose value exceeds the recorded maximum.

A *threshold strategy* is a k-strategy for some choice of k.

The parameter k has to be chosen carefully to balance the tradeoff between the risk of stopping too early and the risk of waiting too long. Below, we will analyze the performance of this strategy for all possible values of k.

▶ Threshold Strategy: Analysis

We now analyze the success probability of the k-strategy by counting the number of permutations σ under which the strategy succeeds.

First, we set up the probability space. Let $\Omega := \mathrm{Perm}(n)$ be the set of all permutations of $[n]$. The probability mass function is uniform, i.e., $p(\sigma) := \frac{1}{n!}$ for all $\sigma \in \Omega$.

Without loss of generality, we assume that $v_1 < v_2 < \cdots < v_n$, so comparing option values is equivalent to comparing their indices. In particular, $v_{\sigma_i} < v_{\sigma_j}$ if and only if $\sigma_i < \sigma_j$, and the best option is v_n.

Let the event A be the set of permutations σ under which the k-strategy succeeds. More precisely, these are the permutations σ such that the following condition holds if the best option is at position $j = \sigma^{-1}(n)$:

- $j > k$, i.e., the best option is not among the first k positions.
- $\max\{\sigma_{k+1}, \ldots, \sigma_{j-1}\} < \max\{\sigma_1, \ldots, \sigma_k\}$, i.e., the options between the k-th position and the best option do not exceed the maximum value among the first k options.

Now we partition the event A into $n-k$ disjoint sets, $A_{k+1}, \ldots, A_n$, where A_j is the set of permutations σ such that $\sigma^{-1}(n) = j$ and the condition for success holds. By the addition principle, we have

$$|A| = \sum_{j=k+1}^{n} |A_j|. \tag{2.82}$$

To count the number of permutations in each A_j, we apply the multiplication principle as follows:

- Choose $j-1$ values that appear before position j: $\binom{n-1}{j-1}$ choices.
- Among positions $1, \ldots, k$, choose where the largest of those $j-1$ values appears: k choices.
- Arrange the remaining $j-2$ selected values in the other $j-2$ slots before position j: $(j-2)!$ choices.
- Fix the best option at position j: 1 choice.
- Arrange the remaining $n-j$ values after position j: $(n-j)!$ choices.

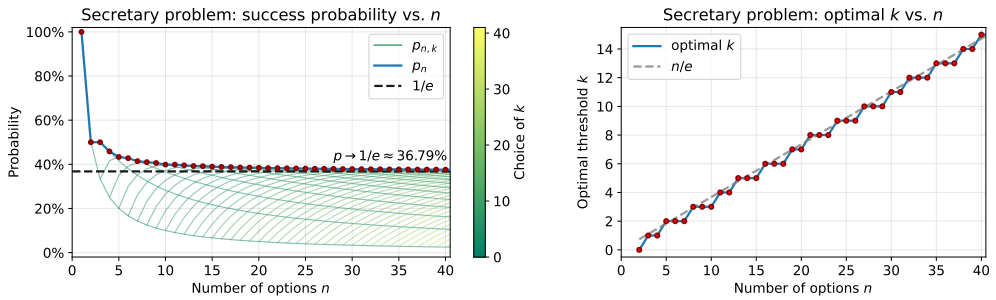
Hence,

$$|A_j| = \binom{n-1}{j-1} \cdot k \cdot (j-2)! \cdot 1 \cdot (n-j)! \tag{2.83}$$

$$= \frac{(n-1)!}{(j-1)!\,(n-j)!} \cdot k \cdot (j-2)! \cdot (n-j)! = n! \cdot \frac{k}{n} \cdot \frac{1}{j-1}. \tag{2.84}$$

Substituting into Equation (2.82), we obtain the following expression for the success probability of the k-strategy with n options:

$$p_{n,k} := \Pr[A] = \frac{|A|}{|\Omega|} = \frac{k}{n} \sum_{j=k+1}^{n} \frac{1}{j-1} = \frac{k}{n} \sum_{i=k}^{n-1} \frac{1}{i}. \tag{2.85}$$



(a) Success probability p_n (b) Optimal threshold k^*

Figure 2.4: Secretary problem: (a) the optimal success probability p_n of the threshold strategy and (b) the optimal threshold k as functions of n.

Consequently, by choosing k carefully, the optimal success probability that the threshold strategy can achieve for n options is

$$p_n := \max_{1 \le k < n} \left\{ \frac{k}{n} \sum_{i=k}^{n-1} \frac{1}{i} \right\}. \tag{2.86}$$

▶ Optimal Threshold Strategy

Although we have derived the success probability Equation (2.86) analytically, it is still somewhat difficult to understand intuitively. How large is this success probability really? Does it vanish as n goes to infinity?

To answer this question, we first plot the numerical values of p_n and k for $1 \le n \le 40$ in Figure 2.4. Two observations are immediately apparent:

- As n increases, the optimal success probability approaches a constant;
- The optimal threshold k increases almost linearly with n.

Interestingly, the natural constant $e \approx 2.718$ (Euler's number) naturally appears. The following theorem formalizes these observations:

Theorem 2.45. For $n \ge 2$, the following two statements hold:

1. the optimal success probability satisfies $p_n = 1/e + \mathcal{O}(\frac{1}{n})$;
2. the optimal threshold satisfies $k^* = n/e + \mathcal{O}(1)$.

So for large n, the optimal threshold strategy is:

Skip the first $\frac{1}{e} \approx 37\%$ options, then accept the first record high.

Where does the natural constant e come from? It turns out that e comes from the harmonic number approximation of the sum in Equation (2.85).

Definition 2.46. The n-th harmonic number H_n is defined as

$$H_n := \sum_{k=1}^{n} \frac{1}{k}. \tag{2.87}$$

Proposition 2.47 (Asymptotic expansion of Harmonic numbers)**.** There exists a constant $\gamma \approx 0.5772$, called the *Euler–Mascheroni constant*, such that for all $n \geq 1$,

$$H_n = \ln n + \gamma + \mathcal{O}\left(\frac{1}{n}\right).$$

We omit the proof of this proposition here. For interested readers, a key idea is to approximate the sum of $\frac{1}{k}$ by the integral of $\frac{1}{x}$ and then bound the error term using Euler's summation formula. A detailed derivation can be found in Section 9.6 of Concrete Mathematics [GKP94].

Now we illustrate the proof of Statement 1 of Theorem 2.45. We leave the proof of Statement 2 to the reader.

Proof of Statement 1 of Theorem 2.45. First, we rewrite the success probability Equation (2.85) with harmonic numbers:

$$p_{n,k} = \frac{k}{n}(H_{n-1} - H_{k-1}). \tag{2.88}$$

By Proposition 2.47, for $n \geq 2$ and $1 \leq k \leq n-1$, we have

$$H_{n-1} = H_n - \frac{1}{n} = \ln n + \gamma + \mathcal{O}\left(\frac{1}{n}\right), \tag{2.89}$$

$$H_{k-1} = H_k - \frac{1}{k} = \ln k + \gamma + \mathcal{O}\left(\frac{1}{k}\right). \tag{2.90}$$

Substituting these into Equation (2.88), we get

$$p_{n,k} = \frac{k}{n}\left(\left(\ln n + \gamma + \mathcal{O}\left(\frac{1}{n}\right)\right) - \left(\ln k + \gamma + \mathcal{O}\left(\frac{1}{k}\right)\right)\right) \tag{2.91}$$

$$= \frac{k}{n}\left(\ln \frac{n}{k} + \mathcal{O}\left(\frac{1}{k}\right)\right) \tag{2.92}$$

$$= \frac{k}{n} \ln \frac{n}{k} + \mathcal{O}\left(\frac{1}{n}\right). \tag{2.93}$$

Let $f(x) := x \ln \frac{1}{x}$ for $x \in (0,1)$ (see Figure 2.5 for a visualization). Then $f'(x) = \ln \frac{1}{x} - 1$. So $f(x)$ is maximized at $x = e^{-1}$, and it is strictly increasing on $(0, e^{-1})$ and strictly decreasing on $(e^{-1}, 1)$. This implies an upper bound on the optimal success probability:

$$p_n = \max_{1 \leq k \leq n-1} p_{n,k} = \max_{1 \leq k \leq n-1} \left\{ \frac{k}{n} \ln \frac{n}{k} + \mathcal{O}\left(\frac{1}{n}\right) \right\} \leq f(e^{-1}) + \mathcal{O}\left(\frac{1}{n}\right). \tag{2.94}$$

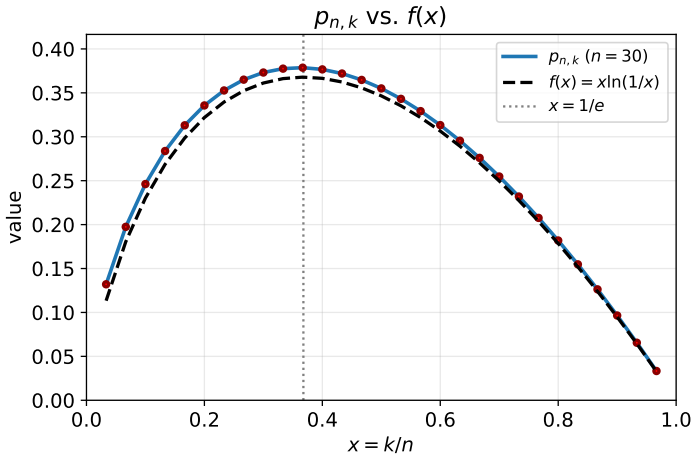


Figure 2.5: The exact curve $p_{n,k}$ as a function of $x = k/n$ when $n = 30$, together with the approximation $f(x) = x\ln(1/x)$ used in the proof.

To show that this upper bound is tight, we take $k = \lceil n/e \rceil$ and obtain

$$p_n \geq f\left(\frac{k}{n}\right) + \mathcal{O}\left(\frac{1}{n}\right) = f\left(e^{-1} + \mathcal{O}\left(\frac{1}{n}\right)\right) + \mathcal{O}\left(\frac{1}{n}\right) \tag{2.95}$$

$$= e^{-1} + \mathcal{O}\left(\frac{1}{n^2}\right) + \mathcal{O}\left(\frac{1}{n}\right) \tag{2.96}$$

$$= e^{-1} + \mathcal{O}\left(\frac{1}{n}\right), \tag{2.97}$$

where the second equality follows from a Taylor expansion of $f(x)$ at $x = e^{-1}$. More specifically, as $\delta \to 0$,

$$f(e^{-1} + \delta) = f(e^{-1}) + f'(e^{-1})\delta + \mathcal{O}(\delta^2) = e^{-1} + \mathcal{O}(\delta^2). \tag{2.98}$$

Combining the lower and upper bounds, we get $p_n = 1/e + \mathcal{O}(\frac{1}{n})$. □

Finally, a natural question is whether the threshold strategy is optimal among *all* possible strategies for the secretary problem. The answer is yes. This can be proved rigorously by formulating the problem as a dynamic programming problem and solving the corresponding recursive equations [Lin61]. See also [Fre83, Fer89] for a historical overview of the secretary problem.

2.3.3 Example: The Locker Puzzle

▶ Problem Statement

Suppose n students attend U.S. visa interviews together, and all students have distinct names. Each student has one bag, and the student's own name tag is attached to that bag. They are not allowed to bring personal belongings into the embassy, so everyone stores one bag in one of n lockers. After they come

out, everyone forgets which locker contains their own bag. So we have a random one-to-one assignment between name tags and lockers.

Students search one by one (not simultaneously): when one student finishes, all opened lockers are closed again and the locker state is restored before the next student starts. Each student may open at most $n/2$ lockers (assume n is an even number), and no communication is allowed during the search. The goal is to design a strategy that maximizes the probability that *all* students succeed in finding their own bags.

At first sight, this success probability may seem tiny, because everyone must succeed simultaneously. Indeed, if every student just opens $n/2$ lockers uniformly at random, then the success probability is $(1/2)^n$, which is exponentially small in n. But surprisingly, there is a strategy that maintains a constant success probability even for large n.

Remark 2.48. From the perspective of data structures, each locker is a memory cell, and bags are the items stored in these cells. Each student corresponds to a query for one item (their own name tag), and opening a locker is a probe that reveals the item stored in a cell. In this view, the puzzle essentially asks the following question: given read-only access to a random permutation of n distinct items stored in memory, and a target item, can we locate the item using at most $n/2$ probes, regardless of which item is queried? Historically, this question arises in the study of succinct data structures [GM03], where the goal is to understand how efficiently can we answer queries when the data structure uses very little extra space beyond the information-theoretic minimum.

▶ Warmup: A Static Strategy

Let us first ask: how can we improve upon random guessing? We stay within *static strategies*, where each student commits in advance to a fixed set of $n/2$ lockers and never changes this plan based on observations. This is a natural starting point since it is unclear how observing other students' name tags can help one's own search.

We start with $n = 2$, where each student can open only one locker. If both students guess randomly, the probability that both succeed is $(1/2)\cdot(1/2) = 1/4$. But they should coordinate: student 1 always opens locker 1, and student 2 always opens locker 2. Then either both succeed or both fail, each with equal probability. As a result, the success probability is $1/2$, which is better than random guessing.

In general, we can split students into two equal groups. Students in the first group only open lockers $\{1, \ldots, n/2\}$, and students in the second group only open lockers $\{n/2 + 1, \ldots, n\}$. Under this rule, everyone succeeds if and only if all first-group name tags lie in the first half (equivalently, all second-group tags

lie in the second half). Hence

$$\Pr[\text{success}] = \frac{(n/2)!(n/2)!}{n!} = \frac{1}{\binom{n}{n/2}}. \tag{2.99}$$

This success probability is still tiny, but it is already much larger than that of random guessing, which is $\frac{1}{2^n}$.

Why is this better than random guessing? Under random guessing, each student succeeds with probability $1/2$ independently, so the overall success probability is 2^{-n}. In contrast, under the above strategy, the success of different students are strongly correlated. Once all name tags belonging to the first group are located within the first half of lockers, the second group automatically succeeds as well. Therefore, when computing the overall success probability, it suffices to enforce success only for the first $n/2$ students, which leads to a much higher probability than 2^{-2}.

Can we do even better? For instance, could we partition into more groups, or design more sophisticated fixed subsets of lockers for each student to open?

The answer is *no*: this two-group strategy is already optimal among all static strategies. To see why, represent a static strategy by a bipartite graph G: left vertices are students, right vertices are lockers, and an edge (s, ℓ) means student s is allowed to open locker ℓ. Each student has degree $n/2$. For a random locker permutation, global success occurs exactly when the induced placement corresponds to a perfect matching in G. Hence

$$\Pr[\text{success}] = \frac{\#\{\text{perfect matchings of } G\}}{n!}.$$

However, the number of perfect matchings in G by $((n/2)!)^2$ based on the fact that the degree of each student is $n/2$. We refer interested readers to check Bregman's theorem for the proof. Therefore, every static strategy satisfies

$$\Pr[\text{success}] \leq \frac{((n/2)!)^2}{n!} = \frac{1}{\binom{n}{n/2}},$$

and this upper bound is attained by the two-group strategy.

▶ Pointer Chasing Strategy

To achieve a constant success probability, we must move beyond static strategies and introduce *adaptivity*. In an adaptive strategy, the lockers opened by a student are not predetermined, but are selected according to the outcomes of previous checks.

Why can adaptivity help? To see this, we now present an adaptive approach known as the *pointer chasing strategy*.

Before opening any locker, students discuss and agree on a bijection between student names and $[n]$,

$$\phi : \{\text{student names}\} \to [n],$$

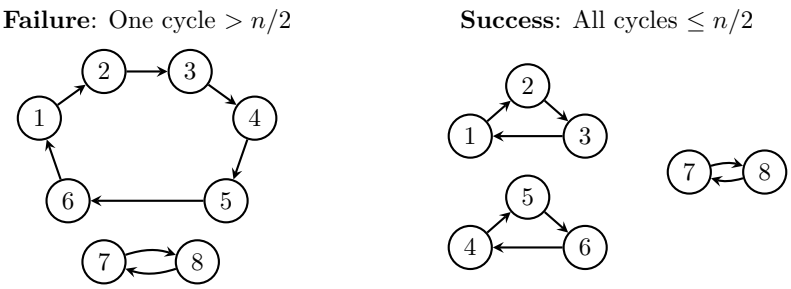


Figure 2.6: An illustration of cycles in two examples of permutations.

which assigns each student a unique ID in $[n]$.

For each student s, let $x \leftarrow \phi(s)$. The search rule is:

- open locker x;
- if the revealed name tag is s, stop (success);
- otherwise, let $x \leftarrow \phi(t)$, where t is the revealed student name, and continue.

If no success occurs, stop after $n/2$ opens (failure).

▶ Success Probability Analysis

To understand why this strategy works, let us rewrite it in permutation language. For each locker $k \in [n]$, if t is the name tag inside locker k, we define

$$\sigma(k) := \phi(t).$$

Since each name tag appears exactly once, σ is a permutation on $[n]$. Moreover, the assignment of name tags to lockers is uniformly random, so the permutation σ is also uniformly random.

If we create a directed graph G with n vertices, where the k-th vertex is pointing to the $\sigma(k)$-th vertex via a directed edge, then each student's search corresponds to a walk in G starting from their own vertex, and the student succeeds if and only if the walk returns to their own vertex.

What does this graph look like? It must be a collection of cycles, since each vertex has exactly one incoming edge and one outgoing edge. Therefore, a student succeeds if and only if the student's ID is in a cycle of length at most $n/2$, and the whole team succeeds if and only if every cycle has length at most $n/2$. See Figure 2.6 for an illustration.

Having all cycles of length at most $n/2$ sounds significantly more likely than 2^{-n}, the probability of random guessing. But how much more likely? The following theorem states the exact success probability of the pointer chasing strategy and its limiting behavior as $n \to \infty$.

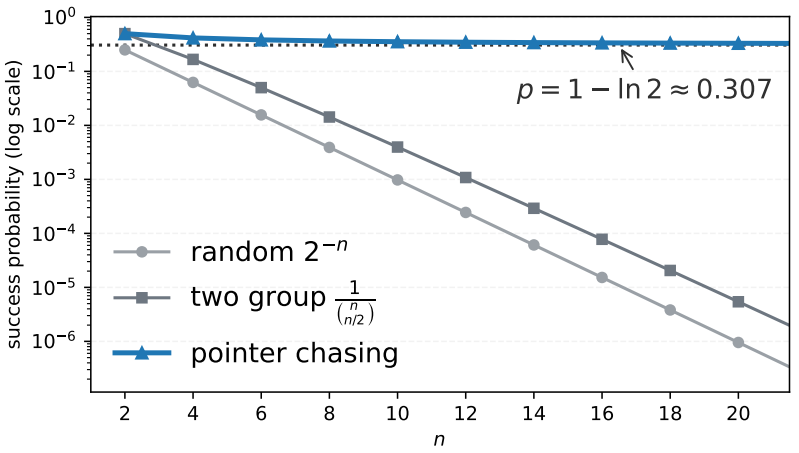


Figure 2.7: Success probability comparison for three locker-puzzle strategies.

Theorem 2.49. For all even $n \geq 2$, the success probability of the pointer chasing strategy is

$$\Pr[\text{success}] = 1 - \sum_{j=n/2+1}^{n} \frac{1}{j} = 1 - \ln 2 + \mathcal{O}\left(\frac{1}{n}\right) \approx 0.307. \tag{2.100}$$

For large n, this constant-level success probability is dramatically larger than random guessing 2^{-n}, as illustrated in Figure 2.7.

Now let us prove the theorem.

Proof. Let $\Omega := \mathrm{Perm}(n)$ be the set of all permutations on $[n]$, with uniform probability $p(\sigma) = 1/n!$ for all $\sigma \in \Omega$. From the previous discussion, we define the success event A as

$$A := \{\sigma \in \Omega : \text{every cycle length of } \sigma \text{ is at most } n/2\}. \tag{2.101}$$

Its complement is the failure event $\bar{A}$:

$$\bar{A} = \{\sigma \in \Omega : \sigma \text{ has a cycle of length } j > n/2\}. \tag{2.102}$$

For each $j > n/2$, let

$$F_j := \{\sigma \in \Omega : \sigma \text{ has a cycle of length } j\}. \tag{2.103}$$

Because $j > n/2$, a permutation cannot contain two different such long cycles, so F_j are disjoint. By the addition principle,

$$|\bar{A}| = \sum_{j=n/2+1}^{n} |F_j|, \qquad \Pr[\bar{A}] = \frac{|\bar{A}|}{|\Omega|}. \tag{2.104}$$

Now count $|F_j|$ by multiplication principle:

- choose the j elements on that cycle: $\binom{n}{j}$ ways;

- arrange them into one cycle: $(j-1)!$ ways;
- permute the remaining $n-j$ elements: $(n-j)!$ ways.

So the size of F_j is

$$|F_j| = \binom{n}{j}(j-1)!(n-j)! = \frac{n!}{j!(n-j)!} \cdot (j-1)!(n-j)! = \frac{n!}{j}. \tag{2.105}$$

Therefore,

$$\Pr[\bar{A}] = \frac{1}{n!} \sum_{j=n/2+1}^{n} \frac{n!}{j} = \sum_{j=n/2+1}^{n} \frac{1}{j} = H_n - H_{n/2}, \tag{2.106}$$

where H_n and $H_{n/2}$ are the n-th and $n/2$-th harmonic numbers, respectively.

By the asymptotic expansion of harmonic numbers in Proposition 2.47,

$$\Pr[\bar{A}] = \left(\ln n + \gamma + \mathcal{O}(\tfrac{1}{n})\right) - \left(\ln(n/2) + \gamma + \mathcal{O}(\tfrac{2}{n})\right) = \ln 2 + \mathcal{O}(\tfrac{1}{n}). \tag{2.107}$$

Hence, the success probability of the pointer chasing strategy is

$$\Pr[A] = 1 - \Pr[\bar{A}] = 1 - \ln 2 + \mathcal{O}(\tfrac{1}{n}) \approx 0.307, \tag{2.108}$$

which proves the theorem. □

Finally, since the pointer chasing strategy may at first seem somewhat arbitrary, a natural question is: *can any strategy beat this $H_n - H_{n/2}$ probability?* The answer is surprisingly *no*: this is already optimal for all n.

A useful proof hint is to compare with a relaxed game: (1) after a student opens a locker, the locker will not be closed again; (2) each student can see all the lockers that have been opened by other students and can only open lockers that have not been opened yet; (3) A student cannot open any new lockers after the studnet's name tag has been found. Still, the students succeed if and only if every student finds their own name tag within $n/2$ opens.

Any strategy feasible for the original puzzle can also be made feasible in this relaxed game with fewer locker opens, so the relaxed game's success probability is an upper bound for the original one. However, no matter what strategy is used, the relaxed game's success probability is exactly the probability of having all cycles of length at most $n/2$ in a uniformly random permutation, which is exactly $H_n - H_{n/2}$. For a full proof, see [WC06].

2.4 Intersection and Union of Events

In the previous section, we repeatedly described success/failure through events (for example, in the locker puzzle), and then computed probabilities by counting those events. To handle more complex problems, we now need basic operations that combine events and tools that bound probabilities of combined events.

2.4.1 Definitions

We start with two basic operations on events. For events $A, B \subseteq \Omega$:

- **Intersection**: The intersection of two events A and B, denoted by $A \cap B$, is the event that both A and B happen.
- **Union**: The union of two events A and B, denoted by $A \cup B$, is the event that at least one of A, B happens.

As events are subsets of the sample space Ω, the operations $\cap$ and $\cup$ are the same as set operations.

We say two events are *mutually exclusive* if their intersection is empty, i.e.,

$$A \cap B = \varnothing. \tag{2.109}$$

For mutually exclusive events, their probabilities are additive:

$$\Pr[A \cup B] = \Pr[A] + \Pr[B]. \tag{2.110}$$

For general events, overlap may be nonempty, so we need a correction term.

Lemma 2.50. For any events $A, B \subseteq \Omega$,

$$\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]. \tag{2.111}$$

Proof. Decompose A as $A = (A \setminus B) \sqcup (A \cap B)$, and also B as $B = (B \setminus A) \sqcup (A \cap B)$. Hence,

$$\Pr[A] = \Pr[A \setminus B] + \Pr[A \cap B], \tag{2.112}$$
$$\Pr[B] = \Pr[B \setminus A] + \Pr[A \cap B]. \tag{2.113}$$

Moreover, $A \cup B = (A \setminus B) \sqcup (B \setminus A) \sqcup (A \cap B)$, so we have

$$\Pr[A \cup B] = \Pr[A \setminus B] + \Pr[B \setminus A] + \Pr[A \cap B] \tag{2.114}$$
$$= (\Pr[A] - \Pr[A \cap B]) + (\Pr[B] - \Pr[A \cap B]) + \Pr[A \cap B] \tag{2.115}$$
$$= \Pr[A] + \Pr[B] - \Pr[A \cap B], \tag{2.116}$$

which proves the lemma. □

Example 2.51. Roll one fair die. Let $A = \{\text{outcome is even}\} = \{2, 4, 6\}$, $B = \{\text{outcome} \geq 4\} = \{4, 5, 6\}$. Then

$$\Pr[A] = \frac{3}{6} = \frac{1}{2}, \quad \Pr[B] = \frac{3}{6} = \frac{1}{2}, \quad \Pr[A \cap B] = \frac{2}{6} = \frac{1}{3}, \tag{2.117}$$

so

$$\Pr[A \cup B] = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} = \frac{2}{3}. \tag{2.118}$$

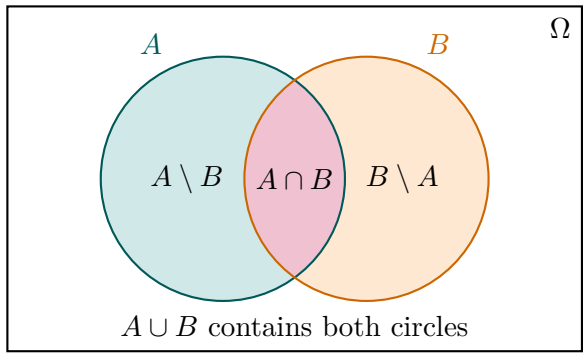


Figure 2.8: Venn diagram for event intersection and union. The overlap is $A \cap B$, and $A \cup B$ is the union of both regions.

2.4.2 Union Bound

When we analyze probability of failure, we often define many bad events and ask for the probability that *at least one* of them happens. The union bound gives a clean upper bound for exactly this situation.

Theorem 2.52 (Union bound)**.** For any two events $A, B \subseteq \Omega$,

$$\Pr[A \cup B] \leq \Pr[A] + \Pr[B]. \tag{2.119}$$

Proof. The result follows from Lemma 2.50 and the fact that $\Pr[A \cap B] \geq 0$. □

It is straightforward to generalize the union bound to multiple events.

Theorem 2.53 (Union bound for multiple events)**.** For any sequence of events $A_1, \ldots, A_m \subseteq \Omega$ with $m \geq 2$,

$$\Pr\left[\bigcup_{i=1}^{m} A_i\right] \leq \sum_{i=1}^{m} \Pr[A_i]. \tag{2.120}$$

Proof. We use induction on m. The base case $m = 2$ is exactly Theorem 2.52. Assuming the claim holds for $m-1$ events, i.e., $\Pr[\bigcup_{i=1}^{m-1} A_i] \leq \sum_{i=1}^{m-1} \Pr[A_i]$, we apply Lemma 2.50 to $A = \bigcup_{i=1}^{m-1} A_i$ and $B = A_m$:

$$\Pr\left[\bigcup_{i=1}^{m} A_i\right] = \Pr\left[\left(\bigcup_{i=1}^{m-1} A_i\right) \cup A_m\right] \tag{2.121}$$

$$\leq \Pr\left[\bigcup_{i=1}^{m-1} A_i\right] + \Pr[A_m] \tag{2.122}$$

$$\leq \sum_{i=1}^{m-1} \Pr[A_i] + \Pr[A_m] = \sum_{i=1}^{m} \Pr[A_i], \tag{2.123}$$

which proves the theorem. □

In algorithm analysis, we often define many "bad events" and bound each one separately, then combine them by the union bound above.

2.4.3 Example: Hashing Collision

▶ Background: Dictionary and Hashing

Given a set $S = \{(k_i, v_i)\}_{i=1}^{n}$ of n key-value pairs, a *dictionary* is a data structure that stores these pairs and supports queries of the form: given a key k, return its corresponding value v (if it exists). We assume that all the given keys are distinct and are taken from a key space K. Each key is represented as an integer in the range $\{0, 1, \ldots, |K| - 1\}$.

For example, a university management system may store student records in a dictionary that maps a student's Chinese resident ID number to their name.

Computer memory does not directly implement a dictionary. Instead, memory is organized as discrete addressable cells, and any data structure ultimately stores its information in such cells. Each cell is indexed by an integer within a bounded range. For example, a laptop with 12GB of memory contains about 12×2^{30} bytes, each of which can store an integer between 0 and 255. Thus, the valid indices for accessing these bytes are $\{0, 1, \ldots, 12 \times 2^{30} - 1\}$.

There are two simple ways to implement a dictionary:

- **Searching**: Store the key-value pairs in an array. To answer a query for a key k, scan the array sequentially until the matching key is found. This requires $\mathcal{O}(n)$ time per query and $\Theta(n)$ space for storing the dictionary. If we preprocess the array by sorting the key-value pairs by key, then we can use binary search to reduce the query time to $\mathcal{O}(\log n)$ per query.
- **Direct Addressing**: Allocate an array of size $|K|$ and store the i-th key-value pair (k_i, v_i) at position k_i. To answer a query for a key k, simply look up the value at index k. This yields $\mathcal{O}(1)$ query time but requires $\Theta(|K|)$ space for storing the dictionary.

While direct addressing is more efficient in terms of query time, it requires $\Theta(|K|)$ space for storing the dictionary, which is impractical for large m. In the example of storing student records, each key is a 18-digit integer, where 17 digits are the numeric part and the last digit is a checksum. This would require about 10^{17} cells. Even if we use one byte per cell to store student names, that is about 10^{17} bytes (roughly 100,000,000 GB), far beyond realistic memory budgets.

A practical solution is to compress the huge key space into a manageable index range by *hashing*. In general, a *hash function* is a function that maps data from a large space to a fixed-size range. Based on hash functions, we can implement a dictionary as follows:

- **Hash Table**: Choose a hash function $h : K \to \{0, 1, \ldots, m - 1\}$ and allocate an array of m *buckets* to store the key-value pairs, where the i-th key-value pair is stored at the $h(k_i)$-th bucket. To answer a query for a key

k, compute $h(k)$ and look up the value at the $h(k)$-th bucket. This yields $\mathcal{O}(1)$ query time while using only $\Theta(m)$ space for storing the dictionary.

The data structure described above is the most basic form of a *hash table*, which provides a better time-space trade-off compared to the binary searching and direct addressing approaches. However, this hash table may not always be able to answer queries correctly, since the hash function h may map multiple keys to the same bucket. When two distinct keys k_i and k_j satisfy $h(k_i) = h(k_j)$, we say that a *hash collision* occurs. Ideally, we would like to choose a hash function that minimizes hashing collisions as much as possible.

▶ Background: Universal Hashing

Here, we specifically consider *linear hash functions*, introduced by Carter and Wegman [CW79].

$$h_{a,b,p,m} : K \to \mathbb{Z}_m, \quad x \mapsto ((ax + b) \bmod p) \bmod m, \tag{2.124}$$

where a, b are integers, p is a large prime number such that $K \subseteq \mathbb{Z}_p$, and m is a positive integer with $m \leq p$, not necessarily prime. Here $\mathbb{Z}_m$ is the integers modulo m, i.e., $\mathbb{Z}_m = \{0, 1, \ldots, m-1\}$. The operation $z \bmod p$ denotes the remainder of z modulo p, which takes the value in $\mathbb{Z}_p$. Since the final result is taken modulo m, the hash function always produces a value in $\mathbb{Z}_m$.

The corresponding *linear hash family* is defined as follows:

$$\mathcal{H}_{p,m} := \{h_{a,b,p,m} : a \in \mathbb{Z}_p \setminus \{0\}, b \in \mathbb{Z}_p\}. \tag{2.125}$$

That is, $\mathcal{H}_{p,m}$ is a family of linear hash functions with fixed prime p and bucket number m. This linear hash family satisfies the following nice property:

Proposition 2.54. For any two distinct integers $k, k' \in K$, if a hash function h is sampled uniformly at random from $\mathcal{H}_{p,m}$, then

$$\Pr[h(k) = h(k')] \leq \frac{1}{m}. \tag{2.126}$$

We leave the proof as an exercise.

Equation (2.126) is a desirable property for hashing, as the probability of a collision is bounded by $\frac{1}{m}$, the probability of getting the same bucket index if the two hash values are instead chosen uniformly at random from $\mathbb{Z}_m$. In general, we define the following notion of *universal hashing*, where linear hash families can be seen as a particular instantiation.

Definition 2.55. A hash family $\mathcal{H}$ is a set of hash functions $h : K \to \{0, 1, \ldots, m-1\}$. We say that $\mathcal{H}$ is *universal* if for any two distinct keys $k, k' \in K$, when h is sampled uniformly at random from $\mathcal{H}$,

$$\Pr[h(k) = h(k')] \leq \frac{1}{m}, \tag{2.127}$$

Remark 2.56. When Equation (2.126) does not strictly hold but instead the inequality $\Pr[h(k) = h(k')] \leq \epsilon$ holds for some $\epsilon > 0$, we say that $\mathcal{H}$ is *ϵ-almost-universal*. Many hash families are almost universal. For example, if we fix $b = 0$ in $\mathcal{H}_{p,m}$, then $\mathcal{H}_{p,m}$ is $\frac{2}{m}$-almost-universal.

▶ Problem Statement

Given a universal hash family $\mathcal{H}$, suppose we sample a hash function $h \in \mathcal{H}$ uniformly at random. Let C be the event that a collision occurs among n distinct keys $k_1, \ldots, k_n \in K$. Can we upper bound the probability $\Pr[C]$?

▶ Solution

This problem is analogous to the birthday paradox: "birthdays" correspond to bucket indices, and "same birthday" corresponds to a hash collision. A key difference is that in the birthday paradox, each student independently gets a random birthday, whereas in hashing, we first choose one hash function h and all hash values are generated by that same function. As a result, directly counting the number of collision-free hash functions becomes difficult. Nevertheless, applying the union bound provides a clean upper bound on the collision probability.

Theorem 2.57. For any universal hash family $\mathcal{H}$ and any n distinct keys $k_1, \ldots, k_n \in K$, the collision probability admits the upper bound

$$\Pr[C] \leq \frac{1}{m}\binom{n}{2}. \tag{2.128}$$

Proof. Let $C_{i,j}$ be the event that $h(k_i) = h(k_j)$ for $i \neq j$. Then

$$C = \bigcup_{1 \leq i < j \leq n} C_{i,j}. \tag{2.129}$$

By the union bound, we have

$$\Pr[C] \leq \sum_{1 \leq i < j \leq n} \Pr[C_{i,j}]. \tag{2.130}$$

By universality, for every $i \neq j$, $\Pr[C_{i,j}] \leq \frac{1}{m}$. Therefore,

$$\Pr[C] \leq \sum_{1 \leq i < j \leq n} \frac{1}{m} = \frac{1}{m}\binom{n}{2}, \tag{2.131}$$

which proves the theorem. □

Theorem 2.57 provides a clean upper bound on the collision probability for any universal hash family. It also provides a simple recipe for choosing the number of buckets m to ensure a low collision probability: simply choose m to be larger than $\binom{n}{2}$. In particular, to achieve a collision probability of at most $\delta \in (0, 1)$, it

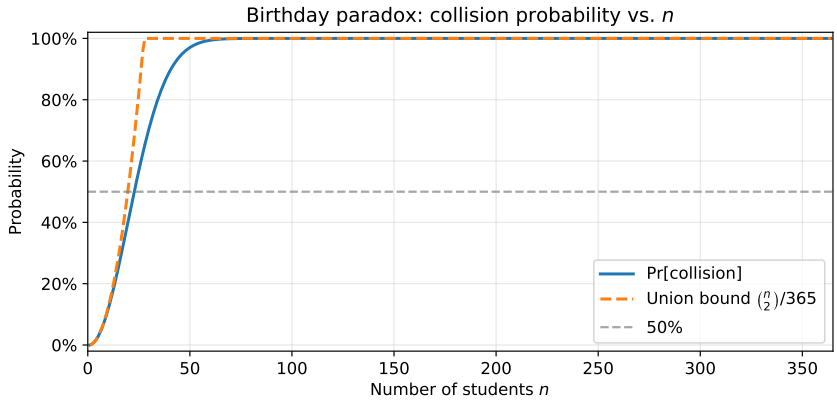


Figure 2.9: The exact collision probability and its union-bound upper bound for the birthday paradox.

suffices to choose $m \geq \frac{1}{\delta}\binom{n}{2}$. This leads to $\Theta(n^2/\delta)$ space for storing a dictionary of n keys.

While this already provides a better space-time trade-off than the binary searching and direct addressing approaches for large m, we can do better by using a more sophisticated hash table implementation. We will revisit this topic in Section 2.6.7 and Section 3.1.3.

▶ Revisiting the Birthday Paradox with Union Bound

In the setting of birthday paradox, every student independently gets a random birthday. This is analogous to hashing n keys into m buckets with the following universal hashing family:

$$\mathcal{H}_m := \Big\{\text{all functions from } K \text{ to } \{0, 1, \ldots, m-1\}\Big\}. \tag{2.132}$$

This hash family is impractical because specifying h requires $\Omega(|K|)$ space, which is essentially as expensive as the direct addressing approach. Nevertheless, it is still interesting to see how the collision probability behaves in this setting.

By the same argument as in the birthday paradox, we can derive the following formula for the collision probability:

$$\Pr[C] = 1 - \prod_{k=0}^{n-1}\left(1 - \frac{k}{m}\right). \tag{2.133}$$

By the union bound, we have $\Pr[C] \leq \frac{1}{m}\binom{n}{2}$. This bound can also be derived from the following inequality, which holds for any $a_1, \ldots, a_n \in [0, 1]$:

$$\prod_{k=0}^{n-1}(1 - a_k) \geq 1 - \sum_{k=0}^{n-1} a_k. \tag{2.134}$$

Applying this inequality with $a_k = \frac{k}{m}$, we obtain

$$\Pr[C] \leq \sum_{k=0}^{n-1} \frac{k}{m} = \frac{1}{m}\binom{n}{2}. \tag{2.135}$$

We also plot the exact collision probability together with its union-bound upper bound in Figure 2.9. When n is small, the union bound is remarkably tight. This should not be surprising, since $\prod_{k=0}^{n-1}(1-a_k) = 1-\sum_{k=0}^{n-1} a_k+\mathcal{O}\left(\sum_{i<j}|a_i||a_j|\right)$ when all a_k are small and we can ignore the higher-order terms. As n grows, the bound becomes increasingly loose. However, once n is large enough that the collision probability is already close to 1, obtaining a sharp upper bound becomes less meaningful.

2.4.4 Example: Ramsey Numbers via the Probabilistic Method

▶ Background

Frank Plumpton Ramsey (1903–1930) was a British philosopher, mathematician and economist. He made significant contributions to all these fields before his untimely death from illness at the age of 26.

In 1928, Ramsey wrote a paper *On a problem in formal logic*, published later in 1930 [Ram30]. In that work, Ramsey studied the *Entscheidungsproblem* (the German word for "decision problem"). This was a challenge posed by David Hilbert and Wilhelm Ackermann in 1928 [HA28]: *Is there an algorithm (called "decision procedure" at that time) that can determine whether a given logical formula is valid?*

If such an algorithm exists, we would solve many important problems very easily. For example, we can write Goldbach's conjecture as a logical formula and input it to the algorithm to determine whether it is valid. Looking back, this hope seems to be too good to be true: in 1936, Alonzo Church and Alan Turing independently proved that no such general algorithm exists [Chu36, Tur37].

Ramsey's work [Ram30] was among the first attempts to address the Entscheidungsproblem. He studied a class of first-order formulas with only universal quantifiers ($\forall$) and no existential quantifiers ($\exists$), i.e., $(\forall x_1)\cdots(\forall x_n)\phi(x_1,\ldots,x_n)$, where ϕ is a quantifier-free formula. When the domain of each variable is finite, the validity of such a formula can be decided simply by enumerating all possible assignments and checking whether the formula holds for each of them. The problem becomes more interesting when the domain may be infinite, which motivated Ramsey to ask whether a formula that holds on a sufficiently large domain must necessarily contain certain regular substructures. Surprisingly, the answer is *yes*.

This insight was later developed into *Ramsey theory*, which broadly studies the emergence of regular substructures from sufficiently large objects. Interestingly, this field has had a far greater impact in combinatorics and theoretical computer science than Ramsey's original contribution to the Entscheidungsproblem.

Remark 2.58. An unrelated but interesting story is that Church published his result first, while Turing was still working on his paper. After Turing learned of Church's result, he had to rush to complete and publish his own work. These two works took very different approaches, though. Church based his argument on the lambda calculus, while Turing introduced what we now call Turing machines. A takeaway from this story is that computer scientists have a long tradition of "scooping" each other. So if you have some good ideas, implement and write down your results as soon as possible!

▶ Warm-up: Theorem on friends and strangers

The simplest nontrivial Ramsey statement is the following theorem.

> **Theorem 2.59** (Theorem on friends and strangers)**.** Among any 6 people, there always exist either 3 mutual friends or 3 mutual strangers.

Proof. Pick any person v. Among the other 5 people, each person is either a friend of v or a stranger to v. By the pigeonhole principle, at least 3 of these 5 people are in the same category. So one of the following two cases must hold:

- **Case 1:** at least 3 of them are friends of v;
- **Case 2:** at least 3 of them are strangers to v.

By symmetry of friend/stranger, it is enough to analyze Case 1. Choose three friends a, b, c of v, as illustrated in Figure 2.10a. Now inspect the relationships among a, b, c:

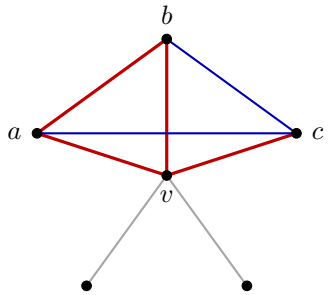
- **Subcase 1:** if at least one pair among $(a, b), (b, c), (c, a)$ are friends, then together with v we get 3 mutual friends;
- **Subcase 2:** if none of $(a, b), (b, c), (c, a)$ are friends, then a, b, c are 3 mutual strangers.

In either subcase, the required triple exists. □

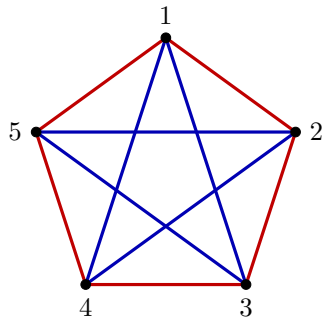
In graph language, each person is a vertex, each pair is an edge, and we color edges red/blue (friend/stranger). The theorem above becomes: every red/blue coloring of K_6 contains a monochromatic K_3. Here K_n denotes the complete graph on n vertices (every pair of vertices is connected), and we say that a complete graph K_n is *monochromatic* if all its edges have the same color. A complete subgraph K_n of a graph G is also called a *clique* of G.

▶ Ramsey number

The warm-up case captures a general phenomenon proved by Ramsey: a graph with sufficiently many vertices unavoidably contains a monochromatic clique. Now we state the general version of Ramsey's theorem for graphs.



(a) Upper bound $R(3) \leq 6$.



(b) Lower bound witness for $R(3) > 5$.

Figure 2.10: **(a)** An illustration of the proof of $R(3) \leq 6$; **(b)** A 2-coloring of K_5 with no monochromatic K_3, thereby proving $R(3) > 5$.

Theorem 2.60 (Ramsey's theorem for graphs)**.** For every integer $k \geq 2$, there exists an integer $N > 0$ such that every red/blue coloring of the edges of K_N contains a monochromatic K_k.

Definition 2.61 (Ramsey number $R(k)$)**.** The Ramsey number $R(k)$ is the smallest integer N such that every red/blue coloring of the edges of K_N contains a monochromatic K_k.

We leave the proof of existence as an exercise. In fact, you will prove the following upper bound on $R(k)$:

$$R(k) \leq \binom{2k-2}{k-1} < 4^k. \tag{2.136}$$

Ramsey's original theorem is in fact more general than the above graph form. We now state the general version and refer interested readers to textbooks on Ramsey theory for further details, such as [GRS90, Prö13].

Theorem 2.62 (Ramsey's theorem for hypergraphs)**.** For any integers $r \geq 1$ and $k \geq m \geq 1$, there exists an integer N such that for every coloring of the m-element subsets of an N-element set with r colors, there exists a k-element subset whose m-element subsets are all assigned the same color.

▶ Lower bound on Ramsey numbers

Can we exactly compute the Ramsey number $R(k)$ for a given k? This requires us to derive both upper bounds on $R(k)$ such as Equation (2.136) and lower bounds on $R(k)$. Here, lower bounding a Ramsey number $R(k)$ essentially means showing

that there exists a red/blue coloring of the edges of K_N with no monochromatic K_k for some N. We start by proving a lower bound on $R(k)$ for the case $k = 3$.

Theorem 2.63 (Lower bound on $R(3)$)**.** There exists a red/blue coloring of the edges of K_5 with no monochromatic K_3. Consequently,

$$R(3) > 5. \tag{2.137}$$

Proof. Use the coloring in Figure 2.10b: color the 5 cycle edges red and the other 5 diagonals blue. There is no monochromatic triangle in this coloring. □

Combining this lower bound with the upper bound from Theorem 2.59 ($R(3) \leq 6$), we obtain

$$R(3) = 6. \tag{2.138}$$

Now we turn to general k. Can we construct a red/blue coloring of K_N with no monochromatic K_k for some N? This becomes substantially more difficult than the case $k = 3$. In fact, it appears that any meaningful progress requires exploiting nontrivial algebraic or recursive structures.

We now introduce a simple lower bound on $R(k)$ by making use of probability theory, even though the definition of Ramsey number itself does not seem to be related to probability at all. This lower bound was proved by Paul Erdős in 1947 [Erd47], and the proof method is now known as the *probabilistic method*.

▶ The Probabilistic Method

The overall idea of the probabilistic method is simple but powerful. Suppose we would like to construct a combinatorial structure with certain desired properties. Instead of constructing it explicitly, we define a random experiment that generates a structure uniformly at random from a large family of candidates. We then define a "good" event to be the event that the sampled structure satisfies the desired properties. If we can show that the probability of this good event is positive, then such a structure must exist. That is,

$$\Pr[\text{good}] > 0 \implies \exists \text{ a structure with the desired properties.} \tag{2.139}$$

In our setting, the random experiment is as follows: we color each edge of K_N independently red or blue with equal probability. The "good" event is that the resulting coloring contains no monochromatic K_k. If we prove that this event has positive probability, then we conclude that there exists a red/blue coloring of K_N with no monochromatic K_k.

Theorem 2.64 ([Erd47])**.** For all $k \geq 3$,

$$R(k) > 2^{k/2}. \tag{2.140}$$

Proof. Let $N := \lfloor 2^{k/2} \rfloor$, and label the vertices by $[N] := \{1, 2, \ldots, N\}$. Color each edge of K_N independently red/blue with probability $1/2$.

Let A be the event that there exists a monochromatic K_k. For each k-subset $S \subseteq [N]$, define A_S as the event that S induces a monochromatic K_k. Then

$$A = \bigcup_{|S|=k} A_S. \tag{2.141}$$

For fixed $S \subseteq [N]$, there are $\binom{k}{2}$ edges inside S. The probability that all these edges are red is $2^{-\binom{k}{2}}$, and the same for all blue. These two cases are disjoint, so

$$\Pr[A_S] = 2^{1-\binom{k}{2}}.$$

By the union bound,

$$\Pr[A] = \Pr\left[\bigcup_{|S|=k} A_S\right] \leq \sum_{|S|=k} \Pr[A_S] = \binom{N}{k} 2^{1-\binom{k}{2}}. \tag{2.142}$$

Using $\binom{N}{k} \leq \frac{N^k}{k!}$ and $N \leq 2^{k/2}$,

$$\Pr[A] \leq \frac{2^{k^2/2}}{k!} \cdot 2^{1-k(k-1)/2} = \frac{2^{1+k/2}}{k!} < 1, \tag{2.143}$$

where the last inequality can be proved for $k \geq 3$. This implies that the complement event $\bar{A}$ has positive probability,

$$\Pr[\bar{A}] > 0. \tag{2.144}$$

Therefore there exists a coloring of K_N with no monochromatic K_k, and thus $R(k) > N = \lfloor 2^{k/2} \rfloor$. Since $R(k)$ is an integer, this implies $R(k) > 2^{k/2}$. □

Combining this lower bound with the upper bound from Equation (2.136),

$$2^{k/2} < R(k) < 4^k. \tag{2.145}$$

Therefore, the Ramsey number grows exponentially with k:

$$R(k) = 2^{\Theta(k)}. \tag{2.146}$$

Remark 2.65. Be careful with the notation $2^{\Theta(k)}$. Strictly speaking, since $\Theta(k)$ denotes a set of functions that may include negative multiples of k, the expression $2^{\Theta(k)}$ could also include terms such as 2^{-k}. However, in theoretical computer science, $2^{\Theta(k)}$ is typically understood to mean $2^{f(k)}$ for some non-negative function $f(k) = \Theta(k)$. Explicitly state the meaning of such expressions if you feel that your readers may be confused.

Remark 2.66 (Ramsey theory implies data structure lower bounds)**.** A natural question is: why do computer scientists need to care about Ramsey numbers? A classical example appears in Yao's famous paper *Should Tables Be Sorted?* [Yao81]. The task is to design a data structure that can answer static membership queries: store a set S of n numbers, then answer queries "is $x \in S$?" Yao introduced the cell-probe model, where each probe can read or write a single

cell of memory, and the query cost is measured by the number of probes in the worst case. Sorting followed by binary search requires about $\log_2 n$ probes per query. *Can we do better?* Of course, we can use hashing to achieve $\mathcal{O}(1)$ query cost, but this requires additional space. *What if we restrict the space to exactly what is needed to store the n elements, with no redundancy?* Unfortunately,



Yao proved that when the universe is large enough, this logarithmic worst-case cost is unavoidable. The key combinatorial ingredient in the proof is Ramsey theory. Roughly speaking, for any data structure that stores n elements, when the universe is large enough, there exists a large subset of elements such that all n-element subsets of this subset are stored in the same order in the table. This structural regularity is then exploited to derive the lower bound.

2.5 Conditional Probability and Independence

In the previous section, we mainly computed probabilities of events directly, such as $\Pr[A]$ and $\Pr[A \cap B]$. In many problems, however, we first observe partial information and then need to update our probability of interest. Even when no new information is observed, conditioning is often a convenient way to break a complicated event into simpler cases. This leads to a basic question: if we already know that event B has occurred, what is the probability that A also occurs?

2.5.1 Conditional Probability

We start from the definition of conditional probability, which is the probability of an event A given that another event B has happened.

Definition 2.67 (Conditional Probability)**.** Let A, B be events with $\Pr[B] > 0$. The conditional probability of A given B is defined as

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}. \tag{2.147}$$

We leave $\Pr[A \mid B]$ undefined if $\Pr[B] = 0$.

After conditioning on B, we restrict attention to outcomes in B and renormalize probabilities inside this reduced sample space. Rearranging the definition, we have the following formula:

$$\Pr[A \cap B] = \Pr[B] \Pr[A \mid B], \tag{2.148}$$

which is the probability analogue of the multiplication principle in counting: first satisfy B, then satisfy A under that condition. In counting we multiply numbers of choices, while here we multiply the probabilities of B and A given B.

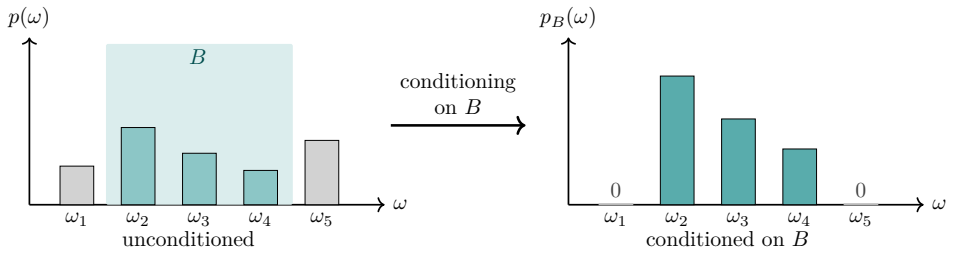


Figure 2.11: Conditioning on event B can be viewed as restricting the probability mass to the event B and renormalizing it so that the total probability is 1.

Example 2.68. From a standard 52-card deck, draw one card uniformly at random. Let A be the event that the card is a queen (Q), and B be the event that the card is a face card (J, Q, K). Then $\Pr[B] = 12/52$ and $\Pr[A \cap B] = 4/52$, so $\Pr[A \mid B] = \frac{4/52}{12/52} = \frac{1}{3}$. That is, among face cards, one third are queens.

Remark 2.69. Another way to understand conditional probability is to think of conditioning on an event B as a new probability space $\mathcal{P}_B := (\Omega, p_B)$ where the sample space is still Ω but the probability mass function is changed to

$$p_B(\omega) = \begin{cases} \frac{p(\omega)}{\Pr[B]} & \text{if } \omega \in B, \\ 0 & \text{if } \omega \notin B. \end{cases} \tag{2.149}$$

Then for any event A, $\Pr[A \mid B]$ can be viewed as the probability of A in $\mathcal{P}_B$:

$$\Pr[A \mid B] = \sum_{\omega \in A} p_B(\omega). \tag{2.150}$$

Beyond discrete probability spaces, conditioning on an event B in a general probability space can be understood in the same way: we keep the original sample space, restrict the probability mass to the event B, and renormalize it so that the total probability is 1. In measure-theoretic probability theory, this idea is formalized using the Radon–Nikodym theorem.

2.5.2 Independence of Events

Conditional probability describes how probabilities change after conditioning on information. Independence is the special case where this conditioning does not change the probability.

Definition 2.70 (Independence)**.** Two events A, B are said to be *independent* if

$$\Pr[A \cap B] = \Pr[A] \Pr[B]. \tag{2.151}$$

If $\Pr[B] > 0$, this is equivalent to

$$\Pr[A \mid B] = \Pr[A]. \tag{2.152}$$

When there are n events, we say they are *pairwise independent* if any two of them are independent. However, this does not mean they are "fully" independent. A stronger notion, called *mutual independence*, is defined below.

Definition 2.71 (Mutual independence)**.** Events $A_1, \ldots, A_m$ are *mutually independent* if for every non-empty subset $I \subseteq [m]$,

$$\Pr\left[\bigcap_{i \in I} A_i\right] = \prod_{i \in I} \Pr[A_i]. \tag{2.153}$$

Example 2.72. Let A, B be two independent events, each with probability $1/2$. Let C be the event that exactly one of A and B happens. Then $\Pr[C] = 1/2 \cdot 1/2 + 1/2 \cdot 1/2 = 1/2$. The events A, B, C are pairwise independent, since A and B are independent, and

$$\Pr[A \cap C] = \Pr[A] \cdot (1 - \Pr[B]) = 1/4 = \Pr[A] \Pr[C], \tag{2.154}$$

$$\Pr[B \cap C] = \Pr[B] \cdot (1 - \Pr[A]) = 1/4 = \Pr[B] \Pr[C]. \tag{2.155}$$

But they are *not* mutually independent, since

$$\Pr[A \cap B \cap C] = \Pr[\varnothing] = 0. \tag{2.156}$$

2.5.3 Chain Rule for Events

A generalization of Equation (2.148) is the chain rule, which states that the probability of the intersection of multiple events is the product of the probabilities of the events given the previous events.

Theorem 2.73 (Chain rule)**.** For $m \geq 2$, let $A_1, \ldots, A_m$ be events with all relevant conditional probabilities well-defined. Then

$$\Pr\left[\bigcap_{i=1}^{m} A_i\right] = \Pr[A_1] \prod_{i=2}^{m} \Pr\left[A_i \,\middle|\, \bigcap_{j=1}^{i-1} A_j\right]. \tag{2.157}$$

Proof. We prove by induction on m. The base case $m = 2$ is exactly

$$\Pr[A_1 \cap A_2] = \Pr[A_1] \Pr[A_2 \mid A_1]. \tag{2.158}$$

Assuming the claim holds for $m - 1$ events, we have

$$\Pr\left[\bigcap_{i=1}^{m-1} A_i\right] = \Pr[A_1] \prod_{i=2}^{m-1} \Pr\left[A_i \,\middle|\, \bigcap_{j=1}^{i-1} A_j\right]. \tag{2.159}$$

By Equation (2.148), we have

$$\Pr\left[\bigcap_{i=1}^{m} A_i\right] = \Pr\left[\bigcap_{i=1}^{m-1} A_i\right] \cdot \Pr\left[A_m \,\middle|\, \bigcap_{j=1}^{m-1} A_j\right]. \tag{2.160}$$

Substituting the induction hypothesis, we get

$$\Pr\left[\bigcap_{i=1}^{m} A_i\right] = \left(\Pr[A_1]\prod_{i=2}^{m-1}\Pr\left[A_i \;\middle|\; \bigcap_{j=1}^{i-1} A_j\right]\right)\cdot \Pr\left[A_m \;\middle|\; \bigcap_{j=1}^{m-1} A_j\right]. \tag{2.161}$$

Rearranging the terms, we get the desired formula. □

Example 2.74 (Revisiting the birthday paradox with the chain rule)**.** Suppose n students' birthdays are uniformly randomly chosen from 365 days. Let E_i be event that student i does not collide with students $1, \ldots, i-1$. Then

$$\Pr[\text{no collision}] = \Pr\left[\bigcap_{i=1}^{n} E_i\right] \tag{2.162}$$

$$= \prod_{i=1}^{n} \Pr\left[E_i \mid E_1 \cap \cdots \cap E_{i-1}\right] \tag{2.163}$$

$$= \prod_{i=1}^{n} \frac{365 - i + 1}{365}. \tag{2.164}$$

Here the first equality holds because no collision happens if and only if each student does not collide with the previous students. The second equality is due to the chain rule. The thrid equality holds because conditioned on $E_1 \cap \cdots \cap E_{i-1}$, the first $i-1$ birthdays are all distinct, leaving $365 - i + 1$ available days for student i to choose without causing a collision. This results in the same formula as in Equation (2.9).

2.5.4 The Law of Total Probability

Sometimes the sample space can be partitioned into mutually exclusive events $B_1, \ldots, B_m$, and then for a given event A, it is easier to compute the conditional probabilities $\Pr[A \mid B_i]$ for all i than to compute $\Pr[A]$ directly. The following theorem provides a fundamental rule in probability theory, called the *law of total probability*, that provides a tool to compute $\Pr[A]$ in this way.

Theorem 2.75 (Law of total probability)**.** Let $B_1, \ldots, B_m$ be mutually exclusive events that satisfy $\bigcup_{i=1}^{m} B_i = \Omega$ and $\Pr[B_i] > 0$ for all $i \in [m]$. Then for any event A,

$$\Pr[A] = \sum_{i=1}^{m} \Pr[A \mid B_i]\Pr[B_i]. \tag{2.165}$$

Proof. Decompose A as $A = \bigcup_{i=1}^{m}(A \cap B_i)$. Since $B_1, \ldots, B_m$ are mutually exclusive, the events $A \cap B_i$ are also mutually exclusive. Therefore, we can add

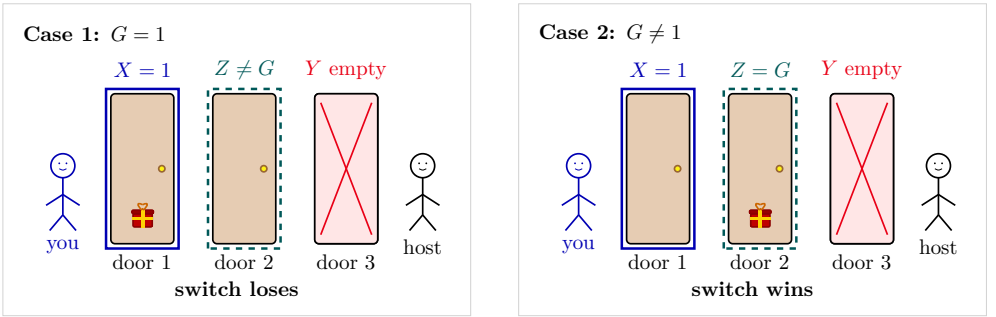


Figure 2.12: Two cases of the Monty Hall problem: (1) If the prize is behind door 1, then switching loses; (2) If the prize is behind door 2 or 3, then the host will open the empty door and leave the prize behind the other unopened door, so switching wins.

up the probabilities of the events $A \cap B_i$ to get the probability of A.

$$\Pr[A] = \sum_{i=1}^{m} \Pr[A \cap B_i] = \sum_{i=1}^{m} \Pr[A \mid B_i] \Pr[B_i], \tag{2.166}$$

where the last equality is due to Equation (2.148). □

Example 2.76 (Monty Hall problem)**.** There are 3 closed doors. Exactly one door has a prize behind it, say door G, and the other two doors are empty. You are invited to choose one door, say door X. You initially have no information, so each door is equally likely to contain the prize. After you choose X, the host opens another door Y such that $Y \neq X$ and door Y is empty.

Now two doors remain: the original choice X, and the other unopened door Z. *Do you want to keep X, or switch to Z?*

Without loss of generality, assume you initially picked door $X = 1$. Let A be the event that you win by switching to Z. We consider two cases:

- if $G \neq 1$, then $Y \notin \{1, G\}$. There are only three doors in total, so the remaining door must be G, and thus $\Pr[A \mid G \neq 1] = 1$.
- if $G = 1$, then $Z \neq 1 = G$ and $\Pr[A \mid G = 1] = 0$.

By the law of total probability, we have

$$\Pr[A] = \Pr[A \mid G \neq 1] \Pr[G \neq 1] + \Pr[A \mid G = 1] \Pr[G = 1] \tag{2.167}$$

$$= 1 \cdot \frac{2}{3} + 0 \cdot \frac{1}{3} \tag{2.168}$$

$$= \frac{2}{3}. \tag{2.169}$$

Therefore switching wins with probability 2/3, while staying wins with probability 1/3. It is always better to switch.

Example 2.77 (Revisiting the Secretary Problem)**.** Let A be the event that the k-strategy in the secretary problem selects the best option. We apply the law of total probability by conditioning on the position of the best option. For $j \in [n]$, let B_j be the event that the best option appears at position j. Then $\Pr[B_j] = 1/n$, and

$$\Pr[A] = \sum_{j=1}^{n} \Pr[A \mid B_j] \Pr[B_j] = \frac{1}{n} \sum_{j=1}^{n} \Pr[A \mid B_j]. \tag{2.170}$$

If $j \leq k$, success is impossible, so $\Pr[A \mid B_j] = 0$. If $j > k$, the strategy succeeds exactly when the best option among positions $1, \ldots, j-1$ appears within the first k positions, so $\Pr[A \mid B_j] = \frac{k}{j-1}$. Therefore

$$\Pr[A] = \frac{1}{n} \sum_{j=k+1}^{n} \frac{k}{j-1} = \frac{k}{n} \sum_{i=k}^{n-1} \frac{1}{i}. \tag{2.171}$$

This is the same formula as derived by counting in Section 2.3.2.

2.6 Random Variables

In previous sections, we mainly worked with events. We now introduce *random variables*. Consider again the example of rolling two dice in Section 2.1.2. Suppose we want to analyze the sum of the two dice. One way to describe this is by defining a collection of events such as "the sum is 2", "the sum is 3", ..., "the sum is 12". However, this is not a very convenient way, since we need to define many separate events to describe this single quantity. A more natural approach is to view the sum as a *random variable*, a concept that we now introduce.

2.6.1 Definitions

Mathematically, a random variable is a function that assigns a real number to every possible outcome $\omega \in \Omega$ in the probability space.

Definition 2.78 (Random variable)**.** Let $\mathcal{P} = (\Omega, p)$ be a discrete probability space. A *random variable* is a function

$$X : \Omega \to \mathbb{R}. \tag{2.172}$$

It is important to note that a random variable is not "random" as a function. The randomness comes from the random outcome $\omega \in \Omega$ in the probability space. Once ω is sampled, $X(\omega)$ is a deterministic number.

Since a random variable is a function, all the usual operations on functions apply to random variables. We can define the sum, difference, product and quotient of two random variables by adding, subtracting, multiplying and dividing the functions respectively. For example, the sum of two random variables X and Y is a random variable Z defined by $Z(\omega) = X(\omega) + Y(\omega)$ for all $\omega \in \Omega$.

A random variable defined on a probability space induces a *probability distribution* on the real numbers. Here, the probability distribution of X is defined as a function that assigns a probability to each subset of possible values of X. That is, for any subset $A \subseteq \mathbb{R}$,

$$\Pr[X \in A] = \sum_{\omega \in \Omega} \mathbb{1}_{[X(\omega) \in A]}\, p(\omega), \tag{2.173}$$

where $\mathbb{1}_{[C]}$ denotes the indicator function defined as follows.

Definition 2.79 (Indicator function)**.** The indicator function of a condition C is:

$$\mathbb{1}_{[C]} = \begin{cases} 1 & \text{if the condition } C \text{ holds,} \\ 0 & \text{otherwise.} \end{cases} \tag{2.174}$$

For a set $A \subseteq U$, where U is some universe, the indicator function of A is

$$\mathbb{1}_A : U \to \{0,1\}, \quad \mathbb{1}_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{otherwise.} \end{cases} \tag{2.175}$$

Remark 2.80. Different authors may use different notations for the indicator function. The following table lists some of the most common ones.

Notation	No Bracket	Parentheses	Square Brackets	Curly Braces
Condition only	/	(C)	$[C]$	/
After $\mathbb{1}$	/	/	$\mathbb{1}[C]$	$\mathbb{1}\{C\}$
Subscript of $\mathbb{1}$	$\mathbb{1}_C$	/	$\mathbb{1}_{[C]}$	$\mathbb{1}_{\{C\}}$

In particular, (C) and $[C]$ are also called the *Iverson bracket*. Sometimes the double brackets are also used, written as $[\![C]\!]$. In some places, 1, $\mathbf{1}$ or $\mathbb{I}$ are used instead of $\mathbb{1}$. So the notation may be written as $1_{[C]}$, $\mathbf{1}_{[C]}$, or $\mathbb{I}[C]$.

Since we only consider discrete probability spaces, the probability distribution of X can be fully described by the *probability mass function* defined as follows:

Definition 2.81 (Probability mass function)**.** For a random variable X, its probability mass function $p_X : \mathbb{R} \to [0,1]$ is defined as

$$p_X(x) := \Pr[X = x], \quad \forall x \in \mathbb{R}. \tag{2.176}$$

2.6.2 Expectation

To understand the typical behavior of a given random variable, it is natural to consider the "average" value of the random variable. This is captured by the concept of *expectation*.

Definition 2.82 (Expectation)**.** Let X be a random variable on a discrete probability space (Ω, p). The *expectation*, *expected value* or *mean* of X is defined as

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) p(\omega). \tag{2.177}$$

Equivalently, we can write

$$\mathbb{E}[X] = \sum_{x} x \Pr[X = x] = \sum_{x} x \cdot p_X(x). \tag{2.178}$$

The expectation $\mathbb{E}[X]$ can be thought of as a weighted average, where each value is weighted by how likely it is. This interpretation comes from the following scenario. Suppose we repeat the random experiment n times for a very large number n. Then we expect roughly $n \Pr[X = x]$ outcomes in which X takes the value x. If we average all these observed values, the result should be close to $\mathbb{E}[X]$. We will rigorously justify this intuition in Chapter 3. For now, you may view it as a high-level explanation of why the expectation represents the "average" value of a random variable.

Remark 2.83. For an event A, you may view it as a random variable $\mathbb{1}_A$ that takes the value 1 if $\omega \in A$ and 0 otherwise. Then $\mathbb{E}[\mathbb{1}_A] = \Pr[A]$.

Example 2.84. Back to our example of rolling two fair dice in Section 2.1.2. The sample space is $\Omega := \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}$. and the probability mass function is uniform over Ω. We can define the random variable X to be the sum of the two dice. That is,

$$X(\omega) = i + j, \quad \forall \omega = (i, j) \in \Omega. \tag{2.179}$$

The probability mass function of X is shown in the following table:

x	2	3	4	5	6	7	8	9	10	11	12
$p_X(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Therefore, the expectation of X is

$$\begin{aligned} \mathbb{E}[X] &= \sum_{x=2}^{12} x \cdot p_X(x) \\ &= 2 \cdot \frac{1}{36} + 3 \cdot \frac{2}{36} + 4 \cdot \frac{3}{36} + 5 \cdot \frac{4}{36} + 6 \cdot \frac{5}{36} + 7 \cdot \frac{6}{36} \\ &\quad + 8 \cdot \frac{5}{36} + 9 \cdot \frac{4}{36} + 10 \cdot \frac{3}{36} + 11 \cdot \frac{2}{36} + 12 \cdot \frac{1}{36} \\ &= 7. \end{aligned} \tag{2.180}$$

2.6.3 Linearity of Expectation

A beautiful property of the expectation is that it is linear.

Theorem 2.85 (Linearity of expectation)**.** For any random variables $X_1, \ldots, X_n$ and constants $c_1, \ldots, c_n \in \mathbb{R}$,

$$\mathbb{E}\left[\sum_{i=1}^{n} c_i X_i\right] = \sum_{i=1}^{n} c_i \mathbb{E}[X_i]. \tag{2.181}$$

Proof. By definition of expectation,

$$\mathbb{E}\left[\sum_{i=1}^{n} c_i X_i\right] = \sum_{\omega \in \Omega} \left(\sum_{i=1}^{n} c_i X_i(\omega)\right) p(\omega) = \sum_{i=1}^{n} c_i \sum_{\omega \in \Omega} X_i(\omega) p(\omega) \tag{2.182}$$

$$= \sum_{i=1}^{n} c_i \mathbb{E}[X_i], \tag{2.183}$$

where the second equality follows by interchanging the order of two sums. □

Example 2.86. In our example of rolling two fair dice, we can decompose X into the sum of the outcomes of the two dice, denoted as X_1 and X_2. By linearity of expectation, we have $\mathbb{E}[X] = \mathbb{E}[X_1] + \mathbb{E}[X_2]$. It is easy to verify that $\mathbb{E}[X_1] = \mathbb{E}[X_2] = 3.5$. Therefore, $\mathbb{E}[X] = 3.5 + 3.5 = 7$.

Note that the linearity of expectation does *not* require independence. Even when $X_1, \ldots, X_n$ are not independent, Theorem 2.85 still holds. The following is an application of this observation.

Example 2.87 (Expected number of cycles in a random permutation)**.** Let σ be a uniformly random permutation on $[n]$, and let X be the number of cycles in σ. We want to compute $\mathbb{E}[X]$.

For each $i \in [n]$, define L_i as the length of the cycle containing i. Then $X = \sum_{i=1}^{n} \frac{1}{L_i}$. By linearity,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}\left[\tfrac{1}{L_i}\right]. \tag{2.184}$$

For each i, we compute the probability mass function of L_i by the multiplication principle: choose the $\ell - 1$ elements on the cycle, then arrange them into one cycle, and finally permute the remaining $n - \ell$ elements.

$$\Pr[L_i = \ell] = \frac{\binom{n-1}{\ell-1}(\ell-1)!(n-\ell)!}{n!} = \frac{(n-1)!}{n!} = \frac{1}{n}. \tag{2.185}$$

So the expectation of $\frac{1}{L_i}$ is $\mathbb{E}[\frac{1}{L_i}] = \sum_{\ell=1}^{n} \frac{1}{\ell} \cdot \Pr[L_i = \ell] = \frac{1}{n} \sum_{\ell=1}^{n} \frac{1}{\ell} = \frac{1}{n} H_n$, where H_n is the n-th harmonic number. Therefore, the expectation of X is $\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[\frac{1}{L_i}] = H_n = \Theta(\log n)$.

2.6.4 Tail-Sum Formula

The tail-sum formula is a useful tool for computing the expectation of a non-negative integer-valued random variable. It states that the expectation of a random variable X equals the sum of the probabilities of X being greater than each non-negative integer.

Theorem 2.88 (Tail-sum formula)**.** If X is a random variable that takes values in non-negative integers, then

$$\mathbb{E}[X] = \sum_{t \geq 0} \Pr[X > t]. \tag{2.186}$$

Proof. Write $X = \sum_{t \geq 0} \mathbb{1}_{[X > t]}$. By linearity of expectation,

$$\mathbb{E}[X] = \sum_{t \geq 0} \mathbb{E}[\mathbb{1}_{[X > t]}] = \sum_{t \geq 0} \Pr[X > t], \tag{2.187}$$

where the last equality uses the fact that $\mathbb{E}[\mathbb{1}_{[A]}] = \Pr[A]$ for any event A. □

In situations where it is easier to compute the tail probabilities $\Pr[X > t]$ for non-negative integers t than to compute the probability distribution of X, the tail-sum formula is a powerful tool.

Example 2.89 (Expected minimum of multiple random variables)**.** Let $X_1, \ldots, X_n$ be independent uniform random variables over $\{1, 2, \ldots, U\}$, and define $M = \min\{X_1, \ldots, X_n\}$. We want to compute $\mathbb{E}[M]$.

By tail-sum formula, we have

$$\mathbb{E}[M] = \sum_{t \geq 0} \Pr[M > t] = \sum_{t=0}^{U-1} \Pr\Big[X_i > t \text{ for all } i \in [n]\Big]. \tag{2.188}$$

For each $t \in \{0, \ldots, n-1\}$, $\Pr[X_i > t] = \frac{U-t}{U}$ is the same for all i. Since the random variables are independent, we have

$$\Pr\Big[X_i > t \text{ for all } i \in [n]\Big] = \left(\frac{U-t}{U}\right)^n. \tag{2.189}$$

Therefore,

$$\mathbb{E}[M] = \sum_{t=0}^{U-1} \left(\frac{U-t}{U}\right)^n = \sum_{k=1}^{U} \frac{k^n}{U^n}. \tag{2.190}$$

When $n = 3$, this formula simplifies to

$$\mathbb{E}[M] = \sum_{k=1}^{U} \frac{k^3}{U^3} = \frac{1}{U^3} \cdot \frac{1}{4}(U^4 + 2U^3 + U^2) = \frac{1}{4}U + \frac{1}{2} + \frac{1}{4U}. \tag{2.191}$$

2.6.5 Conditional Expectation

Similar to the conditional probability, conditional expectation is the average value of a random variable after we reveal partial information.

Definition 2.90 (Conditional expectation given an event)**.** Let X be a random variable and B be an event with $\Pr[B] > 0$. The conditional expectation of X given B is defined as

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[\mathbb{1}_B \cdot X]}{\Pr[B]}. \tag{2.192}$$

Remark 2.91. Following the interpretation of conditional probability in Remark 2.69, we can also define the conditional expectation of X given B as the expectation of X in the probability space $\mathcal{P}_B = (\Omega, p_B)$, where the probability mass function p_B is defined in Equation (2.149).

We can also define the conditional expectation of a random variable X given another random variable Y, written as $\mathbb{E}[X \mid Y]$.

Definition 2.92 (Conditional expectation given a random variable)**.** Let X and Y be random variables. The conditional expectation of X given Y is defined as the following random variable:

$$\mathbb{E}[X \mid Y] : \Omega \to \mathbb{R}, \quad \omega \mapsto \mathbb{E}[X \mid Y = Y(\omega)]. \tag{2.193}$$

Note that $\mathbb{E}[X \mid Y]$ is a random variable rather than a number. For a fixed value y, the quantity $\mathbb{E}[X \mid Y = y]$ is clearly a number defined as in Definition 2.90. However, when conditioning on the random variable Y, the value $Y(\omega)$ depends on ω. Therefore, $\mathbb{E}[X \mid Y](\omega) = \mathbb{E}[X \mid Y = Y(\omega)]$ varies with ω, and is thus a random variable.

Similar as the law of total probability (Theorem 2.75), we have the law of total expectation for conditional expectation.

Theorem 2.93 (Law of total expectation, event form)**.** Let X be a random variable and $B_1, \ldots, B_m$ be mutually exclusive events that satisfy $\bigcup_{i=1}^m B_i = \Omega$ and $\Pr[B_i] > 0$ for all $i \in [m]$. Then

$$\mathbb{E}[X] = \sum_{i=1}^{m} \mathbb{E}[X \mid B_i] \Pr[B_i]. \tag{2.194}$$

Proof. Write X as $X = \sum_{i=1}^m \mathbb{1}_{B_i} \cdot X$. By linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m} \mathbb{E}[\mathbb{1}_{B_i} \cdot X] = \sum_{i=1}^{m} \mathbb{E}[X \mid B_i] \Pr[B_i], \tag{2.195}$$

where the last equality is by the definition of conditional expectation. □

Theorem 2.94 (Law of total expectation, random variable form)**.** Let X and Y be random variables. Then

$$\mathbb{E}[X] = \mathbb{E}\Big[\mathbb{E}[X \mid Y]\Big]. \tag{2.196}$$

Proof. Let B_y be the event that $Y = y$. By Theorem 2.93, we have

$$\mathbb{E}\Big[\mathbb{E}[X \mid Y]\Big] = \sum_y \mathbb{E}\Big[\mathbb{E}[X \mid Y] \;\Big|\; B_y\Big] \Pr[B_y]. \tag{2.197}$$

For each y, we simplify the expectation $\mathbb{E}\Big[\mathbb{E}[X \mid Y] \;\Big|\; B_y\Big]$ as follows:

$$\mathbb{E}\Big[\mathbb{E}[X \mid Y] \;\Big|\; B_y\Big] = \frac{\mathbb{E}[\mathbb{1}_{B_y} \cdot \mathbb{E}[X \mid Y]]}{\Pr[B_y]} \tag{2.198}$$

$$= \frac{\Pr[B_y] \cdot \mathbb{E}[X \mid Y = y]}{\Pr[B_y]} \tag{2.199}$$

$$= \mathbb{E}[X \mid Y = y], \tag{2.200}$$

where the first equality is by the definition of conditional expectation, and the second equality is due to the fact that $\mathbb{E}[X \mid Y]$ is a constant function on B_y.

Substituting this back into the equation, we get

$$\mathbb{E}\Big[\mathbb{E}[X \mid Y]\Big] = \sum_y \mathbb{E}[X \mid B_y] \Pr[B_y] = \mathbb{E}[X], \tag{2.201}$$

where the last equality follows from Theorem 2.93. □

Example 2.95. A drunkman starts at the origin and takes a random walk on the integer line $\{\dots, -2, -1, 0, 1, 2, \dots\}$. At each step, he moves one step to the left or right with equal probability. Let X_t be the position of the drunkman after t steps. *What is* $\mathbb{E}[X_t]$*?*

Let X_t be the position of the drunkman after t steps. Then conditioned on X_{t-1}, X_t is a random variable that takes values in $\{X_{t-1}-1, X_{t-1}+1\}$ with equal probability. So $\mathbb{E}[X_t \mid X_{t-1}] = X_{t-1}$.

By the law of total expectation, we have

$$\mathbb{E}[X_t] = \mathbb{E}\Big[\mathbb{E}[X_t \mid X_{t-1}]\Big] = \mathbb{E}[X_{t-1}]. \tag{2.202}$$

By induction, we have

$$\mathbb{E}[X_t] = \mathbb{E}[X_{t-1}] = \cdots = \mathbb{E}[X_0] = 0. \tag{2.203}$$

The law of total expectation is an important tool for analyzing random variables that are defined recursively. We will see further applications of this law in the analysis of Markov chains in **??** and stochastic gradient descent in Chapter 7.

2.6.6 Independence of Random Variables

Similar to the independence of events, we say two random variables X and Y are *independent* if the events $X \in A$ and $Y \in B$ are independent for all sets $A, B \subseteq \mathbb{R}$.

Definition 2.96 (Independence of random variables)**.** We say that random variables X and Y are *independent* if for all sets $A, B \subseteq \mathbb{R}$,

$$\Pr[X \in A,\, Y \in B] = \Pr[X \in A] \Pr[Y \in B]. \tag{2.204}$$

Equivalently, we say that X and Y are *independent* if for all $a, b \in \mathbb{R}$,

$$\Pr[X = a,\, Y = b] = \Pr[X = a] \Pr[Y = b]. \tag{2.205}$$

While linearity of expectation holds for any random variables, the expectation of the product of independent random variables $\mathbb{E}[XY]$ is in general not equal to the product of the expectations $\mathbb{E}[X]\mathbb{E}[Y]$. However, if X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Theorem 2.97 (Expectation of the product of independent random variables)**.** If X, Y are independent random variables, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]. \tag{2.206}$$

Proof. By the definition of expectation and independence,

$$\begin{aligned}
\mathbb{E}[XY] &= \sum_{a,b} ab \Pr[X = a,\, Y = b] && (2.207)\\
&= \sum_{a,b} ab \Pr[X = a] \Pr[Y = b] && (2.208)\\
&= \left(\sum_{a} a \Pr[X = a]\right)\left(\sum_{b} b \Pr[Y = b]\right) && (2.209)\\
&= \mathbb{E}[X]\mathbb{E}[Y], && (2.210)
\end{aligned}$$

which proves the claim. □

When there are n random variables, we say they are *pairwise independent* if any two of them are independent. Similar to the mutual independence of events, a stronger notion of independence for multiple random variables is called *mutual independence*, which is defined as follows.

Definition 2.98 (Mutual independence of random variables)**.** We say that random variables $X_1, \ldots, X_n$ are *mutually independent* if for all sets $A_1, \ldots, A_n \subseteq \mathbb{R}$,

$$\Pr[X_1 \in A_1, \ldots, X_n \in A_n] = \prod_{i=1}^{n} \Pr[X_i \in A_i]. \tag{2.211}$$

Equivalently, we say that $X_1, \ldots, X_n$ are *mutually independent* if for any sequence of values $a_1, \ldots, a_n \in \mathbb{R}$,

$$\Pr[X_1 = a_1, \ldots, X_n = a_n] = \prod_{i=1}^{n} \Pr[X_i = a_i]. \tag{2.212}$$

Similar to the case of two random variables, mutual independence implies that the expectation of the product of the random variables is the product of the expectations. We will see an application of this proposition in deriving the Chernoff bound in Section 3.3.

Theorem 2.99 (Expectation of the product of mutually independent random variables)**.** If $X_1, \ldots, X_n$ are mutually independent random variables, then

$$\mathbb{E}\left[\prod_{i=1}^{n} X_i\right] = \prod_{i=1}^{n} \mathbb{E}[X_i]. \tag{2.213}$$

There is an important gap between pairwise and mutual independence. Mutual independence implies pairwise independence, but the converse is false. The next hashing example gives a concrete counterexample.

Example 2.100. Let p be an odd prime and define the following linear hash family:

$$\mathcal{H} = \{h_{a,b}(x) = (ax + b) \bmod p : a, b \in \mathbb{Z}_p\}. \tag{2.214}$$

Let x_1, x_2, x_3 be three distinct keys among $\mathbb{Z}_p \setminus \{0\}$. When h is sampled uniformly from $\mathcal{H}$, the random variables $h(x_1), h(x_2), h(x_3)$ are pairwise independent This is can be proved as follows. For any two distinct keys $x_i \neq x_j$ and for any $u, v \in \mathbb{Z}_p$, there is a unique pair $(a, b) \in \mathbb{Z}_p \times \mathbb{Z}_p$ such that $h_{a,b}(x_i) = u$ and $h_{a,b}(x_j) = v$, given by

$$a \equiv \alpha(u - v) \pmod{p} \quad \text{and} \quad b \equiv \alpha(x_i v - x_j u) \pmod{p}, \tag{2.215}$$

where α is the multiplicative inverse of $x_i - x_j$ modulo p. So

$$\Pr[h(x_i) = u,\, h(x_j) = v] = \frac{1}{p^2} = \Pr[h(x_i) = u] \Pr[h(x_j) = v]. \tag{2.216}$$

Now let $x_3 = 2^{-1}(x_1 + x_2)$, where 2^{-1} is the multiplicative inverse of 2 modulo p. Then

$$h(x_1) + h(x_2) \equiv 2h(x_3) \pmod{p}, \tag{2.217}$$

so once $h(x_1)$ and $h(x_2)$ are known, $h(x_3)$ is uniquely determined. This implies that the random variables $h(x_1), h(x_2), h(x_3)$ are not mutually independent.

Therefore, the hash values produced by this hash family are pairwise independent but not 3-wise independent.

Finally, we introduce the concept of independent and identically distributed random variables. These are just a special case of mutually independent random variables where all the random variables have the same probability distribution.

Definition 2.101 (Independent and identically distributed random variables)**.** We say that random variables $X_1, \ldots, X_n$ are *independent and identically distributed*, abbreviated *i.i.d.*, if

- $X_1, \ldots, X_n$ are mutually independent, and
- $X_1, \ldots, X_n$ have the same probability distribution.

We have already seen the examples of i.i.d. random variables many times. For example, if we roll two fair dice, the outcomes of the two dice are i.i.d. random variables. In the birthday paradox, we model the birthdays of n people as i.i.d. random variables. By contrast, values produced by a hash function are usually only pairwise independent, and thus not i.i.d. in general. If you roll a fair die and a fair coin, then the outcomes of the die and the coin are independent, but not i.i.d., since they have different probability distributions.

2.6.7 Example: Hashing with Chaining

▶ Background

In Section 2.4.3, we introduced three methods to implement a dictionary: searching, direct addressing, and hash table. These three approaches provide different trade-offs between time and space. In particular, a hash table reduces the query time to $\mathcal{O}(1)$ while using much less space than direct addressing. However, in the simple hashing scheme we introduced, a key-value pair (k_i, v_i) is directly stored at position $h(k_i)$, and a single hash collision $h(k_i) = h(k_j)$ can cause the dictionary to return incorrect answers. To keep the collision probability at most δ, we therefore need $\Theta(n^2/\delta)$ space.

A more robust way to implement a hash table is to use *chaining*, where collisions only increase the lookup time slightly and do not affect correctness. In chaining, we allocate an array of m buckets, where each bucket j stores a linked list of key-value pairs that hash to bucket j.

More specifically, each linked list consists of zero or more nodes, where each node stores a key-value pair (k_i, v_i) and a pointer `next` that points to the next node in the list. Bucket j stores a pointer `head[j]` to the first node of the linked list, and starting from this pointer, we can traverse the entire linked list through the `next` pointers. See Figure 2.13 for an illustration.

At query time, for a given key k, we first compute its hash value $h(k)$ to determine the bucket j that k hashes to. We then start from `head[j]` and follow the `next` pointers until we either find the key k or reach the end of the list. If we find the key k, we return the corresponding value v. Otherwise, we return a special value `NULL` to indicate that the key is not present in the dictionary.

Table 2.2: Time-space trade-off of different dictionary implementations.

Method	Query Time		Space	Pr[failure]
	Expected	Worst-case		
Searching	$\mathcal{O}(\log n)$	$\mathcal{O}(\log n)$	$\mathcal{O}(n)$	0
Direct Addressing	$\mathcal{O}(1)$	$\mathcal{O}(1)$	$\mathcal{O}(\lvert K\rvert)$	0
Simple Hash Table	$\mathcal{O}(1)$	$\mathcal{O}(1)$	$\mathcal{O}(n^2/\delta)$	δ
Hash Table with Chaining	$\mathcal{O}(1)$	$\mathcal{O}(n)$	$\mathcal{O}(n)$	0

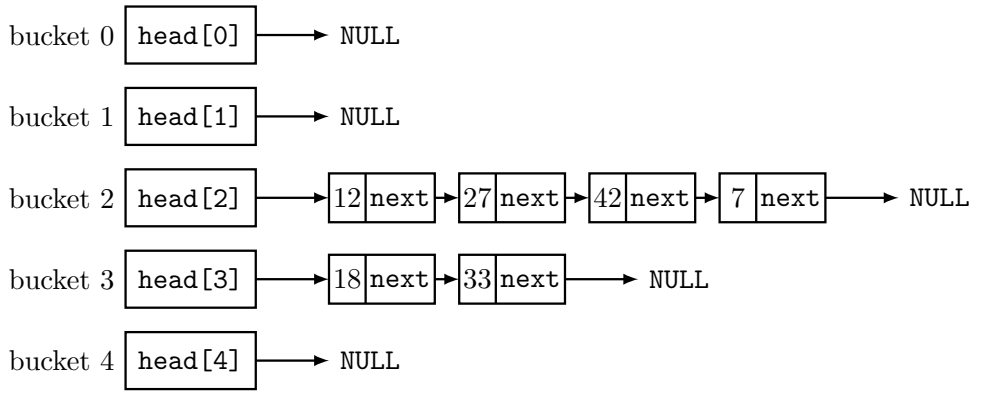


Figure 2.13: A hash table with $m = 5$ buckets.

▶ Problem Statement

We implement a dictionary containing n key-value pairs $(k_1, v_1), \ldots, (k_n, v_n)$ using a hash table with chaining. Let $\mathcal{H}$ be a universal hash family that consists of hash functions mapping from the key space K to $\{0, 1, \ldots, m-1\}$. We sample a hash function $h \in \mathcal{H}$ uniformly at random from $\mathcal{H}$.

The hash table consists of m buckets, where each bucket stores a linked list of key-value pairs. For each (k_i, v_i) in the dictionary, we store the pair in the bucket $h(k_i)$.

For each key k_i, we define X_i to be the query time for finding the key k_i. Give an upper bound on $\mathbb{E}[X_i]$.

▶ Solution

Let Z_i be the number of key-value pairs that hash to the same bucket as k_i (including k_i itself). Then there exists a constant $C > 0$ such that $X_i \leq CZ_i$. We only need to bound Z_i to get an upper bound on X_i.

Lemma 2.102. The expectation of Z_i admits the following upper bound:

$$\mathbb{E}[Z_i] \leq 1 + \frac{n-1}{m}. \tag{2.218}$$

Proof. We write Z_i as a sum of indicator random variables:

$$Z_i = \sum_{j=1}^{n} \mathbb{1}_{[h(k_i)=h(k_j)]}. \tag{2.219}$$

By linearity of expectation,

$$\mathbb{E}[Z_i] = \sum_{j=1}^{n} \mathbb{E}[\mathbb{1}_{[h(k_i)=h(k_j)]}] = \sum_{j=1}^{n} \Pr[h(k_i) = h(k_j)]. \tag{2.220}$$

By universality of $\mathcal{H}$, for any two distinct keys k_i and k_j, $\Pr[h(k_i) = h(k_j)] \leq \frac{1}{m}$. So we have

$$\mathbb{E}[Z_i] = 1 + \sum_{j \in [n] \setminus \{i\}} \Pr[h(k_i) = h(k_j)] \leq 1 + \frac{n-1}{m}, \tag{2.221}$$

which proves the lemma. □

By applying Lemma 2.102, we can conclude that for all $n \geq 1$ and $m \geq 1$,

$$\mathbb{E}[X_i] \leq C \cdot \mathbb{E}[Z_i] \leq C \left(1 + \frac{n-1}{m}\right) = \mathcal{O}\left(1 + \frac{n}{m}\right). \tag{2.222}$$

▶ Parameter Selection

From Equation (2.222), we see that the expected query time for hash tables with chaining is $\mathcal{O}(1 + \frac{n}{m})$, while the space complexity is $\mathcal{O}(n + m)$. This suggests choosing $m = \Theta(n)$ to balance time and space: the expected query time is $\mathcal{O}(1)$ and the space complexity is $\mathcal{O}(n)$. See Table 2.2 for a summary of the time-space trade-off of different dictionary implementations.

However, this choice does not change the worst-case query time. In the worst case, $\Theta(n)$ key-value pairs may hash to the same bucket, so a query can still take $\mathcal{O}(n)$ time. In Section 3.1.3, we will see a more sophisticated hash table construction, called *FKS perfect hashing*, which achieves $\mathcal{O}(1)$ worst-case query time while using $\mathcal{O}(n)$ space.

Nevertheless, the current implementation of hash tables with chaining is already useful enough for real-world applications. Below we give an interesting application of hashing in data deduplication.

2.6.8 Example: Data Deduplication via Hashing

▶ Background

Large language models (LLMs) are trained on large amounts of text data collected from various sources, most importantly from the Internet. A common way to obtain Internet data is through large-scale web crawls, such as those provided by the non-profit organization Common Crawl[2]. However, raw data is often

[2] https://commoncrawl.org/

very noisy, so various data cleaning steps are required before it can be used for training LLMs.

Data deduplication is a crucial data cleaning step that aims to remove duplicate documents that can arise naturally on the web. For example, many websites act as aggregators or mirrors of other webpages, and many pages are generated from templates that contain largely identical content. In addition, duplicate pages may even be introduced by the crawler itself when different URLs point to the same webpage. Empirical studies have shown that removing duplicated content can reduce memorization of training data and improve model performance.

There are two major challenges in data deduplication:

- The number of documents is extremely large: each monthly Common Crawl snapshot contains billions of web pages and hundreds of trillions of tokens. Therefore, a direct pairwise comparison of all documents is not feasible, and the deduplication method must run in time roughly linear in the number of documents.
- Exact deduplication is not enough. If we only needed to remove documents that are byte-for-byte identical, hashing would provide a straightforward solution. In practice, however, two pages may contain almost the same content while differing by only minor edits. For LLM data curation, *fuzzy* deduplication is therefore more appropriate: we need to identify documents that are "almost the same," not just exact matches. This setting is substantially more challenging, since standard hash functions are highly sensitive to even small content changes.

▶ Problem Formulation

Given a corpus of n documents $D_1, \ldots, D_n$, we need to determine which pairs of documents are fuzzy duplicates. To this end, we split each document into words and extract all its 5-grams, where a 5-*gram* is a contiguous sequence of 5 words in the document.

For example, as shown in Figure 2.14, if a document is "I really like this movie because I really like this movie theater.", then its 5-grams are: (I, really, like, this, movie), (really, like, this, movie, because), (like, this, movie, because, I), (this, movie, because, I, really), (movie, because, I, really, like), (because, I, really, like, this), (really, like, this, movie, theater). Note that (I, really, like, this, movie) is repeated twice but only one copy is kept.

Let $S(D)$ be the set of 5-grams of document D. Define *Jaccard similarity* between two documents D_1 and D_2 as

$$J(D_1, D_2) = \frac{|S(D_1) \cap S(D_2)|}{|S(D_1) \cup S(D_2)|}. \tag{2.223}$$

That is, Jaccard similarity is the ratio of the number of common 5-grams to the total number of 5-grams in the two documents.

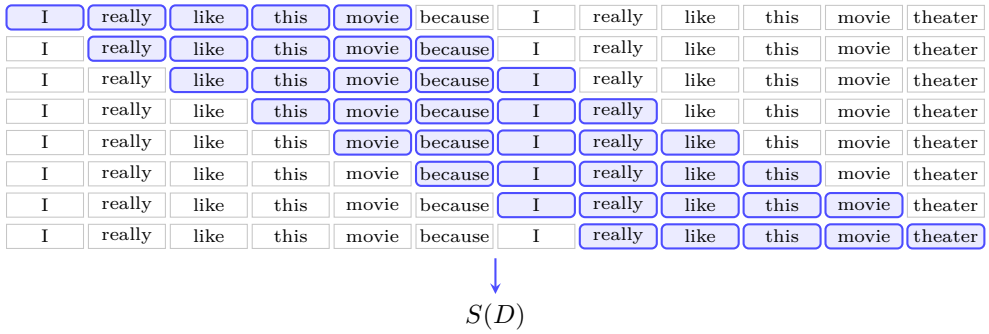


Figure 2.14: A 5-*gram* is obtained by sliding a length-5 window across the document one word at a time. The set $S(D)$ contains all 5-grams of the document.

Fix a threshold $\tau \in (0,1)$ and a tolerance $\epsilon \in (0,1)$. Given a corpus of n documents $D_1, \ldots, D_n$, we build a graph where each document is a node. For any two documents D_1 and D_2, we compute their Jaccard similarity $J(D_1, D_2)$ and discuss the following three cases:

- If $J(D_1, D_2) \geq \tau + \epsilon$, we add an edge between D_1 and D_2.
- If $J(D_1, D_2) \leq \tau - \epsilon$, we do nothing.
- If $\tau - \epsilon < J(D_1, D_2) < \tau + \epsilon$, it is both ok to add an edge between D_1 and D_2 and not to add an edge.

We then keep one representative from each connected component, i.e., each maximal set of documents that are connected by paths in the graph.

A natural question is: *Can the above deduplication procedure be implemented in a near-linear time in the number of documents?*

▶ Key Tool: MinHash

MinHash is a standard technique for fuzzy deduplication. It was introduced by Broder [Bro97] in 1997 to deduplicate web pages for web search. In the context of LLM training, OpenAI used MinHash to deduplicate the training data for the GPT-3 model [BMR+20], and it has since become a standard practice [PMH+23, PKa+24, Olm25].

The high-level idea of MinHash is to use hashing to group similar documents together. Each document is mapped to one or several hash values, and documents that share the same hash value are placed in the same bucket. The hash functions are constructed so that documents with high Jaccard similarity are more likely to collide. In particular, if $J(D, D') \geq \tau + \epsilon$, then D and D' are likely to share at least one hash value.

The basic form of MinHash is as follows. First, we collect all 5-grams appearing in the n documents in the corpus, and denote this set by $\mathcal{G}$. Next, we sample a

random permutation on $\mathcal{G}$, and construct a hash function h by mapping each 5-gram to its rank in the permutation. Finally, for each document D, the MinHash value of D is defined as

$$\text{MinHash}(D; h) = \min\{h(x) : x \in S(D)\}, \tag{2.224}$$

In other words, we keep only one number from a document, namely the minimum permuted rank among its 5-grams $S(D)$. If two documents share many 5-grams, they are more likely to share this minimum value. More precisely, we have the following theorem that characterizes the probability that two documents have the same MinHash value.

Theorem 2.103. For any two documents D, D' in the corpus,

$$\Pr[\text{MinHash}(D; h) = \text{MinHash}(D'; h)] = J(D, D'). \tag{2.225}$$

Proof. Let $U = S(D) \cup S(D')$. Under a random permutation, each element of U is equally likely to be the minimum-ranked element. The two MinHash values are equal exactly when this winner lies in $S(D) \cap S(D')$. Hence

$$\Pr[\text{MinHash}(D; h) = \text{MinHash}(D'; h)] = \frac{|S(D) \cap S(D')|}{|S(D) \cup S(D')|}, \tag{2.226}$$

which is exactly the Jaccard similarity between D and D'. □

▶ The Banding Method

However, this probability is only linear with the Jaccard similarity. Ideally, we would like a sharper behavior: the probability should be close to 0 when the Jaccard similarity is below the threshold τ, and close to 1 when it is above τ.

To achieve this, we combine multiple MinHash functions together. Specifically, we choose parameters b and r, and sample $b \cdot r$ i.i.d. random hash functions

$$h_j^{(i)} : \mathcal{G} \to \mathbb{Z}, \quad \text{for } i \in [b] \text{ and } j \in [r].$$

We group these hash functions into b *bands*, and each band contains r hash functions. Each hash function $h_j^{(i)}$ is obtained by sampling a random permutation of the 5-grams and mapping each 5-gram to its rank in the permutation.

For each document D, we compute the MinHash values of D for all hash functions $h_j^{(i)}$ and group them into b bands, each containing r values.

$$H(D) = \left(H^{(1)}(D), H^{(2)}(D), \ldots, H^{(b)}(D)\right) \in \mathbb{Z}^{br}, \tag{2.227}$$

$$H^{(i)}(D) = \left(H_1^{(i)}(D), H_2^{(i)}(D), \ldots, H_r^{(i)}(D)\right) \in \mathbb{Z}^r \quad \text{for } i \in [b], \tag{2.228}$$

$$H_j^{(i)}(D) = \text{MinHash}(D; h_j^{(i)}) \in \mathbb{Z} \quad \text{for } i \in [b] \text{ and } j \in [r]. \tag{2.229}$$

We call this vector $H(D)$ the *MinHash signature* of D.

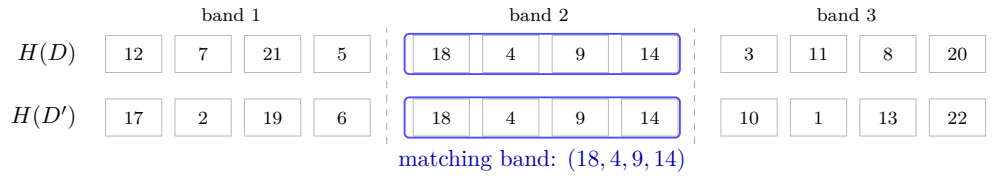


Figure 2.15: Two MinHash signatures collide when there exists a band whose entries match exactly. This happens if and only if the two documents are mapped to the same bucket in the hash table $\mathcal{T}_i$ for that band.

For two documents D and D', we say that their MinHash signatures *collide* if there exists an index $i \in [b]$ such that

$$H^{(i)}(D) = H^{(i)}(D').$$

We write $H(D) \sim H(D')$ when the two signatures collide. See Figure 2.15 for an illustration. The following theorem characterizes the probability that two MinHash signatures collide.

Theorem 2.104. For any two documents D, D' in the corpus,

$$\Pr[H(D) \sim H(D')] = 1 - \left(1 - J(D, D')^r\right)^b. \tag{2.230}$$

Proof. Let A_i be the event that $H^{(i)}(D) = H^{(i)}(D')$. Since the hash functions in different bands are sampled independently, the events $A_1, \ldots, A_b$ are mutually independent. Therefore,

$$\Pr[H(D) \sim H(D')] = \Pr[A_1 \cup \cdots \cup A_b] = 1 - \Pr\left[\bar{A}_1 \cap \cdots \cap \bar{A}_b\right] \tag{2.231}$$

$$= 1 - \prod_{i=1}^{b} \Pr[\bar{A}_i]. \tag{2.232}$$

Since A_i is the event that $H(D; h_j^{(i)}) = H(D'; h_j^{(i)})$ holds for all $j \in [r]$, and the hash functions $h_1^{(i)}, \ldots, h_r^{(i)}$ are sampled independently, we have

$$\Pr[A_i] = \prod_{j=1}^{r} \Pr\left[H(D; h_j^{(i)}) = H(D'; h_j^{(i)})\right] = J(D, D')^r. \tag{2.233}$$

Therefore, $\Pr[H(D) \sim H(D')] = 1 - \left(1 - J(D, D')^r\right)^b$ □

▶ Algorithm

Now we introduce the algorithm for data deduplication using MinHash. The high-level idea is to construct b hash tables, where each hash table corresponds to a band of the MinHash signature. By placing b copies of each document into these b hash tables according to their MinHash values for each band, we can group documents with colliding MinHash signatures together, and then we can identify the connected components of the graph of documents.

More specifically, the algorithm works as follows.

- Compute the MinHash signature $H(D)$ for each document D.
- Build an empty graph G with n vertices.
- For each band $i \in [b]$:
  - Build an empty hash table $\mathcal{T}_i$ with $m = \Theta(n)$ buckets.
  - For each document D:
    - If the key $H^{(i)}(D)$ is already present in $\mathcal{T}_i$, select one stored document D' and add an edge between D and D' in G.
    - Otherwise, insert D into $\mathcal{T}_i$ with key $H^{(i)}(D)$.
- Identify all connected components of G using breadth-first search (BFS).
- Return the documents grouped by these connected components.

The following theorem establishes the correctness of the algorithm.

Theorem 2.105. If two documents D, D' satisfy $H(D) \sim H(D')$, then they lie in the same connected component of G.

Proof. Suppose $H(D) \sim H(D')$. Then there exists a band $i \in [b]$ such that $H^{(i)}(D) = H^{(i)}(D')$. During the processing of band i, all documents with this key are stored in the same bucket of $\mathcal{T}_i$. Whenever a new document D with this key is encountered, the algorithm adds an edge between D and one previously stored document with the same key. Therefore all documents with this key form a connected subgraph of G, thus lying in the same connected component. □

▶ Parameter Selection

The above analysis only establishes a partial correctness guarantee: if two signatures collide, then the corresponding documents are placed in the same connected component. However, our original goal is stronger: we want documents with Jaccard similarity at least $\tau + \epsilon$ to be grouped together, while documents with Jaccard similarity at most $\tau - \epsilon$ should be filtered out. Achieving this separation requires a careful choice of the parameters r and b.

For a document pair with Jaccard similarity s, by Theorem 2.104, the collision probability is

$$P_{r,b}(s) = \Pr[H(D) \sim H(D')] = 1 - (1 - s^r)^b. \tag{2.234}$$

When s is small, s^r is tiny, so the probability stays low. When s is large, one minus a small number to the b-th power becomes close to 1. So the curve of $P_{r,b}(s)$ is of "S-shape". Some examples of $P_{r,b}(s)$ are plotted in Figure 2.16.

A general rule of thumb is to choose r large enough and $b \sim \tau^{-r}$. With this choice, the transition of $P_{r,b}(s)$ occurs near $s = \tau$, which can be seen by the

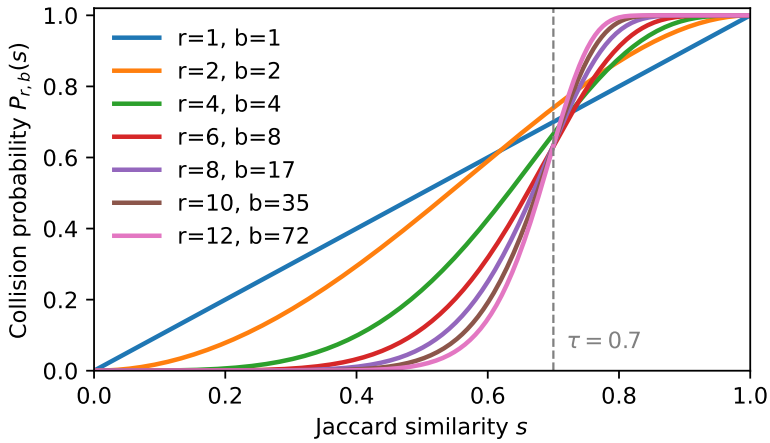


Figure 2.16: Collision probability $P_{r,b}(s) = 1 - (1 - s^r)^b$ for various r and b. For $\tau = 0.7$, when $b \approx \tau^{-r}$, the curve becomes increasingly sharp around $s = \tau$ as r increases.

following approximation:

$$P_{r,b}(s) \approx 1 - \exp(-bs^r) \approx 1 - \exp(-(s/\tau)^r). \tag{2.235}$$

As $r \to \infty$, $P_{r,b}(s)$ approaches the following threshold function:

$$P_{r,b}(s) \to \begin{cases} 0 & \text{if } s < \tau, \\ 1 & \text{if } s > \tau, \end{cases} \tag{2.236}$$

A more careful analysis can be carried out to determine suitable values of r and b such that $P_{r,b}(\tau - \epsilon) \leq \eta$ and $P_{r,b}(\tau + \epsilon) \geq 1 - \eta$, where η is a small error probability. But we omit the details here.

Beyond data deduplication, the MinHash approach we introduced above can be seen as a special case of a more general concept, *locality-sensitive hashing* (LSH), which is also useful for approximate nearest neighbor search in high-dimensional spaces. We refer the reader to [LRU20] for more details.

Exercises

2.1. Prove Equation (2.29) in Lemma 2.9: $\mathcal{O}(f(n))+\mathcal{O}(g(n)) = \mathcal{O}(f(n)+g(n))$.

2.2. Prove Equation (2.32) in Lemma 2.9: $\mathcal{O}(f(n)g(n)) = f(n) \cdot \mathcal{O}(g(n))$.

2.3. Prove that if $g(n) > 0$ for all $n \in \mathbb{N}$, then $f(n) = \mathcal{O}(g(n))$ for all $n \in \mathbb{N}$ is equivalent to $f(n) = \mathcal{O}(g(n))$ as $n \to \infty$.

2.4. Let $f : D \to \mathbb{R}$ be an arbitrary function defined on a domain D with finite number of elements. Show that $f(n) = \mathcal{O}(g(n))$ for all $n \in D$, as long as the function $g : D \to \mathbb{R}$ satisfies $g(n) > 0$ for all $n \in D$.

(So the time complexity of sorting at most $10^{998244353}$ numbers is $\mathcal{O}(1)$.)

2.5. Prove that the two definitions (Definitions 2.6 and 2.24) of big-$\mathcal{O}$ notation are equivalent when $g : \mathbb{N} \to \mathbb{R}$ is a function that is positive for all $n \in \mathbb{N}$.

2.6. Prove that $\sum_{k=1}^{n} k^2 = \Theta(n^3)$. If you use $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$, you must also prove this identity.

2.7. For the alien birthday paradox, prove that Theorem 2.31 holds for all $n \geq 1$.

Important Note: Be careful with asymptotic notation when splitting into cases. It is wrong to split into $n = \mathcal{O}(\sqrt{m})$ vs. $n = \omega(\sqrt{m})$. This is because assuming $n = \mathcal{O}(\sqrt{m})$ means that $n \leq C\sqrt{m}$ for some constant $C > 0$, and subsequent arguments depend on this particular constant C. However, a value such as $n = (C + 1)\sqrt{m}$ is neither $\leq C\sqrt{m}$ nor $\omega(\sqrt{m})$. Instead, a valid split could be $n \leq C\sqrt{m}$ vs. $n > C\sqrt{m}$ for some constant $C > 0$, and later hide the constant C by using $\mathcal{O}(\sqrt{m})$ vs. $\Omega(\sqrt{m})$. (The threshold $\sqrt{m}$ here is used for illustration and you may use a different one if you want.)

2.8. For the alien birthday paradox, derive the probability of the following event: there are exactly $n - 2$ distinct birthdays among n aliens, where exactly one birthday is shared by three aliens. Then, approximate this probability as a simple function of n and m with error $o(1)$ for all $n \geq 1$ as $m \to \infty$.

2.9. Give an example of a function $f : \mathbb{N}^2 \to \mathbb{R}$ such that $f(m, n) = \mathcal{O}(g(m, n))$ as $m, n \to \infty$ for some function $g : \mathbb{N}^2 \to \mathbb{R}$ with $g(m, n) > 0$ but the sum $\sum_{k=1}^{n} f(m, k)$ is not $\mathcal{O}(\sum_{k=1}^{n} g(m, k))$.

2.10. In a New Year's party with $2n$ people, k random names are picked to receive gifts. Assume that these $2n$ people are actually n couples. Let $r_{n,k}$ be the probability that there exists at least one couple such that both people in the couple receive gifts.

(a) Express $r_{n,k}$ as a mathematical formula of n and k.

(b) Determine the value of $\lim_{n\to\infty} r_{n,\lfloor\sqrt{n}\rfloor}$.

2.11. Prove Statement 2 of Theorem 2.45, where $k^* := \arg\max_{1\le k<n} \{p_{n,k}\}$. You may receive partial credit by proving $k^* = n/e + o(n)$.

2.12. For the locker puzzle, let h_n be the success probability of the pointer chasing strategy when every student is allowed to open at most $n/3$ lockers (n is a multiple of 3). Determine the value of $\lim_{n\to\infty} h_n$.

Note: As the answer may not be expressible as an elementary function, simplify your answer as much as possible. Also provide a numerical approximation rounded to 2 decimal places.

2.13. Let p be a prime number. In number theory, one definition of a prime number is as follows: a positive integer $p > 1$ is called a prime number if for all integers a and b, if $p \mid ab$, then $p \mid a$ or $p \mid b$. Here $p \mid a$ means that p divides a (or equivalently, a is a multiple of p).

Prove the following statements for all $a \in \mathbb{Z}_p \setminus \{0\} = \{1, 2, \ldots, p-1\}$:

(a) The congruence

$$ax \equiv 0 \pmod{p},$$

has exactly one solution x among $\{0, 1, \ldots, p-1\}$, which is $x = 0$.

Hint: Use the definition of prime numbers and the fact that $ax \equiv 0 \pmod{p}$ is equivalent to $p \mid ax$.

(b) The following set equality holds:

$$\{ax \bmod p : x \in \mathbb{Z}_p \setminus \{0\}\} = \mathbb{Z}_p \setminus \{0\}.$$

(c) The following congruence holds, known as Fermat's little theorem:

$$a^{p-1} \equiv 1 \pmod{p}.$$

Hint: Multiply all the elements in $\{ax \bmod p : x \in \mathbb{Z}_p \setminus \{0\}\}$ together.

(d) There exists a unique $b \in \mathbb{Z}_p \setminus \{0\}$ such that $ab \equiv 1 \pmod{p}$.

(After proving this, you can use the notation a^{-1} to denote this unique number b and call it the *multiplicative inverse* of a modulo p.)

Background on number theory: If you find this exercise difficult or just want to learn more about number theory, you may refer to Chapter 4 (Number Theory) of the book *Concrete Mathematics* [GKP94]. The definition of prime numbers used there is different from the one we give here, but the two definitions are equivalent. Our definition is in fact consistent with the more general notion of *prime elements* in ring theory.

2.14. Prove Proposition 2.54.

Hint: Do the previous exercise first.

2.15. The *polynomial rolling hash* family is a popular hash family for hashing strings. A string in computer science is typically represented as a sequence of characters from a finite alphabet, such as the ASCII or Unicode character set. Each character is mapped to a positive number in $[u]$ for some integer $u \geq 2$, and a string of length ℓ is then represented as a sequence of positive numbers $(x_0, x_1, \ldots, x_{\ell-1}) \in [u]^\ell$. Note that strings only contain positive numbers, since zero is usually used as the end of string marker. A polynomial rolling hash function is defined as

$$h_{\alpha,p}((x_i)_{i=0}^{\ell-1}) = \left(\sum_{i=0}^{\ell-1} \alpha^{\ell-1-i} x_i\right) \bmod p,$$

where p is a prime number with $p > u$.

In Python, this hash function can be implemented as:

```
def polynomial_rolling_hash(x, a, p):
    h = 0
    for i in range(len(x)):
        h = (h * a + x[i]) % p
    return h
```

Based on $h_{\alpha,p}$, we define two hash functions that map from $[u]^\ell$ to a smaller hash space $\mathbb{Z}_m$, where m can be much smaller than p and may not be prime.

$$h^{(1)}_{\alpha,p,m}((x_i)_{i=0}^{\ell-1}) := h_{\alpha,p}((x_i)_{i=0}^{\ell-1}) \bmod m,$$
$$h^{(2)}_{\alpha,a,b,p,m}((x_i)_{i=0}^{\ell-1}) := \left(\left(a \cdot h_{\alpha,p}((x_i)_{i=0}^{\ell-1}) + b\right) \bmod p\right) \bmod m.$$

Let $\mathcal{H}^{(1)}_{p,m}$ and $\mathcal{H}^{(2)}_{p,m}$ be the corresponding hash families.

$$\mathcal{H}^{(1)}_{p,m} := \left\{h^{(1)}_{\alpha,p,m} : \alpha \in \mathbb{Z}_p \setminus \{0\}\right\},$$
$$\mathcal{H}^{(2)}_{p,m} := \left\{h^{(2)}_{\alpha,a,b,p,m} : \alpha \in \mathbb{Z}_p \setminus \{0\}, a \in \mathbb{Z}_p \setminus \{0\}, b \in \mathbb{Z}_p\right\}.$$

For each of the two hash families, try to find a universal constant $c > 0$ such that the following holds. For any integers p, m, L satisfying $u < m \leq p$ and $L \geq 1$, and for any two distinct strings $(x_i)_{i=0}^{\ell-1}$ and $(x'_i)_{i=0}^{\ell'-1}$ with $\ell, \ell' \leq L$,

$$\Pr\left[h((x_i)_{i=0}^{\ell-1}) = h((x'_i)_{i=0}^{\ell'-1})\right] \leq \frac{cL}{m},$$

where h is sampled uniformly at random from the corresponding hash family. If you cannot find such a constant, give a counterexample.

Note: The term *universal constant* refers to a constant that does not depend on any of the parameters, such as u, p, m, L, ℓ, ℓ'.

Hint: You can use the following fact: If p is a prime and $f(x)$ is a d-degree polynomial with integer coefficients, then either (1) all the coefficients of $f(x)$ are divisible by p, or (2) the congruence $f(x) \equiv 0 \pmod p$ has at most d solutions among $x \in \{0, 1, \ldots, p-1\}$.

2.16. Let $R(k, \ell)$ be the least number N such that every red/blue coloring of the edges of K_N contains a red K_k or a blue K_ℓ.

(a) Prove that $R(k, \ell)$ exists for all $k, \ell \geq 1$ and satisfies the following recurrence relation:

$$R(k, \ell) \leq R(k-1, \ell) + R(k, \ell-1).$$

Hint: Read the proof of Theorem 2.59 for inspiration.

(b) Prove the following upper bound on $R(k, k)$ for all $k \geq 2$:

$$R(k, k) \leq \binom{2k-2}{k-1} < 4^k$$

2.17. Denote by $R_r^{(m)}(k)$ the least number N such that Theorem 2.62 holds. Prove that $R_r^{(m)}(k) > r^{\frac{1}{k}\left(\binom{k}{m}-1\right)}$ for all $r, k, m \geq 2$.

2.18. Let $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ be n vectors in $\mathbb{R}^d$ with $\|\boldsymbol{x}_i\|_2 \leq 1$ for all $i \in [n]$. Prove that there exist $\epsilon_1, \epsilon_2, \ldots, \epsilon_n \in \{-1, +1\}$ such that the following holds:

$$\left\| \sum_{i=1}^{n} \epsilon_i \boldsymbol{x}_i \right\|_2 \leq \sqrt{n}.$$

Hint: Wait, why is this an exercise in probability theory?

2.19. Let A and B be two events in a probability space $\mathcal{P} = (\Omega, p)$, each with non-zero probability. we say that A *attracts* B if $\Pr[B \mid A] > \Pr[B]$. That is, A makes B more likely to happen. Conversely, we say that A *repels* B if $\Pr[B \mid A] < \Pr[B]$. That is, A makes B less likely to happen.

(a) Prove that A attracts B if and only if B attracts A.

(b) In a poker game of two players, each player samples a random hand of 5 cards from a standard deck of 52 cards. Let A be the event that the first player has a royal flush, and B be the event that the second player has a royal flush. Here, a royal flush means you get a hand of A, K, Q, J, 10 of the same suit (e.g., Spade). There are only 4 possible royal flushes in a deck. Prove that A attracts B.

Takeaway: If you get a royal flush, don't get too excited, because your opponent is also more likely to get a royal flush!

2.20. Let A, B, C, D be four events in a probability space $\mathcal{P} = (\Omega, p)$. For example, A is "admitted to Tsinghua", B is "good at biology", C is "good at Chinese", D is "born in Dongbei". Suppose we know the following:

- $\Pr[A \mid B \cap D] > \Pr[A \mid C \cap D]$.

  In our example, this means that among Dongbei students, those who are good at biology are more likely to be admitted to Tsinghua than those who are good at Chinese.

- $\Pr[A \mid B \cap \bar{D}] > \Pr[A \mid C \cap \bar{D}]$.
  In our example, this means that among non-Dongbei students, those who are good at biology are more likely to be admitted to Tsinghua than those who are good at Chinese.

Does this imply that $\Pr[A \mid B] > \Pr[A \mid C]$? If yes, prove this inequality. If no, give a counterexample.

Note: This is called *Simpson's paradox* in statistics.

2.21. Let X and Y be two random variables in a probability space $\mathcal{P} = (\Omega, p)$. Suppose $X(\omega) \leq Y(\omega)$ for all $\omega \in \Omega$. Prove that $\mathbb{E}[X] \leq \mathbb{E}[Y]$.

2.22. A *rooted tree* is a connected (undirected) graph with no cycles together with a distinguished vertex called the *root*. The *descendants* of a vertex v are the vertices u such that v lies on the path from u to the root.

Consider the following random process. We start with a rooted tree T with n vertices, where 1 is the root. At each step, we choose uniformly at random a vertex v and delete v and all its descendants from the tree. The process stops when the tree becomes empty. Let X be the number of steps we make before the process stops. Compute $\mathbb{E}[X]$.

Note: In your expression for $\mathbb{E}[X]$, you may use $d(v, u)$ to denote the distance between vertices v and u in the tree T before the process starts.

2.23. Suppose there are n different types of coupons. Each draw independently returns a coupon whose type is uniformly distributed over these n types.

(a) Let T_i be the number of draws until you obtain a coupon of type i for the first time. Compute $\mathbb{E}[T_i]$.

(b) Let T be the number of draws until you obtain all n types of coupons for the first time. That is, $T = \max_{i \in [n]} T_i$. Compute $\mathbb{E}[T]$ and express the result as a simple function of n up to $\mathcal{O}(1)$ error.

Note: Since T_i can be arbitrarily large, the underlying probability space is infinite. Although this chapter focuses on finite discrete probability spaces, all the definitions and results extend naturally to the infinite discrete case. If you feel uncomfortable to work with an infinite discrete probability space, you may instead assume that all coupons are automatically obtained at the M-th draw, derive the result under this assumption, and then take $M \to \infty$.

2.24. Let $\mathrm{Perm}(2n)$ be the set of all $(2n)!$ permutations of $[2n]$. For any $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_{2n}) \in \mathrm{Perm}(2n)$, we say that a pair of positions (i, j) with $i < j$ is an *inversion* of σ if $\sigma_i > \sigma_j$. For example, the permutation $(2, 4, 1, 3)$ has three inversions: $(1, 3)$, $(2, 3)$, and $(2, 4)$.

For any $\sigma \in \mathrm{Perm}(2n)$, let $f(\sigma)$ be the permutation obtained from σ by sorting the sublist of *odd* positions in increasing order. For example, if $\sigma = (3, 8, 2, 5, 6, 7, 1, 4)$, then $f(\sigma) = (1, 8, 2, 5, 3, 7, 6, 4)$.

Let X be the number of inversions of $f(\sigma)$ when σ is chosen uniformly at random from $\mathrm{Perm}(2n)$. Compute $\mathbb{E}[X]$.

Hint: For $n = 1, 2, 3$, the values of $\mathbb{E}[X]$ are $1/2, 13/6, 5$, respectively.

2.25. Let $\beta = 0.7$. For every $n \geq 10$, show that there exists a probability space $\mathcal{P} = (\Omega, p)$ and n random variable $X_1, X_2, \ldots, X_n$ on $\mathcal{P}$ such that

- $\Pr[X_i > X_{i+1}] \geq \beta$ for all $i = 1, 2, \ldots, n-1$;
- $\Pr[X_n > X_1] \geq \beta$.

Furthermore, for each $n \geq 10$, determine the largest possible value of β for which such a construction is possible.

2.26. Let X and Y be two random variables and $f : \mathbb{R} \to \mathbb{R}$ be a function. Simplify the following two expressions that might confuse you:

(a) $\mathbb{E}[\mathbb{E}[X \mid f(Y)] \mid Y]$;
(b) $\mathbb{E}[\mathbb{E}[X \mid Y] \mid f(Y)]$.

Note: The result is usually referred to as the *tower property* of conditional expectation, and is also known as the *tower rule*.

2.27. Let $\Omega = \{2^k : k \in \{0, 1, 2, \ldots, m\}\}$, where $m = 10^5$. Alice chooses a random number $X \in \Omega$ uniformly at random. Since Alice is very rich, she puts X RMB in one envelope and $2X$ RMB in the other envelope. Bob chooses one envelope uniformly at random and opens it to see the amount of money inside. Bob can then decide to keep the money or switch to the other envelope.

Let Y be the amount of money inside Bob's envelope, Z be the amount of money inside the other envelope, and M be the final amount of money Bob gets according to his decision. Bob wants to maximize the expected value of M, i.e., $\mathbb{E}[M \mid Y]$.

Bob thinks for a while and then concludes that he should switch to the other envelope no matter what Y he sees. The reason is as follows:

- There is equal probability that $Y = X$ and $Y = 2X$.
- If $Y = X$, then $Z = 2Y$. If $Y = 2X$, then $Z = Y/2$.
- By the law of total expectation, $\mathbb{E}[Z \mid Y] = \frac{1}{2} \cdot 2Y + \frac{1}{2} \cdot \frac{Y}{2} = 1.25Y > Y$.
- Therefore, setting $M = Z$ leads to a higher expected value of M.

Is Bob's conclusion correct? If yes, just say yes. If no, compute $\mathbb{E}[Z \mid Y]$.

Note: This is a variant of the *two envelopes problem* in probability theory.

2.28. Let $X_1, X_2, \ldots, X_n$ be n i.i.d. random variables that are drawn uniformly at random from $\{-1, 1\}$. Are the following random variables also i.i.d.? Prove your answer.

$$Y_1 = X_1, \quad Y_2 = X_1 X_2, \quad \ldots, \quad Y_n = X_1 X_2 \cdots X_n.$$

2.29. For any two binary vectors $x, y \in \{0,1\}^d$, define the Hamming distance as

$$\mathrm{dist}_{\mathrm{H}}(x,y) = \sum_{i=1}^{d} \mathbb{1}_{[x_i \neq y_i]}.$$

Construct a family $\mathcal{H}$ of hash functions $h : \{0,1\}^d \to \{0,1\}$ such that for any two binary vectors $x, y \in \{0,1\}^d$, when h is sampled uniformly at random from $\mathcal{H}$,

$$\Pr[h(x) = h(y)] = 1 - \frac{1}{d}\mathrm{dist}_{\mathrm{H}}(x,y).$$

Chapter 3

Sampling and Concentration

So far, we have built a probabilistic language for describing randomness. We introduced probability spaces, events, random variables, and expectation, and we learned how to compute these quantities within the model. But in practice, we do not interact with a probability space directly. What we actually observe are samples drawn from the underlying distribution.

For example, imagine standing beside a table and watching a fair coin being tossed again and again. After 10,000 tosses, you already know the rough answer before doing any counting: the number of heads should be close to 5,000. This intuition is immediate. What is less immediate is how much confidence we should place in such a prediction. Should we regard 5,000 as a very sharp prediction, or just as a very rough estimate? How surprising would it be to see 5,500 heads, or even 6,000 heads?

This leads to the central question of the chapter: *If probability theory tells us that some quantity has a certain expectation or variance, what does that imply about the outcomes we actually observe?* This question is fundamental not only in probability theory but also in the design and analysis of algorithms. In many randomized algorithms, randomness enters through sampled inputs or through random decisions made during execution. To analyze such algorithms, we must understand how the behavior predicted by the probabilistic model relates to the outcomes produced by actual samples.

Concentration inequalities provide essential tools for this purpose. Let X be a random variable. Concentration inequalities provide mathematical bounds on the probability that X deviates significantly from its expectation $\mathbb{E}[X]$, or from some other specified value. In other words, concentration inequalities aim to bound probabilities of the form

$$\Pr\Big[X \text{ is far from } \mathbb{E}[X]\Big]. \tag{3.1}$$

We begin with *Markov's inequality*, which gives a first tail bound using only expectation and the assumption of non-negativity. Building on this, we will introduce *Chebyshev's inequality* and the *Chernoff bound*, which strengthen the

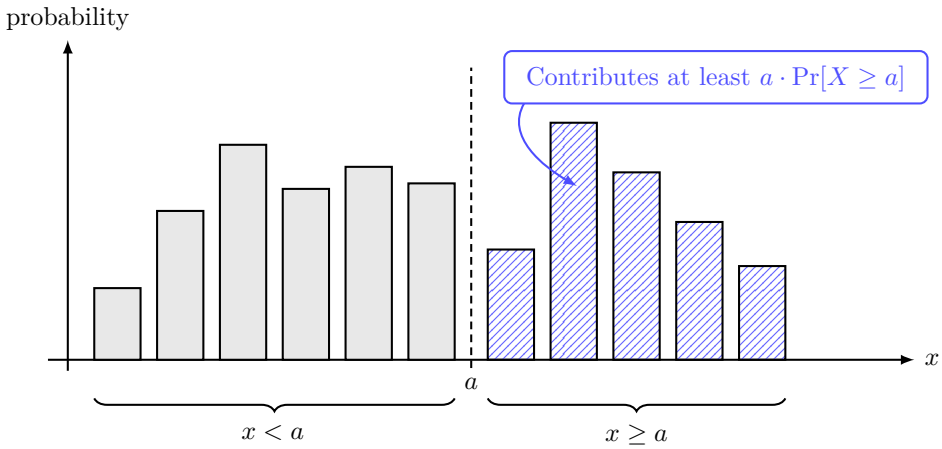


Figure 3.1: Key insight behind Markov's inequality. The shaded area to the right of a is the probability that $X \geq a$. In this tail, X takes values at least a, so it alone already contributes at least $a \cdot \Pr[X \geq a]$ to the expectation.

bounds by incorporating information about higher moments of the distribution. Finally, we will discuss several important extensions and instantiations of Chernoff bounds, including *Hoeffding's inequality* and *Bernstein's inequality*.

3.1 Markov's Inequality

The first concentration inequality we will study is *Markov's inequality*. We start by stating the inequality and proving it, and then discuss some applications.

3.1.1 Statement and Proof

Markov's inequality provides bounds on the concentration of probability mass around the expectation using a weakest kind of information one can reasonably hope to know about a random variable: it is non-negative.

The key insight is that if X is non-negative and its expectation is small, then X cannot spend too much probability mass on very large values. Otherwise, its expectation would be too large. More formally, we have the following theorem.

Theorem 3.1 (Markov's inequality)**.** If X is a non-negative random variable, i.e., $\Pr[X \geq 0] = 1$, then for every $a > 0$,

$$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}. \tag{3.2}$$

Proof. We use the law of total expectation to split the expectation into two parts according to whether $X \geq a$ or $X < a$:

$$\mathbb{E}[X] = \Pr[X \geq a]\,\mathbb{E}[X \mid X \geq a] + \Pr[X < a]\,\mathbb{E}[X \mid X < a]. \tag{3.3}$$

When $X \geq a$, the random variable X is at least a, so

$$\mathbb{E}[X \mid X \geq a] \geq a. \tag{3.4}$$

No matter whether $X \geq a$ or $X < a$, the random variable X is non-negative, so

$$\mathbb{E}[X \mid X < a] \geq 0. \tag{3.5}$$

Therefore,

$$\mathbb{E}[X] \geq a \Pr[X \geq a]. \tag{3.6}$$

Rearranging the inequality, we get $\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$. □

There is also a useful geometric picture behind the proof. See Figure 3.1. Imagine the distribution of X as spreading probability mass over the non-negative line. If a significant fraction of that mass sits to the right of a, then every point in that region contributes at least a to the expectation. Markov's inequality just says that this contribution cannot exceed the expectation itself, $\mathbb{E}[X]$.

The following is a direct corollary of Markov's inequality.

Corollary 3.2. If X is a non-negative random variable, then for every $c > 0$,

$$\Pr[X \geq \alpha \, \mathbb{E}[X]] \leq \frac{1}{\alpha}. \tag{3.7}$$

Proof. Apply Markov's inequality (Theorem 3.1) with $a = \alpha \, \mathbb{E}[X]$. □

This rescaled form of Markov's inequality conveys a simple and useful message: the probability that a non-negative random variable is at least c times larger than the mean is at most $1/c$. For example, being 10 times larger than the mean has probability at most $1/10$, being 100 times larger than the mean has probability at most $1/100$, and so on. The following example illustrates how this bound can be used to analyze randomized algorithms.

Example 3.3 (Should we worry about very long chains in hash tables?)**.** In Section 2.6.7, we saw that hashing with chaining gives expected $\mathcal{O}(1)$ query time. A natural concern is whether a particular lookup might still encounter a very long chain.

Set $m = n - 1$ for our hash table. Let Z_i be the number of key-value pairs that hash to the same bucket as k_i. Lemma 2.102 gives the following upper bound on the expectation of Z_i:

$$\mathbb{E}[Z_i] \leq 1 + \frac{n-1}{m} = 2. \tag{3.8}$$

By Markov's inequality, for every $\delta > 0$,

$$\Pr[Z_i \geq 2/\delta] \leq \Pr[Z_i \geq \mathbb{E}[Z_i]/\delta] \leq \delta. \tag{3.9}$$

So extremely long chains cannot happen too often.

Remark 3.4. In Markov's inequality, the non-negativity assumption on X is crucial. If you do not see why, consider the following example, known as a *Rademacher random variable*:

$$X := \begin{cases} 1 & \text{with probability } 1/2, \\ -1 & \text{with probability } 1/2, \end{cases} \tag{3.10}$$

Then $\mathbb{E}[X] = 0$, but $\Pr[X \geq 1] = 1/2$.

3.1.2 Example: Turning Las Vegas into Monte Carlo

The chaining example above reveals a distinction that appears often in randomized algorithms. Sometimes randomness affects only how long the algorithm runs. Sometimes it affects whether the final answer is correct.

In our hash-table example, if we keep scanning the entire chain, then the lookup always returns the correct answer. What randomness changes is only the time it takes, because the chain length depends on the hash function and on the stored keys. This algorithm belongs to a more general class of algorithms called *Las Vegas algorithms*, which are algorithms that are guaranteed to return a correct answer, but the running time is a random variable.

There is also a different possibility. An algorithm may be required to finish within a specific time bound, but be allowed to make an error with small probability. *Monte Carlo algorithms* are algorithms that are guaranteed to return an answer within a prescribed time bound, but may return an incorrect answer with small probability.

The difference between Las Vegas and Monte Carlo algorithms is that for a Las Vegas algorithm, the uncertainty lies in the running time, whereas for a Monte Carlo algorithm, the uncertainty lies in the correctness of the output.

The same chaining example also explains how Las Vegas algorithms can be turned into Monte Carlo algorithms. Let T denote the running time of a Las Vegas algorithm. If we impose a cutoff $T_{\max}$ and force the algorithm to stop after $T_{\max}$ steps, then the new algorithm always finishes within time $T_{\max}$. But the new algorithm is no longer Las Vegas and can fail when $T > T_{\max}$. By Markov's inequality, the failure probability is at most

$$\Pr[T > T_{\max}] \leq \frac{\mathbb{E}[T]}{T_{\max}}. \tag{3.11}$$

Therefore, a Las Vegas algorithm with expected running time $\mathcal{O}(f(n))$ for some function $f(n)$ can be turned into a Monte Carlo algorithm that always finishes within time $\mathcal{O}(f(n)/\delta)$ and fails with probability at most δ.

Remark 3.5. For anyone who wonders what Las Vegas and Monte Carlo mean, they are not mathematical terms at all! In fact, they are the names of two places. Las Vegas is a city located in Nevada in the United States, and Monte Carlo is a district of Monaco (a microstate in Western Europe). Both places are known

for their casinos. Search on the Internet to find out who coined these algorithm classes and gave them these funny names.

3.1.3 Example: FKS Perfect Hashing

Now we continue with our discussion on the efficient implementation of a dictionary. We already know that simple hashing without collision resolution gives worst-case $\mathcal{O}(1)$ query time but uses $\Theta(n^2)$ space. Hashing with chaining gives expected $\mathcal{O}(1)$ query time and uses $\mathcal{O}(n)$ space, but the worst-case query time can be as large as $\mathcal{O}(n)$. Now we ask for a stronger guarantee: can every lookup run in worst-case $\mathcal{O}(1)$ time while the total space remains $\mathcal{O}(n)$?

As mentioned in Remark 2.66, Yao [Yao81] proved that when the universe is sufficiently large, this is impossible if we use no additional space beyond what is needed to store the keys. However, if we are allowed to increase the space by a constant factor, a breakthrough result of Fredman, Komlós, and Szemerédi [FKS84] shows that this goal can in fact be achieved by a two-level hashing scheme, which is called *FKS perfect hashing*.

▶ Construction of FKS Perfect Hashing

The key idea is to combine the two hashing schemes we have seen so far. See Figure 3.2. At the first level, we use the same idea as chaining: a hash function distributes the keys into buckets. But instead of using a linked list to handle collisions, at the second level, we give each bucket its own private hash table with sufficient hash space to remove collisions entirely, which is achieved by scaling the hash space size quadratically with the number of keys in the bucket. As a result, lookups in the second-level table take deterministic worst-case $\mathcal{O}(1)$ time, leading to an overall deterministic worst-case lookup time of $\mathcal{O}(1)$.

More specifically, the construction of FKS perfect hashing proceeds as follows. Let $(k_1, v_1), \ldots, (k_n, v_n)$ be a set of n key-value pairs to be stored in the hash table, where each key k_i lies in the key space K. At the first level, we set the hash space size to be exactly the number of keys to be stored, which is n. We sample a hash function $h^{(1)} : K \to \{0, 1, \ldots, n-1\}$ from a universal hash family $\mathcal{H}^{(1)}$ to distribute the keys into n buckets $B_0, \ldots, B_{n-1}$ according to the hash values. That is, for each bucket j,

$$B_j := \left\{ k_i : h^{(1)}(k_i) = j \right\}. \tag{3.12}$$

Then we build the second-level hash tables for each bucket B_j. Let $n_j := |B_j|$ be the number of keys in bucket B_j. We set the hash space size to be the square of the number of keys in the bucket, which is $m_j := n_j^2$, and sample a hash function $h^{(2,j)} : B_j \to \{0, 1, \ldots, m_j - 1\}$ from a universal hash family $\mathcal{H}^{(2,j)}$ to distribute the keys in B_j into m_j cells according to the hash values.

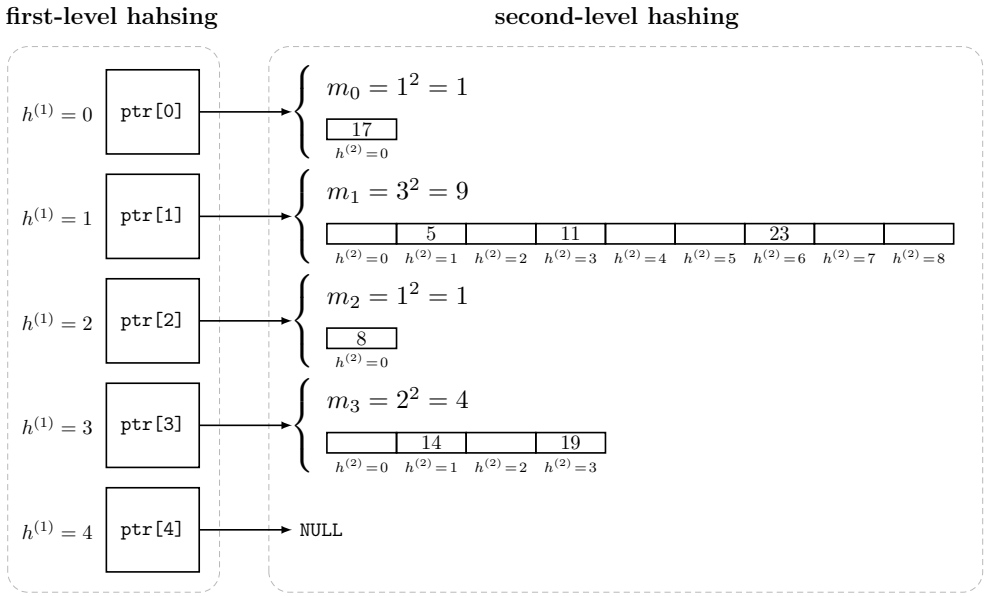


Figure 3.2: An illustration of FKS perfect hashing. Each first-level bucket points to its own second-level hash table, each of which is large enough to contain all the keys in the bucket without any collisions.

The total number of cells consumed by the construction is

$$M := \sum_{j=1}^{n} m_j = \sum_{j=1}^{n} n_j^2. \tag{3.13}$$

If $M \geq 4n$, we resample the hash functions and repeat the construction process until $M < 4n$.

Finally, for each key k_i in bucket B_j, we store the key-value pair (k_i, v_i) in cell $h^{(2,j)}(k_i)$ of the second-level table for bucket j.

At the query time, given a query key k, we first compute $j = h^{(1)}(k)$ to locate the corresponding first-level bucket. Then we compute the second-level address $h^{(2,j)}(k)$ inside bucket j. Finally, we check if the cell at address $h^{(2,j)}(k)$ matches the query key k. If so, we return the corresponding value v_i.

▶ Analysis of FKS Perfect Hashing

It is obvious that the query time is deterministic worst-case $\mathcal{O}(1)$, because each query only involves one first-level access and one second-level access. It is also obvious that the space complexity is $\mathcal{O}(n)$, because $M < 4n$ is ensured by the construction process. The only worry is that even after a large number of trials, M is still at least $4n$. If this is the case, the construction will never terminate.

To eliminate this worry, we need to analyze the following probability:

$$\Pr[M \geq 4n]. \tag{3.14}$$

It is not clear how to give an upper bound on this probability directly. But it is easy to compute the expectation of M, then Markov's inequality can be applied to give a bound on this probability.

Let Z_i be the number of key-value pairs that hash to the same bucket as k_i in the first-level hash table. Since M can also be interpreted as the total number of collisions in the first-level hash table, we can write M as the sum of all Z_i:

$$M = \sum_{i=1}^{n} Z_i. \tag{3.15}$$

By linearity of expectation,

$$\mathbb{E}[M] = \mathbb{E}\left[\sum_{i=1}^{n} Z_i\right] = \sum_{i=1}^{n} \mathbb{E}[Z_i]. \tag{3.16}$$

By Lemma 2.102, we have the following upper bound on the expectation of Z_i:

$$\mathbb{E}[Z_i] \leq 1 + \frac{n-1}{n} < 2. \tag{3.17}$$

Plugging this into the expectation of M, we get

$$\mathbb{E}[M] = \sum_{i=1}^{n} \mathbb{E}[Z_i] < n \cdot 2 = 2n. \tag{3.18}$$

Now it is time to apply Markov's inequality to bound the probability of $M \geq 4n$:

$$\Pr[M \geq 4n] \leq \frac{\mathbb{E}[M]}{4n} < \frac{2n}{4n} = 1/2. \tag{3.19}$$

Therefore, with probability at least $1/2$, $M < 4n$ in a single trial. By resampling until this condition holds, we obtain a suitable first-level hash function after only a constant expected number of trials. Thus, the construction procedure is a Las Vegas algorithm that always terminates in constant expected time.

Remark 3.6. Thanks for your attention to this cute remark. This is the last hashing example in this course, so farewell, hashing! Well, not quite: in Chapter 6, we will still meet hashing's close cousin — random projection.

3.1.4 Beyond the Basic Form

The more ways we can adapt a theorem, the more useful it becomes. The real strength of Markov's inequality is that it imposes very little structure on the random variable itself. As long as the quantity we feed into the theorem is non-negative, Markov applies.

This means that we do not have to apply it only to X. Even when X is not always non-negative, we can instead apply Markov's inequality to a carefully chosen function $f(X)$ that is non-negative. You can think of f as a "wrapper" that transforms X into a more suitable form. We can then bound the probability that $f(X)$ is large, which in turn can be used to control X itself when f preserves order (for example, when f is monotone).

More formally, we have the following corollary.

Corollary 3.7. Let X be a random variable and $f : \mathbb{R} \to \mathbb{R}$ be a function. If $\Pr[f(X) \geq 0] = 1$, then for any $a > 0$,

$$\Pr[f(X) \geq a] \leq \frac{\mathbb{E}[f(X)]}{a}. \tag{3.20}$$

Proof. Apply Markov's inequality to the random variable $f(X)$. □

This corollary suggests a general strategy: once we identify a non-negative quantity related to the event we care about, Markov's inequality becomes available. The simplest choice is to choose a linear function of X. If we know that $X \geq u$, then shifting by u turns $X - u$ into a non-negative random variable, so Markov immediately gives an upper-tail bound $\Pr[X - u \geq a] \leq \frac{\mathbb{E}[X]-u}{a}$.

By symmetry, if $X \leq u$, the same idea also gives a lower-tail bound.

Corollary 3.8. If $\Pr[X \leq u] = 1$ for some $u \in \mathbb{R}$, then for any $a < u$,

$$\Pr[X \leq a] \leq \frac{u - \mathbb{E}[X]}{u - a}. \tag{3.21}$$

Proof. Let $f(X) := u - X$. Then the event $X \leq a$ is equivalent to the event $f(X) \geq u - a$. Since $\Pr[f(X) \geq 0] = 1$, by Markov's inequality, we have

$$\Pr[X \leq a] = \Pr[f(X) \geq u - a] \leq \frac{\mathbb{E}[f(X)]}{u - a} = \frac{u - \mathbb{E}[X]}{u - a}, \tag{3.22}$$

which is the desired inequality. □

Example 3.9 (Exam scores)**.** Suppose

$$X = \text{score of a random student}, \qquad X \leq 100, \qquad \mathbb{E}[X] = 90. \tag{3.23}$$

Then the previous inequality gives

$$\Pr[X \leq 80] \leq \frac{100 - 90}{100 - 80} = \frac{1}{2}. \tag{3.24}$$

Therefore, if the average score is 90, then at most half of the students score less than 80.

This simple example illustrates how choosing a wrapper function $f(X)$ broadens the applicability of Markov's inequality. But the function $f(X)$ does not have to be linear, as we will see in the next few sections.

3.2 Chebyshev's Inequality

The previous subsection introduced Markov's inequality and several of its applications. Let us return to the coin-tossing example: we toss a fair coin 10,000

times and ask how likely it is that the number of heads deviates significantly from 5,000. *Can Markov's inequality help us answer this question?*

More generally, suppose $X_1, \ldots, X_n$ are i.i.d. random variables drawn from the same distribution. Let μ be the expected value of X_i for any $i \in [n]$. It does not matter which X_i we choose, because all X_i have the same distribution.

Let X be the sample average of $X_1, \ldots, X_n$, defined as

$$X := \frac{1}{n} \sum_{i=1}^{n} X_i. \tag{3.25}$$

Our goal is to understand the probability that X deviates significantly from μ. That is, given any $\epsilon > 0$, we want a concentration inequality of the form

$$\Pr\Big[|X - \mu| \geq \epsilon\Big] \leq ? \tag{3.26}$$

By linearity of expectation, it is easy to derive that $\mathbb{E}[X] = \mu$ (see Equation (3.44) below). So the question is essentially asking us to bound $\Pr[|X - \mathbb{E}[X]| \geq \epsilon]$.

However, Markov's inequality in its original form does not directly help us here. It only controls the probability that a non-negative random variable is much larger than its expectation. In particular, it can tell us that, with high probability, X is not too large compared with μ (Corollary 3.2). But this guarantee is too weak for our purpose: it still permits $\Pr[X \geq 2\mu]$ to be as large as $1/2$, even though 2μ is already quite far from the mean μ.

Following the discussion in Section 3.1.4, one natural idea is to carefully choose a wrapper function $f(X)$ before applying Markov's inequality. Indeed, this is exactly the idea that leads to *Chebyshev's inequality*. To obtain this inequality, we simply apply Markov's inequality to the following wrapper function:

$$f(X) := (X - \mu)^2, \tag{3.27}$$

which measures the squared deviation from the mean. In this section, we will introduce Chebyshev's inequality in detail.

3.2.1 Variance

First, we define *variance*, which is a key quantity that appears in Chebyshev's inequality. It is exactly the expectation of the wrapper function (Equation (3.27)) introduced above.

Definition 3.10 (Variance)**.** The *variance* of a random variable X is defined as

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]. \tag{3.28}$$

The notation $\mathbb{E}[(X - \mathbb{E}[X])^2]$ may look confusing at first. Here $\mathbb{E}[X]$ is the expectation of X, which is a deterministic quantity and does not depend on any randomness. Thus, the expression $\mathbb{E}[(X - \mathbb{E}[X])^2]$ simply means that we first define $\mu := \mathbb{E}[X]$, and then compute the expectation of $(X - \mu)^2$.

Variance is a quantification of the uncertainty of a random variable. More precisely, it captures how much the random variable deviates from its mean on average (in a squared sense). You may wonder why we do not instead consider $\mathbb{E}[|X - \mathbb{E}[X]|]$. This is certainly a reasonable alternative, and it is sometimes called the *mean absolute deviation* (MAD). However, the squared form $\mathbb{E}[(X - \mathbb{E}[X])^2]$ is much easier to work with mathematically. For example, you will soon see that it interacts nicely with linearity of expectation and has convenient algebraic properties.

In calculations, the following equivalent formula is often more convenient.

Proposition 3.11. For any random variable X,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2. \tag{3.29}$$

Proof. Expanding the square gives

$$\begin{aligned} \mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2] && (3.30)\\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + \mathbb{E}[X]^2 && (3.31)\\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2, && (3.32) \end{aligned}$$

where the second line follows from the linearity of expectation. □

Example 3.12 (Bernoulli Random Variables)**.** A *Bernoulli random variable* models a single yes/no trial. There are only two possible outcomes: 0 and 1. For any $p \in [0, 1]$, we write $X \sim \mathrm{Bernoulli}(p)$ to mean

$$\Pr[X = 1] = p \quad \text{and} \quad \Pr[X = 0] = 1 - p. \tag{3.33}$$

The mean of X is

$$\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p. \tag{3.34}$$

Since $X^2 = X$ whenever $X \in \{0, 1\}$, the variance of X is

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = p - p^2 = p(1 - p). \tag{3.35}$$

This matches our intuition: the value of X is more predictable when p is closer to 0 or 1.

A special case is when $p = 1/2$. Then $X \sim \mathrm{Bernoulli}(1/2)$ is sometimes called a *standard Bernoulli random variable*. For example, the outcome of a fair coin flip can be modeled as a standard Bernoulli random variable. Its expectation and variance are

$$\mathbb{E}[X] = 1/2 \quad \text{and} \quad \mathrm{Var}(X) = 1/4. \tag{3.36}$$

3.2.2 Statement and Proof

Now we are ready to state Chebyshev's inequality.

Theorem 3.13 (Chebyshev's inequality)**.** For any random variable X and any $\epsilon > 0$,

$$\Pr\Big[|X - \mathbb{E}[X]| \geq \epsilon\Big] \leq \frac{\mathrm{Var}(X)}{\epsilon^2}. \tag{3.37}$$

Proof. Note that $|X - \mathbb{E}[X]| \geq \epsilon$ is equivalent to $(X - \mathbb{E}[X])^2 \geq \epsilon^2$. So let

$$f(X) := (X - \mathbb{E}[X])^2. \tag{3.38}$$

Since $f(X)$ is always non-negative, applying Markov's inequality to $f(X)$ gives

$$\Pr\Big[|X - \mathbb{E}[X]| \geq \epsilon\Big] = \Pr[f(X) \geq \epsilon^2] \leq \frac{\mathbb{E}[f(X)]}{\epsilon^2}. \tag{3.39}$$

Finally, by the definition of variance,

$$\mathbb{E}[f(X)] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathrm{Var}(X). \tag{3.40}$$

Substituting this into the previous inequality proves the theorem. □

The proof is just Markov's inequality applied to the squared deviation from the mean, but the conclusion is very useful. It tells us that the probability of deviating by ϵ from the mean is scaling with $1/\epsilon^2$, and the scaling factor is exactly the variance. To see intuitively how large the deviation typically is, we rewrite the inequality based on the concept of *standard deviation*.

Definition 3.14 (Standard deviation)**.** The *standard deviation* of a random variable X is defined as

$$\sigma(X) = \sqrt{\mathrm{Var}(X)}. \tag{3.41}$$

Corollary 3.15. For any random variable X and any $\alpha > 0$,

$$\Pr\Big[|X - \mathbb{E}[X]| \geq \alpha \cdot \sigma(X)\Big] \leq \frac{1}{\alpha^2}. \tag{3.42}$$

As the name suggests, the standard deviation is a measure of the typical deviation from the mean, and Chebyshev's inequality more specifically upper bounds the probability of deviating by more than α standard deviations from the mean by $1/\alpha^2$. Standard deviation therefore gives a natural unit for measuring the deviation of a random variable from its mean. See also Figure 3.3.

3.2.3 Chebyshev's Inequality for Sample Averages

We now return to the problem posed at the beginning of this section: how far can the sample average deviate from its expected value? Chebyshev's inequality is designed for exactly this kind of question. To use it, we need one more ingredient, which is the exact value of the variance of the sample average.

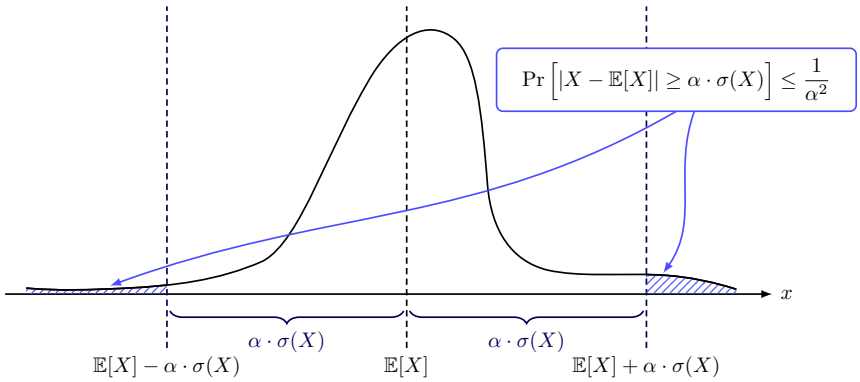


Figure 3.3: Chebyshev's inequality implies that the shaded tails together have probability at most $1/\alpha^2$.

Suppose $X_1, \ldots, X_n$ are i.i.d. random variables. Let $\mu := \mathbb{E}[X_i]$ be the expected value of X_i and $\sigma^2 := \mathrm{Var}(X_i)$ be the variance of X_i, where i is an arbitrary index in $[n]$. Now define the sample average X as

$$X := \frac{1}{n} \sum_{i=1}^{n} X_i. \tag{3.43}$$

By linearity of expectation,

$$\mathbb{E}[X] = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[X_i] = \frac{1}{n} \cdot n\mu = \mu. \tag{3.44}$$

So the sample average X has the same expected value as each individual random variable X_i. To apply Chebyshev's inequality, what remains is to compute the variance of the sample average.

Lemma 3.16 (Variance of a sample average)**.** If $X_1, \ldots, X_n$ are i.i.d. random variables with variance σ^2, then

$$\mathrm{Var}\left(\frac{1}{n} \sum_{i=1}^{n} X_i\right) = \frac{\sigma^2}{n}. \tag{3.45}$$

It would be quite cumbersome to prove this proposition by brute force, for example by directly expanding the definition of variance. In fact, variance admits the following simple and useful properties.

Theorem 3.17. For any random variable X and any constant $c \in \mathbb{R}$,

$$\mathrm{Var}(cX) = c^2 \mathrm{Var}(X). \tag{3.46}$$

Theorem 3.18. If X and Y are independent random variables, then

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y). \tag{3.47}$$

We leave the proofs of these two theorems as exercises. Based on them, we can now prove the lemma on the variance of the sample average.

Proof of Lemma 3.16. By Theorem 3.17, we have

$$\mathrm{Var}\left(\frac{1}{n}X_i\right) = \frac{1}{n^2}\mathrm{Var}(X_i) = \frac{\sigma^2}{n^2}. \tag{3.48}$$

Since $X_1, \ldots, X_n$ are independent, the random variables $\frac{1}{n}X_1, \ldots, \frac{1}{n}X_n$ are also independent. Applying Theorem 3.18 repeatedly, we obtain

$$\mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \mathrm{Var}\left(\frac{1}{n}X_1 + \frac{1}{n}X_2 + \cdots + \frac{1}{n}X_n\right) \tag{3.49}$$

$$= \sum_{i=1}^{n} \mathrm{Var}\left(\frac{1}{n}X_i\right). \tag{3.50}$$

Plugging in Equation (3.48), we get

$$\mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \frac{\sigma^2}{n^2} = \frac{\sigma^2}{n}, \tag{3.51}$$

which is exactly the desired result. □

This calculation how averaging reduces uncertainty. When averaging over n i.i.d. random variables, the variance of the sample average is reduced by a factor of n, or equivalently the standard deviation is reduced by a factor of $\sqrt{n}$. This matches our intuition that the sample average becomes more stable as the number of samples grows.

Now we can finally answer the original question: how far can the sample average deviate from its expected value?

Theorem 3.19 (Chebyshev for sample averages)**.** Let $X := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average of n i.i.d. random variables $X_1, \ldots, X_n$, each with mean μ and variance σ^2. Then for every $\epsilon > 0$,

$$\Pr\Big[|X - \mu| \geq \epsilon\Big] \leq \frac{\sigma^2}{n\epsilon^2}. \tag{3.52}$$

Proof. By Lemma 3.16, we have $\mathrm{Var}(X) = \frac{\sigma^2}{n}$. Applying Chebyshev's inequality to the random variable X gives

$$\Pr\Big[|X - \mu| \geq \epsilon\Big] \leq \frac{\mathrm{Var}(X)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}, \tag{3.53}$$

which is exactly the desired bound. □

Equivalently, we can rewrite the bound as

$$\Pr\left[|X - \mu| \geq \alpha \cdot \frac{\sigma}{\sqrt{n}}\right] \leq \frac{1}{\alpha^2}, \tag{3.54}$$

where $\alpha > 0$ is an arbitrary constant. In other words, the scale of the deviation of X from its mean is on the order of $\frac{\sigma}{\sqrt{n}}$, in the sense that the probability of deviating by α times this quantity is at most $1/\alpha^2$.

Example 3.20. Now let us revisit the coin-tossing example. We toss a fair coin 10,000 times and ask how likely it is that the number of heads deviates significantly from 5,000.

Let X_i be 1 if the i-th coin flip is heads and 0 otherwise. Then $X_i \sim$ Bernoulli(1/2). According to our calculation in Example 3.12, $\mathbb{E}[X_i] = 1/2, \mathrm{Var}(X_i) = 1/4$ and $\sigma(X_i) = 1/2$. By Chebyshev's inequality, the sample average X satisfies

$$\Pr\left[|X - 0.5| \geq \alpha \cdot \frac{1/2}{100}\right] \leq \frac{1}{\alpha^2}. \tag{3.55}$$

Setting $\alpha = 10$, we get

$$\Pr\left[|X - 0.5| \geq 0.05\right] \leq 1\%. \tag{3.56}$$

Therefore, it is unlikely to see a sample average deviate by more than even 0.05 from the mean, or equivalently, it is unlikely to see more than 5,500 heads or fewer than 4,500 heads.

A natural question is whether the bound $\Pr\left[|X - 0.5| \geq 0.05\right] \leq 1\%$ is tight or not. In fact, it is not! Next, we will introduce Chernoff bounds, which are able to show that this probability is in fact at most 3.9×10^{-22}.

3.3 Chernoff Bounds

In the previous section, we derived Chebyshev's inequality by applying Markov's inequality to a quadratic wrapper function $f(X) = (X - \mathbb{E}[X])^2$. Of course, once we start thinking in terms of wrapper functions, there is nothing special about the exponent 2. In principle, we could also consider using an arbitrary exponent m:

$$f(X) := (X - \mathbb{E}[X])^m. \tag{3.57}$$

In probability theory, its expectation $\mathbb{E}[(X - \mathbb{E}[X])^m]$ is called the *central m-th moment* of X. A related quantity is the *m-th raw moment*, or simple the *m-th moment* of X, which is defined as $\mathbb{E}[X^m]$. From this persepctive, Markov's inequality is based on the first raw moment, namely the expectation $\mathbb{E}[X]$; Chebyshev's inequality is based on the second central moment, namely the variance $\mathrm{Var}(X)$. More generally, for any even integer $m \geq 2$, applying Markov's inequality to $f(X) = (X - \mathbb{E}[X])^m$ gives the following concentration inequality:

$$\Pr\Big[|X - \mathbb{E}[X]| \geq \epsilon\Big] \leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^m]}{\epsilon^m}. \tag{3.58}$$

A notable feature of this inequality is that the tail decays as $\mathcal{O}(\epsilon^{-m})$ for large ϵ, which is much faster than the $\mathcal{O}(\epsilon^{-2})$ decay given by Chebyshev's inequality.

However, as m increases, the central m-th moment also increases significantly. As a result, a larger value of m does not automatically lead to a better bound, and we may need to carefully choose m to obtain the best possible bound.

Another difficulty is that, unlike the variance, the central m-th moment of the sample average X of n i.i.d. random variables is often not easy to compute. You will see some concrete examples in the exercises.

Is there a better way to choose the wrapper function? A more aggressive choice is to use an exponential function $f(X) := e^{tX}$ for some $t \in \mathbb{R}$. By Taylor expansion, the exponential function simultaneously incorporates all powers, each with an appropriate weight tuned by the parameter t:

$$e^{tX} = 1 + tX + \frac{t^2}{2!}X^2 + \frac{t^3}{3!}X^3 + \cdots . \tag{3.59}$$

Its expectation $\mathbb{E}[e^{tX}]$ is called the *moment generating function* (MGF) of X, which naturally arises when we apply Markov's inequality to $f(X) = e^{tX}$.

By passing from powers to exponentials, we are no longer focusing on a single moment such as the variance. Instead, we package all moments together into a single object. This idea leads to a powerful class of concentration inequalities known as *Chernoff bounds* [Che52].

3.3.1 Moment Generating Functions

First, we introduce the concept of the *moment generating function* (MGF), which will be used to derive Chernoff bounds.

Definition 3.21 (Moment generating function)**.** The *moment generating function* (MGF) of a random variable X is defined as

$$M_X(t) := \mathbb{E}[e^{tX}]. \tag{3.60}$$

The term "generating function" reflects the fact that MGF can "generate" all the moments of X through its Taylor expansion. By linearity of the expectation,

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{t^k}{k!}X^k\right] = \sum_{k=0}^{\infty} \frac{t^k}{k!}\mathbb{E}[X^k]. \tag{3.61}$$

Therefore, the k-th moment of X can be recovered from the coefficient of t^k in the Taylor expansion of $M_X(t)$. We will revisit this idea in Chapter 4.

The following two properties of the MGF will be used later.

Proposition 3.22 (Properties of the MGF)**.** Let X be a random variable. Then its MGF $M(t)$ satisfies the following properties:

1. $M(0) = 1$;
2. $M(t) \geq e^{t\mathbb{E}[X]}$ for all $t \in \mathbb{R}$.

Proof. The first property follows from

$$M(0) = \mathbb{E}[e^{0X}] = \mathbb{E}[1] = 1. \tag{3.62}$$

For the second property, note that $f(X) = e^{tX}$ is a convex function for any $t \in \mathbb{R}$. Jensen's inequality implies that $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$ for any convex function f (see Exercise 3.2). Applying this to $f(X) = e^{tX}$, we get

$$M(t) = \mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) = e^{t\mathbb{E}[X]}, \tag{3.63}$$

which obtains the desired inequality. □

Example 3.23 (MGF of Bernoulli Random Variables)**.** In Example 3.12, we have computed the mean and variance of a Bernoulli random variable $X \sim \text{Bernoulli}(p)$:

$$\mathbb{E}[X] = p \quad \text{and} \quad \text{Var}(X) = p(1-p). \tag{3.64}$$

The MGF of X can be computed by averaging the two exponential terms e^t and e^0 associated with the two possible values 1 and 0:

$$M_X(t) = \mathbb{E}[e^{tX}] = (1-p)e^0 + pe^t = 1 - p + pe^t. \tag{3.65}$$

So the Bernoulli MGF is just a weighted average of 1 and e^t.

Example 3.24 (Binomial Random Variables)**.** A *binomial random variable* X, denoted by $X \sim \text{Binomial}(n, p)$, counts the number of successes in $n \geq 0$ independent yes/no trials, when each trial succeeds with probability $p \in [0, 1]$. Its probability mass function is given by

$$\Pr[X = k] = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for} \quad k = 0, 1, 2, \ldots, n. \tag{3.66}$$

The random variable X can be written as the sum $\sum_{i=1}^{n} X_i$ of n independent random variables $X_1, \ldots, X_n$, where each $X_i \sim \text{Bernoulli}(p)$. By independence, we can add up the expectations and variances:

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = np \quad \text{and} \quad \text{Var}(X) = \sum_{i=1}^{n} \text{Var}(X_i) = np(1-p). \tag{3.67}$$

To compute the MGF, we use the fact that under independence, the expectation of a product factorizes. Thus, the MGF of a sum of independent random variables is the product of their MGFs:

$$M_X(t) = \mathbb{E}[e^{t(X_1 + \cdots + X_n)}] = \mathbb{E}[e^{tX_1} \cdots e^{tX_n}] = \prod_{i=1}^{n} \mathbb{E}[e^{tX_i}] \tag{3.68}$$

$$= (1 - p + pe^t)^n, \tag{3.69}$$

where the last equality follows from the MGF of each Bernoulli random variable X_i is $M_{X_i}(t) = 1 - p + pe^t$. We will use this factorization property of the MGF again to derive the Chernoff bound.

Example 3.25 (Rademacher Random Variables)**.** A *Rademacher random variable* X is the symmetric two-point random variable taking the values $+1$ and -1 with equal probability:

$$\Pr[X = 1] = \frac{1}{2} \quad \text{and} \quad \Pr[X = -1] = \frac{1}{2}. \tag{3.70}$$

This is similar to a standard Bernoulli random variable, but with the outcome 0 replaced by -1. We have discussed in Remark 3.4 that Markov's inequality is not applicable to Rademacher random variables.

The mean and variance of a Rademacher random variable X are easy to compute:

$$\mathbb{E}[X] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot (-1) = 0 \quad \text{and} \quad \operatorname{Var}(X) = \mathbb{E}[X^2] = 1. \tag{3.71}$$

The MGF of X can be computed by averaging the two exponential terms e^t and e^{-t} associated with the two possible values 1 and -1:

$$M_X(t) = \mathbb{E}[e^{tX}] = \frac{e^t + e^{-t}}{2} = \cosh t, \tag{3.72}$$

where $\cosh t := \frac{e^t + e^{-t}}{2}$ is the hyperbolic cosine function.

Example 3.26 (Geometric Random Variables)**.** A *geometric random variable*, denoted by $X \sim \text{Geometric}(p)$, is the number of trials needed until the first success in a sequence of independent trials, where each trial succeeds with probability $p \in (0, 1]$. The probability of $X = k$ can be computed by multiplying the probability of $k - 1$ consecutive failures followed by one success. Thus, the probability mass function of X is

$$\Pr[X = k] = (1 - p)^{k-1} p \quad \text{for} \quad k \geq 1. \tag{3.73}$$

The name *geometric* comes from the fact that these probabilities form a geometric sequence. The mean and variance of X are given by

$$\mathbb{E}[X] = \tfrac{1}{p} \quad \text{and} \quad \operatorname{Var}(X) = \tfrac{1-p}{p^2}. \tag{3.74}$$

The MGF of X can be computed by summing the geometric series:

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_{k \geq 1} e^{tk} (1 - p)^{k-1} p = p e^t \sum_{k \geq 1} \left(e^t (1 - p)\right)^{k-1}. \tag{3.75}$$

This is a geometric series, so it converges when $e^t(1 - p) < 1$, or equivalently when $t < -\log(1 - p)$. Therefore,

$$M_X : (-\infty, -\log(1 - p)) \to (0, \infty), \quad t \mapsto \tfrac{p e^t}{1 - (1 - p) e^t}. \tag{3.76}$$

Unlike the previous examples defined on finite probability spaces, the geometric random variable is defined on an infinite probability space, where expectations may not always exist. As a result, its MGF is defined only on a half-line containing 0, rather than for all $t \in \mathbb{R}$. We will discuss infinite probability spaces in more detail in Chapter 5.

3.3.2 Generic Chernoff Bounds

Chernoff bounds [Che52] are derived by applying Markov's inequality to $f(X) = e^{tX}$, where the MGF $M_X(t)$ naturally appears. The role of the MGF in Chernoff bounds is analogous to the role of the variance in Chebyshev's inequality.

Theorem 3.27 (Generic Chernoff bound (upper tail))**.** Let X be a random variable with mean μ and MGF $M(t)$. Then for any $a \geq \mu$,

$$\Pr[X \geq a] \leq \inf_{t \in \mathbb{R}} \left\{ e^{-ta} M(t) \right\}. \tag{3.77}$$

Proof. It suffices to prove that $\Pr[X \geq a] \leq e^{-ta} M(t)$ for all $t \in \mathbb{R}$.

- If $t \leq 0$, then by Proposition 3.22, the bound is trivial:
$$e^{-ta} M(t) \geq e^{-ta} \cdot e^{t\mu} = e^{-t(a-\mu)} \geq 1 \geq \Pr[X \geq a]. \tag{3.78}$$
- If $t > 0$, let $f(X) := e^{tX}$. Then $\mathbb{E}[f(X)] = \mathbb{E}[e^{tX}] = M(t)$. Since $f(X)$ is always non-negative, we can apply Markov's inequality to obtain
$$\Pr[X \geq a] = \Pr[e^{tX} \geq e^{ta}] \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = e^{-ta} M(t). \tag{3.79}$$

Since t is arbitrary, taking the infimum over $t \in \mathbb{R}$ yields the desired result. □

Similarly, we can prove the following bound for the lower tail.

Theorem 3.28 (Generic Chernoff bound (lower tail))**.** Let X be a random variable with mean μ and MGF $M(t)$. Then for any $a \leq \mu$,

$$\Pr[X \leq a] \leq \inf_{t \in \mathbb{R}} \left\{ e^{-ta} M(t) \right\}. \tag{3.80}$$

Proof. The proof is similar to the proof of Theorem 3.27.

- If $t \geq 0$, then by Proposition 3.22,
$$e^{-ta} M(t) \geq e^{-ta} \cdot e^{t\mu} = e^{t(\mu-a)} \geq 1 \geq \Pr[X \leq a]. \tag{3.81}$$
- If $t < 0$, applying Markov's inequality to $f(X) = e^{tX}$ gives
$$\Pr[X \leq a] = \Pr[e^{tX} \geq e^{ta}] \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = e^{-ta} M(t), \tag{3.82}$$
where the first equality follows from the fact that e^{tX} is a decreasing function of X for $t < 0$.

Since t is arbitrary, taking the infimum over $t \in \mathbb{R}$ yields the desired result. □

As we have seen, the quantity $e^{-ta} M(t)$ appears in both the upper and lower tail bounds. It is therefore natural to study this quantity more closely. This leads to the notion of the *rate function*, which is essentially the negative logarithm of $e^{-ta} M(t)$ optimized over t.

Definition 3.29 (Rate function)**.** The *rate function* of a random variable X is defined as

$$I_X(a) := \sup_{t \in \mathbb{R}} \{ta - \log M_X(t)\}, \tag{3.83}$$

with the convention that $I_X(a) = +\infty$ if the supremum is unbounded.

Theorem 3.30 (Generic Chernoff bound in terms of the rate function)**.** Let X be a random variable with mean μ and rate function $I(a)$. Then

$$\forall a \geq \mu: \quad \Pr[X \geq a] \leq e^{-I(a)}, \tag{3.84}$$

$$\forall a \leq \mu: \quad \Pr[X \leq a] \leq e^{-I(a)}. \tag{3.85}$$

Proof. Simply apply Theorem 3.27 and note the following identity:

$$e^{-ta} M(t) = e^{-(ta - \log M(t))}. \tag{3.86}$$

Taking the infimum over $t \in \mathbb{R}$ gives $\inf_{t \in \mathbb{R}} \{e^{-ta} M(t)\} = e^{-I(a)}$. □

One may wonder what happens to the rate function at $a = \mu$. In fact, $I(a)$ attains its minimum value 0 at $a = \mu$. At this point, $e^{-I(a)} = 1$, so both the upper and lower tail bounds become trivial:

$$\Pr[X \geq \mu] \leq 1 \qquad \text{and} \qquad \Pr[X \leq \mu] \leq 1. \tag{3.87}$$

See Figure 3.4 for some examples of discrete distributions together with their log MGFs and rate functions. The rate function $I_X(a)$ is finite only between the minimum and maximum values that X can take. Outside this domain, one can show that $I_X(a) = +\infty$, and hence $e^{-I_X(a)} = 0$. Within the domain, the rate function is convex and always non-negative, and its value becomes exactly 0 at $a = \mu$. We summarize these observations in the following proposition, whose proof is left as an exercise.

Proposition 3.31 (Properties of the rate function)**.** Let X be a random variable with mean μ and rate function $I(a)$. Let ℓ and u be the minimum and maximum values that X can take, and assume $\ell < u$. Then

1. $I(a) = +\infty$ if $a < \ell$ or $a > u$;
2. $0 \leq I(a) < +\infty$ for all $a \in [\ell, u]$;
3. $I(\mu) = 0$.
4. $I(a)$ is convex on the interval $[\ell, u]$;

Remark 3.32. As our current focus is on finite discrete probability spaces, random variables are always bounded. In general, however, random variables can be unbounded, such as the geometric random variable in Example 3.26. In such cases, one may set $\ell = -\infty$ and/or $u = +\infty$, and interpret the interval $[\ell, u]$ accordingly as $(-\infty, u]$, $[\ell, +\infty)$, or $(-\infty, +\infty)$.

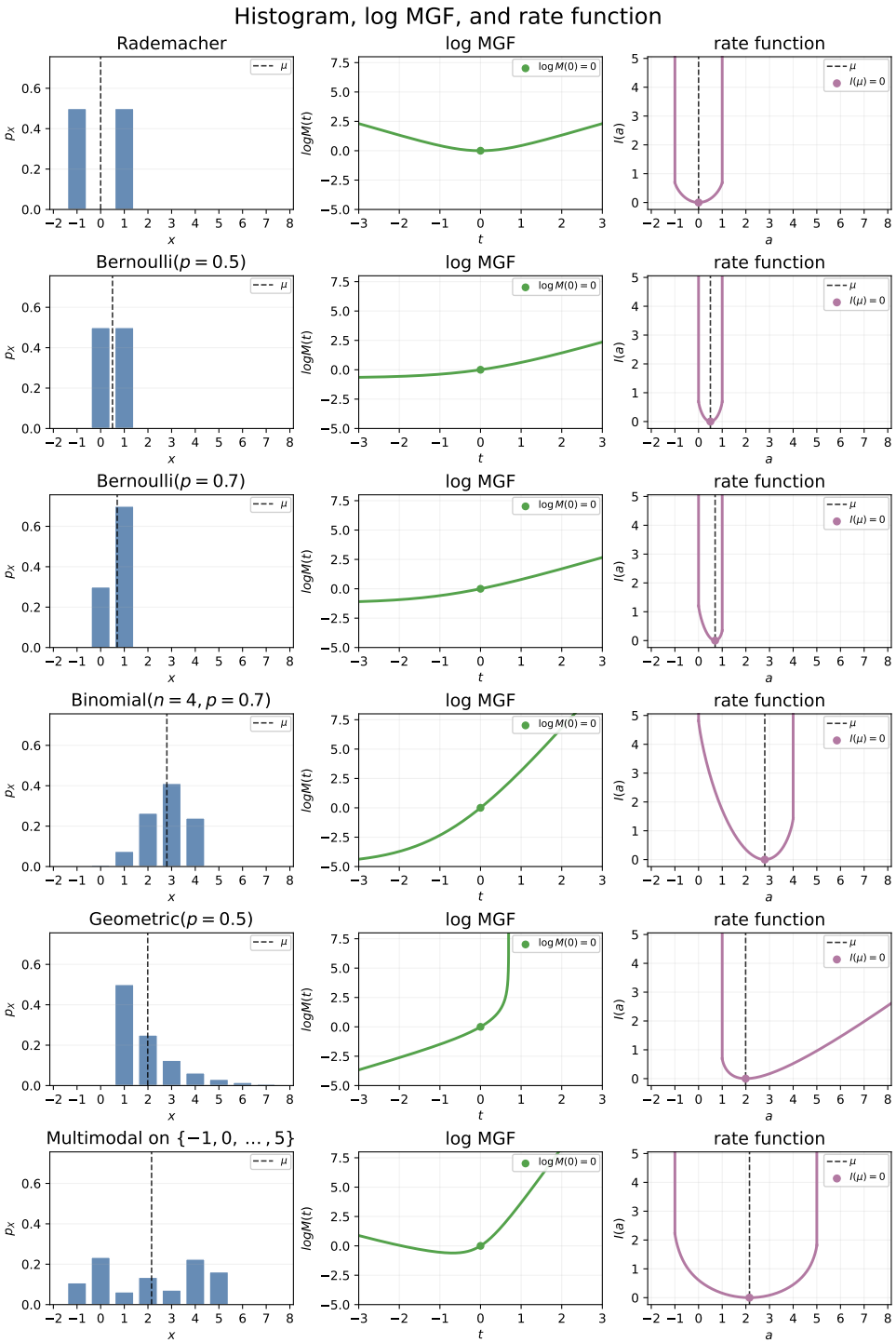


Figure 3.4: Several discrete distributions together with their log MGFs and rate functions. The vertical solid lines indicate that the rate function becomes $+\infty$ outside the support.

Remark 3.33. If $\ell = u$, then the random variable is a constant: $X = \ell = u$ with probability 1. In this case, the log MGF is $M_X(t) = e^{tu}$, and the rate function is

$$I_X(a) = \sup_{t\in\mathbb{R}} \left\{ ta - \log e^{tu} \right\} = \sup_{t\in\mathbb{R}} \{ta - tu\} = \begin{cases} +\infty & \text{if } a < u, \\ 0 & \text{if } a = u, \\ +\infty & \text{if } a > u. \end{cases} \tag{3.88}$$

3.3.3 Chernoff Bounds for Sample Averages

At this point, the introduction of the moment generating function and the rate function may seem to have only made things more complicated than Chebyshev's inequality, which relies solely on the variance. We now return to the problem of bounding the deviation of the sample average X of n i.i.d. random variables from its mean μ. This is where the new framework begins to pay off: the resulting bound turns out to be remarkably simple. Roughly speaking, the only change is that the error probability improves from $e^{-I(a)}$ to $e^{-nI(a)}$, which is exponentially smaller than the $\mathcal{O}(\frac{1}{n})$ decay given by Chebyshev's inequality.

Theorem 3.34 (Generic Chernoff bound for sample averages)**.** Let $X := \frac{1}{n}\sum_{i=1}^n X_i$ be the sample average of n i.i.d. random variables $X_1, \ldots, X_n$, each with mean μ and rate function $I(a)$. Then

$$\forall a \geq \mu: \quad \Pr[X \geq a] \leq e^{-nI(a)}, \tag{3.89}$$

$$\forall a \leq \mu: \quad \Pr[X \leq a] \leq e^{-nI(a)}. \tag{3.90}$$

This theorem is simply due to the fact that the rate function of the sample average X is just $nI(a)$.

Lemma 3.35 (Rate function of the sample average)**.** Let $X := \frac{1}{n}\sum_{i=1}^n X_i$ be the sample average of n i.i.d. random variables $X_1, \ldots, X_n$, each with rate function $I(a)$. Then

$$I_X(a) = nI(a). \tag{3.91}$$

Proof. For any $t \in \mathbb{R}$, the MGF of the sample average $X = \frac{1}{n}\sum_{i=1}^n X_i$ is

$$\mathbb{E}[e^{tX}] = \mathbb{E}\left[e^{\frac{t}{n}\sum_{i=1}^n X_i}\right]. \tag{3.92}$$

Since $X_1, \ldots, X_n$ are independent, the expectation of the exponential factorizes:

$$\mathbb{E}\left[e^{\frac{t}{n}\sum_{i=1}^n X_i}\right] = \mathbb{E}\left[\prod_{i=1}^n e^{\frac{t}{n}X_i}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{\frac{t}{n}X_i}\right] = \left(M(\tfrac{t}{n})\right)^n. \tag{3.93}$$

By the definition of the rate function, we have

$$I_X(a) = \sup_{t\in\mathbb{R}} \{ta - \log M_X(t)\} = \sup_{t\in\mathbb{R}} \{ta - n\log M(\tfrac{t}{n})\} \tag{3.94}$$

$$= n \cdot \sup_{t\in\mathbb{R}} \{\tfrac{t}{n}a - \log M(\tfrac{t}{n})\}. \tag{3.95}$$

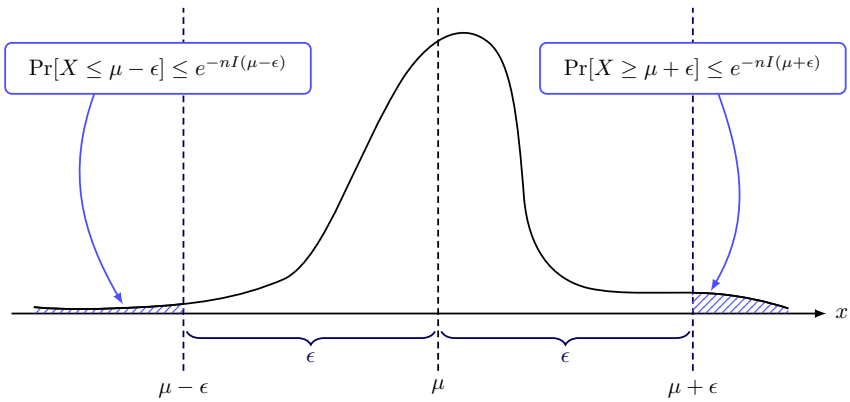


Figure 3.5: An illustration of Chernoff bounds for the lower and upper tails of the sample average of n i.i.d. random variables with rate function $I(a)$.

Now make the substitution $t' = \frac{t}{n}$. Then $t = nt'$, and hence

$$I_X(a) = n \cdot \sup_{t' \in \mathbb{R}} \{t'a - \log M(t')\} = nI(a), \tag{3.96}$$

where the last equality follows from the definition of $I(a)$. □

Proof of Theorem 3.34. Applying Theorem 3.30 to X, we obtain

$$\Pr[X \ge a] \le e^{-I_X(a)}. \tag{3.97}$$

Substituting $I_X(a) = nI(a)$ yields the desired bound. □

Let us pause for a moment and digest what the theorem is really saying. Suppose we want to understand the sample average

$$X = \frac{1}{n} \sum_{i=1}^{n} X_i, \tag{3.98}$$

where $X_1, \ldots, X_n$ are i.i.d. with mean μ. At first this may look like a new problem involving n random variables. But the theorem shows that the heart of the problem is still the original one-variable rate function $I(a)$. Once that function is understood, the bound for sample averages follows immediately, because the only change is that the exponent gets multiplied by n.

In particular, for every $\epsilon > 0$ we can combine the upper and lower tails into the two-sided estimate

$$\Pr\Big[X - \mu \ge \epsilon\Big] \le e^{-nI(\mu+\epsilon)}, \tag{3.99}$$

$$\Pr\Big[X - \mu \le -\epsilon\Big] \le e^{-nI(\mu-\epsilon)}, \tag{3.100}$$

$$\Pr\Big[|X - \mu| \ge \epsilon\Big] \le e^{-nI(\mu+\epsilon)} + e^{-nI(\mu-\epsilon)} \le 2e^{-n \min\{I(\mu+\epsilon), I(\mu-\epsilon)\}}. \tag{3.101}$$

So once a deviation level ϵ is fixed, the probability of seeing an error of that size decays exponentially fast in the sample size n. This should be compared with

the Chebyshev bound from the previous section:

$$\Pr\Big[|X-\mu| \geq \epsilon\Big] \leq \frac{\sigma^2}{n\epsilon^2}. \tag{3.102}$$

There, the decay is only of order $1/n$.

In fact, the following theorem by Harald Cramér [Cra38, CT18] tells us that Chernoff bounds are asymptotically optimal for bounding the sample average at any fixed deviation level.

Theorem 3.36 (Cramér's theorem)**.** Let $\mathcal{D}$ be a probability distribution with mean μ and rate function $I(a)$. Let $X^{(n)} := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average of n i.i.d. random variables $X_1, \ldots, X_n$, drawn from $\mathcal{D}$. Then as $n \to \infty$,

$$\text{for any fixed } a \geq \mu: \quad \Pr[X^{(n)} \geq a] = e^{-n(I(a)+o(1))}, \tag{3.103}$$

$$\text{for any fixed } a \leq \mu: \quad \Pr[X^{(n)} \leq a] = e^{-n(I(a)+o(1))}. \tag{3.104}$$

Remark 3.37. If you are curious why Chernoff bounds turn out to be tight, even though the choice of the wrapper function $f(X) = e^{tX}$ may at first seem arbitrary, be patient! After learning generating functions in Chapter 4, we will give you an exercise that guides you to prove this theorem in an elegant way.

3.3.4 Chernoff Bounds for Small Deviations

There is still something we have not understood. For any fixed deviation level $a - \mu = \Theta(1)$, the Chernoff bound tells us that the corresponding tail probability is exponentially small in n. So as long as the deviation is fixed, the picture is already quite clear. Chebyshev's inequality, however, suggested that the most typical fluctuations should happen on the much smaller scale $1/\sqrt{n}$. From the Chernoff bound alone, it is not immediately clear why this should be the right scale, or what happens in that regime.

To understand this, we should no longer treat $a - \mu$ as a fixed constant, but instead let it shrink with n. In other words, we consider

$$a = \mu + \mathcal{O}\left(\frac{1}{\sqrt{n}}\right). \tag{3.105}$$

Recall that the rate function $I(a)$ attains its minimum at $a = \mu$, with $I(\mu) = 0$. The key question is how $I(a)$ grows as we move away from μ. If the distribution is regular enough so that $I(a)$ is $\mathcal{C}^\infty$ (i.e., infinitely differentiable) in a neighborhood of μ (as is always the case for bounded non-constant random variables), then a Taylor expansion around μ gives

$$I(\mu+\epsilon) = I(\mu) + I'(\mu)\epsilon + \tfrac{1}{2}I''(\mu)\epsilon^2 + \mathcal{O}(\epsilon^3) \tag{3.106}$$

$$= \tfrac{1}{2}I''(\mu)\epsilon^2 + \mathcal{O}(n^{-3/2}), \tag{3.107}$$

where we used $I(\mu) = 0$, $I'(\mu) = 0$, and $\epsilon = \mathcal{O}(\frac{1}{\sqrt{n}})$.

Substituting $u = \epsilon$ into the Chernoff bound yields

$$\Pr[X \geq \mu + \epsilon] \leq e^{-nI(\mu+\epsilon)} \leq e^{-\frac{1}{2}I''(\mu)n\epsilon^2 + \mathcal{O}(n^{-1/2})}, \tag{3.108}$$

$$\Pr[X \leq \mu - \epsilon] \leq e^{-nI(\mu-\epsilon)} \leq e^{-\frac{1}{2}I''(\mu)n\epsilon^2 + \mathcal{O}(n^{-1/2})}. \tag{3.109}$$

Therefore, near μ, the tail probability decays like $e^{-\Omega(n\epsilon^2)}$. In particular, when $\epsilon = \Theta(\frac{1}{\sqrt{n}})$, the exponent $n\epsilon^2$ is of constant order, so the tail probability is no longer exponentially small in n, but remains bounded away from zero. On the other hand, increasing ϵ by a constant factor immediately makes the exponent grow linearly in n, causing the tail probability to decay exponentially fast. This shows that the transition between "likely" and "unlikely" deviations happens precisely at the scale $\epsilon = \Theta(\frac{1}{\sqrt{n}})$, which is exactly the fluctuation scale predicted by Chebyshev's inequality.

3.4 Hoeffding's Inequality for Bounded Variables

So far, we have treated the rate function in an abstract way. To apply Chernoff bounds in concrete settings, we need to understand the explicit form of the rate function for specific distributions. In this section, we specifically focus on *bounded* random variables, which are random variables that take values in a finite interval. For this class of random variables, Hoeffding's inequality shows that the rate function is always lower bounded by a quadratic function.

3.4.1 Case Study: Symmetric Bernoulli Variables

Before discussing general bounded random variables, let us first look at the cleanest possible example: a symmetric Bernoulli random variable.

Lemma 3.38. Let $X \sim \text{Bernoulli}(\frac{1}{2})$. Then for all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{t(X-\mathbb{E}[X])}] \leq e^{t^2/8}. \tag{3.110}$$

Proof. Let $Y := X - \mathbb{E}[X] = X - 1/2$ denote the centered version of X. Since $X \sim \text{Bernoulli}(1/2)$, the random variable Y takes values in $\{-1/2, 1/2\}$ with equal probability. Therefore,

$$\mathbb{E}[e^{tY}] = \frac{1}{2}e^{-t/2} + \frac{1}{2}e^{t/2} = \cosh\frac{t}{2}, \tag{3.111}$$

where $\cosh x := \frac{e^x + e^{-x}}{2}$ is the hyperbolic cosine function.

It remains to prove that $\cosh(t/2) \leq e^{t^2/8}$. See Figure 3.6(a) for a plot of the two functions. The inequality indeed looks plausible.

Now note that the power series expansion of $\cosh x$ is given by

$$\cosh x = \frac{1}{2}\sum_{j\geq 0}\frac{x^j}{j!} + \frac{1}{2}\sum_{j\geq 0}(-1)^j\frac{x^j}{j!} = \sum_{k\geq 0}\frac{x^{2k}}{(2k)!}. \tag{3.112}$$

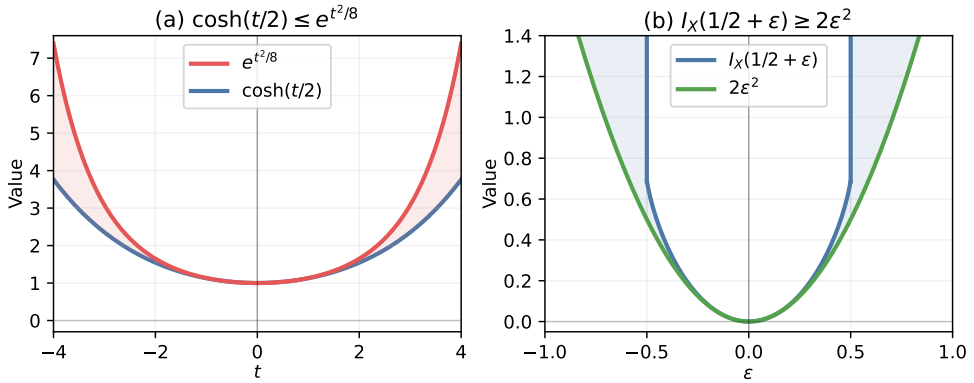


Figure 3.6: The MGF and rate function of symmetric Bernoulli variables. (a) The exact centered MGF $\cosh(t/2)$ lies below the upper bound $e^{t^2/8}$; (b) The rate function is lower bounded by $I_X(1/2+\epsilon) \geq 2\epsilon^2$.

For each $k \geq 0$, we can bound each term in the sum by the corresponding term in the power series expansion of $e^{x^2/2}$:

$$\frac{x^{2k}}{(2k)!} \leq \frac{(x^2/2)^k}{k!}. \tag{3.113}$$

This is because $(2k)! \geq 2 \cdot 4 \cdots (2k) = 2^k k!$, where we extract a factor of 2 from each even factor in $(2k)!$. Summing over k gives

$$\cosh x \leq \sum_{k \geq 0} \frac{(x^2/2)^k}{k!} = e^{x^2/2}. \tag{3.114}$$

Applying this with $x = t/2$, we obtain

$$\mathbb{E}[e^{tY}] = \cosh \frac{t}{2} \leq e^{t^2/8}, \tag{3.115}$$

which completes the proof. □

This MGF bound immediately turns into a lower bound on the rate function.

Lemma 3.39. Let $X \sim \text{Bernoulli}(\frac{1}{2})$. Then for all $\epsilon \in \mathbb{R}$,

$$I_X(1/2+\epsilon) \geq 2\epsilon^2. \tag{3.116}$$

Proof. By Lemma 3.38,

$$M_X(t) = \mathbb{E}[e^{tX}] = e^{t\mathbb{E}[X]}\mathbb{E}[e^{t(X-\mathbb{E}[X])}] \leq e^{t/2}e^{t^2/8} = e^{t/2+t^2/8}. \tag{3.117}$$

Therefore, by the definition of the rate function of X,

$$I_X(1/2+\epsilon) = \sup_{t \in \mathbb{R}} \{(1/2+\epsilon)t - \log M_X(t)\} \geq \sup_{t \in \mathbb{R}} \left\{\epsilon t - t^2/8\right\}. \tag{3.118}$$

The quadratic on the right is maximized at $t = 4\epsilon$. Substituting this value into the quadratic proves the lemma. □

Now we can feed this lower bound back into the generic Chernoff bounds.

Theorem 3.40. Let $X := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average of n independent random variables $X_1, \dots, X_n \sim \text{Bernoulli}(1/2)$. Then for all $\epsilon \geq 0$,

$$\Pr[X \geq 1/2 + \epsilon] \leq e^{-2n\epsilon^2}, \tag{3.119}$$

$$\Pr[X \leq 1/2 - \epsilon] \leq e^{-2n\epsilon^2}. \tag{3.120}$$

Proof. By Lemma 3.39, the rate function $I(a)$ of a single $X_i \sim \text{Bernoulli}(1/2)$ satisfies $I(1/2 + \epsilon) \geq 2\epsilon^2$ and $I(1/2 - \epsilon) \geq 2\epsilon^2$. Substituting this into Theorem 3.34 gives

$$\Pr\Big[X \geq 1/2 + \epsilon\Big] \leq e^{-nI_X(1/2+\epsilon)} \leq e^{-2n\epsilon^2}, \tag{3.121}$$

$$\Pr\Big[X \leq 1/2 - \epsilon\Big] \leq e^{-nI_X(1/2-\epsilon)} \leq e^{-2n\epsilon^2}, \tag{3.122}$$

which completes the proof. □

As usual, combining the two one-sided bounds gives the two-sided estimate

$$\Pr\Big[|X - 1/2| \geq \epsilon\Big] \leq 2e^{-2n\epsilon^2}. \tag{3.123}$$

Example 3.41. Continuing our discussion of fair coin tossing from Example 3.20, consider $n = 10{,}000$ tosses. Let X be the fraction of heads. Applying Equation (3.123) yields

$$\Pr\Big[|X - 1/2| \geq \epsilon\Big] \leq 2e^{-20000\epsilon^2}. \tag{3.124}$$

Setting $\epsilon = 0.05$, we get

$$\Pr\Big[|X - 0.5| \geq 0.05\Big] \leq 2e^{-20000\times 0.05^2} = 2e^{-50} \approx 3.9 \times 10^{-22}. \tag{3.125}$$

This is dramatically smaller than the Chebyshev bound from the previous section, which is at most 1%. See Figure 3.7 for a comparison of the exact tail probability and the Chernoff bound.

3.4.2 Hoeffding's Inequality

What makes the symmetric Bernoulli case important is that the same kind of bound continues to hold for all random variables bounded in an interval $[\ell, u]$. The following lemma is due to Wassily Hoeffding [Hoe63].

Lemma 3.42 (Hoeffding's Lemma)**.** Let X be a random variable taking values in $[\ell, u]$ for some $\ell \leq u$. Then for all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{t(X - \mathbb{E}[X])}] \leq e^{t^2(u-\ell)^2/8}. \tag{3.126}$$

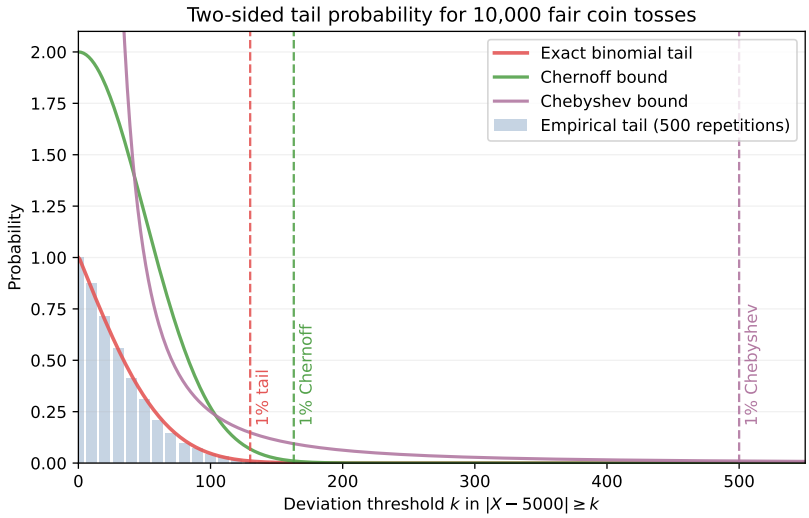


Figure 3.7: For fair coin tossing, the empirical tail probability matches the exact binomial tail closely, and the Chernoff bound gives the correct exponential scale.

We will not prove Hoeffding's lemma here, but it is helpful to understand the ideas behind it. One standard approach is to identify the extremal case: among all random variables supported on $[a, b]$ with a fixed mean, the centered MGF is maximized by a scaled Bernoulli random variable taking values in $\{a, b\}$. The bound then follows from an explicit, though somewhat tedious, calculation for Bernoulli variables.

Another approach comes from a deeper understanding of moment generating functions. After learning more about generating functions in Chapter 4, you will find Hoeffding's lemma almost obvious. Be patient!

Similar to Lemma 3.39, we turn the MGF bound into a lower bound on the rate function.

Lemma 3.43. Let X be a random variable taking values in $[\ell, u]$ for some $\ell \le u$. Then for all $\epsilon \in \mathbb{R}$,

$$I_X(\mathbb{E}[X] + \epsilon) \ge 2\epsilon^2/(u - \ell)^2. \tag{3.127}$$

Proof. By Hoeffding's lemma,

$$M_X(t) = \mathbb{E}[e^{tX}] = e^{t\mathbb{E}[X]}\mathbb{E}[e^{t(X-\mathbb{E}[X])}] \le e^{t\mathbb{E}[X]+t^2(u-\ell)^2/8}. \tag{3.128}$$

Therefore, by the definition of the rate function of X,

$$I_X(\mathbb{E}[X] + \epsilon) = \sup_{t\in\mathbb{R}} \{(\mathbb{E}[X] + \epsilon)t - \log M_X(t)\} \tag{3.129}$$

$$\ge \sup_{t\in\mathbb{R}} \{\epsilon t - t^2(u - \ell)^2/8\}. \tag{3.130}$$

The quadratic on the right is maximized at $t = 4\epsilon/(u - \ell)^2$. Substituting this value into the quadratic proves the lemma. □

The final step is to feed this lower bound back into the generic Chernoff theorem for sample averages. The resulting bound is known as *Hoeffding's inequality*.

Theorem 3.44 (Hoeffding's inequality, i.i.d. case)**.** Let $X := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average of n i.i.d. random variables $X_1, \dots, X_n$ taking values in $[\ell, u]$ for some $\ell \le u$. Then for all $\epsilon \ge 0$,

$$\Pr[X \ge \mathbb{E}[X] + \epsilon] \le e^{-2n\epsilon^2/(u-\ell)^2}, \tag{3.131}$$

$$\Pr[X \le \mathbb{E}[X] - \epsilon] \le e^{-2n\epsilon^2/(u-\ell)^2}. \tag{3.132}$$

Proof. Just plug the lower bound on the rate function into Theorem 3.34. □

As before, combining the two one-sided bounds gives the two-sided estimate

$$\Pr\Big[|X - \mathbb{E}[X]| \ge \epsilon\Big] \le 2e^{-2n\epsilon^2/(u-\ell)^2}. \tag{3.133}$$

By setting $\ell = 0$ and $u = 1$, we recover Equation (3.123) for the symmetric Bernoulli case, where $\mathbb{E}[X] = 1/2$.

Therefore, the bound $\Pr[|X - \mathbb{E}[X]| \ge \epsilon] \le 2e^{-2n\epsilon^2}$ found in the symmetric Bernoulli case is not special to Bernoulli variables. Hoeffding's inequality shows that, after introducing the factor $1/(u-\ell)^2$ for general bounded random variables, the same exponential tail bound continues to hold for the sample average.

As in Chebyshev's inequality, Hoeffding's inequality implies that the typical deviation scale is $\mathcal{O}(\frac{1}{\sqrt{n}})$, since the tail probability decreases rapidly when the deviation is multiplied by a constant factor. But unlike Chebyshev's inequality, the tail probability now decays exponentially rather than polynomially. We have already seen in Example 3.41 that exponential tail bounds can rule out tail events that may appear likely under Chebyshev's inequality.

Remark 3.45. Everything we have studied so far lives on a finite discrete probability space. In this setting, every random variable takes only finitely many values, so every random variable is automatically bounded. Therefore, Hoeffding's inequality applies to all random variables in finite discrete probability spaces. Once we move to infinite probability spaces, Hoeffding's inequality may not hold anymore. For instance, in Example 3.26 we saw that a geometric random variable is not bounded, and the rate function does not seem to grow quadratically as we move away from the mean in Figure 3.4. On the other hand, boundedness is not a necessary condition to obtain a similar kind of bound. In Chapter 5, we will meet the Gaussian distribution: it is certainly not bounded, but the sample average of Gaussian random variables still satisfies $\Pr[|X - \mathbb{E}[X]| \ge \epsilon] \le 2e^{-n\epsilon^2/2}$.

In fact, Hoeffding's inequality does not require the random variables to be i.i.d., but only independent. The following version makes this explicit.

Theorem 3.46 (Hoeffding's inequality)**.** Let $X := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average of n independent random variables $X_1, \ldots, X_n$ taking values in intervals $[\ell_1, u_1], \ldots, [\ell_n, u_n]$, respectively. Let s^2 be the mean squared interval length:

$$s^2 := \frac{1}{n}\sum_{i=1}^{n}(u_i - \ell_i)^2. \tag{3.134}$$

Then for all $\epsilon \geq 0$,

$$\Pr[X \geq \mathbb{E}[X] + \epsilon] \leq e^{-2n\epsilon^2/s^2}, \tag{3.135}$$

$$\Pr[X \leq \mathbb{E}[X] - \epsilon] \leq e^{-2n\epsilon^2/s^2}. \tag{3.136}$$

Proof. The basic idea is still to compute the rate function of X, then plug it into the generic Chernoff bound. However, to compute the rate function of X, we cannot directly apply Lemma 3.35 because the random variables are not necessarily identically distributed. Still, we can factorize the MGF of X into the MGFs of the individual random variables, and then apply Lemma 3.42 to each individual random variable.

Without loss of generality, we may assume $\mathbb{E}[X_1] = \mathbb{E}[X_2] = \cdots = \mathbb{E}[X_n] = \mathbb{E}[X] = 0$. Otherwise, we can always shift the random variables by their means to make them have zero mean. Following Lemma 3.35, we have for all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{tX}] = \mathbb{E}\left[e^{\frac{t}{n}\sum_{i=1}^{n} X_i}\right] = \prod_{i=1}^{n} \mathbb{E}[e^{\frac{t}{n}X_i}], \tag{3.137}$$

where the last equality follows from the independence of $X_1, \ldots, X_n$.

Applying Theorem 3.44 to $X_1, \ldots, X_n$, we obtain that for all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{tX}] = \mathbb{E}[e^{\frac{t}{n}\sum_{i=1}^{n} X_i}] = \prod_{i=1}^{n} \mathbb{E}[e^{\frac{t}{n}X_i}] \leq \prod_{i=1}^{n} e^{(t/n)^2(u_i-\ell_i)^2/8} \tag{3.138}$$

$$= \exp\left(\sum_{i=1}^{n}(t/n)^2(u_i - \ell_i)^2/8\right) \tag{3.139}$$

$$= e^{t^2 s^2/(8n)}. \tag{3.140}$$

Then, repeating the argument in the proof of Lemma 3.43 with $(u-\ell)^2$ replaced by s^2/n, we obtain that for all $\epsilon \in \mathbb{R}$,

$$I_X(\mathbb{E}[X] + \epsilon) \geq 2\epsilon^2 n/s^2. \tag{3.141}$$

Plugging this lower bound into Theorem 3.30 yields the desired bounds. □

Back to the i.i.d. case, all the random variables have the same interval length $u - \ell$, so the concentration is controlled by $s = u - \ell$, and the bound reduces to Theorem 3.44. When the random variables have different intervals, variables with larger possible fluctuations contribute more to s and thus the tail bound, while variables with smaller ranges contribute less. When the interval lengths

are different, variables with larger fluctuations contribute more to s^2 and thus the tail bound, while variables with smaller fluctuations contribute less. This is intuitive, since a random variable with much larger noise than others can dominate the overall signal.

Alternatively, Theorem 3.46 can be written in the following form for the sum of independent random variables.

Theorem 3.47. Let $X := \sum_{i=1}^n \Delta_i$ be the sum of n independent random variables $\Delta_1, \dots, \Delta_n$ taking values in intervals $[\tilde{\ell}_1, \tilde{u}_1], \dots, [\tilde{\ell}_n, \tilde{u}_n]$, respectively. Let L^2 be the sum of the squared interval lengths:

$$L^2 := \sum_{i=1}^n (\tilde{u}_i - \tilde{\ell}_i)^2. \tag{3.142}$$

Then for all $\epsilon \geq 0$,

$$\Pr[X \geq \mathbb{E}[X] + \epsilon] \leq e^{-2\epsilon^2/L^2}, \tag{3.143}$$

$$\Pr[X \leq \mathbb{E}[X] - \epsilon] \leq e^{-2\epsilon^2/L^2}. \tag{3.144}$$

Proof. Of course, we can apply Theorem 3.46 to the random variables $X_i := n\Delta_i$ for $i \in [n]$. But let us prove it directly for the sum $X = \sum_{i=1}^n \Delta_i$.

Without loss of generality, we may assume $\mathbb{E}[\Delta_1] = \mathbb{E}[\Delta_2] = \cdots = \mathbb{E}[\Delta_n] = \mathbb{E}[X] = 0$. Following Lemma 3.35, we have for all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{tX}] = \mathbb{E}\left[e^{t\sum_{i=1}^n \Delta_i}\right] = \prod_{i=1}^n \mathbb{E}[e^{t\Delta_i}], \tag{3.145}$$

Applying Theorem 3.44 to $\Delta_1, \dots, \Delta_n$, we obtain that for all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{tX}] = \mathbb{E}[e^{t\sum_{i=1}^n \Delta_i}] = \prod_{i=1}^n \mathbb{E}[e^{t\Delta_i}] \leq \prod_{i=1}^n e^{t^2(\tilde{u}_i - \tilde{\ell}_i)^2/8} = e^{t^2 L^2/8}. \tag{3.146}$$

Then, we repeat the argument in Lemma 3.43 to obtain $I_X(\mathbb{E}[X] + \epsilon) \geq 2\epsilon^2/L^2$. Plugging this lower bound into Theorem 3.30 yields the desired bounds. □

Remark 3.48. To apply Theorem 3.47 to the sample average of $X_1, \dots, X_n$, set $\Delta_i = \frac{1}{n} X_i$, $\tilde{\ell}_i = \ell_i/n$, and $\tilde{u}_i = u_i/n$. Then it is easy to verify that $L^2 = s^2/n$.

A typical application of Theorem 3.47 is to analyze the weighted averages of independent random variables. In machine learning, a common example is the exponential moving average (EMA), which is widely used to smooth time series data. For T random variables $X_1, \dots, X_T$, the EMA with decay factor $\beta \in [0, 1]$ is defined as

$$Y_0 = 0, \quad Y_t = \beta Y_{t-1} + (1 - \beta) X_t \quad \text{for } t \in [T]. \tag{3.147}$$

Or equivalently,

$$Y_t = \sum_{\tau=1}^t (1 - \beta)\beta^{t-\tau} X_\tau \quad \text{for } t \in [T]. \tag{3.148}$$

For instance, X_t may represent the training loss of a neural network at step t, which is inherently noisy because it is evaluated only on a mini-batch of data. The EMA suppresses random fluctuations across steps by averaging all past observations, while assigning exponentially larger weights to more recent ones. As a result, $(Y_t)_{t=1}^T$ typically reveals the underlying training trend more clearly than the raw sequence $(X_t)_{t=1}^T$.

Example 3.49 (Concentration of Exponential Moving Average)**.** Suppose $X_1, \dots, X_T$ are T i.i.d. random variables taking values in $[0, 1]$. Applying Hoeffding's inequality gives that for all $t \in [T]$ and $\epsilon \geq 0$,

$$\Pr[|Y_t - \mathbb{E}[Y_t]| \geq \epsilon] \leq 2e^{-2\epsilon^2/L_t^2}, \tag{3.149}$$

where

$$L_t^2 := \sum_{\tau=1}^{t} (1-\beta)^2 \beta^{2(t-\tau)} \leq (1-\beta)^2 \cdot \frac{1}{1-\beta^2} = \frac{1-\beta}{1+\beta}. \tag{3.150}$$

This upper bound becomes nearly tight when t is large.

As $\beta \to 1$, we have $L_t^2 \to 0$, so the tail bound exponentially decays to zero. This suggests that the EMA becomes increasingly concentrated to the true mean. In practice, a typical choice is $\beta = 0.9$, in which case $L_t^2 \leq \frac{1}{19}$. Compared with Hoeffding's inequality for sample averages, this means that Y_t concentrates as if it were the average of about 19 independent observations with uniform weights.

3.4.3 Azuma–Hoeffding Inequality for Martingales

Hoeffding's inequality comes from two key ingredients. First, each summand has a bounded range, which allows us to control its MGF. Second, the summands are independent, which allows us to factorize the MGF of the sum into the product of the MGFs of each individual summand:

$$\mathbb{E}[e^{tX}] = \mathbb{E}[e^{t\sum_{i=1}^n \Delta_i}] = \prod_{i=1}^n \mathbb{E}[e^{t\Delta_i}]. \tag{3.151}$$

Throughout the entire proof, this is the only place where independence is used. Although this identity is generally false for dependent random variables, exact factorization is not always necessary if our goal is only to upper bound the MGF of the sum.

In fact, a similar bound can still be obtained when the random variables are not independent, but instead form a *martingale*. Now we introduce the concept of martingales, and show how to obtain a similar bound for them.

Definition 3.50 (Martingale)**.** Let $Z_0, Z_1, \dots, Z_n$ be random variables. We say that $(Z_i)_{i=0}^n$ is a *martingale* if for every $i \in [n]$,

$$\mathbb{E}[Z_i \mid Z_0, \dots, Z_{i-1}] = Z_{i-1}. \tag{3.152}$$

In other words, once we know the entire past, the conditional expectation of the next value is exactly the current value. A useful way to picture a martingale is to imagine a gambler whose fortune evolves over time. At time $i-1$, the gambler's current fortune is Z_{i-1}. The gambler may adjust their strategy based on past experience, for example by betting more aggressively after winning many times in a row. But if the game is fair (well, assuming that the casino owner is not trying to make money), then even after conditioning on the entire history, the expected fortune at time i is still Z_{i-1}.

For convenience, if $(Z_0, Z_1, \ldots, Z_n)$ is a sequence of random variables, we write

$$Z_{<i} := (Z_0, Z_1, \ldots, Z_{i-1}), \quad \text{and} \quad Z_{\le i} := (Z_0, Z_1, \ldots, Z_i). \tag{3.153}$$

Under this notation, the martingale condition can be written as

$$\mathbb{E}[Z_i \mid Z_{<i}] = Z_{i-1}. \tag{3.154}$$

It is often helpful to rewrite a martingale in terms of its increments. The increment at step i is defined as

$$\Delta_i := Z_i - Z_{i-1}, \qquad i = 1, \ldots, n, \tag{3.155}$$

then Z_n can be written as the accumulated sum of the increments:

$$Z_n = Z_0 + \sum_{i=1}^{n} \Delta_i. \tag{3.156}$$

The martingale condition says that each increment has conditional mean zero:

$$\mathbb{E}[\Delta_i \mid Z_{<i}] = 0. \tag{3.157}$$

The following example connects sample averages with martingales.

Example 3.51. Let $X_1, \ldots, X_n$ be independent random variables with mean μ. If we define the partial sums

$$Z_i := \sum_{k=1}^{i} \Delta_i, \quad \Delta_i := \frac{1}{n}(X_i - \mu), \quad \forall i \in \{1, \ldots, n\}, \tag{3.158}$$

with the convention $Z_0 := 0$, then $(Z_i)_{i=0}^n$ is a martingale. This is because

$$\begin{aligned} \mathbb{E}[Z_i \mid Z_{<i}] &= \mathbb{E}[Z_i \mid X_{<i}] && (3.159)\\ &= \mathbb{E}\Big[Z_{i-1} + \tfrac{1}{n}(X_i - \mu) \;\Big|\; X_{<i}\Big] && (3.160)\\ &= Z_{i-1} + \mathbb{E}[\tfrac{1}{n}(X_i - \mu)] && (3.161)\\ &= Z_{i-1}, && (3.162) \end{aligned}$$

where the first equality follows from the fact that $Z_0, \ldots, Z_{i-1}$ are deterministic functions of $X_1, \ldots, X_{i-1}$ and vice versa.

Note that it is important to define the increments Δ_i as $X_i - \mu$ rather than X_i itself. Otherwise, the conditional mean of the increment would be $\mathbb{E}[X_i] = \mu$, rather than zero.

In the above example, $\Delta_1, \dots, \Delta_n$ are independent with each other. In general, these increments Δ_i need not be independent. The following is a classical example due to George Pólya, which appears repeatedly in many statistical models.

Example 3.52 (Pólya urn)**.** Consider an urn that initially contains b black balls and w white balls, so the total number of balls is $n := b + w$. At each step, we draw a ball uniformly at random and then return it together with one additional ball of the same color. Equivalently, you can think of a population of n people, in which each newcomer has to make a choice between two options. To do this, the newcomer talks to a random existing member and then adopts the same choice.

Let Z_t be the fraction of black balls in the urn after t steps. Then $(Z_t)_{t\geq 0}$ is a martingale. To see this, write

$$Z_t = \frac{(n+t-1)Z_{t-1} + X_t}{n+t}, \tag{3.163}$$

where $X_t \sim \text{Bernoulli}(Z_{t-1})$ indicates whether the ball drawn at step t is black. Conditioned on the past, the probability of drawing a black ball is exactly the current black fraction Z_{t-1}, so

$$\mathbb{E}[X_t \mid Z_{<t}] = Z_{t-1}. \tag{3.164}$$

Therefore,

$$\mathbb{E}[Z_t \mid Z_{<t}] = \frac{(n+t-1)Z_{t-1} + Z_{t-1}}{n+t} = Z_{t-1}. \tag{3.165}$$

So even though the composition of the urn changes over time, the expected next black fraction is still the current one.

For concentration, the martingale condition by itself is not enough. But once the increments are bounded, Hoeffding's inequality can be generalized to martingales. To see this, we first present the following lemma due to Kazuoki Azuma [Azu67], which generalizes Equation (3.146).

Lemma 3.53 (Azuma's lemma)**.** Let $(Z_i)_{i=0}^n$ be a martingale. Assume that for each $i \in [n]$, the increment $\Delta_i := Z_i - Z_{i-1}$ always lies in an interval $[\tilde{\ell}_i, \tilde{u}_i]$ for some $\tilde{\ell}_i, \tilde{u}_i \in \mathbb{R}$. Let L^2 be the sum of the squared interval lengths:

$$L^2 := \sum_{i=1}^{n} (\tilde{u}_i - \tilde{\ell}_i)^2. \tag{3.166}$$

Then for every $t \in \mathbb{R}$,

$$\mathbb{E}[e^{t(Z_n - Z_0)}] \leq e^{t^2 L^2/8}. \tag{3.167}$$

Proof. Since $(Z_i)_{i=1}^n$ is a martingale, each increment has conditional mean zero:

$$\mathbb{E}[\Delta_i \mid Z_{<i}] = 0. \tag{3.168}$$

We view the conditional probability as a new probability space following Remark 2.69 and apply Lemma 3.42 to obtain:

$$\mathbb{E}[e^{t\Delta_i} \mid Z_{<i}] \le e^{t^2(\tilde{u}_i-\tilde{\ell}_i)^2/8}. \tag{3.169}$$

Now we apply this inequality iteratively to bound the MGF of $Z_n - Z_0$. First, by the law of total expectation,

$$\mathbb{E}[e^{t(Z_n-Z_0)}] = \mathbb{E}\left[e^{t\sum_{i=1}^{n}\Delta_i}\right] = \mathbb{E}\left[e^{t\sum_{i=1}^{n-1}\Delta_i}\cdot e^{t\Delta_n}\right] \tag{3.170}$$

$$= \mathbb{E}\left[\mathbb{E}\left[e^{t\sum_{i=1}^{n-1}\Delta_i}\cdot e^{t\Delta_n} \mid Z_{<n}\right]\right]. \tag{3.171}$$

Since $\Delta_1, \dots, \Delta_{n-1}$ are deterministic functions of $Z_0, \dots, Z_{n-1}$, the only randomness in the inside expectation is $e^{t\Delta_n}$. Applying Equation (3.169), we have

$$\mathbb{E}\left[e^{t\sum_{i=1}^{n-1}\Delta_i}\cdot e^{t\Delta_n} \;\middle|\; Z_{<n}\right] = e^{t\sum_{i=1}^{n-1}\Delta_i}\mathbb{E}[e^{t\Delta_n} \mid Z_{<n}] \tag{3.172}$$

$$\le e^{t\sum_{i=1}^{n-1}\Delta_i}e^{t^2(\tilde{u}_n-\tilde{\ell}_n)^2/8}. \tag{3.173}$$

Plugging this back into Equation (3.171), we obtain

$$\mathbb{E}[e^{t(Z_n-Z_0)}] \le \mathbb{E}\left[e^{t\sum_{i=1}^{n-1}\Delta_i}\right]e^{t^2(\tilde{u}_n-\tilde{\ell}_n)^2/8}. \tag{3.174}$$

Repeating this argument iteratively for $\Delta_{n-1}, \Delta_{n-2}, \dots, \Delta_1$ yields

$$\mathbb{E}[e^{t(Z_n-Z_0)}] \le \prod_{i=1}^{n} e^{t^2(\tilde{u}_i-\tilde{\ell}_i)^2/8} = e^{t^2L^2/8}, \tag{3.175}$$

where $L^2 := \sum_{i=1}^{n}(\tilde{u}_i - \tilde{\ell}_i)^2$. □

This upper bound on the MGF of the martingale naturally yields a lower bound on the rate function, which in turn leads to the following Chernoff bound for martingales, known as the *Azuma–Hoeffding inequality*.

Theorem 3.54 (Azuma–Hoeffding inequality)**.** Let $(Z_i)_{i=0}^{n}$ be a martingale. Assume that for each $i \in [n]$, the increment $\Delta_i := Z_i - Z_{i-1}$ always lies in an interval $[\tilde{\ell}_i, \tilde{u}_i]$ for some $\tilde{\ell}_i, \tilde{u}_i \in \mathbb{R}$. Let L^2 be the sum of the squared interval lengths:

$$L^2 := \sum_{i=1}^{n}(\tilde{u}_i - \tilde{\ell}_i)^2. \tag{3.176}$$

Then for all $\epsilon \ge 0$,

$$\Pr[Z_n \ge Z_0 + \epsilon] \le e^{-2\epsilon^2/L^2}, \tag{3.177}$$

$$\Pr[Z_n \le Z_0 - \epsilon] \le e^{-2\epsilon^2/L^2}. \tag{3.178}$$

Proof. Based on Lemma 3.53, we repeat the argument in Lemma 3.43 to obtain $I_{Z_n-Z_0}(\epsilon) \ge 2\epsilon^2/L^2$. Then plug this lower bound into Theorem 3.30. □

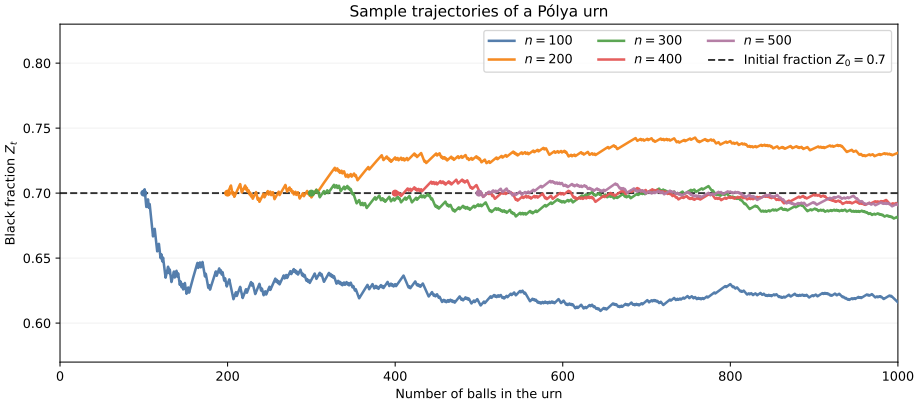


Figure 3.8: Trajectories of Z_t in a Pólya urn with initial condition $Z_0 = 0.7$.

This theorem should be read as the martingale version of Hoeffding's inequality. The quantity $u_i - \ell_i$ measures how much uncertainty can be introduced by the increment at step i. If every step is tightly controlled, then the whole process is tightly concentrated. The bound shares the same L^2 as in Theorem 3.47, which is defined as the sum of the squared interval lengths.

Example 3.55 (Concentration for a Pólya urn)**.** Let us return to the Pólya urn example. We already know that $(Z_t)_{t\geq 0}$ is a martingale, where Z_t is the fraction of black balls after t steps. To apply Azuma–Hoeffding, we only need to bound the increments $\Delta_t := Z_t - Z_{t-1}$.

From Equation (3.163), we have

$$\Delta_t = \frac{(n+t-1)Z_{t-1} + X_t}{n+t} - Z_{t-1} = \frac{X_t - Z_{t-1}}{n+t}, \tag{3.179}$$

which is always bounded in the interval $[-\frac{1}{n+t}, \frac{1}{n+t}]$. Applying Azuma–Hoeffding inequality, we obtain

$$\Pr\Big[|Z_t - Z_0| \geq \epsilon\Big] \leq 2e^{-2\epsilon^2/L^2} \quad \text{where} \quad L^2 = \sum_{\tau=1}^{t} \left(\frac{2}{n+\tau}\right)^2. \tag{3.180}$$

To get a more concrete bound, we need to compute L^2. Since

$$\sum_{\tau=1}^{t} \frac{1}{(n+\tau)^2} \leq \int_0^t \frac{1}{(n+\tau)^2} d\tau = -\frac{1}{n+\tau}\Big|_0^t \leq \frac{1}{n}, \tag{3.181}$$

we can upper bound L^2 by $\frac{4}{n}$, and thus obtain the tail bound

$$\Pr\Big[|Z_t - Z_0| \geq \epsilon\Big] \leq 2e^{-2\epsilon^2/(4/n)} = 2e^{-\epsilon^2 n/2}. \tag{3.182}$$

So the typical deviation scale is on the order of $\mathcal{O}(\frac{1}{\sqrt{n}})$. In particular, if the urn starts with many balls, then each subsequent step changes the black fraction only a little, and even after many dependent updates the fraction still stays close to its initial value $Z_0 = b/(b+w)$. See Figure 3.8.

We can relax Azuma–Hoeffding inequality in several ways. First, in our definition of the martingale $(Z_i)_{i=0}^n$ (Definition 3.50), we require that

$$\mathbb{E}[Z_i \mid Z_{<i}] = Z_{i-1}. \tag{3.183}$$

However, in many applications, $(Z_i)_{i=0}^n$ itself is constructed from another sequence of random variables $(X_i)_{i=0}^n$, where Z_i is a deterministic function of $X_0, \dots, X_i$. In such cases, conditioning on $X_{<i}$ is more convenient than conditioning on $Z_{<i}$. We will soon see an example of this in the proof of McDiarmid's inequality. Therefore, we extend the definition of a martingale as follows:

Definition 3.56 (Martingale)**.** Let $(X_i)_{i=0}^n$ be a sequence of random variables, and let $(Z_i)_{i=0}^n$ be another sequence of random variables such that each Z_i is a deterministic function of $X_0, \dots, X_i$. We say that $(Z_i)_{i=0}^n$ is a *martingale* with respect to the information revealed by $(X_i)_{i=0}^n$ if for every $i \in [n]$,

$$\mathbb{E}[Z_i \mid X_{<i}] = Z_{i-1}. \tag{3.184}$$

By the law of total expectation, or more precisely, the tower rule (Exercise 2.26), conditioning on $X_{<i}$ and then on a function of $X_{<i}$ is equivalent to conditioning directly on that function. Therefore,

$$\mathbb{E}[Z_i \mid Z_{<i}] = \mathbb{E}\big[\mathbb{E}[Z_i \mid X_{<i}] \,\big|\, Z_{<i}\big] = \mathbb{E}[Z_{i-1} \mid Z_{<i}] = Z_{i-1}. \tag{3.185}$$

Therefore, any martingale defined according to Definition 3.56 is also a martingale in the sense of Definition 3.50, and hence it automatically satisfies the Azuma–Hoeffding inequality.

Further, we note that the interval $[\tilde{\ell}_i, \tilde{u}_i]$ that bounds the increment $\Delta_i := Z_i - Z_{i-1}$ can be allowed to depend on the previous values $Z_0, \dots, Z_{i-1}$, since Hoeffding's lemma in Equation (3.169) is always applied conditionally on $Z_{<i}$. In fact, for a martingale defined according to Definition 3.56, the interval $[\tilde{\ell}_i, \tilde{u}_i]$ can even depend on $X_0, \dots, X_{i-1}$ instead of $Z_0, \dots, Z_{i-1}$, since we can replace the conditioning variable $Z_{<i}$ by $X_{<i}$ throughout the proof of the Azuma–Hoeffding inequality. Now we state the following variant of Azuma–Hoeffding inequality without proof.

Theorem 3.57 (Azuma–Hoeffding inequality)**.** Let $(Z_i)_{i=0}^n$ be a martingale with respect to the information revealed by $(X_i)_{i=0}^n$. Assume that for each $i \in [n]$, the increment $\Delta_i := Z_i - Z_{i-1}$ always lies in an interval of length at most c_i when conditioned on $X_{<i}$. That is, there exist functions $\tilde{\ell}_i, \tilde{u}_i : \mathbb{R}^i \to \mathbb{R}$ such that with probability 1,

$$\tilde{\ell}_i(X_{<i}) \le \Delta_i \le \tilde{u}_i(X_{<i}), \qquad \tilde{u}_i(X_{<i}) - \tilde{\ell}_i(X_{<i}) \le c_i. \tag{3.186}$$

Let $L^2 := \sum_{i=1}^n c_i^2$. Then for all $\epsilon \ge 0$,

$$\Pr[Z_n \ge Z_0 + \epsilon] \le e^{-2\epsilon^2/L^2}, \tag{3.187}$$

$$\Pr[Z_n \le Z_0 - \epsilon] \le e^{-2\epsilon^2/L^2}. \tag{3.188}$$

3.4.4 From Averages to Functions: McDiarmid's Inequality

An interesting martingale to apply Azuma–Hoeffding is the *Doob martingale*, named after Joseph L. Doob [Doo40].

Definition 3.58 (Doob martingale)**.** Let $X_1, \dots, X_n$ be independent random variables, and let $f : \mathbb{R}^n \to \mathbb{R}$ be a function. The *Doob martingale* associated with f and $X_1, \dots, X_n$ is the sequence $(Z_i)_{i=0}^n$ defined by

$$Z_i := \mathbb{E}[f(X_1, \dots, X_n) \mid X_1, \dots, X_i]. \tag{3.189}$$

Intuitively, the Doob martingale describes a process that reveals the values of $X_1, \dots, X_n$ one by one. At each step, it updates the conditional expectation of f based on the revealed values. It forms a martingale for a very simple reason: at each step, Z_i is basically *defined* as the conditional expectation of the next step, so very unsurprisingly, the conditional expectation of the next step is just Z_i.

Theorem 3.59. The Doob martingale $(Z_i)_{i=0}^n$ is a martingale with respect to the information revealed by $(X_i)_{i=1}^n$. That is, for all $i \in [n]$,

$$\mathbb{E}[Z_i \mid X_{<i}] = Z_{i-1}. \tag{3.190}$$

Proof. By the definition of Z_i, we have

$$\mathbb{E}[Z_i \mid X_{<i}] = \mathbb{E}[\mathbb{E}[f(X_1, \dots, X_n) \mid X_{\le i}] \mid X_{<i}]. \tag{3.191}$$

By the tower rule, this simplifies to

$$\mathbb{E}[Z_i \mid X_{<i}] = \mathbb{E}[f(X_1, \dots, X_n) \mid X_{<i}] = Z_{i-1}, \tag{3.192}$$

which verifies the martingale condition. □

Although the Azuma–Hoeffding inequality appears to only bound sums or sample averages, applying it to the Doob martingale leads to a striking consequence: as long as a function f has bounded influence on each coordinate, it concentrates around its mean under independent sampling of the coordinates. This is known as *McDiarmid's inequality*, named after Colin McDiarmid [McD89].

Theorem 3.60 (McDiarmid's inequality)**.** Let $X_1, \dots, X_n$ be independent random variables, and let $f : \mathbb{R}^n \to \mathbb{R}$ be a function. Assume that for each $i = 1, \dots, n$ there is a constant $c_i \ge 0$ such that whenever two inputs differ only in coordinate i,

$$|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x_i', \dots, x_n)| \le c_i. \tag{3.193}$$

Let $L^2 := \sum_{i=1}^n c_i^2$. Then for all $\epsilon \ge 0$,

$$\Pr\Big[f(X_1, \dots, X_n) \ge \mathbb{E}[f(X_1, \dots, X_n)] + \epsilon\Big] \le e^{-2\epsilon^2/L^2}, \tag{3.194}$$

$$\Pr\Big[f(X_1, \dots, X_n) \le \mathbb{E}[f(X_1, \dots, X_n)] - \epsilon\Big] \le e^{-2\epsilon^2/L^2}. \tag{3.195}$$

Proof. Let $F := f(X_1, \ldots, X_n)$ and define the Doob martingale

$$Z_i := \mathbb{E}[F \mid X_1, \ldots, X_i]. \tag{3.196}$$

Fix i and condition on $X_1, \ldots, X_{i-1}$. For each possible value x_i of the i-th coordinate, define

$$g(x_i) := \mathbb{E}\big[F \mid X_{<i}, X_i = x_i\big], \tag{3.197}$$

where we omit the dependence of the function g on $X_1, \ldots, X_{i-1}$ because they have already been conditioned on.

By definition, the increment Δ_i is given by

$$\begin{aligned}
\Delta_i := Z_i - Z_{i-1} &= \mathbb{E}[F \mid X_{\le i}] - \mathbb{E}[F \mid X_{<i}] && (3.198)\\
&= \mathbb{E}[F \mid X_{\le i}] - \mathbb{E}[\mathbb{E}[F \mid X_{\le i}] \mid X_{<i}] && (3.199)\\
&= g(X_i) - \mathbb{E}[g(X_i) \mid X_{<i}]. && (3.200)
\end{aligned}$$

We claim that Δ_i lies in an interval of length at most c_i when conditioned on $X_{<i}$. To see this, let x_i^* be the value of X_i that minimizes $g(x_i)$. By Equation (3.193),

$$f(X_1, \ldots, X_i, \ldots, X_n) \le f(X_1, \ldots, x_i^*, \ldots, X_n) + c_i. \tag{3.201}$$

Taking conditional expectation on $X_{\le i}$, we get $g(X_i) \le g(x_i^*) + c_i$. Therefore,

$$g(x_i^*) - \mathbb{E}[g(X_i) \mid X_{<i}] \le \Delta_i \le g(x_i^*) + c_i - \mathbb{E}[g(X_i) \mid X_{<i}], \tag{3.202}$$

Both the lower and upper bounds are functions of $X_{<i}$, and hence Δ_i lies in an interval of length at most c_i when conditioned on $X_{<i}$.

Applying Theorem 3.57 to the martingale $(Z_i)_{i=0}^n$, we obtain

$$\Pr\Big[Z_n \ge Z_0 + \epsilon\Big] \le e^{-2\epsilon^2/L^2}, \quad \Pr\Big[Z_n \le Z_0 - \epsilon\Big] \le e^{-2\epsilon^2/L^2}. \tag{3.203}$$

Substituting $Z_n = F$ and $Z_0 = \mathbb{E}[F]$, we obtain the desired bounds. □

The proof is worth comparing with the sample-average case. For a sample average, changing one coordinate can change the function value by at most $(u - \ell)/n$. McDiarmid's inequality shows that the exact form of the arithmetic mean is not essential. What really matters is Equation (3.193): changing any single coordinate only has a limited effect on the final result. This property is known as the *bounded difference* condition, and McDiarmid's inequality is also referred to as the *bounded difference inequality* in the literature.

Remark 3.61. Although we state McDiarmid's inequality for real-valued input variables $X_1, \ldots, X_n$, it is easy to see McDiarmid's inequality can be generalized to allow for arbitrary input types, not just real numbers. The only important assumption is that the function f has bounded differences.

The following example illustrates how McDiarmid's inequality applies to a function that is far more complicated and computationally hard than the arithmetic mean. It may seem surprising at first that one can still obtain sharp concentration bounds without even computing the expectation of the function.

Example 3.62 (Bin packing)**.** The bin packing problem is a classic problem in computer science, which asks how many fixed-size bins are needed to pack a given set of items. It captures practical tasks such as packing files into a minimum number of fixed-capacity storage blocks, consolidating virtual machines onto as few physical servers as possible in cloud platforms (such as Aliyun or Volcengine), or grouping data chunks into fixed-size packets for network transmission.

Now let us consider the bin packing problem with random item sizes. Let $X_1, \ldots, X_n$ be i.i.d. random variables taking values in $\{0, 1, \ldots, m\}$. Each X_i represents the size of an item to be packed into a bin. Each bin can hold items of total size at most m. Let $B^{(n)} = f(X_1, \ldots, X_n)$ be the minimum number of bins needed to store all the items into bins without exceeding the capacity of any bin.

Computing $B^{(n)}$ for a given set of item sizes is *NP-hard* in general. Even so, McDiarmid's inequality still implies that $B^{(n)}$ concentrates around its mean:

$$\Pr\Big[|B^{(n)} - \mathbb{E}[B^{(n)}]| \geq \epsilon\Big] \leq 2e^{-2\epsilon^2/L^2} = 2e^{-2\epsilon^2/n}, \tag{3.204}$$

where $L^2 = \sum_{i=1}^{n} 1 = n$ because changing one item size can change the bin count by at most 1. So the typical deviation scale is of order $\mathcal{O}(\sqrt{n})$.

Finally, a key limitation of Hoeffding's inequality and all these extensions is that it relies on worst-case bounds such as $X \in [\ell, u]$ and ignores finer information such as the variance. In contrast, Chebyshev's inequality provides smaller bounds for random variables with smaller variance. Can we develop a more refined concentration inequalities that incorporate both the variance and the worst-case bounds on the random variables? The answer is yes, and we will return to this question later in Section 6.2.

Exercises

3.1. Is it possible that for some random variable X that takes non-negative integer values, Markov's inequality (Theorem 3.1) only looses by a constant factor from being tight for all integers $a > 0$? That is, is there a random variable X and a constant $c > 0$ such that the following inequality holds for all integers $a > 0$?

$$\frac{c\,\mathbb{E}[X]}{a} \leq \Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

Of course, the inequality holds in the trivial case where X always takes value 0. So the question is whether there exists any other random variable X that satisfies this inequality.

Note: Although we have not yet introduced infinite probability spaces, you may assume that the probability mass function $p_X : \mathbb{N} \to \mathbb{R}$ of X can be any non-negative function that sums to 1. This function also has to ensure that $\mathbb{E}[X]$ exists and is finite.

3.2. A function $f : \mathbb{R} \to \mathbb{R}$ is called *convex* if for all $x, y \in \mathbb{R}$ and $t \in [0, 1]$,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y).$$

For example, $|x|, x^2, e^x, -\log x$ and $\log(1+e^x)$ are all convex functions, but x^3 and $\sin x$ are not. A convex function is always continuous. It may not be differentiable, but for any convex function f, the left-derivative $f'_-(x) := \lim_{h\to 0^-} \frac{f(x+h)-f(x)}{h}$ and the right-derivative $f'_+(x) := \lim_{h\to 0^+} \frac{f(x+h)-f(x)}{h}$ exist at every point $x \in \mathbb{R}$.

(a) Prove the following statement: If $f : \mathbb{R} \to \mathbb{R}$ is a convex function, then for any $y \in \mathbb{R}$, there exists a factor $\alpha \in \mathbb{R}$ such that for all $x \in \mathbb{R}$,

$$f(x) \geq f(y) + \alpha(x - y).$$

That is, the function f is lower bounded by a line with slope α that passes through the point $(y, f(y))$.

Hint: What happens when $t \to 0^+$?

(b) Let X be a random variable and $f : \mathbb{R} \to \mathbb{R}$ be a convex function. Prove *Jensen's inequality* for random variables:

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

Note: For this problem, you should not use any advanced results, such as some alternative version of Jensen's inequality that you may have seen elsewhere. Instead, prove the above statements directly from the definition and the basic properties of convexity listed above. By the way, the length of this note is already longer than the proof you are expected to write.

3.3. Prove Theorem 3.17 and Theorem 3.18.

3.4. Let $X_1, \ldots, X_n$ be i.i.d. zero-mean random variables, i.e., $\mathbb{E}[X_i] = 0$ for all $i \in [n]$. Let $\mu_m := \mathbb{E}[X_i^m]$ denote the m-th raw moment of an arbitrary X_i. Let X be the sample average of $X_1, \ldots, X_n$. Express $\mathbb{E}[X^3]$, $\mathbb{E}[X^4]$, and $\mathbb{E}[X^5]$ in terms of $\mu_2, \mu_3, \mu_4, \mu_5$.

Note: This exercise together with the previous one are meant to illustrate that higher-order moments are harder to compute than lower-order ones.

3.5. Define the L^p-norm of a random variable X by

$$\|X\|_p := \mathbb{E}[|X|^p]^{1/p}.$$

Use Jensen's inequality (Exercise 3.2) to prove the following inequalities for any real numbers $p, q > 0$ and any pair of random variables X, Y defined on the same finite discrete probability space:

(a) **Lyapunov's inequality** (L^p-norm increases with p):

$$\|X\|_p \leq \|X\|_q \quad \text{if} \quad p \leq q.$$

(b) **Hölder's inequality** ("product $\leq$ product"):

$$\|XY\|_1 \leq \|X\|_p \|Y\|_q \quad \text{if} \quad 1/p + 1/q = 1.$$

(c) **Young's inequality** ("product $\leq$ sum"):

$$\|XY\|_1 \leq \tfrac{1}{p}\|X\|_p^p + \tfrac{1}{q}\|Y\|_q^q \quad \text{if} \quad 1/p + 1/q = 1.$$

(d) **Minkowski's inequality** ("sum $\leq$ sum"):

$$\|X + Y\|_p \leq \|X\|_p + \|Y\|_p \quad \text{if} \quad p \geq 1.$$

Note: It is generally impossible to have "sum $\leq$ product," so you only have four inequalities to prove.

Note: Although these inequalities can also be proved by other methods you may have seen elsewhere, such as calculus, the point of this exercise is to practice applying Jensen's inequality. Finding the right way to apply Jensen's inequality here is quite challenging, but once you work it out after a sufficient amount of struggle, you will never forget these fundamental inequalities you proved. For convenience, you may simply state that any function to which you apply Jensen's inequality is convex, without proving it from the definition.

Note: You may have encountered the following special cases elsewhere:

- **Cauchy-Schwarz inequality** (Hölder's inequality with $p = q = 2$):

$$\|XY\|_1 \leq \|X\|_2 \|Y\|_2$$

- **Triangle inequality** (Minkowski's inequality with $p = 2$):

$$\|X + Y\|_2 \leq \|X\|_2 + \|Y\|_2$$

Hint: The following are hints for each part of the problem:

(a) The proof will be a one-liner if you find the right convex function to apply Jensen's inequality;

(b) There are too many variables: shift the probability mass in the probability space so that the inequality is only about X;

(c) Make each of p and q to appear at most once on the right-hand side of the inequality (for this part, it is ok to prove the inequality for deterministic X, Y first);

(d) Now it is your turn: combine several of the ideas above to complete the proof.

3.6. Let X be a random variable.

(a) Prove that for all $a \geq 0$,

$$\mathbb{E}[X \mathbb{1}_{[X>a]}] \leq \mathbb{E}[X^2]^{1/2} \Pr[X > a]^{1/2}.$$

(b) Suppose X is non-negative and $\mathbb{E}[X^2] \neq 0$. Prove that for all $\theta \in [0, 1]$,

$$\Pr[X > \theta \mathbb{E}[X]] \geq (1-\theta)^2 \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}.$$

Note: This inequality is known as the Paley–Zygmund inequality. It is similar to Markov's inequality, but it provides an *anti-concentration* bound rather than a concentration bound.

(c) Suppose $\sigma(X) > 0$. Prove that for all $\alpha \in [0, 1]$,

$$\Pr\Big[|X - \mathbb{E}[X]| > \alpha \cdot \sigma(X)\Big] \geq \frac{(1-\alpha^2)^2}{\kappa(X)}.$$

where

$$\kappa(X) := \frac{\mathbb{E}[|X - \mathbb{E}[X]|^4]}{\sigma(X)^4}$$

is called the *kurtosis* of X.

Note: Compared with Corollary 3.15, this inequality gives a non-trivial probability lower bound exactly in the regime where Chebyshev's inequality becomes vacuous.

3.7. Consider a random walk on a circle with vertices $0, 1, \ldots, n-1$. Initially, the drunkman is at vertex 0. At each step, a drunkman moves from the current vertex i to either $(i+1) \bmod n$ or $(i-1) \bmod n$ with equal probability. Prove that there exist universal constants $A > 0$ and $c > 0$ such that the probability that the drunkman has visited all vertices after t steps is at least $1 - Ae^{-ct/n^2}$ for all $t \geq 0$.

Hint: One proof strategy is to first show that for some $T = \mathcal{O}(n^2)$, this probability is at least 0.01 after T steps.

3.8. Prove Proposition 3.31. For convenience, you can only need to consider the case of $a \geq \mu$. That is, you only need to prove the following:

(a) $I(a) = +\infty$ if $a > u$;

(b) $0 \leq I(a) < +\infty$ for all $a \in [\mu, u]$;

(c) $I(\mu) = 0$.

(d) $I(a)$ is convex on the interval $[\mu, u]$: for any $a, b \in [\mu, u]$ and $\lambda \in [0, 1]$,

$$I(\lambda a + (1-\lambda)b) \le \lambda I(a) + (1-\lambda) I(b).$$

3.9. Let P be a math problem whose answer is either yes or no. Suppose you have trained an LLM that is good at problem solving. But P is very hard, so your LLM can only solve it with probability $1/2 + \epsilon$. To increase the probability of success, you can run the LLM multiple times and take the majority vote. That is, you run the LLM n times and answer yes if and only if at least half of the answers are yes. Prove that for some $n_{\min} = \mathcal{O}(\frac{1}{\epsilon^2} \log \frac{1}{\delta})$ as $(\epsilon, \delta) \to (0, 0)$, if you run the LLM $n \ge n_{\min}$ times, then the majority vote is correct with probability at least $1 - \delta$.

3.10. Let G be a random graph drawn from the Erdős–Rényi model $G(n, p)$. That is, G has n vertices, and for each pair of distinct vertices $i \ne j$, between them an undirected edge is included with probability p. A coloring of G is an assignment of each vertex with a color such that no two adjacent vertices have the same color. The *chromatic number* $\chi(G)$ is the minimum number of colors needed to color the graph G. Show that for all $\alpha \ge 0$,

$$\Pr\Big[|\chi(G) - \mathbb{E}[\chi(G)]| \ge \alpha\sqrt{n}\Big] \le 2e^{-2\alpha^2}.$$

Note: As $n \to \infty$, $\mathbb{E}[\chi(G)] \sim \frac{n}{2 \log_b n}$, where $b = 1/(1-p)$ [Bol88]. But the fluctuation scale is only of order $\mathcal{O}(\sqrt{n})$.

3.11. Following McDiarmid's inequality, let $X_1, \dots, X_n$ be independent random variables, and let $f : \mathbb{R}^n \to \mathbb{R}$ be a function. Assume that for each $i = 1, \dots, n$ there is a constant $c_i \ge 0$ such that whenever two inputs differ only in coordinate i,

$$|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x_i', \dots, x_n)| \le c_i. \tag{3.205}$$

Let $L^2 := \sum_{i=1}^n c_i^2$. Now, instead of giving a high-probability tail bound, prove the following upper bound on the variance of $f(X_1, \dots, X_n)$:

$$\mathrm{Var}[f(X_1, \dots, X_n)] \le \frac{1}{4} L^2.$$

3.12. Let $p(x)$ be a probability mass function supported on a finite set $\mathcal{X}$. That is, $p(x) > 0$ for all $x \in \mathcal{X}$. In information theory, the *entropy* of p is defined as

$$H(p) := \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{1}{p(x)}.$$

(a) Prove the following inequality:

$$0 \le H(p) \le \log_2 |\mathcal{X}|.$$

(b) Let $X_1, \dots, X_n$ be n i.i.d. random variables drawn from p. For any $(x_1, \dots, x_n) \in \mathcal{X}^n$, let $P(x_1, \dots, x_n)$ be the probability of the event

$\{X_1 = x_1, \ldots, X_n = x_n\}$. Prove that there exists $C > 0$ such that for all $\epsilon > 0$ and $n \geq 1$,

$$\Pr\Big[2^{-n(H(p)+\epsilon)} \leq P(X_1, \ldots, X_n) \leq 2^{-n(H(p)-\epsilon)}\Big] \geq 1 - 2e^{-C\epsilon^2 n}.$$

That is, when n is large, typically $P(X_1, \ldots, X_n) \approx 2^{-nH(p)}$. This is called the *asymptotic equipartition property* (AEP).

Note: In information theory, the *ϵ-typical set* is defined as the set of all sequences $x_1, \ldots, x_n$ such that

$$2^{-n(H(p)+\epsilon)} \leq P(x_1, \ldots, x_n) \leq 2^{-n(H(p)-\epsilon)}.$$

So $(X_1, \ldots, X_n)$ is typically in the ϵ-typical set.

3.13. Let $p(x)$ be a probability mass function supported on a finite set $\mathcal{X}$. Alice randomly draws n i.i.d. random variables $X_1, \ldots, X_n$ from p and wants to send the values of these random variables to Bob. However, Alice's network is very slow, so she wants to minimize the number of bits she needs to send.

A *coding scheme* is a pair of encoding and decoding functions (f, g), where $f : \mathcal{X}^n \to \{0,1\}^*$ encodes the random variables $X_1, \ldots, X_n$ into a binary string $s \in \{0,1\}^*$ of arbitrary length, and $g : \{0,1\}^* \to \mathcal{X}^n$ decodes the binary string s back into the random variables $X_1, \ldots, X_n$.

A coding scheme is *lossless* if for all $(x_1, \ldots, x_n) \in \mathcal{X}^n$,

$$g(f(x_1, \ldots, x_n)) = (x_1, \ldots, x_n).$$

Design a lossless coding scheme such that

$$\mathbb{E}[|f(X_1, \ldots, X_n)|] \leq nH(p) + \mathcal{O}(\sqrt{n \log n}), \quad \text{as } n \to \infty,$$

where $|f(X_1, \ldots, X_n)|$ is the length of the binary string $f(X_1, \ldots, X_n)$.

3.14. Prove the *data processing inequality* for entropy: if X is a random variable with probability mass function p_X, and $Y = f(X)$ is a function of X, then the probability mass function of Y, denoted as p_Y, satisfies

$$H(p_Y) \leq H(p_X).$$

Then, use the data processing inequality to prove that in the setting of the previous problem, it is impossible to design a lossless coding scheme (f, g) such that

$$\mathbb{E}[|f(X_1, \ldots, X_n)|] < nH(p) - \log_2(2 + n \log_2 |\mathcal{X}|), \quad \text{as } n \to \infty.$$

Hint: First, argue that if the maximum length of $f(X_1, \ldots, X_n)$ exceeds $\lceil n \log_2 |\mathcal{X}| \rceil$, we can always find a coding scheme with a shorter expected length. Then, if the expected length of $f(X_1, \ldots, X_n)$ is L, what is the maximum possible value of its entropy?

Note: Together with the previous problem, you have proved Shannon's source coding theorem: the minimum expected length for encoding n i.i.d. random variables is $\sim nH(p)$ bits. In fact, for *prefix-free* coding schemes,

one can use *Kraft's inequality* to show that the minimum expected length lies in the interval $[nH(p), nH(p) + 1)$. Interested readers may refer to any textbook on information theory, such as [CT05].

Chapter 4

Generating Functions

If a problem looks difficult, maybe we just haven't found the right way to look at the problem. A notable example of this idea is the use of *generating functions* to study sequences. The basic idea works as follows. Suppose we have a sequence of numbers $(a_0, a_1, a_2, \dots)$. Instead of studying them directly, we *encode* them as a single object called the generating function, which is a power series of the form

$$A(x) = \sum_{k \geq 0} a_k x^k. \tag{4.1}$$

At first glance, this may seem like an unnecessary change of language. In fact, it even introduces some seemingly meaningless expressions. For example, if $(a_n)_{n \geq 0}$ counts the number of permutations $\sigma \in \mathrm{Perm}(n)$ whose maximum cycle length exceeds 50%, then what does $A(10)$ even mean? Why encoding these numbers as a function if we can study them directly?

One obvious reason is that we often do not know these numbers explicitly. Instead, we may have to derive them indirectly, for example by relating them to other sequences. Once we encode a sequence as a generating function, these relations can become operations on functions, such as addition and multiplication. This allows us to first derive the generating function of the target sequence from these relations, and then recover, or "generate," the numbers we want from it.

Another important reason is that encoding a sequence as a function allows us to study its global properties. In fact, we have already seen such an example in Chapter 3: we can encode the moments $\mathbb{E}[X^k]$ of a random variable as the moment generating function (MGF),

$$M_X(t) := \mathbb{E}[e^{tX}] = \sum_{k \geq 0} \frac{t^k}{k!} \mathbb{E}[X^k]. \tag{4.2}$$

From this single function, we can derive many useful concentration properties of the underlying random variable X.

In this chapter, we give a brief introduction to generating functions, for both sequences and random variables.

4.1 Generating Functions of Sequences

In this section, we introduce generating functions for study sequences $(a_k)_{k\geq 0}$.

4.1.1 Ordinary Generating Functions

When the object we want to study is a sequence $(a_k)_{k\geq 0} = (a_0, a_1, a_2, \dots)$, the most direct generating function is the *ordinary generating function*.

Definition 4.1 (Ordinary generating function)**.** The *ordinary generating function* (OGF) of a sequence of real numbers $(a_k)_{k\geq 0}$ is the power series

$$A(x) := \sum_{k\geq 0} a_k x^k. \tag{4.3}$$

The coefficient of x^k records the k-th term of the sequence, so the entire sequence is stored in a single expression. While the variable x may look mysterious, it is simply a placeholder that marks the index k. Once we encode the sequence in this way, algebraic manipulations of $A(x)$ often translate into useful identities for the coefficients a_k.

Example 4.2 (Geometric series)**.** If $a_k = \beta^k$ for all $k \geq 0$, then

$$A(x) = \sum_{k\geq 0} \beta^k x^k = \frac{1}{1-\beta x}. \tag{4.4}$$

In particular, when $a_k = 1$ for all $k \geq 0$, then

$$A(x) = \frac{1}{1-x}. \tag{4.5}$$

When $a_k = (-1)^k$ for all $k \geq 0$, then

$$A(x) = \frac{1}{1+x}. \tag{4.6}$$

Example 4.3 (Binomial coefficients)**.** If $a_k = \binom{n}{k}$ for all $0 \leq k \leq n$, then by the binomial theorem,

$$A(x) = \sum_{k=0}^{n} \binom{n}{k} x^k = (1+x)^n. \tag{4.7}$$

More generally, Newton's generalized binomial theorem states that for any $r \in \mathbb{R}$,

$$(1+x)^r = \sum_{k\geq 0} \binom{r}{k} x^k, \quad \text{where} \quad \binom{r}{k} = \frac{r(r-1)\cdots(r-k+1)}{k!}. \tag{4.8}$$

This corresponds to the sequence $a_k = \binom{r}{k}$.

Given a generating function $A(x)$, we can recover the corresponding sequence by taking its Taylor expansion at 0:

$$A(x) = \sum_{k\geq 0} \frac{A^{(k)}(0)}{k!} x^k \quad \Rightarrow \quad a_k = \frac{A^{(k)}(0)}{k!}. \tag{4.9}$$

Here $A^{(k)}$ denotes the k-th order derivative of A.

Example 4.4 (Exponential function and factorials)**.** The Taylor expansion of the exponential function e^x is

$$e^x = \sum_{k\geq 0} \frac{1}{k!} x^k. \tag{4.10}$$

Therefore, when viewed as a generating function, the corresponding sequence is $a_k = \frac{1}{k!}$.

Remark 4.5. If the sequence is $a_k = k!$, what is its generating function? What a silly question! Of course it is ...

$$A(x) = \sum_{k\geq 0} k!\, x^k. \tag{4.11}$$

Wait — does this even converge? In fact,



It does not converge for any $x \neq 0$. This illustrates that not all sequences give rise to generating functions that converge in a neighborhood of 0. A more robust definition of generating functions is through *formal power series*, where operations such as addition, multiplication, and differentiation are defined directly by operations on coefficients. That being said, in most applications we focus on generating functions that converge in some neighborhood of 0, so you do not need to worry too much about this issue for this course.

4.1.2 Example: Tile Covering

▶ Problem Setup

Consider a $2 \times n$ rectangle. We want to tile it using 1×2 dominoes, placed either horizontally or vertically, without overlap and without leaving any square uncovered. Let a_n denote the number of such tilings. The first two values are:

$$a_0 = 1, \qquad a_1 = 1. \tag{4.12}$$

What is the value of a_n for general n?

▶ Recurrence

Now consider a tiling of a $2\times n$ rectangle, and inspect its right end. As illustrated in Figure 4.1, there are only two possibilities. Either the last column is covered

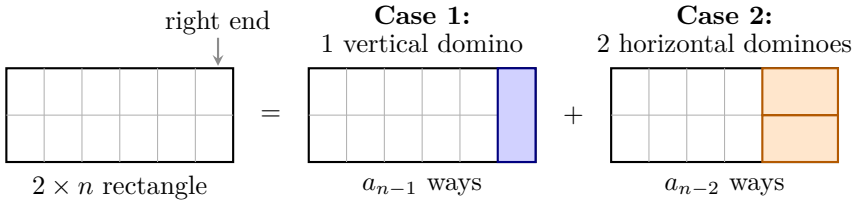


Figure 4.1: A tiling of a $2 \times n$ rectangle must end either with one vertical domino or with two horizontal dominoes.

by a single vertical domino, or the last two columns are covered by two horizontal dominoes. These two cases are disjoint and cover all possibilities.

In the first case, after removing the last vertical domino, the remaining region is a $2 \times (n-1)$ rectangle, so there are a_{n-1} tilings. In the second case, after removing the last two horizontal dominoes, the remaining region is a $2 \times (n-2)$ rectangle, so there are a_{n-2} tilings. Therefore,

$$a_n = a_{n-1} + a_{n-2}, \qquad n \geq 2. \tag{4.13}$$

So the sequence $(a_n)_{n \geq 0}$ satisfies the same recurrence as the Fibonacci numbers.

▶ Generating Function

We now encode the whole sequence into one power series:

$$A(x) := \sum_{n \geq 0} a_n x^n. \tag{4.14}$$

Using Equation (4.13), we compute

$$A(x) - a_0 - a_1 x = \sum_{n \geq 2} a_n x^n = \sum_{n \geq 2} a_{n-1} x^n + \sum_{n \geq 2} a_{n-2} x^n. \tag{4.15}$$

These two sums look like $A(x)$ with index shifted by 1 and 2, respectively. More precisely, we can rewrite them as

$$\sum_{n \geq 2} a_{n-1} x^n = x \sum_{n \geq 2} a_{n-1} x^{n-1} = x \sum_{n \geq 1} a_n x^n = x(A(x) - a_0), \tag{4.16}$$

$$\sum_{n \geq 2} a_{n-2} x^n = x^2 \sum_{n \geq 2} a_{n-2} x^{n-2} = x^2 \sum_{n \geq 0} a_n x^n = x^2 A(x). \tag{4.17}$$

Therefore,

$$A(x) - a_0 - a_1 x = x(A(x) - a_0) + x^2 A(x). \tag{4.18}$$

Substituting $a_0 = a_1 = 1$, we obtain

$$A(x) - 1 - x = x(A(x) - 1) + x^2 A(x), \tag{4.19}$$

which simplifies to

$$A(x) = \frac{1}{1 - x - x^2}. \tag{4.20}$$

So the entire sequence can be encoded by one rational function, $A(x) = \frac{1}{1-x-x^2}$.

▶ Closed-form Formula

Of course, one could say that a_n is simply "the coefficient of the nth-order term in the power series expansion of $A(x) = \frac{1}{1-x-x^2}$." But this characterization is far from satisfactory, because it does not express a_n in a form that is sufficiently "familiar" to us.

In general, we call a formula a *closed-form formula*, or simply a *closed formula*, if it involves only a bounded number of terms built from constants, variables, and functions or operations that are considered "basic." The precise set of "basic" mathematical objects depends on the context, but it typically includes π, e, arithmetic operations, exponentiation, logarithms, trigonometric functions, etc.

We now turn to deriving a closed-form formula for a_n from its generating function $A(x)$. It is time to leverage our algebraic skills. We factor the denominator:

$$1 - x - x^2 = (1 - \varphi x)(1 - \psi x), \tag{4.21}$$

where φ and ψ are the reciprocals of the two roots of the quadratic equation $1 - x - x^2 = 0$. Solving this equation, we obtain

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618, \qquad \psi = \frac{1-\sqrt{5}}{2} \approx -0.618. \tag{4.22}$$

Then we can decompose $A(x)$ into partial fractions:

$$A(x) = \frac{\alpha}{1 - \varphi x} + \frac{\beta}{1 - \psi x} \tag{4.23}$$

where (α, β) is the solution to $\alpha(1 - \psi x) + \beta(1 - \varphi x) = 1$:

$$\alpha = \frac{\sqrt{5}+1}{2\sqrt{5}}, \qquad \beta = \frac{\sqrt{5}-1}{2\sqrt{5}}. \tag{4.24}$$

Expanding each term as a geometric series gives

$$A(x) = \sum_{n\geq 0} \alpha\varphi^n x^n + \sum_{n\geq 0} \beta\psi^n x^n = \sum_{n\geq 0} (\alpha\varphi^n + \beta\psi^n) x^n. \tag{4.25}$$

Therefore, we obtain that the number of domino tilings of a $2 \times n$ rectangle is

$$a_n = \alpha\varphi^n + \beta\psi^n = \frac{\sqrt{5}+1}{2\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n + \frac{\sqrt{5}-1}{2\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n. \tag{4.26}$$

Or more compactly,

$$a_n = \frac{1}{\sqrt{5}} \left(\varphi^{n+1} - \psi^{n+1}\right). \tag{4.27}$$

An important lesson here is: algebraic manipulations of generating functions can actually help us solve combinatorial problems!

4.1.3 Basic Manipulations

As illustrated in the above example, one can write operations on sequences as operations on generating functions. For example, we shift the index of a_n by 2 by

Table 4.1: Basic manipulations of ordinary generating functions. Throughout this table, $A(x) = \sum_{n\geq 0} a_n x^n$ and $B(x) = \sum_{n\geq 0} b_n x^n$.

Generating Functions	Power Series
$x^m A(x)$	$\sum_{n\geq m} a_{n-m} x^n$
$x^{-m}(A(x) - a_0 - a_1 x - \cdots - a_{m-1}x^{m-1})$	$\sum_{n\geq 0} a_{n+m} x^n$
$\alpha A(x) + \beta B(x)$	$\sum_{n\geq 0} (\alpha a_n + \beta b_n) x^n$
$\alpha A(x) - \beta B(x)$	$\sum_{n\geq 0} (\alpha a_n - \beta b_n) x^n$
$A(x)B(x)$	$\sum_{n\geq 0} \left(\sum_{k=0}^{n} a_k b_{n-k}\right) x^n$
$\frac{1}{1-x} A(x)$	$\sum_{n\geq 0} \left(\sum_{k=0}^{n} a_k\right) x^n$
$A(cx)$	$\sum_{n\geq 0} c^n a_n x^n$
$xA'(x)$	$\sum_{n\geq 0} n a_n x^n$
$\int_0^x \frac{1}{t}(A(t) - a_0)\, dt$	$\sum_{n\geq 1} \frac{1}{n} a_n x^n$

multiplying $A(x)$ by x^2. Now in Table 4.1, we present a more comprehensive list of how operations on generating functions correspond to operations on sequences.

The first two rows explain how to shift the entire sequence right or left. To shift the sequence right by m indices, we can multiply $A(x)$ by x^m. To shift the sequence left by m indices, we can multiply $A(x)$ by x^{-m}, but we have to subtract the first m terms before doing so.

The next two rows state that elementwise addition and subtraction of sequences become ordinary addition and subtraction of generating functions. This is useful when we want to linearly combine two sequences.

When we multiply two generating functions, the underlying sequences are not simply multiplied elementwise. Instead, the resulting sequence is the *convolution* of the two sequences:

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}. \tag{4.28}$$

This is because the coefficient of x^n in $A(x)B(x)$ has to come from the product of $a_k x^k$ and $b_{n-k}x^{n-k}$ for some k. Here the word "convolution" originally means to roll things together. The convolution of two sequences can also be viewed this way: reverse one sequence, sweep it across the other, and at each position add the products of the overlapping entries. See Figure 4.2 for an illustration.

A special case is when the sequence b is a constant sequence with value $b_n = 1$ for all n, i.e., its generating function is $B(x) = \sum_{n\geq 0} x^n = \frac{1}{1-x}$, then the convolution of a and b results in the sequence holding the prefix sums of a:

$$\frac{1}{1-x} A(x) = \sum_{n\geq 0} \left(\sum_{k=0}^{n} a_k\right) x^n. \tag{4.29}$$

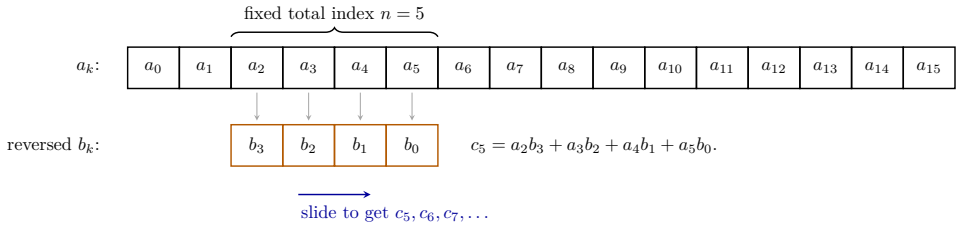


Figure 4.2: To compute a convolution term c_n, reverse the sequence b and line up the pairs whose indices add to n. The overlapping entries contribute the products $a_k b_{n-k}$.

Can we perform some elementwise operations on the sequence? The rest of the table lists some of the most common operations. If we substitute cx for x in $A(x)$, we obtain the sequence $c^n a_n$. That is, we multiply a_n elementwise by a geometric sequence $(c^n)_{n\geq 0}$.

The next row shows that differentiation lets us multiply the sequence elementwise by n. More specifically, differentiating $A(x)$ gives

$$A'(x) = \sum_{n\geq 0} n a_n x^{n-1}. \tag{4.30}$$

Multiplying by x gives

$$xA'(x) = \sum_{n\geq 0} n a_n x^n. \tag{4.31}$$

Therefore, $A(x) \mapsto xA'(x)$ multiplies the sequence elementwise by n.

Conversely, integrating $A(x)$ gives

$$\int_0^x A(t)\,dt = \sum_{n\geq 0} \frac{a_n}{n+1} x^{n+1}. \tag{4.32}$$

Shifting the index left by 1 before integrating, we obtain

$$\int_0^x \frac{1}{t}(A(t) - a_0)\,dt = \sum_{n\geq 0} \frac{a_n}{n} x^n. \tag{4.33}$$

Therefore, $A(x) \mapsto \int_0^x \frac{1}{t}(A(t) - a_0)\,dt$ divides the sequence elementwise by n.

Having introduced these operations, we now turn to more advanced examples.

4.1.4 Example: Parenthesizing Expressions

▶ Problem Setup

Let a_n be the number of ways to fully parenthesize the following expression:

$$x_1 + x_2 + \cdots + x_n \tag{4.34}$$

Here, "fully parenthesized" means that every addition operation is enclosed in parentheses. Equivalently, a_n is the number of ways to evaluate the expression using only binary additions.

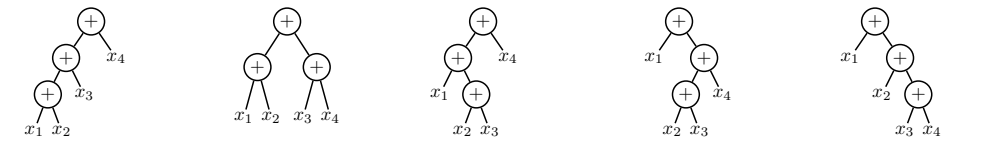


Figure 4.3: The five binary expression trees associated with the five full parenthesizations of $x_1 + x_2 + x_3 + x_4$.

For instance, $a_1 = 1$, because with only one term there is no choice to make. When $n = 4$, we have $a_4 = 5$ since there are 5 possible parenthesizations:

$$(((x_1 + x_2) + x_3) + x_4), \tag{4.35}$$
$$((x_1 + x_2) + (x_3 + x_4)), \tag{4.36}$$
$$((x_1 + (x_2 + x_3)) + x_4), \tag{4.37}$$
$$(x_1 + ((x_2 + x_3) + x_4)), \tag{4.38}$$
$$(x_1 + (x_2 + (x_3 + x_4))). \tag{4.39}$$

See Figure 4.3 for a visualization.

▶ Recurrence

To derive a recurrence, look at the last addition performed. At the final step, the expression must be split into two already-parenthesized blocks,

$$(x_1 + \cdots + x_k) \qquad \text{and} \qquad (x_{k+1} + \cdots + x_n) \tag{4.40}$$

for some $k \in \{1, 2, \ldots, n-1\}$. The first block can be parenthesized in a_k ways, and the second block can be parenthesized in a_{n-k} ways. Since these two choices are independent, a fixed split point k contributes $a_k a_{n-k}$ possibilities. Therefore,

$$a_1 = 1, \qquad a_n = \sum_{k=1}^{n-1} a_k a_{n-k} \qquad \text{for all } n \geq 2. \tag{4.41}$$

This is a natural example of convolution: each term a_n is obtained by summing products $a_k a_{n-k}$ over all possible ways to split the expression. In the view of convolution, the above recurrence simply states that for every $n \geq 2$, a_n is the n-th element of the convolution of $(a_k)_{k\geq 1}$ with itself.

▶ Generating Function

Now define the ordinary generating function of $(a_n)_{n\geq 1}$ as

$$A(x) := \sum_{n\geq 1} a_n x^n. \tag{4.42}$$

Here we start at $n = 1$ because a_1 is the first nontrivial term. Translating the recurrence into generating-function language, we get

$$A(x) - a_1 x = \sum_{n\geq 2} a_n x^n = \sum_{n\geq 2} \sum_{k=1}^{n-1} a_k a_{n-k} x^n = A(x)^2. \tag{4.43}$$

This can be further simplified as the following quadratic equation:

$$A(x)^2 - A(x) + x = 0. \tag{4.44}$$

Solving this quadratic equation gives

$$A(x) = \frac{1 \pm \sqrt{1-4x}}{2}. \tag{4.45}$$

There are two possible solutions, but since $A(0) = \sum_{n\geq 1} a_n 0^n = 0$, we must choose the minus sign:

$$A(x) = \frac{1 - \sqrt{1-4x}}{2}. \tag{4.46}$$

▶ Closed-form Formula

Now we turn to deriving a closed-form formula for a_n. By Newton's generalized binomial theorem (Example 4.3),

$$(1-4x)^{1/2} = \sum_{n\geq 0} \binom{1/2}{n} (-4x)^n. \tag{4.47}$$

Then we can write $A(x)$ as

$$A(x) = \frac{1 - \sqrt{1-4x}}{2} = \frac{1}{2} - \frac{1}{2} \sum_{n\geq 0} \binom{1/2}{n} (-4x)^n. \tag{4.48}$$

Therefore, we obtain the following formula for a_n:

$$a_n = -\frac{1}{2}(-4)^n \binom{1/2}{n} \quad \text{for } n \geq 1. \tag{4.49}$$

To further simplify the formula, we expand the binomial coefficient:

$$\binom{1/2}{n} = \frac{1/2(1/2-1)\cdots(1/2-n+1)}{n!}. \tag{4.50}$$

Now multiplying the numerator and denominator by 2^n, we get

$$\binom{1/2}{n} = \frac{1 \cdot (-1) \cdot (-3) \cdots (-(2n-3))}{2^n n!} \tag{4.51}$$

Then, we extract a factor of -1 from each negative term in the numerator, and multiply the numerator and denominator by the missing even numbers in the numerator. This gives

$$\binom{1/2}{n} = \frac{(-1)^{n-1} \cdot 1 \cdot 2 \cdot 3 \cdot 4 \cdots (2n-3) \cdot (2n-2)}{(2 \cdot 4 \cdot 6 \cdots (2n-2)) \cdot 2^n n!} \tag{4.52}$$

$$= \frac{(-1)^{n-1}(2n-2)!}{2^{n-1}(n-1)! \cdot 2^n n!} \tag{4.53}$$

$$= \frac{(-1)^{n-1}}{2^{2n-1}} \cdot \frac{(2n-2)!}{(n-1)!n!}. \tag{4.54}$$

Plugging this back into the formula for a_n, we get

$$a_n = -\frac{1}{2}(-4)^n \cdot \frac{(-1)^{n-1}}{2^{2n-1}} \cdot \frac{(2n-2)!}{(n-1)!n!} = \frac{(2n-2)!}{(n-1)!n!}. \tag{4.55}$$

▶ Catalan Numbers

In combinatorics, *Catalan numbers*, named after the French mathematician Eugène Catalan, are a sequence of natural numbers that appear in many counting problems. These numbers are defined as:

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{n!(n+1)!} \tag{4.56}$$

Comparing with the formula for a_n, we see that for all $n \geq 1$,

$$a_n = C_{n-1}. \tag{4.57}$$

As the formula is so clean and only involves one binomial coefficient, one may naturally wonder if there is a more combinatorial way to derive this formula. Interestingly, the answer is *yes*.

First, we introduce a new encoding for each parenthesization, which is essentially the *postfix notation* of the expression. We assign each variable x_i with an empty string $\phi(x_i) =$ `""`. Then, whenever we perform an addition $(a + b)$, where a and b are two fully parenthesized expressions, we define $\phi((a + b)) = \phi(a) +$ `"("` $+ \phi(b) +$ `")"`. For example,

$$(((x_1 + x_2) + x_3) + x_4) \quad \mapsto \quad \texttt{"()()()"}, \tag{4.58}$$
$$((x_1 + x_2) + (x_3 + x_4)) \quad \mapsto \quad \texttt{"()(())"}, \tag{4.59}$$
$$((x_1 + (x_2 + x_3)) + x_4) \quad \mapsto \quad \texttt{"(())()"}, \tag{4.60}$$
$$(x_1 + ((x_2 + x_3) + x_4)) \quad \mapsto \quad \texttt{"(()())"}, \tag{4.61}$$
$$(x_1 + (x_2 + (x_3 + x_4))) \quad \mapsto \quad \texttt{"((()))"}. \tag{4.62}$$

This encoding is in fact a bijection between the set of all parenthesizations for n variables and the set of valid bracket sequences with $n-1$ pairs of brackets. Here we say that a bracket sequence is *valid* if left and right brackets are matched correctly. Or more precisely, every prefix of the sequence has at least as many left brackets as right brackets.

It is straightforward to verify that every parenthesization is encoded as a valid bracket sequence. Conversely, every valid bracket sequence uniquely determines a parenthesization. To see this, we can reverse the encoding process: find the rightmost matched pair of brackets and split the sequence into two parts, then construct the corresponding two parenthesizations recursively and add them together. Another way is to imagine a stack initialized with x_1 and a counter $i = 1$. As we read the bracket sequence from left to right, whenever we see a left bracket "(", increment i by 1 and push x_i onto the stack. Whenever we see a right bracket ")", pop the top two elements and push their sum back onto the stack. A full parenthesization is obtained when we finish reading the sequence.

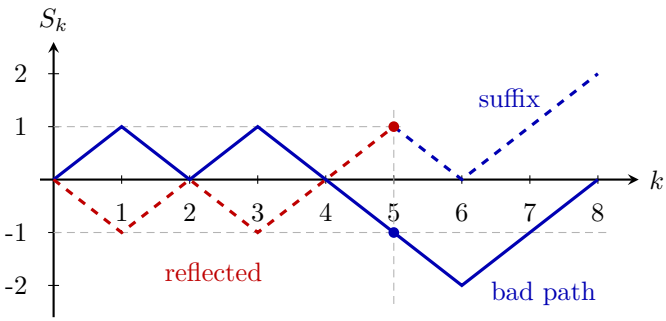


Figure 4.4: A bad bracket path: it reaches -1 for the first time at time $t = 5$. We reflect the initial segment across the horizontal axis. The endpoint at time t moves from -1 to $+1$, and the later steps are shifted up by 2.

Now we encode each left bracket "(" as $+1$, and each right bracket ")" as -1. For a sequence of $n-1$ pairs of brackets, this gives a sequence

$$\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_{2n-2} \in \{+1, -1\} \tag{4.63}$$

with exactly $n-1$ entries equal to $+1$ and exactly $n-1$ entries equal to -1. For every $1 \le k \le 2n-2$, let

$$S_k := \varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_k \tag{4.64}$$

be the k-th prefix sum. This quantity records the running excess of left brackets over right brackets among the first k symbols. For example, $S_k = 2$ means that we have seen two more left brackets than right brackets in the first k positions, while $S_k = -1$ means that at that point we have used one more right bracket than left bracket.

Then it is easy to see that a bracket sequence is *valid* if and only if

$$S_{2n-2} = 0 \qquad \text{and} \qquad S_k \ge 0 \quad \text{for all } 1 \le k \le 2n-2. \tag{4.65}$$

That is, the total numbers of left and right brackets are equal, and in every prefix, the number of left brackets is at least the number of right brackets.

Therefore, we can reduce the problem of counting parenthesizations to the following problem: how many ± 1 sequences start at 0, end at 0, and never let the prefix sums go below 0?

If we ignore the prefix condition for the moment, then the problem becomes counting the number of ± 1 sequences with $n-1$ plus signs and $n-1$ minus signs. For this, we only need to choose which $n-1$ positions carry $+1$, and the rest will be minus signs. The number of such sequences is given by

$$\binom{2n-2}{n-1}. \tag{4.66}$$

Among these sequences, some are valid and some are not. So we will count the *bad* ones and subtract them from the total.

Take a bad sequence, and let t be the first index with $S_t = -1$. Because t is the first time the prefix sum becomes negative, the path stays at height at least 0 before time t, and then lands at -1 exactly at time t. Now reflect the first t steps by changing each $+1$ to -1 and each -1 to $+1$. Geometrically, if we draw the prefix sums as a lattice path, this reflects the initial segment across the horizontal axis. See Figure 4.4 for an illustration.

Why does this help? The sum of the first t steps was originally $S_t = -1$, so after reflection it becomes $+1$. The later steps are left unchanged, so the whole reflected sequence has total sum $+2$ instead of 0, which means it contains n plus signs and $n-2$ minus signs. Moreover, the time t is exactly the first time the prefix sum becomes $+1$.

Conversely, start with any sequence containing n plus signs and $n-2$ minus signs. Its total sum is $+2$, so as we read from left to right, there must be a first time t at which the prefix sum becomes $+1$. Reflecting the initial segment up to t changes the sum of that segment from $+1$ to -1, then we can recover a bad sequence whose first negative prefix sum occurs exactly at time t. In other words, reflection gives a bijection between bad sequences and sequences with n entries equal to $+1$ and $n-2$ entries equal to -1.

Therefore, the number of bad sequences is exactly the number of sequences with n plus signs, which is given by

$$\binom{2n-2}{n}. \tag{4.67}$$

Subtracting these bad sequences from the total gives

$$\begin{aligned} a_n &= \binom{2n-2}{n-1} - \binom{2n-2}{n} && (4.68)\\ &= \frac{(2n-2)!}{(n-1)!(n-1)!} - \frac{(2n-2)!}{n!(n-2)!} && (4.69)\\ &= \frac{(2n-2)!}{(n-1)!n!}\,(n-(n-1)) && (4.70)\\ &= \frac{(2n-2)!}{(n-1)!n!} && (4.71)\\ &= C_{n-1}, && (4.72) \end{aligned}$$

which is exactly the same as the result we obtained from generating functions.

This argument is elegant, but it relies on the clever idea of reformulating the problem and applying the reflection trick at exactly the right place. In contrast, generating functions reach the same answer through a more systematic algebraic procedure.

Just as some people prefer to apply a series of clever tricks to find the minimum of a function, while others prefer to use calculus by taking derivatives and setting them to zero, the two approaches here reflect different styles of thinking.

Combinatorial arguments can be elegant and intuitive, but they often rely on non-trivial insights that are not always easy to discover. Generating functions may feel more mechanical and even a bit clumsy, but they provide a reliable way to derive the answer through routine calculations.

Ultimately, it is up to you to decide which method to use.

4.1.5 Example: Summing Variables with Reordering

▶ Problem Setup

Before we move on to more difficult problems, let us consider a simple variant of the parenthesizing problem.

In addition to inserting parentheses, we now also allow *shuffling* the order of the operands. At the same time, we treat addition as *commutative*, meaning that the order of the two operands in each addition does not matter.

For example, when $n = 3$, if we do not allow shuffling, there are only two ways to parenthesize:

$$(x_1 + (x_2 + x_3)) \quad \text{and} \quad ((x_1 + x_2) + x_3). \tag{4.73}$$

If we allow shuffling, there is one more way to parenthesize:

$$(x_2 + (x_1 + x_3)). \tag{4.74}$$

Here we regard expressions such as $((x_2 + x_3) + x_1)$ and $(x_1 + (x_2 + x_3))$ as the same, since the order within each addition does not matter.

In other words, we ask: Given n variables $x_1, \ldots, x_n$, how many essentially different ways are there to compute their sum by repeatedly adding two terms at a time, where the order of operands in each addition is ignored?

▶ Recurrence

Although this problem looks similar to the original parenthesizing problem, it requires a slightly different approach to derive the correct answer.

Let a_n be the number of ways to compute the sum of n variables. Following the same reasoning as in the parenthesizing problem, we check the last addition performed. At the final step, the expression must be split into two blocks, and the number of variables in the two blocks sum to n.

Let k be the number of variables in the first block. Then the first block can be computed in a_k ways, and the second block can be computed in a_{n-k} ways. Therefore, the total number of ways to compute the sum of n variables is

$$a_n = \sum_{k=1}^{n-1} a_k a_{n-k}. \tag{4.75}$$

Is this correct?



Note that a_k is the number of ways to compute the sum of $x_1, \ldots, x_k$. However, in the first block, the variables are not necessarily $x_1, \ldots, x_k$. We have to choose which k variables to put in the first block. Therefore, the number of ways to compute the sum of n variables is

$$a_n = \sum_{k=1}^{n-1} \binom{n}{k} a_k a_{n-k}. \tag{4.76}$$

Is this correct now?



In fact, we have to divide the total number of ways by 2 because the order of the two blocks is not important. In the above formula, each pair of variable assignments is counted twice. Therefore, the correct formula is

$$a_n = \frac{1}{2} \sum_{k=1}^{n-1} \binom{n}{k} a_k a_{n-k}. \tag{4.77}$$

Is this finally correct?



▶ Generating Function

Now define the ordinary generating function of $(a_n)_{n\geq 1}$ as

$$A(x) = \sum_{n\geq 1} a_n x^n. \tag{4.78}$$

It is time to write the recurrence relation as an equation for $A(x)$. After that, the problem should be over — a few lines of algebra will solve the equation for $A(x)$ with ease!

But wait, something is wrong. The right-hand side of Equation (4.77) is not the ordinary convolution that OGFs can turn into a product. Instead, it is a convolution with binomial coefficients, called a *binomial convolution*. Given two sequences $(a_n)_{n\geq 0}$ and $(b_n)_{n\geq 0}$, their *binomial convolution* is the sequence $(c_n)_{n\geq 0}$ defined by

$$c_n := \sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}. \tag{4.79}$$

If we extend our sequence by setting $a_0 := 0$, then the right-hand side of Equation (4.77) is precisely $\frac{1}{2}c_n$, where $(c_n)_{n\geq 0}$ is the binomial convolution of $(a_n)_{n\geq 0}$ with itself.

Can we write a binomial convolution as the product of two generating functions? In fact, yes, but not with ordinary generating functions. The trick is to switch to the *exponential generating function*, where each term in the power series is scaled by a factorial:

$$\widehat{A}(x) = \sum_{n\geq 1} a_n \frac{x^n}{n!}. \tag{4.80}$$

This turns out to be exactly what we need. The factorials align perfectly with the recurrence, since the binomial coefficients themselves are defined in terms of factorials. More specifically,

$$\sum_{n\geq 2} a_n \frac{x^n}{n!} = \frac{1}{2} \sum_{n\geq 2} \left(\sum_{k=1}^{n-1} \binom{n}{k} a_k a_{n-k} \right) \frac{x^n}{n!} \tag{4.81}$$

$$= \frac{1}{2} \sum_{n\geq 2} \left(\sum_{k=1}^{n-1} \frac{n!}{k!(n-k)!} a_k a_{n-k} \right) \frac{x^n}{n!} \tag{4.82}$$

$$= \frac{1}{2} \sum_{n\geq 2} \left(\sum_{k=1}^{n-1} a_k \frac{x^k}{k!} \cdot a_{n-k} \frac{x^{n-k}}{(n-k)!} \right). \tag{4.83}$$

Now we really have an ordinary convolution. We can write it as

$$\widehat{A}(x) - a_1 x = \frac{1}{2} \widehat{A}(x)^2. \tag{4.84}$$

Since $a_1 = 1$, we can rewrite the equation as

$$\widehat{A}(x)^2 - 2\widehat{A}(x) + 2x = 0. \tag{4.85}$$

Solving this quadratic equation, we obtain

$$\widehat{A}(x) = 1 \pm \sqrt{1 - 2x}. \tag{4.86}$$

There are two possible solutions, but since $\widehat{A}(0) = \sum_{n\geq 1} a_n \frac{0^n}{n!} = 0$, we must choose the minus sign:

$$\widehat{A}(x) = 1 - \sqrt{1 - 2x}. \tag{4.87}$$

▶ Closed-form Formula

As usual, we expand the square root as a power series using Newton's generalized binomial theorem:

$$(1-2x)^{1/2} = \sum_{n\geq 0} \binom{1/2}{n} (-2x)^n. \tag{4.88}$$

Same as our derivation in the parenthesizing problem, we have

$$\binom{1/2}{n} = \frac{1 \cdot (-1) \cdot (-3) \cdots (-(2n-3))}{2^n n!} \tag{4.89}$$

Plugging this back into the power series expansion, we obtain

$$\widehat{A}(x) = 1 - \sum_{n\geq 0} \binom{1/2}{n} (-2x)^n \tag{4.90}$$

$$= 1 - \sum_{n\geq 0} \frac{1 \cdot (-1) \cdot (-3) \cdots (-(2n-3))}{2^n n!} (-2x)^n \tag{4.91}$$

$$= 1 - \sum_{n\geq 0} \frac{(-1)^{n-1} \cdot (2n-3)!!}{2^n n!} \cdot (-1)^n \cdot 2^n \cdot x^n \tag{4.92}$$

$$= 1 + \sum_{n\geq 0} (2n-3)!! \, \frac{x^n}{n!}, \tag{4.93}$$

where $m!!$ denotes the *double factorial* of m, defined as

$$m!! := \begin{cases} m \cdot (m-2) \cdot (m-4) \cdots 3 \cdot 1 & \text{if } m \text{ is odd,} \\ m \cdot (m-2) \cdot (m-4) \cdots 4 \cdot 2 & \text{if } m \text{ is even.} \end{cases} \tag{4.94}$$

Therefore, the number of ways to compute the sum of n variables is

$$a_n = (2n-3)!!. \tag{4.95}$$

▶ Combinatorial Proof

In hindsight, this result can also be derived from a combinatorial argument. This proof takes a different perspective: we build the expression incrementally by inserting the variables $x_1, \ldots, x_n$ one by one.

Inserting the first variable x_1 is trivial. When inserting the n-th variable x_n with $n \geq 2$, there is already an expression describing how to compute the sum of the first $n-1$ variables. This expression can be viewed as a process where the $n-1$ variables are gradually merged into a single result. There are $2n-3$ possible choices to insert x_n into this process:

- Initially, there are $n-1$ variables, and we may merge x_n with any one of them before they are merged with others.
- During the process of merging the $n-1$ variables, there are $n-2$ internal merges, and whenever a new merge is performed, we may merge x_n with the result of the merge before it is merged with others.

By the multiplication principle, the total number of ways is

$$a_n = 1 \cdot 3 \cdot 5 \cdots (2n-3) = (2n-3)!!. \tag{4.96}$$

This is exactly the same as the result we obtained from the generating function.

4.1.6 Exponential Generating Functions

The previous example shows where ordinary generating functions (OGFs) stop being the natural tool. That recurrence involved binomial coefficients, so the OGF did not turn it into a clean product. What made the calculation work was

the factorial in the exponential generating function (EGF), which absorbs those combinatorial factors and turns the recurrence into a clean convolution.

This is the general principle behind EGFs. When two *unlabeled* objects are combined, only their sizes matter, so the counting leads to an ordinary convolution, and OGFs are the natural encoding. When two *labeled* objects are combined, we must also decide which labels go to which part. That extra choice produces binomial coefficients, and EGFs are designed to handle exactly this situation.

We now introduce EGFs systematically. We first define them and record their basic manipulation rules. After that, we will return to examples where this encoding works more naturally than OGF.

Definition 4.6 (Exponential generating function)**.** The *exponential generating function* (EGF) of a sequence of real numbers $(a_k)_{k\geq 0}$ is the power series

$$\widehat{A}(x) := \sum_{k\geq 0} a_k \frac{x^k}{k!}. \tag{4.97}$$

The name comes from the same factorial pattern that appears in the power series expansion of the exponential function. In particular, for the constant sequence $a_n = 1$, its EGF is

$$\widehat{A}(x) = \sum_{n\geq 0} \frac{x^n}{n!} = e^x. \tag{4.98}$$

More generally, for the geometric series $a_n = \beta^n$, its EGF is

$$\widehat{A}(x) = \sum_{n\geq 0} \beta^n \frac{x^n}{n!} = e^{\beta x}. \tag{4.99}$$

This should be compared with Example 4.2, where we have demonstrated that the OGF of the same geometric series is $\frac{1}{1-\beta x}$.

Remark 4.7. At first sight, you may find the following fact surprising: dividing by $n!$ makes the coefficients smaller, yet the EGF of a geometric sequence is the exponential function $e^{\beta x}$, which appears to grow faster than the rational function $\frac{1}{1-\beta x}$. The point is that the OGF has already blown up once $x > 1/\beta$, because the power series $\sum_{n\geq 0} \beta^n x^n$ then diverges. By contrast, the exponential function always remains finite for every real x.

We present in Table 4.2 a list of basic manipulations of exponential generating functions. It is best read side by side with the earlier OGF table, Table 4.1.

Part of this table should look familiar. For example, the rows for addition and subtraction with another sequence, as well as elementwise multiplication by c^n, n, and $1/n$, look exactly like their OGF counterparts. The reason is simple: if

we write $c_n := a_n/n!$, then

$$\widehat{A}(x) = \sum_{n\geq 0} c_n x^n, \tag{4.100}$$

so an EGF is simply the OGF of the rescaled coefficient sequence $(c_n)_{n\geq 0}$. In these rows, the operations on $(a_n)_{n\geq 0}$ induce exactly the same operations on $(c_n)_{n\geq 0}$, which is why the formulas remain unchanged.

The first real change appears when we shift the sequence. For OGFs, this is done by multiplying or dividing by powers of x. But for EGFs, the corresponding operations are differentiation and integration. In particular, differentiating $\widehat{A}(x)$ term by term gives

$$\widehat{A}'(x) = \sum_{n\geq 1} a_n \frac{x^{n-1}}{(n-1)!} = \sum_{n\geq 0} a_{n+1} \frac{x^n}{n!}. \tag{4.101}$$

So differentiation removes the first coefficient and shifts the sequence one step to the left. Repeating this process m times gives

$$\widehat{A}^{(m)}(x) = \sum_{n\geq 0} a_{n+m} \frac{x^n}{n!}. \tag{4.102}$$

Similarly, integrating $\widehat{A}(x)$ term by term gives

$$\int_0^x \widehat{A}(t)\ \mathrm{d}t = \sum_{n\geq 0} a_n \int_0^x \frac{t^n}{n!}\ \mathrm{d}t = \sum_{n\geq 0} a_n \frac{x^{n+1}}{(n+1)!} = \sum_{n\geq 1} a_{n-1} \frac{x^n}{n!}, \tag{4.103}$$

so integration shifts the sequence one step to the right. Repeating this process m times gives

$$\underbrace{\int_0^x \int_0^{t_m} \cdots \int_0^{t_2}}_{m \text{ times}} \widehat{A}(t_1)\ \mathrm{d}t_1 \cdots \mathrm{d}t_m = \sum_{n\geq m} a_{n-m} \frac{x^n}{n!}. \tag{4.104}$$

Another key difference appears in multiplication. For OGFs, multiplying two generating functions produces the ordinary convolution of their coefficient sequences. For EGFs, the calculation becomes

$$\widehat{A}(x)\widehat{B}(x) = \left(\sum_{k\geq 0} a_k \frac{x^k}{k!}\right)\left(\sum_{\ell\geq 0} b_\ell \frac{x^\ell}{\ell!}\right) \tag{4.105}$$

$$= \sum_{k,\ell\geq 0} a_k b_\ell \frac{x^{k+\ell}}{k!\ell!} \tag{4.106}$$

$$= \sum_{n\geq 0} \left(\sum_{k=0}^{n} \frac{a_k b_{n-k}}{k!(n-k)!}\right) x^n \tag{4.107}$$

$$= \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}\right) \frac{x^n}{n!}, \tag{4.108}$$

where the last step follows from $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. So EGF multiplication produces the *binomial convolution* that we already encountered in Section 4.1.5.

Table 4.2: Basic manipulations of exponential generating functions. Throughout this table, $\widehat{A}(x) = \sum_{n\geq 0} a_n \frac{x^n}{n!}$ and $\widehat{B}(x) = \sum_{n\geq 0} b_n \frac{x^n}{n!}$.

Generating Functions	Power Series
$\widehat{A}^{(m)}(x)$	$\sum_{n\geq 0} a_{n+m} \frac{x^n}{n!}$
$\underbrace{\int_0^x \int_0^{t_m} \cdots \int_0^{t_2}}_{m \text{ times}} \widehat{A}(t_1)\, \mathrm{d}t_1 \cdots \mathrm{d}t_m$	$\sum_{n\geq m} a_{n-m} \frac{x^n}{n!}$
$\alpha\widehat{A}(x) + \beta\widehat{B}(x)$	$\sum_{n\geq 0} (\alpha a_n + \beta b_n) \frac{x^n}{n!}$
$\alpha\widehat{A}(x) - \beta\widehat{B}(x)$	$\sum_{n\geq 0} (\alpha a_n - \beta b_n) \frac{x^n}{n!}$
$\widehat{A}(x)\widehat{B}(x)$	$\sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}\right) \frac{x^n}{n!}$
$a_0 e^x + \int_0^x \widehat{A}'(t) e^{x-t}\, \mathrm{d}t$	$\sum_{n\geq 0} \left(\sum_{k=0}^{n} a_k\right) \frac{x^n}{n!}$
$\widehat{A}(cx)$	$\sum_{n\geq 0} c^n a_n \frac{x^n}{n!}$
$x\widehat{A}'(x)$	$\sum_{n\geq 0} n a_n \frac{x^n}{n!}$
$\int_0^x \frac{1}{t}(\widehat{A}(t) - a_0)\, \mathrm{d}t$	$\sum_{n\geq 1} \frac{1}{n} a_n \frac{x^n}{n!}$

We now turn to prefix sums. For OGFs, prefix sums are produced by multiplying the generating function with $\frac{1}{1-x}$. For EGFs, the answer is still clean, but it no longer comes from multiplying by a fixed power series. Instead, we start from the recurrence and derive a differential equation for the EGF of the prefix sums. Define $s_n := \sum_{k=0}^{n} a_k$, then s_n satisfies the recurrence $s_{n+1} = s_n + a_{n+1}$. The corresponding EGF is

$$\widehat{S}(x) := \sum_{n\geq 0} s_n \frac{x^n}{n!}. \tag{4.109}$$

Differentiating this equation term by term gives

$$\widehat{S}'(x) = \sum_{n\geq 0} s_{n+1} \frac{x^n}{n!} = \sum_{n\geq 0} s_n \frac{x^n}{n!} + \sum_{n\geq 0} a_{n+1} \frac{x^n}{n!} = \widehat{S}(x) + \widehat{A}'(x). \tag{4.110}$$

So $\widehat{S}$ satisfies the linear differential equation

$$\widehat{S}'(x) - \widehat{S}(x) = \widehat{A}'(x), \qquad \widehat{S}(0) = s_0 = a_0. \tag{4.111}$$

Multiplying by e^{-x} on both sides gives

$$\frac{\mathrm{d}}{\mathrm{d}x}(\widehat{S}(x)e^{-x}) = \widehat{A}'(x)e^{-x}. \tag{4.112}$$

Integrating both sides from 0 to x, we obtain

$$\widehat{S}(x)e^{-x} = a_0 + \int_0^x \widehat{A}'(t)e^{-t}\, \mathrm{d}t, \tag{4.113}$$

where we used the initial condition $\widehat{S}(0) = a_0$. Therefore, the EGF of the prefix sums is

$$\widehat{S}(x) = a_0 e^x + \int_0^x \widehat{A}'(t)e^{x-t}\, \mathrm{d}t. \tag{4.114}$$

Equivalently, by integrating by parts, we can also write this as

$$\widehat{S}(x) = \widehat{A}(x) + \int_0^x \widehat{A}(t)e^{x-t}\,\mathrm{d}t. \tag{4.115}$$

Finally, it is also worth noting that an EGF can be converted back into the corresponding OGF by a Laplace transform. The *Laplace transform* of a function f is defined by

$$\mathcal{L}[f](s) := \int_0^\infty f(t)e^{-st}\,\mathrm{d}t. \tag{4.116}$$

If the OGF and EGF of a sequence $(a_n)_{n\geq 0}$ are given by

$$A(z) := \sum_{n\geq 0} a_n z^n \qquad \text{and} \qquad \widehat{A}(t) := \sum_{n\geq 0} a_n \frac{t^n}{n!}, \tag{4.117}$$

then

$$\mathcal{L}\Big[\widehat{A}\Big](s) = \int_0^\infty \left(\sum_{n\geq 0} a_n \frac{t^n}{n!}\right) e^{-st}\,\mathrm{d}t \tag{4.118}$$

$$= \sum_{n\geq 0} \frac{a_n}{n!} \int_0^\infty t^n e^{-st}\,\mathrm{d}t. \tag{4.119}$$

Here we swapped the order of integration and summation. Strictly speaking, this step requires justification in the analytic setting. However, if we treat generating functions as formal power series, then operations such as integration and differentiation are *defined* term by term, and this interchange is purely symbolic.

With the change of variables $u = st$, we can rewrite the integral in terms of the Gamma function $\Gamma(z) := \int_0^\infty u^{z-1}e^{-u}\,\mathrm{d}u$:

$$\int_0^\infty t^n e^{-st}\,\mathrm{d}t = \frac{1}{s^{n+1}} \int_0^\infty u^n e^{-u}\,\mathrm{d}u = \frac{1}{s^{n+1}}\Gamma(n+1) = \frac{n!}{s^{n+1}}, \tag{4.120}$$

where the last step uses the fact that $\Gamma(n+1) = n!$ for all integers $n \geq 0$.

Plugging this back into the Laplace transform, we obtain

$$\mathcal{L}\Big[\widehat{A}\Big](s) = \sum_{n\geq 0} \frac{a_n}{n!} \cdot \frac{n!}{s^{n+1}} = \frac{1}{s} A\left(\frac{1}{s}\right). \tag{4.121}$$

Setting $s = \frac{1}{z}$ gives

$$\frac{1}{z}\mathcal{L}\Big[\widehat{A}\Big]\left(\frac{1}{z}\right) = A(z). \tag{4.122}$$

Therefore, the Laplace transform can be used to convert an EGF into the corresponding OGF, up to a simple rescaling of the variable.

Remark 4.8. If the Laplace transform feels unfamiliar, compare it with the moment generating function (MGF) that we already encountered in Chapter 3. We will discuss continuous random variables and densities more systematically in Chapter 5, but if a non-negative continuous random variable X has density $f_X(x)$ on $[0, \infty)$, then

$$M_X(t) := \mathbb{E}[e^{tX}] = \int_0^\infty f_X(x)e^{tx}\,\mathrm{d}x. \tag{4.123}$$

This is exactly the Laplace transform of f_X evaluated at $s = -t$. In other words, $M_X(t) = \mathcal{L}[f_X](-t)$.

4.1.7 Example: Sum of Powers

▶ Problem Setup

We now look at a different kind of problem. In Example 2.89, when we applied the tail-sum formula to the minimum of several independent uniform random variables, we naturally encountered sums of the form

$$S_k(n) := 1^k + 2^k + \cdots + n^k. \tag{4.124}$$

Also in the study of birthday paradox, we encountered the sum of the first n squares $S_2(n) = 1^2 + 2^2 + \cdots + n^2$. More generally, sums of this form arise naturally across combinatorics and probability, which makes it worthwhile to study them systematically.

You may already be familiar with these sums for small values of k:

$$S_1(n) = \frac{n(n+1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n, \tag{4.125}$$

$$S_2(n) = \frac{n(n+1)(2n+1)}{6} = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n, \tag{4.126}$$

$$S_3(n) = \left(\frac{n(n+1)}{2}\right)^2 = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2. \tag{4.127}$$

In fact, these formulas admit nice combinatorial interpretations, which we will not discuss here. But what if we want to understand $S_k(n)$ for general k? Even for $k = 4$, the formula already looks rather mysterious:

$$S_4(n) = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30} \tag{4.128}$$

$$= \frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n. \tag{4.129}$$

At this point, it becomes hard to imagine a simple combinatorial argument that explains such expressions. So instead of chasing combinatorial interpretations, let us take a different route: can we use generating functions to derive a general formula for $S_k(n)$ for all k?

▶ Generating Function

Fix n. We now encode the numbers $S_0(n), S_1(n), S_2(n), \ldots$ by an EGF:

$$\widehat{F}_n(x) := \sum_{k\geq 0} S_k(n)\frac{x^k}{k!}. \tag{4.130}$$

Since $S_k(n) = \sum_{j=1}^{n} j^k$, we may interchange the order of summation and obtain

$$F_n(x) = \sum_{k\geq 0}\left(\sum_{j=1}^{n} j^k\right)\frac{x^k}{k!} = \sum_{k\geq 0}\sum_{j=1}^{n}\frac{(jx)^k}{k!} = \sum_{j=1}^{n}\sum_{k\geq 0}\frac{(jx)^k}{k!} = \sum_{j=1}^{n} e^{jx}. \tag{4.131}$$

Now this is just a sum of finite geometric series:

$$F_n(x) = e^x + e^{2x} + \cdots + e^{nx} = \frac{e^{(n+1)x} - e^x}{e^x - 1}. \tag{4.132}$$

The only question remaining is how to extract the coefficients $S_k(n)$ from this expression.

▶ Bernoulli Numbers

Of course, we can rewrite $F_n(x)$ as

$$F_n(x) = \frac{e^{nx} - 1}{x} \cdot \frac{xe^x}{e^x - 1}. \tag{4.133}$$

Since $\frac{xe^x}{e^x-1} = \frac{x}{1-e^{-x}}$, we can further rewrite $F_n(x)$ as

$$F_n(x) = \frac{e^{nx} - 1}{x} \cdot \frac{x}{1 - e^{-x}}. \tag{4.134}$$

The coefficients of the first term $\frac{e^{nx}-1}{x}$ are easy to extract:

$$\frac{e^{nx} - 1}{x} = \frac{1}{x} \sum_{k\geq 0} \left(n^k - \mathbb{1}_{[k=0]}\right) \frac{x^k}{k!} = \sum_{k\geq 0} \frac{n^{k+1}}{k+1} \frac{x^k}{k!}. \tag{4.135}$$

So if we can also extract the coefficients of the second term $\frac{x}{1-e^{-x}}$, then we can recover the coefficients of $F_n(x)$ by performing a binomial convolution with these two coefficient sequences.

However, it is not immediately clear how to extract the coefficients of the second term $\frac{x}{1-e^{-x}}$. While this function is known as the *Sigmoid Linear Unit* (SiLU) or *Swish* in machine learning [HG16, RZL17, EUD18], for some reason in mathematics the reflection of this function about the y-axis is more commonly studied:

$$\widehat{B}(x) = \frac{x}{e^x - 1}. \tag{4.136}$$

It is easy to check that

$$\frac{x}{1 - e^{-x}} = \frac{-x}{e^{-x} - 1} = \widehat{B}(-x). \tag{4.137}$$

See also Figure 4.5 for the graphs of $\frac{x}{1-e^{-x}}$ and $\widehat{B}(x)$.

This transformation reduces the problem to extracting the coefficients of $\widehat{B}(x)$. However, this is still very difficult, and it is so difficult that mathematicians have given it a name: the *Bernoulli numbers*. We define the *Bernoulli numbers* $(B_m)_{m\geq 0}$ as the coefficients of the EGF $\widehat{B}(x)$:

$$\frac{x}{e^x - 1} = \sum_{m\geq 0} B_m \frac{x^m}{m!}. \tag{4.138}$$

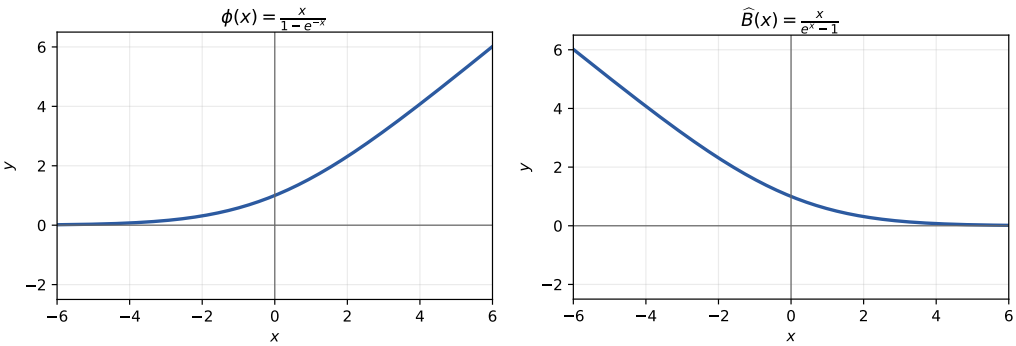
For reference, the first few values are

$$B_0 = 1, \qquad B_1 = -\frac{1}{2}, \qquad B_2 = \frac{1}{6}, \qquad B_3 = 0, \qquad B_4 = -\frac{1}{30}. \tag{4.139}$$

See also Table 4.3 for the first 20 Bernoulli numbers.

Table 4.3: The first 20 Bernoulli numbers.

B_0	B_1	B_2	B_3	B_4	B_5	B_6	B_7	B_8	B_9
1	$-\frac{1}{2}$	$\frac{1}{6}$	0	$-\frac{1}{30}$	0	$\frac{1}{42}$	0	$-\frac{1}{30}$	0
B_{10}	B_{11}	B_{12}	B_{13}	B_{14}	B_{15}	B_{16}	B_{17}	B_{18}	B_{19}
$\frac{5}{66}$	0	$-\frac{691}{2730}$	0	$\frac{7}{6}$	0	$-\frac{3617}{510}$	0	$\frac{43867}{798}$	0



(a) The graph of $\phi(x) = \frac{x}{1-e^{-x}}$. (b) The graph of $\widehat{B}(x) = \frac{x}{e^x-1}$.

Figure 4.5: Two closely related functions: $\phi(x) = \frac{x}{e^{-x}-1}$ and $\widehat{B}(x) = \frac{x}{e^x-1}$.

Once we have the Bernoulli numbers, we can rewrite $F_n(x)$ as

$$F_n(x) = \frac{e^{nx}-1}{x} \cdot \widehat{B}(-x) \tag{4.140}$$

$$= \left(\sum_{k\geq 0} \frac{n^k}{k+1}\frac{x^k}{k!}\right)\left(\sum_{m\geq 0}(-1)^m B_m \frac{x^m}{m!}\right). \tag{4.141}$$

Therefore, the k-th coefficient of $F_n(x)$, which is $S_k(n)$, is given by the binomial convolution of the underlying coefficient sequences. This leads to the following formula, known as *Faulhaber's formula*.

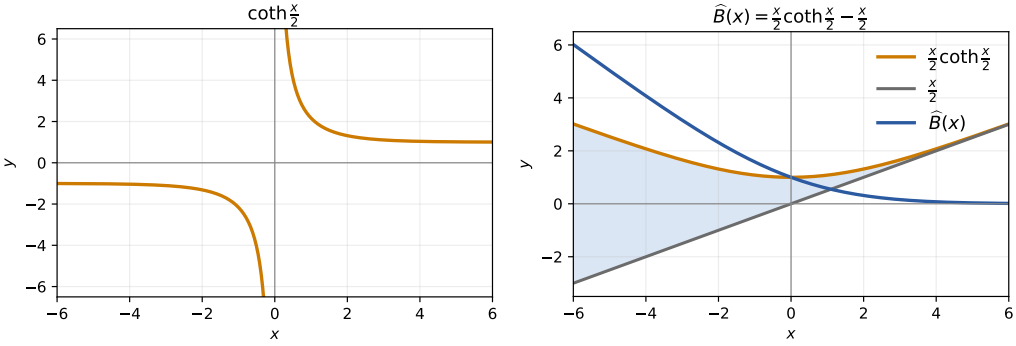
Theorem 4.9 (Faulhaber's formula)**.** For all $k \geq 0$ and $n \geq 1$,

$$S_k(n) = \frac{1}{k+1}\sum_{j=0}^{k}(-1)^j \binom{k+1}{j} B_j\, n^{k+1-j}. \tag{4.142}$$

Proof. By the binomial convolution, we have

$$S_k(n) = \sum_{j=0}^{k}(-1)^j \binom{k}{j} \frac{n^{k+1-j}}{k+1-j} B_j. \tag{4.143}$$

This only has a minor difference from the formula in the theorem. To recover



(a) The graph of $\coth\frac{x}{2}$.

(b) The shaded difference is $\widehat{B}(x)$.

Figure 4.6: The decomposition $\widehat{B}(x) = \frac{x}{2}\coth\frac{x}{2} - \frac{x}{2}$.

the formula in the theorem, we just need to note

$$\binom{k}{j}\frac{1}{k+1-j} = \frac{k!}{j!(k+1-j)!} = \binom{k+1}{j}\frac{1}{k+1}. \tag{4.144}$$

Replacing $\binom{k}{j}\frac{1}{k+1-j}$ with $\binom{k+1}{j}\frac{1}{k+1}$ gives the formula in the theorem. □

If you look more closely at the first few Bernoulli numbers in Equation (4.139), you may find that B_3 is zero. In fact, it is not an coincidence that $B_3 = 0$: all the odd Bernoulli numbers after B_1 are zero.

Lemma 4.10. If $m > 1$ and m is odd, then $B_m = 0$.

Proof. Rewrite $\widehat{B}(x)$ using the hyperbolic cotangent function $\coth z := \frac{e^z+e^{-z}}{e^z-e^{-z}}$:

$$\widehat{B}(x) = \frac{x}{e^x-1} = \frac{xe^{-x/2}}{e^{x/2}-e^{-x/2}} = \frac{\frac{x}{2}e^{x/2}+\frac{x}{2}e^{-x/2}}{e^{x/2}-e^{-x/2}} - \frac{x}{2} = \frac{x}{2}\coth\frac{x}{2} - \frac{x}{2}. \tag{4.145}$$

See also Figure 4.6. You may notice that $\coth\frac{x}{2}$ diverges to $\pm\infty$ as $x \to 0$. To fix this issue, define $\psi(x) := \frac{x}{2}\coth\frac{x}{2}$ for $x \neq 0$ and then define $\psi(0)$ by continuity:

$$\psi(0) := \lim_{x\to 0}\frac{x}{2}\coth\frac{x}{2} = 1. \tag{4.146}$$

As a result, one can check that $\psi(x)$ is now a well-defined and infinitely differentiable function in a neighborhood of $x = 0$. In fact, the original EGF $\widehat{B}(x) = \frac{x}{e^x-1}$ has the same issue at $x = 0$, so there we also have to define $\widehat{B}(0) = 1$ by continuity. This is also why $B_0 = 1$.

Since coth is an odd function, after multiplying by $\frac{x}{2}$ yields an even function, which is $\psi(x)$. Therefore, the power series expansion of $\psi(x)$ does not contain odd-degree terms. Since $\widehat{B}(x) = \psi(x) - x/2$, we can conclude that all the odd-degree coefficients of $\widehat{B}(x)$ are zero except for the linear term. □

As a result, the $(-1)^j$ factor in Faulhaber's formula is almost useless, except when $j = 1$. Therefore, we can rewrite Faulhaber's formula as

$$S_k(n) = \frac{1}{k+1}n^{k+1} + \frac{1}{2}n^k + \frac{1}{k+1}\sum_{d=1}^{\lfloor k/2 \rfloor}\binom{k+1}{2d}B_{2d}\,n^{k+1-2d} \tag{4.147}$$

$$= \frac{1}{k+1}n^{k+1} + \frac{1}{2}n^k + \frac{k}{12}n^{k-1} + \mathcal{O}(n^{k-3}). \tag{4.148}$$

Remark 4.11. Yes, Bernoulli random variables and Bernoulli numbers are named after the same Swiss mathematician, Jacob Bernoulli (1655–1705). In fact, the natural constant $e := \lim_{n\to\infty}(1 + 1/n)^n \approx 2.71828$ was also discovered by Jacob Bernoulli in his study of compound interest (if my money grows at an interest rate of $1/n$ per year, how much will I have after n years?). Therefore, e is also a "Bernoulli number" in some sense. However, the central role of e in analysis was not fully recognized until another Swiss mathematician, Leonhard Euler (1707–1783), who was Jacob Bernoulli's student's student, discovered how naturally the natural constant e arises in dealing with exponential functions, for example $(e^x)' = e^x$. So e is sometimes called the Euler's number.

Having settled the odd Bernoulli numbers, we now turn to computing the even Bernoulli numbers. Before doing so, we need to introduce a new tool to help us: *complex analysis*.

4.2 Complex Analysis: A Roller Coaster Ride

We now introduce complex analysis to solve a key question in the theory of generating functions: once a generating function has been identified, how can we recover information about its coefficients from the function itself?

Complex analysis provides a systematic way to do this. More importantly, it does not only solve a single isolated problem, but rather solves a whole class of problems that at first glance appear unrelated. For example:

1. How can we actually compute the Bernoulli numbers hidden in $\frac{x}{e^x-1}$?
2. How can we compute the improper integral $\int_{-\infty}^{\infty}\frac{1}{1+x^4}\,\mathrm{d}x$?
3. If $s(x) := \frac{1}{1+e^{-x}}$ is the sigmoid function, how far from 0 does its power series remain convergent? In other words, what is the largest radius R such that the series converges for all $|x| < R$?

The point of introducing complex analysis is not that it is especially useful in computer science. In fact, it is less directly useful than the concentration inequalities we introduced in the previous chapter. Rather, complex analysis is a beautiful and elegant theory, and an excellent example of what a good theory can look like. Of course, there is no single standard for what a good theory should look like. With that in mind, here is my personal view:

A good theory should make difficult problems trivial.

The theory of generating functions itself is an example of good theory: it makes many combinatorial problems trivial. Instead of trying to come up with a clever combinatorial argument, one can often simply find the generating function through some tedious algebra, after which the problem reduces to extracting its coefficients. The line of work from Markov, Chernoff, to Hoeffding, Azuma, and McDiarmid reduces the problem of deriving concentration inequalities to computing a few interval lengths. This is another example of how theory can make difficult problems trivial.

Complex analysis will be another example of good theory. As you will see, although complex numbers are in some sense "more complex" than real numbers, working in the complex plane offers a fundamentally new perspective, which may reshape your mind and allow you to approach real analysis problems in a completely new way. This perspective is so powerful that it often makes difficult problems surprisingly simple, including all the examples above.

Before we begin, let me make one pedagogical choice explicit. In this section, we will sometimes trade full rigor for a cleaner view of the main ideas. For example, we may differentiate or integrate functions without checking every analytic condition in detail. The goal here is to show the beauty and power of complex analysis. If you want a fully rigorous treatment, you may refer to standard textbooks, such as [APP11, Ahl79, TW06].

4.2.1 Complex Numbers

You may have encountered quadratic equations with no solutions. For example,

$$x^2 + 1 = 0. \tag{4.149}$$

But what if we simply assume that this equation *does* have a solution, and then explore the consequences? This is the starting point of complex analysis.

We introduce a new symbol, the *imaginary unit* i, and impose the rule

$$i^2 = -1. \tag{4.150}$$

We then manipulate expressions involving i by the usual algebra rules, and every time i^2 appears we replace it by -1.

Definition 4.12 (Complex numbers)**.** A *complex number* is a number of the form

$$z = x + iy, \qquad x, y \in \mathbb{R}, \tag{4.151}$$

where i is the imaginary unit. The number x is called the *real part* of z, and y is called the *imaginary part*. We write

$$\Re(z) = x, \qquad \Im(z) = y. \tag{4.152}$$

The set of all complex numbers is denoted by $\mathbb{C}$.

In essence, a complex number can be viewed as a pair of real numbers (x, y), or equivalently as a point in a two-dimensional plane, called the *complex plane*. Un-

der this viewpoint, complex numbers are real pairs equipped with the following rules for addition and multiplication:

$$(x+iy)+(a+ib) = (x+a)+i(y+b) \tag{4.153}$$
$$\iff (x,y)+(a,b) = (x+a, y+b), \tag{4.154}$$
$$(x+iy)\cdot(a+ib) = (xa-yb)+i(xb+ya) \tag{4.155}$$
$$\iff (x,y)\cdot(a,b) = (xa-yb, xb+ya). \tag{4.156}$$

Here the multiplication rule follows from expanding brackets and simplifying i^2 to -1:

$$(x+iy)(a+ib) = xa+ixb+iya+i^2yb = (xa-yb)+i(xb+ya). \tag{4.157}$$

Subtraction is defined as the inverse of addition:

$$(x+iy)-(a+ib) = (x-a)+i(y-b). \tag{4.158}$$

Division is defined as the inverse of multiplication, but the formula is more complicated. To write it in a cleaner form, we need to introduce the conjugate and modulus.

Definition 4.13. If $z = x+iy$, then its *conjugate* is

$$\overline{z} = x-iy, \tag{4.159}$$

and its *absolute value* or *modulus* is

$$|z| := \sqrt{x^2+y^2}. \tag{4.160}$$

A useful fact is that multiplying a complex number by its conjugate gives the square of its modulus:

$$z\overline{z} = (x+iy)(x-iy) = x^2-(iy)^2 = x^2+y^2 = |z|^2. \tag{4.161}$$

This is useful because the result is an ordinary real number. Based on this fact, we can derive a clean formula for reciprocals: for any non-zero complex number z, its reciprocal is given by

$$\frac{1}{z} = \frac{\overline{z}}{|z|^2}. \tag{4.162}$$

So if $z = a+ib$, then

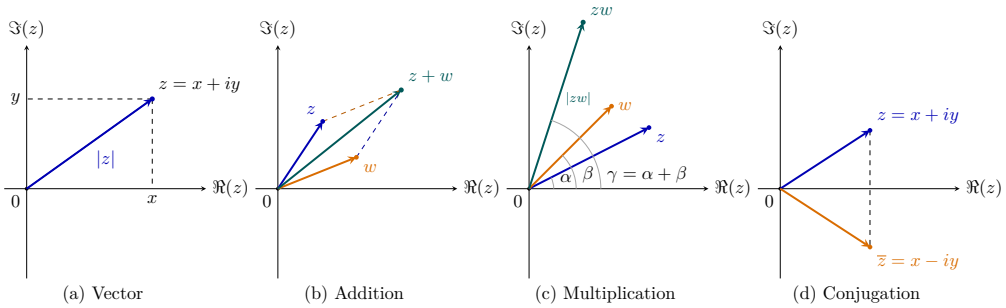
$$\frac{1}{z} = \frac{a-ib}{a^2+b^2}. \tag{4.163}$$

The same idea works for general division. If $a+ib \neq 0$, then

$$\frac{x+iy}{a+ib} = \frac{(x+iy)(a-ib)}{|a+ib|^2} = \frac{(x+iy)(a-ib)}{a^2+b^2}. \tag{4.164}$$

In other words, to divide by a complex number, we multiply by its conjugate and then divide by its squared modulus.

These arithmetic operations on complex numbers also have a clear geometric meaning. Geometrically, the number $z = x+iy$ is a vector from the origin to the point (x,y). Its modulus is the length of this vector. Addition is just the usual vector addition. Conjugation reflects a complex number across the real axis.



(a) Vector (b) Addition (c) Multiplication (d) Conjugation

Figure 4.7: Four basic geometric pictures for complex numbers, from left to right: a complex number as a vector, addition, multiplication, and conjugation.

Multiplication is harder to understand from Cartesian coordinates, so it helps to change coordinates. For any non-zero complex number, we may describe the same point instead by its distance from the origin and its direction. If $z \neq 0$, then for some angle $\theta \in (-\pi, \pi]$,

$$z = r(\cos\theta + i\sin\theta), \qquad \text{where } r = |z|. \tag{4.165}$$

This is called the *polar form* of z, which separates the magnitude from the direction. The angle θ is called the *argument* of z, denoted as $\arg(z)$. In this form, multiplication becomes much simpler:

$$z = r(\cos\theta + i\sin\theta), \quad w = s(\cos\phi + i\sin\phi) \tag{4.166}$$

$$\Rightarrow zw = rs(\cos(\theta+\phi) + i\sin(\theta+\phi)). \tag{4.167}$$

That is, in the polar form, the modulus of the product is the product of the moduli rs, and the argument of the product is the sum of the arguments $\theta + \phi$. This can be verified directly from the angle addition formulas for sine and cosine, but soon we will see how to prove this easily with Euler's formula.

Similarly, division becomes much simpler in the polar form:

$$z = r(\cos\theta + i\sin\theta), \quad w = s(\cos\phi + i\sin\phi) \tag{4.168}$$

$$\Rightarrow z/w = (r/s)(\cos(\theta-\phi) + i\sin(\theta-\phi)). \tag{4.169}$$

Example 4.14. Here are some basic examples of complex arithmetic:

$$(2+i) + (1-i) = 3, \tag{4.170}$$

$$(2+i) - (1-i) = 1 + 2i, \tag{4.171}$$

$$(2+i)(1-i) = 3 - i. \tag{4.172}$$

For division, we multiply the numerator and denominator by the conjugate of the denominator:

$$\frac{2+i}{1-i} = \frac{(2+i)(1+i)}{(1-i)(1+i)} = \frac{1+3i}{2} = \frac{1}{2} + \frac{3}{2}i. \tag{4.173}$$

Based on the above geometric interpretations, one can easily verify the follow-

ing identities and inequalities on modulus:

$$||z| - |w|| \leq |z + w| \leq |z| + |w|, \tag{4.174}$$

$$|zw| = |z||w|, \tag{4.175}$$

$$\left|\frac{z}{w}\right| = \frac{|z|}{|w|}. \tag{4.176}$$

4.2.2 Complex Functions and Power Series

So far we have only discussed individual complex numbers. Similar to real numbers, we can also define functions on complex numbers.

Definition 4.15 (Complex functions)**.** Let $S \subseteq \mathbb{C}$ be a set of complex numbers. A *complex function* on S is a function $f : S \to \mathbb{C}$.

We plot several complex functions in Figure 4.8. Unlike functions on the real line, complex functions are harder to visualize because both the input and the output lie in two-dimensional planes, which yields four real dimensions in total. Here we use a standard visualization technique called *domain coloring*. This technique maps function values to colors, where the hue encodes the argument of the function value, and the brightness encodes the magnitude $|f(z)|$.

Among these functions, it is clear how $f(z) = z$, $f(z) = z^2$, $f(z) = \frac{1}{1+z^2}$, $f(z) = \overline{z}$ and $f(z) = \frac{z^4+1}{(z-1-i)^2(z^2+4)}$ are defined, since these functions are compositions of complex arithmetic operations. But what about $f(z) = e^z$, $f(z) = \sin z$, $f(z) = \cos z$? In fact, these functions are defined by *power series*, which we now define formally.

Definition 4.16 (Power series)**.** A *power series* is an infinite sum of the form

$$\sum_{n\geq 0} a_n z^n, \tag{4.177}$$

where $a_n \in \mathbb{C}$ is the coefficient of the n-th term z^n. More generally, a power series with respect to a center $z_0 \in \mathbb{C}$ is an infinite sum of the form

$$\sum_{n\geq 0} a_n (z - z_0)^n. \tag{4.178}$$

More specifically, $\sum_{n\geq 0} a_n z^n$ means the limit of the partial sums $\sum_{n=0}^{N} a_n z^n$ as $N \to \infty$, provided that this limit exists in $\mathbb{C}$. That is,

$$\lim_{N\to\infty} \sum_{n=0}^{N} a_n z^n, \tag{4.179}$$

A stronger condition is that the series converges *absolutely*, which means that the series of absolute values converges to finite value:

$$\lim_{N\to\infty} \sum_{n\geq 0} |a_n z^n| < \infty. \tag{4.180}$$

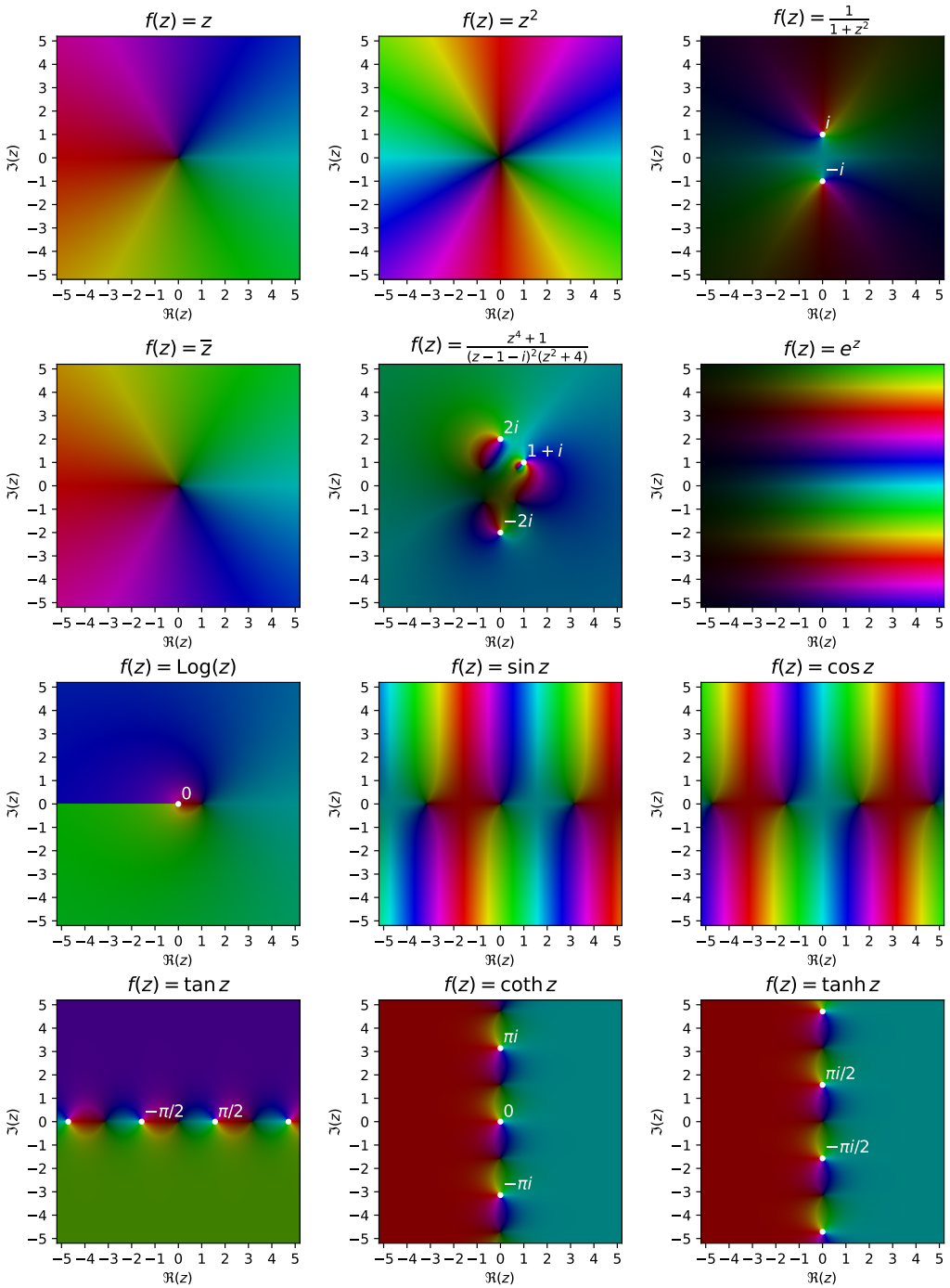


Figure 4.8: Domain coloring of twelve complex functions. Hue encodes the argument of the function value, and brightness encodes the magnitude $|f(z)|$.

If a series converges absolutely, then it converges, but the converse may not be true. For example, the series $\sum_{n\geq 0}(-1)^n z^n$ converges for all $z \in \mathbb{C}$, but it does not converge absolutely for $z = 1$ or $z = -1$.

Note that we have not yet defined what it means for a complex sequence to converge. Interestingly, this is just defined as the most natural generalization of the real case.

Definition 4.17 (Limits of complex sequences)**.** We say that a sequence $(z_n)_{n\geq 0}$ of complex numbers converges to $A \in \mathbb{C}$ if for every $\epsilon > 0$ there exists N such that

$$|z_n - A| < \epsilon \qquad \text{for all } n \geq N. \tag{4.181}$$

Comparing this with the definition of a real sequence limit, we see that the only difference is that distance is now measured by the modulus of the complex difference. Since the modulus is just the Euclidean distance in the complex plane, this is the same definition as for sequences in $\mathbb{R}^2$. The definition of a function limit is completely analogous: we say that $\lim_{z\to z_0} f(z) = L$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that $|f(z) - L| < \epsilon$ whenever $|z - z_0| < \delta$.

Back to our question of how to define e^z, $\sin z$, and $\cos z$. In fact, these functions are defined by directly extending their power series expansions from the real case. By Taylor expansion, for all $x \in \mathbb{R}$, we have

$$e^x = \sum_{n\geq 0} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots, \tag{4.182}$$

$$\sin x = \sum_{m\geq 0} \frac{(-1)^m}{(2m+1)!} x^{2m+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots, \tag{4.183}$$

$$\cos x = \sum_{m\geq 0} \frac{(-1)^m}{(2m)!} x^{2m} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots. \tag{4.184}$$

We then define these functions for complex numbers by simply replacing $x \in \mathbb{R}$ with $z \in \mathbb{C}$:

$$e^z := \sum_{n\geq 0} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \frac{z^5}{5!} + \cdots, \tag{4.185}$$

$$\sin z := \sum_{m\geq 0} \frac{(-1)^m}{(2m+1)!} z^{2m+1} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots, \tag{4.186}$$

$$\cos z := \sum_{m\geq 0} \frac{(-1)^m}{(2m)!} z^{2m} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots. \tag{4.187}$$

Do these series converge for all $z \in \mathbb{C}$? Similar to the real case, we have the following theorem that characterizes the radius of convergence of a power series.

Theorem 4.18 (Cauchy–Hadamard theorem)**.** Let $\sum_{n\geq 0} a_n z^n$. Define

$$R := \frac{1}{\limsup_{n\to\infty}|a_n|^{1/n}} \in [0,+\infty)\cup\{+\infty\}. \tag{4.188}$$

Then, for all $z \in \mathbb{C}$,

1. if $|z| < R$, then the series converges absolutely;
2. if $|z| > R$, then the series does not converge.

This R is called the *radius of convergence* of the power series.

This theorem shows that the region of convergence of a power series is always a disk centered at the origin with radius R. This disk is called the *disk of convergence* of the power series. It does not, however, determine whether the series converges on the boundary of the disk $|z| = R$.

The exact value of R is given by Equation (4.188). A more intuitive way to write this R is:

$$R = \sup\left\{r > 0 : |a_n| = \mathcal{O}(r^{-n}) \text{ as } n \to \infty\right\}. \tag{4.189}$$

The proof is essentially the same as the proof of the real case, with the only difference being that the modulus is now used instead of the absolute value.

Proof. First assume that $|z| < R$. Choose a number $r \in (|z|, R)$ and let $\rho = \frac{|z|}{r}$, which must lie in $(0,1)$. Then $|a_n| = \mathcal{O}(r^{-n})$ as $n \to \infty$, and thus

$$|a_n z^n| = \mathcal{O}(r^{-n}) \cdot |z|^n = \mathcal{O}(\rho^n). \tag{4.190}$$

Summing over all n, we get

$$\sum_{n=0}^{N}|a_n z^n| = \sum_{n=0}^{N}\mathcal{O}(\rho^n) = \mathcal{O}\left(\sum_{n=0}^{N}\rho^n\right) = \mathcal{O}\left(\frac{1}{1-\rho}\right) = \mathcal{O}(1). \tag{4.191}$$

Therefore, the series converges absolutely.

Now assume that $|z| > R$. Then $|z|/R > 1$. Choose a number $r \in (R, |z|)$ and let $\rho = \frac{|z|}{r}$, which must satisfy $\rho > 1$. By definition,

$$R^{-1} = \limsup_{n\to\infty}|a_n|^{1/n}. \tag{4.192}$$

So the inequality

$$|a_n|^{1/n} > r^{-1} = \frac{\rho}{|z|} \tag{4.193}$$

must hold for infinitely many n. For those n we get

$$|a_n z^n| > \rho^n. \tag{4.194}$$

Since $\rho > 1$, we can conlude that $|a_n z^n|$ does not converge to 0 as $n \to \infty$. However, if the series converges, then $\lim_{n\to\infty}|a_n z^n| = 0$ must hold. □

Following Cauchy–Hadamard theorem, we can compute the radius of convergence of e^z, $\sin z$, and $\cos z$. It is easy to see that the radius of convergence is

infinite for all these functions, since $\frac{1}{n!} = \mathcal{O}(r^{-n})$ for all $r > 0$. Therefore, these functions are well-defined for all $z \in \mathbb{C}$.

Moreover, one can check that these functions still satisfy the usual identities in the real case, such as $e^{z+w} = e^z e^w$, $e^{-z} = \frac{1}{e^z}$, $\sin(z+w) = \sin z \cos w + \cos z \sin w$, $\cos(z+w) = \cos z \cos w - \sin z \sin w$, and other trigonometric identities.

Comparing the power series expansions of e^z, $\sin z$, and $\cos z$ leads to a key identity in complex analysis: Euler's formula.

Theorem 4.19 (Euler's formula)**.** For every $x \in \mathbb{R}$,

$$e^{ix} = \cos x + i \sin x. \tag{4.195}$$

Proof. Using the power series definition of the exponential,

$$e^{ix} = \sum_{n\geq 0} \frac{(ix)^n}{n!} = \sum_{m\geq 0} \frac{(ix)^{2m}}{(2m)!} + \sum_{m\geq 0} \frac{(ix)^{2m+1}}{(2m+1)!} \tag{4.196}$$

$$= \sum_{m\geq 0} \frac{(-1)^m x^{2m}}{(2m)!} + i \sum_{m\geq 0} \frac{(-1)^m x^{2m+1}}{(2m+1)!}. \tag{4.197}$$

These two series are exactly the power series expansions of $\cos x$ and $\sin x$, respectively. Therefore, we have $e^{ix} = \cos x + i \sin x$. □

A direct consequence of Euler's formula is *Euler's identity*:

Corollary 4.20 (Euler's identity)**.** The following identity holds:

$$e^{i\pi} + 1 = 0. \tag{4.198}$$

This follows immediately from Euler's formula by setting $x = \pi$.

Remark 4.21. Although this identity is simple to derive, it is often cited as an example of mathematical beauty for the following reasons:

- It involves 3 basic operations: addition, multiplication, and expoentiation.
- It brings together 5 fundamental mathematical constants: $0, 1, \pi, e, i$.

It is remarkable that such a simple identity connects all of them!

Based on Euler's formula, we can rewrite the polar form of a complex number $z = r(\cos\theta + i \sin\theta)$ into a more compact notation:

$$z = re^{i\theta}. \tag{4.199}$$

This is called the *exponential form* of z. This exponential form also gives a very elegant explanation of the multiplication rule for complex numbers: If z and w admit the following exponential forms:

$$z = re^{i\theta} \qquad \text{and} \qquad w = se^{i\phi}, \tag{4.200}$$

then

$$zw = re^{i\theta} se^{i\phi} = (rs)e^{i(\theta+\phi)}. \tag{4.201}$$

Therefore, the modulus of the product is the product of the moduli, and the argument of the product is the sum of the arguments.

4.2.3 Continuous Functions

So far, we have seen complex functions that are built from the usual arithmetic operations on complex numbers, or defined by power series. We now turn to more general complex functions and ask how to develop calculus for them.

As in the real case, we begin by defining continuity as the property that infinitesimal changes in the input lead to infinitesimal changes in the output.

Definition 4.22 (Continuity)**.** Let $f : S \to \mathbb{C}$ be a complex function. We say that f is *continuous* at $z_0 \in S$ if the limit of $f(z)$ as $z \in S$ approaches z_0 exists and is equal to $f(z_0)$. That is,

$$\lim_{z \to z_0} f(z) = f(z_0). \tag{4.202}$$

We say that f is *continuous* on S if it is continuous at every point of S.

Unlike the real case, now z can approach z_0 from every direction. In fact, this is exactly the same definition as for functions on $\mathbb{R}^2$.

Example 4.23. In Figure 4.8, the function $f(z) = z^2$ is continuous everywhere on $\mathbb{C}$. Indeed, for any $z_0 \in \mathbb{C}$, as $z \to z_0$, we have

$$z^2 = z_0^2 + 2z_0(z - z_0) + (z - z_0)^2 \to z_0^2. \tag{4.203}$$

The function $f(z) = \frac{1}{1+z^2}$ is also continuous, but we need to be careful about its domain. This function is not defined at $z = \pm i$, where the denominator becomes zero. Thus we restrict its domain to $\mathbb{C} \setminus \{\pm i\}$, on which one can check that this function is continuous. In fact, every polynomial or rational function is continuous whenever it is defined.

Example 4.24 (Exponential function)**.** The exponential function e^z is continuous because as $z \to z_0$, we have

$$e^z = e^{z_0} + e^{z_0}(e^{z-z_0} - 1) \tag{4.204}$$

$$= e^{z_0} + e^{z_0} \sum_{n \geq 1} \frac{1}{n!}(z - z_0)^n \to e^{z_0}. \tag{4.205}$$

Most functions in Figure 4.8 are continuous wherever they are defined, but the logarithm is an exception.

Example 4.25 (Complex logarithm)**.** How about the inverse of the exponential function, the logarithm? If you think about it carefully, you will find that the main obstacle is that it is unclear how to recover the input from the output of the exponential function. If $e^w = z$, then for every $k \in \mathbb{Z}$, Euler's formula implies

$$e^{w+2k\pi i} = e^w e^{2k\pi i} = e^w = z. \tag{4.206}$$

Therefore, the exponential function is not injective, and thus its inverse cannot be uniquely defined.

To resolve this issue, we define the inverse of the exponential function similarly to what we have done for the inverse of sine and cosine functions: we select a principal value. For every non-zero complex number $z \in \mathbb{C} \setminus \{0\}$, define $\operatorname{Log} z$ as the *principal logarithm* of z, given by

$$\operatorname{Log} z = \log |z| + i \arg(z), \tag{4.207}$$

where $\arg(z) \in (-\pi, \pi]$ is the argument of z. One can verify that $e^{\operatorname{Log} z} = z$. However, as we can see from Figure 4.8, this function is not continuous along $(-\infty, 0)$.

4.2.4 Holomorphic Functions

While continuous functions are common, just like real functions, complex functions often have more structure. Recall that in real calculus, continuity is just the weakest regularity condition to satisfy. A stronger regularity condition is differentiability. More specifically, we say that a function is *differentiable* if the function admits a derivative at every point. Beyond that, we can also differentiate a function as many times as we want. If we are always allowed to do this, we say that the function is *infinitely differentiable*. Finally, if we can use these derivatives to rewrite the function as a power series via Taylor expansion, we say that the function is *analytic*.

These four regularity conditions form a hierarchy, since each condition is stronger than the previous one. For example, a function may be continuous but not differentiable. Another function may be differentiable but not infinitely differentiable. Yet another function may be infinitely differentiable but not analytic. See also Figure 4.10 below.

Did you feel very *frustrated* to distinguish between these four different notions when you were learning real calculus? Will complex analysis just make the picture even more complicated? The answer is



In fact, differentiability is a very strong condition in complex analysis. It is so strong that it implies all the other three conditions! As shown in Figure 4.9, these

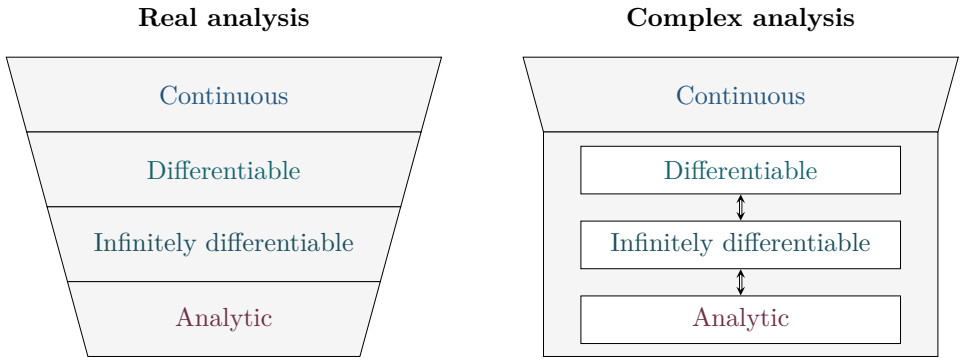


Figure 4.9: A conceptual picture of regularity. In real analysis, continuity, differentiability, infinite differentiability, and analyticity are genuinely different notions. For complex functions, once differentiability holds on an open set, the higher two layers come for free.

four notions are genuinely different in real analysis. In complex analysis, however, there are only two levels of regularity: the first level consists of continuous functions, while the second contains the other three, now equivalent notions, namely differentiable, infinitely differentiable, and analytic functions.

▶ Formal Definitions

Now let us dive into the details of these four notions. We will restrict our attention to functions defined or restricted to open sets in the complex plane. This is because differentiability at boundary points of a set (e.g., a, b in $[a, b]$) is trickier to handle correctly, but open sets do not contain their boundary at all (e.g., $a, b \notin (a, b)$) and thus we only need to consider interior points. First, we need to define open sets in the complex plane.

Definition 4.26 (Open and closed sets)**.** Let $S \subseteq \mathbb{C}$ be a set. We say that S is *open* if for every $z_0 \in S$, there exists some $\delta > 0$ such that for all $z \in \mathbb{C}$ with $|z - z_0| < \delta$, we have

$$z \in S. \tag{4.208}$$

We say that S is *closed* if its complement $\mathbb{C} \setminus S$ is open. An equivalent definition is that S is closed if for every convergent sequence $(z_n)_{n \geq 0}$ in S, the limit $\lim_{n \to \infty} z_n$ is also in S.

This definition is the same as that for open sets on $\mathbb{R}^2$. Now we proceed to define differentiability.

Definition 4.27 (Differentiability)**.** Let $f : S \to \mathbb{C}$ be a complex function defined on an open set $S \subseteq \mathbb{C}$. We say that f is *differentiable* at $z_0 \in S$ if the limit

$$f'(z_0) := \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \tag{4.209}$$

exists. Further, we say that f is *differentiable* on S if it is differentiable at every point of S.

The formula for derivative looks similar to that in real calculus, but it is much stronger than the differentiability for real functions. It is in fact more similar to the differentiability for real functions on $\mathbb{R}^2$. There, if we interpret the function f as a function from $\mathbb{R}^2$ to $\mathbb{R}^2$:

$$F(x, y) = (u(x, y), v(x, y)) \quad \text{if } f(x + iy) = u(x, y) + iv(x, y), \tag{4.210}$$

then the derivative is given by the Jacobian matrix of f:

$$J_F = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix}. \tag{4.211}$$

By Taylor expansion, we then have

$$F(x_0 + \Delta x, y_0 + \Delta y) \approx F(x_0, y_0) + J_F(x_0, y_0) \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}. \tag{4.212}$$

However, as you can see, the division in Equation (4.209) is a complex division! Having a complex derivative actually means the following local approximation:

$$f((x_0 + \Delta x) + (y_0 + \Delta y)i) \approx f(x_0 + y_0 i) + f'(x_0 + y_0 i)(\Delta x + \Delta y\, i). \tag{4.213}$$

That is, if we perturb the input by a small amount $\Delta x + \Delta y\, i$, then the perturbed output changes linearly with the perturbation, but this linarity is characterized by a complex number $f'(x_0 + y_0 i)$ rather than a real 2×2 matrix.

This is in fact a stronger condition than the differentiability for real functions on $\mathbb{R}^2$: Jacobian matrices have 4 degrees of freedom, but complex derivatives only allow for 2 degrees of freedom. In particular, multiplying by a complex derivative is a special kind of linear transformation. It can only scale and rotate a perturbation, but not shear it, and therefore preserves angles and orientation.

Now we demonstrate a few examples of differentiable and non-differentiable complex functions.

Example 4.28. Not very surprisingly, the quadratic function $f(z) = z^2$ is differentiable at every point, with derivative

$$f'(z_0) = 2z_0. \tag{4.214}$$

This is due to the fact that

$$\frac{f(z) - f(z_0)}{z - z_0} = \frac{z^2 - z_0^2}{z - z_0} = z + z_0, \tag{4.215}$$

which tends to $2z_0$ as $z \to z_0$.

In general, a power series $f(z) = \sum_{n\geq 0} a_n z^n$ is differentiable whenever z lies inside the radius of convergence of the series. The derivative is given by

$$f'(z) = \sum_{n\geq 1} n a_n z^{n-1}. \tag{4.216}$$

We will not prove this here, but it follows from the fact that a power series converges uniformly on every closed disk inside its radius of convergence, which allows term-by-term differentiation.

Example 4.29. Perhaps surprisingly, the complex conjugate function $f(z) = \overline{z}$ is continuous everywhere, but it is not differentiable! Consider the following limit at $z_0 = 0$:

$$\lim_{z\to z_0} \frac{f(z) - f(z_0)}{z - z_0}. \tag{4.217}$$

Write $z = re^{i\theta}$. Then the difference quotient is

$$\frac{\overline{z} - 0}{z - 0} = \frac{re^{-i\theta}}{re^{i\theta}} = e^{-2i\theta}. \tag{4.218}$$

This depends on the direction θ but not on the distance r. If $\theta = 0$, then the quotient is 1. If $\theta = \frac{\pi}{2}$, then it is -1. So the limit depends on the direction of approach and therefore does not exist.

You may get a feeling that differentiability is a very strong condition from the above example: even a simple function like the complex conjugate is not differentiable. Now we proceed to introduce the two seemingly stronger conditions that are actually equivalent to differentiability: *infinite differentiability* and *analyticity*.

Definition 4.30 (Infinite differentiability)**.** Let $f : S \to \mathbb{C}$ be a complex function defined on an open set $S \subseteq \mathbb{C}$. We say that f is *infinitely differentiable* at $z_0 \in S$ if all the higher order derivatives of f,

$$f'(z_0), f''(z_0), f^{(3)}(z_0), f^{(4)}(z_0), \ldots \tag{4.219}$$

exist at z_0. Further, we say that f is *infinitely differentiable* on S if it is infinitely differentiable at every point of S.

Taylor expansion tells us that if a real function is infinitely differentiable at a point $x_0 \in \mathbb{R}$, then for every $N \geq 0$, it can be expanded around x_0 as

$$f(x) = \sum_{n=0}^{N} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \mathcal{O}\Big((x - x_0)^{N+1}\Big). \tag{4.220}$$

However, it is not always true that the function $f(x)$ can be written as the corresponding power series:

$$f(x) = \sum_{n\geq 0} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n. \tag{4.221}$$

This is because the remainder term $\mathcal{O}\Big((x-x_0)^{N+1}\Big)$ may not converge to 0 as $N \to \infty$. Therefore, in real analysis, representing a function by a power series is a strictly stronger condition than being infinitely differentiable. This stronger condition is called *analyticity*. In complex analysis, the same notion can be similarly defined.

Definition 4.31 (Analyticity)**.** Let $f : S \to \mathbb{C}$ be a complex function defined on an open set $S \subseteq \mathbb{C}$. We say that f is *analytic* at $z_0 \in S$ if there exists a sequence of complex numbers $(a_n)_{n\geq 0}$ such that for some $\delta > 0$, for all $z \in S$ with $|z - z_0| < \delta$,

$$f(z) = \sum_{n\geq 0} a_n (z - z_0)^n. \tag{4.222}$$

Further, we say that f is *analytic* on S if it is analytic at every point of S.

▶ Three Notions Collapse to One

The next theorem tells us something a bit embarrassing: sorry for wasting your time, but all the distinctions we just worked so hard to make between differentiability, infinite differentiability, and analyticity completely disappear in complex analysis. They are all the same.

Theorem 4.32. Let $f : S \to \mathbb{C}$ be a complex function defined on an open set $S \subseteq \mathbb{C}$. Then the following properties are equivalent:

(i) f is differentiable on S;

(ii) f is infinitely differentiable on S;

(iii) f is analytic on S.

We will not prove this theorem right now since a key technique is Cauchy's integral formula, which we have not introduced yet.

Now let us look at some examples that separate these three notions in the real case, and see why they no longer separate in the complex case.

Example 4.33 (Differentiable vs. infinitely differentiable)**.** The function $f(x) := x|x|$ is differentiable everywhere, but it is not infinitely differentiable at 0. This is because the first derivative is

$$f'(x) = \begin{cases} 2x & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -2x & \text{if } x < 0. \end{cases} \tag{4.223}$$

So the left derivative of $f'(x)$ at 0 is -2 but the right derivative of $f'(x)$ at 0 is 2. Therefore $f''(0)$ does not exist.

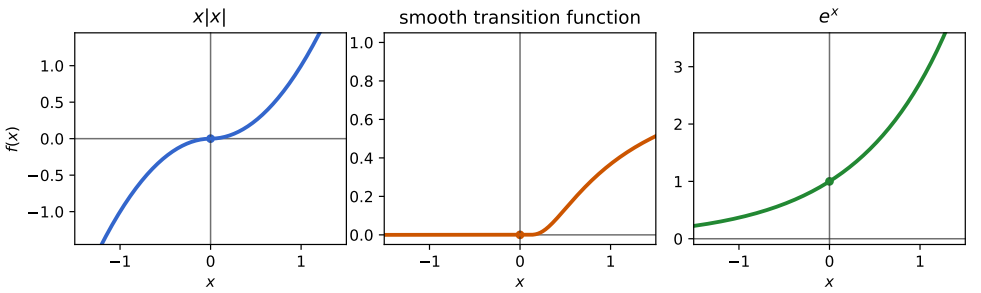


Figure 4.10: Three examples of real functions: (1) $x|x|$ is differentiable but not infinitely differentiable at 0 (Example 4.33); (2) the smooth transition function is infinitely differentiable but not analytic at 0 (Example 4.34); (3) e^x is analytic (Example 4.35).

Example 4.34 (Infinitely differentiable vs. analytic)**.** The *smooth transition function*

$$f(x) := \begin{cases} e^{-1/x} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0, \end{cases} \tag{4.224}$$

is infinitely differentiable everywhere. Too see this, we only need to check $x = 0$ since it is the only non-trivial point. We will not present the entire proof here, but the key idea is as follows. For $x > 0$, one can show by induction that for some polynomial P_n of degree n,

$$f^{(n)}(x) = P_n(1/x)e^{-1/x}, \tag{4.225}$$

But polynomial functions grow much slower than exponential functions. As $x \to 0^+$, we have $f^{(n)}(x) \to 0$. Combining this with the fact that for $x \leq 0$, all derivatives are 0, we can conclude that the derivative exists at 0 and equals 0. This proves the infinite differentiability of f at 0.

However, the function $f(x)$ is not analytic at 0. If we Taylor expand it around 0, we get

$$\sum_{n\geq 0} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n\geq 0} \frac{0}{n!} x^n = 0, \tag{4.226}$$

which does not agree with $f(x)$ for $x > 0$.

The smooth transition function above is infinitely differentiable but not analytic at 0. In contrast, the basic building blocks of the functions, such as the exponential function, are analytic.

Example 4.35 (Exponential function is analytic)**.** The function $f(x) := e^x$ is analytic since it admits the power series expansion

$$e^x = \sum_{n\geq 0} \frac{x^n}{n!}. \tag{4.227}$$

See also Figure 4.10 for a visualization of these three examples. Now let us return to the complex plane and see why these examples no longer separate differentiability from infinite differentiability, or infinite differentiability from analyticity.

Example 4.36 (Example 4.33 fails in the complex plane)**.** A natural extension of the real function $x|x|$ to the complex plane is

$$f(z) = z|z| = z\sqrt{z\overline{z}}. \tag{4.228}$$

The problematic part is $\overline{z}$ in the formula. We have shown in Example 4.29 that the complex conjugate function is not differentiable. Therefore, we cannot just differentiate the function through the chain rule.

In fact, one can rigorously show that $f(z)$ is not differentiable at any non-zero point. The only exception is at $z = 0$, where $f'(z) = 0$. This is therefore not an example of a complex function that is differentiable but not infinitely differentiable.

Example 4.37 (Example 4.34 fails in the complex plane)**.** Consider the smooth transition function $f(x)$ from Example 4.34, which equals $e^{-1/x}$ for $x > 0$ and 0 for $x \leq 0$. The main obstacle to make this example work in the complex plane is that it is unclear what "$z > 0$" means in the complex plane. If we just define the function as

$$f(z) := \begin{cases} e^{-1/z} & \text{if } \Re(z) > 0, \\ 0 & \text{if } \Re(z) \leq 0. \end{cases} \tag{4.229}$$

Then $f(z)$ is not even continuous at any point on the imaginary axis. For example, if we take $z = x + iy$ with $y \neq 0$, then as $x \to 0^+$, we have

$$f(x + iy) = e^{-1/(x+iy)} \to e^{-1/(iy)} = e^{i/y} \neq 0. \tag{4.230}$$

Therefore, $f(z)$ is not an example of a complex function that is infinitely differentiable but not analytic.

Example 4.38 (A variant of Example 4.34 also fails)**.** A closely related variant of Example 4.34 is

$$f(z) := \begin{cases} e^{-1/z^2} & \text{if } z \neq 0, \\ 0 & \text{if } z = 0. \end{cases} \tag{4.231}$$

One can show that as a real function, $f(z)$ is infinitely differentiable but not analytic at 0. While this function can be defined on the complex plane in the same way, it is not even continuous at $z = 0$. Indeed, $f(z) \to 0$ as $z \to 0$ along the real axis, but along the imaginary axis, as $y \to 0$,

$$f(iy) = e^{-1/(iy)^2} = e^{1/y^2} \to +\infty. \tag{4.232}$$

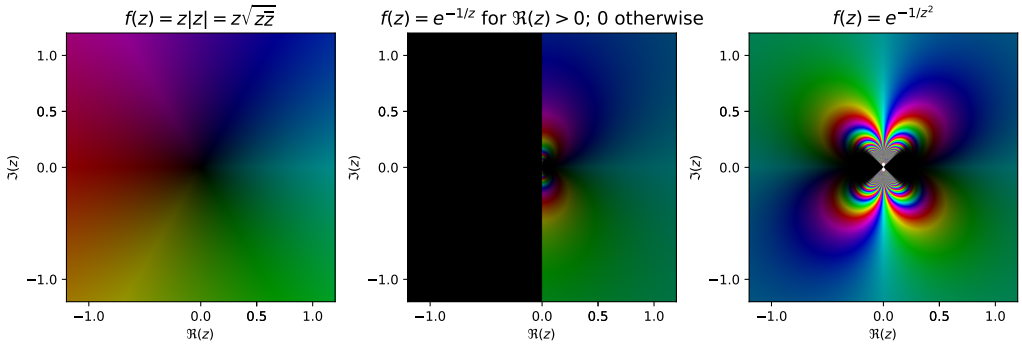


Figure 4.11: Domain coloring for real function examples after moving to the complex plane. See Example 4.36, Example 4.37, and Example 4.38 for details.

▶ Cauchy–Riemann Equations

We can now return to the point we postponed earlier. The reason complex differentiability implies infinite differentiability and analyticity is that the difference quotient in the derivative

$$\frac{f(z) - f(z_0)}{z - z_0} \tag{4.233}$$

does not treat the real and imaginary parts separately. Instead, it forces them to fit together in a very rigid way. This is expressed by the following Cauchy–Riemann equations.

Theorem 4.39 (Cauchy–Riemann equations)**.** Let f be differentiable at $z_0 = x_0 + iy_0$, and write

$$f(x + iy) = u(x, y) + iv(x, y). \tag{4.234}$$

Then the following four partial derivatives exist at (x_0, y_0):

$$\frac{\partial u}{\partial x}, \qquad \frac{\partial u}{\partial y}, \qquad \frac{\partial v}{\partial x}, \qquad \frac{\partial v}{\partial y}.$$

Furthermore, they satisfy the Cauchy–Riemann equations:

$$\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0), \qquad \frac{\partial u}{\partial y}(x_0, y_0) = -\frac{\partial v}{\partial x}(x_0, y_0). \tag{4.235}$$

Proof Sketch. We will not prove this theorem in full detail, but the key idea is as follows. Approaching z_0 along the real direction, we have

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} = \frac{\partial u}{\partial x}(x_0, y_0) + i\frac{\partial v}{\partial x}(x_0, y_0). \tag{4.236}$$

Approaching z_0 along the imaginary direction, we have

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + ih) - f(z_0)}{ih} = \frac{\partial v}{\partial y}(x_0, y_0) - i\frac{\partial u}{\partial y}(x_0, y_0). \tag{4.237}$$

These two directional limits must agree. □

Another way to think about the Cauchy–Riemann equations is to view f as a vector field in the complex plane, given by $F(x, y) = (u(x, y), v(x, y))$. Then the Jacobian matrix of F must have the following form:

$$J_F(x, y) = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}. \tag{4.238}$$

This is perhaps a more direct way to see how complex differentiability forces special structures on a complex function.

If you love physics, please continue reading. If you hate physics, skip this part! Let us reflect the vector field across the real axis and write

$$G(x, y) := (G_1(x, y), G_2(x, y)) \tag{4.239}$$

$$\text{where} \quad g_1(x, y) = u(x, y) \text{ and } g_2(x, y) = -v(x, y), \tag{4.240}$$

then the Cauchy–Riemann equations essentially state that the vector field G is both divergence-free and curl-free:

$$\nabla \cdot G = \frac{\partial g_1}{\partial x} + \frac{\partial g_2}{\partial y} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = 0, \tag{4.241}$$

$$\nabla \times G = \frac{\partial g_1}{\partial y} - \frac{\partial g_2}{\partial x} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = 0. \tag{4.242}$$

This is essentially saying that the vector field F after reflection is a conservative vector field and has no sources or sinks! Soon we will see that this can yield a more interesting and beautiful consequence when combined with Green's theorem: the complex integral of f around any closed curve is zero.

▶ Holomorphic Functions

Note that Theorem 4.32 holds when the function is defined on an open set. If the function is only differentiable at a single point, it is not necessarily infinitely differentiable or analytic. Therefore, in complex analysis, a more important notion than being differentiable at a single point is being *differentiable in an open neighborhood*. But this phrase is too long! Let us introduce a new term.

Definition 4.40 (Holomorphic function)**.** Let $f : S \to \mathbb{C}$ be a complex function defined on an open set $S \subseteq \mathbb{C}$. We say that f is *holomorphic* at $z_0 \in S$ if it is differentiable in an open set containing z_0. Further, we say that f is *holomorphic* on S if it is differentiable at every point of S.

If a function is holomorphic at a point or on an open set, it is automatically differentiable, infinitely differentiable, and analytic at that point or on that set.

Remark 4.41. The term *holomorphic* is not a daily English word. It was made up by Charles Briot and Jean-Claude Bouquet in 1875 by combining the Greek words ὅλος (*hólos*), meaning "whole," and μορφή (*morphé*), meaning "form," "appearance," or "type." So the term *holomorphic* literally means that a function exhibits a consistent "form" throughout a "whole" region.

From now on, we will simply use the term *holomorphic* in most cases and avoid saying that a function is differentiable, infinitely differentiable, or analytic.

4.2.5 Complex Integrals

Having understood the differentiability of complex functions, we are now ready to move on to the next step: *complex integrals*.

Just like in real calculus, we can integrate functions over intervals.

Definition 4.42 (Integral over an interval)**.** Let $f : [a, b] \to \mathbb{C}$ be a continuous function. If $f(t) = u(t) + iv(t)$, then the integral of f over the interval $[a, b]$ is defined by

$$\int_a^b f(t)\ \mathrm{d}t := \int_a^b u(t)\ \mathrm{d}t + i \int_a^b v(t)\ \mathrm{d}t. \tag{4.243}$$

This definition reduces the problem to real integrals applied to the real and imaginary parts separately. As a result, many familiar properties from the real case continue to hold. For example, if $f(t)$ admits an antiderivative $F(t)$, namely $F'(t) = f(t)$ for all t, then

$$\int_a^b f(t)\ \mathrm{d}t = F(b) - F(a). \tag{4.244}$$

However, this perspective does not yet reveal anything essentially new about complex functions. In multivariable calculus, you may recall that we can integrate functions along curves. Such an integral is called a *line integral*. We can do the same in the complex plane, and as we will soon see, this leads to many genuinely new phenomena.

▶ Curves and Contours

To define such integrals, we first need to clarify what we mean by a *curve*.

Definition 4.43 (Curve)**.** A *curve* in $\mathbb{C}$ is a continuous function $\gamma : [a, b] \to \mathbb{C}$. Furthermore,

- We say that γ is *smooth* if it is continuously differentiable and its derivative γ' is nowhere zero.
- We say that γ is *simple* if it is injective. That is, γ does not cross itself.
- We say that γ is *closed* if $\gamma(a) = \gamma(b)$. That is, γ returns to its starting point at the very end.
- We say that γ is a *simple closed curve* if it is closed and is injective on $[a, b)$. That is, γ does not cross itself except at the endpoints.

When we connect multiple smooth curves together, we obtain a *contour*.

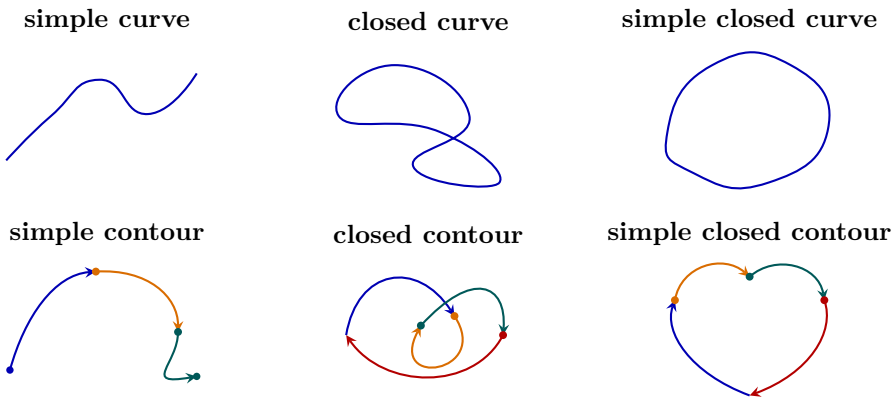


Figure 4.12: Examples of curves and contours classified by simplicity and closedness. All the curves in the figure are smooth.

Definition 4.44 (Contour)**.** A *contour* is a sequence of smooth curves

$$\gamma = \{\gamma_1, \gamma_2, \ldots, \gamma_n\} \tag{4.245}$$

such that the end point of γ_i is the starting point of γ_{i+1} for $i \in [n-1]$.

A curve can be seen as a contour consisting of a single curve. Conversely, a contour can be viewed as a single curve obtained by concatenating the smooth curves in the sequence. We write "+" to denote the concatenation of curves together to form a new curve. Thus, if $\gamma = \{\gamma_1, \gamma_2, \ldots, \gamma_n\}$ is a contour, we may instead write

$$\gamma = \gamma_1 + \gamma_2 + \cdots + \gamma_n. \tag{4.246}$$

We say that the contour is *simple*, *closed*, or *simple closed* if the corresponding curve is simple, closed, or simple closed.

▶ Line Integrals

A line integral is defined as an integral along a curve or a contour.

Definition 4.45 (Line integral)**.** Let $f : S \to \mathbb{C}$ be a complex function defined on an open set $S \subseteq \mathbb{C}$. If $\gamma : [a, b] \to S$ is a smooth curve in S, then the *integral of f along γ* is defined by

$$\int_\gamma f(z)\ \mathrm{d}z := \int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t. \tag{4.247}$$

If $\gamma = \gamma_1 + \cdots + \gamma_n$ is a contour, then the *contour integral of f along γ* is defined as the sum of the integrals of f along the smooth curves that make up the contour:

$$\int_\gamma f(z)\ \mathrm{d}z := \sum_{i=1}^{n} \int_{\gamma_i} f(z)\ \mathrm{d}z. \tag{4.248}$$

To understand the line integral along a curve or a contour γ geometrically, we may break γ into many tiny line segments with endpoints $z_0, z_1, \ldots, z_n$ on the curve, which the j-th segment connects z_{j-1} to z_j. Let $\Delta z_j := z_j - z_{j-1}$ denote the increment along the j-th segment. Then the integral along γ is approximately

$$\int_\gamma f(z)\ \mathrm{d}z \approx \sum_j f(z_j) \cdot \Delta z_j. \tag{4.249}$$

As we make the segments shorter and shorter, the approximation becomes more and more accurate. In the limit, we obtain the exact value of the integral.

Below, we mainly discuss integrals along smooth curves, since the generalization to contours is straightforward. The real usefulness of contour integrals will become clear later in this section.

Example 4.46. Let γ be the real interval $[a, b]$ with parameterization $\gamma(t) = t$ for $t \in [a, b]$. Then by definition, we have

$$\int_\gamma f(z)\ \mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t = \int_a^b f(t)\ \mathrm{d}t, \tag{4.250}$$

which is the usual integral of f over the interval $[a, b]$.

Example 4.47. Let γ be a circle of radius R centered at the origin with parameterization $\gamma(t) = Re^{it}$ for $t \in [0, 2\pi]$. Then by definition, for any integer $n \neq -1$, the integral of z^n along γ is

$$\int_\gamma z^n\ \mathrm{d}z = \int_0^{2\pi} (Re^{it})^n\, iRe^{it}\ \mathrm{d}t = \int_0^{2\pi} iR^{n+1}e^{it(n+1)}\ \mathrm{d}t \tag{4.251}$$

$$= \frac{R^{n+1}}{n+1} e^{it(n+1)}\Big|_{t=0}^{t=2\pi} = 0. \tag{4.252}$$

The only exception is when $n = -1$, in which case the integral is $2\pi i$:

$$\int_\gamma z^{-1}\ \mathrm{d}z = \int_0^{2\pi} R^{-1}e^{-it}\, iRe^{it}\ \mathrm{d}t = \int_0^{2\pi} i\ \mathrm{d}t = 2\pi i. \tag{4.253}$$

Example 4.48. *How does the integral of z^n along γ change with the choice of γ?* In fact, only the starting and ending points of γ matter. Let γ be an arbitrary smooth curve in the complex plane. Then for any integer $n \neq -1$, the integral of z^n along γ is

$$\int_\gamma z^n\ \mathrm{d}z = \int_a^b \gamma(t)^n\gamma'(t)\ \mathrm{d}t = \frac{\gamma(t)^{n+1}}{n+1}\Big|_{t=a}^{t=b} \tag{4.254}$$

$$= \frac{\gamma(b)^{n+1} - \gamma(a)^{n+1}}{n+1}. \tag{4.255}$$

If the curve is closed, then this integral is exactly zero.

Example 4.49. In the above example, we only considered the case of $n \neq -1$. *How about the integral of z^{-1} along γ?* It is tempting to write:

$$\int_\gamma z^{-1}\,\mathrm{d}z = \int_a^b \frac{\gamma'(t)}{\gamma(t)}\,\mathrm{d}t = \log\gamma(t)\Big|_{t=a}^{t=b}. \tag{4.256}$$

However, this is wrong! This is because log cannot be properly defined for complex numbers. While we can use the principal logarithm $\operatorname{Log} z$ instead (see Example 4.25), the derivative of $\operatorname{Log} z$ is z^{-1} only if $\arg(z) \in (-\pi, \pi)$. If $\arg(z) = \pi$, $\operatorname{Log} z$ is not even continuous at z.

To handle this integral correctly, a more graceful way is to write

$$\gamma(t) = R(t)e^{i\theta(t)}, \tag{4.257}$$

with $\theta(t)$ chosen to be continuous along the curve. This can be done by letting $\theta(t)$ be the total angle traversed along γ, starting from any initial value. Then,

$$\int_\gamma z^{-1}\,\mathrm{d}z = \int_a^b \frac{R'(t)e^{i\theta(t)} + R(t)i\theta'(t)e^{i\theta(t)}}{R(t)e^{i\theta(t)}}\,\mathrm{d}t \tag{4.258}$$

$$= \int_a^b \left(\frac{R'(t)}{R(t)} + i\theta'(t)\right)\mathrm{d}t \tag{4.259}$$

$$= \log R(t)\Big|_{t=a}^{t=b} + i\theta(t)\Big|_{t=a}^{t=b}. \tag{4.260}$$

In particular, if γ is closed, then the integral is $2\pi i$ times the number of times γ winds around the origin, counted counterclockwise.

▶ Reparameterization and Reversal of Curves

How does the value of the integral along a curve depend on how we parameterize the same geometric curve? The following theorem states that reparameterizing the curve does not change the integral.

Theorem 4.50 (Reparameterization and reversal of direction)**.** Let $f : S \to \mathbb{C}$ be a complex function defined on an open set $S \subseteq \mathbb{C}$. Let $\gamma_1 : [a, b] \to S$ and $\gamma_2 : [c, d] \to S$ be two smooth curves. If there exists an increasing continuously differentiable function $\phi : [c, d] \to [a, b]$ such that $\phi(c) = a$, $\phi(d) = b$, and $\gamma_2(t) = \gamma_1(\phi(t))$ for all $t \in [c, d]$, then

$$\int_{\gamma_2} f(z)\,\mathrm{d}z = \int_{\gamma_1} f(z)\,\mathrm{d}z. \tag{4.261}$$

Proof. Assume first that $\phi : [c, d] \to [a, b]$ is increasing and that $\gamma_2(t) = \gamma_1(\phi(t))$ for all $t \in [c, d]$. Then

$$\int_{\gamma_2} f(z)\,\mathrm{d}z = \int_c^d f(\gamma_2(t))\gamma_2'(t)\,\mathrm{d}t = \int_c^d f(\gamma_1(\phi(t)))\gamma_1'(\phi(t))\phi'(t)\,\mathrm{d}t. \tag{4.262}$$

Now change variables from t to $\tau = \phi(t)$. Since $\phi(c) = a$ and $\phi(d) = b$, we obtain

$$\int_c^d f(\gamma_1(\phi(t)))\gamma_1'(\phi(t))\phi'(t)\ \mathrm{d}t = \int_a^b f(\gamma_1(\tau))\gamma_1'(\tau)\ \mathrm{d}\tau, \tag{4.263}$$

which is exactly $\int_{\gamma_1} f(z)\ \mathrm{d}z$. □

Instead, if we reverse the direction in which we traverse the curve, the value of the integral is multiplied by -1. More formally, let $-\gamma$ denote the curve obtained by traversing γ in the opposite direction, defined by $(-\gamma)(t) = \gamma(a + b - t)$ for $t \in [a, b]$. Then we have

$$\int_{-\gamma} f(z), \mathrm{d}z = -\int_{\gamma} f(z), \mathrm{d}z. \tag{4.264}$$

Here the minus sign acts on the curve itself, not on the function value.

Example 4.51. Consider the integral of z^{-1} along a circle of radius R centered at 0. Regardless of the parametrization, the value of the integral is $2\pi i$ if the circle is traversed counterclockwise, and $-2\pi i$ if it is traversed clockwise.

▶ ML Inequality

If we do not know how to compute an integral, we can at least bound its absolute value. For a smooth curve $\gamma : [a, b] \to \mathbb{C}$, we define its length by

$$L(\gamma) := \int_a^b |\gamma'(t)|\ \mathrm{d}t, \tag{4.265}$$

which is the same as the usual definition of length in multivariable calculus.

Theorem 4.52 (ML Inequality)**.** Let $f : S \to \mathbb{C}$ be a complex function defined on an open set $S \subseteq \mathbb{C}$. If $\gamma : [a, b] \to S$ is a smooth curve in S and $|f(\gamma(t))| \le M$ for all $t \in [a, b]$, then

$$\left|\int_\gamma f(z)\ \mathrm{d}z\right| \le M \cdot L(\gamma). \tag{4.266}$$

The point of this inequality is very simple: along each tiny piece of the curve, the integrand has size at most M, so the total contribution cannot exceed M times the total distance traveled along the curve.

Proof. The proof is straightforward based on the definition of the integral and the properties of the absolute value:

$$\left|\int_\gamma f(z)\ \mathrm{d}z\right| = \left|\int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t\right| \le \int_a^b |f(\gamma(t))|\,|\gamma'(t)|\ \mathrm{d}t \tag{4.267}$$

$$\le \int_a^b M\,|\gamma'(t)|\ \mathrm{d}t = M \cdot L(\gamma), \tag{4.268}$$

which is the desired inequality. □

▶ Closed Contours and Holomorphic Functions

In the examples above, we already saw that $\int_\gamma z^n \, \mathrm{d}z = 0$ on every closed curve as long as $n \neq -1$. The next theorem shows that this is not a special feature of monomials. Once a function is holomorphic, its integral around a simple closed contour must vanish.

Theorem 4.53 (Cauchy–Goursat theorem)**.** Let $f : S \to \mathbb{C}$ be a holomorphic function defined on an open set $S \subseteq \mathbb{C}$. Let γ be a simple closed contour such that the region enclosed by γ is contained in S. Then the contour integral of f along γ is zero. That is,

$$\int_\gamma f(z) \, \mathrm{d}z = 0. \tag{4.269}$$

If you love physics, this result may not seem surprising at all. It is just a combination of Cauchy–Riemann equations (Theorem 4.39) and your favorite Green's theorem. We omit the details here, since you may enjoy working them out yourself if you really love physics.

If you also love mathematics, however, you may not be fully satisfied with this argument. The issue is that Green's theorem requires the function to be continuously differentiable. Although we have mentioned that holomorphic functions are in fact infinitely differentiable, using that fact here would lead to a "chicken-or-egg" problem: in complex analysis, we have to establish this theorem first, and only then can we deduce the infinite differentiability of holomorphic functions.

Therefore, below we provide a proof sketch of the Cauchy–Goursat theorem using only the definition of complex differentiability.

Proof sketch. The first idea is to divide the region enclosed by γ into many small triangles. This is called a *triangulation*, a notion that you will see or may have already seen in a computer graphics course.

For simplicity, let us first consider the case where γ is a polygon, that is, each piece of the contour is a line segment. The general case can then be obtained by approximating a smooth contour by a polygon.

We also assume that γ goes counterclockwise around the region, since otherwise we can reverse the direction of the contour. For each triangle, let its boundary go counterclockwise around the triangle. When we sum the contour integrals over all these triangles, every interior edge appears twice with opposite orientations, and hence the contributions along these edges cancel. As a result, only the integral along the outer boundary γ remains.

So if we can show that the contour integral around each small triangle is 0, then after summing over all triangles we will get that the contour integral around the outer boundary γ is also 0. Therefore it remains to understand what happens for one triangle.

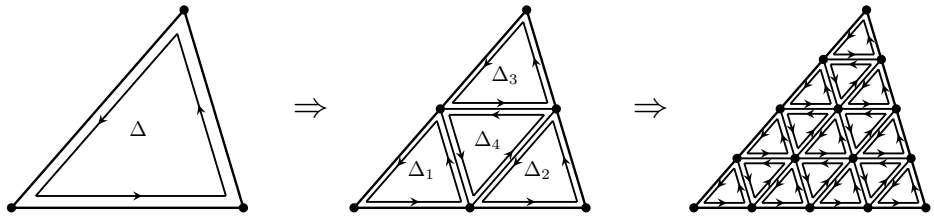


Figure 4.13: Two rounds of triangulation refinement. At each step, we join the midpoints of the three sides of every triangle, thereby dividing each triangle into four smaller triangles. Repeating this refinement makes the diameters and perimeters of all triangles tend to 0.

Let us now focus on the case where γ is the boundary of a single triangle Δ and explain why its contour integral must vanish. Let I be the contour integral around γ:

$$I := \int_\gamma f(z)\ \mathrm{d}z. \tag{4.270}$$

Assume to the contrary that $I \neq 0$. We will show that this would lead to a contradiction.

Trianglation Refinement. A key idea is to refine the triangle by joining the midpoints of its three sides, which divides it into four smaller triangles. We then repeat this refinement process iteratively. At each round, for every triangle in the current triangulation, we join the midpoints of its three sides and divide it into four smaller triangles. These four new triangles are referred to as the *children* of the original triangle. See Figure 4.13 for an illustration.

Each time we refine the triangulation,

$$\text{integral over original triangle} = \text{sum of integrals over its children.} \tag{4.271}$$

This is because every newly created interior edge is traversed twice in opposite directions, so the corresponding contributions cancel out in the sum.

We now recursively choose a sequence of triangles as follows. After the first refinement, at least one of the four smaller triangles must have boundary $\gamma^{(1)}$ such that

$$\left|\int_{\gamma^{(1)}} f(z)\ \mathrm{d}z\right| \geq \frac{|I|}{4}. \tag{4.272}$$

We then refine that triangle and choose one of its four children whose boundary integral has the largest absolute value. Continuing in this way, we obtain recursively a nested sequence of triangles

$$\Delta \supseteq \Delta^{(1)} \supseteq \Delta^{(2)} \supseteq \cdots \tag{4.273}$$

with boundaries $\gamma^{(n)}$ satisfying

$$\left|\int_{\gamma^{(n)}} f(z)\ \mathrm{d}z\right| \geq \frac{|I|}{4^n}. \tag{4.274}$$

Local Approximation. Geometrically, the triangles $\{\Delta^{(n)}\}_{n\geq 1}$ are collapsing. At each refinement the diameter and the perimeter are divided by 2. So for the n-th triangle, the diameter and perimeter are both $\mathcal{O}(2^{-n})$.

Therefore, the diameters of these triangles tend to 0. Since they are nested, it can be shown that their intersection consists of exactly one point, which we call z_0. More formally, there exists a unique point z_0 such that for every neighborhood U of z_0, there exists an N such that for all $n \geq N$, $\Delta^{(n)} \subseteq U$.

Since f is holomorphic on S, including the point z_0, the function can be locally approximated by a linear function near z_0. More precisely, by the definition of the complex derivative, we can write

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \epsilon(z), \tag{4.275}$$

for some function $\epsilon(z)$ with $|\epsilon(z)| = o(|z - z_0|)$ as $z \to z_0$.

Now we come to the key step, where we really make use of the special algebra of contour integrals: the constant and linear terms contribute nothing.

$$\int_{\gamma^{(n)}} \mathrm{d}z = 0, \qquad \int_{\gamma^{(n)}} (z - z_0)\ \mathrm{d}z = 0. \tag{4.276}$$

So only the higher-order error term matters in the integral:

$$\int_{\gamma^{(n)}} f(z)\ \mathrm{d}z = \int_{\gamma^{(n)}} \epsilon(z)\ \mathrm{d}z. \tag{4.277}$$

Now we argue that this integral is fact decays faster than Equation (4.274). Let $M^{(n)}$ be the maximum value of $|\epsilon(z)|$ on $\gamma^{(n)}$. By ML inequality, we have

$$\left|\int_{\gamma^{(n)}} f(z)\ \mathrm{d}z\right| \leq M^{(n)} \cdot L(\gamma^{(n)}). \tag{4.278}$$

As $n \to 0$, $M^{(n)}$ is bounded by a little-o of the diameter of the n-th triangle, since $\epsilon(z) = o(|z - z_0|)$ as $z \to z_0$. So $M^{(n)} = o(2^{-n})$. The length of $\gamma^{(n)}$, which is the perimeter of the n-th triangle, is bounded by $\mathcal{O}(2^{-n})$. Therefore,

$$\left|\int_{\gamma^{(n)}} f(z)\ \mathrm{d}z\right| \leq o(2^{-n}) \cdot \mathcal{O}(2^{-n}) = o(4^{-n}). \tag{4.279}$$

This leads to a contradiction, since Equation (4.274) implies that the integral is bounded by $\Omega(4^{-n})$. Therefore, $I = 0$. □

4.2.6 Integrals Around Singularities

Cauchy–Goursat Theorem implies that the integral of a holomorphic function around a closed contour is somewhat boring: it is always 0. But why do we get a non-zero value if we integrate the function $f(z) = z^{-1}$ around any circle centered at the origin (Example 4.47)?

The point is that although $f(z) = z^{-1}$ is holomorphic on $\mathbb{C} \setminus \{0\}$, the Cauchy–Goursat theorem requires more: the entire region enclosed by γ must lie in the domain of f. However, any circle centered at the origin encloses the origin, where f is not defined. Therefore, the theorem does not apply.

This example reveals an important principle: in complex analysis, when integrating a function around a contour, what matters is not the points that behave nicely, but the bad points enclosed by the contour. Points at which the function is not holomorphic, or not even defined, play a decisive role. This motivates us to conduct a more careful study of such points, which we call *singularities*.

▶ Singularities

Now we formally define what a singularity means for a complex function.

Definition 4.54 (Singularity)**.** Let $f : S \to \mathbb{C}$ be a complex function. We say that z_0 is a *singular point*, or a *singularity* of f if f is either not defined at z_0 or not holomorphic at z_0, but for every $\epsilon > 0$, there exists a complex number $z \in S$ such that $|z - z_0| < \epsilon$ and f is holomorphic at z.

In plain language, z_0 is a singularity of f if f breaks at z_0, but looks fine if you stay very close to z_0. A stronger notion is isolated singularity, which is the most common case that we will encounter in this course.

Definition 4.55 (Isolated singularity)**.** Let $f : S \to \mathbb{C}$ be a complex function. We say that z_0 is an *isolated singularity* of f if f is either not defined at z_0 or not holomorphic at z_0, but there exists an $\epsilon > 0$ such that f is holomorphic on the punctured disk $\{z \in \mathbb{C} : 0 < |z - z_0| < \epsilon\}$.

If you only have the capacity to remember one definition, just remember this isolated singularity definition and forget about the general one, since we will not use the general one again in this course.

Example 4.56. The function $f(z) = \frac{1}{1+z^2}$ is defined and holomorphic on $\mathbb{C} \setminus \{\pm i\}$. When $z = \pm i$, the denominator $1 + z^2$ vanishes, and thus f has two isolated singularities: $\pm i$.

In general, if $g(z) = \frac{f(z)}{P(z)}$ for some holomorphic function $f : S \to \mathbb{C}$ and some polynomial $P(z)$ on an open set $S \subseteq \mathbb{C}$, then g is not defined at the zeros of $P(z)$, but g is holomorphic on the rest of the points in S. By the fundamental theorem of algebra (which we will not prove here), a polynomial has finitely many zeros, so these points must be isolated. Therefore, the zeros of $P(z)$ are the isolated singularities of g.

▶ Cauchy's Integral Formula

Let $f : S \to \mathbb{C}$ be a holomorphic function on an open set $S \subseteq \mathbb{C}$. Fix $z \in S$ and consider the function

$$g(w) = \frac{f(w)}{w - z}. \tag{4.280}$$

Then g is holomorphic on $S \setminus \{z\}$, but at z, it is not even defined. Therefore, z is an isolated singularity of g.

What if we integrate g around a simple closed contour γ enclosing z? Here we say that γ *encloses* z if the region enclosed by γ contains z, where the region enclosed by γ is the open set that contains all points inside γ. If γ does not enclose z, then the integral of g around γ is 0, according to the Cauchy–Goursat theorem. For the case that γ encloses z, we have the following formula, known as *Cauchy's integral formula*.

Theorem 4.57 (Cauchy's integral formula)**.** Let $f : S \to \mathbb{C}$ be a holomorphic function on an open set $S \subseteq \mathbb{C}$. Let γ be a simple closed counterclockwise contour such that the region enclosed by γ is contained in S. Then for every z enclosed by γ,

$$f(z) = \frac{1}{2\pi i} \int_\gamma \frac{f(w)}{w - z} \, \mathrm{d}w. \tag{4.281}$$

This is an integral along a contour that encloses the isolated singularity z. Very surprisingly, taking such an integral directly gives us the value of f at z. This is not a phenomenon one easily encounters in real calculus: a local function value can be recovered from a global integral.

Before we proceed, we define the following notations for convenience:

Definition 4.58 (Circles and Disks)**.** A circle centered at z with radius r is the following curve $C_r(t; z)$:

$$C_r(t; z) = z + re^{it}, \quad \forall t \in [0, 2\pi]. \tag{4.282}$$

An open disk centered at z with radius r is the following set $D_r(z)$:

$$D_r(z) = \{w \in \mathbb{C} : |w - z| < r\}. \tag{4.283}$$

When the center z is clear from the context, we may omit it and simply write C_r and D_r for brevity.

Proof. Fix a point z enclosed by γ. Let C_r be the circle centered at z with radius r, and let D_r be the closed disk enclosed by C_r. Choose r small enough so that C_r and D_r are both contained in the region enclosed by γ.

The proof proceeds in two steps. First, we deform the contour γ to a small circle C_r centered at z, and show that this deformation does not change the integral value. See Figure 4.14 for an illustration. Then, we compute the integral along C_r explicitly using a local approximation of f around z.

Deformation. Note that the function

$$g(w) := \frac{f(w)}{w - z} \tag{4.284}$$

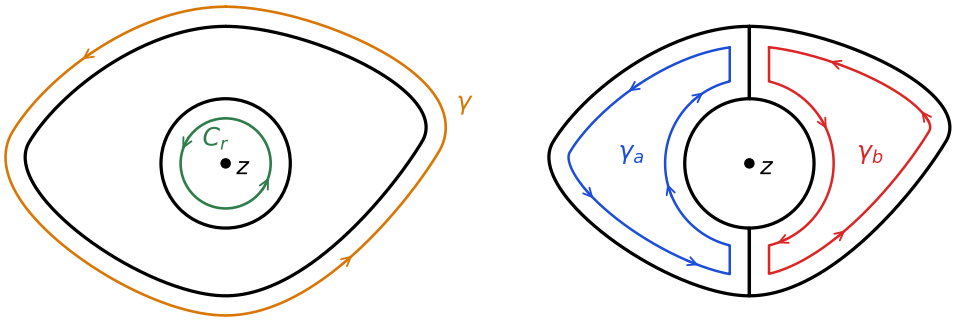


Figure 4.14: An illustration of the contour deformation in the proof of Cauchy's integral formula. To show that the integral of g along γ equals the integral of g along C_r, we cut the annular region between γ and C_r along two segments joining the two curves. This produces two regions with boundaries γ_a and γ_b. The integrals along γ_a and γ_b are exactly zero, and adding them together gives the difference between the integral along γ and the integral along C_r.

is holomorphic on the annular region between γ and C_r, since its only singularity is at $w = z$, which lies inside C_r.

Now we draw two segments joining γ to C_r, as shown in Figure 4.14. These segments split the region into two pieces, each without a hole. Let γ_a and γ_b be the boundaries of the two regions, oriented counterclockwise. Then by Cauchy–Goursat theorem,

$$\int_{\gamma_a} g(w)\,\mathrm{d}w = 0, \qquad \int_{\gamma_b} g(w)\,\mathrm{d}w = 0. \tag{4.285}$$

Adding these two integrals, we get

$$0 = \int_{\gamma_a} g(w)\,\mathrm{d}w + \int_{\gamma_b} g(w)\,\mathrm{d}w = \int_{\gamma} g(w)\,\mathrm{d}w - \int_{C_r} g(w)\,\mathrm{d}w. \tag{4.286}$$

Here, the two cuts are traversed once in each direction, so their contributions cancel. The outer boundary γ is traversed counterclockwise, whereas the inner boundary C_r is traversed clockwise. This is why the sum of the two integrals is the difference between the integral around γ and the integral around C_r.

So now we know that the integral of g around γ is the same as the integral of g around C_r:

$$\int_{\gamma} \frac{f(w)}{w-z}\,\mathrm{d}w = \int_{C_r} \frac{f(w)}{w-z}\,\mathrm{d}w. \tag{4.287}$$

Local Approximation. Since f is holomorphic at z, by definition of the complex derivative, we can approximate f around z by a linear function:

$$f(w) = f(z) + (w-z)f'(z) + \epsilon(w), \tag{4.288}$$

where $\epsilon(w)$ is a function such that $|\epsilon(w)| = o(|w-z|)$ as $w \to z$. Therefore,

$$\int_{C_r} \frac{f(w)}{w-z}\,\mathrm{d}w = \int_{C_r} \frac{f(z)+(w-z)f'(z)+\epsilon(w)}{w-z}\,\mathrm{d}w \tag{4.289}$$

$$= \underbrace{\int_{C_r} \frac{f(z)}{w-z}\,\mathrm{d}w}_{=:I_1} + \underbrace{\int_{C_r} f'(z)\,\mathrm{d}w}_{=:I_2} + \underbrace{\int_{C_r} \frac{\epsilon(w)}{w-z}\,\mathrm{d}w}_{=:I_3}. \tag{4.290}$$

The first term I_1 is similar to the integral of z^{-1} around origin, which we have seen in Example 4.47. More specifically,

$$\int_{C_r} \frac{f(z)}{w-z}\,\mathrm{d}w = \int_0^{2\pi} \frac{f(z)}{C_r(t)-z}\,iC_r'(t)\,\mathrm{d}t = \int_0^{2\pi} \frac{f(z)}{re^{it}}\,ire^{it}\,\mathrm{d}t \tag{4.291}$$

$$= 2\pi i\, f(z). \tag{4.292}$$

The second term I_2 is exactly zero, since $f'(z)$ is a constant and the constant function is holomorphic on any open set.

For the third term I_3, since the length of C_r is $2\pi r$, the ML inequality gives

$$\left|\int_{C_r} \frac{\epsilon(w)}{w-z}\,\mathrm{d}w\right| \le M_r \cdot L(C_r) = 2\pi r M_r, \tag{4.293}$$

where M_r is the maximum value of $|\frac{\epsilon(w)}{w-z}|$ on C_r. Since $\epsilon(w) = o(|w-z|)$, we have $M_r = o(1)$ as $r \to 0$. Therefore,

$$\left|\int_{C_r} \frac{\epsilon(w)}{w-z}\,\mathrm{d}w\right| = 2\pi r \cdot o(1) = o(r). \tag{4.294}$$

Putting everything together, we get

$$\int_\gamma \frac{f(w)}{w-z}\,\mathrm{d}w = \int_{C_r} \frac{f(w)}{w-z}\,\mathrm{d}w = 2\pi i\, f(z) + 0 + o(r). \tag{4.295}$$

Taking the limit as $r \to 0$, we get

$$\int_\gamma \frac{f(w)}{w-z}\,\mathrm{d}w = 2\pi i\, f(z). \tag{4.296}$$

Dividing by $2\pi i$ on both sides, we get Cauchy's integral formula. □

▶ Taylor Series Expansion

Cauchy's integral formula can be used to justify a point that we have mentioned earlier: holomorphic functions are analytic.

Recall that Cauchy's integral formula enables us to rewrite the function f as an integral along a fixed contour γ, where the integrand is $\frac{f(w)}{w-z}$. As a function of w, this integrand can be quite complicated because $f(w)$ can be arbitrary, but as a function of z, this is extremely simple. It has very nice properties: it is continuous, differentiable, infinitely differentiable, and analytic. Therefore, we can first expand $\frac{f(w)}{w-z}$ as a power series, then apply the integral to get a power series expansion of $f(z)$. This is summarized in the following theorem.

Theorem 4.59 (Holomorphic functions are analytic)**.** Let $f : S \to \mathbb{C}$ be a holomorphic function on an open set $S \subseteq \mathbb{C}$. Fix $z_0 \in S$ and $r > 0$ such that the circle C_r centered at z_0 and its interior D_r are both contained in S. Then for every $z \in D_r$,

$$f(z) = \sum_{n \geq 0} a_n (z - z_0)^n, \quad \text{where} \quad a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(w)}{(w - z_0)^{n+1}} \, \mathrm{d}w. \tag{4.297}$$

Since analyticity implies differentiability and infinite differentiability, a direct corollary of this theorem is Theorem 4.32: on an open set, differentiability, infinite differentiability, and analyticity are all the same.

The formula of a_n is commonly referred to as the *Cauchy integral formula for derivatives*, since it directly gives us the value of the n-th order derivative of f at z_0. Let us write it down explicitly.

Theorem 4.60 (Cauchy integral formula for derivatives)**.** Let $f : S \to \mathbb{C}$ be a holomorphic function on an open set $S \subseteq \mathbb{C}$. Let γ be a simple closed counterclockwise contour such that the region enclosed by γ is contained in S. Then for every z enclosed by γ and every integer $n \geq 0$,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z)^{n+1}} \, \mathrm{d}w. \tag{4.298}$$

Proof. Just need to note that taking the n-th order derivative of a power series means extracting the n-th coefficient of the power series and multiplying it by $n!$. Another way to see this is to take the n-th order derivative on both sides of $f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} \, \mathrm{d}w$. □

Putting everything together, we can conclude that any holomorphic function admits the following power series expansion locally around each point z_0 in its domain, known as the *Taylor series* of f around z_0:

$$f(z) = \sum_{n \geq 0} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n. \tag{4.299}$$

Note that there is a minor difference between Theorem 4.59 and Theorem 4.60: the former enforces the contour to be a circle, while the latter does not. It is easy to see that we can deform a circle to an arbitrary simple closed contour γ for computing the integral of $\frac{f(w)}{(w-z)^{n+1}}$, similar to what we did in the proof of Cauchy's integral formula. However, why do we need to restrict the contour to be a circle in Theorem 4.59? The reason for this difference will be clear in the following proof.

Proof of Theorem 4.59. By Cauchy's integral formula, we have

$$f(z) = \frac{1}{2\pi i} \int_{C_r} \frac{f(w)}{w - z} \, \mathrm{d}w. \tag{4.300}$$

We can expand the integrand as a geometric series:

$$\frac{f(w)}{w-z} = \frac{f(w)}{w-z_0} \cdot \frac{1}{1-\frac{z-z_0}{w-z_0}} = \frac{f(w)}{w-z_0} \sum_{n\geq 0} \left(\frac{z-z_0}{w-z_0}\right)^n \tag{4.301}$$

$$= \sum_{n\geq 0} \frac{f(w)}{(w-z_0)^{n+1}} (z-z_0)^n. \tag{4.302}$$

This series converges because

$$\left|\frac{z-z_0}{w-z_0}\right| = \frac{|z-z_0|}{|w-z_0|} = \frac{|z-z_0|}{r} < 1. \tag{4.303}$$

Substituting this expansion into the integral, we get

$$f(z) = \frac{1}{2\pi i} \int_{C_r} \sum_{n\geq 0} \frac{f(w)}{(w-z_0)^{n+1}} (z-z_0)^n \, \mathrm{d}w. \tag{4.304}$$

Swapping the order of integration and summation, we get

$$f(z) = \sum_{n\geq 0} \left(\frac{1}{2\pi i} \int_{C_r} \frac{f(w)}{(w-z_0)^{n+1}} \, \mathrm{d}w\right) (z-z_0)^n, \tag{4.305}$$

which is the desired power series expansion of $f(z)$. □

The above proof is straightforward except that we have to check if the power series converges or not. The key step is Equation (4.303). If the integral is not along the circle C_r but along an arbitrary simple closed contour γ, then we have to ensure that every point w on γ satisfies $\left|\frac{z-z_0}{w-z_0}\right| < 1$. That is, the set of points z that can make the power series converge is the largest open disk that is entirely contained in the region enclosed by γ. Therefore, it is better to directly choose the contour to be a circle C_r for simplicity.

▶ Identity Theorem

An interesting consequence of Taylor series expansion is the following identity theorem: once two holomorphic functions agree on an open set, they agree on the entire domain.

Theorem 4.61 (Identity theorem)**.** Let $S \subseteq \mathbb{C}$ be a connected open set, and let $f, g : S \to \mathbb{C}$ be holomorphic. If there exists a non-empty open set $U \subseteq S$ such that $f(z) = g(z)$ for every $z \in U$, then $f(z) = g(z)$ for every $z \in S$.

Proof. Let $h = f - g$. Then h is holomorphic on S. It suffices to show that h is identically zero on S.

Let z_0 be a point in S such that $h(z_0) = 0$. Since h is holomorphic at z_0, there exists some $r > 0$ such that for all $z \in D_r(z_0)$,

$$h(z) = \sum_{n\geq 0} \frac{h^{(n)}(z_0)}{n!} (z-z_0)^n. \tag{4.306}$$

We argue that there are only two possible cases:

1. h is identically zero on $D_r(z_0)$ and all its derivatives at z_0 are zero.
2. z_0 is an isolated zero of h, i.e., there exists some $\epsilon > 0$ such that $h(z) \neq 0$ for all $z \in D_\epsilon(z_0) \setminus \{z_0\}$.

To see this, suppose that h is not identically zero on $D_r(z_0)$. Then there exists some $m > 0$ such that $h^{(m)}(z_0) \neq 0$. Then we can write h as

$$h(z) = (z - z_0)^m q(z), \quad \text{where} \quad q(z) := \sum_{n \geq 0} \frac{h^{(m+n)}(z_0)}{(m+n)!}(z - z_0)^n. \tag{4.307}$$

The function value of q is non-zero at z_0 because $q(z_0) = \frac{h^{(m)}(z_0)}{m!} \neq 0$. By continuity, $q(z) \neq 0$ near z_0, so h has no zeros near z_0 other than z_0 itself.

Based on this observation, let E be the set of zeros of the first type, i.e.,

$$E = \{z \in S : h = 0 \text{ on } D_r(z) \text{ for some } r > 0\}. \tag{4.308}$$

Since $U \subseteq E$, the set E is non-empty. It suffices to show that E is both open and closed. Then combining this with the fact that S is connected, we will have $E = S$.

It is easy to see why E is open. If $h = 0$ on $D_r(z)$ for some $z \in S$ and some $r > 0$, then $h = 0$ on $D_{r/2}(z')$ for every $z' \in D_{r/2}(z)$.

To see why E is also closed, assume to the contrary that E is not closed. Then there exists a sequence $(z_n)_{n \geq 0}$ in E that converges to some $z \in S \setminus E$. By continuity of h, we have $h(z) = \lim_{n \to \infty} h(z_n) = 0$. But since $z \notin E$, z must be an isolated zero of h. This means that there exists some $\epsilon > 0$ such that $h(z) \neq 0$ for all $z \in D_\epsilon(z) \cap E$, which implies that $D_\epsilon(z) \cap E = \varnothing$ by definition of E.

But since $z_n \to z$, there exists some $n \geq 0$ such that $z_n \in D_\epsilon(z)$. As $z_n \in E$ for all $n \geq 0$, this contradicts the fact that $D_\epsilon(z) \cap E = \varnothing$. Therefore, E must be closed, and hence $E = S$. □

Finally, we note that real analytic functions also satisfy the identity theorem. However, in in complex analysis, holomorphicity alone already suffices to guarantee this property.

▶ Radius of Convergence of Taylor Series

For every choice of a circle C_r centered at z_0, Theorem 4.59 guarantees that the Taylor series expansion around z_0 converges to $f(z)$ for every $z \in D_r$. However, this region D_r is not necessarily the largest region on which this power series converges. A natural next question is: *what is exactly the region on which this power series converges?*

From Cauchy–Hadamard theorem, we know that the region of convergence is a disk centered at z_0 with radius R, so we only need to determine this radius R.

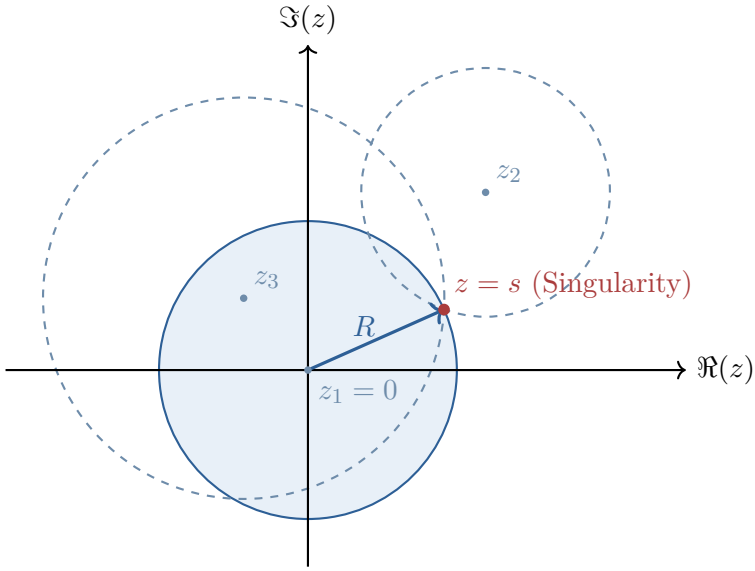


Figure 4.15: As we deviate from a point, the Taylor series expansion continues to hold until it meets the nearest singularity.

To determine R, there is a surprisingly simple method:

$$R = \text{distance from } z_0 \text{ to the nearest singularity.} \tag{4.309}$$

More precisely, the singularity here has to be a "real" singularity. Note that not all isolated singularities are equally bad. Some can be "repaired" by assigning the right value at the missing point. For example, 0 is a singularity of $\frac{\sin z}{z}$, but setting the value at 0 to be 1 makes the function holomorphic there. By contrast, 0 is also a singularity of $\frac{1}{z}$, but there is no way to repair it.

Mathematically, we classify the singularities into two types: *removable* and *non-removable*.

Definition 4.62 (Removable and non-removable singularities)**.** Let $f : S \to \mathbb{C}$ be a complex function with an isolated singularity at z_0. We say that z_0 is a *removable singularity* of f if there exists a complex number $w \in \mathbb{C}$ such that assigning the value w to $f(z_0)$ makes f holomorphic at z_0. We say that z_0 is a *non-removable singularity* of f if it is not removable.

For computing the radius of convergence, removable singularities are harmless, because after ignoring them, the Taylor series continues to agree with the rest of the function. However, a non-removable singularity is a genuine problem, as shown in the following theorem.

Theorem 4.63. Let $f : S \to \mathbb{C}$ be a holomorphic function on an open set $S \subseteq \mathbb{C}$. Fix $z_0 \in S$. If the following holds for some $R > 0$:

1. f is holomorphic on the open disk D_R centered at z_0 with radius R;
2. f has a non-removable singularity at some point s with $|s - z_0| = R$,

Then the radius of convergence of the Taylor series of f around z_0,

$$\sum_{n\geq 0} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n, \tag{4.310}$$

is exactly R.

Remark 4.64. In this course, non-removable singularities are defined only for isolated singularities, so the theorem is stated only for isolated singularities. In general, one can also define non-removable singularities for non-isolated singularities, and the theorem still holds.

Proof. Let ρ be the radius of convergence of the Taylor series of f at z_0. First, we show that $\rho \geq R$. For every $z \in D_R$, we can find a radius $r \in (|z - z_0|, R)$. Then the circle C_r and its interior D_r are both contained in S. By Theorem 4.59,

$$f(z) = \sum_{n\geq 0} a_n(z - z_0)^n, \quad \text{where} \quad a_n = \frac{1}{2\pi i}\int_{C_r} \frac{f(w)}{(w - z_0)^{n+1}}\, \mathrm{d}w. \tag{4.311}$$

Therefore, the Taylor series of f around z_0 converges to $f(z)$, and thus $\rho \geq R$.

Now we show that $\rho \leq R$. If $\rho > R$, then the Taylor series

$$g(z) := \sum_{n\geq 0} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n \tag{4.312}$$

defines a holomorphic function on the disk D_ρ. By the identity theorem, f and g must be the same function on $D_\rho \cap S$. In particular, on a punctured open disk $D_{r_0}(s) \setminus \{s\}$ for some $r_0 > 0$, we must have $f(z) = g(z)$. This implies that assigning $f(s) = g(s)$ makes f holomorphic at s. However, this contradicts the fact that s is a non-removable singularity of f. Therefore, $\rho \leq R$.

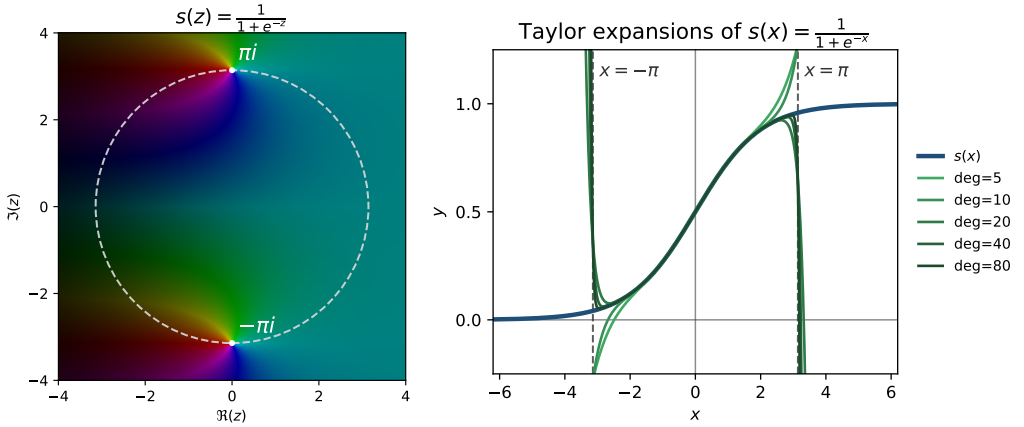
Combining the two parts, we conclude that $\rho = R$. □

An important application of this theorem is to study Taylor series of real functions via complex analysis. Suppose a real function is defined near a point $x_0 \in \mathbb{R}$, and suppose that the same formula also defines a holomorphic function f on an open subset of $\mathbb{C}$ containing x_0. If the nearest non-removable singularity of f is at distance R from x_0, then the Taylor series of f around x_0 converges in the disk of convergence

$$x \in D_R(x_0). \tag{4.313}$$

Restricting this disk of convergence back to the real axis, we obtain the interval of convergence for the original real function:

$$x \in (x_0 - R,\ x_0 + R). \tag{4.314}$$



(a) Domain coloring (b) Taylor expansions along the real axis

Figure 4.16: The sigmoid function $s(x) = \frac{1}{1+e^{-x}}$. In the complex plane, the nearest non-removable singularities of s to the origin are at $\pm\pi i$. Therefore, the radius of convergence of the Taylor series of s around 0 is π. In particular, along the real axis, no matter how high the degree of the Taylor polynomial is, it cannot approximate $s(x)$ once $|x| > \pi$.

To illustrate this point, let us go back to the third motivating question from the beginning of the chapter.

Example 4.65 (The sigmoid function)**.** Consider the sigmoid function

$$s(z) := \frac{1}{1+e^{-z}}, \tag{4.315}$$

where we extend its domain from $\mathbb{R}$ to $\mathbb{C}$. Its singularities are the points where the denominator vanishes. That is, $1 + e^{-z} = 0$. The roots of this equation are the points

$$z = (2k+1)\pi i, \qquad k \in \mathbb{Z}, \tag{4.316}$$

since $e^{-(2k+1)\pi i} = \cos((2k+1)\pi) + i\sin((2k+1)\pi) = -1$ for all $k \in \mathbb{Z}$.

So the singularities of $s(z)$ are located at odd multiples of πi. The nearest ones to the origin are $\pm\pi i$, both at distance π. See Figure 4.16. Therefore, the Taylor series of the sigmoid function around 0 has radius of convergence

$$R = \pi. \tag{4.317}$$

More generally, if we expand around a point $x_0 \in \mathbb{R}$, the radius of convergence is

$$R = \sqrt{\pi^2 + x_0^2}. \tag{4.318}$$

Without complex analysis, this answer would be quite surprising. But with complex analysis, the radius of convergence can be derived very intuitively from the nearest singularities.

4.2.7 Expansion Around Isolated Singularities

We have seen that near a holomorphic point z_0, a complex function f can be expanded into a Taylor series in powers of $(z - z_0)$. More ambitiously, *can we still expand a function near an isolated singularity?*

The answer is



▶ Laurent Series

The correct replacement of Taylor series near an isolated singularity is a *Laurent series*, which may contain not only non-negative powers, but also negative powers of $(z - z_0)$.

Theorem 4.66 (Laurent's theorem)**.** Let f be holomorphic on a punctured disk $D_r(z_0) \setminus \{z_0\}$ for some $z_0 \in \mathbb{C}$ and some $r > 0$. Then there exists a unique sequence $(a_n)_{n\geq 0}$ and a unique sequence $(b_n)_{n\geq 1}$ such that

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b_n (z - z_0)^{-n} \tag{4.319}$$

for every $z \in D_r(z_0) \setminus \{z_0\}$.

This theorem is a consequence of Cauchy's integral formula, which we leave as an exercise.

More conveniently, we can view $(a_n)_{n\geq 0}$ and $(b_n)_{n\geq 1}$ as a single sequence $(a_n)_{n\in\mathbb{Z}}$, where $a_{-n} = b_n$ for all $n \geq 1$. Then we can write $f(z)$ in the following form, known as the *Laurent series* of f around z_0:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n. \tag{4.320}$$

Here $\sum_{n=-\infty}^{\infty}$ is just a convenient notation for the sum of two series, one with non-negative powers and the other with negative powers.

The negative-power part is called the *principal part* of the Laurent series, which records how the function fails to be holomorphic at the singularity. Laurent's theorem essentially says that although isolated singularities are "bad" points, they are still highly structured. Near such a point, the function cannot fail in an arbitrary way: it is always given by an ordinary Taylor series together with some negative-power terms. In other words, the only obstruction to holomorphicity at an isolated singularity is the presence of negative powers of $(z - z_0)$:

$$(z - z_0)^{-1},\ (z - z_0)^{-2},\ (z - z_0)^{-3},\ \ldots. \tag{4.321}$$

In the following, we classify the singularities by the highest negative power in the Laurent series.

Definition 4.67. Let z_0 be an isolated singularity of f with Laurent series expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n \tag{4.322}$$

- If $a_j = 0$ for all $j < 0$, then z_0 is a *removable singularity* of f.
- If $a_{-m} \neq 0$ for some positive integer m and $a_j = 0$ for all $j < -m$, then we say that z_0 is a *pole of order* m of f. A pole of order 1 is called a *simple pole*.
- If $a_j \neq 0$ for infinitely many negative integers j, then we say that z_0 is an *essential singularity* of f.

Example 4.68. Consider

$$f(z) = \frac{1}{1+z^2} = \frac{1}{(z-i)(z+i)}. \tag{4.323}$$

Around $z = i$, we write

$$z + i = 2i + (z - i). \tag{4.324}$$

Then

$$f(z) = \frac{1}{(z-i)(2i+(z-i))} = \frac{1}{2i(z-i)} \cdot \frac{1}{1+\frac{z-i}{2i}} \tag{4.325}$$

$$= \frac{1}{2i(z-i)} \sum_{n\geq 0} \left(-\frac{z-i}{2i}\right)^n \tag{4.326}$$

$$= \sum_{n\geq 0} \frac{1}{2i} (i/2)^n (z-i)^{n-1}. \tag{4.327}$$

Therefore, the Laurent series of f around $z = i$ is

$$f(z) = \frac{1}{2i}(z-i)^{-1} + \frac{1}{4} + \frac{i}{8}(z-i) - \frac{1}{16}(z-i)^2 + \cdots \tag{4.328}$$

Therefore, the lowest power is $(z-i)^{-1}$, and thus $z = i$ is a simple pole of f. By symmetry, $z = -i$ is also a simple pole.

Example 4.69. The point $z = 0$ is an essential singularity of the function

$$f(z) = e^{1/z}. \tag{4.329}$$

This is because

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = \sum_{n=-\infty}^{\infty} \mathbb{1}_{[n\leq 0]} \frac{1}{(-n)!} z^n. \tag{4.330}$$

Note that you can apply your generating function skills to check if a singularity is a pole of order m or not.

Proposition 4.70. Let z_0 be an isolated singularity of f. Then z_0 is a pole of order at most m if and only if the following limit exists:

$$\lim_{z \to z_0} (z - z_0)^m f(z). \tag{4.331}$$

In this case, the limit equals a_{-m}. If this limit is non-zero, then z_0 is a pole of order exactly m.

Proof. If $f(z)$ is a pole of order at least m, then after multiplying the Laurent series by $(z - z_0)^m$, we get

$$(z - z_0)^m f(z) = a_{-m} + a_{-m+1}(z - z_0) + a_{-m+2}(z - z_0)^2 + \cdots, \tag{4.332}$$

which is a power series. Taking the limit as $z \to z_0$, we get

$$a_{-m} = \lim_{z \to z_0} (z - z_0)^m f(z). \tag{4.333}$$

If this limit is non-zero, then z_0 is a pole of order exactly m by definition.

In contrast, if $f(z)$ is not a pole of order at least m, then after multiplying the Laurent series by $(z - z_0)^m$, the Laurent series still has negative powers of $(z - z_0)$, which prevents the limit $\lim_{z \to z_0}(z - z_0)^m f(z)$ from existing. □

▶ Cauchy's Residue Theorem

We now introduce the most important theorem in this section: *Cauchy's residue theorem*. This theorem answers the following question: *how to compute the integral of a function around a closed contour that encloses a singularity?*

If there is no singularity inside the contour, then the integral is zero by Cauchy–Goursat theorem. If we artificially create a singularity by dividing a holomorphic function $f(z)$ by $z - z_0$ for some z_0 inside the contour, then the integral is $2\pi i\, f(z_0)$ by Cauchy's integral formula. More generally, Cauchy's residue theorem provides a systematic way to compute the integral of a function around a closed contour that encloses one or more singularities.

Before presenting Cauchy's residue theorem, we need to introduce one more concept: *residue*.

Definition 4.71 (Residue)**.** Let z_0 be an isolated singularity of f that admits a Laurent series expansion around z_0:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n \tag{4.334}$$

Then the coefficient a_{-1} of $(z - z_0)^{-1}$ is called the *residue* of f at z_0, which is denoted by $\mathrm{Res}(f, z_0)$.

Cauchy's residue theorem states that the integral of a function around a closed contour can be computed by a discrete sum of the residues at the singularities inside the contour.

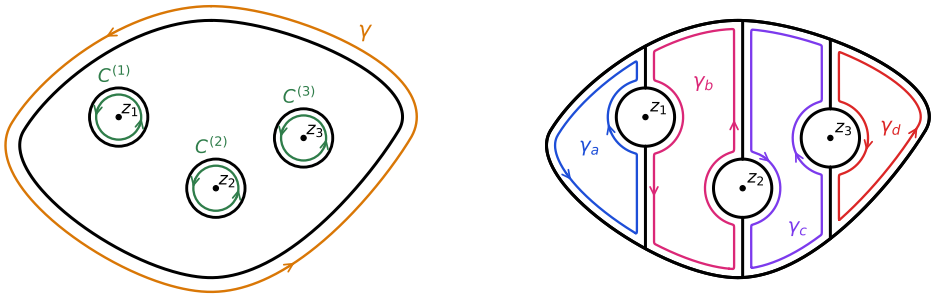


Figure 4.17: An illustration of the contour deformation in the proof of Cauchy's residue theorem. To show that the integral of f along γ equals the sum of the integrals along the small circles $C^{(1)}, \dots, C^{(m)}$, we cut the multiply punctured region along short segments joining the outer contour γ to the inner circles. This produces finitely many regions with boundaries $\gamma_1, \dots, \gamma_\ell$.

Theorem 4.72 (Cauchy's residue theorem)**.** Let f be a holomorphic function on an open set $S \subseteq \mathbb{C}$. Let γ be a simple closed counterclockwise contour such that the region enclosed by γ is contained in S except for m isolated singularities $z_1, \dots, z_m$. Then

$$\frac{1}{2\pi i} \int_\gamma f(z)\, \mathrm{d}z = \sum_{j=1}^{m} \operatorname{Res}(f, z_j). \tag{4.335}$$

Proof. Similar to the proof of Cauchy's integral formula, the proof proceeds in two steps. First, we deform the contour γ to the union of small circles around the singularities, and show that this does not change the integral value. See Figure 4.17 for an illustration. Then, we compute the integral along each small circle using the Laurent series near that singularity.

Deformation. Choose m disjoint small circles $C^{(1)}, \dots, C^{(m)}$ around the singularities $z_1, \dots, z_m$, all contained in the region enclosed by γ. Now draw short segments joining γ to these circles, as shown in Figure 4.17. These cuts split the region into ℓ pieces, each without a hole. Let $\gamma_1, \dots, \gamma_\ell$ be the boundaries of these pieces, oriented counterclockwise. Then by Cauchy–Goursat theorem,

$$\int_{\gamma_k} f(z)\, \mathrm{d}z = 0, \qquad \forall k \in [\ell]. \tag{4.336}$$

Adding these integrals together, we get

$$0 = \sum_{k=1}^{\ell} \int_{\gamma_k} f(z)\, \mathrm{d}z = \int_\gamma f(z)\, \mathrm{d}z - \sum_{j=1}^{m} \int_{C^{(j)}} f(z)\, \mathrm{d}z. \tag{4.337}$$

Here, each cut is traversed once in each direction, so the contributions from the cuts cancel. The outer boundary γ is traversed counterclockwise, whereas each inner circle $C^{(j)}$ is traversed clockwise. This is why the sum of the boundary

integrals is the integral around γ minus the sum of the integrals around the small circles. Therefore, we can deform the contour γ to the union of small circles around the singularities:

$$\int_\gamma f(z)\,\mathrm{d}z = \sum_{j=1}^{m} \int_{C^{(j)}} f(z)\,\mathrm{d}z. \tag{4.338}$$

Local Expansion. Now fix one singularity z_j. Inside the small circle $C^{(j)}$, the Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n^{(j)} (z - z_j)^n \tag{4.339}$$

converges to $f(z)$. So we may integrate term by term:

$$\int_{C^{(j)}} f(z)\,\mathrm{d}z = \sum_{n=-\infty}^{\infty} a_n^{(j)} \int_{C^{(j)}} (z - z_j)^n\,\mathrm{d}z. \tag{4.340}$$

As we have seen in Example 4.47, every term vanishes except when $n = -1$:

$$\int_{C^{(j)}} (z - z_j)^n\,\mathrm{d}z = \begin{cases} 0, & n \neq -1, \\ 2\pi i, & n = -1. \end{cases} \tag{4.341}$$

Therefore

$$\int_{C^{(j)}} f(z)\,\mathrm{d}z = 2\pi i\, a_{-1}^{(j)} = 2\pi i\, \operatorname{Res}(f, z_j). \tag{4.342}$$

Summing over $j = 1, \ldots, m$ gives the theorem. □

Example 4.73. Although Cauchy's residue theorem is built on Laurent and Taylor series, and hence ultimately on Cauchy's integral formula, let us now go in the reverse direction and derive Cauchy's integral formula from the residue theorem.

Consider the function $g(z) = \frac{f(z)}{z - z_0}$, where f is holomorphic. Let γ be a simple closed counterclockwise contour that encloses z_0. Then f is holomorphic on the region enclosed by γ, except for z_0. By Cauchy's residue theorem,

$$\int_\gamma g(z)\,\mathrm{d}z = 2\pi i\, \operatorname{Res}(g, z_0). \tag{4.343}$$

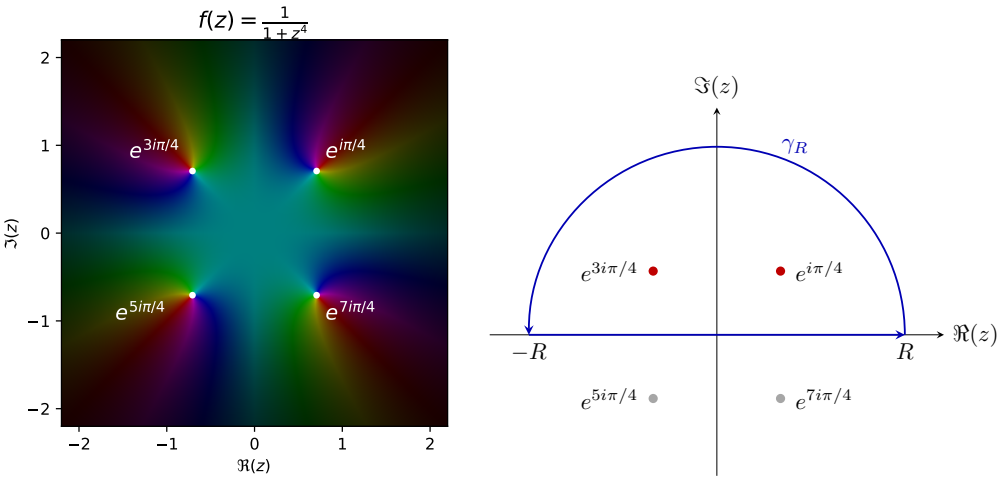
Note that as $z \to z_0$,

$$\lim_{z \to z_0} (z - z_0) g(z) = \lim_{z \to z_0} f(z) = f(z_0). \tag{4.344}$$

So z_0 is either a removable singularity or a simple pole of g. Therefore, the residue is given by

$$\operatorname{Res}(g, z_0) = f(z_0), \tag{4.345}$$

which implies $\int_\gamma \frac{f(z)}{z - z_0}\,\mathrm{d}z = 2\pi i\, f(z_0)$.



(a) Domain coloring of $f(z) = \frac{1}{1+z^4}$. (b) The contour γ_R in the upper half-plane.

Figure 4.18: The domain coloring plot of $f(z) = \frac{1}{1+z^4}$ and an illustration of the contour γ_R in the upper half-plane.

▶ Complex Analysis Helps Real Integration

Before turning to coefficient extraction, let us return to the second motivating question from the beginning of the section: how can we compute the following improper integral?

$$\int_{-\infty}^{\infty} \frac{1}{1+x^4}\, \mathrm{d}x. \tag{4.346}$$

The first step is to view the integral as the integral of a complex function

$$f(z) = \frac{1}{1+z^4} \tag{4.347}$$

along the real interval $[-R, R]$, where $R \to +\infty$.

To leverage the tools that we have developed, we take an integral along a simple closed contour that contains the real interval $[-R, R]$, and then subtract the part that lies outside the interval. More specifically, for each $R > 0$, let Γ_R be the upper semicircle of radius R, and let γ_R be the contour formed by concatenating the interval $[-R, R]$ with Γ_R, oriented counterclockwise. See Figure 4.18 for an illustration.

Then we have

$$\int_{\gamma_R} f(z)\, \mathrm{d}z = \int_{-R}^{R} \frac{1}{1+x^4}\, \mathrm{d}x + \int_{\Gamma_R} f(z)\, \mathrm{d}z. \tag{4.348}$$

By Cauchy's residue theorem,

$$\int_{\gamma_R} f(z)\, \mathrm{d}z = 2\pi i \sum_{z_0} \mathrm{Res}(f, z_0), \tag{4.349}$$

where z_0 runs over the isolated singularities of f inside γ_R.

Therefore, we can rewrite the original integral as

$$\int_{-R}^{R} \frac{1}{1+x^4}\,\mathrm{d}x = 2\pi i \sum_{z_0} \operatorname{Res}(f, z_0) - \int_{\Gamma_R} f(z)\,\mathrm{d}z. \tag{4.350}$$

So the whole problem reduces to two tasks:

1. compute the residues of f at the isolated singularities inside γ_R;
2. show that the integral along the semicircle $\int_{\Gamma_R} f(z)\,\mathrm{d}z$ tends to 0 as $R \to \infty$.

Computing Residues. First, we provide a proposition that summarizes how to compute residues at various types of singularities.

Proposition 4.74 (How to compute residues)**.** Let z_0 be an isolated singularity of f.

1. If $\lim_{z\to z_0}(z-z_0)f(z)$ exists, then
$$\operatorname{Res}(f, z_0) = \lim_{z\to z_0}(z-z_0)f(z). \tag{4.351}$$
Moreover, z_0 is a simple pole if $\operatorname{Res}(f, z_0) \neq 0$, or a removable singularity if $\operatorname{Res}(f, z_0) = 0$.
2. If $f(z) = \frac{g(z)}{h(z)}$ for some holomorphic functions g and h near z_0, with $h(z_0) = 0$ and $h'(z_0) \neq 0$, then
$$\operatorname{Res}(f, z_0) = \frac{g(z_0)}{h'(z_0)}, \tag{4.352}$$
Moreover, z_0 is a simple pole if $\operatorname{Res}(f, z_0) \neq 0$, or a removable singularity if $\operatorname{Res}(f, z_0) = 0$.
3. If z_0 is a pole of order at most m, then
$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z\to z_0} \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}\Big[(z-z_0)^m f(z)\Big]. \tag{4.353}$$

Proof. The first claim is straightforward due to Proposition 4.70. For the second claim, we can just use L'Hôpital's rule for the limit $\lim_{z\to z_0}(z-z_0)f(z)$:

$$\operatorname{Res}(f, z_0) = \lim_{z\to z_0}(z-z_0)f(z) = \lim_{z\to z_0} \frac{(z-z_0)g(z)}{h(z)} \tag{4.354}$$

$$= \lim_{z\to z_0} \frac{g(z) + (z-z_0)g'(z)}{h'(z_0)} \tag{4.355}$$

$$= \frac{g(z_0)}{h'(z_0)}. \tag{4.356}$$

For the last claim, we note that for a pole of order at most m, multiplying its Laurent series by $(z-z_0)^m$ gives a power series, so the coefficient of $(z-z_0)^{m-1}$ is the residue. □

Now we return to the original problem. The singularities of f are the zeros of the denominator $h(z) := 1 + z^4$. By writing complex numbers in exponential form, we can see that these are the following four points:

$$z_1 := e^{i\pi/4}, \quad z_2 := e^{3i\pi/4}, \quad z_3 := e^{5i\pi/4}, \quad z_4 := e^{7i\pi/4}. \tag{4.357}$$

Only z_1 and z_2 lie in the upper half-plane, so only those two lie inside γ_R.

Now for $k \in \{1, 2\}$, we compute the residue of f at z_k. Since the denominator $h(z)$ satisfies $h(z_k) = 0$ and $h'(z_k) = 4z_k^3 \neq 0$, by Proposition 4.74, z_k is a simple pole of f and

$$\operatorname{Res}(f, z_k) = \frac{1}{4z_k^3} = -\frac{z_k}{4}. \tag{4.358}$$

Therefore, the sum of the residues is

$$\sum_{k=1,2} \operatorname{Res}(f, z_k) = -\frac{z_1}{4} - \frac{z_2}{4} = -\frac{\sqrt{2}}{4}i. \tag{4.359}$$

Integral over the Semicircle. To compute the integral over the semicircle, we only need to show that it tends to 0 as $R \to \infty$. This is a simple consequence of the ML inequality. On Γ_R, we have $|z| = R$, so

$$|1 + z^4| \geq \big||z|^4 - 1\big| = R^4 - 1. \tag{4.360}$$

Therefore,

$$\left|\frac{1}{1 + z^4}\right| \leq \frac{1}{R^4 - 1}. \tag{4.361}$$

Since $L(\Gamma_R) = \pi R$, the ML inequality gives

$$\left|\int_{\Gamma_R} \frac{1}{1 + z^4}\, \mathrm{d}z\right| \leq \frac{\pi R}{R^4 - 1} = \mathcal{O}(R^{-3}) \to 0. \tag{4.362}$$

Conclusion. Putting everything together, we obtain

$$\int_{-\infty}^{\infty} \frac{1}{1 + x^4}\, \mathrm{d}x = 2\pi i \cdot \left(-\frac{\sqrt{2}}{4}i\right) = \frac{\sqrt{2}}{2}\pi. \tag{4.363}$$

Remark 4.75 (How to compute integrals)**.** In most cases, computing an integral of f along a simple closed contour γ follows the following steps:

(1) Find the singularities of f and decide which ones lie inside γ.

(2) Compute the residues at those singularities, for example by using Proposition 4.74.

(3) Add them up and multiply by $2\pi i$.

If the goal is a real integral, there is usually one extra step: adding some extra curves to constitute a simple closed contour. In the end, either show that these extra curves are negligible in the limit, or find a way to compute the integral along them explicitly.

4.2.8 Example: Bernoulli Numbers Revisited

We now return to the first motivating question from the beginning of the section: *how can we extract the Bernoulli numbers from their generating function?*

Recall that the Bernoulli numbers are defined by the generating function

$$\widehat{B}(z) := \frac{z}{e^z - 1} = \sum_{m \geq 0} B_m \frac{z^m}{m!}. \tag{4.364}$$

While this generating function is originally defined for real numbers, extending it to the complex plane makes it possible to leverage complex analysis to bound the Bernoulli numbers and even derive exact formulas for them.

▶ The Singularities of $\widehat{B}(z)$

We begin by understanding the singularities of $\widehat{B}(z) := \frac{z}{e^z-1}$. These points are the The denominator of this function is

$$h(z) := e^z - 1. \tag{4.365}$$

Its roots are exactly the points

$$z = 2\pi i k, \qquad k \in \mathbb{Z}, \tag{4.366}$$

since $e^{2\pi i k} = \cos(2\pi k) + i \sin(2\pi k) = 1$ for all $k \in \mathbb{Z}$.

These points are the singularities of $\widehat{B}(z)$. For every $k \in \mathbb{Z}$, the denominator $h(z)$ satisfies

$$h(2\pi i k) = 0 \qquad \text{and} \qquad h'(2\pi i k) = e^{2\pi i k} = 1 \neq 0, \tag{4.367}$$

so by Proposition 4.74,

$$\operatorname{Res}(\widehat{B}, 2\pi i k) = \frac{2\pi i k}{h'(2\pi i k)} = 2\pi i k. \tag{4.368}$$

Therefore, $\widehat{B}(z)$ has a removable singularity at 0, and simple poles at the points $2\pi i k$ for every $k \in \mathbb{Z} \setminus \{0\}$, with residue $2\pi i k$ at each pole.

▶ Bounding the Growth of Bernoulli Numbers

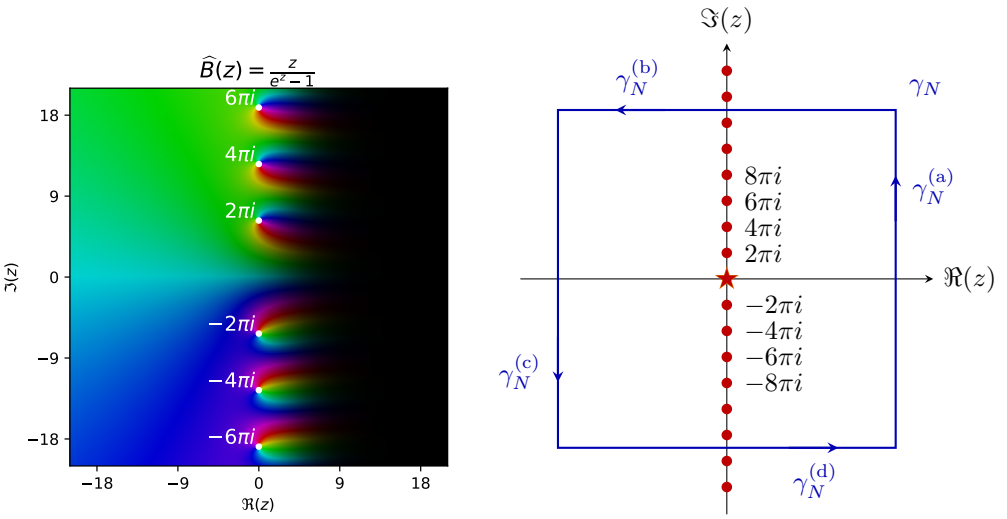
Having identified the singularities, we can now derive a tight bound on the growth of the Bernoulli numbers from the radius of convergence of the Taylor series.

As shown in Figure 4.19(a), the nearest poles to the origin are at $\pm 2\pi i$, both with distance 2π to the origin. Therefore, the Taylor series of $\widehat{B}(z)$ at 0 has radius of convergence

$$R = 2\pi. \tag{4.369}$$

On the other hand, the Taylor series of $\widehat{B}(z)$ at 0 is given by

$$\widehat{B}(z) = \sum_{m \geq 0} \frac{B_m}{m!} z^m, \tag{4.370}$$



(a) Domain coloring of $\widehat{B}(z) = \frac{z}{e^z-1}$. (b) The contour γ_N and the poles it encloses.

Figure 4.19: The domain coloring plot of $\widehat{B}(z) = \frac{z}{e^z-1}$ and the square contour γ_N used in the residue computation.

so by the Cauchy–Hadamard theorem (Theorem 4.18),

$$\sup\{r > 0 : |B_m| = \mathcal{O}(r^{-m})\} = 2\pi. \tag{4.371}$$

A direct consequence is that for every $r < 2\pi$,

$$|B_m| = \mathcal{O}\left(\frac{m!}{r^m}\right). \tag{4.372}$$

Since odd Bernoulli numbers are zero, we can write the bound only for even $m = 2d$:

$$|B_{2d}| = \mathcal{O}\left(\frac{(2d)!}{r^{2d}}\right) \qquad \text{for every } r < 2\pi. \tag{4.373}$$

Note that this does *not* mean $|B_m| = \mathcal{O}(\frac{m!}{(2\pi)^m})$ because the constant hidden in the big-$\mathcal{O}$ notation may depend on r. Still, this result is already very strong. Using complex analysis and some simple geometric information about the poles, we can easily derive an upper bound on the Bernoulli numbers. Without this perspective, it would be much more difficult to see why π should naturally appear in the bound.

▶ Exact Formula for Bernoulli Numbers

Now we turn to derive an exact formula for the Bernoulli numbers by extracting the coefficients of the power series expansion of $\widehat{B}(z)$ at 0.

First, we introduce the following lemma that reduces the problem of coefficient extraction to computing residues.

Proposition 4.76. If f is holomorphic at 0, then for every $n \geq 0$,

$$\frac{f^{(n)}(0)}{n!} = \operatorname{Res}\left(\frac{f(z)}{z^{n+1}}, 0\right). \tag{4.374}$$

Proof. Since f is holomorphic at 0, it has a Taylor series expansion at 0:

$$f(z) = \sum_{n\geq 0} \frac{f^{(n)}(0)}{n!} z^n. \tag{4.375}$$

Dividing both sides by z^{n+1}, we get

$$\frac{f(z)}{z^{n+1}} = \sum_{k\geq -n} \frac{f^{(k+n)}(0)}{(k+n)!} z^k. \tag{4.376}$$

Then the residue of $\frac{f(z)}{z^{n+1}}$ at 0 is the coefficient of z^{-1}, which is $\frac{f^{(n)}(0)}{n!}$. □

Now fix $m \geq 0$ and let $g_m(z) := \frac{\widehat{B}(z)}{z^{m+1}} = \frac{1}{z^m(e^z-1)}$. As the m-th Bernoulli number B_m is simply the m-th order derivative of $\widehat{B}(z)$ at 0, by Proposition 4.76, we have

$$B_m = \widehat{B}^{(m)}(0) = m! \cdot \operatorname{Res}(g_m, 0). \tag{4.377}$$

But it is still unclear how to compute the residue of g_m at 0, since its Laurent series expansion around 0 is not obvious. Of course, we can compute the first two terms explicitly:

$$g_m(z) = \frac{1}{z^m(e^z-1)} = \frac{1}{z^m(z + \frac{1}{2}z^2 + \mathcal{O}(z^3))} \tag{4.378}$$

$$= \frac{1}{z^{m+1}} \cdot \frac{1}{1 + (\frac{1}{2}z + \mathcal{O}(z^2))} \tag{4.379}$$

$$= \frac{1}{z^{m+1}} \left(1 - (\tfrac{1}{2}z + \mathcal{O}(z^2)) + \mathcal{O}((\tfrac{1}{2}z + \mathcal{O}(z^2))^2)\right) \tag{4.380}$$

$$= \frac{1}{z^{m+1}} \left(1 - \tfrac{1}{2}z + \mathcal{O}(z^2)\right). \tag{4.381}$$

Therefore,

$$g_m(z) = z^{-(m+1)} - \frac{1}{2}z^{-m} + \mathcal{O}(z^{-(m-1)}). \tag{4.382}$$

This implies that the first two Bernoulli numbers are:

$$B_0 = 0! \cdot \operatorname{Res}(g_0, 0) = 1, \tag{4.383}$$

$$B_1 = 1! \cdot \operatorname{Res}(g_1, 0) = -\frac{1}{2}. \tag{4.384}$$

However, this method cannot be easily extended to compute further Bernoulli numbers, since it is difficult to compute the entire Laurent series of g_m at 0.

To get around this difficulty, a key idea is to use Cauchy's residue theorem to prove that the sum of residues of g_m at all its singularities is zero.

Lemma 4.77. For every $m \geq 2$,

$$\lim_{N\to\infty} \sum_{k=-N}^{N} \operatorname{Res}(g_m, 2\pi i k) = 0. \tag{4.385}$$

Proof. Let γ_N be a square contour that encloses the square $(-L_N, L_N)^2$, where $L_N := (2N+1)\pi$. More specifically, $\gamma_N = \gamma_N^{(\mathrm{a})} + \gamma_N^{(\mathrm{b})} + \gamma_N^{(\mathrm{c})} + \gamma_N^{(\mathrm{d})}$, where

$$\gamma_N^{(\mathrm{a})} : [-L_N, L_N] \to \mathbb{C}, \quad t \mapsto L_N + it, \tag{4.386}$$

$$\gamma_N^{(\mathrm{b})} : [-L_N, L_N] \to \mathbb{C}, \quad t \mapsto -t + L_N i, \tag{4.387}$$

$$\gamma_N^{(\mathrm{c})} : [-L_N, L_N] \to \mathbb{C}, \quad t \mapsto -L_N - it, \tag{4.388}$$

$$\gamma_N^{(\mathrm{d})} : [-L_N, L_N] \to \mathbb{C}, \quad t \mapsto t - L_N i. \tag{4.389}$$

See Figure 4.19 for a visualization of γ_N.

Contour Integration. Since g_m is obtained by dividing $\widehat{B}(z)$ by z^{m+1}, every non-zero pole $2\pi i k$ of $\widehat{B}(z)$ remains a pole of g_m. At $z = 0$, the situation changes: $\widehat{B}(z)$ has a removable singularity there, but dividing by z^{m+1} turns that removable singularity into a pole of g_m. Therefore, the singularities of g_m are exactly the points

$$z = 2\pi i k, \qquad k \in \mathbb{Z}. \tag{4.390}$$

For each N, the singularities enclosed by γ_N are exactly the points

$$z = 2\pi i k, \qquad k \in \{-N, -N+1, \ldots, N-1, N\}. \tag{4.391}$$

By Cauchy's residue theorem,

$$\sum_{k=-N}^{N} \operatorname{Res}(g_m, 2\pi i k) = \frac{1}{2\pi i} \int_{\gamma_N} g_m(z)\, \mathrm{d}z. \tag{4.392}$$

So it remains to show that as $N \to \infty$,

$$\int_{\gamma_N} g_m(z)\, \mathrm{d}z \to 0. \tag{4.393}$$

Bounding the Integral. To show that the integral tends to 0, we only need to prove that for some constant $C > 0$, it holds for all z on the contour γ_N that

$$|e^z - 1| \geq C. \tag{4.394}$$

Once this is shown, we can upper bound $g_m(z)$ by

$$|g_m(z)| \leq \frac{1}{|z|^m \cdot C} \leq \mathcal{O}(\tfrac{1}{L_N^m}) = \mathcal{O}(N^{-m}), \tag{4.395}$$

and then use the ML inequality to conclude that

$$\int_{\gamma_N} g_m(z)\, \mathrm{d}z \leq \mathcal{O}(N^{-m}) \cdot L(\gamma_N) = \mathcal{O}(N^{-m}) \cdot \mathcal{O}(N) = \mathcal{O}(N^{-m+1}), \tag{4.396}$$

which proves that the integral tends to 0.

Now we turn to bounding $|e^z - 1|$. For the two vertical sides, the real part of z is given by $\Re(z) = \pm L_N = \pm(2N+1)\pi$, so we have

$$|e^z - 1| \geq \big||e^z| - 1\big| \geq \min\{1 - e^{-L_N}, e^{L_N} - 1\}, \tag{4.397}$$

which is at least $1/2$ for all $N \geq 1$.

For the two horizontal sides, the imaginary part of z is given by $\Im(z) = \pm L_N = \pm(2N+1)\pi$, so by Euler's formula,

$$e^z = e^{\Re(z)}(\cos(\Im(z)) + i\sin(\Im(z))) = -e^{\Re(z)} \in \{x \in \mathbb{R} : x < 0\}. \tag{4.398}$$

Then we can bound $|e^z - 1|$ by

$$|e^z - 1| = e^{\Re(z)} + 1 \geq 1. \tag{4.399}$$

Therefore, in both cases, we have $|e^z - 1| \geq \frac{1}{2}$, and thus $\sum_{k=-N}^{N} \operatorname{Res}(g_m, 2\pi ik) = \int_{\gamma_N} g_m(z)\, \mathrm{d}z \to 0$ as $N \to \infty$. □

A direct consequence of Lemma 4.77 and the fact that $B_m = m! \cdot \operatorname{Res}(g_m, 0)$ is that the Bernoulli numbers can be computed as follows:

$$B_m = -(m!) \lim_{N\to\infty} \sum_{k=-N}^{N} \mathbb{1}_{[k\neq 0]} \operatorname{Res}(g_m, 2\pi ik). \tag{4.400}$$

Fortunately, we can use Proposition 4.74 to deduce that for every $k \neq 0$, $2\pi ik$ is a simple pole of g_m and has residue

$$\operatorname{Res}(g_m, 2\pi ik) = \lim_{z\to 2\pi ik} \frac{z - 2\pi ik}{z^m(e^z - 1)} = \frac{1}{(2\pi ik)^m} \lim_{z\to 2\pi ik} \frac{z - 2\pi ik}{e^z - 1} \tag{4.401}$$

$$= \frac{1}{(2\pi ik)^m} \lim_{z\to 2\pi ik} \frac{1}{e^z} \tag{4.402}$$

$$= \frac{1}{(2\pi ik)^m}, \tag{4.403}$$

where the second line uses the L'Hôpital's rule to compute the limit. Substituting this into Equation (4.400), we get

$$B_m = -(m!) \lim_{N\to\infty} \sum_{k=-N}^{N} \mathbb{1}_{[k\neq 0]} \frac{1}{(2\pi ik)^m} \tag{4.404}$$

$$= -\frac{m!}{(2\pi i)^m} \sum_{k\geq 1} \left(\frac{1}{k^m} + \frac{1}{(-k)^m} \right). \tag{4.405}$$

If m is odd, then the terms for k and $-k$ cancel, so for every integer $d \geq 1$,

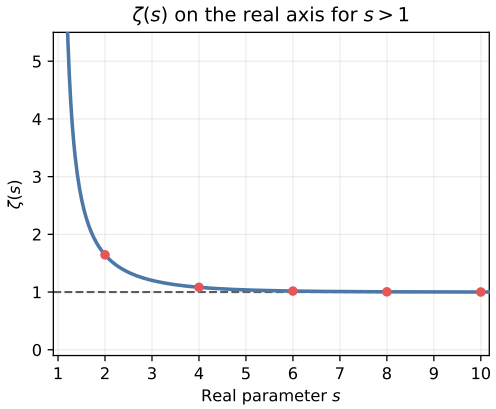
$$B_{2d+1} = 0, \tag{4.406}$$

which is consistent with Lemma 4.10.

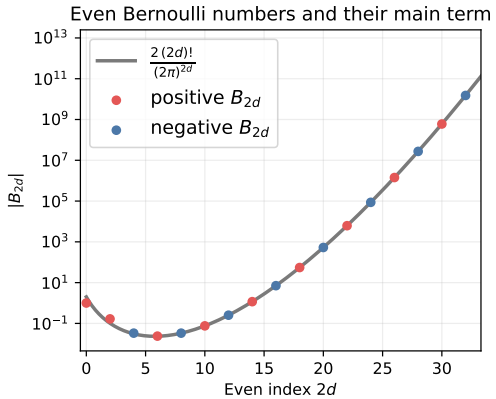
If m is even, then for every integer $d \geq 1$,

$$B_{2d} = -\frac{(2d)!}{(2\pi i)^{2d}} \sum_{k\geq 1} \frac{2}{k^{2d}} = (-1)^{d+1} \frac{2(2d)!}{(2\pi)^{2d}} \sum_{k\geq 1} \frac{1}{k^{2d}}. \tag{4.407}$$

The last factor $\sum_{k\geq 1} \frac{1}{k^{2d}}$ occurs repeatedly in many areas of mathematics, and is known as the *Riemann zeta function*. Its formal definition is as follows.



(a) $\zeta(s)$ on $s > 1$.



(b) Even Bernoulli numbers.

Figure 4.20: **(a)** The graph of $\zeta(s)$ for real $s > 1$. **(b)** The even Bernoulli numbers $|B_{2d}|$ compared with the approximation $\frac{2(2d)!}{(2\pi)^{2d}}$. Here $(2d)!$ is interpreted as $\Gamma(2d+1)$ when $2d$ is not an integer.

Definition 4.78 (Riemann zeta function)**.** For every real number $s > 1$, the *Riemann zeta function* is defined as

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}. \tag{4.408}$$

With this definition, we can now rewrite even Bernoulli numbers in a more concise form. For every integer $d \geq 1$,

$$B_{2d} = (-1)^{d+1} \frac{2(2d)!}{(2\pi)^{2d}} \zeta(2d). \tag{4.409}$$

Example 4.79 (Computing the first few Bernoulli numbers)**.** For $d = 1$,

$$B_2 = \frac{2 \cdot 2!}{(2\pi)^2} \zeta(2) = \frac{4}{4\pi^2} \cdot \frac{\pi^2}{6} = \frac{1}{6}. \tag{4.410}$$

For $d = 2$,

$$B_4 = -\frac{2 \cdot 4!}{(2\pi)^4} \zeta(4) = -\frac{48}{16\pi^4} \cdot \frac{\pi^4}{90} = -\frac{1}{30}. \tag{4.411}$$

Although $\zeta(s)$ does not admit a closed-form expression for general s, when $s \to +\infty$, it converges to 1 very quickly:

$$\zeta(s) = 1 + \frac{1}{2^s} + \sum_{n \geq 3} \frac{1}{n^s} = 1 + \frac{1}{2^s} + \frac{1}{2^s} \sum_{n \geq 3} \left(\frac{2}{n}\right)^s = 1 + \mathcal{O}(2^{-s}). \tag{4.412}$$

Therefore, for large d, the Bernoulli number B_{2d} is approximately

$$B_{2d} = (-1)^{d+1} \frac{2(2d)!}{(2\pi)^{2d}} \left(1 + \mathcal{O}(2^{-2d})\right). \tag{4.413}$$

This is consistent with the upper bound in Equation (4.373), which already showed that $B_{2d} = \mathcal{O}(\frac{(2d)!}{r^{2d}})$ for every $r < 2\pi$. But this formula is much sharper, as it identifies the sign, the leading constant 2, and even specifies a relative error bound $\mathcal{O}(2^{-2d})$.

Remark 4.80 (How to extract coefficients from a generating function)**.** The Bernoulli example shows that coefficient extraction can often be handled by contour integration. Let f be holomorphic at 0, and suppose that all singularities of f away from 0 are isolated. Write $f(z) = \sum_{n\geq 0} a_n z^n$ for the Taylor series of f around 0. To recover a_n, a common strategy is as follows:

(1) Define $g_n(z) := \frac{f(z)}{z^{n+1}}$. By Proposition 4.76, $a_n = \operatorname{Res}(g_n, 0)$. This reduces the problem to computing the residue of g_n at 0.

(2) Choose a large contour γ_R that encloses 0, where R is a parameter that controls its size (e.g., the radius of a circle, or the side length of a square). If as $R \to \infty$,

$$|g_n(z)| = o(R^{-1}) \quad \text{and} \quad L(\gamma_R) = \mathcal{O}(R), \tag{4.414}$$

then by the ML inequality, the contour integral along γ_R tends to 0:

$$\int_{\gamma_R} g_n(z)\, \mathrm{d}z \to 0. \tag{4.415}$$

Note that such a contour is likely to exist because $g_n(z)$ is obtained by dividing $f(z)$ by z^{n+1}, which makes $g_n(z)$ small when z is far away from 0.

(3) By Cauchy's residue theorem, this implies

$$\lim_{R\to\infty} \sum_{z_0 \in S_R} \operatorname{Res}(g_n, z_0) = 0, \tag{4.416}$$

where S_R denotes the set of singularitites of g_n enclosed by γ_R.

(4) Identify the singularities in $S_R \backslash \{0\}$ and compute their residues, for example by using Proposition 4.74. Then

$$\operatorname{Res}(g_n, 0) = -\lim_{R\to\infty} \sum_{z_0 \in S_R \backslash \{0\}} \operatorname{Res}(g_n, z_0). \tag{4.417}$$

The reason that this may be easier than computing the residue at 0 directly is that 0 is a pole of g_n of order at least $n+1$, but the other singularities are usually of lower order.

▸ Faulhaber's Formula Revisited

If you are interested in what Faulhaber's formula looks like when the Bernoulli numbers are replaced by their exact formula in terms of the Riemann zeta function, here it is. Recall that in Equation (4.147), we derived the following form of Faulhaber's formula: for $S_k(n) = 1^k + 2^k + \cdots + n^k$, we have

$$S_k(n) = \frac{1}{k+1} n^{k+1} + \frac{1}{2} n^k + \frac{1}{k+1} \sum_{d=1}^{\lfloor k/2 \rfloor} \binom{k+1}{2d} B_{2d}\, n^{k+1-2d} \tag{4.418}$$

Substituting the expression for B_{2d} into this formula, we get

$$S_k(n) = \frac{1}{k+1}n^{k+1} + \frac{1}{2}n^k \tag{4.419}$$

$$+ \frac{1}{k+1}\sum_{d=1}^{\lfloor k/2 \rfloor}\binom{k+1}{2d}(-1)^{d+1}\frac{2(2d)!}{(2\pi)^{2d}}\zeta(2d)\,n^{k+1-2d} \tag{4.420}$$

Or equivalently, we can expand the binomial coefficient $\binom{k+1}{2d}$ and obtain

$$S_k(n) = \frac{1}{k+1}n^{k+1} + \frac{1}{2}n^k \tag{4.421}$$

$$+ 2\,k!\sum_{d=1}^{\lfloor k/2 \rfloor}\frac{(-1)^{d+1}}{(k+1-2d)!\,(2\pi)^{2d}}\zeta(2d)\,n^{k+1-2d}. \tag{4.422}$$

It is not clear whether this formula is more useful than the original one, or whether it can be derived without using the fancy machinery of generating functions and complex analysis that we have developed in this chapter, but we hope you find it interesting.

4.3 Generating Functions of Random Variables

In Chapter 3, we already met the moment generating function and the rate function as tools for Chernoff bounds. We now look at the same objects from a different angle. The question here is no longer how to bound a tail probability, but how to package the information of a random variable into a generating function that we can manipulate.

Conversely, we will also study how to extract information from a generating function to understand the random variable. This requires the tools from complex analysis that we have just learned, and it will help us understand the famous *Central Limit Theorem* in Chapter 6.

4.3.1 Probability Generating Functions

Let us forget about moment generating functions for a moment. Let X be a random variable taking values in non-negative integers, i.e., $X \in \{0, 1, 2, \dots\}$. *What is the most natural way to define a generating function for X?*

Obviously, we can view the probabilities $\Pr[X = 0], \Pr[X = 1], \Pr[X = 2], \dots$ as a sequence, and define the generating function as the ordinary generating function of this sequence. This is called the *probability generating function* (PGF).

Definition 4.81 (Probability generating function)**.** Let X be a random variable taking values in non-negative integers. Its *probability generating function* (PGF) is defined as

$$G_X(z) := \sum_{k\geq 0}\Pr[X = k]\,z^k. \tag{4.423}$$

In other words, the PGF is just the ordinary generating function of the probability mass function $p_X(k) = \Pr[X = k]$ written as a sequence. Note that this is not an arbitrary sequence, but a sequence of probabilities. When summing over all probabilities, we get 1. So the PGF satisfies

$$G_X(1) = \sum_{k \geq 0} \Pr[X = k] = 1. \tag{4.424}$$

Moreover, the first-order and second-order derivatives of the PGF at $z = 1$ reveal information about the expectation and variance of X.

Theorem 4.82 (Expectation and variance from the PGF)**.** Let X be a random variable taking values in non-negative integers. Then

$$\mathbb{E}[X] = G'_X(1), \tag{4.425}$$

$$\mathbb{E}[X^2] = G''_X(1) + G'_X(1), \tag{4.426}$$

$$\mathrm{Var}(X) = G''_X(1) + G'_X(1) - \left(G'_X(1)\right)^2. \tag{4.427}$$

Proof. Differentiating the series term by term gives

$$G'_X(z) = \sum_{n \geq 1} n \Pr[X = n] z^{n-1}, \tag{4.428}$$

so evaluating at $s = 1$ gives

$$G'_X(1) = \sum_{n \geq 1} n \Pr[X = n] = \mathbb{E}[X], \tag{4.429}$$

which is exactly Equation (4.425). Differentiating $G'_X(z)$ once more gives

$$G''_X(z) = \sum_{n \geq 2} n(n-1) \Pr[X = n] z^{n-2}, \tag{4.430}$$

so evaluating at $z = 1$ gives

$$G''_X(1) = \sum_{n \geq 2} n(n-1) \Pr[X = n] = \mathbb{E}[X(X-1)]. \tag{4.431}$$

Since $X^2 = X(X-1) + X$, we obtain

$$\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = G''_X(1) + G'_X(1), \tag{4.432}$$

which is exactly Equation (4.426). Finally, to prove Equation (4.427), we just need to note that $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$ by Proposition 3.11. □

Example 4.83 (Bernoulli PGF)**.** Let $X \sim \mathrm{Bernoulli}(p)$. Then

$$G_X(z) = \Pr[X = 0] + \Pr[X = 1] z = 1 - p + pz. \tag{4.433}$$

One can check that $G'_X(1) = p$ and $G''_X(1) = 0$, so

$$\mathbb{E}[X] = G'_X(1) = p, \tag{4.434}$$

$$\mathrm{Var}(X) = G''_X(1) + G'_X(1) - \left(G'_X(1)\right)^2 = 0 + p - p^2 = p(1-p). \tag{4.435}$$

This matches the mean and variance obtained in Example 3.12.

A more interesting property of the PGF is that when we add two independent random variables together, the PGF of the sum is the product of the PGFs of the two random variables. This is because adding two random variables is equivalent to performing a convolution of the probability mass functions.

Theorem 4.84. Let X and Y be independent random variables taking values in non-negative integers. Then

$$G_{X+Y}(z) = G_X(z)G_Y(z). \tag{4.436}$$

Proof. For every $k \geq 0$,

$$\Pr[X + Y = k] = \sum_{j=0}^{k} \Pr[X = j] \Pr[Y = k - j], \tag{4.437}$$

which is the convolution of $(\Pr[X = j])_{j\geq 0}$ and $(\Pr[Y = j])_{j\geq 0}$. As we have seen in Section 4.1.3, this means that the OGF of $(\Pr[X + Y = k])_{k\geq 0}$ is the product of the OGFs of $(\Pr[X = j])_{j\geq 0}$ and $(\Pr[Y = j])_{j\geq 0}$. Therefore,

$$G_{X+Y}(z) = G_X(z)G_Y(z), \tag{4.438}$$

which proves the claim. □

By induction, we can prove a similar formula for the PGF of the sum of multiple independent random variables $X_1, \ldots, X_n$:

$$G_{X_1+\cdots+X_n}(z) = \prod_{k=1}^{n} G_{X_k}(z). \tag{4.439}$$

While PGFs enjoy so many nice properties, a key limitation is that they only work for random variables that take values in non-negative integers. In general, random variables can take any values, and then it is not clear how to view their probabilities as a sequence.

4.3.2 Moment Generating Functions

The *moment generating function* (MGF) is more general than the PGF because it works for random variables that take any values. We have already defined the MGF in Definition 3.21, but let us repeat it here for convenience.

Definition 4.85 (Moment generating function)**.** The *moment generating function* (MGF) of a random variable X is defined as

$$M_X(t) := \mathbb{E}[e^{tX}]. \tag{4.440}$$

By Taylor expansion of the exponential function, we have

$$e^{tX} = \sum_{k\geq 0} \frac{(tX)^k}{k!} = \sum_{k\geq 0} X^k \frac{t^k}{k!}. \tag{4.441}$$

Taking expectations term by term, we can see that the MGF is the EGF of the raw moments $(\mathbb{E}[X^k])_{k\geq 0}$:

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_{k\geq 0} \frac{t^k}{k!}\mathbb{E}[X^k]. \tag{4.442}$$

To relate the MGF to the PGF, suppose X takes values in the non-negative integers, then the PGF can be written as

$$G_X(z) = \sum_{k\geq 0} \Pr[X = k]\, z^k = \mathbb{E}[z^X]. \tag{4.443}$$

Now taking $z = e^t$, we obtain

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}[z^X] = G_X(e^t). \tag{4.444}$$

Therefore, the key difference is how the index k is encoded: in the PGF, $\Pr[X = k]$ is attached to the monomial z^k, while in the MGF it is weighted by an exponential function, e^{tk}.

We have already listed some basic properties of the MGF in Proposition 3.22. Examples 3.23 to 3.26 further provide some concrete examples of MGFs.

Similar to the PGF, a key property of the MGF is that the MGF of the sum of independent random variables is the product of the MGFs of the individual random variables. We have used this property repeatedly in Chapter 3, for example in Example 3.24 and Lemma 3.35. Below we state this property formally.

Theorem 4.86. Let $X_1, \ldots, X_n$ be independent random variables, and let X be a weighted sum of these random variables:

$$X := \sum_{k=1}^{n} c_k X_k, \tag{4.445}$$

where $c_1, \ldots, c_n \in \mathbb{R}$ are constants. Then

$$M_X(t) = \prod_{k=1}^{n} M_{X_k}(c_k t). \tag{4.446}$$

Proof. By the definition of the MGF, we have

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\exp\left(\sum_{k=1}^{n} tc_k X_k\right)\right] \tag{4.447}$$

Since $X_1, \ldots, X_n$ are independent, the expectation of the exponential factorizes to the product of the expectations of the individual exponential terms:

$$\mathbb{E}\left[\exp\left(\sum_{k=1}^{n} tc_k X_k\right)\right] = \mathbb{E}\left[\prod_{k=1}^{n} \exp(tc_k X_k)\right] = \prod_{k=1}^{n} \mathbb{E}\big[\exp(tc_k X_k)\big]. \tag{4.448}$$

Again by the definition of the MGF, we have $M_{X_k}(c_k t) = \mathbb{E}[\exp(t c_k X_k)]$. Therefore, we obtain

$$M_X(t) = \prod_{k=1}^{n} M_{X_k}(c_k t), \tag{4.449}$$

which proves the claim. □

Note that we used two different strategies to analyze the PGF and MGF of the sum of independent random variables (Theorem 4.84 and Theorem 4.86). For the PGF case, we express the probability mass function of $X + Y$ as a convolution of the probability mass functions of X and Y, then associate the convolution with the product of the PGFs. We can also use the same proof strategy for the MGF case. If X and Y are independent, then by the binomial theorem, the raw moments of $X + Y$ is the binomial convolution of the raw moments of X and Y:

$$\mathbb{E}[(X+Y)^k] = \mathbb{E}\left[\sum_{j=0}^{k} \binom{k}{j} X^j Y^{k-j}\right] = \sum_{j=0}^{k} \binom{k}{j} \mathbb{E}[X^j]\mathbb{E}[Y^{k-j}]. \tag{4.450}$$

This is exactly the coefficient of t^k in $M_X(t)M_Y(t)$.

Conversely, in the MGF case, we factorize the expectation of the product of independent random variables into the product of the expectations of the individual random variables. The same proof strategy can be applied to the PGF case. If X and Y are independent, then

$$G_{X+Y}(z) = \mathbb{E}[z^{X+Y}] = \mathbb{E}[z^X z^Y] = \mathbb{E}[z^X]\mathbb{E}[z^Y] = G_X(z)G_Y(z), \tag{4.451}$$

which proves Theorem 4.84 in one line.

4.3.3 Recovering Distributions from MGFs

While the moment generating function is called a *generating function*, it is not clear whether it stores enough information about a random variable to recover the distribution. Of course, if the random variable takes values in the non-negative integers, then from MGF we can recover the PGF, and then the probability distribution of X. However, *what if the random variable can take any real values?*

More precisely, we are interested in the following question: if X is a random variable on a finite discrete probability space, can we recover its probability distribution from its MGF $M_X(t) = \mathbb{E}[e^{tX}]$?

The answer is



The first step is to write the MGF as a sum of exponential terms:

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_x p_X(x) e^{tx}, \tag{4.452}$$

where the sum is over all possible values of X. As usual, $p_X(x) = \Pr[X = x]$.

From this form, it is natural to expect that $p_X(x)$ can be recovered from $M_X(t)$, since the function is expressed as a linear combination of exponential terms with distinct exponents, and it is hard to believe that such a linear combination can be rewritten as a linear combination of a different set of exponential terms. For example, if we have $M_X(t) = \frac{1}{3}e^{-t/2} + \frac{2}{3}e^{t}$, then we may have a feeling that it is hard to rewrite $M_X(t)$ as $\sum_k a_k e^{tx_k}$ for any set $\{(a_k, x_k)\}_{k\geq 0} \neq \{(\frac{1}{3}, -\frac{1}{2}), (\frac{2}{3}, 1)\}$.

In fact, we had the same feeling when dealing with power series: if two power series are equal in function values, then their coefficients must be equal. One way to see this is to take derivatives and evaluate at $t = 0$. Another way is to use the Cauchy integral formula $\int_\gamma \frac{f(z)}{z^{n+1}}\,\mathrm{d}z$ to isolate the coefficient of z^n (Theorem 4.60). In both cases, the key idea is the same: we apply an operation that depends only on the function values, but its effect on a monomial $a_n z^n$ is to keep its coefficient unchanged for a specific n and discard the others.

We face a similar situation for MGFs, but now the function is a sum of exponential terms. To extract the coefficients, we need to design an operation that depends only on the function values but can isolate a single exponential term. The following lemma provides the key technical tool for designing such an operation.

Lemma 4.87. Let $c > 0$. For every $\delta \in \mathbb{R}$, we have

$$\lim_{T\to+\infty} \frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{e^{\delta z}}{z}\,\mathrm{d}z = \begin{cases} 0 & \text{if } \delta < 0, \\ 1/2 & \text{if } \delta = 0, \\ 1 & \text{if } \delta > 0, \end{cases} \tag{4.453}$$

where $\int_{c-iT}^{c+iT}$ means integrating along the line segment from $c - iT$ to $c + iT$, i.e., $\gamma_{c,T}(t) = c + it$ for $t \in [-T, T]$.

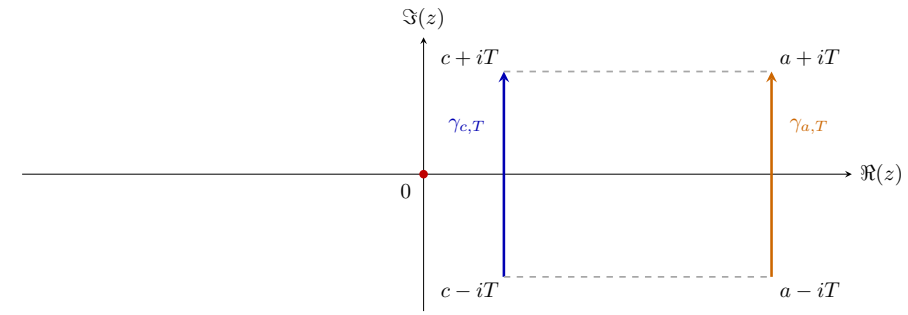
Proof Sketch. Here we provide a proof sketch and leave the details as an exercise (see Exercise 4.16). Let $f(z) = \frac{e^{\delta z}}{z}$. Then $f(z)$ is holomorphic on $\mathbb{C} \setminus \{0\}$, and 0 is a simple pole with residue 1.

For all $a \in \mathbb{R} \setminus \{0\}$, we claim that we can move the curve $\gamma_{c,T}$ to $\gamma_{a,T}$ with a minor change in the integral value. More specifically, as $T \to +\infty$,

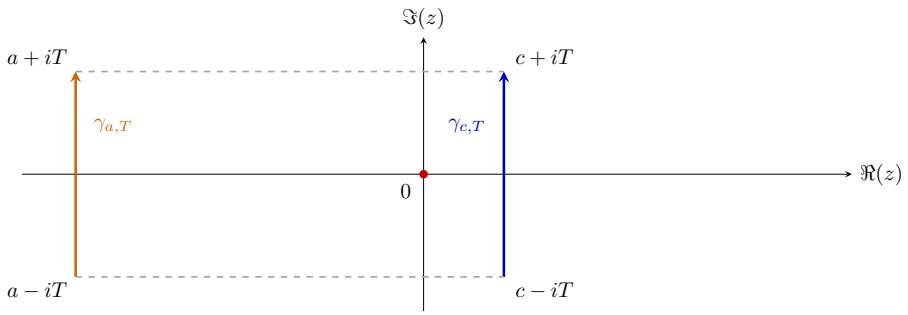
$$\frac{1}{2\pi i}\underbrace{\int_{\gamma_{c,T}} \frac{e^{\delta z}}{z}\,\mathrm{d}z}_{=:I_c(T)} = \mathbb{1}_{[a<0]} + \frac{1}{2\pi i}\underbrace{\int_{\gamma_{a,T}} \frac{e^{\delta z}}{z}\,\mathrm{d}z}_{=:I_a(T)} + \mathcal{O}\Big(\tfrac{1}{T}|(e^{\delta a} - e^{\delta c})/\delta|\Big). \tag{4.454}$$

This can be proved by showing the following two facts:

- By Cauchy's residue theorem, the contour integral along the rectangle formed by connecting $c-iT$, $c+iT$, $a+iT$ and $a-iT$ is $2\pi i \operatorname{Res}(f, 0) = 2\pi i$ if it encloses 0, and 0 otherwise.
- The integral along the upper and lower sides of the rectangle can be upper bounded by $\mathcal{O}\Big(\frac{1}{T}|(e^{\delta a} - e^{\delta c})/\delta|\Big)$.



(a) $\delta < 0$.



(b) $\delta > 0$.

Figure 4.21: **(a)** When $\delta < 0$, we move $\gamma_{c,T}$ to a line segment $\gamma_{a,T}$ on the right. The rectangle does not enclose the pole at 0. **(b)** When $\delta > 0$, we move $\gamma_{c,T}$ to a line segment $\gamma_{a,T}$ on the left. The rectangle encloses the pole at 0.

Now we discuss three cases:

- If $\delta < 0$, we let $a \to +\infty$ and obtain $I_a(T) \to 0$, which implies $I_c(T) \to 0$.
- If $\delta = 0$, we directly integrate $f(z)$ over $\gamma_{c,T}$ and obtain $I_c(T) \to \pi i$.
- If $\delta > 0$, we let $a \to -\infty$ and obtain $I_a(T) \to 0$, which implies $I_c(T) \to 2\pi i$.

Dividing these results by $2\pi i$ proves the lemma. □

To see why Lemma 4.87 is useful, let $x_1 < x_2$ be two real numbers. Construct the following integral:

$$I_{12}(x) := \frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-x_1 z} - e^{-x_2 z}}{z} e^{xz} \, \mathrm{d}z. \tag{4.455}$$

This integral can be split into two integrals:

$$I_{12}(x) = \underbrace{\frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{(x-x_1)z}}{z} \, \mathrm{d}z}_{=: I_1(x)} - \underbrace{\frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{(x-x_2)z}}{z} \, \mathrm{d}z}_{=: I_2(x)}. \tag{4.456}$$

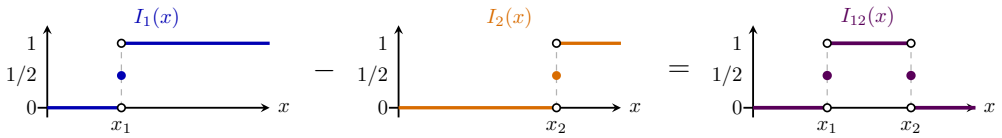


Figure 4.22: A visualization of $I_{12}(x) = I_1(x) - I_2(x)$.

By Lemma 4.87,

$$I_1(x) = \begin{cases} 0 & \text{if } x < x_1, \\ 1/2 & \text{if } x = x_1, \\ 1 & \text{if } x > x_1, \end{cases} \qquad I_2(x) = \begin{cases} 0 & \text{if } x < x_2, \\ 1/2 & \text{if } x = x_2, \\ 1 & \text{if } x > x_2. \end{cases} \tag{4.457}$$

Putting these together, we obtain

$$I_{12}(x) = \begin{cases} 0 & \text{if } x < x_1, \\ 1/2 & \text{if } x = x_1, \\ 1 & \text{if } x_1 < x < x_2, \\ 1/2 & \text{if } x = x_2, \\ 0 & \text{if } x > x_2. \end{cases} \tag{4.458}$$

Since the MGF is a weighted sum of e^{xz} over all possible values x of X, we can apply this integral term by term to filter out the probability mass of X between x_1 and x_2. This immediately implies the following inversion formula that recovers the probability distribution of X from the MGF.

Theorem 4.88 (Inversion formula for the MGF)**.** Let X be a random variable with MGF $M_X(t)$. For every $x_1 < x_2$, define

$$p_X(x_1, x_2) := \frac{1}{2}\Pr[X = x_1] + \Pr[x_1 < X < x_2] + \frac{1}{2}\Pr[X = x_2]. \tag{4.459}$$

Then for every $c > 0$,

$$p_X(x_1, x_2) = \frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-x_1 z} - e^{-x_2 z}}{z} M_X(z)\, \mathrm{d}z. \tag{4.460}$$

Proof. The proof follows from the above discussion. Let

$$Q(x_1, x_2) := \frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-x_1 z} - e^{-x_2 z}}{z} M_X(z)\, \mathrm{d}z. \tag{4.461}$$

By the definition of the MGF,

$$Q(x_1, x_2) = \frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-x_1 z} - e^{-x_2 z}}{z} \sum_x p_X(x) e^{xz}\, \mathrm{d}z \tag{4.462}$$

$$= \sum_x p_X(x) \underbrace{\frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-x_1 z} - e^{-x_2 z}}{z} e^{xz}\, \mathrm{d}z}_{=: I_{12}(x)}. \tag{4.463}$$

By Equation (4.458), only the terms with $x_1 \le x \le x_2$ contribute to the sum. More specifically, we have

$$Q(x_1, x_2) = \sum_x \Pr[X = x] \left(\frac{1}{2} \mathbb{1}_{[x=x_1]} + \mathbb{1}_{[x_1<x<x_2]} + \frac{1}{2} \mathbb{1}_{[x=x_2]} \right) \tag{4.464}$$

$$= \frac{1}{2} \Pr[X = x_1] + \Pr[x_1 < X < x_2] + \frac{1}{2} \Pr[X = x_2], \tag{4.465}$$

which is exactly the definition of $p_X(x_1, x_2)$. □

A direct corollary of the inversion formula is that the MGF uniquely determines the probability mass function.

Proposition 4.89. Let X be a random variable with MGF $M_X(t)$. Then for every $c > 0$,

$$p_X(x) = \frac{1}{2\pi i} \lim_{\epsilon \to 0^+} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-(x-\epsilon)z} - e^{-(x+\epsilon)z}}{z} M_X(z) \, \mathrm{d}z. \tag{4.466}$$

Proof. Note that $p_X(x) = p_X(x - \epsilon, x + \epsilon)$ when ϵ is sufficiently small. Then the proposition follows from the inversion formula with $x_1 = x - \epsilon$ and $x_2 = x + \epsilon$. □

Note that the inversion formula is rarely used to recover the probability distribution, since the integral is usually not easy to compute. In fact, its main use is to show that if two random variables have the same MGF, then they must have the same distribution. For example, when you want to show that two distributions are the same, you can just show that they have the same MGF. We will see some examples of this in Section 4.3.6.

Remark 4.90. One may wonder if we can write a formula that recovers the probability distribution directly from the moments $\mathbb{E}[X^k]$, not through the MGF. For example,

$$p_X(x_1, x_2) \stackrel{?}{=} \sum_{k \ge 0} \frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-x_1 z} - e^{-x_2 z}}{z} \mathbb{E}[X^k] z^k \, \mathrm{d}z. \tag{4.467}$$

However, this manipulation is not valid: for $k \ge 1$, each individual integral does not converge as an ordinary improper integral. Although this course places less emphasis on the regularity conditions needed to interchange integration, limits, and infinite summation, this is a case where the order cannot be changed.

Remark 4.91. In fact, one can show that the inversion formula also holds for $c = 0$. In this case, the integral simplifies to

$$p_X(x_1, x_2) = \frac{1}{2\pi i} \lim_{T \to +\infty} \int_{-T}^{T} \frac{e^{-itx_1} - e^{-itx_2}}{t} M_X(it) \, \mathrm{d}t. \tag{4.468}$$

This means the probability distribution can be recovered from the values of the MGF on the imaginary axis. This is particularly useful because beyond the finite discrete case, the MGF may diverge and may fail to be defined on any

open set in the complex plane. In contrast, its values on the imaginary axis are always well-defined, so the inversion formula can still be applied. For this reason, the MGF restricted to the imaginary axis has a special name, called the *characteristic function* of X, denoted by $\varphi_X(t) := \mathbb{E}[e^{itX}]$. For this reason, the MGF restricted to the imaginary axis is important enough to deserve a special name: the *characteristic function* of X, denoted by $\varphi_X(t) := \mathbb{E}[e^{itX}]$.

Remark 4.92. The characteristic function and its inversion formula (Equation (4.468)) are closely related to the *Fourier transform*, but we will not pursue this connection further here.

4.3.4 Cumulant Generating Functions

The last generating function we will study for random variables is the *cumulant generating function* (CGF). This is not a new object: it was already introduced under the name *log MGF* in Chapter 3. We have also seen how the CGF can be used to derive the rate function and Chernoff bounds.

Definition 4.93 (Cumulant generating function)**.** The *cumulant generating function* (CGF) of a random variable X is defined as

$$K_X(t) := \log M_X(t). \tag{4.469}$$

In this course, we will continue to use "log MGF" and "CGF" interchangeably. However, introducing the name CGF places special emphasis on the fact that its coefficients are themselves meaningful quantities.

$$K_X(t) = \sum_{m \geq 1} \kappa_m(X) \frac{t^m}{m!}. \tag{4.470}$$

The 0-th term of $K_X(t)$ is zero, since $K_X(0) = \log M_X(0) = \log 1 = 0$. For $m \geq 1$, we call the coefficient of $\frac{t^m}{m!}$ the m-th *cumulant* of X, denoted as $\kappa_m(X)$.

The reason why the cumulants are called "cumulants" is that they can be cumulatively summed up for independent random variables. As we have already seen in Exercise 3.4, this is not true for moments. If you had a painful experience calculating high-order moments of the sample average, you will appreciate the simplicity of cumulants.

Theorem 4.94 (Additivity of CGFs and cumulants)**.** Let $X_1, \ldots, X_n$ be independent random variables, and let X be a weighted sum of these random variables:

$$X := \sum_{k=1}^{n} c_k X_k, \tag{4.471}$$

where $c_1, \ldots, c_n \in \mathbb{R}$ are constants. Then

$$K_X(t) = \sum_{k=1}^{n} K_{X_k}(c_k t), \qquad \kappa_m(X) = \sum_{k=1}^{n} \kappa_m(c_k X_k). \tag{4.472}$$

Proof. By Theorem 4.86, we have $M_X(t) = \prod_{k=1}^n M_{X_k}(c_k t)$. Taking logarithms on both sides, we get

$$K_X(t) = \log\left(\prod_{k=1}^n M_{X_k}(c_k t)\right) = \sum_{k=1}^n \log M_{X_k}(c_k t) = \sum_{k=1}^n K_{X_k}(c_k t). \tag{4.473}$$

Then, comparing the coefficients of $\frac{t^m}{m!}$ in the Taylor expansions of $K_X(t)$ and $\sum_{k=1}^n K_{X_k}(c_k t)$, we obtain $\kappa_m(X) = \sum_{k=1}^n \kappa_m(c_k X_k)$. □

The next proposition lists the first few cumulants.

Proposition 4.95. For every random variable X,

$$\kappa_1(X) = \mathbb{E}[X], \tag{4.474}$$
$$\kappa_2(X) = \mathrm{Var}(X), \tag{4.475}$$
$$\kappa_3(X) = \mu_3(X), \tag{4.476}$$
$$\kappa_4(X) = \mu_4(X) - 3\,\mathrm{Var}(X)^2, \tag{4.477}$$
$$\kappa_5(X) = \mu_5(X) - 10\,\mu_3(X)\mathrm{Var}(X), \tag{4.478}$$

where $\mu_k(X) := \mathbb{E}[(X - \mathbb{E}[X])^k]$ is the k-th central moment of X.

We will only prove the formulas for the first two cumulants and omit the proof for the rest. Unlike what we usually do, we will not leave the rest of the proof as an exercise, since Exercise 3.4 might have already made you hate this course.

Proof of the first two cumulants. Since $K_X(t) = \log M_X(t)$, differentiating gives

$$K'_X(t) = \frac{M'_X(t)}{M_X(t)}. \tag{4.479}$$

At $t = 0$, we have $M_X(0) = 1$ and $M'_X(0) = \mathbb{E}[X]$. Therefore, the first cumulant is the mean:

$$K'_X(0) = \frac{M'_X(0)}{M_X(0)} = \mathbb{E}[X]. \tag{4.480}$$

Differentiating $M'_X(t)$ once more,

$$K''_X(t) = \frac{M''_X(t)}{M_X(t)} - \left(\frac{M'_X(t)}{M_X(t)}\right)^2. \tag{4.481}$$

Using $M_X(0) = 1$, $M'_X(0) = \mathbb{E}[X]$, and $M''_X(0) = \mathbb{E}[X^2]$, we obtain

$$K''_X(0) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathrm{Var}(X), \tag{4.482}$$

which proves that the second cumulant is the variance. □

Example 4.96 (Binomial cumulants)**.** Let $X \sim \text{Binomial}(n, p)$. Since X can be written as the sum of n independent Bernoulli(p) random variables, by Theorem 4.94 we have

$$K_X(t) = n \log(1 - p + pe^t). \tag{4.483}$$

Therefore the first five cumulants are

$$\kappa_1(X) = np, \tag{4.484}$$
$$\kappa_2(X) = np(1-p), \tag{4.485}$$
$$\kappa_3(X) = np(1-p)(1-2p), \tag{4.486}$$
$$\kappa_4(X) = np(1-p)(1-6p+6p^2), \tag{4.487}$$
$$\kappa_5(X) = np(1-p)(1-2p)(1-12p+12p^2). \tag{4.488}$$

4.3.5 Exponential Tilting

In Chapter 3, we presented Hoeffding's lemma (Lemma 3.42), which states that if a random variable X takes values in $[\ell, u]$ for some $\ell \leq u$, then for all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{t(X - \mathbb{E}[X])}] \leq e^{t^2(u-\ell)^2/8}. \tag{4.489}$$

Based on this lemma, we further derived Hoeffding's inequality (Theorem 3.44). However, we did not explain why Hoeffding's lemma holds, only verified that this lemma is true for the symmetric Bernoulli case (Section 3.4.1).

Now we have developed sufficient tools from generating functions, and we are almost ready to prove Hoeffding's lemma. As shown in Table 4.4, we have learned how to transform the MGF and CGF of a random variable under various operations. Linear transformations and sums of independent random variables correspond to simple scalings, additions, or multiplications of the MGF and CGF. Statistical quantities such as the mean and variance can be obtained from derivatives of the MGF and CGF at $t = 0$. To recover the probability distribution from the MGF or CGF, we can use a line integral on the complex plane.

There is, however, one important operation that we have not yet discussed: *exponential tilting*. Unlike the previous operations, it does not act directly on the random variable itself, but instead reweights its distribution. As we will see, this corresponds to a shift in the argument of the CGF, which allows the derivatives of the CGF at $t = \lambda$ for any arbitrary $\lambda \in \mathbb{R}$ to be interpreted as cumulants of the tilted distribution.

Definition 4.97 (Exponential tilting)**.** Let X be a random variable with probability mass function $p_X(x)$. The λ-tilted distribution of X, denoted by $\mathcal{T}_{X,\lambda}$, is defined by the distribution corresponding to the probability mass function

$$q(x) = \frac{p_X(x)e^{\lambda x}}{M_X(\lambda)}, \quad \forall x \in \mathbb{R}. \tag{4.490}$$

Operation	MGF	CGF
Shift $X + c$	$e^{ct} M_X(t)$	$ct + K_X(t)$
Scaling cX	$M_X(ct)$	$K_X(ct)$
Independent Sum $X + Y$	$M_X(t) M_Y(t)$	$K_X(t) + K_Y(t)$
Expectation $\mathbb{E}[X]$	$M'_X(0)$	$K'_X(0)$
Second Moment $\mathbb{E}[X^2]$	$M''_X(0)$	$K''_X(0) + K'_X(0)^2$
Variance $\operatorname{Var}(X)$	$M''_X(0) - M'_X(0)^2$	$K''_X(0)$
Interval Mass $p_X(x_1, x_2)$	$\frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-x_1 z} - e^{-x_2 z}}{z} M_X(z) \, \mathrm{d}z$	$\frac{1}{2\pi i} \lim_{T \to +\infty} \int_{c-iT}^{c+iT} \frac{e^{-x_1 z} - e^{-x_2 z}}{z} e^{K_X(z)} \, \mathrm{d}z$
Exponential Tilting $Y \sim \mathcal{T}_\lambda(X)$	$\frac{M_X(\lambda + t)}{M_X(\lambda)}$	$K_X(\lambda + t) - K_X(\lambda)$

Table 4.4: Operations on random variables and the corresponding transformations of the MGF and CGF.

When we apply exponential tilting to a random variable, we first multiply the probability mass function by $e^{\lambda x}$ and then normalize it by the MGF of X evaluated at λ, i.e., $M_X(\lambda) = \sum_x p_X(x) e^{\lambda x}$. It is easy to check that $q(x)$ is non-negative and sums to 1, so it is a valid probability mass function.

A key property of exponential tilting is that it shifts the MGF and CGF of the original distribution by λ.

Theorem 4.98. Let X be a random variable and $Y \sim \mathcal{T}_{X,\lambda}$ be a random variable drawn from the λ-tilted distribution of X. Then

$$M_Y(t) = \frac{M_X(\lambda + t)}{M_X(\lambda)}, \tag{4.491}$$

$$K_Y(t) = K_X(\lambda + t) - K_X(\lambda). \tag{4.492}$$

Proof. By definition, the MGF of Y is

$$M_Y(t) = \sum_x \frac{p(x) e^{\lambda x}}{M_X(\lambda)} e^{tx} = \frac{1}{M_X(\lambda)} \sum_x p(x) e^{(\lambda + t)x} = \frac{M_X(\lambda + t)}{M_X(\lambda)}. \tag{4.493}$$

Taking logarithms gives the corresponding formula for the CGF:

$$K_Y(t) = \log M_X(\lambda + t) - \log M_X(\lambda) = K_X(\lambda + t) - K_X(\lambda), \tag{4.494}$$

which proves the theorem. □

By taking derivatives, we can easily derive from Theorem 4.98 that

$$K'_X(\lambda) = K'_Y(0) = \mathbb{E}[Y], \tag{4.495}$$
$$K''_X(\lambda) = K''_Y(0) = \mathrm{Var}(Y). \tag{4.496}$$

In general, for any $m \geq 1$,

$$K_X^{(m)}(\lambda) = K_Y^{(m)}(0) = \kappa_m(Y). \tag{4.497}$$

Therefore, whenever we encounter quantities such as $K'_X(\lambda)$ or $K''_X(\lambda)$ for an arbitrary $\lambda \in \mathbb{R}$, we can use exponential tilting to shift the distribution so that these quantities become cumulants of the tilted distribution, where they admit direct probabilistic interpretations.

We now prove Hoeffding's lemma using this idea. We first perform a Taylor expansion of the CGF around $t = 0$. Then, since the remainder term in the Taylor expansion involves second-order derivatives of the CGF, we use exponential tilting to interpret and bound it as the variance of the tilted distribution.

Proof of Lemma 3.42. Without loss of generality, we assume that $\mathbb{E}[X] = 0$. It suffices to prove that $K_X(t) := \log M_X(t) \leq \frac{t^2(u-\ell)^2}{8}$ for all $t \in \mathbb{R}$. By Taylor expansion, for any $t \in \mathbb{R}$, there exists some $\xi \in \mathbb{R}$ between 0 and t such that

$$K_X(t) = K_X(0) + K'_X(0)t + \frac{K''_X(\xi)}{2}t^2. \tag{4.498}$$

Since $K_X(0) = 0$ and $K'_X(0) = \mathbb{E}[X] = 0$, it suffices to prove that

$$K''_X(\xi) \leq \frac{(u-\ell)^2}{4}. \tag{4.499}$$

Let $Y \sim \mathcal{T}_{X,\xi}$ be a random variable drawn from the ξ-tilted distribution of X. Then $K_Y(t) = K_X(\xi + t) - K_X(\xi)$, and thus

$$K''_X(\xi) = K''_Y(0) = \mathrm{Var}(Y). \tag{4.500}$$

Since exponential tilting does not change the support of the distribution, Y always lies in $[\ell, u]$, just like X. Therefore,

$$\mathrm{Var}(Y) = \mathbb{E}[(Y - \mathbb{E}[Y])^2] \leq \mathbb{E}\left[\left(Y - \frac{u+\ell}{2}\right)^2\right] \leq \frac{(u-\ell)^2}{4}, \tag{4.501}$$

where the first inequality follows from the fact that $\mathbb{E}[(Y-c)^2]$ is minimized at $c = \mathbb{E}[Y]$, and the second inequality follows from the fact that $|y - \frac{u+\ell}{2}| \leq \frac{u-\ell}{2}$ holds for all $y \in [\ell, u]$. This proves the required bound on $K''_X(\xi)$, and hence the lemma. □

Another useful application of exponential tilting is to prove Cramér's theorem (Theorem 3.36). Recall that Cramér's theorem states that for any probability distribution $\mathcal{D}$ with mean μ, MGF $M(t)$, CGF $K(t)$ and rate function $I(a)$, if $X^{(n)} := \frac{1}{n}\sum_{i=1}^n X_i$ is the sample average of n i.i.d. random variables $X_1, \ldots, X_n$,

drawn from $\mathcal{D}$, then as $n \to \infty$,

$$\text{for any fixed } a \geq \mu: \quad \Pr[X^{(n)} \geq a] = e^{-n(I(a)+o(1))}, \tag{4.502}$$

$$\text{for any fixed } a \leq \mu: \quad \Pr[X^{(n)} \leq a] = e^{-n(I(a)+o(1))}. \tag{4.503}$$

At the first glance, it seems that we need to develop some new techniques to lower bound the tail probability of $X^{(n)}$. But actually we just need to combine exponential tilting with concentration inequalities.

Proof sketch of Theorem 3.36. Here we explain the high-level idea and leave the details as an exercise. The idea is very simple: exponential tilting turns a rare event into a typical one. Let $Y_1, \ldots, Y_n$ be i.i.d. random variables drawn from the λ-tilted distribution of $\mathcal{D}$, and let $Y^{(n)} := \frac{1}{n}\sum_{i=1}^{n} Y_i$. Then one can show that $Y^{(n)}$ has the same distribution as the $(n\lambda)$-tilted version of $X^{(n)}$, either by brute force or by using CGFs.

Now choose λ such that $\mathbb{E}[Y^{(n)}] \approx a$. Under this tilted distribution, a is no longer a rare value: by concentration, $Y^{(n)}$ is typically close to a. Therefore, we can understand the distribution of $Y^{(n)}$ very concretely, since it behaves almost like a deterministic variable $Y^{(n)} \approx a$.

To prove Cramér's theorem, we now translate this understanding back to $X^{(n)}$. We transform the distribution of $Y^{(n)}$ back to $X^{(n)}$ by multiplying the probability by $\mathbb{E}[e^{n\lambda X^{(n)}}]e^{-n\lambda Y^{(n)}}$. Since $Y^{(n)} \approx a$, this is just a deterministic factor $\mathbb{E}[e^{n\lambda X^{(n)}}]e^{-n\lambda a}$. Therefore, we have

$$\Pr[X^{(n)} \geq a] \approx \Pr[Y^{(n)} \geq a] \cdot \mathbb{E}[e^{n\lambda X^{(n)}}]e^{-n\lambda a} = \Pr[Y^{(n)} \geq a] \cdot e^{-n(\lambda a - \log M(\lambda))}.$$

We choose λ such that $\mathbb{E}[Y^{(n)}]$ is slightly above a, then $\Pr[Y^{(n)} \geq a] \approx 1$. Meanwhile, $\mathbb{E}[Y^{(n)}] \approx a$ means $K'(\lambda) \approx a$. It turns out that this choice of λ is exactly the one that achieves the supremum in the definition of the rate function

$$I(a) = \sup_{t \in \mathbb{R}} \{ta - K(t)\}.$$

Therefore, for this choice of λ, the tail probability becomes $\approx e^{-nI(a)}$. □

4.3.6 A First Look at the Central Limit Theorem

In Chapter 3, we mainly used MGFs and CGFs to derive inequalities. The main result there was several upper bounds on a tail probability. When we want a matching lower bound, the story becomes more delicate and requires additional ideas, such as the exponential tilting trick behind Cramér's theorem.

This chapter has a different emphasis. Here, generating functions are not merely a tool for proving inequalities. Rather, they are meant to encode the distribution itself as faithfully as possible. With this perspective in mind, we can now ask a new question: *Can we precisely describe the distribution of the sample average of n random variables, especially when $n \to \infty$?*

Needless to say, there is a huge gap between deriving concentration inequalities and precisely describing the distribution. The point of pursuing a precise description is that such kind of description often allows us to directly "see" the answers to many questions. For example:

1. Can the tail bound $\leq e^{-2n\epsilon^2/(u-\ell)^2}$ from Hoeffding's inequality be improved to something like $e^{-\Omega(n\epsilon^3)}$?
2. Hoeffding's inequality only depends on the range $[\ell, u]$ of the random variable. How much tighter can we make the bound if we know more about the distribution?
3. In what situations can the interval length $u-\ell$ be replaced by a significantly smaller quantity?

If you feel that these questions are hard to answer just by guessing, let us do some simple calculations with generating functions!

▶ The Law of Large Numbers

Let $X_1, \ldots, X_n$ be i.i.d. random variables with common CGF $K(t)$, mean μ, and variance σ^2. As usual, we define their sample average as

$$X^{(n)} := \frac{1}{n}\sum_{k=1}^{n} X_k. \tag{4.504}$$

By Theorem 4.94,

$$K_{X^{(n)}}(t) = \sum_{k=1}^{n} K_{\frac{1}{n}X_i}(t) = \sum_{k=1}^{n} K(\tfrac{t}{n}) = nK(\tfrac{t}{n}). \tag{4.505}$$

Now expand $K(s)$ at 0. Since the first two cumulants are μ and σ^2, we have

$$K(s) = \mu s + \frac{\sigma^2}{2}s^2 + \mathcal{O}(s^3). \tag{4.506}$$

Substituting $s = t/n$, then for every fixed $t \in \mathbb{R}$, as $n \to \infty$, we have

$$K_{X^{(n)}}(t) = n\left(\mu\frac{t}{n} + \frac{\sigma^2}{2}\cdot\frac{t^2}{n^2} + \mathcal{O}(n^{-3})\right) = \mu t + \mathcal{O}(n^{-1}). \tag{4.507}$$

As $n \to \infty$, this converges to the following limit:

$$K_{X^{(n)}}(t) \to \mu t =: K^*(t). \tag{4.508}$$

Since the CGF uniquely determines the MGF and thus uniquely determines the distribution, we can understand the distribution of $X^{(n)}$ by viewing the above function $K^*(t)$ as a CGF and then recovering the corresponding distribution from it.

Of course, one can apply the inversion formula in Proposition 4.89 to recover the distribution of $X^{(n)}$. A more direct way is just to note that $K^*(t) = \mu t$ is precisely the CGF of a point mass at μ, i.e., $p(x) = \mathbb{1}_{[x=\mu]}$. Hence, the limiting distribution of $X^{(n)}$ is the point mass at μ.

This phenomenon is generally referred to as the *Law of Large Numbers* (LLN): the sample average of n random variables converges to the mean as $n \to \infty$. While this is intuitive, it does not provide much new insight. Concentration inequalities already tell us that the sample average concentrates around the mean, and even provide quantitative bounds on how fast this convergence occurs.

▶ The Central Limit Theorem

To observe a non-trivial limit, we need to zoom in at the correct scale. Instead of taking the limit directly for $X^{(n)}$, we focus on the deviations of $X^{(n)}$ from the mean μ and rescale them by $\sqrt{n}$.

Note that this scale is not arbitrary. We have already discussed in Sections 3.2.3 and 3.3.4 that a typical deviation from the mean is of order $\mathcal{O}(\frac{1}{\sqrt{n}})$. Therefore, by scaling the deviation by $\sqrt{n}$, we should be able to clearly observe its distribution.

Define the rescaled deviation $Z^{(n)}$ as

$$Z^{(n)} := \sqrt{n}\,(X^{(n)} - \mu) = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} X_k - \sqrt{n}\mu. \tag{4.509}$$

Before doing any further calculations, it is useful to plot the distribution of $Z^{(n)}$ first. For a fixed discrete distribution, Figure 4.23 compares the distributions of $X^{(n)}$ and $Z^{(n)}$. The distribution of $X^{(n)}$ collapses towards the mean, while the distribution of $Z^{(n)}$, which is essentially a rescaled version of $X^{(n)}$, keeps a visible and stable bell-like shape. This suggests that $Z^{(n)}$ should converge to a non-degenerate limiting distribution.

Now we compute the CGF of $Z^{(n)}$. By Theorem 4.94,

$$K_{Z^{(n)}}(t) = \sum_{k=1}^{n} K_{\frac{1}{\sqrt{n}}X_k}(t) - \sqrt{n}\mu t = nK(\tfrac{t}{\sqrt{n}}) - \sqrt{n}\mu t. \tag{4.510}$$

Again we expand $K(s)$ at 0 and obtain

$$K(s) = \mu s + \frac{\sigma^2}{2}s^2 + \mathcal{O}(s^3). \tag{4.511}$$

Substituting $s = \frac{t}{\sqrt{n}}$, then for every fixed $t \in \mathbb{R}$, as $n \to \infty$, we have

$$K_{Z^{(n)}}(t) = n\left(\mu\frac{t}{\sqrt{n}} + \frac{\sigma^2}{2}\cdot\frac{t^2}{n} + \mathcal{O}(n^{-3/2})\right) - \sqrt{n}\mu t \tag{4.512}$$

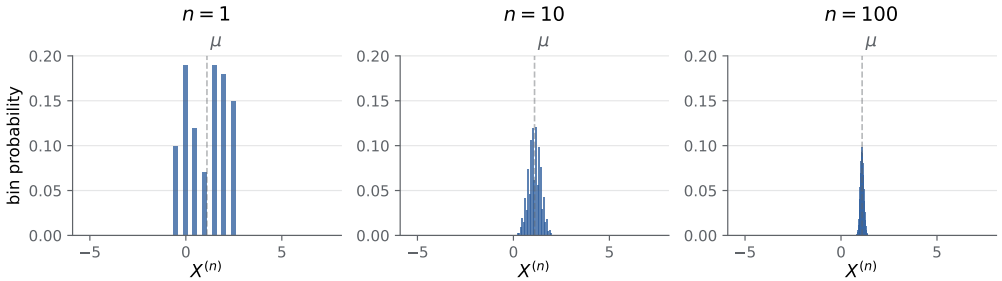
$$= \sqrt{n}\mu t + \frac{1}{2}\sigma^2 t^2 + \mathcal{O}(n^{-1/2}) - \sqrt{n}\mu t \tag{4.513}$$

$$= \frac{1}{2}\sigma^2 t^2 + \mathcal{O}(n^{-1/2}). \tag{4.514}$$

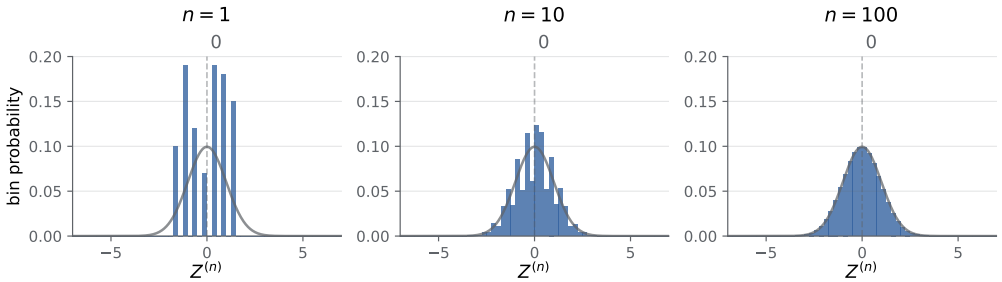
Therefore, as $n \to \infty$, this converges to the following limit:

$$K_{Z^{(n)}}(t) \to \frac{1}{2}\sigma^2 t^2 =: K^*(t), \tag{4.515}$$

which is a quadratic function of t.



(a) LLN scaling.



(b) CLT scaling.

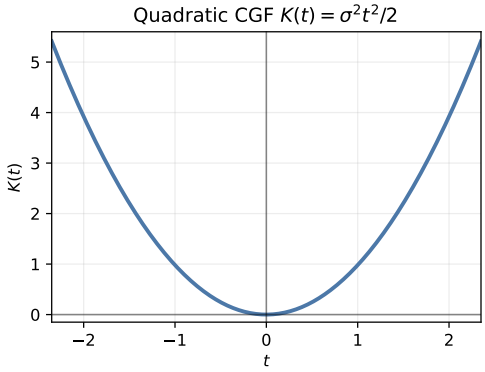
Figure 4.23: The distributions of $X^{(n)}$ and $Z^{(n)}$ for $n = 1, 10, 100$. **(a)** $X^{(n)}$ concentrates around the mean as n grows. **(b)** $Z^{(n)} = \sqrt{n}(X^{(n)} - \mu)$ appears to exhibit a stable bell-shaped profile. The values are first grouped into bins of a fixed width, then the vertical axis records the probability of the random variable falling into each bin. The bin width of $X^{(n)}$ is scaled by $1/\sqrt{n}$ and the bin width of $Z^{(n)}$ is fixed over all n.

This quadratic CGF looks remarkably clean, even though it is not immediately clear what the corresponding distribution is. Recall that when deriving Hoeffding's inequality, we upper bounded the log MGF by a quadratic function and then plugged that bound into Chernoff's method. This quadratic CGF is, in some sense, "ideal" for Hoeffding-type inequalities, since it is already exactly quadratic and no approximation is needed.

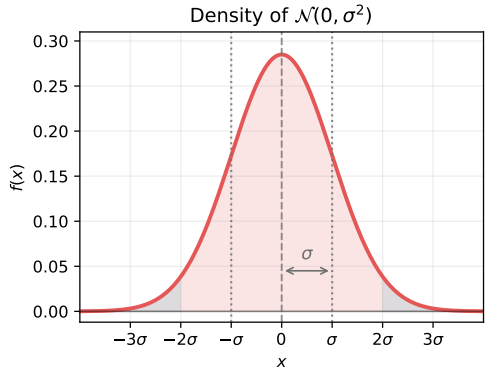
So what is this distribution? In fact, the distribution that can produce this quadratic CGF is the famous *Gaussian distribution* (or *normal distribution*) with mean 0 and variance σ^2, denoted by $\mathcal{N}(0, \sigma^2)$. This distribution is not a discrete distribution, but a continuous distribution with probability density function:

$$f_X(x) := \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}, \qquad x \in \mathbb{R}. \tag{4.516}$$

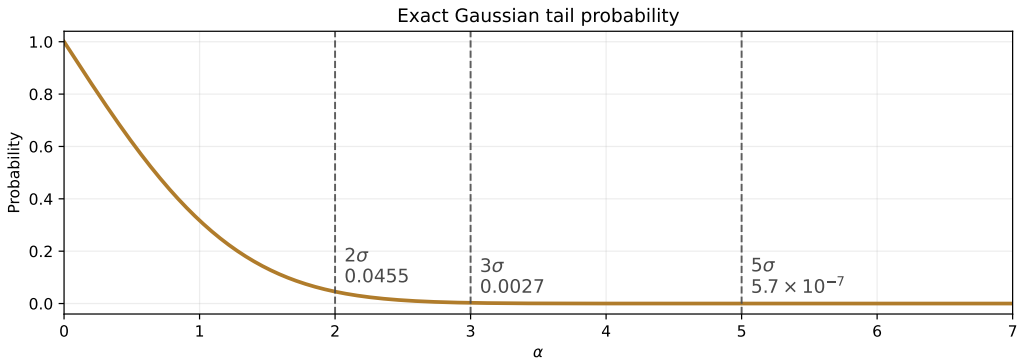
This is our first encounter with a continuous distribution. Unlike a discrete random variable, a Gaussian random variable does not put positive probability on any single point: if $X \sim \mathcal{N}(0, \sigma^2)$, then $\Pr[X = x] = 0$ for every fixed $x \in \mathbb{R}$. Instead, probabilities are assigned to intervals. The probability that X falls into



(a) Quadratic CGF.



(b) Gaussian density.



(c) Tail probability $\Pr[|X| > \alpha \cdot \sigma]$.

Figure 4.24: **(a)** The Gaussian CGF is quadratic, with coefficient determined by the variance σ^2. **(b)** The Gaussian density is bell-shaped and has very little mass beyond a few standard deviations. **(c)** The exact tail probability $\Pr[|X| > \alpha \cdot \sigma]$ decays rapidly as α grows.

an interval (a, b) is the integral of the density over the interval:

$$\Pr[a < X < b] = \int_a^b f_X(x)\, \mathrm{d}x. \tag{4.517}$$

When $\epsilon > 0$ is small, the probability that X falls into the interval $[x, x + \epsilon]$ is approximately

$$\Pr[x < X < x + \epsilon] \approx \epsilon \cdot f_X(x). \tag{4.518}$$

Therefore, $f_X(x)$ describes the local density of X around x, not the exact probability that $X = x$. For this reason, functions like $f_X(x)$ are called *probability density functions*. We will formally build up our language of describing continuous distributions in Chapter 5.

The shape of the Gaussian is bell-like: it is symmetric around 0, attains its maximum at 0, and decays rapidly once $|x|$ is several multiples of σ. Roughly speaking, almost all the probability mass lies within a few standard deviations

of the mean. More precisely, if $X \sim \mathcal{N}(0, \sigma^2)$, then

$$\Pr[|X| > 2\sigma] \approx 4.55\%, \tag{4.519}$$

$$\Pr[|X| > 3\sigma] \approx 0.27\%, \tag{4.520}$$

$$\Pr[|X| > 5\sigma] \approx 0.000057\%. \tag{4.521}$$

Since the CGF uniquely determines the distribution, we can now state the following result, which will be made precise in Chapter 6 and is commonly referred to as the *Central Limit Theorem* (CLT): if $X^{(n)}$ is the sample average of n i.i.d. random variables with mean μ and variance σ^2, then as $n \to \infty$, the distribution of $Z^{(n)} := \sqrt{n}(X^{(n)} - \mu)$ converges to the Gaussian distribution $\mathcal{N}(0, \sigma^2)$.

In general, one can define $\mathcal{N}(\mu, \sigma^2)$ as the Gaussian distribution with mean μ and variance σ^2, whose probability density function is given by

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad x \in \mathbb{R}. \tag{4.522}$$

Then the Central Limit Theorem implies that for large n, around $x \approx \mu$, the sample average $X^{(n)}$ behaves like the following Gaussian distribution:

$$\mathcal{N}\Big(\mu, \frac{\sigma^2}{n}\Big). \tag{4.523}$$

That is, $X^{(n)}$ can be approximated by a Gaussian random variable with mean μ and variance σ^2/n. Consistent with our discussion in Sections 3.2.3 and 3.3.4, this implies that a typical deviation from the mean is of order $\mathcal{O}(\frac{1}{\sqrt{n}})$. In particular, under this Gaussian approximation, the probability that $X^{(n)}$ deviates from the mean by more than $\frac{3\sigma}{\sqrt{n}}$ is approximately 0.27%.

Remark 4.99. Historically, Gaussian distributions became central through the study of measurement errors, especially in astronomy. It is named after the German mathematician Carl Friedrich Gauss (1777–1855), who used this distribution to model measurement errors in his 1823 monograph [Gau23]. But in fact, a version of the Central Limit Theorem (CLT) was proved much earlier by the French mathematician Pierre-Simon Laplace (1749–1827) [Lap12] before Gauss's work. The term "Laplace distribution", however, refers to another distribution, whose probability density function is given by $f_{\text{Laplace}(\mu,b)}(x) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}}$.

▶ Back to the Three Motivating Questions

Let us return to the questions from the beginning of this subsection. The CLT tells us that the quadratic exponent in Hoeffding-type bounds is not an artifact of the proof. At the typical fluctuation scale $\mathcal{O}(\frac{1}{\sqrt{n}})$, the limiting CGF is exactly quadratic. Therefore, Hoeffding's lemma (Lemma 3.42) does not lose much by upper bounding the CGF by a quadratic function:

$$K_{X_k-\mu}(t) \le \frac{(u-\ell)^2}{8} t^2 \quad \Rightarrow \quad K_{Z^{(n)}}(t) \le \frac{(u-\ell)^2}{8} t^2. \tag{4.524}$$

The CLT also tells us what Hoeffding misses. The limiting CGF is $\frac{1}{2}\sigma^2 t^2$, so the constant in the quadratic term should be governed by the variance σ^2. However, Hoeffding's inequality is based on the worst-case interval length $u - \ell$. When X is supported on a long interval but usually fluctuates on a much smaller scale, Hoeffding's inequality may significantly underestimate the concentration.

Therefore, it is impossible to significantly improve the quadratic form of the exponent $2n\epsilon^2/(u-\ell)^2$ in Hoeffding's inequality $\leq e^{-2n\epsilon^2/(u-\ell)^2}$. However, the constant in the exponent can still be improved. The key idea is to retain the quadratic behavior while replacing the crude interval-length parameter with variance information whenever possible. We will see such a refinement in Section 6.2, known as *Bernstein's inequality*.

4.4 A Brief Introduction to the Riemann Hypothesis

We live in an age of rapid AI development.[1] Still, many people wonder whether AI can truly help humans solve important mathematical or scientific problems. Step into almost any panel discussion about AGI (abbreviated from Artificial General Intelligence), and the conversation will quickly drift towards a familiar question: could AI prove the Riemann hypothesis?[2]

What is the Riemann hypothesis, and why should computer scientists care about it? Without knowing the answers to these questions, it is completely meaningless to engage in such a discussion.

In this section, we will give you a sense of the Riemann hypothesis. If you could not understand this section at all, do not worry. The main message is as follows:

- Number theory can be studied through the lens of generating functions.
- The Riemann hypothesis is about a specific generating function, $\zeta(s)$.
- Once we understand this generating function, many open problems in number theory can be solved easily, thanks to the power of generating functions.

4.4.1 Searching for the Next Prime

Suppose we want to find the first prime larger than a given integer n. The most naïve algorithm is:

```
def next_prime(n):
    m = n + 1
    while not is_prime(m):
        m += 1
    return m
```

[1] That's so true! — Codex (who helped revise the lecture notes and generated most of the figures).

[2] https://simons.berkeley.edu/news/sparks-vs-embers

This kind of subroutine appears whenever we want a prime parameter. For example, in hashing, one often chooses a prime table size or a prime modulus to make the modular arithmetic cleaner. If we measure the running time of this search by the number of calls to `is_prime`, then the key question is how long the `while` loop can keep going.

Do we exactly know the time complexity of this algorithm yet?



But Riemann hypothesis will bound the number of calls to `is_prime` as:

$$\mathcal{O}(\sqrt{n}\log n). \tag{4.525}$$

Just we do not know how to prove Riemann hypothesis yet.

4.4.2 The Distribution of Primes

In essence, this is a question about the distribution of primes. If we know how primes are distributed, we can answer the question of how far to search before finding the next prime.

Formally, we define the *prime-counting function* by

$$\pi(x) := |\{p \le x : p \text{ is prime}\}|. \tag{4.526}$$

Then $\pi(b) - \pi(a)$ counts the primes in the interval $(a, b]$. So estimates for $\pi(x)$ describe the distribution of primes, and the next-prime search problem becomes finding the smallest x such that $\pi(x) > \pi(n)$.

That is, if we define the first prime larger than n as

$$P(n) := \min\{p > n : p \text{ is prime}\} \tag{4.527}$$

then

$$P(n) = \min\{x > n : \pi(x) > \pi(n)\}. \tag{4.528}$$

The number of calls to `is_prime` is then bounded by $P(n) - n$.

A classical result on the distribution of primes is the *prime number theorem*.

Theorem 4.100 (Prime number theorem)**.** As $x \to \infty$,

$$\pi(x) \sim \frac{x}{\log x}. \tag{4.529}$$

Recall from Definition 2.26 that $f(x) \sim g(x)$ means $f(x)/g(x) \to 1$ in the limiting regime under discussion. So the prime number theorem says that the number of primes up to x is well approximated by $x/\log x$ as $x \to \infty$. Near size x, this suggests that primes have average spacing about $\log x$.

The prime number theorem already gives a coarse answer to the next-prime search problem. To see this, fix any $\varepsilon > 0$ and compare the number of primes

up to n with the number of primes up to $(1+\varepsilon)n$. The prime number theorem gives

$$\frac{\pi((1+\varepsilon)n)}{\pi(n)} \sim \frac{(1+\varepsilon)n/\log((1+\varepsilon)n)}{n/\log n} \tag{4.530}$$

$$= (1+\varepsilon)\frac{\log n}{\log n + \log(1+\varepsilon)} \to 1+\varepsilon > 1. \tag{4.531}$$

Thus, for all sufficiently large n, we have $\pi((1+\varepsilon)n) > \pi(n)$. Therefore

$$P(n) \le (1+\varepsilon)n, \tag{4.532}$$

or equivalently $P(n) - n \le \varepsilon n$. Since $\varepsilon > 0$ was arbitrary, we get

$$P(n) - n = o(n). \tag{4.533}$$

Although this result is already non-trivial, it is not the tightest bound possible. If we can get a better estimate for $\pi(x)$, we can get a better bound on $P(n) - n$.

Under the Riemann hypothesis, we will prove a much more precise formula:

$$\pi(x) = \sum_{2\le t\le x} \frac{1}{\log t} + \mathcal{O}(\sqrt{x}\log x). \tag{4.534}$$

Asymptotically, $\sum_{2\le t\le x} \frac{1}{\log t} \sim \frac{x}{\log x}$. But if we replace the sum by $\frac{x}{\log x}$, the error term will no longer be $\mathcal{O}(\sqrt{x}\log x)$.

Let us see what this estimate gives for the next-prime search. Take

$$h = C\sqrt{n}\log^2 n \tag{4.535}$$

for a constant $C > 0$ to be chosen later. By the estimate for $\pi(x)$, we have

$$\pi(n+h) - \pi(n) = \underbrace{\sum_{2\le t\le n+h} \frac{1}{\log t} - \sum_{2\le t\le n} \frac{1}{\log t}}_{=:S_1} \tag{4.536}$$

$$+ \underbrace{\mathcal{O}(\sqrt{n+h}\log(n+h)) + \mathcal{O}(\sqrt{n}\log n)}_{=:S_2}. \tag{4.537}$$

Since $h = o(n)$, the main term S_1 satisfies

$$S_1 = \sum_{n<t\le n+h} \frac{1}{\log t} = \Omega\left(\frac{h}{\log n}\right) = \Omega(C\sqrt{n}\log n). \tag{4.538}$$

Since the error term S_2 is clearly bounded by $\mathcal{O}(\sqrt{n}\log n)$, if C is large enough, the main term S_1 dominates the error term S_2, and $\pi(n+h) > \pi(n)$. Thus the interval $(n, n+h]$ contains a prime, and therefore

$$P(n) - n = \mathcal{O}(\sqrt{n}\log^2 n). \tag{4.539}$$

Remark 4.101. This is the bound we will prove in this course. In fact, the Riemann hypothesis is strong enough to imply that $P(n) - n = \mathcal{O}(\sqrt{n}\log n)$, but the proof is more delicate and we will omit it. There is also a much stronger statement, called *Cramér's conjecture*, which predicts $P(n) - n = \mathcal{O}(\log^2 n)$ from a model that assumes the primes are distributed randomly. Even if the Riemann hypothesis is true, it is not known how to prove Cramér's conjecture.

For now, our goal is to understand why the Riemann hypothesis already gives a much sharper picture of the distribution of primes. So what is the Riemann hypothesis? To state it, we first need to introduce the Riemann zeta function.

4.4.3 The Riemann Zeta Function

We already met the Riemann zeta function in Definition 4.78 for real $s > 1$. But to unleash the full power of the Riemann zeta function, we need to consider it as a complex function.

Definition 4.102 (Riemann zeta function)**.** The *Riemann zeta function* is the unique holomorphic function on $\mathbb{C} \setminus \{1\}$ such that

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} \tag{4.540}$$

for every $s \in \mathbb{C}$ with $\Re(s) > 1$.

This definition is hiding a theorem. The displayed series only converges when $\Re(s) > 1$, but it can be extended to be defined on all of $\mathbb{C} \setminus \{1\}$ in the sense that there is exactly one holomorphic function on $\mathbb{C} \setminus \{1\}$ that agrees with the series when $\Re(s) > 1$. The uniqueness is a consequence of the identity theorem for holomorphic functions, but the existence requires a more subtle argument, which we will not prove here.

Below we list some special values of $\zeta(s)$ for fun.

Example 4.103 (Special values of ζ)**.** The point of the definition is that $\zeta(s)$ still has values at many points where the original series no longer converges. The non-positive integer values are closely related to the Bernoulli numbers: for every integer $m \geq 0$,

$$\zeta(-m) = (-1)^m \frac{B_{m+1}}{m+1}. \tag{4.541}$$

Taking $m = 0$ gives $\zeta(0) = B_1 = -\frac{1}{2}$. Taking $m = 1$ gives

$$\zeta(-1) = -\frac{B_2}{2} = -\frac{1}{12}. \tag{4.542}$$

This leads to a famous identity: the sum of all natural numbers is

$$1 + 2 + 3 + \cdots = -\frac{1}{12}. \tag{4.543}$$

Note that the ordinary series $1 + 2 + 3 + \cdots$ still diverges, but the Riemann zeta function assigns a value to this divergent series. Finally, since $B_{2d+1} = 0$ for every $d \geq 1$, the same formula gives

$$\zeta(-2d) = 0. \tag{4.544}$$

These zeros are called the *trivial zeros* of ζ.

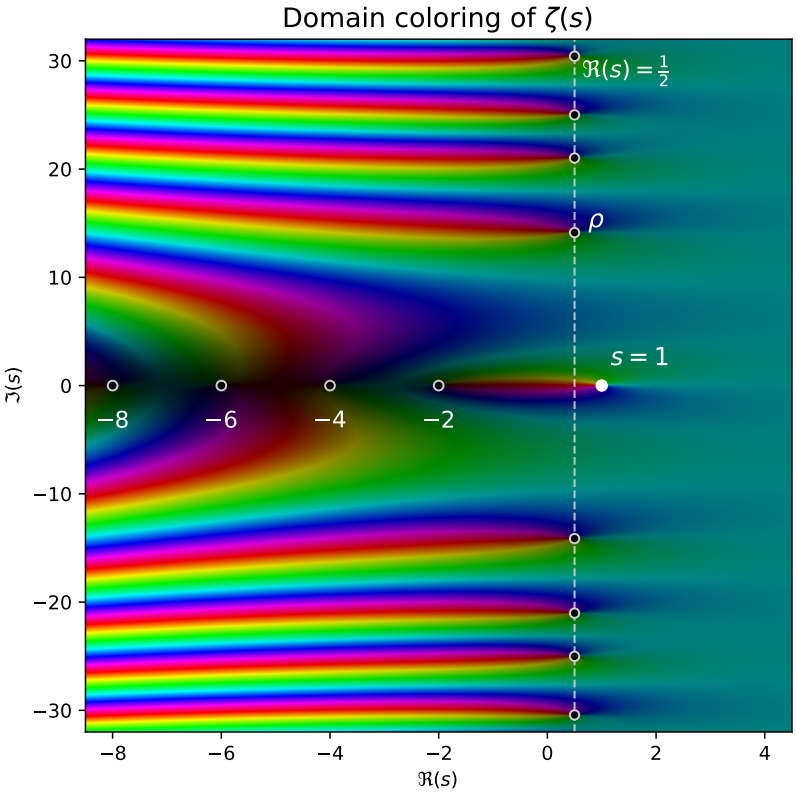


Figure 4.25: Domain coloring of the Riemann zeta function. There is a pole at $s = 1$ and infinitely many trivial zeros at the negative even integers. Riemann Hypothesis predicts that all the non-trivival zeros lie on the critical line $\Re(s) = \frac{1}{2}$.

Before stating the conjecture itself, we state without proof some basic properties of ζ that will be useful for the rest of the argument.

Lemma 4.104 (The singularities and zeros of ζ)**.** The zeta function has the following singularities and zeros:

(i) it has a simple pole at $s = 1$;

(ii) it has simple zeros at the negative even integers

$$-2, -4, -6, \ldots; \tag{4.545}$$

(iii) every other zero lies in the *critical strip*

$$0 < \Re(s) < 1. \tag{4.546}$$

(iv) Let $N(T)$ be the number of zeros ρ of ζ in the critical strip with $|\Im(\rho)| \leq T$, counted with multiplicity. Then

$$N(T) = \mathcal{O}(T \log T). \tag{4.547}$$

We will use this lemma without proof. Here *multiplicity* has the usual meaning

for zeros of holomorphic functions. If a holomorphic function f can be written near a as

$$f(s) = (s-a)^m h(s) \tag{4.548}$$

with $h(a) \neq 0$, then a is a zero of multiplicity m. In particular, a simple zero is a zero of multiplicity 1. When zeros are counted with multiplicity, such a zero is counted m times.

The zeros at the negative even integers are called the *trivial zeros* of ζ. The name is slightly unfair. They are not "trivial" in the sense that they are obvious from the defining series. They are called trivial because their locations are completely regular and are not the mystery in the Riemann hypothesis. The remaining zeros are called the *non-trivial zeros*. By the above lemma, every non-trivial zero lies in the strip

$$0 < \Re(s) < 1, \tag{4.549}$$

which is called the *critical strip*.

The first few non-trivial zeros are known to lie on the middle line of the strip, as shown in Figure 4.25. This middle line is called the *critical line*. The Riemann hypothesis says that this is not an accident.

Hypothesis 4.105 (Riemann hypothesis)**.** Every non-trivial zero ρ of ζ satisfies

$$\Re(\rho) = \frac{1}{2}. \tag{4.550}$$

We will not try to prove this conjecture. If you can prove it, please let us know and we will invite you to teach us for the rest of the course. Instead, our main goal here is to convince you that the Riemann hypothesis is an important conjecture in number theory. In particular, we will explain why it would imply a much sharper estimate for the distribution of primes.

4.4.4 Dirichlet Generating Function

For studying primes, none of the OGFs, EGFs, PGFs, MGFs, or CGFs we have learned so far are the right language. The correct tool is a *Dirichlet generating function*, that is, a series of the form

$$\sum_{n \geq 1} \frac{a_n}{n^s}. \tag{4.551}$$

This should be compared with an ordinary generating function $\sum_{n \geq 0} a_n z^n$. In both cases, the coefficient a_n is attached to the integer n. The difference is the basis function: ordinary generating functions use z^n, while Dirichlet generating functions use n^{-s}.

The zeta function is the simplest example of a Dirichlet generating function:

in the half-plane where the series converges,

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s} \qquad (\Re(s) > 1). \tag{4.552}$$

So $\zeta(s)$ is the Dirichlet generating function of the constant sequence $1, 1, 1, \ldots$.

To see why Dirichlet generating functions are the right tool for primes, we need to understand how they encode the prime factorization of integers. For $\Re(s) > 1$, $\zeta(s)$ can be written as the *Euler product*

$$\zeta(s) = \prod_{p \text{ prime}} \left(1 + p^{-s} + p^{-2s} + \cdots\right) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}. \tag{4.553}$$

Here is the mechanism behind the product. If we multiply only over the first few primes, say $2, 3, 5$, then a typical term in the expanded product is

$$2^{-as}3^{-bs}5^{-cs} = (2^a 3^b 5^c)^{-s}, \tag{4.554}$$

where $a, b, c \geq 0$. Thus choosing one term from each geometric series is the same as choosing the exponent of each prime in a factorization. This means

$$\frac{1}{1-2^{-s}}\frac{1}{1-3^{-s}}\frac{1}{1-5^{-s}} = \sum_{a,b,c\geq 0} (2^a 3^b 5^c)^{-s}. \tag{4.555}$$

However, this sum only includes positive integers whose prime factors are among $2, 3, 5$. If we multiply over all primes, then all prime factors are included. By the unique prime factorization theorem, every positive integer admits a unique prime factorization:

$$n = \prod_p p^{a_p}, \tag{4.556}$$

where the exponents a_p are non-negative integers and only finitely many of them are non-zero. Therefore, the product over all primes gives the sum of n^{-s} over all positive integers n.

To extract prime information from the Euler product, we take a logarithmic derivative. The result is the following Dirichlet generating function associated with the *von Mangoldt function*.

Theorem 4.106 (The logarithmic derivative of ζ)**.** Define the *von Mangoldt function* by

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime } p \text{ and some } k \geq 1, \\ 0 & \text{otherwise.} \end{cases} \tag{4.557}$$

Then, for $\Re(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n\geq 1} \frac{\Lambda(n)}{n^s}. \tag{4.558}$$

The left-hand side is $-\frac{\zeta'(s)}{\zeta(s)}$, which is essentially $\big(-\log \zeta(s)\big)'$. There is only one technical caveat: the complex logarithm is not globally well-defined. This is why

the theorem statement avoids writing $\log \zeta(s)$ and instead uses the logarithmic derivative directly.

In the proof below, we will nevertheless use the notation log informally. This is only meant to help you understand the high-level idea more quickly. If we want to make the proof fully rigorous, we need to define the logarithm only in the local region where it is used, but it is a bit technical and we will not go into the details here.

Proof. Taking the logarithm of the Euler product gives

$$-\log \zeta(s) = \sum_{p \text{ prime}} \log(1 - p^{-s}). \tag{4.559}$$

Now expand each logarithm into its power series:

$$\log(1 - p^{-s}) = -\sum_{k \geq 1} \frac{p^{-ks}}{k}. \tag{4.560}$$

Therefore,

$$-\log \zeta(s) = -\sum_{p \text{ prime}} \sum_{k \geq 1} \frac{p^{-ks}}{k}. \tag{4.561}$$

Differentiating term by term gives

$$\Big(-\log \zeta(s)\Big)' = \sum_{p \text{ prime}} \sum_{k \geq 1} (\log p)\, p^{-ks}. \tag{4.562}$$

Recall that the von Mangoldt function is defined by $\Lambda(n) = 0$ for n not a prime power, and $\Lambda(p^k) = \log p$ for a prime p and $k \geq 1$. So we can rewrite the sum as

$$\Big(-\log \zeta(s)\Big)' = \sum_{p \text{ prime}} \sum_{k \geq 1} \frac{\Lambda(p^k)}{(p^k)^s} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}, \tag{4.563}$$

which proves the theorem. □

While this theorem does not directly give a prime-counting estimate, it already isolates prime powers from the rest of the integers. Next, we will see how to use this information to obtain an estimate for the distribution of primes.

4.4.5 From the Zeta Function to Prime Counting

▶ The Main Estimates

A function related to von Mangoldt function is called *the second Chebyshev function*, defined by

$$\psi(x) := \sum_{n=1}^{\lfloor x \rfloor} \Lambda(n), \tag{4.564}$$

which is the sum of the von Mangoldt function up to x. This function counts prime powers with logarithmic weights. It may look less natural than $\pi(x)$, but

it is the quantity that $\zeta(s)$ controls most directly. In this subsection, we will prove the following estimate for $\psi(x)$.

Theorem 4.107 (Estimate for $\psi(x)$)**.** Assume the Riemann hypothesis. Then, as $x \to \infty$,

$$\psi(x) = x + \mathcal{O}(\sqrt{x}\log^2 x). \tag{4.565}$$

Before proving this theorem, we will first show how this estimate for $\psi(x)$ implies an estimate for $\pi(x)$.

Theorem 4.108 (Estimate for $\pi(x)$)**.** Assume the Riemann hypothesis. Then, as $x \to \infty$,

$$\pi(x) = \sum_{2 \le n \le x} \frac{1}{\log n} + \mathcal{O}(\sqrt{x}\log x). \tag{4.566}$$

Proof. We prove this theorem from Theorem 4.107. It suffices to prove the case when x is an integer.

First, we remove the contribution of higher prime powers. Let $\vartheta(x)$ be *the first Chebyshev function*, which is defined by

$$\vartheta(x) := \sum_{n \le x} \mathbb{1}_{[n \text{ prime}]} \log n. \tag{4.567}$$

This is the part of $\psi(x)$ coming from primes themselves. The remaining part comes from prime powers p^k with $k \ge 2$:

$$\psi(x) = \vartheta(x) + \sum_{k \ge 2} \sum_{p^k \le x} \log p. \tag{4.568}$$

For each $k \ge 2$, there are at most $x^{1/k}$ primes p such that $p^k \le x$, which is simply because $p \le x^{1/k}$. Each prime contributes at most $\log p$ to $\psi(x)$, which is smaller than $\log x$. Adding up all the contributions, we get

$$\sum_{k \ge 2} \sum_{p^k \le x} \log p \le \sum_{k \ge 2} x^{1/k} \log x = \mathcal{O}(\sqrt{x}\log x). \tag{4.569}$$

Therefore,

$$\psi(x) = \vartheta(x) + \mathcal{O}(\sqrt{x}\log x). \tag{4.570}$$

Combining this with Theorem 4.107 gives

$$\vartheta(x) = x + \mathcal{O}(\sqrt{x}\log^2 x). \tag{4.571}$$

Now convert from $\vartheta(x)$ to $\pi(x)$. Note that the only difference between $\vartheta(x)$ and $\pi(x)$ is that $\vartheta(x)$ counts prime powers with logarithmic weights $\log p$, while $\pi(x)$ counts primes with weight 1. Motivated by this, we rewrite $\pi(x)$ as

$$\pi(x) = \sum_{2 \le n \le x} \mathbb{1}_{[n \text{ prime}]} \log n \cdot \frac{1}{\log n} \tag{4.572}$$

Now for all integers $n \geq 1$, define the error term $R(n) := \vartheta(n) - n$, then the estimate above gives

$$R(n) = \mathcal{O}(\sqrt{n} \log^2 n). \tag{4.573}$$

Note that $\mathbb{1}_{[n \text{ prime}]} \log n$ can be written in terms of $R(n)$ as

$$\mathbb{1}_{[n \text{ prime}]} \log n = \vartheta(n) - \vartheta(n-1) = 1 + R(n) - R(n-1). \tag{4.574}$$

Rewriting Equation (4.572) using $R(n)$, we get

$$\pi(x) = \sum_{2 \leq n \leq x} \frac{1}{\log n} + \sum_{2 \leq n \leq x} \frac{R(n) - R(n-1)}{\log n}. \tag{4.575}$$

It remains to show that

$$\sum_{2 \leq n \leq x} \frac{R(n) - R(n-1)}{\log n} = \mathcal{O}(\sqrt{x} \log x). \tag{4.576}$$

To show this, we split the sum into two parts:

$$\sum_{2 \leq n \leq x} \frac{R(n) - R(n-1)}{\log n} = \sum_{2 \leq n \leq x} \frac{R(n)}{\log n} - \sum_{2 \leq n \leq x} \frac{R(n-1)}{\log n} \tag{4.577}$$

$$= \sum_{2 \leq n \leq x} \frac{R(n)}{\log n} - \sum_{1 \leq n \leq x-1} \frac{R(n)}{\log(n+1)}. \tag{4.578}$$

Merging the two sums, we get

$$\sum_{2 \leq n \leq x} \frac{R(n) - R(n-1)}{\log n} = \frac{R(x)}{\log x} + \sum_{2 \leq n \leq x-1} R(n) \left(\frac{1}{\log n} - \frac{1}{\log(n+1)} \right) - \frac{R(1)}{\log 2}. \tag{4.579}$$

The boundary terms are $\mathcal{O}(\sqrt{x} \log x)$. For the remaining sum, we have

$$R(n) = \mathcal{O}(\sqrt{n} \log^2 n), \qquad \frac{1}{\log n} - \frac{1}{\log(n+1)} = \mathcal{O}\left(\frac{1}{n \log^2 n} \right). \tag{4.580}$$

Thus each summand is $\mathcal{O}(n^{-1/2})$. By the summation rule for big-$\mathcal{O}$ terms in Lemma 2.32,

$$\sum_{2 \leq n \leq x-1} \mathcal{O}(n^{-1/2}) = \mathcal{O}\left(\sum_{2 \leq n \leq x-1} n^{-1/2} \right) = \mathcal{O}(\sqrt{x}). \tag{4.581}$$

This proves the required bound. □

Therefore, once we prove Theorem 4.107, we can immediately obtain an estimate for $\pi(x)$. In the rest of this subsection, we will prove Theorem 4.107.

▶ Perron's Formula

We now prove Theorem 4.107. The first step is to express the prefix sum $\psi(x)$ of $\Lambda(n)$ in terms of the Dirichlet generating function of $\Lambda(n)$. This uses the same

vertical-line integral as in Lemma 4.87. For each integer $n \geq 1$, consider

$$\frac{1}{2\pi i} \lim_{T\to+\infty} \int_{c-Ti}^{c+Ti} \frac{(x/n)^s}{s}\, \mathrm{d}s. \tag{4.582}$$

Since $(x/n)^s = e^{(\log(x/n))s}$, Lemma 4.87 says that this value is 1 when $n < x$ and 0 when $n > x$. We can use this to filter out the terms of $\Lambda(n)$ with $n < x$.

Lemma 4.109 (Perron formula for $\psi(x)$)**.** For every $c > 1$ and every non-integer $x > 1$,

$$\psi(x) = \lim_{T\to+\infty} \frac{1}{2\pi i} \int_{c-Ti}^{c+Ti} \left(-\frac{\zeta'(s)}{\zeta(s)}\right) \frac{x^s}{s}\, \mathrm{d}s. \tag{4.583}$$

Proof. Let $I := \frac{1}{2\pi i} \int_{c-Ti}^{c+Ti} \left(-\frac{\zeta'(s)}{\zeta(s)}\right) \frac{x^s}{s}\, \mathrm{d}s$. By Theorem 4.106, we rewrite the integrand as

$$\left(-\frac{\zeta'(s)}{\zeta(s)}\right) \frac{x^s}{s} = \sum_{n\geq 1} \frac{\Lambda(n)}{n^s} \frac{x^s}{s} = \sum_{n\geq 1} \Lambda(n) \frac{(x/n)^s}{s}. \tag{4.584}$$

Now we swap the order of the sum and the integral:

$$I = \sum_{n\geq 1} \Lambda(n) \cdot \frac{1}{2\pi i} \int_{c-Ti}^{c+Ti} \frac{(x/n)^s}{s}\, \mathrm{d}s. \tag{4.585}$$

Now fix n. The inner integral is exactly of the form studied in Lemma 4.87, since

$$\frac{(x/n)^s}{s} = \frac{e^{(\log(x/n))s}}{s}. \tag{4.586}$$

Since x is assumed be a non-integer, we have $\log(x/n) \neq 0$ for every integer n. Therefore Lemma 4.87 gives

$$\lim_{T\to+\infty} \frac{1}{2\pi i} \int_{c-Ti}^{c+Ti} \frac{(x/n)^s}{s}\, \mathrm{d}s = \begin{cases} 1 & \text{if } x/n > 1, \\ 0 & \text{if } x/n < 1. \end{cases} \tag{4.587}$$

This directly implies that

$$\lim_{T\to+\infty} I = \sum_{n\geq 1} \Lambda(n) \mathbb{1}_{[x/n>1]} = \sum_{n\leq x} \Lambda(n), \tag{4.588}$$

which is exactly $\psi(x)$. □

▶ Explicit Formula

Perron's formula writes $\psi(x)$ as a vertical-line integral. To compute this integral, we close the vertical line into a large rectangle and apply Cauchy's residue theorem. After the integrals over the extra sides vanish in the limit, the answer is the sum of the residues inside the rectangle. This gives an explicit formula for $\psi(x)$ in terms of the zeros of $\zeta(s)$.

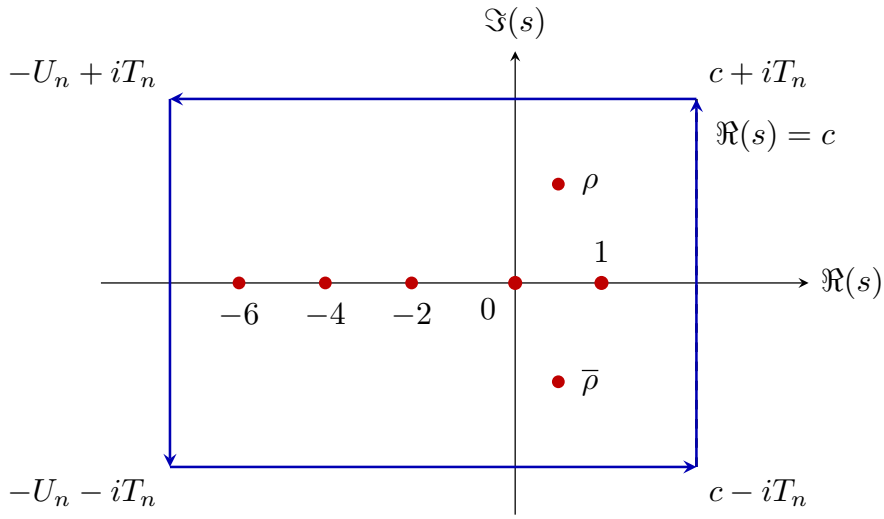


Figure 4.26: A visualization of the proof of the explicit formula for $\psi(x)$.

Theorem 4.110 (Explicit formula for $\psi(x)$)**.** Let $x > 1$ be non-integer. Then

$$\psi(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2}), \tag{4.589}$$

where the sum runs over the non-trivial zeros ρ of $\zeta(s)$, counted with multiplicity, and is understood symmetrically along the imaginary axis:

$$\sum_{\rho} \frac{x^\rho}{\rho} := \lim_{T \to +\infty} \sum_{|\Im(\rho)| \le T} \frac{x^\rho}{\rho}. \tag{4.590}$$

Proof sketch. Define

$$F_x(s) := \left(-\frac{\zeta'(s)}{\zeta(s)}\right) \frac{x^s}{s}. \tag{4.591}$$

Fix any $c > 1$. By Lemma 4.109, $\psi(x)$ is the vertical-line integral of F_x on the line $\Re(s) = c$.

$$\psi(x) = \lim_{T \to +\infty} \frac{1}{2\pi i} \int_{c-Ti}^{c+Ti} F_x(s)\, \mathrm{d}s. \tag{4.592}$$

Singularities. Let us identify the singularities of F_x. Looking at the formula for $F_x(s)$, there are three possible sources:

- **Singularity of $\zeta'(s)$**: the numerator of $F_x(s)$ contains $\zeta'(s)$, so any singularity of $\zeta'(s)$ becomes a singularity of $F_x(s)$. By Lemma 4.104, the only singularity of $\zeta(s)$ is the pole at $s = 1$. Differentiating does not create singularities at new points, so this source only gives one point:

$$s = 1. \tag{4.593}$$

- **Zeros of $\zeta(s)$ in the denominator**: the denominator of $F_x(s)$ contains $\zeta(s)$, so zeros of $\zeta(s)$ become singularities of $F_x(s)$:
$$\{\rho \in \mathbb{C} : \zeta(\rho) = 0\}. \tag{4.594}$$
- **The factor s in the denominator**: the denominator of $F_x(s)$ also contains s, so this factor yields a singularity at
$$s = 0. \tag{4.595}$$

Contour Integration. Let $(T_N)_{N\geq 1}$ and $(U_N)_{N\geq 1}$ be sequences of positive real numbers that tend to $+\infty$. Let γ_N be the contour that encloses the rectangle $[-U_N, c] \times [-T_N, T_N]$. More specifically, $\gamma_N = \gamma_N^{(\mathrm{a})} + \gamma_N^{(\mathrm{b})} + \gamma_N^{(\mathrm{c})} + \gamma_N^{(\mathrm{d})}$, where
$$\gamma_N^{(\mathrm{a})} : [-T_N, T_N] \to \mathbb{C}, \qquad t \mapsto c + it, \tag{4.596}$$
$$\gamma_N^{(\mathrm{b})} : [-c, U_N] \to \mathbb{C}, \qquad t \mapsto -t + iT_N, \tag{4.597}$$
$$\gamma_N^{(\mathrm{c})} : [-T_N, T_N] \to \mathbb{C}, \qquad t \mapsto -U_N - it, \tag{4.598}$$
$$\gamma_N^{(\mathrm{d})} : [-U_N, c] \to \mathbb{C}, \qquad t \mapsto t - iT_N. \tag{4.599}$$
We choose the sequences $(T_N)_{N\geq 1}$ and $(U_N)_{N\geq 1}$ so that the boundary γ_N stays away from the singularities of F_x. By Cauchy's residue theorem,
$$\frac{1}{2\pi i}\int_{\gamma_N} F_x(s)\ \mathrm{d}s = \sum_{z_0} \mathrm{Res}(F_x, z_0), \tag{4.600}$$
where z_0 runs over the singularities of F_x inside γ_N. A careful choice of T_N and U_N can enable the following bound on the top and bottom sides of γ_N:
$$\frac{\zeta'(s)}{\zeta(s)} = \mathcal{O}(\log^2 T_N), \tag{4.601}$$
and the following bound on the left side:
$$\frac{\zeta'(s)}{\zeta(s)} = \mathcal{O}(\log(U_N + T_N)). \tag{4.602}$$
Using these bounds, together with the factors x^s and $1/s$ in $F_x(s)$, the ML inequality shows that the integrals over the top side, bottom side, and left side of γ_N all tend to 0 as $N \to \infty$. Therefore, only the right side remains in the limit:
$$\psi(x) = \lim_{N\to+\infty} \sum_{z_0} \mathrm{Res}(F_x, z_0), \tag{4.603}$$
where, for each N, the variable z_0 runs over the singularities of F_x inside γ_N.

Computing Residues. Computing the residues takes some effort, but the results are very clean:
$$\mathrm{Res}(F_x, 1) = x, \tag{4.604}$$
$$\mathrm{Res}(F_x, \rho) = -m\frac{x^\rho}{\rho} \qquad \text{if } \rho \text{ is a zero of } \zeta(s) \text{ with multiplicity } m, \tag{4.605}$$
$$\mathrm{Res}(F_x, 0) = -\log(2\pi). \tag{4.606}$$

Here we only prove the formula for the residue at zeros of $\zeta(s)$. Near a zero ρ of $\zeta(s)$, we may write

$$\zeta(s) = (s-\rho)^m h(s) \tag{4.607}$$

for some holomorphic function $h(s)$ with $h(\rho) \neq 0$. Then

$$-\frac{\zeta'(s)}{\zeta(s)} = \Big(-\log\zeta(s)\Big)' = \Big(m\log(s-\rho) + \log h(s)\Big)' = -\frac{m}{s-\rho} - \frac{h'(s)}{h(s)}. \tag{4.608}$$

Multiplying by x^s/s, which is holomorphic at ρ, gives

$$\operatorname{Res}(F_x, \rho) = -m\frac{x^\rho}{\rho}. \tag{4.609}$$

Now split the zeros of $\zeta(s)$ into the non-trivial zeros and the trivial zeros. Counting the non-trivial zeros with multiplicity, their contribution is

$$-\sum_\rho \frac{x^\rho}{\rho}. \tag{4.610}$$

The trivial zeros are $-2, -4, -6, \ldots$, so they contribute

$$\sum_{m\geq 1} \frac{x^{-2m}}{2m} = -\frac{1}{2}\log(1-x^{-2}). \tag{4.611}$$

Adding all contributions gives

$$\psi(x) = \underbrace{x}_{s=1} \quad \underbrace{-\sum_\rho \frac{x^\rho}{\rho}}_{\text{non-trivial zeros}} \quad \underbrace{-\log(2\pi)}_{s=0} \underbrace{-\frac{1}{2}\log(1-x^{-2})}_{\text{trivial zeros}}, \tag{4.612}$$

which is the desired formula. □

A variant of the explicit formula is its truncated version.

Lemma 4.111 (Truncated explicit formula)**.** Let $x > 1$ be non-integer, and let $\langle x\rangle$ denote the distance from x to the nearest prime power. Then for every $T \geq 2$,

$$\psi(x) = x - \sum_{|\Im(\rho)|\leq T} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1-x^{-2}) \tag{4.613}$$

$$+ \mathcal{O}\left(\frac{x\log^2(xT)}{T} + (\log x)\min\left\{1, \frac{x}{T\langle x\rangle}\right\}\right), \tag{4.614}$$

where the sum runs over the non-trivial zeros ρ of $\zeta(s)$, counted with multiplicity.

The proof is based on the same contour argument as the proof of Theorem 4.110. The difference is that we stop the Perron integral at a finite height T instead of letting $T \to +\infty$. Thus the horizontal sides of the rectangle and the truncation error from Perron's formula no longer disappear. They are estimated explicitly and become the error term above.

▶ Proof of Theorem 4.107

The explicit formula for $\psi(x)$ in Theorem 4.110 and Lemma 4.111 make the role of the zeros completely explicit. After the main term x, the only large contribution is the sum over the non-trivial zeros of $\zeta(s)$. Therefore, the zeros of $\zeta(s)$ control the error term precisely. If the Riemann hypothesis is true, the real parts of the non-trivial zeros ρ are all $\Re(\rho) = \frac{1}{2}$, and every non-trivial zero ρ only contributes a term of size $x^\rho/\rho = \mathcal{O}(x^{1/2})$. In short, the real parts of non-trivial zeros control the order of the error term!

Proof of Theorem 4.107. It is enough to prove the estimate for non-integer x. If x is an integer, then $\psi(x) = \psi(x + 1/2)$, and replacing x by $x + 1/2$ does not change the size of the stated error term.

Start from the truncated explicit formula in Lemma 4.111 with $T = x$:

$$\psi(x) = x - \sum_{|\Im(\rho)|\le x} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2}) + \mathcal{O}(\log^2 x). \tag{4.615}$$

Here we used

$$\frac{x\log^2(x^2)}{x} + (\log x)\min\left\{1, \frac{1}{\langle x\rangle}\right\} = \mathcal{O}(\log^2 x). \tag{4.616}$$

The two explicit terms after the sum over the non-trivial zeros are bounded as $x \to \infty$. Therefore,

$$\psi(x) = x - \sum_{|\Im(\rho)|\le x} \frac{x^\rho}{\rho} + \mathcal{O}(\log^2 x). \tag{4.617}$$

It remains to bound the sum over the non-trivial zeros. Under the Riemann hypothesis, every non-trivial zero has the form

$$\rho = \frac{1}{2} + i\gamma \tag{4.618}$$

for some real number γ. This means $|x^\rho| = x^{1/2}$, and thus each non-trivial zero contributes a term of size

$$\left|\frac{x^\rho}{\rho}\right| = \frac{\sqrt{x}}{|\rho|}. \tag{4.619}$$

To estimate the truncated sum with $|\gamma| \le x$, divide the zeros into blocks, where the k-th block contains the non-trivial zeros with

$$2^k \le |\gamma| \le 2^{k+1}. \tag{4.620}$$

For each non-trivial zero ρ in the k-th block, we have

$$\left|\frac{x^\rho}{\rho}\right| = \frac{\sqrt{x}}{|\rho|} = \mathcal{O}(x^{1/2}2^{-k}). \tag{4.621}$$

By the bound on the number of non-trivial zeros with $|\Im(\rho)| \le t$ in Lemma 4.104, the number of non-trivial zeros in the k-th block is $\mathcal{O}(2^k k)$. Thus the total contribution from the k-th block is at most

$$\mathcal{O}(x^{1/2}2^{-k}) \cdot \mathcal{O}(2^k k) = \mathcal{O}(x^{1/2}k). \tag{4.622}$$

Summing over all blocks with $2^k \le x$ gives

$$\left| \sum_{|\gamma| \le x} \frac{x^\rho}{\rho} \right| \le \sum_{k=0}^{\lfloor \log_2 x \rfloor} \mathcal{O}(x^{1/2} k) = \mathcal{O}(\sqrt{x} \log^2 x). \tag{4.623}$$

The key point is that the exponent $\frac{1}{2}$ in the Riemann hypothesis becomes the factor $\sqrt{x}$, and the two logarithms come from counting the zeros by height. Combining this with the truncated explicit formula gives $\psi(x) = x + \mathcal{O}(\sqrt{x} \log^2 x)$. $\square$

This proves Theorem 4.107, and therefore also establishes the estimate for $\pi(x)$ in Theorem 4.108. The proof is extremely intricate and far exceeds what is expected in a first-year undergraduate course, but we hope you now appreciate the significance of the Riemann hypothesis. If you are curious and willing to explore this topic further, you may consult textbooks on analytic number theory, such as [Dav80, Apo76].

Since you have already chosen computer science as your major, you probably will not pursue number theory as a career. However, if you decide to study artificial intelligence, one day when your AI becomes powerful enough to tackle many challenging problems with ease, try giving the Riemann hypothesis a shot!

Exercises

4.1. Consider the tiling of a $3 \times n$ rectangle using 1×2 dominoes, where n is an even number. Let a_n be the number of ways to tile the rectangle. Note that $a_0 = 1$ and $a_2 = 3$.

(a) Determine a closed-form formula for the generating function of a_n.

$$A(x) = \sum_{n \text{ even}} a_n x^n.$$

(b) Determine a closed-form formula for a_n.

4.2. Let $(a_n)_{n \geq 1}$ be the sequence that satisfies the recurrence

$$a_n = \frac{2}{n} \sum_{k=1}^{n-1} a_k + n \qquad \text{for all } n \geq 1. \tag{4.624}$$

(a) Determine a closed-form formula for the generating function of a_n.

$$A(x) = \sum_{n \geq 1} a_n x^n.$$

(b) Determine a closed-form formula for a_n. Your answer may involve a Harmonic number.

Note: This is the expected running time of the quicksort algorithm on a randomly shuffled array. This can also be understood as the total sum of the depths of all nodes in a random binary search tree.

4.3. The Stirling numbers of the second kind $\left\{ {n \atop k} \right\}$ are the number of ways to partition a set of n elements into k non-empty subsets. The order of numbers in each subset is not important, and the order of subsets is also not important. For example, $\left\{ {3 \atop 2} \right\} = 3$ because the set $\{1, 2, 3\}$ can be partitioned into 2 subsets in 3 ways:

$$\{\{1\}, \{2, 3\}\}, \quad \{\{2\}, \{1, 3\}\}, \quad \{\{3\}, \{1, 2\}\}.$$

Some other special values are $\left\{ {n \atop k} \right\} = 0$ if $n < k$, $\left\{ {n \atop n} \right\} = 1$, $\left\{ {n \atop n-1} \right\} = \binom{n}{2}$, $\left\{ {n \atop 2} \right\} = 2^{n-1} - 1$ and $\left\{ {n \atop 1} \right\} = 1$. Fix $k \geq 1$. Determine a closed-form formula for the exponential generating function of $\left\{ {n \atop k} \right\}$.

$$\widehat{S}_k(x) = \sum_{n \geq 0} \left\{ {n \atop k} \right\} \frac{x^n}{n!}.$$

4.4. Prove the following identities for all $x \in \mathbb{R}$:

(a) $\sinh(ix) = i \sin x$

(b) $\cosh(ix) = \cos x$

(c) $\sinh(x) = -i \sin(ix)$

(d) $\cosh(x) = \cos(ix)$

Here, $\sinh(x) = \frac{e^x - e^{-x}}{2}$ and $\cosh(x) = \frac{e^x + e^{-x}}{2}$.

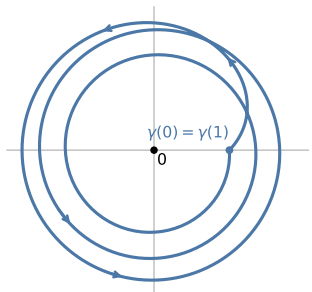
4.5. In Example 4.29, we showed that the complex conjugate function $f(z) = \overline{z}$ is continuous everywhere, but it is not differentiable at 0. Prove that f is not differentiable at any point, not just 0.

4.6. Evaluate the following integrals:

(a) $\int_1^3 (t^2 - i)^2 \, \mathrm{d}t$

(b) $\int_0^{\pi/4} e^{-2it} \, \mathrm{d}t$

(c) Let $\gamma : [0, 1] \to \mathbb{C}, t \mapsto \gamma(t) = (1 + 3t(1 - t))e^{i(t-t^2+6t^3)\pi}$ be a closed curve. Compute $\int_\gamma (z^{-2} + z^{-1} + 1 + z) \, \mathrm{d}z$.



Hint: Try to find an antiderivative whenever possible. See also Examples 4.47 to 4.49.

4.7. Show that the following variant of the identity theorem (Theorem 4.61) is true, or give a counterexample. Let $f : S_1 \to \mathbb{C}$ and $g : S_2 \to \mathbb{C}$ be holomorphic functions on two connected open sets S_1 and S_2. If there exists a non-empty open set $U \subseteq S_1 \cap S_2$ such that $f(z) = g(z)$ for every $z \in U$, then $f(z) = g(z)$ for every $z \in S_1 \cap S_2$.

Note: For this problem, you do not have to be fully rigorous. Just state the result and give a brief explanation.

4.8. Determine the radius of convergence of the Taylor series of each of the following functions around 0:

(a) $f(z) = \dfrac{1}{(z - 5)(z^2 + 4)}$

(b) $\tanh(z) = \dfrac{e^z - e^{-z}}{e^z + e^{-z}}$

(c) $\operatorname{softplus}(z) = \operatorname{Log}(1 + e^z)$

Note: For convenience, you may simply state that a point $z \in \mathbb{C}$ is a singularity of a function, without proving it from the definition.

4.9. The error function $\operatorname{erf} : \mathbb{R} \to \mathbb{R}$ is defined as

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \, \mathrm{d}t.$$

Find a power series $f(z) = \sum_{n \geq 0} a_n z^n$ such that the series converges absolutely for all $z \in \mathbb{C}$ and $f(x) = \operatorname{erf}(x)$ for all $x \in \mathbb{R}$.

Note: This function is useful in probability theory and statistics because it is closely related to the cumulative distribution function of Gaussian distributions.

4.10. In this problem, we will guide you through the proof of Laurent's theorem (Theorem 4.66). Let f be holomorphic on a punctured disk $D_r(z_0) \setminus \{z_0\}$ for some $z_0 \in \mathbb{C}$ and some $r > 0$. Fix $z \in D_r(z_0) \setminus \{z_0\}$. Let $\epsilon > 0$ be an arbitrary positive number such that $z_0 \notin D_\epsilon(z)$.

(a) Prove the following identity:

$$f(z) = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(w)}{w - z} \, \mathrm{d}w - \frac{1}{2\pi i} \int_{C_\epsilon(z_0)} \frac{f(w)}{w - z} \, \mathrm{d}w.$$

(b) Show that for any $w \in D_\epsilon(z_0)$, we have

$$\frac{f(w)}{w - z} = -\sum_{n \geq 1} f(w)(w - z_0)^{n-1}(z - z_0)^{-n}.$$

(c) Let $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 1}$ be the following sequences:

$$a_n = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(w)}{(w - z_0)^{n+1}} \, \mathrm{d}w,$$

$$b_n = \frac{1}{2\pi i} \int_{C_\epsilon(z_0)} f(w)(w - z_0)^{n-1} \, \mathrm{d}w.$$

Prove that

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b_n (z - z_0)^{-n}.$$

(d) Prove that if $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$ for some sequence $(a_n)_{n \in \mathbb{Z}}$, then for all $n \in \mathbb{Z}$,

$$a_n = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(w)}{(w - z_0)^{n+1}} \, \mathrm{d}w.$$

As the right-hand side only depends on f and z_0, this automatically proves the uniqueness of the coefficients in the Laurent series.

Note: For this problem, you may freely swap the order of integration and summation, without providing any justification.

4.11. Let $f : S \to U$, $g : U \to S$ be holomorphic functions on two open subsets S and U of $\mathbb{C}$ such that $f(g(z)) = z$ for all $z \in U$ and $g(f(w)) = w$ for all $w \in S$. That is, f and g are inverse functions. Assume $0 \in S$ and $f(0) = 0$. Prove that there exists $r > 0$ such that for all $z \in D_r(0)$,

$$g(z) = \sum_{n \geq 1} g_n z^n,$$

where $g_n := \frac{1}{n} \operatorname{Res}(h_n, 0)$ and h_n is the following function:

$$h_n(w) = \left(\frac{1}{f(w)} \right)^n.$$

Note: This is known as the *Lagrange inversion theorem*.

Hint: You may want to use integration by parts:

$$\int_a^b F(t)G'(t)\ \mathrm{d}t = [F(t)G(t)]\Big|_a^b - \int_a^b F'(t)G(t)\ \mathrm{d}t.$$

4.12. Let t_n be the number of trees with n vertices labeled from 1 to n. Here, a tree is a connected (undirected) graph that contains no cycles. Two trees are considered the same if for every pair of vertices u and v, either there is an edge between u and v in both trees, or there is no edge between u and v in both trees. Further, let a_n be the number of rooted trees with n vertices labeled from 1 to n. That is, the number of pairs (T, r) where T is a tree and r is a vertex of T, designated as the root. Obviously, $a_n = n \cdot t_n$. Let $\widehat{A}(x)$ be the exponential generating function of a_n:

$$\widehat{A}(x) = \sum_{n \geq 1} a_n \frac{x^n}{n!}.$$

(a) Prove that

$$\widehat{A}(x) = xe^{\widehat{A}(x)}.$$

(b) Determine a closed-form formula for a_n.

Hint: Use the Lagrange inversion theorem.

(c) Determine a closed-form formula for t_n.

Note: The formula for t_n is known as *Cayley's formula*.

4.13. You are designing an AI for a game based on tree search. For each state of the game, there are m possible actions, each of which leads to a new state of the game. Starting from the current state, the algorithm explores the game tree according to the following procedure:

```
def search(state):
    if should_stop(state):
        return estimate_value(state)
    else:
        value = 0
        for action in all_actions(state):
            child_state = apply_action(state, action)
            value = max(value, search(child_state))
        return value
```

That is, at each state, the algorithm either stops and evaluates the current state, or expands it into exactly m child states and then recursively explores each of them.

In practice, the stopping decision may depend on various factors, such as the depth of the search, a time limit, or the quality of the estimate. In this problem, however, we assume a simple randomized model: at each state, the algorithm continues exploring with probability p, and stops with

probability $1 - p$. All decisions are made independently across different states. Let T be the total number of states explored by the algorithm.

(a) Let $G(z)$ be the probability generating function of T. Show that

$$G(z) = z(1 - p + pG(z)^m).$$

(b) Determine a closed-form formula for $\Pr[T = t]$ for every $t \geq 0$.
Hint: Use the Lagrange inversion theorem.

4.14. Use Cauchy's Residue Theorem (see Remark 4.75) to evaluate the following integrals:

(a) $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 4}\, \mathrm{d}x$

(b) $\displaystyle\int_{-\infty}^{\infty} \frac{2x^2 - 1}{x^4 + 5x^2 + 4}\, \mathrm{d}x$

(c) $\displaystyle\int_{-\infty}^{\infty} \frac{e^{x/3}}{e^x + 1}\, \mathrm{d}x$

4.15. For every odd integer $n \geq 1$, a permutation $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \mathrm{Perm}(n)$ is called an *up-down permutation* if

$$\sigma_1 < \sigma_2 > \sigma_3 < \sigma_4 > \cdots < \sigma_{n-1} > \sigma_n.$$

For example, the permutation $(2, 5, 4, 7, 1, 6, 3)$ is an up-down permutation, but the permutation $(1, 3, 2, 4, 6, 5, 7)$ is not. Let a_n be the number of up-down permutations of $[n]$. Then $a_1 = 1$ and $a_3 = 2$.

(a) Determine a closed-form formula for the exponential generating function of a_n.

$$A(x) = \sum_{n \text{ odd}} a_n \frac{x^n}{n!}.$$

(b) Use Cauchy's Residue Theorem (see Remark 4.80) to express a_n as a simple function of the Riemann zeta function.

4.16. Write a complete proof of Lemma 4.87.

4.17. In this problem, we will guide you through the proof of Cramér's theorem (Theorem 3.36). For any probability mass function p, we write $p^{(\lambda)}$ for the probability mass function of the λ-tilted distribution of p. That is,

$$p^{(\lambda)}(x) = \frac{p(x)e^{\lambda x}}{\sum_y p(y)e^{\lambda y}}, \quad \forall x \in \mathbb{R}.$$

Let $\mathcal{D}$ be a probability distribution with probability mass function p, minimum value ℓ, maximum value u, mean μ, MGF $M(t)$ and rate function $I(a)$. Let $X_1, \ldots, X_n$ be i.i.d. random variables drawn from $\mathcal{D}$, and let $X^{(n)} := \frac{1}{n}\sum_{i=1}^n X_i$ be their sample average.

For simplicity, we restrict attention to the case $a \in (\mu, u)$. Fix such an a, and our goal is to show that as $n \to \infty$,

$$\Pr[X^{(n)} \geq a] = e^{-n(I(a)+o(1))}.$$

(a) Let $\lambda \in \mathbb{R}$ be a parameter. Let $Y_1, \ldots, Y_n$ be i.i.d. random variables, each with distribution $q = p^{(\lambda)}$. Let p_n be the probability mass function of $X^{(n)} := \frac{1}{n}\sum_{i=1}^n X_i$ and q_n be the probability mass function of $Y^{(n)} := \frac{1}{n}\sum_{i=1}^n Y_i$. Show that for all $x \in \mathbb{R}$,

$$q_n(x) = p_n^{(n\lambda)}(x).$$

(b) Prove that for any $\lambda \in \mathbb{R}$, the corresponding $Y^{(n)}$ satisfies

$$\Pr[X^{(n)} \geq a] = M(\lambda)^n \cdot \mathbb{E}\left[\mathbb{1}_{[Y^{(n)} \geq a]} e^{-n\lambda Y^{(n)}}\right].$$

(c) For every $\epsilon > 0$ such that $a + \epsilon < u$, show that it is possible to find a value $\lambda = \lambda(\epsilon) > 0$ such that $\mathbb{E}[Y^{(n)}] = a + \epsilon$. Then show that

$$\lim_{n\to\infty} \Pr[a \leq Y^{(n)} \leq a + \epsilon + \epsilon/\lambda] = 1.$$

Note: This means $Y^{(n)}$ concentrates around $a + \epsilon$, and thus it is almost trivial to give a tight lower bound for $\mathbb{E}\left[\mathbb{1}_{[Y^{(n)} \geq a]} e^{-n\lambda Y^{(n)}}\right]$.

(d) Prove Cramér's theorem for $a \in (\mu, u)$. Since Chernoff bounds already give us the upper bound $\Pr[X^{(n)} \geq a] \leq e^{-nI(a)}$, you only need to show the following lower bound:

$$\log \Pr[X^{(n)} \geq a] \geq -n(I(a) + o(1)), \tag{4.625}$$

or equivalently,

$$\liminf_{n\to\infty} \frac{1}{n} \log \Pr[X^{(n)} \geq a] \geq -I(a). \tag{4.626}$$

Hint: First show that for every $\epsilon > 0$,

$$\liminf_{n\to\infty} \frac{1}{n} \log \Pr[X^{(n)} \geq a] \geq -I(a + \epsilon) - \epsilon.$$

4.18. Let $X_1, \ldots, X_n$ be i.i.d. random variables drawn from a distribution with mean μ and variance σ^2. Consider the following limit for every $\alpha \in \mathbb{R}$:

$$\lim_{n\to\infty} K_{Z_\alpha^{(n)}}(t), \quad \text{where} \quad Z_\alpha^{(n)} := \frac{1}{n^\alpha} \sum_{i=1}^n (X_i - \mu).$$

Determine the range of α for which the limit exists and is finite.

4.19. Let $X^{(n)} \sim \text{Binomial}(n, p_n)$, where $p_n = \lambda/n$ for some fixed $\lambda > 0$. Directly compute the following two limits and show that they are the same:

(a) $\lim_{n\to\infty} K_{X^{(n)}}(t)$; (b) $\log\left(\sum_{k\geq 0} e^{kt} \lim_{n\to\infty} \Pr[X^{(n)} = k]\right)$.

Note: The limiting distribution is known as the *Poisson distribution*, which describes the number of rare events occurring in a large number of independent trials. See also Exercise 5.3.

Chapter 5

Continuous Probability

We began with discrete probability because it captures many of the first examples one meets in probability theory. Coin flips, dice rolls, hashing, and many randomized algorithms are essentially discrete in nature. In this setting, probabilities are computed by a finite sum over discrete points, which makes it easy to illustrate many basic concepts, such as linearity of expectation and concentration inequalities.

However, in many other cases, the probability space is better described as continuous rather than discrete. Neural network weights, embedding vectors, and regression outputs all live in spaces built from real numbers, and their possible values cannot be listed as discrete points. Instead, we need to talk about probabilities of intervals, regions, and high-dimensional sets, and we need expectations defined by integrals rather than finite sums.

This chapter will develop the language needed for continuous probability. We will begin with probability density functions and the Gaussian distribution, which will serve as our main example in both one dimension and higher dimensions. Working through these examples will help us transfer the intuition we developed for discrete probability to the continuous setting. After that, we will introduce a little bit of measure theory, which will allow us to describe discrete and continuous probability spaces within a unified framework.

5.1 Leaving the Finite World

In a finite discrete probability space, probabilities and expectations are computed by sums over individual outcomes ω. More specifically, in Definition 2.1, we define a finite discrete probability space as a pair (Ω, p) consisting of a finite sample space Ω together with a probability mass function p on Ω. If $A \subseteq \Omega$ is an event, then by Definition 2.3,

$$\Pr[A] = \sum_{\omega \in A} p(\omega). \tag{5.1}$$

For a random variable X on the same probability space, by Definition 2.82,

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) p(\omega). \tag{5.2}$$

When Ω is finite, these sums always contain only finitely many terms, so there is no convergence issue. The first question is what happens when we keep the discrete probability space viewpoint but allow the sample space to be infinite.

5.1.1 Example: Waiting Time Until the First Head

A basic example is the number of coin tosses until the first heads appears. Flip a fair coin repeatedly, and let T be the number of tosses until the first heads appears. Then the possible values of T are

$$1, 2, 3, 4, \ldots, \tag{5.3}$$

and for every $k \geq 1$,

$$\Pr[T = k] = 2^{-k}. \tag{5.4}$$

This is exactly a geometric random variable, which has also appeared in Example 3.26 and Exercise 2.23. Here the sample space is infinite, but probabilities can still be computed by summing over the listed outcomes. For example, the probability that the first heads appears on an even toss is

$$\Pr[T \text{ is even}] = \sum_{j \geq 1} \Pr[T = 2j] = \sum_{j \geq 1} (1/4)^j = \frac{1/4}{1 - 1/4} = \frac{1}{3}. \tag{5.5}$$

The expectation of T is given by an infinite sum:

$$\mathbb{E}[T] = \sum_{k \geq 1} k 2^{-k}. \tag{5.6}$$

To evaluate this sum, we use a trick: subtract 2 times the series by itself, then

$$2\mathbb{E}[T] - \mathbb{E}[T] = \sum_{k \geq 1} k 2^{-k+1} - \sum_{k \geq 1} k 2^{-k} \tag{5.7}$$

$$= \sum_{k \geq 0} (k+1) 2^{-k} - \sum_{k \geq 1} k 2^{-k} \tag{5.8}$$

$$= 1 \cdot 2^0 + \sum_{k \geq 1} 2^{-k} \tag{5.9}$$

$$= 1 + 1 = 2. \tag{5.10}$$

Wait, do we really need to get so complicated? Actually, the tail-sum formula (Theorem 2.88) also holds when the sample space is infinite:

$$\mathbb{E}[T] = \sum_{t \geq 0} \Pr[T > t] = \sum_{t \geq 0} 2^{-t} = 2. \tag{5.11}$$

As you can see, most of your skills for dealing with finite discrete probability spaces can easily transfer to the infinite case.

However, an important difference is that the expectation of a random variable may be infinite! For example, let $X = 2^T$. Then

$$\mathbb{E}[X] = \sum_{t\geq 0} 2^t \Pr[T > t] = \sum_{t\geq 0} 2^t 2^{-t} = \sum_{t\geq 0} 1 = +\infty. \tag{5.12}$$

A more tricky example is $X = (-2)^T$. Then

$$\mathbb{E}[X] = \sum_{t\geq 0} (-2)^t \Pr[T > t] = \sum_{t\geq 0} (-2)^t 2^{-t} = \sum_{t\geq 0} (-1)^t = \text{undefined}. \tag{5.13}$$

Here, we leave the expectation undefined because the infinite series $\sum_{t\geq 0}(-1)^t$ is divergent. If we insist on assigning a value to this sum, say S, then different algebraic manipulations lead to incompatible answers. On the one hand,

$$S = (1-1) + (1-1) + (1-1) + \cdots = \sum_{j\geq 0} (1-1) = 0 \quad \Rightarrow \quad S = 0. \tag{5.14}$$

On the other hand,

$$S = 1 - (1 - 1 + 1 - 1 + 1 - \cdots) = 1 - \sum_{t\geq 0} (-1)^t = 1 - S \quad \Rightarrow \quad S = \frac{1}{2}. \tag{5.15}$$

So there is no way to assign a proper value to $\sum_{t\geq 0}(-1)^t$.

5.1.2 Discrete Probability Spaces Beyond the Finite Case

A key reason why this example is easy to work with is that the sample space is "discrete": all possible outcomes can be listed as a sequence $k = 1, 2, 3, \ldots$. Thus, we can sum over all possible outcomes as usual, except that the sum now contains infinitely many terms rather than finitely many. In general, we say that a set is *countable* if its elements can be listed as a sequence.

Definition 5.1 (Countable and uncountable sets)**.** A set Ω is *countable* if it is finite, or if there exists a sequence $(a_n)_{n\geq 0}$ such that

$$\Omega = \{a_0, a_1, a_2, a_3, \ldots\}. \tag{5.16}$$

A set that is not countable is *uncountable*.

Example 5.2. The natural numbers $\mathbb{N}$ are countable, since they can be listed as

$$0, 1, 2, 3, \ldots. \tag{5.17}$$

The even natural numbers are countable, since they can be listed as

$$0, 2, 4, 6, \ldots. \tag{5.18}$$

The integers $\mathbb{Z}$ are countable, since they can be listed as

$$0, 1, -1, 2, -2, 3, -3, \ldots. \tag{5.19}$$

Example 5.3 (Tuples are countable)**.** For every fixed integer $k \geq 1$, the set of all k-tuples of natural numbers,

$$\mathbb{N}^k := \{(x_1, x_2, \ldots, x_k) : x_1, \ldots, x_k \in \mathbb{N}\}, \tag{5.20}$$

is countable.

One way to enumerate its elements is to first list all tuples whose coordinates sum to 0, then those whose coordinates sum to 1, then those whose coordinates sum to 2, and so on. For each fixed sum, there are only finitely many such tuples.

Note that it is wrong to first enumerate all tuples starting with $x_1 = 0$, since there are infinitely many such tuples and the enumeration would never move on to tuples with $x_1 \neq 0$.

Example 5.4 (Rational numbers are countable)**.** The set of rational numbers $\mathbb{Q} := \{\frac{a}{b} : a \in \mathbb{Z}, b \in \mathbb{Z} \setminus \{0\}\}$ is countable. This can be proved based on two facts:

- Both $\mathbb{Z}$ and $\mathbb{Z} \setminus \{0\}$ can be listed as sequences, so every rational number can be encoded as a pair (or 2-tuple) in $\mathbb{N}^2$.
- Any subset of a countable set is countable.

Example 5.5 (Real numbers are uncountable)**.** Not every set is countable. A famous counterexample is that the set of real numbers is uncountable. To see this, it is enough to show that the interval $(0, 1)$ cannot be listed. Assume to the contrary that $(0, 1)$ can be listed as a sequence,

$$(0, 1) = \{a_1, a_2, a_3, \ldots\}. \tag{5.21}$$

Write each a_n in decimal form. For example, $\frac{1}{7} = 0.142857142857\cdots$. Note that some numbers have two decimal forms: $0.15 = 0.149999999\cdots$. In this case, we choose the one that does not end with infinitely many 9s.

Now construct a number $x = 0.b_1b_2b_3\cdots$ by choosing each b_n to be a number in $\{1, 2\}$ that is different from the n-th decimal digit of a_n.

$$a_1 = 0.d_{11}d_{12}d_{13}\cdots \tag{5.22}$$
$$a_2 = 0.d_{21}d_{22}d_{23}\cdots \tag{5.23}$$
$$a_3 = 0.d_{31}d_{32}d_{33}\cdots \tag{5.24}$$
$$\vdots \quad \vdots \tag{5.25}$$

Then x differs from a_n in the n-th decimal digit for every n, so x is not on the list. But this contradicts the fact that $x \in (0, 1)$. Therefore, the set $\mathbb{R}$ of real numbers is uncountable. This proof technique is known as *Cantor diagonalization* [Can91].

We now return to probability spaces. If the sample space is countable, then its outcomes can be listed as

$$\Omega = \{\omega_1, \omega_2, \omega_3, \dots\}. \tag{5.26}$$

Thus, we can naturally extend the definition of a finite discrete probability space to countable sample spaces: the only change is that the finite sum over outcomes is replaced by an infinite sum. Readers may compare the following definitios with Definition 2.1 and Definition 2.3.

Definition 5.6 (Discrete probability space)**.** A *discrete probability space* is a pair $\mathcal{P} = (\Omega, p)$ consisting of:

- A non-empty countable set Ω, called the *sample space*, whose elements represent all possible outcomes of an experiment.
- A function $p : \Omega \to [0, 1]$, called the *probability mass function*, which assigns probabilities to outcomes and satisfies $\sum_{\omega\in\Omega} p(\omega) = 1$.

Definition 5.7 (Event)**.** An event A is a subset of Ω ($A \subseteq \Omega$), and its probability is defined as

$$\Pr[A] := \sum_{\omega\in A} p(\omega). \tag{5.27}$$

For both the finite case and the countable case, we list the outcomes and assign a non-negative probability mass to each one. The only change is how we interpret the sum $\sum_{\omega\in A}$. Since A is countable, we may list its elements as a sequence $(\omega_n)_{n\geq 0}$ and define

$$\sum_{\omega\in A} p(\omega) := \sum_{n\geq 0} p(\omega_n). \tag{5.28}$$

Note that this definition does not depend on the particular order in which we list the elements of A. This is because for any sum $\sum_{n\geq 0} a_n$ of non-negative values $a_n \geq 0$, we may rewrite the sum in a form that does not depend on the order:

$$\sum_{n\geq 0} a_n = \sup_{\substack{F\subseteq\mathbb{N} \\ F \text{ finite}}} \sum_{n\in F} a_n. \tag{5.29}$$

That is, an infinite sum of a sequence of non-negative values can be seen as the supremum of all finite sums of the sequence. No matter how we permute the sequence $(a_n)_{n\geq 0}$, the formula on the right-hand side always gives the same value. This ensures that we can get the same probability of an event regardless of the order in which we list the elements of the event. We will see this idea again when we introduce Lebesgue integration for continuous probability spaces.

Example 5.8. For the coin tossing example above, the sample space can be the set of all `01` sequences ending with a `1`:

$$\Omega = \{\texttt{0}^n\texttt{1} : n \geq 0\}, \tag{5.30}$$

where $\texttt{0}^n$ means n consecutive tails and `1` means a head. For any string $s \in \Omega$, the probability mass function is

$$p(s) = 2^{-|s|}, \tag{5.31}$$

where $|s|$ is the length of the string s. The waiting time T is defined as the following random variable:

$$T : \Omega \to \mathbb{R}, \qquad T(s) = |s|. \tag{5.32}$$

The event that T is even is thus given by

$$A = \{s \in \Omega : |s| \text{ is even}\}. \tag{5.33}$$

The probability of this event is

$$\Pr[A] = \sum_{s \in A} p(s) = \sum_{n \geq 0} 2^{-2n} = \frac{1}{3}. \tag{5.34}$$

While the probability of an event is always well-defined, this is not the case for the expectation of a random variable. We have already seen in the coin tossing example that the expectation of 2^T is infinite, and the expectation of $(-2)^T$ is undefined.

If a random variable is non-negative, then it is easy to make its expectation well-defined: we just need to allow the expectation to be $+\infty$. More formally, for a non-negative random variable X, we define its expectation as

$$\mathbb{E}[X] := \sum_{\omega \in \Omega} X(\omega) p(\omega) \in \mathbb{R} \cup \{+\infty\}. \tag{5.35}$$

In general, a random variable can take both positive and negative values, such as $(-2)^T$. It is not appropriate to just leave all such sums undefined, since obviously $\mathbb{E}[(-1)^T]$ should be set to $\sum_{k \geq 1} (-1)^k 2^{-k} = -\frac{1}{3}$.

In probability theory, a standard way is to separate the positive and negative parts of a random variable, and then define the expectation as the difference of the expectations of the positive and negative parts. More formally, for any real number x, write

$$[x]_+ := \max\{x, 0\}, \qquad [x]_- := \max\{-x, 0\}. \tag{5.36}$$

Then the expectation of X is defined as follows.

Definition 5.9 (Expectation)**.** Let X be a random variable on a discrete probability space (Ω, p). The expectation of X is defined as

$$\mathbb{E}[X] := \sum_{\omega\in\Omega} [X(\omega)]_+ p(\omega) - \sum_{\omega\in\Omega} [X(\omega)]_- p(\omega), \tag{5.37}$$

provided the two sums are not both $+\infty$. If both sums are finite, then $\mathbb{E}[X]$ is finite. If exactly one of them is $+\infty$, then $\mathbb{E}[X]$ is $+\infty$ or $-\infty$. If both are $+\infty$, then the expectation is left undefined.

Note that if the expectation of X is well-defined, then

$$\mathbb{E}[X] = \sum_{\omega\in\Omega} X(\omega)p(\omega), \tag{5.38}$$

regardless of the order in which we list the elements of Ω. This means that in most cases, we should just use the formula $\sum_{\omega\in\Omega} X(\omega)p(\omega)$, but we need to keep in mind that in some rare cases, even if we obtain a finite value from this formula, the expectation may still be undefined.

Example 5.10. To compute the expectation of $(-2)^T$, we need to separate the positive and negative parts of $(-2)^T$:

$$\mathbb{E}[(-2)^T] = \sum_{\substack{k\geq 1\\ k \text{ even}}} [(-2)^k]_+ 2^{-k} - \sum_{\substack{k\geq 1\\ k \text{ odd}}} [(-2)^k]_- 2^{-k} \tag{5.39}$$

$$= \sum_{\substack{k\geq 1\\ k \text{ even}}} 1 - \sum_{\substack{k\geq 1\\ k \text{ odd}}} 1 \tag{5.40}$$

$$= (+\infty) - (+\infty). \tag{5.41}$$

So the expectation of $(-2)^T$ is undefined.

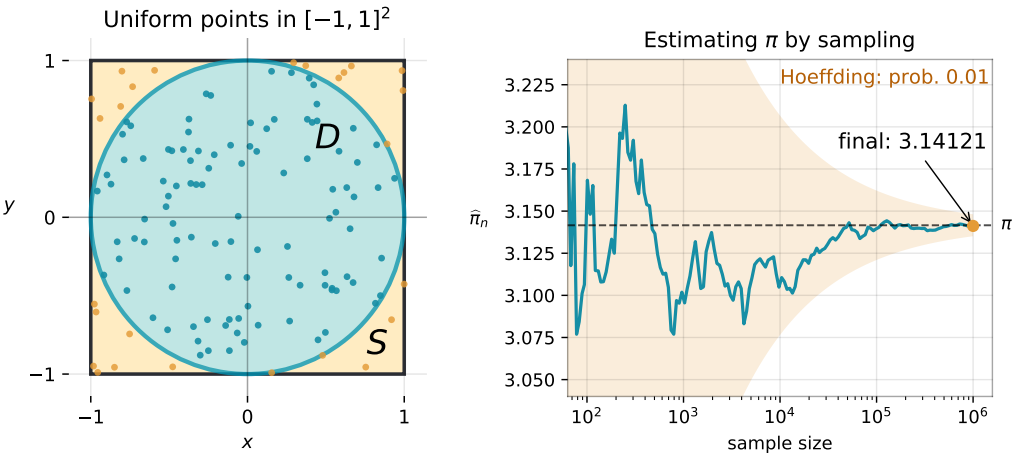
In summary, discrete probability spaces extend naturally from finite sample spaces to countably infinite ones, but we just need to be careful that the expectation of a random variable may be $+\infty$, $-\infty$, or undefined.

5.1.3 Example: Monte Carlo Estimation of Area

Now we go beyond the discrete case and begin with our first example of a continuous probability space.

▶ Problem Statement

Suppose you want to estimate the area of a region A inside a square $S = [-1, 1]^2$ in the plane. The region A could be as simple as the unit disk $D := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$, or a lake or a country drawn on a map. Sometimes it is not easy to compute the area of A directly. Or you know that the area of the unit disk is π but you just want to verify that $\pi \approx 3.14$. Then you can use the following Monte Carlo method to estimate the area of A.



(a) Uniform samples in the square. (b) The estimate of π from up to 10^6 samples.

Figure 5.1: Monte Carlo area estimation for the unit disk. The probability of lying in the disk is $\mu(D)/\mu(S) = \frac{\pi}{4}$.

First, you sample n points uniformly from S, say $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n \in S$. Then, we count the number of points that fall into A, say there are m such points. Finally, we estimate the area of A as

$$\mu(A) \approx \hat{\mu} := \frac{m}{n} \cdot \mu(S), \tag{5.42}$$

where we write $\mu(A)$ for the true area of A.

▶ Analysis

Why is this a good estimate? Let E_i be the event that the i-th point $\boldsymbol{z}_i$ falls into A. Then although we have not defined the probability of an event in the continuous case, intuitively the probability of E_i should be

$$\Pr[E_i] = \frac{\mu(A)}{\mu(S)}. \tag{5.43}$$

Now we define $X_i := \mathbb{1}_{E_i}$ as the indicator random variable of E_i. Then X_i is a Bernoulli random variable with parameter $p = \frac{\mu(A)}{\mu(S)}$, and $X := \frac{m}{n}$ is the sample average of n i.i.d. Bernoulli random variables $X_1, \ldots, X_n$.

Applying Hoeffding's inequality (Theorem 3.44) to the sample average X gives

$$\Pr\left[|X - p| \geq \epsilon\right] \leq 2e^{-2n\epsilon^2}. \tag{5.44}$$

Equivalently,

$$\Pr\left[|X \cdot \mu(S) - p \cdot \mu(S)| \geq \epsilon \cdot \mu(S)\right] \leq 2e^{-2n\epsilon^2}. \tag{5.45}$$

Since $X \cdot \mu(S) = \hat{\mu}$ is the estimated area and $p \cdot \mu(S) = \mu(A)$ is the true area,

$$\Pr\Big[|\hat{\mu} - \mu(A)| \geq \epsilon \cdot \mu(S)\Big] \leq 2e^{-2n\epsilon^2}. \tag{5.46}$$

This means that the estimated area has error at most $\epsilon \cdot \mu(S)$ with probability at least $1 - 2e^{-2n\epsilon^2}$. If you still remember the definition of Monte Carlo methods from Section 3.1.2, this error bound explains why the method is called "Monte Carlo" estimation: the algorithm always runs in a fixed time, and with high probability, we can get a good estimate of the area.

▶ A Technical Issue

While the above analysis looks quite straightforward, one question remains: *how can we describe the sample points $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n$ formally in probability theory?*

Clearly, these points can take arbitrary values in the square S. According to our discussion in Example 5.5, the square S is uncountable, so we cannot list all possible values of the sample points and assign a probability mass to each of the possible values.

Even if we ignore this issue, we will quickly run into trouble when we try to reason as in the discrete case. For any fixed point $\boldsymbol{z} \in S$, we should have

$$\Pr[\boldsymbol{z}_i = \boldsymbol{z}] = 0, \tag{5.47}$$

since no individual point should have positive area. If we compute $\Pr[\boldsymbol{z}_i \in A]$ by summing over all points in A, we get

$$\Pr[\boldsymbol{z}_i \in A] \stackrel{?}{=} \sum_{\boldsymbol{z} \in A} \Pr[\boldsymbol{z}_i = \boldsymbol{z}] = \sum_{\boldsymbol{z} \in A} 0 = 0, \tag{5.48}$$

which is clearly wrong when A has positive area.

This is a technical issue, but it is important to understand that despite the simplicity of Monte Carlo estimation, we need a fundamentally different kind of probability space to describe and analyze this method.

5.1.4 Continuous Probability Spaces

The term *continuous probability space* is an informal phrase that refers to a probability space in which probabilities are computed by integrals rather than sums. Now we introduce the most common way to build such a space: specifying a *probability density function* (PDF).

Definition 5.11 (Probability density function)**.** A function $f : \mathbb{R}^d \to [0, \infty)$ is called a *probability density function* (PDF) on $\mathbb{R}^d$ if

$$\int_{\mathbb{R}^d} f(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x} = 1. \tag{5.49}$$

Here the sample space is $\mathbb{R}^d$, the d-dimensional Euclidean space. An event is naturally described as a subset $A \subseteq \mathbb{R}^d$. Note that $f(\boldsymbol{x})$ is not the probability that the outcome ω equals $\boldsymbol{x}$. Instead, it should be interpreted as follows: if B is a small region around $\boldsymbol{x}$, then

$$\Pr[B] \approx f(\boldsymbol{x}) \cdot \mathrm{vol}(B), \tag{5.50}$$

where $\mathrm{vol}(B)$ is the volume of B. So $f(\boldsymbol{x})$ is the probability "mass" $\Pr[B]$ divided by the "volume" of the region B, which is analogous to density in physics.

Whenever we need to compute the probability of an event A, we take an integral rather than a sum of the density over A:

$$\Pr[A] := \int_A f(x)\ \mathrm{d}x, \tag{5.51}$$

where the integral $\int_A$ is defined as the integral of f over the region A:

$$\int_A f(x)\ \mathrm{d}x := \int_{\mathbb{R}^d} \mathbb{1}_{[x\in A]} f(x)\ \mathrm{d}x \tag{5.52}$$

Similarly, a random variable X is naturally described as a function $X : \mathbb{R}^d \to \mathbb{R}$. The expectation of X is defined as the integral of X with respect to the PDF:

$$\mathbb{E}[X] := \int_{\mathbb{R}^d} X(x) f(x)\ \mathrm{d}x, \tag{5.53}$$

Example 5.12. To setup a probability space for Section 5.1.3, we define $S = [-1, 1]^2$ and

$$\Omega = \mathbb{R}^2, \qquad f(x, y) = \begin{cases} \frac{1}{\mu(S)} & \text{if } (x, y) \in S, \\ 0 & \text{otherwise.} \end{cases} \tag{5.54}$$

It is easy to verify that $\int_{\mathbb{R}^2} f(x, y)\ \mathrm{d}x\ \mathrm{d}y = 1$, so f is a PDF. The values of x and y stand for the coordinates of a sample point in the plane. The event that the sample point lies in A is simply

$$A \subseteq \Omega. \tag{5.55}$$

Then the probability of A is

$$\Pr[A] = \int_A f(x, y)\ \mathrm{d}x\ \mathrm{d}y = \frac{\mu(A)}{\mu(S)}. \tag{5.56}$$

If we draw n samples $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n \in S$, then the probability space should be defined as

$$\Omega = \mathbb{R}^{2n}, \tag{5.57}$$

$$f(x_1, y_1, \ldots, x_n, y_n) = \begin{cases} \frac{1}{\mu(S)^n} & \text{if } (x_1, y_1), \ldots, (x_n, y_n) \in S, \\ 0 & \text{otherwise.} \end{cases} \tag{5.58}$$

Here x_i and y_i stand for the coordinates of the i-th sample point.

It looks like we only need to replace $\sum$ with $\int$ and $p(\omega)$ with $f(x)$ to get the continuous version of the discrete formulas. *Is there anything missing here?*

Yes, and the missing part is so important and unimportant that we have to include a cartoon here:



The important point is that not every function is integrable. For example, the function $\mathbb{1}_{\mathbb{Q}}$ is not Riemann integrable on any interval. You may argue that we should switch to more general definitions of integration, such as Lebesgue integration. This indeed fixes many problems, including the example $\mathbb{1}_{\mathbb{Q}}$. However, no matter which definition we use, there will still be some functions that are not integrable. We will get back to this issue later. Therefore, in a continuous probability space, we can only talk about the probabilities of some events, but not all events.

This issue is also unimportant in practice because most functions we encounter are integrable. As a computer scientist, you will almost never encounter a naturally arising function for which this integral is not well-defined. In 99.999% of computer science papers, just as the equal sign is often abused in big-$\mathcal{O}$ notation (Remark 2.8), authors often use probability theory as if all the relevant integrals were well-defined.

Therefore, to keep our discussion simple, we will pretend that all the functions we encounter are integrable. For anyone who care deeply about mathematical rigor, sorry!

5.1.5 Probability Distributions

Now it is a good time to introduce the probability distribution of a random variable more formally.

We have already seen the simplest version in the discrete setting. A random variable is a function $X : \Omega \to \mathbb{R}$ that assigns a real number to each outcome. When the possible values of X can be listed, its probability distribution can be described by the probability mass function.

In general, we define discrete random variables as follows:

Definition 5.13 (Discrete random variables)**.** A random variable X is *discrete* if it takes countably many values $\{x_1, x_2, \dots\}$. We define the *probability mass function* (PMF) of X as

$$p_X(x) := \Pr[X = x]. \tag{5.59}$$

When the underlying probability space is continuous, a random variable may still be discrete. For example, if $\boldsymbol{z}$ is sampled uniformly from the square $[-1, 1]^2$ and $D := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$, then

$$X := \mathbb{1}_{[\boldsymbol{z} \in D]} \tag{5.60}$$

is a Bernoulli random variable. On the other hand, many random variables defined on continuous probability spaces are genuinely continuous. For example, the x-coordinate of a sample point drawn uniformly from the square $[-1, 1]^2$ is distributed uniformly on $[-1, 1]$ and cannot be described by a PMF. To handle this case, we need to introduce the concept of continuous random variables.

Definition 5.14 (Continuous random variables)**.** A random variable X is *continuous* if there exists a function $f_X : \mathbb{R} \to [0, +\infty)$ such that

$$\int_{\mathbb{R}} f_X(x) \, \mathrm{d}x = 1, \tag{5.61}$$

and for every $a < b$,

$$\Pr[a < X < b] = \int_a^b f_X(x) \, \mathrm{d}x. \tag{5.62}$$

The function f_X is called the *probability density function* (PDF) of X.

Thus some random variables are naturally described by PMFs, and some are naturally described by PDFs. But these two descriptions do not cover everything. For example, if with probability $1/2$, we set $X = 0$, and with probability $1/2$, we set X to be uniformly distributed on $[0, 1]$, then this random variable has a discrete atom at 0 and a continuous part on $[0, 1]$. It is not described well by either a PMF or a PDF alone.

A more unified language is to directly describe the probability that X lies in each set of values.

Definition 5.15 (Probability distribution)**.** The *probability distribution*, or *law*, of X is defined as a function P_X that maps sets to probabilities:

$$P_X(A) := \Pr[X \in A], \tag{5.63}$$

In general, a probability distribution refers to any function $\mathcal{D}$ that maps sets to probabilities, and we write $X \sim \mathcal{D}$ if $P_X = \mathcal{D}$. A probability distribution $\mathcal{D}$ is *discrete* if $X \sim \mathcal{D}$ for some discrete random variable X, and *continuous* if $X \sim \mathcal{D}$ for some continuous random variable X.

Again, we deliberately ignore the case where $\Pr[X \in A]$ is not well-defined.

This is just for simplicity. Otherwise, you would have to deal with one thousand technical issues the first time you encounter continuous probability. Still, you have to understand what kind of mathematical object a probability distribution really is, and what we mean by writing $X \sim \mathcal{D}$.

Example 5.16 (Bernoulli distribution)**.** The Bernoulli distribution with parameter p, denoted by Bernoulli(p), is the discrete probability distribution that assigns probability p to 1 and probability $1-p$ to 0. Equivalently, as a function on sets, it is given by

$$\text{Bernoulli}(p)(A) = p\mathbb{1}_{[1\in A]} + (1-p)\mathbb{1}_{[0\in A]}. \tag{5.64}$$

Example 5.17 (Uniform distribution)**.** Let $a < b$. The following four continuous probability distributions are all the same:

- The uniform distribution on $[a, b]$: $f(x) = \mathbb{1}_{[x\in[a,b]]}/(b-a)$;
- The uniform distribution on $[a, b)$: $f(x) = \mathbb{1}_{[x\in[a,b)]}/(b-a)$;
- The uniform distribution on $(a, b]$: $f(x) = \mathbb{1}_{[x\in(a,b]]}/(b-a)$;
- The uniform distribution on (a, b): $f(x) = \mathbb{1}_{[x\in(a,b)]}/(b-a)$.

This is because they all correspond to the same assignment of probabilities to sets:

$$\mathcal{D}(A) = \int_A \frac{\mathbb{1}_{[x\in[a,b]]}}{b-a}\,\mathrm{d}x = \int_{A\cap[a,b]} \frac{1}{b-a}\,\mathrm{d}x. \tag{5.65}$$

A discrete probability distribution uniquely determines the PMF. However, the PDF of a continuous probability distribution may not be unique. In fact, you can change the PDF on countably many points without changing the probability distribution. Nevertheless, if f, f' are two PDFs of the same distribution, then $\Pr[f(X) = f'(X)] = 1$.

Finally, we note that to check whether two distributions are the same, it is enough to check whether they assign the same probability to every interval.

Theorem 5.18. Let P_X and P_Y be the probability distributions of two random variables X and Y. Then the following conditions are equivalent to $P_X = P_Y$:

1. For every $a < b$, $P_X((a, b)) = P_Y((a, b))$.
2. For every $a < b$, $P_X([a, b]) = P_Y([a, b])$.
3. For every $a \in \mathbb{R}$, $P_X((-\infty, a)) = P_Y((-\infty, a))$.
4. For every $a \in \mathbb{R}$, $P_X((-\infty, a]) = P_Y((-\infty, a])$.

The theorem is very intuitive: it is hard to imagine two different distributions that share the same probabilities on all intervals. But proving this theorem

requires defining probability distributions more formally first, so we will not give the proof here.

Another useful way to check whether two distributions are the same is to check whether they have the same MGF.

Theorem 5.19. Let P_X and P_Y be the probability distributions of two random variables X and Y. Suppose that the MGFs of X and Y, denoted as M_X and M_Y, are defined on some open interval $(-\epsilon, \epsilon)$. Then $P_X = P_Y$ if and only if $M_X(t) = M_Y(t)$ for all $t \in (-\epsilon, \epsilon)$.

Equivalently, we can replace the MGF M_X, M_Y with the CGF K_X, K_Y in the above theorem, and the theorem still holds.

We omit the proof details here, but the high-level idea is to first extend the MGFs to the complex plane, then use the inversion formula (Proposition 4.89) to show that the probability of X and Y on every interval are the same, and finally use Theorem 5.18 to conclude that the distributions are the same.

Note that beyond the finite discrete case, the MGF may not be defined on any open interval. Then one has to rely on the function values on the imaginary axis, where the MGF is always defined. As we have discussed in Remark 4.91, using the values of the MGF on the imaginary axis is also sufficient to recover the probability distribution.

5.2 Gaussian Distribution

After a stretch of definitions and general concepts, we can finally return to the Gaussian distribution and study it in detail. At first glance, the Gaussian distribution may look rather complicated. But once we become familiar with it, we will find that it is surprisingly simple and easy to compute with. Understanding the Gaussian distribution will also help you study statistics and machine learning, since it appears very frequently in these fields.

5.2.1 The Standard Gaussian Distributions

The Gaussian distribution with mean μ and variance σ^2 is usually denoted by $\mathcal{N}(\mu, \sigma^2)$. We have already seen its corresponding PDF in Equation (4.522). For simplicity, let us just focus on $\mathcal{N}(0, 1)$ first, where the mean is 0 and the variance is 1. This special case is called the *standard Gaussian distribution*.

Definition 5.20 (Standard Gaussian distribution)**.** The *standard Gaussian distribution* is the continuous probability distribution with PDF

$$\varphi(x) := \frac{1}{\sqrt{2\pi}} e^{-x^2/2}. \tag{5.66}$$

We denote the standard Gaussian distribution by $\mathcal{N}(0, 1)$.

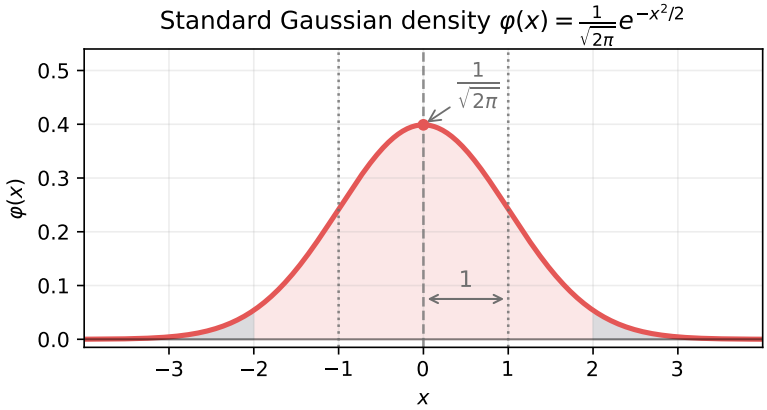


Figure 5.2: The PDF of the standard Gaussian distribution $\mathcal{N}(0,1)$.

This means for every set $A \subseteq \mathbb{R}$,

$$\mathcal{N}(0,1)(A) = \int_A \varphi(x)\, \mathrm{d}x. \tag{5.67}$$

This PDF is bell-shaped and symmetric around 0. It attains its maximum value $\frac{1}{\sqrt{2\pi}}$ at 0 and decays very quickly as $|x|$ grows. See Figure 5.2.

First, let us verify that φ is indeed a probability density function. There are two conditions to check:

1. $\varphi(x) \geq 0$ for all $x \in \mathbb{R}$. This is clear since $e^{-x^2/2}$ is always non-negative.
2. $\int_{-\infty}^{+\infty} \varphi(x)\, \mathrm{d}x = 1$. How can we compute this integral?

In principle, as long as the integral

$$I := \int_{-\infty}^{+\infty} e^{-x^2/2}\, \mathrm{d}x \tag{5.68}$$

is finite, we can always normalize $e^{-x^2/2}$ by dividing it by I to obtain a probability density function. So the interesting question is: why does I equal $\sqrt{2\pi}$? There is obviously no circle in the definition of the Gaussian PDF, so why is there a π?

Perhaps a first thought is to find the antiderivative of $e^{-x^2/2}$ and then evaluate it at $x = \pm\infty$. However, as you may have discovered after struggling for a while in Exercise 4.9, $\int e^{-x^2/2}\, \mathrm{d}x$ is not an elementary function.

A second thought is to extend the function to the complex plane and use Cauchy's residue theorem to evaluate the integral... Stop, stop, stop! As a real human, you should not apply complex analysis everywhere! It will only make your life more complex than it already is!

There is a clever trick to evaluate this integral: *integrating one Gaussian is too hard, but integrating two Gaussians is easy.* If we multiply the PDFs of two Gaussians (without the normalization factor $\frac{1}{\sqrt{2\pi}}$), we obtain:

$$e^{-x^2/2} e^{-y^2/2} = e^{-(x^2+y^2)/2}. \tag{5.69}$$

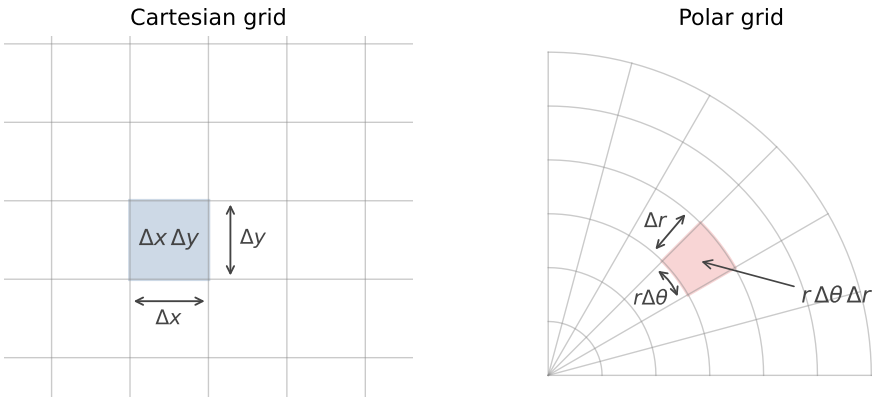


Figure 5.3: In Cartesian coordinates, a small grid cell has area approximately $\Delta x\,\Delta y$. In polar coordinates, a small grid cell at radius r has radial thickness Δr and arc length approximately $r\Delta\theta$, so its area is approximately $r\,\Delta\theta\,\Delta r$.

Integrating on both sides over $\mathbb{R}^2$ gives:

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} e^{-x^2/2}e^{-y^2/2}\,\mathrm{d}x\,\mathrm{d}y = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} e^{-(x^2+y^2)/2}\,\mathrm{d}x\,\mathrm{d}y. \tag{5.70}$$

For the left-hand side, since the inner integral does not depend on y, we can pull it out of the outer integral:

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} e^{-x^2/2}e^{-y^2/2}\,\mathrm{d}x\,\mathrm{d}y = \int_{-\infty}^{+\infty}\left(\int_{-\infty}^{+\infty} e^{-x^2/2}\,\mathrm{d}x\right)\cdot e^{-y^2/2}\,\mathrm{d}y \tag{5.71}$$

$$= \left(\int_{-\infty}^{+\infty} e^{-x^2/2}\,\mathrm{d}x\right)\left(\int_{-\infty}^{+\infty} e^{-y^2/2}\,\mathrm{d}y\right) \tag{5.72}$$

$$= I^2. \tag{5.73}$$

For the right-hand side, we note that the integrand $e^{-(x^2+y^2)/2}$ is rotationally symmetric, so it should be easier to integrate in polar coordinates (r,θ):

$$x = r\cos\theta, \qquad y = r\sin\theta. \tag{5.74}$$

In general, the following proposition shows how to transform an integral over Cartesian coordinates into polar coordinates:

Proposition 5.21. Let $f(x,y)$ be a continuous function on $\mathbb{R}^2$. Then

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_0^{2\pi}\int_0^{+\infty} f(r\cos\theta, r\sin\theta)\,r\,\mathrm{d}r\,\mathrm{d}\theta. \tag{5.75}$$

"Proof". You should have learned a rigorous proof of this proposition in the calculus course. If not, here we give a proof with physics-level rigor.

The integral of f over Cartesian coordinates can be understood by cutting the plane into tiny rectangular cells and summing up the contributions $f(x,y)\Delta x\Delta y$ over all cells, as on the left side of Figure 5.3.

Now we still compute the integral of f, but we cut the plane into a polar grid instead. A small polar cell at radius r_i and angle θ_j has radial thickness Δr and angular width $\Delta\theta$. As shown in the right side of Figure 5.3, this cell is the difference between two circular sectors. Its area is approximately

$$\frac{1}{2}(r_i + \Delta r)^2 \Delta\theta - \frac{1}{2} r_i^2 \Delta\theta = r_i\, \Delta r\, \Delta\theta + \frac{1}{2}(\Delta r)^2 \Delta\theta \approx r_i\, \Delta r\, \Delta\theta, \tag{5.76}$$

where the approximation is valid because $(\Delta r)^2$ is negligible compared to Δr when Δr is small. The value of the function in this cell is approximately

$$f(r_i \cos\theta_j, r_i \sin\theta_j). \tag{5.77}$$

Hence the total contribution from the polar grid is approximately

$$\sum_{i,j} f(r_i \cos\theta_j, r_i \sin\theta_j)\, r_i\, \Delta r\, \Delta\theta. \tag{5.78}$$

Now we make the polar grid finer and finer, so that $\Delta r, \Delta\theta \to 0$. Then we get

$$\sum_{i,j} f(r_i \cos\theta_j, r_i \sin\theta_j)\, r_i\, \Delta r\, \Delta\theta \longrightarrow \int_0^{2\pi} \int_0^{+\infty} f(r\cos\theta, r\sin\theta)\, r\ \mathrm{d}r\ \mathrm{d}\theta. \tag{5.79}$$

This is exactly the formula for polar coordinates. □

Now we return to the right-hand side of Equation (5.70). Transforming the integral using the proposition above, we get

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-(x^2+y^2)/2}\ \mathrm{d}x\, \mathrm{d}y = \int_0^{2\pi} \int_0^{+\infty} e^{-r^2/2}\, r\ \mathrm{d}r\, \mathrm{d}\theta. \tag{5.80}$$

It is easy to find the antiderivative of $e^{-r^2/2}\, r$:

$$\int e^{-r^2/2}\, r\ \mathrm{d}r = -e^{-r^2/2} + C. \tag{5.81}$$

Then

$$\int_0^{\infty} e^{-r^2/2}\, r\ \mathrm{d}r = \left(-e^{-r^2/2}\right)\Big|_0^{+\infty} = 1. \tag{5.82}$$

Putting it back into the integral, we get

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-(x^2+y^2)/2}\ \mathrm{d}x\, \mathrm{d}y = \int_0^{2\pi} 1\, \mathrm{d}\theta = 2\pi. \tag{5.83}$$

This is where the π appears!

Now combining Equation (5.73) and Equation (5.83), we get

$$I^2 = 2\pi. \tag{5.84}$$

Therefore, $I = \sqrt{2\pi}$, and the right normalization factor for standard Gaussian distribution is indeed $\frac{1}{\sqrt{2\pi}}$.

5.2.2 Mean and Variance of the Standard Gaussian

Now let us prove that the standard Gaussian distribution $\mathcal{N}(0,1)$ has mean 0 and variance 1. That is,

$$\mathbb{E}[X] = 0, \qquad \mathrm{Var}(X) = 1, \qquad \text{when } X \sim \mathcal{N}(0,1). \tag{5.85}$$

▶ Mean

For $X \sim \mathcal{N}(0,1)$, the expectation of X is

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x e^{-x^2/2} \, \mathrm{d}x. \tag{5.86}$$

Before doing any further calculation, remember that the standard Gaussian distribution is symmetric around 0, meaning that X and $-X$ appear with equal probability density. If we group the contributions from X and $-X$ together, we get 0. Integrating over all possible values of $|X|$ still gives 0. Therefore, $\mathbb{E}[X]$ has to be 0.

To write a rigorous proof, we split the integral into two parts: the positive part and the negative part.

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} x e^{-x^2/2} \, \mathrm{d}x + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 x e^{-x^2/2} \, \mathrm{d}x. \tag{5.87}$$

For the second integral, we substitute $u = -x$, so that $\mathrm{d}u = -\mathrm{d}x$. Then the integral becomes

$$\int_{-\infty}^0 x e^{-x^2/2} \, \mathrm{d}x = \int_{-\infty}^0 (-x) e^{-x^2/2} \cdot (-1) \, \mathrm{d}x = \int_{+\infty}^0 u e^{-u^2/2} \, \mathrm{d}u \tag{5.88}$$

$$= -\int_0^{+\infty} u e^{-u^2/2} \, \mathrm{d}u. \tag{5.89}$$

Therefore, we get

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} x e^{-x^2/2} \, \mathrm{d}x - \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} u e^{-u^2/2} \, \mathrm{d}u = 0. \tag{5.90}$$

This proves that $\mathbb{E}[X] = 0$.

▶ Variance

For $X \sim \mathcal{N}(0,1)$, the variance is

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2]. \tag{5.91}$$

It suffices to compute $\mathbb{E}[X^2]$, which is given by the following integral:

$$\mathbb{E}[X^2] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^2 e^{-x^2/2} \, \mathrm{d}x. \tag{5.92}$$

We use integration by parts on the symmetric interval $[-T, T]$ and then let $T \to +\infty$:

$$\int_{-T}^T x^2 e^{-x^2/2} \, \mathrm{d}x = \int_{-T}^T (-x) \cdot (e^{-x^2/2})' \, \mathrm{d}x \tag{5.93}$$

$$= (-x) \cdot e^{-x^2/2} \Big|_{-T}^T - \int_{-T}^T (-x)' \cdot e^{-x^2/2} \, \mathrm{d}x \tag{5.94}$$

$$= -x e^{-x^2/2} \Big|_{-T}^T + \int_{-T}^T e^{-x^2/2} \, \mathrm{d}x. \tag{5.95}$$

For the first term, note that $xe^{-x^2/2} \to 0$ as $x \to \pm\infty$ because $e^{-x^2/2}$ decays much faster than x grows. So the first term tends to 0:

$$\lim_{T\to+\infty}(-x)e^{-x^2/2}\Big|_{-T}^{T} = 0. \tag{5.96}$$

For the second term, when $T \to +\infty$, it becomes the integral that we have already computed:

$$\int_{-\infty}^{+\infty} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}. \tag{5.97}$$

Therefore,

$$\mathbb{E}[X^2] = \frac{1}{\sqrt{2\pi}}\left(0 + \sqrt{2\pi}\right) = 1, \tag{5.98}$$

and hence $\mathrm{Var}(X) = 1$.

5.2.3 General Gaussian Distributions

If we shift and scale a standard Gaussian random variable, then the resulting distribution still "looks like" a standard Gaussian, so in principle it should also be called a Gaussian distribution. In general, if $Z \sim \mathcal{N}(0,1)$ is a standard Gaussian, then

$$X = \mu + \sigma Z \tag{5.99}$$

is a random variable with mean μ and variance σ^2. This is because:

$$\mathbb{E}[X] = \mathbb{E}[\mu + \sigma Z] = \mu + \sigma\mathbb{E}[Z] = \mu, \tag{5.100}$$

$$\mathrm{Var}(X) = \mathrm{Var}(\mu + \sigma Z) = \mathrm{Var}(\sigma Z) = \sigma^2\mathrm{Var}(Z) = \sigma^2, \tag{5.101}$$

where we used the linearity of expectation and the scaling and additivity properties of variance. These properties were only proved for finite discrete random variables in Theorems 2.85, 3.17 and 3.18, but they also hold in general for all random variables. Based on this observation, we define the general Gaussian distribution as follows:

Definition 5.22 (General Gaussian distribution)**.** Let $\mu \in \mathbb{R}$ and $\sigma \geq 0$. The *Gaussian distribution* with mean μ and variance σ^2 is defined as the distribution of random variable

$$X = \mu + \sigma Z, \tag{5.102}$$

where $Z \sim \mathcal{N}(0,1)$. We denote this distribution by $\mathcal{N}(\mu, \sigma^2)$.

As one may wonder what this distribution really looks like. If $\sigma = 0$, then X is in fact a discrete random variable, whose value is always μ. This degenerate case is not very interesting, so let us focus on $\sigma > 0$.

When $\sigma > 0$, X is a continuous random variable, and we can compute its PDF. First, we state how the PDF will change if we shift or scale an arbitrary continuous random variable.

Theorem 5.23. Let X be a continuous random variable with PDF $f_X(x)$. Let $\alpha \in \mathbb{R} \setminus \{0\}$ and $\beta \in \mathbb{R}$ be constants. Then:

- **Scaling:** αX has PDF $\frac{1}{|\alpha|} f_X\left(\frac{x}{\alpha}\right)$.
- **Shift:** $X + \beta$ has PDF $f_X(x - \beta)$.
- **Scaling and Shift:** $\alpha X + \beta$ has PDF $\frac{1}{|\alpha|} f_X\left(\frac{x-\beta}{\alpha}\right)$.

Proof. We prove the first two claims. The third claim follows directly by composing the first two.

Shift. For every $a < b$,

$$\Pr[a < X + \beta < b] = \Pr\left[a - \beta < X < b - \beta\right] = \int_{a-\beta}^{b-\beta} f_X(x)\,\mathrm{d}x. \tag{5.103}$$

We substitute $u = x + \beta$. Then the integral becomes

$$\int_{a-\beta}^{b-\beta} f_X(x)\,\mathrm{d}x = \int_{a-\beta}^{b-\beta} f_X(u-\beta)\,\mathrm{d}x = \int_a^b f_X(u-\beta)\,\mathrm{d}u. \tag{5.104}$$

Therefore, $X + \beta$ has PDF $f_X(x - \beta)$.

Scaling. For every $a < b$, if $\alpha > 0$, then

$$\Pr[a < \alpha X < b] = \Pr\left[\frac{a}{\alpha} < X < \frac{b}{\alpha}\right] = \int_{a/\alpha}^{b/\alpha} f_X(x)\,\mathrm{d}x. \tag{5.105}$$

We substitute $u = \alpha x$, so that $\mathrm{d}u = \alpha\,\mathrm{d}x$. Then the integral becomes

$$\int_{a/\alpha}^{b/\alpha} f_X(x)\,\mathrm{d}x = \int_{a/\alpha}^{b/\alpha} \frac{1}{\alpha} f_X\left(\frac{u}{\alpha}\right) \cdot \alpha\,\mathrm{d}x = \int_a^b \frac{1}{\alpha} f_X(u)\,\mathrm{d}u. \tag{5.106}$$

Therefore, αX has PDF $\frac{1}{\alpha} f_X\left(\frac{x}{\alpha}\right)$ when $\alpha > 0$. If $\alpha < 0$, then

$$\Pr[a < \alpha X < b] = \Pr\left[\frac{b}{\alpha} < X < \frac{a}{\alpha}\right] = \int_{b/\alpha}^{a/\alpha} f_X(x)\,\mathrm{d}x. \tag{5.107}$$

Again we substitute $u = \alpha x$, so that $\mathrm{d}u = \alpha\,\mathrm{d}x$. Then the integral becomes

$$\int_{b/\alpha}^{a/\alpha} f_X(x)\,\mathrm{d}x = \int_{b/\alpha}^{a/\alpha} \frac{1}{\alpha} f_X\left(\frac{u}{\alpha}\right) \cdot \alpha\,\mathrm{d}x = \int_b^a \frac{1}{\alpha} f_X(u)\,\mathrm{d}u. = \int_a^b \frac{1}{-\alpha} f_X(u)\,\mathrm{d}u. \tag{5.108}$$

Therefore, αX has PDF $\frac{1}{-\alpha} f_X\left(\frac{x}{\alpha}\right)$ when $\alpha < 0$. Combining the two cases, we conclude that αX has PDF $\frac{1}{|\alpha|} f_X\left(\frac{x}{\alpha}\right)$. □

The PDF of a general Gaussian distribution then follows easily from the above theorem.

Theorem 5.24 (Gaussian PDF)**.** If $X \sim \mathcal{N}(\mu, \sigma^2)$ for some $\mu \in \mathbb{R}$ and $\sigma > 0$, then

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{5.109}$$

is a PDF of X.

Proof. Let $X \sim \mathcal{N}(\mu, \sigma^2)$. We can write $X = \mu + \sigma Z$, where $Z \sim \mathcal{N}(0, 1)$. Then X has the following PDF:

$$\frac{1}{\sigma}\varphi\Big(\frac{x-\mu}{\sigma}\Big) \tag{5.110}$$

Simplifying the expression, we get

$$\frac{1}{\sigma}\varphi\Big(\frac{x-\mu}{\sigma}\Big) = \frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} e^{-(\frac{x-\mu}{\sigma})^2/2} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \tag{5.111}$$

which proves the theorem. □

This is the same as Equation (4.522), but now, with the context above, this PDF should feel less like a random mess of mysterious symbols and more like something you can actually understand.

5.2.4 Moment Generating Functions

Back to the standard Gaussian distribution, let us compute its moment generating function (MGF). For $X \sim \mathcal{N}(0, 1)$, the MGF of X is given by

$$\begin{aligned} M_X(t) := \mathbb{E}[e^{tX}] &= \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{tx} e^{-x^2/2} \, \mathrm{d}x \\ &= \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2 - 2tx}{2}\right) \mathrm{d}x. \end{aligned} \tag{5.112}$$

Taking a closer look at the exponent, we can see that $x^2 - 2tx$ is almost a perfect square:

$$x^2 - 2tx = x^2 - 2tx + t^2 - t^2 = (x-t)^2 - t^2. \tag{5.113}$$

Therefore, we can rewrite the integral as

$$M_X(t) = \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-t)^2 - t^2}{2}\right) \mathrm{d}x \tag{5.114}$$

$$= e^{t^2/2} \underbrace{\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-t)^2}{2}} \, \mathrm{d}x}_{=:I}. \tag{5.115}$$

What is this integral I? Does it look familiar to you? Have you seen the integrand $\frac{1}{\sqrt{2\pi}} e^{-\frac{(x-t)^2}{2}}$ somewhere before?



Yes, the integrand is just the PDF of $\mathcal{N}(t, 1)$, and thus the integral is asking us to compute the integral of the PDF of $\mathcal{N}(t, 1)$ over the entire real line. Of course, the answer is $I = 1$.

Therefore, we have

$$M_X(t) = e^{t^2/2}. \tag{5.116}$$

And the cumulant generating function (CGF) of X is

$$K_X(t) = \log M_X(t) = \frac{1}{2}t^2. \tag{5.117}$$

More generally, the MGF and CGF of a general Gaussian distribution are given by the following theorem:

Theorem 5.25 (Gaussian MGF and CGF)**.** If $X \sim \mathcal{N}(\mu, \sigma^2)$ is a Gaussian random variable with mean μ and variance σ^2, then

$$M_X(t) = e^{\mu t + \sigma^2 t^2/2}, \qquad K_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2. \tag{5.118}$$

Proof. We can write X as $X = \sigma Z + \mu$, where $Z \sim \mathcal{N}(0, 1)$. Then we apply the rules in Table 4.4 to obtain:

$$M_X(t) = M_{\sigma Z+\mu}(t) = M_Z(\sigma t)e^{\mu t} = e^{\sigma^2 t^2/2 + \mu t}, \tag{5.119}$$

$$K_X(t) = K_{\sigma Z+\mu}(t) = K_Z(\sigma t) + \mu t = \frac{1}{2}\sigma^2 t^2 + \mu t, \tag{5.120}$$

which proves the theorem. □

This explains why the CGF of a Gaussian random variable is a quadratic function, a question that we left unanswered in Section 4.3.6.

5.2.5 Sum of Independent Gaussians

If we have two random variables X_1 and X_2, then their sum $Y = X_1 + X_2$ is also a random variable, and we can compute its mean and variance as

$$\mathbb{E}[Y] = \mathbb{E}[X_1] + \mathbb{E}[X_2], \qquad \mathrm{Var}(Y) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2). \tag{5.121}$$

However, if we further ask what the distribution of Y really is, the answer may not be as straightforward.

For example, if X_1 and X_2 are independent uniform random variables on the interval $[-1, 1]$, then Y is a triangular distribution with support on the interval $[-2, 2]$. See Figure 5.4.

Gaussian random variables behave much more rigidly. If X_1 and X_2 are independent Gaussian random variables, then $Y = X_1 + X_2$ is still Gaussian. Since a Gaussian distribution is determined by its mean and variance, this means that we only need to add the means and variances of X_1 and X_2 to identify the distribution of Y. This makes Gaussian random variables easy to compute with.

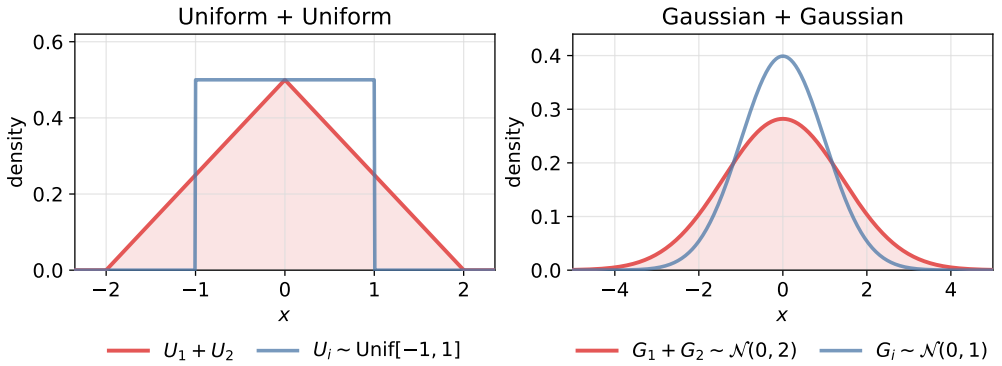


Figure 5.4: Adding two uniform random variables changes the shape of the distribution, while adding two independent Gaussian random variables still yields a Gaussian.

Theorem 5.26 (Sum of two independent Gaussians)**.** Let X_1 and X_2 be independent Gaussian random variables. Suppose

$$X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), \qquad X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2). \tag{5.122}$$

Then

$$X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2). \tag{5.123}$$

Proof. By Table 4.4,

$$K_{X_1+X_2}(t) = K_{X_1}(t) + K_{X_2}(t). \tag{5.124}$$

By Theorem 5.25,

$$K_{X_1}(t) = \mu_1 t + \frac{1}{2}\sigma_1^2 t^2, \quad K_{X_2}(t) = \mu_2 t + \frac{1}{2}\sigma_2^2 t^2. \tag{5.125}$$

Therefore,

$$\begin{aligned} K_{X_1+X_2}(t) &= \left(\mu_1 t + \frac{1}{2}\sigma_1^2 t^2\right) + \left(\mu_2 t + \frac{1}{2}\sigma_2^2 t^2\right) \\ &= (\mu_1 + \mu_2)t + \frac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2. \end{aligned} \tag{5.126}$$

This is the CGF of $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Since the CGF determines the distribution, $X_1 + X_2$ has the claimed Gaussian distribution. □

A direct consequence of this theorem is the following result:

Theorem 5.27 (Gaussian linearity)**.** Let $X_1, \ldots, X_n$ be independent Gaussian random variables with

$$X_i \sim \mathcal{N}(\mu_i, \sigma_i^2), \qquad \text{for all } i \in [n]. \tag{5.127}$$

For any constants $c_1, \ldots, c_n \in \mathbb{R}$,

$$\sum_{i=1}^{n} c_i X_i \sim \mathcal{N}\left(\sum_{i=1}^{n} c_i \mu_i, \sum_{i=1}^{n} c_i^2 \sigma_i^2\right). \tag{5.128}$$

Proof. We write $X_i = \sigma_i Z_i + \mu_i$, where $Z_i \sim \mathcal{N}(0,1)$. Then

$$c_i X_i = c_i(\sigma_i Z_i + \mu_i) = (c_i \sigma_i) Z_i + c_i \mu_i \sim \mathcal{N}(c_i \mu_i, c_i^2 \sigma_i^2). \tag{5.129}$$

The random variables $c_1 X_1, \ldots, c_n X_n$ are still independent. Applying Theorem 5.26 repeatedly gives

$$\sum_{i=1}^{n} c_i X_i \sim \mathcal{N}\left(\sum_{i=1}^{n} c_i \mu_i, \sum_{i=1}^{n} c_i^2 \sigma_i^2\right). \tag{5.130}$$

This proves the theorem. □

Thus whenever we take a linear combination of independent Gaussian random variables, the result is still Gaussian. Once we know its mean and variance, we know the whole distribution.

5.2.6 Gaussian Tail Bounds

Suppose $X_1, \ldots, X_n$ are i.i.d. samples from the same Gaussian distribution

$$X_i \sim \mathcal{N}(\mu, \sigma^2). \tag{5.131}$$

How sharply does the sample average $X := \frac{1}{n}\sum_{i=1}^{n} X_i$ concentrate around μ?

For Gaussian distributions, we can answer this question exactly. By Theorem 5.27, we can exactly compute the distribution of X:

$$X := \frac{1}{n}\sum_{i=1}^{n} X_i \sim \mathcal{N}\left(\sum_{i=1}^{n} \frac{1}{n}\mu, \sum_{i=1}^{n} \frac{1}{n^2}\sigma^2\right) = \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right). \tag{5.132}$$

Therefore, we can exactly compute the tail of X as follows. Write $X = \mu + \frac{\sigma}{\sqrt{n}} Z$, where $Z \sim \mathcal{N}(0,1)$. Then for every $\epsilon \geq 0$,

$$\Pr[X - \mu \geq \epsilon] = \Pr\left[Z \geq \frac{\epsilon}{\sigma}\sqrt{n}\right] = \bar{\Phi}\left(\frac{\epsilon}{\sigma}\sqrt{n}\right), \tag{5.133}$$

where $\bar{\Phi}$ is conventionally defined as the tail probability of the standard Gaussian distribution:

$$\bar{\Phi}(x) := \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, \mathrm{d}u. \tag{5.134}$$

This characterization of the tail probability is a bit abstract, though. Since $\bar{\Phi}(x)$ is not an elementary function, we still do not know how fast the tail probability

decays as $x \to +\infty$. However, based on some tricks with integration by parts, one can show that

$$\bar{\Phi}(x) \sim \frac{1}{\sqrt{2\pi}} \frac{e^{-x^2/2}}{x} \quad \text{as } x \to +\infty. \tag{5.135}$$

We leave the proof of this approximation as an exercise.

By symmetry, we can conclude with the following theorem.

Theorem 5.28 (Exact tail probability for Gaussian sample averages)**.** Let $X := \frac{1}{n}\sum_{i=1}^n X_i$ be the sample average of n i.i.d. Gaussian random variables $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. Then for every $\epsilon \geq 0$, as $n \to \infty$,

$$\Pr[X \geq \mu + \epsilon] = \Pr[X \leq \mu - \epsilon] = \bar{\Phi}\left(\frac{\epsilon}{\sigma}\sqrt{n}\right) \tag{5.136}$$

$$\sim \frac{\sigma}{\epsilon\sqrt{2\pi n}} e^{-n\epsilon^2/(2\sigma^2)}. \tag{5.137}$$

Let us compare this with the Chernoff bound. For a single Gaussian varable, we have the following lemma.

Lemma 5.29 (Gaussian Chernoff bound)**.** Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then, for every $\epsilon \geq 0$,

$$I_X(\mu + \epsilon) = I_X(\mu - \epsilon) = \frac{\epsilon^2}{2\sigma^2}, \tag{5.138}$$

$$\Pr[X \geq \mu + \epsilon] \leq e^{-\epsilon^2/(2\sigma^2)}, \quad \Pr[X \leq \mu - \epsilon] \leq e^{-\epsilon^2/(2\sigma^2)}. \tag{5.139}$$

Proof. Let $G = X - \mu$. Then $G \sim \mathcal{N}(0, \sigma^2)$. By Theorem 5.25,

$$M_G(t) = e^{\sigma^2 t^2/2}, \qquad K_G(t) = \frac{1}{2}\sigma^2 t^2. \tag{5.140}$$

By the definition of the rate function in Definition 3.29, for every $a \in \mathbb{R}$,

$$I_G(a) = \sup_{t\in\mathbb{R}}\{ta - K_G(t)\} = \sup_{t\in\mathbb{R}}\left\{ta - \frac{\sigma^2 t^2}{2}\right\} = \frac{a^2}{2\sigma^2}, \tag{5.141}$$

where the last equality follows from the fact that the maximum of the quadratic function $ta - \frac{\sigma^2 t^2}{2}$ is attained at $t = a/\sigma^2$. The Chernoff bound in Theorem 3.30 then gives the desired result. □

Applying the above lemma to the Gaussian sample average gives the following theorem.

Theorem 5.30 (Gaussian Chernoff bound for sample averages)**.** Let $X := \frac{1}{n}\sum_{i=1}^n X_i$ be the sample average of n i.i.d. Gaussian random variables $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. Then, for every $\epsilon \geq 0$,

$$\Pr[X \geq \mu + \epsilon] \leq e^{-n\epsilon^2/(2\sigma^2)}, \quad \Pr[X \leq \mu - \epsilon] \leq e^{-n\epsilon^2/(2\sigma^2)}. \tag{5.142}$$

Proof. Let $G = X - \mu$. Since $X \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$, the distribution of G is $G \sim \mathcal{N}(0, \frac{\sigma^2}{n})$. By Equation (5.141),

$$I_G(a) = \frac{a^2}{2\sigma^2/n} = \frac{na^2}{2\sigma^2}. \tag{5.143}$$

The Chernoff bound in Theorem 3.30 then gives

$$\Pr[X \geq \mu + \epsilon] = \Pr[G \geq \epsilon] \leq e^{-I_G(\epsilon)} = e^{-n\epsilon^2/(2\sigma^2)}, \tag{5.144}$$

$$\Pr[X \leq \mu - \epsilon] = \Pr[G \leq -\epsilon] \leq e^{-I_G(-\epsilon)} = e^{-n\epsilon^2/(2\sigma^2)}, \tag{5.145}$$

which completes the proof. □

Comparing the exact tail probability with the Chernoff bound, we see that the Chernoff bound gets the correct exponential rate $e^{-n\epsilon^2/(2\sigma^2)}$, just as what Cramér's theorem predicts. However, the prefactor is different. The exact factor in front of the exponential is $\frac{\sigma}{\epsilon\sqrt{2\pi n}} = \Theta(\frac{1}{\sqrt{n}})$, while the Chernoff bound has a constant factor 1. This means Chernoff bound overestimates the tail probability.

Nevertheless, the exponential term dominates the decay rate. The Chernoff bound ignores the $\Theta(\frac{1}{\sqrt{n}})$ factor, but this does not significantly affect the overall estimate. Therefore, Chernoff bounds are accurate enough for Gaussians.

5.2.7 Gaussian Vectors

So far, we have mostly considered random variables, which are random objects whose value is a real number. In high-dimensional probability, we also study random objects whose values are vectors. We call such objects *random vectors*.

Definition 5.31 (Random vector)**.** A *random vector* in $\mathbb{R}^d$ is a vector

$$\boldsymbol{x} = (X_1, \ldots, X_d) \tag{5.146}$$

whose coordinates $X_1, \ldots, X_d$ are random variables. The *probability distribution* of a random vector $\boldsymbol{x} \in \mathbb{R}^d$, or equivalently the *joint distribution* of $X_1, \ldots, X_d$, is a function that assigns probabilities to regions in $\mathbb{R}^d$:

$$P_{\mathbf{x}}(A) := \Pr[\boldsymbol{x} \in A]. \tag{5.147}$$

When referring to the distribution of a single coordinate X_i, we use the term *marginal distribution of* X_i to emphasize that we are considering the probability distribution of this coordinate alone.

As usual, we say that a random vector $\boldsymbol{x}$ or its probability distribution $P_{\mathbf{x}}$ is *continuous* if there exists a probability density function $f_{\mathbf{x}}$ such that for every region $A \subseteq \mathbb{R}^d$,

$$P_{\mathbf{x}}(A) = \int_A f_{\mathbf{x}}(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x}. \tag{5.148}$$

Since a random vector consists of multiple random variables, we may want to use more than one real number to describe its mean and variance.

Definition 5.32 (Mean vector)**.** The *mean* of a random vector $\boldsymbol{x} = (X_1, \ldots, X_d)$ is the vector

$$\mathbb{E}[\boldsymbol{x}] := \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \vdots \\ \mathbb{E}[X_d] \end{bmatrix}. \tag{5.149}$$

Definition 5.33 (Covariance matrix)**.** The *covariance matrix* of a random vector $\boldsymbol{x} = (X_1, \ldots, X_d)$ is the matrix

$$\mathrm{Cov}(\boldsymbol{x}) := \mathbb{E}[(\boldsymbol{x} - \mathbb{E}[\boldsymbol{x}])(\boldsymbol{x} - \mathbb{E}[\boldsymbol{x}])^\top], \tag{5.150}$$

where the expectation is taken entrywise. More explicitly,

$$\mathrm{Cov}(\boldsymbol{x}) := \begin{bmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) & \cdots & \mathrm{Cov}(X_1, X_d) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) & \cdots & \mathrm{Cov}(X_2, X_d) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_d, X_1) & \mathrm{Cov}(X_d, X_2) & \cdots & \mathrm{Cov}(X_d, X_d) \end{bmatrix}, \tag{5.151}$$

where $\mathrm{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$ is defined as the *covariance* between X and Y.

The mean of a random vector is made by just putting the means of all its coordinates into a vector. This is a natural extension of the mean of a random variable.

But why does variance become a matrix? Why can't we just put the variances of all the coordinates into a vector? We can surely do that. In fact, the diagonal entries of the covariance matrix are exactly the variances of the coordinates. This is because for any random variable X,

$$\mathrm{Cov}(X, X) = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big] = \mathrm{Var}(X). \tag{5.152}$$

However, if we only keep the diagonal entries, we lose information about the correlations between different coordinates. For example, consider the following two random vectors:

$$\boldsymbol{x}_1 = (X_1, X_1, X_1, \ldots, X_1), \tag{5.153}$$

$$\boldsymbol{x}_2 = (X_1, X_2, X_3, \ldots, X_d), \tag{5.154}$$

where $X_1, X_2, \ldots, X_d$ are i.i.d. random variables. Both vectors have the same diagonal entries, because the variances of all the coordinates are the same. But the second random vector $\boldsymbol{x}_2$ obviously looks more random than the first one, since the coordinates are more likely to be different from each other. To capture this information, we need to keep the off-diagonal entries of the covariance matrix:

$$\mathrm{Cov}(X_i, X_j) = \mathbb{E}\Big[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])\Big]. \tag{5.155}$$

If X_i and X_j are independent, then

$$\mathrm{Cov}(X_i, X_j) = \mathbb{E}\Big[(X_i - \mathbb{E}[X_i])\Big]\mathbb{E}\Big[(X_j - \mathbb{E}[X_j])\Big] = 0 \cdot 0 = 0. \tag{5.156}$$

Therefore, if we keep the off-diagonal entries, we can tell apart the two random vectors: the covariance matrix of $\boldsymbol{x}_1$ is a matrix filled with the same value $\mathrm{Var}(X_1)$, while the covariance matrix of $\boldsymbol{x}_2$ is a diagonal matrix.

In general, we can tell how correlated the coordinates are with each other by looking at the off-diagonal entries of the covariance matrix. If the off-diagonal entries are almost all zero, then the coordinates are almost independent. An important caveat is that $\mathrm{Cov}(X, Y) = 0$ does not imply that X and Y are independent, though.

Now let us define Gaussian random vectors. The most basic Gaussian random vector is obtained by putting independent standard Gaussian random variables in each coordinate. Such a random vector is called a *standard Gaussian vector*.

Definition 5.34 (Standard Gaussian vector)**.** A random vector $\boldsymbol{x} = (X_1, \ldots, X_d) \in \mathbb{R}^d$ is called a *standard Gaussian vector* if $X_1, \ldots, X_d$ are independent and

$$X_i \sim \mathcal{N}(0, 1), \qquad \text{for all } i \in [d]. \tag{5.157}$$

The distribution of this random vector is denoted by $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$.

The two parameters in $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$ represent the mean vector and the covariance matrix, respectively. Here $\mathbf{0}$ is the zero vector in $\mathbb{R}^d$, and $\boldsymbol{I}$ is the d-dimensional identity matrix.

It is easy to see that the mean of the standard Gaussian vector is the zero vector, since $\mathbb{E}[X_i] = 0$ for all $i \in [d]$. For the covariance matrix, note that for all $i, j \in [d]$,

$$\mathrm{Cov}(X_i, X_j) = \begin{cases} \mathrm{Var}(X_i) = 1 & \text{if } i = j, \\ \mathrm{Cov}(X_i, X_j) = \mathbb{E}[X_i]\mathbb{E}[X_j] = 0 & \text{if } i \neq j. \end{cases} \tag{5.158}$$

Thus $\mathrm{Cov}(\boldsymbol{x}) = \boldsymbol{I}_d$ as expected.

Similar to how we define general Gaussian random variables, we define general Gaussian vectors by linearly transforming a standard Gaussian vector.

Definition 5.35 (Gaussian vector)**.** A random vector $\boldsymbol{x} \in \mathbb{R}^d$ is called a *Gaussian vector* if there exists a vector $\boldsymbol{\mu} \in \mathbb{R}^d$, a matrix $\boldsymbol{A} \in \mathbb{R}^{d \times r}$, and a r-dimensional standard Gaussian vector $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ such that

$$\boldsymbol{x} = \boldsymbol{\mu} + \boldsymbol{A}\boldsymbol{z}. \tag{5.159}$$

The distribution of this random vector is denoted by $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} = \boldsymbol{A}\boldsymbol{A}^\top$.

Here $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the mean vector and covariance matrix of $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, respectively. One point needs more explanation. In the above definition, if there are two random vectors:

$$\boldsymbol{x}_1 = \boldsymbol{\mu} + \boldsymbol{A}_1 \boldsymbol{z}_1, \tag{5.160}$$

$$\boldsymbol{x}_2 = \boldsymbol{\mu} + \boldsymbol{A}_2 \boldsymbol{z}_2, \tag{5.161}$$

where $\boldsymbol{z}_1$ is a r_1-dimensional standard Gaussian vector, and $\boldsymbol{z}_2$ is a r_2-dimensional standard Gaussian vector, then the distributions of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are denoted by the same $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, as long as $\boldsymbol{A}_1 \boldsymbol{A}_1^\top = \boldsymbol{A}_2 \boldsymbol{A}_2^\top$. *Do they really have the same distribution?*

Here we explain the reason briefly. Similar to random variables, we can define MGFs and CGFs for random vectors, but now both the MGF and CGF are functions of a vector parameter $\boldsymbol{t} \in \mathbb{R}^d$:

$$M_{\boldsymbol{x}}(\boldsymbol{t}) := \mathbb{E}[e^{\boldsymbol{t}^\top \boldsymbol{x}}], \qquad K_{\boldsymbol{x}}(\boldsymbol{t}) := \log M_{\boldsymbol{x}}(\boldsymbol{t}). \tag{5.162}$$

One can generalize Theorem 5.19 to random vectors:

Theorem 5.36. Let $P_{\mathbf{x}}$ and $P_{\mathbf{y}}$ be the probability distributions of two random vectors $\boldsymbol{x}$ and $\boldsymbol{y}$. Suppose that the MGFs of $\boldsymbol{x}$ and $\boldsymbol{y}$, denoted as $M_{\mathbf{x}}$ and $M_{\mathbf{y}}$, are defined on some open neighborhood of the origin. Then $P_{\mathbf{x}} = P_{\mathbf{y}}$ if and only if $M_{\mathbf{x}}(\boldsymbol{t}) = M_{\mathbf{y}}(\boldsymbol{t})$ for all $\boldsymbol{t}$ in this neighborhood.

Equivalently, we can replace the MGF M_X, M_Y with the CGF K_X, K_Y in the above theorem, and the theorem still holds.

The next step is to compute the MGF and CGF of a Gaussian vector, and show that they only depend on the mean vector and covariance matrix.

Theorem 5.37 (Gaussian MGF and CGF)**.** Let $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then the MGF and CGF of $\boldsymbol{x}$ are given by

$$M_{\mathbf{x}}(\boldsymbol{t}) = e^{\boldsymbol{t}^\top \boldsymbol{\mu} + \boldsymbol{t}^\top \boldsymbol{\Sigma} \boldsymbol{t}/2}, \qquad K_{\mathbf{x}}(\boldsymbol{t}) = \boldsymbol{t}^\top \boldsymbol{\mu} + \frac{1}{2} \boldsymbol{t}^\top \boldsymbol{\Sigma} \boldsymbol{t}. \tag{5.163}$$

Proof. If $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, then $\boldsymbol{t}^\top \boldsymbol{z} \sim \mathcal{N}(0, \|\boldsymbol{t}\|_2^2)$ by Theorem 5.27. So the MGF and CGF of $\boldsymbol{z}$ are given by

$$M_{\mathbf{z}}(\boldsymbol{t}) = \mathbb{E}[e^{\boldsymbol{t}^\top \boldsymbol{z}}] = M_{(\boldsymbol{t}^\top \boldsymbol{z})}(1) = e^{\|\boldsymbol{t}\|_2^2/2}, \qquad K_{\mathbf{z}}(\boldsymbol{t}) = \frac{1}{2} \|\boldsymbol{t}\|_2^2. \tag{5.164}$$

For $\boldsymbol{x} = \boldsymbol{\mu} + \boldsymbol{A}\boldsymbol{z}$, we can compute the CGF as follows:

$$K_{\mathbf{x}}(\boldsymbol{t}) = \log \mathbb{E}[e^{\boldsymbol{t}^\top \boldsymbol{x}}] = \log \mathbb{E}[e^{\boldsymbol{t}^\top (\boldsymbol{\mu} + \boldsymbol{A}\boldsymbol{z})}] \tag{5.165}$$

$$= \log \mathbb{E}[e^{\boldsymbol{t}^\top \boldsymbol{\mu}} e^{(\boldsymbol{A}^\top \boldsymbol{t})^\top \boldsymbol{z}}] \tag{5.166}$$

$$= \langle \boldsymbol{t}, \boldsymbol{\mu} \rangle + K_{\mathbf{z}}(\boldsymbol{A}^\top \boldsymbol{t}). \tag{5.167}$$

Since $K_{\mathbf{z}}(\boldsymbol{A}^\top \boldsymbol{t}) = \frac{1}{2}\|\boldsymbol{A}^\top \boldsymbol{t}\|_2^2 = \frac{1}{2}\boldsymbol{t}^\top \boldsymbol{A}\boldsymbol{A}^\top \boldsymbol{t} = \frac{1}{2}\boldsymbol{t}^\top \boldsymbol{\Sigma}\boldsymbol{t}$, we can conclude that

$$K_{\mathbf{x}}(\boldsymbol{t}) = \langle \boldsymbol{t}, \boldsymbol{\mu} \rangle + \frac{1}{2}\boldsymbol{t}^\top \boldsymbol{\Sigma}\boldsymbol{t}, \tag{5.168}$$

and the MGF is just the exponential of the CGF. □

Therefore, $K_{\mathbf{x}_1}$ and $K_{\mathbf{x}_2}$ only depend on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. This means that as long as $\boldsymbol{A}_1\boldsymbol{A}_1^\top = \boldsymbol{A}_2\boldsymbol{A}_2^\top$, the distributions of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are the same.

It is possible to directly write down the PDF of a Gaussian vector, where the formula indeed only depends on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. This PDF is very complicated and is not often used directly, though. Still, let us write it down for completeness.

Theorem 5.38 (Gaussian PDF)**.** Let $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\det(\boldsymbol{\Sigma}) > 0$. Then

$$f_{\mathbf{x}}(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^d \det(\boldsymbol{\Sigma})}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}. \tag{5.169}$$

is a PDF of $\boldsymbol{x}$.

An important special case is $\mathcal{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I})$, whose PDF only depends on the norm of the vector:

$$f_{\mathbf{x}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{d/2}\sigma^d} e^{-\|\boldsymbol{x}\|_2^2/(2\sigma^2)}. \tag{5.170}$$

The following theorem shows that a linear transformation of a Gaussian vector is still a Gaussian vector, which generalizes the Gaussian linearity in Theorem 5.27 to the case of multiple dimensions.

Theorem 5.39 (Gaussian linearity)**.** Let $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then for any fixed matrix $\boldsymbol{S} \in \mathbb{R}^{r \times d}$,

$$\boldsymbol{S}\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{S}\boldsymbol{\mu}, \boldsymbol{S}\boldsymbol{\Sigma}\boldsymbol{S}^\top). \tag{5.171}$$

Proof. Since $\boldsymbol{x} = \boldsymbol{\mu} + \boldsymbol{A}\boldsymbol{z}$ for some $\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{\Sigma}$ and $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, we have

$$\boldsymbol{S}\boldsymbol{x} = \boldsymbol{S}(\boldsymbol{\mu} + \boldsymbol{A}\boldsymbol{z}) = \boldsymbol{S}\boldsymbol{\mu} + \boldsymbol{S}\boldsymbol{A}\boldsymbol{z}. \tag{5.172}$$

By the definition of a Gaussian vector, the distribution of $\boldsymbol{S}\boldsymbol{x}$ is

$$\mathcal{N}(\boldsymbol{S}\boldsymbol{\mu}, \boldsymbol{S}\boldsymbol{A}\boldsymbol{A}^\top\boldsymbol{S}^\top). \tag{5.173}$$

Noting that $\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{\Sigma}$ proves the theorem. □

A direct consequence of the above theorem is that if $\boldsymbol{x} = (X_1, X_2, \ldots, X_d) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then for every $i \in [d]$,

$$X_i \sim \mathcal{N}(\mu_i, \Sigma_{ii}), \tag{5.174}$$

where Σ_{ii} is the i-th diagonal entry of $\boldsymbol{\Sigma}$.

Finally, we note that the sum of two independent Gaussian vectors is still a Gaussian vector.

Theorem 5.40 (Sum of two independent Gaussian vectors)**.** Let $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ be independent Gaussian vectors in $\mathbb{R}^d$. Suppose

$$\boldsymbol{x}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1), \qquad \boldsymbol{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2). \tag{5.175}$$

Then

$$\boldsymbol{x}_1 + \boldsymbol{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2). \tag{5.176}$$

Proof. Note that $\boldsymbol{x}_1 + \boldsymbol{x}_2$ is a linear transformation of the Gaussian vector

$$\begin{bmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \end{bmatrix} \in \mathbb{R}^{2d}. \tag{5.177}$$

By Gaussian linearity in Theorem 5.39, $\boldsymbol{x}_1 + \boldsymbol{x}_2$ is still a Gaussian vector. Its mean vector is

$$\mathbb{E}[\boldsymbol{x}_1 + \boldsymbol{x}_2] = \mathbb{E}[\boldsymbol{x}_1] + \mathbb{E}[\boldsymbol{x}_2] = \boldsymbol{\mu}_1 + \boldsymbol{\mu}_2. \tag{5.178}$$

And its covariance matrix is

$$\mathrm{Cov}(\boldsymbol{x}_1 + \boldsymbol{x}_2) = \mathrm{Cov}(\boldsymbol{x}_1) + \mathrm{Cov}(\boldsymbol{x}_2) = \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2. \tag{5.179}$$

Therefore, $\boldsymbol{x}_1 + \boldsymbol{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)$. □

In summary, when moving from Gaussian random variables to Gaussian vectors, many one-dimensional results extend naturally. The main difference is that the mean becomes a mean vector and the variance becomes a covariance matrix, so you need to upgrade your real number arithmetic to linear algebra.

5.3 The Strange Geometry of High Dimensions

We live in a three-dimensional world. Your favorite cartoon characters may live in a two-dimensional world, whose number of dimensions is even less than three. In fact, you may have never seen *high-dimensional* objects with your naked eye, but many objects that you may encounter in computer science live in high-dimensional spaces.

5.3.1 Example: Embedding Vectors

Whenever you input a search query, the search engine compares it with the documents in its database. Similarly, when you browse a Xiaohongshu note or a Douyin video, the recommendation system estimates your interests and treats your interests as a query to find items you might like.

In both cases, we need to represent documents, notes, or short videos in a form that computers can easily understand and use for comparison. If we only use the raw text of documents or the raw content of short videos, to a computer these are just sequences of bytes. User queries are also just another sequence of bytes. How can a computer tell when two sequences of bytes are similar?

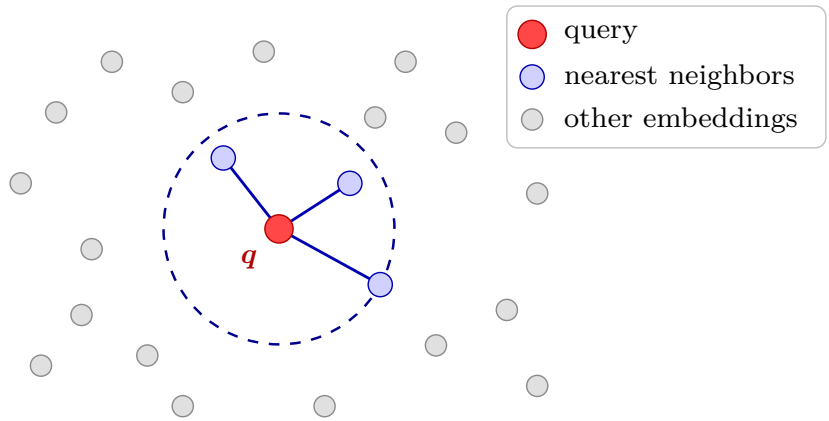


Figure 5.5: An example of nearest neighbor search in an embedding space.

Here is where machine learning comes in. A common approach is to use a machine learning model, such as a neural network, to *embed* each item (document, note, or short video) into a vector in a high-dimensional space $\mathbb{R}^d$. For each item x, the embedding model outputs a vector

$$f(x) \in \mathbb{R}^d. \tag{5.180}$$

Such vectors are called *embedding vectors*, or simply *embeddings*. The model to produce embeddings is called an *embedding model*. An embedding model is usually trained on a large amount of data to learn the semantic meaning of the items and to make similar items have similar embeddings. Mathematically, if two items x and x' are similar, the model is expected to produce embeddings that are close to each other:

$$\text{if } x \text{ and } x' \text{ are similar for humans} \quad \Rightarrow \quad \|f(x) - f(x')\|_2 \text{ is small.} \tag{5.181}$$

For example, the model is expected to embed two short videos of the same Internet meme into nearby vectors in the embedding space. Here we choose the Euclidean distance as the distance metric. But in general, one can use any distance metric that is easily computable.

Whenever a query comes in, the search engine or recommendation system embeds the query into a vector $\boldsymbol{q}$ and compares it with the embeddings in its database. The embeddings with the top k smallest Euclidean distances to $\boldsymbol{q}$ are identified as the most similar items. This step is called *nearest neighbor search*. These items are then returned to the user after additional filtering and ordering steps, commonly referred to as *pre-ranking* and *ranking*.

To make the whole system work, the embeddings need to be high-dimensional to capture the complexity of the items. It is hard to imagine that a 2-dimensional embedding could fully represent all the relevant information in a document, such as its topic, key entities, main claims. However, if we have a few hundred or even a few thousand dimensions, maybe one vector is enough? Well, since some of you may already be addicted to short videos, it seems that one vector is enough.

It is perfectly fine if you do not fully understand the details of the above example on embeddings. If you are interested in this topic, you could start with a machine learning course. To really understand how the Google search engine or Xiaohongshu/Douyin recommendation systems work, you would need a more specialized course on information retrieval.

It is ok if you could not fully understand the above example on embeddings, since the point is just to give you an idea of how high-dimensional vectors naturally arise in machine learning. But if you are interested in this topic, you can take a machine learning course first. If you really want to understand the Google search engine or Xiaohongshu/Douyin recommendation systems, you need to take a more specialized course on information retrieval.

5.3.2 High-dimensional Gaussians, Spheres, and Balls

You have never seen high-dimensional objects with your naked eye. Are you sure that your intuition for low-dimensional objects is still valid in high dimensions?

To build up your intuition for high-dimensional objects, let us start with three fundamental distributions:

1. Gaussian distribution $\mathcal{G} = \mathcal{N}(\mathbf{0}, \boldsymbol{I}/d)$:

   The Gaussian distribution with mean $\mathbf{0} \in \mathbb{R}^d$ and covariance matrix $\boldsymbol{I}/d \in \mathbb{R}^{d\times d}$. The scaling factor $1/d$ here is intentional. For $\boldsymbol{x} = (X_1, \ldots, X_d) \sim \mathcal{G}$, we have $\mathbb{E}[\|\boldsymbol{x}\|_2^2] = \sum_{i=1}^d \mathbb{E}[X_i^2] = d\cdot(1/d) = 1$. Therefore, this scaling keeps the expected squared norm equal to 1, independent of d.

2. Uniform sphere distribution $\mathcal{U}_{\mathbb{S}}$:

   The uniform distribution on the unit sphere $\mathbb{S}^{d-1} = \{\boldsymbol{x} \in \mathbb{R}^d : \|\boldsymbol{x}\|_2 = 1\}$. The norm of a point on the unit sphere is always 1.

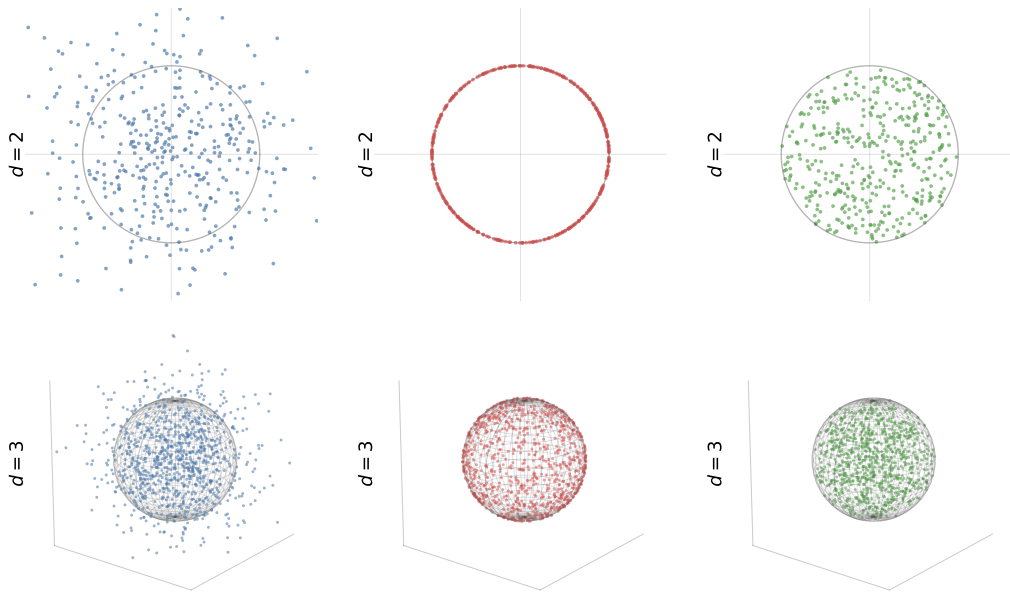
3. Uniform ball distribution $\mathcal{U}_B$:

   The uniform distribution on the unit ball $B^d = \{\boldsymbol{x} \in \mathbb{R}^d : \|\boldsymbol{x}\|_2 \le 1\}$. The norm of a point in the unit ball is always upper bounded by 1.

It can be seen that all three distributions are *isotropic* distributions. In general, a distribution is said to be *isotropic* if $\boldsymbol{x} \sim \mathcal{D}$ has the same distribution as $\boldsymbol{A}\boldsymbol{x}$ for any orthogonal matrix $\boldsymbol{A} \in \mathbb{R}^{d\times d}$. It is see to see that $\mathcal{U}_{\mathbb{S}}$ and $\mathcal{U}_B$ are isotropic, since orthogonal transformations preseve the norm: $\|\boldsymbol{A}\boldsymbol{x}\|_2^2 = \boldsymbol{x}^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{I}\boldsymbol{x} = \|\boldsymbol{x}\|_2^2$. For the Gaussian distribution $\mathcal{G}$, by Theorem 5.39, $\boldsymbol{A}\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{A}(\boldsymbol{I}/d)\boldsymbol{A}^\top) = \mathcal{N}(\mathbf{0}, \boldsymbol{A}\boldsymbol{A}^\top/d) = \mathcal{N}(\mathbf{0}, \boldsymbol{I}/d)$, so it is also isotropic.

In the radial direction, however, the three distributions are different: The Gaussian fills all of $\mathbb{R}^d$, the sphere distribution lives exactly on $\mathbb{S}^{d-1}$, and the ball distribution fills B^d.

From Figure 5.6, we can see that the three distributions indeed look very different in $\mathbb{R}^2$ and $\mathbb{R}^3$. However, we will show that if you want to understand the high-dimensional geometry of these distributions, this picture is very misleading!



(a) $\mathcal{G}$: Gaussian $\mathcal{N}(\mathbf{0}, \boldsymbol{I}/d)$. (b) $\mathcal{U}_{\mathbb{S}}$: Uniform on $\mathbb{S}^{d-1}$. (c) $\mathcal{U}_B$: Uniform on B^d.

Figure 5.6: Three isotropic distributions in $\mathbb{R}^2$ and $\mathbb{R}^3$.

▶ Observation 1: Concentration Around the Surface

First, we show that the uniform ball distribution is not so different from the uniform sphere distribution in high dimensions. This is simply due to the following observation:

Most of the volume of a high-dimensional object is near its surface.

To make this observation precise, let us introduce some notations. In the following, we use $\mu(A) := \int_A \mathrm{d}\boldsymbol{x}$ to denote the volume of a set $A \subseteq \mathbb{R}^d$. For any $r \in \mathbb{R}$, let $rA := \{r\boldsymbol{x} : \boldsymbol{x} \in A\}$ be the set obtained by scaling A by a factor of r. Then we have the following theorem:

Theorem 5.41. For any $A \subseteq \mathbb{R}^d$ with $\mu(A) > 0$ and any $\epsilon \in (0, 1)$,

$$\frac{\mu((1-\epsilon)A)}{\mu(A)} = (1-\epsilon)^d \leq e^{-\epsilon d}. \tag{5.182}$$

Proof. Integration by substitution with $\boldsymbol{u} = \frac{1}{1-\epsilon}\boldsymbol{x}$ gives

$$\mu((1-\epsilon)A) = \int_{(1-\epsilon)A} \mathrm{d}\boldsymbol{x} = \int_A (1-\epsilon)^d \,\mathrm{d}\boldsymbol{u} = (1-\epsilon)^d \mu(A). \tag{5.183}$$

Noting that $(1-\epsilon)^d \leq e^{-\epsilon d}$, the theorem follows. □

This theorem implies that even if you shrink the set A only by a small amount $\epsilon > 0$, the volume of the shrunk set will be significantly smaller in high dimen-

sions. If the set A is *star-shaped* around the origin, which refers to the property that $rA \subseteq A$ for every $r \in [0,1]$, then $(1-\epsilon)A \subseteq A$. In this case, the theorem implies that most of the volume of A is near its surface:

$$\frac{\mu(A \setminus (1-\epsilon)A)}{\mu(A)} \geq 1 - e^{-\epsilon d}. \tag{5.184}$$

Applying Theorem 5.41 to the uniform ball distribution implies the following:

Theorem 5.42 (Uniform ball distribution concentrates near the sphere)**.** Let $\boldsymbol{x} \sim \mathcal{U}_B$. Then, for every $\epsilon \in (0,1)$,

$$\Pr\Big[1 - \epsilon < \|\boldsymbol{x}\|_2 \leq 1\Big] \geq 1 - e^{-\epsilon d}. \tag{5.185}$$

Proof. Just need to note that $B^d \setminus (1-\epsilon)B^d = \{\boldsymbol{x} \in \mathbb{R}^d : 1 - \epsilon < \|\boldsymbol{x}\|_2 \leq 1\}$. □

Imagine that your mom steps into your room with an orange. Since you are busy working on your homework, your mom may peel the orange for you and give you the juicy flesh inside so that you can eat it directly. This only happens in low dimensions. If we were high-dimensional creatures, after your mom peeled the orange, there would be almost nothing left inside.

▶ Observation 2: Concentration of the Norm

Next, we show that the Gaussian distribution is also not so different from the uniform sphere distribution in high dimensions. While Gaussian vectors in low dimensions may look like they can spread all over the place, in high dimensions they almost have a deterministic norm:

$$\textit{If } \boldsymbol{x} \sim \mathcal{N}(0, \boldsymbol{I}/d), \textit{ then } \|\boldsymbol{x}\|_2 \textit{ concentrates near } 1.$$

This is a simple consequence of Chernoff bound, which we state as the following theorem:

Theorem 5.43 (Gaussian concentrates around the sphere)**.** Let $\boldsymbol{x} \sim \mathcal{G}$. Then, for every $\epsilon \in (0,1)$,

$$\Pr\Big[\Big|\|\boldsymbol{x}\|_2^2 - 1\Big| \geq \epsilon\Big] \leq 2e^{-d\epsilon^2/8}. \tag{5.186}$$

In particular,

$$\Pr\Big[1 - \epsilon < \|\boldsymbol{x}\|_2 < 1 + \epsilon\Big] \geq 1 - 2e^{-d\epsilon^2/8}. \tag{5.187}$$

Proof. Write $\boldsymbol{x} = \frac{1}{\sqrt{d}}(Z_1, \ldots, Z_d)$, where $Z_1, \ldots, Z_d$ are i.i.d. standard Gaussians. Then $|\boldsymbol{x}|^2$ is the sample average of $Z_1^2, \ldots, Z_d^2$:

$$\|\boldsymbol{x}\|_2^2 = \frac{1}{d}\sum_{i=1}^{d} Z_i^2. \tag{5.188}$$

Let $Z \sim \mathcal{N}(0,1)$. Then Z^2 has the same distribution as $Z_1^2, \dots, Z_d^2$. It is easy to see that $\mathbb{E}[Z^2] = 1$. To apply Chernoff bound, we need to compute the MGF and rate function of Z^2. Integrating the Gaussian PDF over $\mathbb{R}$, we have

$$M_{Z^2}(t) = \mathbb{E}[e^{tZ^2}] = \int_{\mathbb{R}} e^{tx^2} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \, \mathrm{d}x \tag{5.189}$$

$$= \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-(1-2t)x^2/2} \, \mathrm{d}x. \tag{5.190}$$

If $t \geq 1/2$, then $e^{-(1-2t)x^2/2}$ does not decay to 0 as $x \to \infty$, and thus the integral is infinite. If $t < 1/2$, then we can compute the integral by comparing the integrand with the PDF of $\mathcal{N}(0, \frac{1}{1-2t})$ from Theorem 5.24. Therefore, we have

$$M_{Z^2}(t) = \begin{cases} \frac{1}{\sqrt{1-2t}} & \text{if } t < 1/2, \\ +\infty & \text{if } t \geq 1/2. \end{cases} \tag{5.191}$$

Next, we compute the rate function (Definition 3.29):

$$I_{Z^2}(a) = \sup_{t \in \mathbb{R}} \{ta - \log M_{Z^2}(t)\} = \sup_{t < 1/2} \left\{ ta + \frac{1}{2}\log(1-2t) \right\}. \tag{5.192}$$

Solving the optimization problem by setting the derivative to zero, we get

$$I_{Z^2}(a) = ta + \frac{1}{2}\log(1-2t)\Big|_{t=(1-1/a)/2} = \frac{1}{2}(a - 1 - \log a). \tag{5.193}$$

See Figure 5.7 for a visualization of the MGF and rate function of Z^2.

By the generic Chernoff bound for sample averages (Theorem 3.34), we have

$$\forall a \geq 1: \quad \Pr[\|\boldsymbol{x}\|_2^2 \geq a] \leq e^{-dI_{Z^2}(a)}, \tag{5.194}$$

$$\forall a \leq 1: \quad \Pr[\|\boldsymbol{x}\|_2^2 \leq a] \leq e^{-dI_{Z^2}(a)}. \tag{5.195}$$

For every $\epsilon \in (0,1)$, we have

$$I_{Z^2}(1+\epsilon) = \frac{1}{2}(\epsilon - \log(1+\epsilon)) \geq \frac{\epsilon^2}{8}, \tag{5.196}$$

$$I_{Z^2}(1-\epsilon) = \frac{1}{2}(-\epsilon - \log(1-\epsilon)) \geq \frac{\epsilon^2}{8}, \tag{5.197}$$

where we use the inequality $\ln(1+x) \leq x - \frac{1}{4}x^2$ for $x \in (-1,1)$. Combining these with the Chernoff bound and the union bound, we get

$$\Pr\Big[\Big|\|\boldsymbol{x}\|_2^2 - 1\Big| \geq \epsilon\Big] \leq 2e^{-d\epsilon^2/8}. \tag{5.198}$$

Finally, since $\big|\|\boldsymbol{x}\|_2^2 - 1\big| = \big|\|\boldsymbol{x}\|_2 - 1\big| \cdot \big|\|\boldsymbol{x}\|_2 + 1\big|$, the condition $\big|\|\boldsymbol{x}\|_2 - 1\big| \geq \epsilon$ implies the condition $\big|\|\boldsymbol{x}\|_2^2 - 1\big| \geq \epsilon$. Therefore,

$$\Pr\Big[1 - \epsilon < \|\boldsymbol{x}\|_2 < 1 + \epsilon\Big] = 1 - \Pr\Big[\Big|\|\boldsymbol{x}\|_2 - 1\Big| \geq \epsilon\Big] \geq 1 - 2e^{-d\epsilon^2/8}. \tag{5.199}$$

This proves the concentration of the norm itself. □

In fact, one can easily generalize the above proof to a more general setting. If $Z_1, \dots, Z_d$ are i.i.d. random variables, and $\boldsymbol{x} = \frac{1}{\sqrt{d}}(Z_1, \dots, Z_d)$ is a normalized

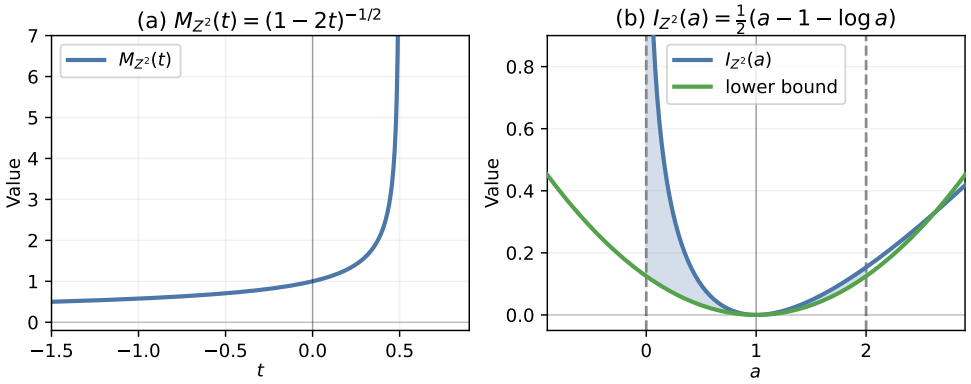


Figure 5.7: The MGF and rate function of Z^2 when $Z \sim \mathcal{N}(0,1)$. (a) The MGF of Z^2 is finite only for $t < 1/2$ and equals $(1-2t)^{-1/2}$ there; (b) The rate function is $I_{Z^2}(a) = \frac{1}{2}(a - 1 - \log a)$ for $a > 0$, which is lower bounded by $\frac{1}{8}(a-1)^2$ when $a \in (0, 2)$.

random vector formed by them, then the squared norm of $\boldsymbol{x}$ is the sample average of $Z_1^2, \ldots, Z_d^2$:

$$\|\boldsymbol{x}\|_2^2 = \frac{1}{d}\sum_{i=1}^{d} Z_i^2. \tag{5.200}$$

Whenever the squared coordinates Z_i^2 have a good concentration bound for their sample average, the norm of $\boldsymbol{x}$ concentrates near its expected value, or equivalently, $\boldsymbol{x}$ concentrates near a sphere.

▶ Observation 3: Concentration Around the Equator

Back to our discussion on high-dimensional oranges. After realizing that peeling the skin of an orange may leave you with almost nothing to eat, your mom buys a watermelon and brings it into your room instead. She cuts the watermelon into two halves through the middle, but during the cut, a very thin slice of thickness ϵ around the middle is lost.

In low dimensions, this is not a big deal: after removing this thin middle slice, there is still plenty of watermelon left on both sides. However, in high dimensions, the situation is completely different:

Most of the volume of a high-dimensional ball is concentrated near its equator.

Therefore, most of the watermelon is in fact in the ϵ-thin middle slice that was lost during the cut. Again, you have nothing to eat.

More formally, we fix a unit vector $\boldsymbol{u} \in \mathbb{R}^d$. The equator perpendicular to $\boldsymbol{u}$ is the set of points $\boldsymbol{x}$ satisfying $\langle \boldsymbol{u}, \boldsymbol{x} \rangle = 0$ on the unit sphere. The equatorial plane perpendicular to $\boldsymbol{u}$ is the set of points $\boldsymbol{x}$ satisfying $\langle \boldsymbol{u}, \boldsymbol{x} \rangle = 0$ in the unit ball. The following theorem shows that the uniform ball distribution concentrates near the equatorial plane.

Theorem 5.44 (Uniform ball distribution concentrates near the equatorial plane)**.** Let $d \geq 3$. Fix a unit vector $\boldsymbol{u} \in \mathbb{R}^d$, and let $\boldsymbol{x} \sim \mathcal{U}_B$. Then, for every $\epsilon \in (0, 1)$,

$$\Pr[|\langle \boldsymbol{u}, \boldsymbol{x} \rangle| < \epsilon] \geq 1 - 6e^{-(d-1)\epsilon^2/2}. \tag{5.201}$$

The above theorem is a direct consequence of the following lemma, which bounds the inner product $\langle \boldsymbol{u}, \boldsymbol{x} \rangle$ from only one side.

Lemma 5.45. Let $d \geq 3$. Fix a unit vector $\boldsymbol{u} \in \mathbb{R}^d$, and let $\boldsymbol{x} \sim \mathcal{U}_B$. Then, for every $\epsilon \in (0, 1)$,

$$\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq \epsilon] \leq 3e^{-(d-1)\epsilon^2/2}, \quad \Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \leq -\epsilon] \leq 3e^{-(d-1)\epsilon^2/2}. \tag{5.202}$$

Proof of Theorem 5.44. Add the two inequalities in Lemma 5.45 together, then

$$\Pr[|\langle \boldsymbol{u}, \boldsymbol{x} \rangle| < \epsilon] = \Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq \epsilon] + \Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \leq -\epsilon] \leq 6e^{-(d-1)\epsilon^2/2}. \tag{5.203}$$

Taking the complement gives the desired bound. □

This lemma implies that if you cut a ball into two parts near the equatorial plane, then even a small deviation from the equatorial plane can lead to a highly uneven distribution of volume: one part contains almost all the volume, while the other has exponentially small volume. This means that when you cut a watermelon through the middle to share it with someone else, such as your mom, even if no volume is lost during the cut, getting a fair share is still very difficult unless you control the cutting position very precisely.

Why is this the case? To get some quick intuition, we consider the Gaussian case instead. Since we have already established that $\mathcal{U}_B \approx \mathcal{U}_{\mathbb{S}}$ and $\mathcal{G} \approx \mathcal{U}_{\mathbb{S}}$ in the previous two observations, we can analyze $\mathcal{G}$ to gain intuition about $\mathcal{U}_B$.

It is very easy to derive the distribution of $\langle \boldsymbol{u}, \boldsymbol{x} \rangle$ for $\boldsymbol{x} \sim \mathcal{G}$ explicitly. By Gaussian linearity in Theorem 5.39,

$$\langle \boldsymbol{u}, \boldsymbol{x} \rangle \sim \mathcal{N}\left(0, \tfrac{1}{d}\right). \tag{5.204}$$

Applying the Gaussian Chernoff bound in Lemma 5.29, we have

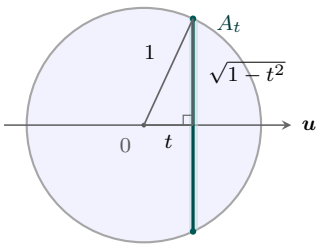
$$\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq \epsilon] \leq e^{-d\epsilon^2/2}, \qquad \Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \leq -\epsilon] \leq e^{-d\epsilon^2/2}. \tag{5.205}$$

Therefore, for every $\epsilon \geq 0$,

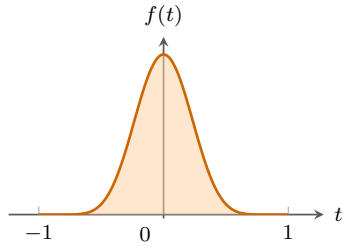
$$\Pr[|\langle \boldsymbol{u}, \boldsymbol{x} \rangle| < \epsilon] \geq 1 - 2e^{-d\epsilon^2/2}. \tag{5.206}$$

This yields almost the same bound as Theorem 5.44, with only minor difference in the exponent and the constant factor.

Now let us prove the real bound for the uniform ball distribution $\mathcal{U}_B$.



(a) A slice A_t.



(b) The PDF of $\langle \boldsymbol{u}, \boldsymbol{x} \rangle$ when $d = 20$.

Figure 5.8: Slicing the unit ball perpendicular to $\boldsymbol{u}$. A slice at signed distance t has radius $\sqrt{1-t^2}$, so the volume of the slice is proportional to $(1-t^2)^{(d-1)/2}$.

Proof of Lemma 5.45. Let V_m denote the volume of the m-dimensional unit ball. For a fixed value $t \in [-1, 1]$, consider the slice

$$A_t := \{\boldsymbol{x} \in B^d : \langle \boldsymbol{u}, \boldsymbol{x} \rangle = t\}. \tag{5.207}$$

This slice lies in an affine hyperplane whose normal vector is $\boldsymbol{u}$. Inside that affine hyperplane, it is a $(d-1)$-dimensional ball of radius $\sqrt{1-t^2}$. Then by Theorem 5.41,

$$\mu(A_t) := \left(\sqrt{1-t^2}\right)^{d-1} \cdot V_{d-1} = (1-t^2)^{(d-1)/2} V_{d-1}, \tag{5.208}$$

where $\mu(A_t)$ denotes the $(d-1)$-dimensional volume of A_t.

For every $a < b$, the probability that $\langle \boldsymbol{u}, \boldsymbol{x} \rangle \in (a, b)$ is the ratio of the volume of $\{\boldsymbol{x} \in B^d : a < \langle \boldsymbol{u}, \boldsymbol{x} \rangle < b\}$ to the volume of B^d. Therefore,

$$\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \in (a, b)] = \frac{\int_a^b \mu(A_t)\ \mathrm{d}t}{\int_{-1}^1 \mu(A_t)\ \mathrm{d}t} = \frac{\int_a^b (1-t^2)^{(d-1)/2} V_{d-1}\ \mathrm{d}t}{\int_{-1}^1 (1-t^2)^{(d-1)/2} V_{d-1}\ \mathrm{d}t} \tag{5.209}$$

$$= \frac{\int_a^b (1-t^2)^{(d-1)/2}\ \mathrm{d}t}{\int_{-1}^1 (1-t^2)^{(d-1)/2}\ \mathrm{d}t}. \tag{5.210}$$

Let

$$I := \int_{-1}^1 (1-t^2)^{(d-1)/2}\ \mathrm{d}t. \tag{5.211}$$

Then $\langle \boldsymbol{u}, \boldsymbol{x} \rangle$ has the PDF:

$$f(t) = \begin{cases} \frac{1}{I}(1-t^2)^{(d-1)/2} & \text{if } t \in [-1, 1], \\ 0 & \text{otherwise.} \end{cases} \tag{5.212}$$

By symmetry, $\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq \epsilon] = \Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \leq -\epsilon]$. It suffices to upper bound the probability $\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq \epsilon]$. Based on the PDF $f(t)$, we can express the probability that $\langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq \epsilon$ as:

$$\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq \epsilon] = \int_\epsilon^{+\infty} f(t)\ \mathrm{d}t. \tag{5.213}$$

Let $\tilde{\varphi}(t) := \frac{\sqrt{d-1}}{\sqrt{2\pi}} e^{-(d-1)t^2/2}$ be the PDF of $\mathcal{N}(0, \frac{1}{d-1})$. Then, for every $t \in [-1, 1]$,

$$f(t) = \frac{1}{I}(1-t^2)^{(d-1)/2} \le \frac{1}{I} e^{-(d-1)t^2/2} = \frac{\sqrt{2\pi}}{I\sqrt{d-1}} \tilde{\varphi}(t). \tag{5.214}$$

For $t \notin [-1, 1]$, we have $f(t) = 0$, so the same upper bound still holds. Integrating over the interval $[\epsilon, +\infty)$, we get

$$\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \ge \epsilon] = \int_{\epsilon}^{+\infty} f(t)\, \mathrm{d}t \le \frac{\sqrt{2\pi}}{I\sqrt{d-1}} \int_{\epsilon}^{+\infty} \tilde{\varphi}(t)\, \mathrm{d}t. \tag{5.215}$$

By Gaussian Chernoff bound in Lemma 5.29,

$$\int_{\epsilon}^{+\infty} \tilde{\varphi}(t)\, \mathrm{d}t \le e^{-(d-1)\epsilon^2/2}. \tag{5.216}$$

Combining the two inequalities, we get

$$\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \ge \epsilon] \le \frac{\sqrt{2\pi}}{I\sqrt{d-1}} e^{-(d-1)\epsilon^2/2}. \tag{5.217}$$

Now we turn to lower bound I. Restricting the integration interval in the definition of I to $[-\frac{1}{\sqrt{d-1}}, \frac{1}{\sqrt{d-1}}]$, we have

$$I = \int_{-1}^{+1} (1-t^2)^{(d-1)/2}\, \mathrm{d}t \ge \int_{-\frac{1}{\sqrt{d-1}}}^{\frac{1}{\sqrt{d-1}}} (1-t^2)^{(d-1)/2}\, \mathrm{d}t \tag{5.218}$$

$$\ge \int_{-\frac{1}{\sqrt{d-1}}}^{\frac{1}{\sqrt{d-1}}} \left(1 - \frac{d-1}{2} t^2\right) \mathrm{d}t \tag{5.219}$$

$$\ge \int_{-\frac{1}{\sqrt{d-1}}}^{\frac{1}{\sqrt{d-1}}} \frac{1}{2}\, \mathrm{d}t = \frac{1}{\sqrt{d-1}}, \tag{5.220}$$

where the second line uses the fact that $(1-x)^\alpha \ge 1 - \alpha x$ for $x \in [0, 1]$ and $\alpha \ge 1$. Putting this back into the inequality, we get

$$\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \ge \epsilon] \le \sqrt{2\pi} e^{-(d-1)\epsilon^2/2} < 3e^{-(d-1)\epsilon^2/2}. \tag{5.221}$$

This proves the upper bound for the upper tail $\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \ge \epsilon]$. The upper bound for the lower tail $\Pr[\langle \boldsymbol{u}, \boldsymbol{x} \rangle \le -\epsilon]$ follows by symmetry. □

The above proof shows that the uniform ball distribution concentrates near the equatorial plane. Since we have already established in Theorem 5.42 that the uniform ball distribution concentrates near the sphere, we can apply the union bound to conclude that the uniform ball distribution concentrates near the equator on the sphere.

Theorem 5.46 (Uniform ball distribution concentrates near the equator)**.** Let $d \ge 3$. Fix a unit vector $\boldsymbol{u} \in \mathbb{R}^d$, and let $\boldsymbol{x} \sim \mathcal{U}_B$. Then, for every $\epsilon \in (0, 1)$,

$$\Pr\left[1 - \tfrac{1}{2}\epsilon^2 < \|\boldsymbol{x}\|_2 \le 1 \text{ and } |\langle \boldsymbol{u}, \boldsymbol{x} \rangle| < \epsilon\right] \ge 1 - 7e^{-(d-1)\epsilon^2/2}. \tag{5.222}$$

Proof. Let A be the event that $1 - \frac{1}{2}\epsilon^2 < \|\boldsymbol{x}\|_2 \le 1$ and B be the event that $|\langle \boldsymbol{u}, \boldsymbol{x}\rangle| < \epsilon$. By the union bound, we have

$$\Pr[\bar{A} \cup \bar{B}] \le \Pr[\bar{A}] + \Pr[\bar{B}] \le e^{-\frac{1}{2}\epsilon^2 d} + 6e^{-(d-1)\epsilon^2/2} \le 7e^{-(d-1)\epsilon^2/2}. \tag{5.223}$$

Taking the complement gives the desired bound. □

Similarly, from Theorem 5.43 and Equation (5.206), we can derive that the Gaussian distribution $\mathcal{G}$ also concentrates near the equator on the sphere.

Theorem 5.47 (Gaussian distribution concentrates near the equator)**.** Fix a unit vector $\boldsymbol{u} \in \mathbb{R}^d$, and let $\boldsymbol{x} \sim \mathcal{G}$. Then, for every $\epsilon \in (0, 1)$,

$$\Pr\Big[1 - 2\epsilon < \|\boldsymbol{x}\|_2 < 1 + 2\epsilon \text{ and } |\langle \boldsymbol{u}, \boldsymbol{x}\rangle| < \epsilon\Big] \ge 1 - 4e^{-d\epsilon^2/2}. \tag{5.224}$$

Proof. Let A be the event that $1 - 2\epsilon < \|\boldsymbol{x}\|_2 < 1 + 2\epsilon$ and B be the event that $|\langle \boldsymbol{u}, \boldsymbol{x}\rangle| < \epsilon$. By the union bound, we have

$$\Pr[\bar{A} \cup \bar{B}] \le \Pr[\bar{A}] + \Pr[\bar{B}] \le 2e^{-d(2\epsilon)^2/8} + 2e^{-d\epsilon^2/2} = 4e^{-d\epsilon^2/2}. \tag{5.225}$$

Taking the complement gives the desired bound. □

▶ Observation 4: Exponentially Many Nearly Orthogonal Directions

Since you failed to eat any orange or watermelon that your mom brought you, you decide to eat some durian instead, a ball-shaped fruit covered with spikes. In low dimensions, the spikes can hurt you, so you have to handle the durian with care.

But in high dimensions, a durain is much more manageable: Even if a durain has exponentially many spikes, the spikes may still point in almost perpendicular directions. Thus the spikes do not crowd together in the way our three-dimensional intuition might suggest. This is all based on the following observation:

Even exponentially many randomly sampled directions are almost orthogonal.

More formally, the following theorem says that even if we sample exponentially many directions from the sphere, there is still a good chance that all pairs of directions are almost orthogonal.

Theorem 5.48 (Random directions are almost orthogonal)**.** Let $d \ge 3$. For any $\epsilon \in (0, 1)$ with $\epsilon = \Omega(\frac{1}{\sqrt{d}})$, there exists $N = e^{\Theta(d\epsilon^2)}$ such that for every $n \le N$, if we sample n i.i.d. random vectors $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$ from $\mathcal{U}_{\mathbb{S}}$, then

$$\Pr\Big[|\langle \boldsymbol{x}_i, \boldsymbol{x}_j\rangle| < \epsilon \text{ for every } i \ne j\Big] \ge 0.99. \tag{5.226}$$

Here $N = e^{\Theta(d\epsilon^2)}$ means that $N = e^{f(d,\epsilon)}$ for some non-negative function $f(d, \epsilon) = \Theta(d\epsilon^2)$, where $\Theta(\,\cdot\,)$ is written for $d \ge 3$ and as $\epsilon \to 0$.

An immediate consequence is that these unit vectors are well-separated in terms of Euclidean distances:

$$\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 = 2 - 2\langle \boldsymbol{x}_i, \boldsymbol{x}_j \rangle > 2 - 2\epsilon. \tag{5.227}$$

Another implication is that it is possible to find $e^{\Theta(d\epsilon^2)}$ unit vectors that are nearly orthogonal, which follows directly from the probabilistic method. This is fundamentally a high-dimensional phenomenon: in low dimensions, it is impossible to find many nearly orthogonal directions. For instance, in $\mathbb{R}^2$, if we require the pairwise inner products to be at most $\cos 80^\circ$, then at most 2 vectors can satisfy this condition.

Now we prove the theorem.

Proof. Let $\delta = 0.01$. Let E_{ij} be the event that $|\langle \boldsymbol{x}_i, \boldsymbol{x}_j \rangle| \geq \epsilon$. It suffices to show that $\Pr[E_{ij}] \leq \delta/n^2$ for every $i \neq j$. Once we have this, by the union bound,

$$\Pr\left[\bigcup_{i<j} E_{ij}\right] \leq n^2 \cdot \frac{\delta}{n^2} \leq \delta. \tag{5.228}$$

To upper bound $\Pr[E_{ij}]$, we sample $\boldsymbol{z}_i, \boldsymbol{z}_j \sim \mathcal{U}_B$ independently and then set $\boldsymbol{y}_i := \frac{\boldsymbol{z}_i}{\|\boldsymbol{z}_i\|_2}$ and $\boldsymbol{y}_j := \frac{\boldsymbol{z}_j}{\|\boldsymbol{z}_j\|_2}$. The direction of a random vector in the ball is uniform on the sphere, so $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ has the same distribution as $(\boldsymbol{y}_i, \boldsymbol{y}_j)$.

By Theorem 5.46, for every $\epsilon_0 \in (0,1)$, with probability at least $1 - 7e^{-(d-1)\epsilon_0^2/2}$,

$$1 - \tfrac{1}{2}\epsilon_0^2 < \|\boldsymbol{z}_i\|_2 \leq 1 \quad \text{and} \quad |\langle \boldsymbol{z}_i, \boldsymbol{y}_j \rangle| < \epsilon_0. \tag{5.229}$$

In this case,

$$|\langle \boldsymbol{y}_i, \boldsymbol{y}_j \rangle| = \frac{1}{\|\boldsymbol{z}_i\|_2} |\langle \boldsymbol{z}_i, \boldsymbol{y}_j \rangle| < \frac{\epsilon_0}{1 - \frac{1}{2}\epsilon_0^2} < 2\epsilon_0. \tag{5.230}$$

Setting $\epsilon_0 = \epsilon/2$, we get

$$\Pr[E_{ij}] = \Pr[|\langle \boldsymbol{y}_i, \boldsymbol{y}_j \rangle| \geq \epsilon] \leq 7e^{-(d-1)\epsilon_0^2/2} = 7e^{-(d-1)\epsilon^2/8}. \tag{5.231}$$

Setting the right-hand side to δ/N^2, we get

$$N = \frac{\sqrt{\delta}}{\sqrt{7e^{-(d-1)\epsilon^2/8}}} = \sqrt{\tfrac{\delta}{7}} e^{(d-1)\epsilon^2/16} \tag{5.232}$$

$$= e^{(d-1)\epsilon^2/16 + \log(\delta/7)/2} = e^{\Theta(d\epsilon^2)}. \tag{5.233}$$

Under this choice of N, we have

$$\Pr[E_{ij}] = \frac{\delta}{N^2} \leq \frac{\delta}{n^2}. \tag{5.234}$$

Taking the union bound over all pairs $i < j$, we obtain the desired bound. □

5.3.3 Random Projection and Johnson–Lindenstrauss Lemma

Now return to the embedding example from the beginning of the section. Suppose you have a large collection of items and have used an embedding model to

map them to vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$. However, the dimension d may be quite large, and you might wonder whether it is possible to reduce the embeddings to a smaller dimension k while approximately preserving the pairwise distances between items. More specifically, can we find a map $F : \mathbb{R}^d \to \mathbb{R}^k$ such that

$$\|F(\boldsymbol{x}_i) - F(\boldsymbol{x}_j)\|_2^2 \approx \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 \tag{5.235}$$

for all $i, j \in [n]$?

You might give a negative answer to this question if you use your intuition from low dimensions. Clearly, if you have many points in 3D, there is no way to map them into 2D while preserving all pairwise distances. However, in high dimensions, the famous *Johnson-Lindenstrauss lemma* (JL lemma) shows that finding such a map is indeed possible. This result is named after William J. Johnson and Joram Lindenstrauss, who proved it in 1984 [JL84].

Theorem 5.49 (Johnson-Lindenstrauss lemma)**.** Let $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$ be n points in $\mathbb{R}^d$, where $n \geq 2$. For every $\epsilon \in (0, 1)$ and any integer k with

$$k \geq \frac{16 \log n}{\epsilon^2}, \tag{5.236}$$

there exists a linear map $F : \mathbb{R}^d \to \mathbb{R}^k$ such that for all $1 \leq i, j \leq n$

$$(1 - \epsilon)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 \leq \|F(\boldsymbol{x}_i) - F(\boldsymbol{x}_j)\|_2^2 \leq (1 + \epsilon)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2. \tag{5.237}$$

A surprising property of the JL lemma is that the dimension k is independent of the dimension d of the input space. This suggests that no matter how high-dimensional the input space is, we can always map vectors to a lower-dimensional space of fixed dimension k while approximately preserving the pairwise distances between points. Due to this good property, the JL lemma is often used as a tool to reduce the dimension of high-dimensional data in machine learning and theoretical computer science, which is commonly referred to as *dimension reduction*.

The dimension k does depend on other factors: it depends logarithmically on the number n of points and inverse quadratically on the error tolerance ϵ. Suppose you want to reduce the storage cost of n vectors in $\mathbb{R}^d$ while still being able to perform nearest neighbor search with a small error tolerance ϵ, then you have to choose the dimension k very carefully according to the bound $k \geq \Omega(\frac{\log n}{\epsilon^2})$ in the JL lemma.

In fact, the construction of the map in the JL lemma is surprisingly simple: just use a random projection whose coefficients are drawn from a Gaussian distribution! To see this, we first prove the following lemma on Gaussian random projection.

Lemma 5.50. Let $\boldsymbol{A} \in \mathbb{R}^{k\times d}$ be a Gaussian random matrix, where every entry of $\boldsymbol{A}$ is sampled independently from $\mathcal{N}(0, 1/k)$. For every fixed vector $\boldsymbol{x} \in \mathbb{R}^d$,

$$\boldsymbol{A}\boldsymbol{x} \sim \mathcal{N}\Big(\boldsymbol{0}, \tfrac{\|\boldsymbol{x}\|_2^2}{k}\boldsymbol{I}\Big). \tag{5.238}$$

Moreover, if $\boldsymbol{x} \neq \boldsymbol{0}$, then for every $\epsilon \in (0, 1)$,

$$\Pr\left[\left|\frac{\|\boldsymbol{A}\boldsymbol{x}\|_2^2}{\|\boldsymbol{x}\|_2^2} - 1\right| \geq \epsilon\right] \leq 2e^{-k\epsilon^2/8}. \tag{5.239}$$

Proof. Let $\boldsymbol{a}_1^\top, \ldots, \boldsymbol{a}_k^\top$ be the k rows of $\boldsymbol{A}$. Then $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_k$ are i.i.d. random vectors from $\mathcal{N}(\boldsymbol{0}, \boldsymbol{I}/k)$. Then $\boldsymbol{A}\boldsymbol{x}$ can be written as

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{a}_1^\top \boldsymbol{x} \\ \vdots \\ \boldsymbol{a}_k^\top \boldsymbol{x} \end{bmatrix}. \tag{5.240}$$

This is a linear transformation of $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_k$. By Gaussian linearity in Theorem 5.39, $\boldsymbol{A}\boldsymbol{x}$ is still a Gaussian vector. Its mean vector is

$$\mathbb{E}[\boldsymbol{A}\boldsymbol{x}] = \mathbb{E}[\boldsymbol{A}]\boldsymbol{x} = \boldsymbol{0}. \tag{5.241}$$

For every $i, j \in [k]$,

$$\begin{aligned} \mathrm{Cov}(\boldsymbol{a}_i^\top \boldsymbol{x}, \boldsymbol{a}_j^\top \boldsymbol{x}) = \mathbb{E}[(\boldsymbol{a}_i^\top \boldsymbol{x})(\boldsymbol{a}_j^\top \boldsymbol{x})] &= \mathbb{E}[\boldsymbol{x}^\top \boldsymbol{a}_i \boldsymbol{a}_j^\top \boldsymbol{x}] && (5.242)\\ &= \boldsymbol{x}^\top \mathbb{E}[\boldsymbol{a}_i \boldsymbol{a}_j^\top]\boldsymbol{x} && (5.243)\\ &= \begin{cases} \boldsymbol{x}^\top (\boldsymbol{I}/k)\boldsymbol{x} = \frac{\|\boldsymbol{x}\|_2^2}{k} & \text{if } i = j, \\ \boldsymbol{x}^\top \boldsymbol{0}\boldsymbol{x} = 0 & \text{if } i \neq j. \end{cases} && (5.244) \end{aligned}$$

Therefore, the distribution of $\boldsymbol{A}\boldsymbol{x}$ is

$$\boldsymbol{A}\boldsymbol{x} \sim \mathcal{N}\Big(\boldsymbol{0}, \tfrac{\|\boldsymbol{x}\|_2^2}{k}\boldsymbol{I}\Big). \tag{5.245}$$

Write $\boldsymbol{A}\boldsymbol{x} = \|\boldsymbol{x}\|_2 \boldsymbol{z}$ for some Gaussian vector $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}/k)$. Applying Theorem 5.43 to $\boldsymbol{z} \in \mathbb{R}^k$ gives

$$\Pr\Big[\Big|\|\boldsymbol{z}\|_2^2 - 1\Big| \geq \epsilon\Big] \leq 2e^{-k\epsilon^2/8}. \tag{5.246}$$

Replacing $\|\boldsymbol{z}\|_2^2$ with $\|\boldsymbol{A}\boldsymbol{x}\|_2^2/\|\boldsymbol{x}\|_2^2$ gives the desired bound. □

Proof of Theorem 5.49. Let $\boldsymbol{A} \in \mathbb{R}^{k\times d}$ be a Gaussian random matrix, where every entry of $\boldsymbol{A}$ is sampled independently from $\mathcal{N}(0, 1/k)$. Define the map $F : \mathbb{R}^d \to \mathbb{R}^k$ by

$$F(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}. \tag{5.247}$$

For every $i, j \in [n]$, by Lemma 5.50, with probability at least $1 - 2e^{-k\epsilon^2/8}$,

$$\left|\frac{\|\boldsymbol{A}(\boldsymbol{x}_i - \boldsymbol{x}_j)\|_2^2}{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2} - 1\right| \leq \epsilon \tag{5.248}$$

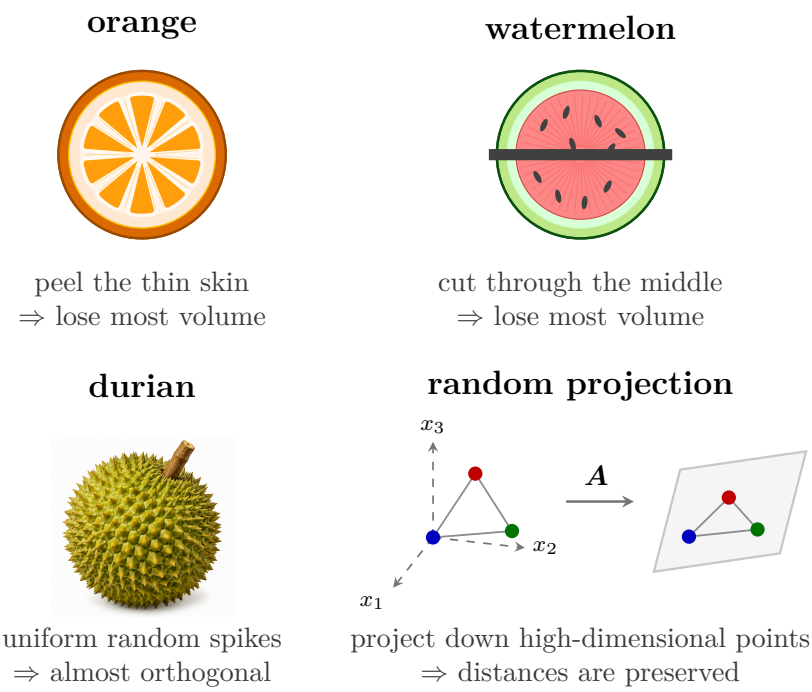


Figure 5.9: Four high-dimensional surprises we met along the way.

Rearranging the inequality gives

$$(1-\epsilon)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 \leq \|\boldsymbol{A}(\boldsymbol{x}_i - \boldsymbol{x}_j)\|_2^2 \leq (1+\epsilon)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2. \tag{5.249}$$

Let E_{ij} be the event that the above inequality does not hold.

Taking the union bound over all pairs $i, j \in [n]$, we get

$$\Pr\left[\bigcup_{i,j\in[n]} E_{ij}\right] \leq \binom{n}{2} \cdot 2e^{-k\epsilon^2/8} < n^2 e^{-k\epsilon^2/8}. \tag{5.250}$$

Setting k to be an integer such that $k \geq \frac{16\log n}{\epsilon^2}$, we have $e^{-k\epsilon^2/8} \leq e^{-2\log n} = \frac{1}{n^2}$, and thus

$$\Pr\left[\bigcup_{i,j\in[n]} E_{ij}\right] < n^2 \cdot \frac{1}{n^2} = 1. \tag{5.251}$$

Therefore, there must exist $\boldsymbol{A}$ such that Equation (5.249) holds for all i, j. □

In summary, from the observations in Section 5.3.2 and Section 5.3.3, we can conclude the following high-dimensional surprises:

- High-dimensional oranges leave you hungry still,
- High-dimensional watermelons can't be sliced well.
- High-dimensional durians' spikes fail to give a thrill.
- Only random projections keep distances chill.

5.4 Probability Distances

In the last section, we argued that the Gaussian distribution $\mathcal{N}(\mathbf{0}, \boldsymbol{I}/d)$, the uniform sphere distribution $\mathcal{U}_{\mathbb{S}}$, and the uniform ball distribution $\mathcal{U}_B$ are similar to each other in high dimensions. But how similar are they really? Besides showing that their norms all concentrate near 1, it would be better to have a unified way to measure the similarity between any pair of probability distributions.

Throughout this section, let P and Q be two probability distributions on $\mathbb{R}^d$. As usual, we write $P(A)$ for the probability of $\boldsymbol{x} \in A$ when $\boldsymbol{x} \sim P$. We also introduce some new notations for convenience. We may also write $P(A)$ as $\Pr_{\boldsymbol{x}\sim P}[A]$ to emphasize that $\boldsymbol{x}$ is the random variable when writing the probability. For expectation, we may write $P(\phi)$ or $\mathbb{E}_{\boldsymbol{x}\sim P}[\phi(\boldsymbol{x})]$ for the expectation of $\phi(\boldsymbol{x})$ when $\boldsymbol{x} \sim P$. We write these notations similarly for the distribution Q.

Remark 5.51. It is easy to see that $P(\phi)$ is linear in ϕ: for any $\alpha, \beta \in \mathbb{R}$ and any functions ϕ_1, ϕ_2,

$$P(\alpha\phi_1 + \beta\phi_2) = \alpha P(\phi_1) + \beta P(\phi_2). \tag{5.252}$$

But note that ϕ itself is a function. So from this point of view, $P(\phi)$ is not an ordinary function of a number or a vector, but a function of a function. In general, a function that maps a function to a number is called a *functional*. A functional that is linear in its argument is called a *linear functional*. Therefore, $P(\phi)$ is a linear functional of ϕ. For anyone who is interested in learning more about linear functionals, consider taking a course on functional analysis.

5.4.1 Total Variation Distance

A classic example of abductive reasoning is the *duck test*:

If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck.

A related idea appears in the *Turing test* for artificial intelligence, which evaluates intelligence based on externally observable behavior:

If a human cannot distinguish a machine from a human through conversation, then the machine is intelligent.

The same idea also appears in how we define the probability distribution: two random variables have the same distribution if their probabilities on all sets are the same, i.e., $\Pr[X \in A] = \Pr[Y \in A]$ for all sets A. We can apply the same philosophy to define a distance between two probability distributions.

Definition 5.52 (Total variation distance)**.** The *total variation distance* between P and Q is

$$d_{\mathrm{TV}}(P, Q) := \sup_{A \subseteq \mathbb{R}^d} \Big| P(A) - Q(A) \Big|. \tag{5.253}$$

In other words, the total variation distance asks the following question: if we are allowed to choose any set of possible values A, how different can the probability of $\boldsymbol{x} \in A$ be under $\boldsymbol{x} \sim P$ and under $\boldsymbol{x} \sim Q$? If for all choices of A, the probabilities of $\boldsymbol{x} \in A$ are almost the same under P and Q, then the two distributions are close in total variation.

For discrete distributions, the total variation distance admits a more explicit formula.

Theorem 5.53. If P and Q are discrete probability distributions with PMFs $p : \mathcal{X} \to [0,1]$ and $q : \mathcal{X} \to [0,1]$, where $\mathcal{X}$ is a finite set of values, then

$$d_{\mathrm{TV}}(P,Q) = \frac{1}{2} \sum_{x \in \mathcal{X}} |p(x) - q(x)|. \tag{5.254}$$

Proof. Let $\delta(x) := p(x) - q(x)$. For every set $A \subseteq \mathcal{X}$,

$$P(A) - Q(A) = \sum_{x \in \mathcal{X}} \mathbb{1}_{[x \in A]} \delta(x) \leq \underbrace{\sum_{x \in \mathcal{X}} [\delta(x)]_+}_{=: I_+}. \tag{5.255}$$

Similarly,

$$P(A) - Q(A) = \sum_{x \in \mathcal{X}} \mathbb{1}_{[x \in A]} \delta(x) \geq -\underbrace{\sum_{x \in \mathcal{X}} [\delta(x)]_-}_{=: I_-}. \tag{5.256}$$

To make the above two inequalities become equalities, we only need to take $A = \{x \in \mathcal{X} : \delta(x) \geq 0\}$ and $A = \{x \in \mathcal{X} : \delta(x) \leq 0\}$, respectively. So it remains to understand the values of I_+ and I_-.

In fact, I_+ and I_- are equal because

$$I_+ - I_- = \sum_{x \in \mathcal{X}} [\delta(x)]_+ - \sum_{x \in \mathcal{X}} [\delta(x)]_- = \sum_{x \in \mathcal{X}} \delta(x) = \sum_{x \in \mathcal{X}} p(x) - \sum_{x \in \mathcal{X}} q(x) \tag{5.257}$$

$$= 1 - 1 = 0. \tag{5.258}$$

Therefore, $d_{\mathrm{TV}}(P,Q) = I_+ = I_-$. Taking the average of I_+ and I_- gives

$$d_{\mathrm{TV}}(P,Q) = \frac{I_+ + I_-}{2} = \sum_{x \in \mathcal{X}} \frac{1}{2} ([\delta(x)]_+ + [\delta(x)]_-) = \sum_{x \in \mathcal{X}} \frac{1}{2} |\delta(x)|, \tag{5.259}$$

which is exactly the formula in the theorem. □

Example 5.54. The total variation distance between Bernoulli(p) and Bernoulli(q) is

$$d_{\mathrm{TV}}\Big(\mathrm{Bernoulli}(p), \mathrm{Bernoulli}(q)\Big) = |p - q|. \tag{5.260}$$

For continuous distributions, the same formula becomes an integral, which we state without proof.

Theorem 5.55. If P and Q are continuous probability distributions with PDFs f and g on $\mathbb{R}^d$, then

$$d_{\mathrm{TV}}(P,Q) = \frac{1}{2}\int_{\mathbb{R}^d} |f(\boldsymbol{x}) - g(\boldsymbol{x})| \, \mathrm{d}\boldsymbol{x}. \tag{5.261}$$

Example 5.56. The total variation distance between the uniform distribution on $[0,1]$ and the uniform distribution on $[0,2]$ is

$$d_{\mathrm{TV}}\left(\mathcal{U}_{[0,1]}, \mathcal{U}_{[0,2]}\right) = \frac{1}{2}\int_0^1 |1 - \tfrac{1}{2}| \, \mathrm{d}x + \frac{1}{2}\int_1^2 |0 - \tfrac{1}{2}| \, \mathrm{d}x \tag{5.262}$$

$$= \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{2}. \tag{5.263}$$

There is another way to interpret the total variation distance. A set $A \subseteq \mathbb{R}^d$ can be represented by its indicator function (Definition 2.79):

$$\mathbb{1}_A(\boldsymbol{x}) = \begin{cases} 1 & \text{if } \boldsymbol{x} \in A, \\ 0 & \text{if } \boldsymbol{x} \notin A. \end{cases} \tag{5.264}$$

Then

$$P(A) - Q(A) = P(\mathbb{1}_A) - Q(\mathbb{1}_A). \tag{5.265}$$

Therefore, we can write the total variation distance as

$$d_{\mathrm{TV}}(P,Q) = \sup_{A \subseteq \mathbb{R}^d} |P(\mathbb{1}_A) - Q(\mathbb{1}_A)|. \tag{5.266}$$

The above formula admits an important interpretation. We may test a distribution by taking the expectation of an indicator function $\mathbb{1}_A$, and distributions P and Q may give different answers. The total variation distance is then defined as the maximum possible difference between these two expectations.

More generally, we can compare P and Q by looking at $P(\phi) - Q(\phi)$ for any test function $\phi : \mathbb{R}^d \to \mathbb{R}$. If we restrict our attention to bounded functions, then the maximum difference is the total variation distance after rescaling.

Theorem 5.57 (Variational form of total variation)**.** Let $\ell < u$. For any two probability distributions P and Q on $\mathbb{R}^d$,

$$d_{\mathrm{TV}}(P,Q) = \frac{1}{u-\ell} \sup_{\phi \in \mathcal{F}_{\ell,u}} \Big| P(\phi) - Q(\phi) \Big|, \tag{5.267}$$

where $\mathcal{F}_{\ell,u}$ denotes the family of all functions $\phi : \mathbb{R}^d \to [\ell, u]$.

To understand why the above theorem is true, we first generalize the tail sum formula from the finite discrete case (Theorem 2.88) to the general case.

Theorem 5.58 (Tail-sum formula)**.** If X is a random variable that takes non-negative real values, then

$$\mathbb{E}[X] = \int_0^\infty \Pr[X \geq t] \, \mathrm{d}t. \tag{5.268}$$

Proof. We write $X = \int_0^{+\infty} \mathbb{1}_{[X \geq t]} \, \mathrm{d}t$. By linearity of expectation,

$$\mathbb{E}[X] = \int_0^{+\infty} \mathbb{E}[\mathbb{1}_{[X \geq t]}] \, \mathrm{d}t = \int_0^{+\infty} \Pr[X \geq t] \, \mathrm{d}t. \tag{5.269}$$

While we have only justified the linearity of expectation in the finite discrete case, it is in fact true for general random variables. We omit the technical details here. □

Now we are ready to prove Theorem 5.57.

Proof of Theorem 5.57. It suffices to prove the case of $[\ell, u] = [0, 1]$, since every $\phi : \mathbb{R}^d \to [\ell, u]$ can be rescaled to $(\phi - \ell)/(u - \ell)$. For every event A, the indicator function $\mathbb{1}_A$ lies in $\mathcal{F}_{0,1}$. Hence

$$\sup_{\phi \in \mathcal{F}_{0,1}} \Big| P(\phi) - Q(\phi) \Big| \geq \sup_{A \subseteq \mathbb{R}^d} \Big| P(\mathbb{1}_A) - Q(\mathbb{1}_A) \Big| = d_{\mathrm{TV}}(P, Q). \tag{5.270}$$

Conversely, fix any $\phi : \mathbb{R}^d \to [0, 1]$. Let $A_t := \{\boldsymbol{x} : \phi(\boldsymbol{x}) \geq t\}$ for $t \in [0, 1]$. By the tail-sum formula (Theorem 5.58),

$$P(\phi) = \int_0^1 P(A_t) \, \mathrm{d}t, \qquad Q(\phi) = \int_0^1 Q(A_t) \, \mathrm{d}t, \tag{5.271}$$

where the upper bound 1 is used because $A_t = \varnothing$ for all $t > 1$. Subtracting the two equations, we get

$$P(\phi) - Q(\phi) = \int_0^1 P(A_t) \, \mathrm{d}t - \int_0^1 Q(A_t) \, \mathrm{d}t = \int_0^1 (P(A_t) - Q(A_t)) \, \mathrm{d}t. \tag{5.272}$$

Taking the absolute value,

$$|P(\phi) - Q(\phi)| \leq \int_0^1 |P(A_t) - Q(A_t)| \, \mathrm{d}t. \tag{5.273}$$

Since $|P(A_t) - Q(A_t)| \leq d_{\mathrm{TV}}(P, Q)$ for all $t \in [0, 1]$, we have

$$|P(\phi) - Q(\phi)| \leq \int_0^1 d_{\mathrm{TV}}(P, Q) \, \mathrm{d}t = d_{\mathrm{TV}}(P, Q). \tag{5.274}$$

Combining this with Equation (5.270) proves $|P(\phi) - Q(\phi)| = d_{\mathrm{TV}}(P, Q)$ for the case of $[\ell, u] = [0, 1]$, and the general case follows by rescaling. □

5.4.2 Wasserstein Distance

An important limitation of total variation distance is that it completely ignores the geometry of the underlying Euclidean space.

To illustrate this, let $\delta_{\boldsymbol{x}}(A) = \mathbb{1}_{[\boldsymbol{x}\in A]}$ denote the point mass at $\boldsymbol{x} \in \mathbb{R}^d$, If $\boldsymbol{a}, \boldsymbol{b}$ are two points in $\mathbb{R}^d$, then as long as $\boldsymbol{a} \neq \boldsymbol{b}$,

$$d_{\mathrm{TV}}(\delta_{\boldsymbol{a}}, \delta_{\boldsymbol{b}}) = 1. \tag{5.275}$$

To see this, take $A = \{\boldsymbol{a}\}$, then $\delta_{\boldsymbol{a}}(A) = 1$ while $\delta_{\boldsymbol{b}}(A) = 0$.

This conclusion does not depend on how close $\boldsymbol{a}$ and $\boldsymbol{b}$ are. Even if $\|\boldsymbol{a}-\boldsymbol{b}\|_2 = 10^{-100}$, total variation distance still says that the two point masses are maximally far apart. This is too harsh in many situations where we want to say that $\delta_{\boldsymbol{a}} \approx \delta_{\boldsymbol{b}}$ when $\boldsymbol{a}$ and $\boldsymbol{b}$ are close.

We can understand this problem from another angle using the variational form of total variation. In this form, instead of choosing a set A directly, we choose a test function ϕ and compare $P(\phi)$ with $Q(\phi)$. But the test function ϕ here is only required to be bounded, which means it can still be extremely sensitive to small changes in the input: even when $\boldsymbol{a}$ and $\boldsymbol{b}$ are very close, ϕ may still assign completely different values to them. This can make $|P(\phi) - Q(\phi)|$ large even when the probability mass has only moved a tiny distance.

A natural fix here is to restrict the test functions ϕ to be "sufficiently continuous". If two inputs are close, then we require the function values to be close as well. This leads to the following definitions of *Lipschitz functions* and *Wasserstein distance*.

Definition 5.59 (Lipschitz function)**.** For $L \geq 0$, a function $\phi : \mathbb{R}^d \to \mathbb{R}$ is L-*Lipschitz* if for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d$,

$$|\phi(\boldsymbol{x}) - \phi(\boldsymbol{y})| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|_2. \tag{5.276}$$

In particular, ϕ is 1-*Lipschitz* if it is L-Lipschitz with $L = 1$.

Definition 5.60 (Wasserstein-1 distance)**.** The *Wasserstein-1 distance* between P and Q is

$$W_1(P, Q) := \sup_{\phi\in \mathrm{Lip}_1} \Big|P(\phi) - Q(\phi)\Big|, \tag{5.277}$$

where Lip_1 denotes the family of all 1-Lipschitz functions $\phi : \mathbb{R}^d \to \mathbb{R}$.

This definition has the same variational form as total variation, but now the test functions are restricted to be 1-Lipschitz, which means they cannot change too quickly. If $\boldsymbol{x}$ and $\boldsymbol{y}$ are close, then a 1-Lipschitz function is forced to satisfy

$$|\phi(\boldsymbol{x}) - \phi(\boldsymbol{y})| \leq \|\boldsymbol{x} - \boldsymbol{y}\|_2. \tag{5.278}$$

Thus a tiny movement of probability mass can only cause a tiny change in each expectation $P(\phi)$.

This fixes the problem that the total variation distance between two point masses is always 1.

Example 5.61 (Point masses)**.** If $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^d$, then

$$W_1(\delta_{\boldsymbol{a}}, \delta_{\boldsymbol{b}}) = \|\boldsymbol{a} - \boldsymbol{b}\|_2. \tag{5.279}$$

To see this, note that for every 1-Lipschitz ϕ,

$$|\delta_{\boldsymbol{a}}(\phi) - \delta_{\boldsymbol{b}}(\phi)| = |\phi(\boldsymbol{a}) - \phi(\boldsymbol{b})| \le \|\boldsymbol{a} - \boldsymbol{b}\|_2. \tag{5.280}$$

If $\boldsymbol{a} = \boldsymbol{b}$, then the inequality holds with equality for any test function. If $\boldsymbol{a} \neq \boldsymbol{b}$, take

$$\phi(\boldsymbol{x}) := \left\langle \frac{\boldsymbol{a} - \boldsymbol{b}}{\|\boldsymbol{a} - \boldsymbol{b}\|_2}, \boldsymbol{x} \right\rangle. \tag{5.281}$$

This function is 1-Lipschitz and satisfies $\phi(\boldsymbol{a}) - \phi(\boldsymbol{b}) = \|\boldsymbol{a} - \boldsymbol{b}\|_2$.

When measuring the distance between the Gaussian and uniform sphere distributions, Wasserstein distance also exhibits a more reasonable behavior than total variation distance.

Example 5.62 (Gaussian and uniform sphere distributions)**.** Let $\mathcal{G} = \mathcal{N}(\mathbf{0}, \boldsymbol{I}/d)$ and let $\mathcal{U}_{\mathbb{S}}$ be the uniform distribution on $\mathbb{S}^{d-1}$. Although we have argued that these two distributions are close in high dimensions, their total variation distance is always $d_{\mathrm{TV}}(\mathcal{G}, \mathcal{U}_{\mathbb{S}}) = 1$. This is because we can take A to be the unit sphere $\mathbb{S}^{d-1}$. Then $\mathcal{G}(A) = 1$ and $\mathcal{U}_{\mathbb{S}}(A) = 0$.

If we use Wasserstein distance instead, we can see that the two distributions become close in high dimensions. Let $\boldsymbol{x} \sim \mathcal{G}$, and write $r := \|\boldsymbol{x}\|_2$ and $\boldsymbol{\theta} := \boldsymbol{x}/\|\boldsymbol{x}\|_2$. Since Gaussian is isotropic, the direction $\boldsymbol{\theta}$ is distributed uniformly on $\mathbb{S}^{d-1}$. Then for any 1-Lipschitz function ϕ,

$$\begin{aligned} |\mathcal{G}(\phi) - \mathcal{U}_{\mathbb{S}}(\phi)| &= |\mathbb{E}[\phi(r\boldsymbol{\theta})] - \mathbb{E}[\phi(\boldsymbol{\theta})]| && (5.282)\\ &\le \mathbb{E}[|\phi(r\boldsymbol{\theta}) - \phi(\boldsymbol{\theta})|] && (5.283)\\ &\le \mathbb{E}[\|r\boldsymbol{\theta} - \boldsymbol{\theta}\|_2] = \mathbb{E}[|r - 1|]\,. && (5.284) \end{aligned}$$

Recall that r^2 can be written as the sample average of $Z_1^2, \ldots, Z_d^2$, where $Z_1, \ldots, Z_d$ are i.i.d. standard Gaussians. Since $\mathbb{E}[Z_i^2] = 1$ and $\mathrm{Var}(Z_i^2) = 2$, we have $\mathbb{E}[r^2] = 1$ and $\mathrm{Var}(r^2) = 2/d$. Then

$$\begin{aligned} \mathbb{E}\Big[\big|r - 1\big|\Big] = \mathbb{E}\left[\tfrac{|r^2-1|}{r+1}\right] &\le \mathbb{E}\Big[\big|r^2 - 1\big|\Big] && (5.285)\\ &\le \sqrt{\mathbb{E}\Big[(r^2-1)^2\Big]} = \sqrt{\mathrm{Var}(r^2)} = \mathcal{O}(\tfrac{1}{\sqrt{d}}). && (5.286) \end{aligned}$$

Taking the supremum over all $\phi \in \mathrm{Lip}_1$ gives

$$W_1(\mathcal{G}, \mathcal{U}_{\mathbb{S}}) \le \mathcal{O}(\tfrac{1}{\sqrt{d}}) \to 0 \quad \text{as } d \to \infty. \tag{5.287}$$

Remark 5.63 (Optimal transport)**.** In optimal transport theory, there is another way to understand Wasserstein distance. A *coupling* of P and Q is the distribution of a pair of random vectors $(\boldsymbol{x}, \boldsymbol{y})$ such that $\boldsymbol{x} \sim P$ and $\boldsymbol{y} \sim Q$. We use $\Gamma(P, Q)$ to denote the set of all couplings of P and Q. We may think of $\boldsymbol{x}$

as the original location of a unit of mass, and $\boldsymbol{y}$ as the location where that unit of mass is sent. The 1-Wasserstein distance can be written as

$$W_1(P,Q) := \inf_{\gamma \in \Gamma(P,Q)} \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim\gamma}[\|\boldsymbol{x}-\boldsymbol{y}\|_2]\,. \tag{5.288}$$

That is, $W_1(P,Q)$ is the smallest possible average distance needed to move mass distributed as P into mass distributed as Q. This is also why the Wasserstein-1 distance is sometimes called the *Earth Mover's Distance* (EMD): it is similar to the minimum amount of work needed to move a pile of dirt with P to a pile of dirt with shape Q. Proving that Equation (5.277) and Equation (5.288) are equivalent is non-trivial and requires some advanced techniques such as the duality theory of infinite-dimensional linear programming, which we will not cover here. More generally, for $p \geq 1$, the p-Wasserstein distance is defined by

$$W_p(P,Q) := \left(\inf_{\gamma \in \Gamma(P,Q)} \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim\gamma}[\|\boldsymbol{x}-\boldsymbol{y}\|_2^p]\right)^{1/p}. \tag{5.289}$$

5.4.3 Example: GAN and Wasserstein GAN

▶ Background

Let us end this chapter with an example from image generation. Modern AI systems can generate images or even videos that look real. But if we go back to around 2014, near the beginning of this wave of deep learning, it was not clear at all how a computer program could generate images that look real.

Suppose we have a large dataset of real images. If each image contains w columns and h rows of pixels, with 3 channels for each pixel (red, green, and blue), then each image can be represented as a point in $\mathbb{R}^d$, where $d = 3wh$. We may assume that these real images are samples from some unknown distribution, called the *data distribution* $\mathcal{D}_{\text{data}}$.

A *generative model* is a model that can generate images. Typically, a generative model inputs a random vector $\boldsymbol{z}$ sampled from a distribution $P_{\mathbf{z}}$ that is easy to sample from, and then performs a deterministic transformation G to generate an image $\boldsymbol{x} = G(\boldsymbol{z})$. The distribution of the generated images is denoted by P_G.

A predominant approach for generative models is to use a neural network to learn the transformation G. We will not explain what a neural network is here, but you can think of it as a parameterized function with vector inputs and vector outputs. The number of parameters in the neural network can range from a few thousand to more than a trillion (10^{12}).

Of course, at the very beginning, the neural network has no idea what a realistic image should look like. We need some way to tell it which generated images are good and which are bad. After repeating this process many times, the neural network might gradually learn to generate images that look real.

But how can we tell if the generated images are good or bad? This is exactly the kind of question we have been studying: *how can we compare two probability*

distributions? One natural answer is to define a distance between the two distributions $\mathcal{D}_{\text{data}}$ and P_G, and then train the neural network G to minimize this distance.

▶ Generative Adversarial Networks

A *Generative Adversarial Network* (GAN) [GPAM+14] was one of the first successful approaches to the image generation problem at the beginning of this wave of deep learning. GAN sets up a two-layer zero-sum game between two neural networks, the *generator* G and the *discriminator* D. In this game, the generator tries to generate images that are similar to the real images, while the discriminator is a classifier $D : \mathbb{R}^d \to (0, 1)$ that tries to distinguish real images from generated images. More specifically, GAN defines the following objective function:

$$\mathcal{L}(G, D) := \mathbb{E}_{\boldsymbol{x} \sim \mathcal{D}_{\text{data}}}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{x} \sim P_G}[\log(1 - D(\boldsymbol{x}))]. \tag{5.290}$$

The discriminator D tries to make $\mathcal{L}(G, D)$ large: it wants $D(\boldsymbol{x})$ to be close to 1 for real images and close to 0 for generated images. The generator G tries to make $\mathcal{L}(G, D)$ small: it wants the generated images to look so real that the discriminator cannot tell them apart from real images. Thus GAN training is formulated as the following minimax optimization problem:

$$\min_G \max_D \mathcal{L}(G, D). \tag{5.291}$$

Why does this minimax optimization problem make sense? A key intuition provided in the original GAN paper [GPAM+14] is that the inner maximization problem has a strong connection to the *Jensen–Shannon divergence* between the data distribution and the generator distribution, which we now explain in detail.

▶ Jensen–Shannon divergence

First, we need to define the Jensen–Shannon divergence.

Definition 5.64 (Jensen–Shannon divergence)**.** Let P and Q be probability distributions on $\mathbb{R}^d$ with PDFs f and g. The *Jensen–Shannon divergence* (JSD) between P and Q is

$$\text{JSD}(P \| Q) := \frac{1}{2} \int_{\mathbb{R}^d} \left(f(\boldsymbol{x}) \log \frac{f(\boldsymbol{x})}{m(\boldsymbol{x})} + g(\boldsymbol{x}) \log \frac{g(\boldsymbol{x})}{m(\boldsymbol{x})} \right) \mathrm{d}\boldsymbol{x}, \tag{5.292}$$

$$\text{where } m(\boldsymbol{x}) := \frac{f(\boldsymbol{x}) + g(\boldsymbol{x})}{2}. \tag{5.293}$$

Here we use the convention that $0 \log 0 = 0$.

The JSD for discrete distributions can be defined similarly by replacing the PDFs with PMFs and the integral with a sum.

Note that the JSD is *not* a distance between probability distributions in the mathematical sense. Formally, a distance function is defined as follows.

Definition 5.65 (Metric space and distance function)**.** A *metric space* is a pair (M, d) where M is a set of elements and $d : M \times M \to \mathbb{R}$ is a *metric*, or a *distance function*, that satisfies the following properties:

- $d(P, P) = 0$ for all $P \in M$.
- Positivity: $d(P, Q) > 0$ for all $P, Q \in M$ with $P \neq Q$.
- Symmetry: $d(P, Q) = d(Q, P)$ for all $P, Q \in M$.
- Triangle inequality: $d(P, Q) \leq d(P, R) + d(R, Q)$ for all $P, Q, R \in M$.

It can be shown that the probability distributions equipped with the total variation distance form a metric space. For the JSD, it is easy to see that $\mathrm{JSD}(P\|P) = 0$ and $\mathrm{JSD}(P\|Q) = \mathrm{JSD}(Q\|P)$ for all P, Q. The positivity property is not obvious, but it simply follows from Jensen's inequality. However, the JSD can violate the triangle inequality. For example, let P, Q, R be three Bernoulli distributions, where $P = \text{Bernoulli}(0)$, $Q = \text{Bernoulli}(1)$, and $R = \text{Bernoulli}(1/2)$. Then

$$\mathrm{JSD}(P\|Q) = \log 2 \approx 0.693, \tag{5.294}$$

$$\mathrm{JSD}(P\|R) = \frac{3}{4}\log\frac{4}{3} \approx 0.216, \tag{5.295}$$

$$\mathrm{JSD}(R\|Q) = \frac{3}{4}\log\frac{4}{3} \approx 0.216, \tag{5.296}$$

so $\mathrm{JSD}(P\|Q) + \mathrm{JSD}(R\|Q) < \mathrm{JSD}(P\|R)$.

In some sense, the JSD is more like a squared distance "$\|x-y\|_2^2$" rather than a distance. In fact, one can show that the probability distributions equipped with the squared root of the JSD, $d_{\mathrm{JS}}(P, Q) = \sqrt{\mathrm{JSD}(P\|Q)}$, form a metric space. We call this distance function $d_{\mathrm{JS}}(P, Q)$ the *Jensen–Shannon distance*.

The next theorem shows that the inner maximization problem in GANs corresponds to the Jensen–Shannon divergence.

Theorem 5.66 (Variational form of Jensen–Shannon divergence)**.** Let $\mathcal{D}_{\text{data}}$ and P_G be continuous probability distributions on $\mathbb{R}^d$. Then

$$\sup_{D:\mathbb{R}^d\to(0,1)} \mathcal{L}(\mathcal{D}_{\text{data}}, P_G) = -\log 4 + 2 \cdot \mathrm{JSD}(\mathcal{D}_{\text{data}}\|P_G). \tag{5.297}$$

Proof. Let f and g be the PDFs of $\mathcal{D}_{\text{data}}$ and P_G, respectively. Then $\mathcal{L}(P, Q)$ can be written as

$$\mathcal{L}(\mathcal{D}_{\text{data}}, P_G) = \int_{\mathbb{R}^d} \left(f(\boldsymbol{x}) \log D(\boldsymbol{x}) + g(\boldsymbol{x}) \log(1 - D(\boldsymbol{x}))\right) \mathrm{d}\boldsymbol{x}. \tag{5.298}$$

To maximize this function, we only need to maximize the integrand separately at each point $\boldsymbol{x}$. Fix a point $\boldsymbol{x}$ and abbreviate $a := f(\boldsymbol{x})$ and $b := g(\boldsymbol{x})$. For $t \in (0, 1)$, consider

$$h(t) := a \log t + b \log(1 - t). \tag{5.299}$$

Taking the derivative of $h(t)$ with respect to t, we get

$$h'(t) = \frac{a}{t} - \frac{b}{1-t}. \tag{5.300}$$

If $a > 0$ and $b > 0$, we can set $h'(t) = 0$ to conclude that $t = \frac{a}{a+b}$ attains the maximum. Then,

$$\sup_{t\in(0,1)} h(t) = a \log \frac{a}{a+b} + b \log \frac{b}{a+b}. \tag{5.301}$$

We can check that this equation also holds when $a = 0$ or $b = 0$, by noting that the maximum is attained at $t = 0$ or $t = 1$, respectively. Therefore, this equation holds for all $a, b \geq 0$.

Since the discriminator can choose $D(\boldsymbol{x})$ separately at each point, we maximize point by point and integrate:

$$\sup_{D:\mathbb{R}^d\to(0,1)} \mathcal{L}(\mathcal{D}_{\text{data}}, P_G) \tag{5.302}$$

$$= \int_{\mathbb{R}^d} \left(f(\boldsymbol{x}) \log \frac{f(\boldsymbol{x})}{f(\boldsymbol{x}) + g(\boldsymbol{x})} + g(\boldsymbol{x}) \log \frac{g(\boldsymbol{x})}{f(\boldsymbol{x}) + g(\boldsymbol{x})} \right) \mathrm{d}\boldsymbol{x} \tag{5.303}$$

$$= \int_{\mathbb{R}^d} \left(f(\boldsymbol{x}) \log \frac{f(\boldsymbol{x})}{2m(\boldsymbol{x})} + g(\boldsymbol{x}) \log \frac{g(\boldsymbol{x})}{2m(\boldsymbol{x})} \right) \mathrm{d}\boldsymbol{x} \tag{5.304}$$

$$= \log \frac{1}{2} \int_{\mathbb{R}^d} (f(\boldsymbol{x}) + g(\boldsymbol{x})) \, \mathrm{d}\boldsymbol{x} + \int_{\mathbb{R}^d} \left(f(\boldsymbol{x}) \log \frac{f(\boldsymbol{x})}{m(\boldsymbol{x})} + g(\boldsymbol{x}) \log \frac{g(\boldsymbol{x})}{m(\boldsymbol{x})} \right) \mathrm{d}\boldsymbol{x} \tag{5.305}$$

$$= -\log 4 + 2 \cdot \mathrm{JSD}(\mathcal{D}_{\text{data}} \| P_G), \tag{5.306}$$

where $m(\boldsymbol{x})$ is defined as $m(\boldsymbol{x}) = \frac{f(\boldsymbol{x})+g(\boldsymbol{x})}{2}$ and the last equality follows from the definition of the JSD. □

The above theorem implies that if the inner optimization problem for the discriminator is optimized to its best, then the minimax optimization in GAN training can be simplified to

$$\min_G \max_D \mathcal{L}(G, D) = \min_G \Big\{ -\log 4 + 2 \cdot \mathrm{JSD}(\mathcal{D}_{\text{data}} \| P_G) \Big\} \tag{5.307}$$

Or simply,

$$\min_G \mathrm{JSD}(\mathcal{D}_{\text{data}} \| P_G). \tag{5.308}$$

Therefore, if the generator and discriminator play their best in this zero-sum game, then the generator should try to make the JSD between the data distribution and the generator distribution as small as possible. As the square root of the JSD is a distance function, this helps the generator to generate images that follow a similar distribution to the real images.

▶ Wasserstein GANs

The connection to the Jensen–Shannon divergence also reveals a weakness of the original GAN objective. The problem is similar to the one we saw for total variation distance: it does not pay attention to the geometry of the underlying Euclidean space. For example, two different point masses are perfectly distinguishable by the discriminator, no matter how close the two points are. But in reality, if we perturb a real image by a small amount, it should still look like a real image.

From the variational point of view, the missing ingredient is some form of restriction, or *regularization*, that makes the discriminator less sensitive to small changes in the input. In general, the word *regularization* refers to any technique that restricts the power or complexity of a model. Here, we specifically need a form of regularization that prevents the discriminator from distinguishing between nearby images.

Based on this intuition, Wasserstein GAN [ACB17] proposes the following objective function:

$$\mathcal{L}(G, D) := \mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}_{\text{data}}}[D(\boldsymbol{x})] - \mathbb{E}_{\boldsymbol{y}\sim P_G}[D(\boldsymbol{y})]. \tag{5.309}$$

The minimax optimization problem is then

$$\min_G \max_D \mathcal{L}(G, D). \tag{5.310}$$

More importantly, WGAN requires the discriminator to be Lipschitz, and the authors of WGAN proposed several practical methods for enforcing this Lipschitzness. If we consider the ideal case where the discriminator $D : \mathbb{R}^d \to \mathbb{R}$ is controlled to be 1-Lipschitz $D \in \text{Lip}_1$, then the minimax optimization problem can be simplified to

$$\min_G \max_D \mathcal{L}(G, D) = \min_G W_1(\mathcal{D}_{\text{data}}, P_G). \tag{5.311}$$

Therefore, if the generator and discriminator play their best in this zero-sum game, then the generator should try to make the Wasserstein-1 distance between the data distribution and the generator distribution as small as possible. This is arguably a more reasonable objective than the JSD objective.

WGAN is a very influential work in the study of GANs. Although WGAN itself was quickly superseded by stronger GAN variants, it helped establish the idea that the discriminator should be carefully regularized. This idea has inspired many follow-up works, most notably *WGAN-GP* [GAA+17] and *spectral normalization* [MKKY18]. These techniques do not exactly make the discriminator Lipschitz, but they were strongly influenced by the idea of WGAN and have been adopted in many GAN training pipelines.

Note that this is one of the most common ways in which mathematical theory influences machine learning practice: there is something fully rigorous and mathematically sound, and then machine learning, like a monster stitched together from many parts, randomly searches for rigorous ideas and attaches them to

itself with a bunch of engineering tricks. The resulting method is no longer rigorous in the original mathematical sense, but the good news is that the machine learning models get better.

You may wonder whether WGAN still influences generative models today. Unfortunately, not as much as before! GANs was a very popular topic in the field of generative models between 2014 and 2020. But around 2020, another paradigm, called *diffusion models*, emerged and quickly became the dominant approach to generative modeling. We will return to this point in Chapter 8, but the direct importance of WGAN has declined as the field has moved from GANs to diffusion models. Still, WGAN remains a good example of how understanding probability distances better can help us design better machine learning methods.

Exercises

5.1. Let A be set that we do not know whether it is countable or uncountable. Prove the following statements:

(a) If there exists an injection $f : A \to B$ from A to a countable set B, then A is countable.

(b) If there exists an surjection $f : A \to B$ from A to an uncountable set B, then A is uncountable.

Here, we say that a function $f : A \to B$ is an *injection* if for every two distinct elements $a_1, a_2 \in A$, we have $f(a_1) \neq f(a_2)$. Similarly, we say that a function $f : A \to B$ is a *surjection* if for every element $b \in B$, there exists an element $a \in A$ such that $f(a) = b$.

5.2. For each of the following sets, determine whether it is countable or uncountable:

(a) The set of all finite tuples of integers: $\cup_{k \geq 1} \mathbb{Z}^k$.

(b) The set of all Python code that can be executed. You can view a Python code as a finite sequence of symbols from a finite alphabet. Some sequences are executable, some are not.

(c) The set of all functions $f : \mathbb{N} \to \{0, 1\}$.

(d) The set of functions $f : \mathbb{N} \to \{0, 1\}$ that can be implemented by a Python program. Here a program implements a function f if for every input $n \in \mathbb{N}$, the program outputs the value of $f(n)$.

Note: The functions in the last set are called *computable functions*. For anyone interested in learning more about them, take a course on the theory of computation!

5.3. The *Poisson distribution* with parameter $\lambda > 0$, denoted by Poisson(λ), is a discrete probability distribution with probability mass function

$$P(k) := \frac{\lambda^k}{k!} e^{-\lambda}, \quad \text{for } k \in \mathbb{N}.$$

(a) Prove that P is a valid probability mass function. That is, prove that $\sum_{k \geq 0} P(k) = 1$.

(b) Let $X \sim \text{Poisson}(\lambda)$. Determine $\mathbb{E}[X]$.

(c) Let $X_1 \sim \text{Poisson}(\lambda_1), X_2 \sim \text{Poisson}(\lambda_2)$ be independent Poisson random variables. Prove that $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

5.4. The *exponential distribution* with parameter $\lambda > 0$, denoted by Exp(λ), is a continuous probability distribution with probability density function

$$f(x) := \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

(a) Prove that f is a valid probability density function. That is, prove that $\int_{-\infty}^{\infty} f(x) \, \mathrm{d}x = 1$.

(b) Let $X \sim \text{Exp}(\lambda)$. Determine $\mathbb{E}[X]$.

(c) Determine the MGF and CGF of $\text{Exp}(\lambda)$.

(d) Let $X_1 \sim \text{Exp}(\lambda_1), X_2 \sim \text{Exp}(\lambda_2)$ be independent exponential random variables. Prove that $X_1 + X_2$ does not follow any exponential distribution.

5.5. For any probability distribution $\mathcal{D}$ on $\mathbb{R}$, define the *cumulative distribution function* (CDF) of $\mathcal{D}$ as

$$F_{\mathcal{D}}(x) := \mathcal{D}((-\infty, x]), \quad \text{for } x \in \mathbb{R}.$$

If $X \sim \mathcal{D}$, then $F_{\mathcal{D}}(x) = \Pr[X \leq x]$, which is the probability that X is less than or equal to x. It can be shown that the CDF is a non-decreasing function: $F_{\mathcal{D}}(x) \leq F_{\mathcal{D}}(y)$ for all $x \leq y$. Another good property is that the CDF is always right-continuous: $\lim_{x \to x_0^+} F_{\mathcal{D}}(x) = F_{\mathcal{D}}(x_0)$ for all $x_0 \in \mathbb{R}$.

Define the *quantile function* of $\mathcal{D}$ as

$$F_{\mathcal{D}}^{-1}(p) := \inf\{x \in \mathbb{R} : F_{\mathcal{D}}(x) \geq p\}, \quad \text{for } p \in (0, 1).$$

In other words, the quantile function gives a threshold value x that tries its best to separate the probability mass of X into a lower part of mass p and an upper part of mass $1 - p$.

(a) Determine the CDF and quantile function of the exponential distribution $\text{Exp}(\lambda)$.

(b) Determine the CDF and quantile function of the geometric distribution $\text{Geometric}(p)$, where $p \in (0, 1]$ (Example 3.26).

5.6. Let $\mathcal{D}$ be a probability distribution on $\mathbb{R}$. If it is not possible to generate samples from $\mathcal{D}$ in a computer program, then $\mathcal{D}$ would be meaningless for computer scientists. It is easy to generate samples from the uniform distribution on $[0, 1)$, which we denote by $\mathcal{U}$ for short. For example, in Python, we can write

```
import random
print(random.random())
```

To generate samples from an arbitrary distribution $\mathcal{D}$, we have to apply a deterministic transformation to the uniform samples.

(a) Let $Z \sim \mathcal{U}$. Determine the probability density function of $\alpha Z + \beta$ for any $\alpha \in \mathbb{R} \setminus \{0\}$ and $\beta \in \mathbb{R}$.

(b) Let $Z \sim \mathcal{U}$ and $F_{\mathcal{D}}^{-1}$ be the quantile function of $\mathcal{D}$. Show that

$$F_{\mathcal{D}}^{-1}(Z) \sim \mathcal{D}.$$

This means that we can generate samples from $\mathcal{D}$ by generating samples from $\mathcal{U}$ and then applying the quantile function of $\mathcal{D}$ to them.

Hint: $F_{\mathcal{D}}(F_{\mathcal{D}}^{-1}(Z)) \geq Z$, but they may not be equal.

Note: Although standard libraries in most programming languages provide built-in functions for sampling from commonly used distributions, it is good for computer scientists to know how computer programs work.

5.7. Let $X \sim \mathcal{N}(0, \sigma^2)$. Compute the moments $\mathbb{E}[X^k]$ for all integers $k \geq 1$.

Hint: Use something we have learned in this chapter, then your solution will be very short.

5.8. Define $\bar{\Phi}(x) := \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} \, \mathrm{d}u$ as the tail probability of the standard Gaussian distribution. Prove that

$$\bar{\Phi}(x) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-x^2/2} - \frac{1}{x^3} e^{-x^2/2} + \int_x^{+\infty} \frac{3}{u^4} e^{-u^2/2} \, \mathrm{d}u \right). \tag{5.312}$$

Note: This explains why $\bar{\Phi}(x) \sim \frac{1}{\sqrt{2\pi}} \frac{e^{-x^2/2}}{x}$ in Theorem 5.28.

5.9. In Definition 2.96, we defined the independence of random variables. Although the definition is stated only for the finite discrete case, it is also the general definition of independence for arbitrary random variables. From this definition, it can be shown that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ whenever X and Y are independent and these expectations are well-defined and finite.

(a) Prove that if $(X, Y) \in \mathbb{R}^2$ is a Gaussian random vector, then X and Y are independent if and only if $\mathrm{Cov}(X, Y) = 0$.

Note: For showing that X and Y are independent, you have to verify that $\Pr[X \in A, Y \in B] = \Pr[X \in A] \Pr[Y \in B]$ for all sets $A, B \subseteq \mathbb{R}$.

(b) Let $(X, Y) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ be a standard Gaussian random vector in $\mathbb{R}^2$. Are the following two random variables independent?

$$X + Y \quad \text{and} \quad X - Y.$$

5.10. In Definition 2.67, we defined the conditional probability of an event A given an event B with $\Pr[B] > 0$, written as $\Pr[A \mid B]$. In Definition 2.90, we defined the conditional expectation of a random variable X given an event B with $\Pr[B] > 0$, written as $\mathbb{E}[X \mid B]$. The same definition works for all probability spaces. For example, if $(X, Y) \in \mathbb{R}^2$ is a Gaussian random vector, we may write:

$$\Pr[X \leq 0 \mid Y \geq 0] := \frac{\Pr[X \leq 0 \cap Y \geq 0]}{\Pr[Y \geq 0]},$$

$$\mathbb{E}[X \mid Y \geq 0] := \frac{\mathbb{E}[\mathbb{1}_{[Y \geq 0]} \cdot X]}{\Pr[Y \geq 0]}.$$

However, there is a subtle issue for continuous random variables. We often want to talk about the conditional probability distribution of X given $Y = y$ for some $y \in \mathbb{R}$, but $\Pr[Y = y] = 0$ when Y is continuous.

In general, let X, Y be two random variables on the same probability space. Let $g(A, y)$ be a function that maps a set $A \subseteq \mathbb{R}$ and a real number $y \in \mathbb{R}$ to a real number. We say that $g(A, y)$ is a *version* of the *conditional*

probability distribution $P_{X|Y}$ if it satisfies that for every $A, B \subseteq \mathbb{R}$ with $\Pr[Y \in B] > 0$,

$$\Pr[X \in A \mid Y \in B] = \mathbb{E}[g(A, Y) \mid Y \in B].$$

Please prove the following statements:

(a) Let $(X, Y) \in \mathbb{R}^2$ be a random vector with PDF $f(x, y)$. Show that the following function is a version of $P_{X|Y}$:

$$g(A, y) := \frac{\int_A f(x, y)\ \mathrm{d}x}{\int_{\mathbb{R}} f(x, y)\ \mathrm{d}x}.$$

Here, when the denominator is zero, we define $g(A, y)$ arbitrarily.

(b) Let $(X, Z) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ be a standard Gaussian random vector in $\mathbb{R}^2$. Let $Y := X + Z$. Determine a version of $P_{X|Y}$.

(c) Similar to probability density functions, the conditional probability distributions may not be unique. In the setting of the previous part, construct another version of $P_{X|Y}$.

Note: In fact, the conditional probability distribution is unique in the following sense: if X and Y are two random variables on the same probability space, and g, g' are two versions of $P_{X|Y}$, then there exists a set $B \subseteq \mathbb{R}$ with $\Pr[Y \in B] = 1$ such that for all $A \subseteq \mathbb{R}$ and $y \in B$,

$$g(A, Y) = g'(A, Y).$$

This statement is a bit of a headache to read, but if you want to learn more, you can take a course on measure theory and learn the concept of *Radon–Nikodym derivative*, or read standard textbooks such as [Dur19, Bil95].

5.11. Let $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$. Show that $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is a valid Gaussian distribution on $\mathbb{R}^d$ if and only if $\boldsymbol{\Sigma}$ is symmetric and positive semi-definite.

Here a matrix $\boldsymbol{\Sigma}$ is called *positive semi-definite* if for all $\boldsymbol{x} \in \mathbb{R}^d$, we have $\boldsymbol{x}^\top \boldsymbol{\Sigma} \boldsymbol{x} \geq 0$. According to our definition in Definition 5.35 (other textbooks may use different definitions), a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is valid if there exists $\boldsymbol{A} \in \mathbb{R}^{d \times r}$ for some r such that $\boldsymbol{\Sigma} = \boldsymbol{A}\boldsymbol{A}^\top$.

Hint: Any real symmetric matrix $\boldsymbol{A}$ can be diagonalized as $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$, where $\boldsymbol{\Lambda}$ is a diagonal matrix and $\boldsymbol{Q}$ is an orthogonal matrix. The diagonal entries of $\boldsymbol{\Lambda}$ are the eigenvalues of $\boldsymbol{A}$.

5.12. Let $B^d := \{\boldsymbol{x} \in \mathbb{R}^d : \|\boldsymbol{x}\|_2 \leq 1\}$ be the unit ball in $\mathbb{R}^d$. Consider an axis-aligned hypercube of the following form:

$$H := [\ell_1, u_1] \times [\ell_2, u_2] \times \cdots \times [\ell_d, u_d], \tag{5.313}$$

where $u_1 - \ell_1 = u_2 - \ell_2 = \cdots = u_d - \ell_d > 0$.

(a) Among all such hypercubes H that are contained in B^d, what is the maximum possible volume of H?

(b) Let V_d be the volume of B^d and U_d be the maximum possible volume of H that is contained in B^d. What is the limit of $\frac{U_d}{V_d}$ as $d \to \infty$?

Hint: You can use the fact that the volume of B^d is

$$V_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)} = \Theta\left(\frac{1}{\sqrt{d}}\left(\frac{2\pi e}{d}\right)^{d/2}\right),$$

where $\Gamma(x) := \int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ is the Gamma function.

5.13. Let $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}/d)$. In Theorem 5.43, we showed that $\left|\|\boldsymbol{x}\|_2^2 - 1\right| \geq \epsilon$ with probability at most $2e^{-d\epsilon^2/8}$, but this only holds for $\epsilon \in (0,1)$.

(a) Prove that for all $d \geq 1$ and for all $\epsilon \geq 1$,

$$\Pr\left[\left|\|\boldsymbol{x}\|_2^2 - 1\right| \geq \epsilon\right] \leq e^{-d\epsilon/8}.$$

(b) Is it possible to improve the above upper bound to $\leq e^{-Cd\epsilon^2}$ for some constant $C > 0$? That is, is there a universal constant $C > 0$ such that for all $d \geq 1$ and for all $\epsilon \geq 1$,

$$\Pr\left[\left|\|\boldsymbol{x}\|_2^2 - 1\right| \geq \epsilon\right] \leq e^{-Cd\epsilon^2}\ ?$$

(c) Prove that for all $d \geq 1$ and for all $\epsilon > 0$,

$$\Pr\left[\left|\|\boldsymbol{x}\|_2 - 1\right| \geq \epsilon\right] \leq 2e^{-d\epsilon^2/8}.$$

Note: Be careful about the difference between $\|\boldsymbol{x}\|_2^2$ and $\|\boldsymbol{x}\|_2$!

5.14. How can we sample from the standard Gaussian distribution $\mathcal{N}(0,1)$? According to Exercise 5.6, it seems that we only need to generate samples from $\mathcal{U}$ and then apply the quantile function of $\mathcal{N}(0,1)$ to them. But the quantile function of $\mathcal{N}(0,1)$ is not an elementary function, so it is not easy to compute it in a computer program. But remember: *integrating one Gaussian is too hard, but integrating two Gaussians is easy.*

Let $Z_1, Z_2 \sim \mathcal{U}$, where $\mathcal{U}$ denotes the uniform distribution on $[0,1)$ as in Exercise 5.6. Construct a function $g : [0,1]^2 \to \mathbb{R}^2$ such that

$$g(Z_1, Z_2) \sim \mathcal{N}(0, \boldsymbol{I}).$$

The function should only involve the following elementary operations and their compositions: addition, subtraction, multiplication, division, sqaure root, exponential, logarithm, and trigonometric functions.

Note: This transformation is known as the *Box–Muller transform*. Its variants have been implemented in the standard libraries of most programming languages to generate Gaussian samples. To practice your skills, do not look up this transform before solving the problem.

5.15. Given n points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$, define the *diameter* of this point cloud as

$$\operatorname{diam}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) := \max_{i,j \in [n]} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2.$$

A naïve algorithm to compute the diameter is to compute the distance between every pair of points, and then take the maximum. This takes

$\mathcal{O}(n^2 d)$ time because there are $\mathcal{O}(n^2)$ pairs of points and each pair takes $\mathcal{O}(d)$ time to compute the distance.

Describe an algorithm that inputs a point cloud $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$ and parameters $\delta \in (0,1), \epsilon \in (0,1)$, and satisfies:

- It always runs in time
$$\mathcal{O}\left(\frac{n(n+d)}{\epsilon^2} \log \frac{n}{\delta}\right).$$
- With probability at least $1-\delta$, it outputs a pair of points $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ such that
$$\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \geq (1-\epsilon)\mathrm{diam}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n).$$

Your algorithm should only involve elementary operations on integers and real numbers. Each elementary operation takes at most $\mathcal{O}(1)$ time. Your algorithm can access a random number generator that generates samples from the uniform distribution $\mathcal{U}$ on $[0,1)$, and each sample takes at most $\mathcal{O}(1)$ time to generate.

Note: If you wonder if this running time is optimal, the answer is no. It can be further reduced to
$$\mathcal{O}\left(nd \log d + \frac{n}{\epsilon^2} \log^3 \frac{n}{\delta} + \frac{n^2}{\epsilon^2} \log \frac{n}{\delta}\right).$$
See [AC06, FHL23] if you are interested.

5.16. Show that the absolute value in the definition of total variation distance is unnecessary. That is, for any two probability distributions P and Q,
$$d_{\mathrm{TV}}(P,Q) = \sup_{A \subseteq \mathbb{R}^d} \Big(\Pr[X \in A] - \Pr[Y \in A]\Big).$$

Note: Similarly, the absolute value in the variational form of total variation distance is also unnecessary.
$$d_{\mathrm{TV}}(P,Q) = \frac{1}{u-\ell} \sup_{\phi \in \mathcal{F}_{\ell,u}} \Big(P(\phi) - Q(\phi)\Big).$$

5.17. Let P be a probability distribution on $\mathbb{R}$ and $X_1, \ldots, X_n$ be n i.i.d. samples from P. Define the *empirical distribution* be the following discrete probability distribution on $\mathbb{R}$:
$$\hat{P}^{(n)}(A) := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{[X_i \in A]}.$$

(a) Let P be a discrete probability distribution with PMF $p(x)$. Define $C(p) := \frac{1}{2} \sum_x \sqrt{p(x)(1-p(x))}$ and assume that $C(p) < +\infty$. Prove the following upper bound on $\mathbb{E}\Big[d_{\mathrm{TV}}(\hat{P}^{(n)}, P)\Big]$:
$$\mathbb{E}\Big[d_{\mathrm{TV}}(\hat{P}^{(n)}, P)\Big] \leq \tfrac{C(p)}{\sqrt{n}}.$$

Hint: Jensen's inequality.

(b) In the setting of the previous part, prove that for any $\epsilon > 0$ and $n \geq 1$,

$$\Pr\Big[d_{\mathrm{TV}}(\hat{P}^{(n)}, P) \leq \tfrac{C(p)}{\sqrt{n}} + \epsilon\Big] \geq 1 - e^{-2\epsilon^2 n}. \tag{5.314}$$

(c) If P is a continuous probability distribution, prove that for all $n \geq 1$, with probability 1,

$$d_{\mathrm{TV}}(\hat{P}^{(n)}, P) = 1.$$

Note: If we use the Wasserstein distance instead of the total variation distance, then for any probability distribution P on $[0, 1]$,

$$\Pr\Big[W_1(\hat{P}^{(n)}, P) \leq \epsilon\Big] \geq 1 - 2e^{-2\epsilon^2 n}.$$

This follows from the Dvoretzky–Kiefer–Wolfowitz inequality. Look it up if you are interested.

5.18. The Gaussian distribution also has an important application in Fourier analysis. In this exercise, we will learn a little bit about the Fourier transform and its inverse. The *Fourier transform* of a function $f : \mathbb{R} \to \mathbb{C}$ is defined as the function $\hat{f} : \mathbb{R} \to \mathbb{C}$ given by

$$\hat{f}(\omega) := \int_{-\infty}^{+\infty} e^{i\omega x} f(x)\, \mathrm{d}x.$$

The *inverse Fourier transform* of a function $\hat{f} : \mathbb{R} \to \mathbb{C}$ is defined as the function $\check{f} : \mathbb{R} \to \mathbb{C}$ given by

$$\check{f}(x) := \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-i\omega x} \hat{f}(\omega)\, \mathrm{d}\omega.$$

We denote the Fourier transform of f by $\mathcal{F}[f]$ and the inverse Fourier transform of $\hat{f}$ by $\mathcal{F}^{-1}[\hat{f}]$. A sufficient condition for the Fourier transform of f to be well-defined is that $f \in L^1(\mathbb{R})$, where $L^1(\mathbb{R})$ is the space of functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$\int_{-\infty}^{+\infty} |f(x)|\, \mathrm{d}x < +\infty.$$

(a) Let φ_σ be the PDF of the Gaussian distribution $\mathcal{N}(0, \sigma^2)$. Compute its Fourier transform $\mathcal{F}[\varphi_\sigma]$ and verify that

$$\mathcal{F}^{-1}\Big[\mathcal{F}[\varphi_\sigma]\Big] = \varphi_\sigma.$$

Hint: Have you seen the integrals involved here elsewhere?

(b) Let $f \in L^1(\mathbb{R})$. For every $\sigma > 0$, define the *Gaussian smoothing* of f with bandwidth σ to be the function $f_\sigma : \mathbb{R} \to \mathbb{R}$ given by

$$f_\sigma(x) := \mathbb{E}[f(x + Z)] = \int_{-\infty}^{+\infty} f(x + z)\varphi_\sigma(z)\, \mathrm{d}z,$$

where $Z \sim \mathcal{N}(0, \sigma^2)$. Prove that $f_\sigma \in L^1(\mathbb{R})$.

(c) Express $\mathcal{F}[f_\sigma]$ in terms of $\mathcal{F}[f]$ and verify that

$$\mathcal{F}^{-1}\Big[\mathcal{F}[f_\sigma]\Big] = f_\sigma.$$

(d) Let $f \in L^1(\mathbb{R})$ be a function such that both the left and right limits of f exist at every point in $\mathbb{R}$:

$$f_+(x) := \lim_{y \to x^+} f(y), \qquad f_-(x) := \lim_{y \to x^-} f(y).$$

Prove that

$$\mathcal{F}^{-1}\Big[\mathcal{F}[f]\Big] = \frac{f_- + f_+}{2}.$$

Hint: Take $\sigma \to 0$. Although this problem looks more rigorous about integration than this course usually requires, we still allow you to swap the order of the limit and integration without justification.

(e) Let X, Y be two random variables with continuous PDFs f_X and f_Y. Prove that if $M_X(it) = M_Y(it)$ for all $t \in \mathbb{R}$, then $f_X = f_Y$.

Note: This provides an alternative proof for Remark 4.91 in the continuous case.

5.19. Let $L \geq 3$. An L-layer ReLU network is a neural network with L linear layers, where each intermediate layer is followed by a ReLU activation function $[x]_+ := \max\{x, 0\}$. When applied to a vector $\boldsymbol{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d$, the ReLU activation function is applied element-wise:

$$[\boldsymbol{x}]_+ = (\max\{x_1, 0\}, \ldots, \max\{x_d, 0\}).$$

More specifically, an L-layer ReLU network is parameterized by a sequence of weight matrices $(\boldsymbol{W}_1, \ldots, \boldsymbol{W}_L)$. Given an input $\boldsymbol{x} \in \mathbb{R}^d$, we write $\boldsymbol{x}^{(0)} := \boldsymbol{x}$, and then for each $\ell \in [L-1]$, the output of the ℓ-th layer is given by

$$\boldsymbol{x}^{(\ell)} := [\boldsymbol{W}_\ell \boldsymbol{x}^{(\ell-1)}]_+.$$

Finally, the output of the L-th layer, which is also the output of the network, is given by

$$f(\boldsymbol{x}) := \boldsymbol{W}_L \boldsymbol{x}^{(L-1)}.$$

The first and last layers are called the *input layer* and *output layer*, respectively. The intermediate layers are called *hidden layers*. Let d and k be the dimensions of the input and output, respectively, and let m be the parameter that denotes the *width* of the network. The weight matrix in the input layer has shape $m \times d$, the weight matrix in the hidden layers have shape $m \times m$, and the weight matrix in the output layer has shape $k \times m$. We use d_ℓ to denote the dimension of $\boldsymbol{x}^{(\ell)}$, so that $d_0 = d$ and $d_\ell = m$ for $\ell \in [L-1]$.

Having introduced all this background, we finally ask the following question: *how should we initialize a deep ReLU network?* In deep learning, weight matrices are usually initialized randomly, and the network is then trained by gradient descent. A typical choice is to sample each entry of each weight matrix $\boldsymbol{W}_\ell$ independently from

$$\mathcal{N}\left(0, c \cdot d_{\ell-1}^{-\alpha}\right),$$

where $c > 0$ and $\alpha > 0$ are constants.

(a) Fix an input $\boldsymbol{x} \in \mathbb{R}^d$. For every $\ell \in [L-1]$, express $\mathbb{E}[\|\boldsymbol{x}^{(\ell)}\|_2^2]$ in terms of $\|\boldsymbol{x}\|_2$, k, d, m, c, α, and ℓ. Here the expectation is taken over the random draw of the weight matrices.

Hint: You may need to prove the following fact: if the distribution of X is symmetric around 0, then $\mathbb{E}[[X]_+^2] = \frac{1}{2}\mathbb{E}[X^2]$.

(b) An ideal initialization should ensure that the outputs of intermediate layers are not too small and not too large. Find a pair (c, α) such that for every fixed input $\boldsymbol{x} \in \mathbb{R}^d \setminus \{0\}$ and every coordinate $i \in [m]$,

$$\mathbb{E}\left[\left(x_i^{(L-1)}\right)^2\right]$$

converges to a non-zero constant as $m \to \infty$ and $L \to \infty$, where $x_i^{(L-1)}$ is the i-th coordinate of $\boldsymbol{x}^{(L-1)}$.

Note: The choice of c and α is unique, and is usually referred to as the *He initialization* or *Kaiming initialization*. This initialization is named after Kaiming He [HZRS15].

Chapter 6

Central Limit Theorem

We introduced the Central Limit Theorem (CLT) in Section 4.3.6: the sample average of n random variables converges to a Gaussian distribution as $n \to \infty$. However, at the time, we had not even formally defined Gaussian distributions.

In this short chapter, we will formally state the central limit theorem and discuss some of its consequences. We will start by defining what we mean by "convergence" for random variables. Then we will state the law of large numbers and the central limit theorem using this notion of convergence. After that, we will introduce confidence intervals and discuss how the CLT gives an approximate error bar for an unknown mean. Finally, as the CLT highlights the importance of variance, we will discuss Bernstein's inequality, which incorporates variance information into tail bounds for sample averages.

6.1 Limiting Behavior of Sample Averages

In this section, we will formally state the law of large numbers and the central limit theorem, two fundamental theorems for characterizing the limiting behavior of sample averages. Before stating these theorems, we will first define what we mean by "convergence" for probability distributions and random variables.

6.1.1 Convergence in Distribution

In Section 5.4, we discussed several ways to compare two probability distributions, and a common idea was to test the distributions by sets or functions. For example, total variation distance compares the probabilities on every possible set and takes the largest difference. Here we borrow this idea to define convergence for probability distributions and random variables.

▶ Convergence in Total Variation

Our first attempt to define convergence is to use total variation distance: if the total variation distance between $\mathcal{D}_n$ and $\mathcal{D}$ goes to 0, then $\mathcal{D}_n$ converges to $\mathcal{D}$.

Definition 6.1 (Convergence in total variation)**.** Let $(\mathcal{D}_n)_{n\geq 1}$ be a sequence of probability distributions on $\mathbb{R}^d$. We say that $\mathcal{D}_n$ *converges in total variation* to a probability distribution $\mathcal{D}$ if

$$d_{\mathrm{TV}}(\mathcal{D}_n, \mathcal{D}) \to 0. \tag{6.1}$$

Equivalently,

$$\sup_A |\mathcal{D}_n(A) - \mathcal{D}(A)| \to 0. \tag{6.2}$$

While this definition is natural, it is too strong for characterizing the limiting behavior of random variables. To see this, consider the following limit in the usual sense: as $n \to \infty$,

$$\tfrac{1}{n} \to 0. \tag{6.3}$$

There is no randomness in this equation, but deterministic values can be viewed as a special case of random variables. We denote by $\delta_a(A) := \mathbb{1}_{[a\in A]}$ the point mass at $a \in \mathbb{R}$. Ideally, we would like to view each $\frac{1}{n}$ as a random variable with distribution $\delta_{1/n}$, and then take the limit:

$$\delta_{1/n} \to \delta_0. \tag{6.4}$$

Does this make sense in total variation distance? In fact, as we have already discussed in Section 5.4.2, the total variation distance tends to ignore the geometry of the underlying space. As a result, for every $n \geq 1$,

$$d_{\mathrm{TV}}(\delta_{1/n}, \delta_0) = 1. \tag{6.5}$$

Therefore, $1/n$ does not converge to 0 in total variation.

▶ Setwise Convergence

Recall what $1/n \to 0$ means in the usual sense: we may take an arbitrary open neighborhood A of 0, and the sequence will eventually enter and stay in A. That is, for any open neighborhood A of 0,

$$\exists N \in \mathbb{N}, \quad \forall n \geq N, \quad \tfrac{1}{n} \in A. \tag{6.6}$$

In terms of point masses, this says

$$\exists N \in \mathbb{N}, \quad \forall n \geq N, \quad \delta_{1/n}(A) = 1 = \delta_0(A). \tag{6.7}$$

Equivalently,

$$\delta_{1/n}(A) \to \delta_0(A) \tag{6.8}$$

This gives us a potential way to generalize this view, which is to consider the convergence of the probability of any set A. But unlike convergence in total variation, we do not take the supremum over all sets.

Definition 6.2 (Setwise Convergence)**.** Let $(\mathcal{D}_n)_{n\geq 1}$ be a sequence of probability distributions on $\mathbb{R}^d$. We say that $\mathcal{D}_n$ *converges setwise* to a probability distribution $\mathcal{D}$ if

$$\mathcal{D}_n(A) \to \mathcal{D}(A) \tag{6.9}$$

for all sets $A \subseteq \mathbb{R}^d$.

This setwise convergence is weaker than convergence in total variation because the set A is fixed before taking the limit. Back to the example of $1/n$, we can see that many fixed sets behave exactly as we want. For example,

$$\delta_{1/n}((0.1,1)) \to 0 = \delta_0((0.1,1)), \tag{6.10}$$

$$\delta_{1/n}((-1,0.1)) \to 1 = \delta_0((-1,0.1)). \tag{6.11}$$

But setwise convergence is still too strong. If $A = \{0\}$, then

$$\delta_{1/n}(A) = 0 \qquad \text{but} \qquad \delta_0(A) = 1. \tag{6.12}$$

Similarly, if $A = (0,2)$, then

$$\delta_{1/n}(A) = 1 \qquad \text{but} \qquad \delta_0(A) = 0. \tag{6.13}$$

Therefore, we still need a weaker notion of convergence than setwise convergence.

▶ Convergence in Distribution

After playing around with the examples above for a while, the issue becomes clearer. The usual limit $1/n \to 0$ controls what happens inside every open set around 0. Therefore, if A contains an open set around 0, then $1/n \in A$ for all sufficiently large n. Similarly, if $\bar{A}$ contains an open set around 0, then $1/n \notin A$ for all sufficiently large n. In both cases, the membership of $1/n$ in A eventually agrees with the membership of 0 in A.

The problematic case is when 0 lies on the boundary of A. Then every open set containing 0 intersects both A and $\bar{A}$. Thus the usual limit $1/n \to 0$ does not decide on which side of A the nearby points $1/n$ should eventually lie. In particular, it does not force the indicators $\mathbb{1}_{[1/n\in A]}$ and $\mathbb{1}_{[0\in A]}$ to agree eventually. This observation motivates the following definition.

Definition 6.3 (Continuity set)**.** Let $\mathcal{D}$ be a probability distribution on $\mathbb{R}^d$. The *boundary* of a set $A \subseteq \mathbb{R}^d$, denoted by ∂A, is the set of points $\boldsymbol{x} \in \mathbb{R}^d$ such that every open set U containing $\boldsymbol{x}$ intersects both A and its complement $\bar{A}$, i.e., $U \cap A \neq \varnothing$ and $U \cap \bar{A} \neq \varnothing$. A set $A \subseteq \mathbb{R}^d$ is called a *$\mathcal{D}$-continuity set* if

$$\mathcal{D}(\partial A) = 0, \tag{6.14}$$

On the real line, the boundary of an interval is just the set of its endpoints. For example,

$$\partial(a,b) = \partial[a,b] = \partial(a,b] = \partial[a,b) = \{a,b\}, \qquad \partial\{a\} = \{a\}. \tag{6.15}$$

For the example above, $(0.1, 1)$ and $(-1, 0.1)$ are δ_0-continuity sets, while $\{0\}$ and $(0, 1)$ are not. This suggests that we should test sets only when their boundaries are invisible to the limiting distribution.

Definition 6.4 (Convergence in distribution)**.** Let $(\mathcal{D}_n)_{n\geq 1}$ be a sequence of probability distributions on $\mathbb{R}^d$. We say that $\mathcal{D}_n$ *converges in distribution* to a probability distribution $\mathcal{D}$ if

$$\mathcal{D}_n(A) \to \mathcal{D}(A) \tag{6.16}$$

for all $\mathcal{D}$-continuity sets A. We write this convergence as

$$\mathcal{D}_n \xrightarrow{d} \mathcal{D}. \tag{6.17}$$

For random variables and vectors, we use the same notation for their distributions. More specifically, we write

$$X_n \xrightarrow{d} X \tag{6.18}$$

to mean that $P_{X_n} \xrightarrow{d} P_X$, where P_{X_n} and P_X are the distributions of X_n and X, respectively.

Example 6.5 (Deterministic limits)**.** Suppose $a_n \to a$ as $n \to \infty$ in the usual sense. Then

$$\delta_{a_n} \xrightarrow{d} \delta_a \qquad \text{and} \qquad a_n \xrightarrow{d} a. \tag{6.19}$$

This is because δ_a-continuity sets are just the sets that contain a in their interior or exterior. Hence, once n is sufficiently large, a_n and a are either both in A or both outside A.

Remark 6.6 (Wasserstein convergence)**.** Recall that we have also introduced the Wasserstein-1 distance W_1 in Equation (5.277), where $W_1(\delta_{a_n}, \delta_a) = \|a_n - a\|_2$. Can we use Wasserstein distance to define a notion of convergence? Yes, but this is just a notion that is slightly stronger than convergence in distribution. More specifically, $W_1(\mathcal{D}_n, \mathcal{D}) \to 0$ is equivalent to the following two conditions: (i) $\mathcal{D}_n \xrightarrow{d} \mathcal{D}$ and (ii) $\mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}_n}[\|\boldsymbol{x}\|_2] \to \mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}}[\|\boldsymbol{x}\|_2]$. We will not go into the details here but a key idea is that when the expected norm converges, each Lipschitz test function in Wasserstein distance can be truncated to a bounded Lipschitz function with a small error.

▶ Equivalent Formulations

The definition above is conceptually useful because it specifies exactly which sets are safe to test. However, it is not always convenient to check all $\mathcal{D}$-continuity sets. On the real line, there is a much simpler criterion using CDFs.

Recall from Exercise 5.5 that the CDF of a probability distribution $\mathcal{D}$ on $\mathbb{R}$ is

$$F_{\mathcal{D}}(x) := \mathcal{D}((-\infty, x]). \tag{6.20}$$

In other words, the CDF of a probability distribution $\mathcal{D}$ tests this distribution on each half-line $(-\infty, x]$. By Theorem 5.18, these half-lines uniquely determine the distribution on $\mathbb{R}$.

Theorem 6.7 (CDF convergence criterion)**.** Let $(\mathcal{D}_n)_{n\geq 1}$ be a sequence of probability distributions on $\mathbb{R}$ with CDFs $F_{\mathcal{D}_n}$ and $F_{\mathcal{D}}$. Then

$$\mathcal{D}_n \xrightarrow{d} \mathcal{D} \tag{6.21}$$

if and only if

$$F_{\mathcal{D}_n}(x) \to F_{\mathcal{D}}(x) \tag{6.22}$$

whenever $F_{\mathcal{D}}$ is continuous at x. Such x is also called a *continuity point* of $F_{\mathcal{D}}$.

We will not prove this theorem here but this theorem is very intuitive. The condition that $F_{\mathcal{D}}$ is continuous at x is equivalent to $\mathcal{D}(\{x\}) = 0$, and thus it is equivalent to saying that the half-line $(-\infty, x]$ is a $\mathcal{D}$-continuity set. Therefore, the theorem says that to check convergence in distribution on $\mathbb{R}$, it is enough to test these very special $\mathcal{D}$-continuity sets. This is because half-lines determine the whole distribution on $\mathbb{R}$: once we know the probabilities of all half-lines, we can recover the probabilities of all intervals, and then approximate each $\mathcal{D}$-continuity set by unions of intervals.

We may also test the distributions with test functions instead of sets. In contrast to the total variation distance where any bounded test function is allowed, here we only allow bounded continuous test functions.

Theorem 6.8 (Test function characterization)**.** Let $(\mathcal{D}_n)_{n\geq 1}$ be a sequence of probability distributions on $\mathbb{R}^d$. Then

$$\mathcal{D}_n \xrightarrow{d} \mathcal{D} \tag{6.23}$$

if and only if

$$\mathcal{D}_n(\phi) \to \mathcal{D}(\phi) \tag{6.24}$$

for every bounded continuous function $\phi : \mathbb{R}^d \to \mathbb{R}$.

Again we will not prove this theorem here, but the main idea is that bounded continuous functions and indicators of $\mathcal{D}$-continuity sets can approximate each other in the right sense. On one hand, the indicator of a $\mathcal{D}$-continuity set can be approximated from above and below by bounded continuous functions. On the other hand, a bounded continuous function can be discretized into a finite sum of indicators of its level sets, and the thresholds can be chosen so that the corresponding level sets are $\mathcal{D}$-continuity sets.

The CDF and test function criteria are still a bit hard to check in practice. For discrete and continuous distributions, a more direct way is to check the convergence of their PMFs and PDFs.

Theorem 6.9 (PMF convergence criterion)**.** Let $(\mathcal{D}_n)_{n\geq 1}$ be a sequence of discrete probability distributions on $\mathcal{X}$ and $\mathcal{D}$ be another discrete probability distribution on $\mathcal{X}$. Let p_n and p be the PMFs of $\mathcal{D}_n$ and $\mathcal{D}$, respectively. Then

$$\mathcal{D}_n \xrightarrow{d} \mathcal{D} \tag{6.25}$$

if and only if

$$p_n(x) \to p(x) \tag{6.26}$$

for all $x \in \mathcal{X}$.

Theorem 6.10 (PDF convergence criterion)**.** Let $(\mathcal{D}_n)_{n\geq 1}$ be a sequence of continuous probability distributions on $\mathbb{R}^d$ and $\mathcal{D}$ be another continuous probability distribution on $\mathbb{R}^d$. Let f_n and f be the PDFs of $\mathcal{D}_n$ and $\mathcal{D}$, respectively. Then if

$$f_n(x) \to f(x) \tag{6.27}$$

for all $x \in \mathbb{R}^d$, then

$$\mathcal{D}_n \xrightarrow{d} \mathcal{D}. \tag{6.28}$$

We will not prove the above two theorems here, but they are easy to derive from the CDF criterion.

When moment generating functions exist near the origin, there is a more algebraic way to check convergence in distribution. This is analogous to using the MGFs to check whether two distributions are the same (Theorem 5.19).

Theorem 6.11 (MGF convergence criterion)**.** Let $(X_n)_{n\geq 1}$ be a sequence of random variables and let X be a random variable. Suppose that the MGFs M_{X_n} and M_X are all defined on some common open interval $(-\epsilon, \epsilon)$. If

$$M_{X_n}(t) \to M_X(t) \tag{6.29}$$

for every $t \in (-\epsilon, \epsilon)$, then

$$X_n \xrightarrow{d} X. \tag{6.30}$$

Once again, we will not prove this theorem here. The main idea is that the CDF can be recovered from the MGF by the inversion formula in Theorem 4.88. After taking the limit in the MGFs, we can use this inversion formula to obtain convergence of the CDFs, and then apply the CDF criterion.

This MGF criterion is especially useful when the limiting distribution has a different nature from the approximating distributions, for example when a sequence of continuous distributions converges to a discrete distribution, or when a sequence of discrete distributions converges to a continuous distribution.

Example 6.12. Let $X_n \sim \mathcal{N}(0, 1/n)$ and $X = 0$. There are two ways to check convergence in distribution. First, we can use the CDF criterion. The CDF of X_n is given by

$$F_n(x) := \Pr[X_n \leq x] = \Pr[Z \leq x\sqrt{n}], \tag{6.31}$$

where $Z \sim \mathcal{N}(0, 1)$. If $x > 0$, then this CDF converges to 1. If $x < 0$, then this CDF converges to 0. Noting that the CDF of X is $F(x) = \mathbb{1}_{[x \geq 0]}$, we can see that $F_n(x) \to F(x)$ for all $x \neq 0$. Since 0 is not a continuity point of F, we can conclude that $X_n \xrightarrow{d} X$.

We may also use the MGF criterion. By Theorem 5.25, the MGF of X_n is

$$M_{X_n}(t) = e^{t^2/(2n)}. \tag{6.32}$$

As $n \to \infty$, this MGF converges to $e^0 = 1 = M_X(t)$. Therefore, $X_n \xrightarrow{d} X$.

6.1.2 The Law of Large Numbers

We have already seen the law of large numbers from the generating function viewpoint in Section 4.3.6. There, the idea was the same as the MGF criterion above: under suitable assumptions, we can study the limiting distribution by studying the limiting MGF or CGF. If $X_1, X_2, \ldots, X_n$ are i.i.d. random variables with CGF $K(t)$ and mean μ, then the CGF of the sample average $X^{(n)}$ satisfies

$$K_{X^{(n)}}(t) = nK(t/n) \to \mu t, \tag{6.33}$$

which is the CGF of the point mass at μ. In the language we have just introduced, this says that $P_{X^{(n)}} \xrightarrow{d} \delta_\mu$, or more compactly, $X^{(n)} \xrightarrow{d} \mu$.

However, MGFs and CGFs are not always defined on a real interval around 0. A more direct way is to apply Chebyshev's inequality and use the CDF criterion. Now we state the law of large numbers more formally.

Theorem 6.13 (Weak law of large numbers)**.** Let $X_1, X_2, \ldots$ be i.i.d. random variables with

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \mathrm{Var}(X_i) = \sigma^2 < +\infty. \tag{6.34}$$

Let $X^{(n)} := \frac{1}{n}\sum_{i=1}^n X_i$ be the sample average. Then, for every $\epsilon > 0$,

$$\Pr\Big[|X^{(n)} - \mu| \geq \epsilon\Big] \to 0. \tag{6.35}$$

Equivalently, as $n \to \infty$,

$$X^{(n)} \xrightarrow{d} \mu. \tag{6.36}$$

Proof. By Lemma 3.16,

$$\mathrm{Var}(X^{(n)}) = \frac{\sigma^2}{n}. \tag{6.37}$$

By Chebyshev's inequality, we get the tail bound:

$$\Pr\Big[|X^{(n)} - \mu| \geq \epsilon\Big] \leq \frac{\sigma^2}{n\epsilon^2} \to 0. \tag{6.38}$$

It remains to show that the tail bound implies convergence in distribution to the point mass δ_μ, whose CDF is $F(x) = \mathbb{1}_{[x \geq \mu]}$. Since the only non-continuity point of δ_μ is μ, we only need to check that for all $\epsilon > 0$,

$$\Pr[X^{(n)} \leq \mu + \epsilon] \to 1 \quad \text{and} \quad \Pr[X^{(n)} \leq \mu - \epsilon] \to 0. \tag{6.39}$$

But both limits follow simply from the tail bound. Therefore, $X^{(n)} \xrightarrow{d} \mu$. □

The convergence in Equation (6.35) is often called *convergence in probability* to μ. More generally, we have the following definition for general random variables.

Definition 6.14 (Convergence in probability)**.** Let $(X_n)_{n \geq 1}$ and X be random variables defined on the same probability space. We say that X_n *converges in probability* to X if for every $\epsilon > 0$,

$$\Pr[|X_n - X| \geq \epsilon] \to 0 \qquad \text{as } n \to \infty. \tag{6.40}$$

We write this convergence as

$$X_n \xrightarrow{\mathrm{P}} X. \tag{6.41}$$

Thus, Equation (6.35) says that $X^{(n)} \xrightarrow{\mathrm{P}} \mu$. Convergence in probability is a stronger notion than convergence in distribution because the former implies the latter, but not vice versa. When the limiting random variable is a constant, which is the case for the weak law of large numbers, these two notions are equivalent.

Remark 6.15. There is also a stronger version called the *strong law of large numbers*. It says that if $\mathbb{E}[X_i] = \mu$ and $\mathbb{E}[|X_i|] < +\infty$, then

$$\Pr\Big[\lim_{n \to \infty} X^{(n)} = \mu\Big] = 1. \tag{6.42}$$

Note that convergence in distribution considers convergence at the distribution level: we repeat the experiment many times, collect the outcomes, and look at the distribution of $X^{(n)}$. In contrast, the strong law of large numbers says that within a single infinite experiment, as we keep sampling more and more, the sample averages themselves converge to μ with probability 1.

6.1.3 The Central Limit Theorem

We already had a first look at the central limit theorem in Section 4.3.6. There, the CGF calculation suggested the following picture: although $X^{(n)}$ itself collapses to the mean μ, the rescaled deviation from the mean

$$\sqrt{n}\,(X^{(n)} - \mu) \tag{6.43}$$

should approach a Gaussian distribution. Now that we have learned about Gaussian distributions and convergence in distribution, we can finally state this theorem precisely.

Theorem 6.16 (Central limit theorem)**.** Let $X_1, X_2, \dots$ be i.i.d. random variables with

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \operatorname{Var}(X_i) = \sigma^2 < +\infty. \tag{6.44}$$

Let $X^{(n)} := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average. Then

$$\sqrt{n}\,(X^{(n)} - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \tag{6.45}$$

Proof when the MGF exists near 0. Assume that the MGF of X_1 is defined and analytic on an open interval around 0. Let $K(t) := \log M_{X_1}(t)$ be its CGF. Since $\mathbb{E}[X_1] = \mu$ and $\operatorname{Var}(X_1) = \sigma^2$, the Taylor expansion of K at 0 is

$$K(s) = \mu s + \frac{\sigma^2}{2}s^2 + \mathcal{O}(s^3). \tag{6.46}$$

Let

$$Z^{(n)} := \sqrt{n}\,(X^{(n)} - \mu). \tag{6.47}$$

By the additivity of CGFs for independent random variables,

$$K_{Z^{(n)}}(t) = nK(\tfrac{t}{\sqrt{n}}) - \sqrt{n}\mu t \tag{6.48}$$

$$= n\left(\mu\frac{t}{\sqrt{n}} + \frac{\sigma^2}{2}\cdot\frac{t^2}{n} + \mathcal{O}(n^{-3/2})\right) - \sqrt{n}\mu t \tag{6.49}$$

$$= \frac{1}{2}\sigma^2 t^2 + \mathcal{O}(n^{-1/2}) \to \frac{\sigma^2 t^2}{2}. \tag{6.50}$$

The limiting CGF $\frac{\sigma^2 t^2}{2}$ is the CGF of $\mathcal{N}(0, \sigma^2)$. Equivalently, the MGFs converge to the MGF of $\mathcal{N}(0, \sigma^2)$. Therefore, by the MGF convergence criterion in Theorem 6.11,

$$Z^{(n)} = \sqrt{n}\,(X^{(n)} - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \tag{6.51}$$

which completes the proof. □

Remark 6.17. The proof above assumes that the MGF exists on a real interval around 0, but it may not always be the case. In general, one can use the same idea on the imaginary axis, where the MGF is always defined. As noted in Remark 4.91, the MGF along the imaginary axis is called the *characteristic function* of X.

For completeness, we also state the multidimensional version of the central limit theorem.

Theorem 6.18 (Multidimensional central limit theorem)**.** Let $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots$ be i.i.d. random vectors in $\mathbb{R}^d$ with

$$\mathbb{E}[\boldsymbol{x}_i] = \boldsymbol{\mu} \qquad \text{and} \qquad \mathrm{Cov}(\boldsymbol{x}_i) = \boldsymbol{\Sigma}. \tag{6.52}$$

Let $\boldsymbol{x}^{(n)} := \frac{1}{n}\sum_{i=1}^n \boldsymbol{x}_i$ be the sample average. Then

$$\sqrt{n}\,(\boldsymbol{x}^{(n)} - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}). \tag{6.53}$$

6.1.4 Confidence Intervals

A direct application of the central limit theorem is the notion of *confidence intervals*. To motivate this, we begin with Bernoulli random variables. Different from Section 3.4.1, here we consider biased Bernoulli variables. Suppose $X_1, \ldots, X_n \sim \mathrm{Bernoulli}(p)$ are n i.i.d. Bernoulli random variables. Let $X := \frac{1}{n}\sum_{i=1}^n X_i$ be the sample average. Then the expectation and variance of X are

$$\mathbb{E}[X] = p \quad \text{and} \quad \mathrm{Var}(X) = \frac{p(1-p)}{n}. \tag{6.54}$$

We want to understand how likely it is that X deviates significantly from p.

This question appears constantly in machine learning experiments. Suppose a classifier is evaluated on n test samples, and each sample is either classified correctly or incorrectly. If we write

$$X_i = \begin{cases} 1 & \text{if the prediction on the } i\text{-th sample is correct,} \\ 0 & \text{otherwise,} \end{cases} \tag{6.55}$$

then the empirical accuracy is exactly the sample average $X = \frac{1}{n}\sum_{i=1}^n X_i$. Now imagine that this accuracy is 80%. *Should we trust this number?* If you deploy a bag of tricks to the training pipeline and improve the accuracy to 81%, *is it really a better model?*

Not necessarily. If the test set is small, then the observed accuracy may be a few percentage points away from the true accuracy. Why not using a very large test set? There are two reasons. First, evaluation itself can be very expensive, especially for large models such as large language models (LLMs). You do not want to spend a significant amount of money just to obtain a slightly more accurate estimate of the model's performance. Second, data is often limited, and when more data is available, we usually prefer to allocate most of it to training rather than testing.

For example, the ImageNet dataset [DDS+09, RDS+15], which played a key role in demonstrating the effectiveness of deep learning methods [KSH12] over classical approaches, contains 1.3 million images for training, but only 50K for validation and 100K for testing. SWE-bench [JYW+24] is a benchmark for evaluating the software engineering capability of LLMs on fixing real-world GitHub issues. It contains only around 2K test cases, yet evaluating all of them can require hundreds of GPU hours.

Hoeffding's inequality provides the following bound for the sample average of Bernoulli random variables:

$$\Pr[X \geq p + \epsilon] \leq e^{-2n\epsilon^2} \quad \text{and} \quad \Pr[X \leq p - \epsilon] \leq e^{-2n\epsilon^2}. \tag{6.56}$$

Equivalently, we can write this as

$$\Pr\Big[p \in (X - \epsilon, X + \epsilon)\Big] \geq 1 - 2e^{-2n\epsilon^2}. \tag{6.57}$$

That is, with high probability, the true probability p lies within the interval $(X - \epsilon, X + \epsilon)$. We call such an interval a *confidence interval* for p.

Definition 6.19 (Confidence interval)**.** A *confidence interval* for a parameter θ is an interval (a, b), where the endpoints a and b are functions of certain random samples related to θ, such that

$$\Pr\Big[\theta \in (a, b)\Big] \geq 1 - \delta, \tag{6.58}$$

We call $1 - \delta$ the *confidence level* of the interval.

In our case, the parameter is the true probability p, and the random samples are the Bernoulli random variables $X_1, \ldots, X_n$. The endpoints are $X - \epsilon$ and $X + \epsilon$, where $X = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the sample average. Equation (6.57) implies that

$(X - \epsilon, X + \epsilon)$ is a confidence interval for p with confidence level $1 - 2e^{-2n\epsilon^2}$.

In particular, we can set $2e^{-2n\epsilon^2} = 1\%$ to conclude that

$(X - \frac{1.63}{\sqrt{n}}, X + \frac{1.63}{\sqrt{n}})$ is a confidence interval for p with confidence level 99%.

This is helpful since it tells us very concretely which range the true probability p is likely to lie in.

However, if we take a closer look at the variance of the sample average in Equation (6.54), we will find that it is much smaller when p is close to 0 or 1 than when p is close to 0.5. Then Chebyshev's inequality suggests that the sample average should fluctuate much less in that regime:

$$\Pr\Big[|X - p| \geq \epsilon\Big] \leq \frac{p(1-p)}{n\epsilon^2}. \tag{6.59}$$

In other words, the possibility of deviating by more than ϵ from the true accuracy is less likely to happen when the true accuracy is closer to 0 or 1 than when the true accuracy is closer to 0.5. See Figure 6.1 for a numerical demonstration. But this phenomenon is not captured by Hoeffding's inequality as it ignores variance and treats all distributions in the same way, which suggests that there is still much room for improvement.

Based on the central limit theorem, we do not have to worry about the tightness of the concentration inequalities we derived earlier. Instead, we can exactly derive the tail probabilities for the sample average as $n \to \infty$.

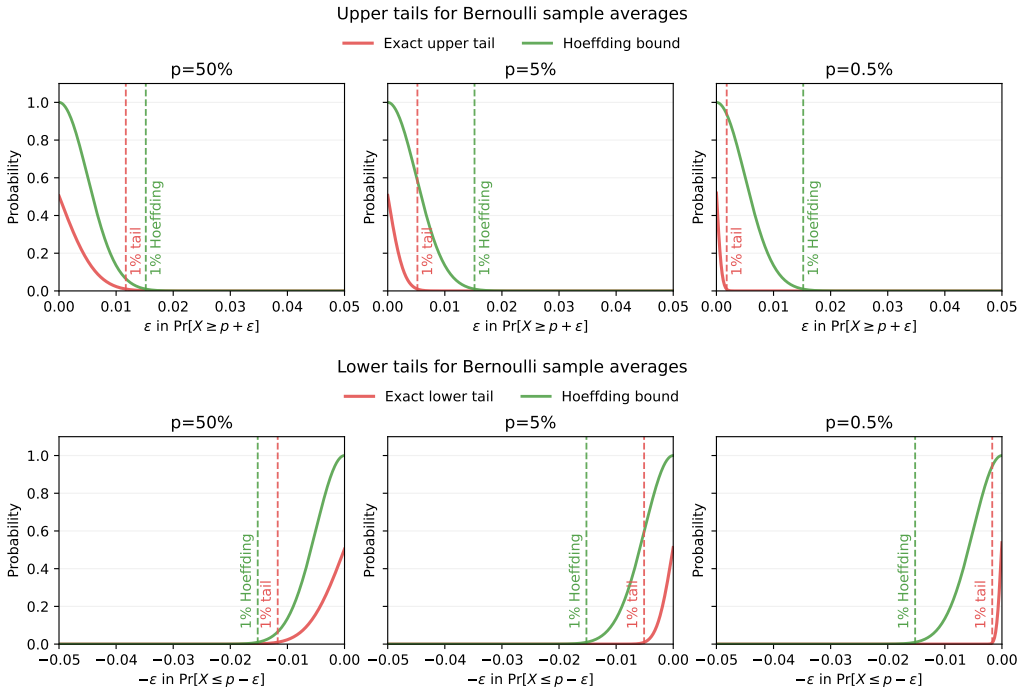


Figure 6.1: For biased Bernoulli sample averages, both the lower and upper tails are often much smaller than the Hoeffding bound as p moves away from 50%. Here we fix $n = 10{,}000$ and vary p from 50% to 5% and 0.5%.

Theorem 6.20. Let $X_1, \ldots, X_n$ be i.i.d. random variables with

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \mathrm{Var}(X_i) = \sigma^2 \in (0, +\infty). \tag{6.60}$$

Let $X^{(n)} := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average. Then for every $\alpha > 0$, as $n \to \infty$,

$$\Pr\left[\left|X^{(n)} - \mu\right| < \alpha \cdot \frac{\sigma}{\sqrt{n}}\right] \to 1 - 2\bar{\Phi}(\alpha), \tag{6.61}$$

where $\bar{\Phi}(x) := \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du$ is the tail probability of the standard Gaussian distribution.

Remark 6.21. For anyone who still remembers the error function in Exercise 4.9, recall that it is defined as $\mathrm{erf}(x) := \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$. In fact, one can use integration by substitution to show that $1 - 2\bar{\Phi}(\alpha) = \mathrm{erf}(\frac{\alpha}{\sqrt{2}})$. So we can write this limit more compactly as

$$\Pr\left[\left|X^{(n)} - \mu\right| < \alpha \cdot \frac{\sigma}{\sqrt{n}}\right] \to \mathrm{erf}\left(\frac{\alpha}{\sqrt{2}}\right). \tag{6.62}$$

Proof. By Theorem 6.16,

$$\sqrt{n}\,(X^{(n)} - \mu) \xrightarrow{d} \sigma Z, \qquad Z \sim \mathcal{N}(0, 1). \tag{6.63}$$

Now fix $\alpha > 0$. Since the interval $(-\alpha\sigma, \alpha\sigma)$ is a continuity set for the standard Gaussian distribution, convergence in distribution gives

$$\Pr\Big[\Big|X^{(n)} - \mu\Big| < \alpha \cdot \frac{\sigma}{\sqrt{n}}\Big] = \Pr\Big[-\alpha\sigma < \sqrt{n}\,(X^{(n)} - \mu) < \alpha\sigma\Big] \tag{6.64}$$

$$\to \Pr\Big[-\alpha\sigma < \sigma Z < \alpha\sigma\Big] \tag{6.65}$$

$$= \Pr\Big[-\alpha < Z < \alpha\Big]. \tag{6.66}$$

The last probability can be simplified as

$$\Pr\Big[-\alpha < Z < \alpha\Big] = 1 - \Pr[Z \geq \alpha] - \Pr[-Z \geq \alpha] = 1 - 2\bar{\Phi}(\alpha). \tag{6.67}$$

Substituting back to Equation (6.66) proves the theorem. □

Based on the above theorem, we can now conclude with the following proposition on the confidence interval for an unknown mean μ.

Proposition 6.22. Let $X_1, \ldots, X_n$ be i.i.d. random variables with

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \mathrm{Var}(X_i) = \sigma^2 \in (0, +\infty). \tag{6.68}$$

Let $X^{(n)} := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average. For every $\delta \in (0,1)$, let $\alpha(\delta)$ be the $(1-\delta/2)$-quantile of the standard Gaussian distribution, or equivalently, the solution to the equation

$$2\bar{\Phi}(\alpha(\delta)) = \delta, \tag{6.69}$$

Then the interval

$$\Big(X - \alpha(\delta) \cdot \frac{\sigma}{\sqrt{n}}, X + \alpha(\delta) \cdot \frac{\sigma}{\sqrt{n}}\Big) \tag{6.70}$$

is a confidence interval for μ with confidence level $1 - \delta + o(1)$, where $o(1)$ is a term that converges to 0 as $n \to \infty$.

Proof. By Theorem 6.20,

$$\Big(X - \alpha(\delta) \cdot \frac{\sigma}{\sqrt{n}}, X + \alpha(\delta) \cdot \frac{\sigma}{\sqrt{n}}\Big) = \Pr\Big[\Big|X^{(n)} - \mu\Big| < \alpha \cdot \frac{\sigma}{\sqrt{n}}\Big] \tag{6.71}$$

$$\to 1 - 2\bar{\Phi}(\alpha(\delta)) = 1 - \delta. \tag{6.72}$$

Therefore, the interval $(X - \alpha(\delta) \cdot \frac{\sigma}{\sqrt{n}}, X + \alpha(\delta) \cdot \frac{\sigma}{\sqrt{n}})$ is a confidence interval with confidence level $1 - \delta + o(1)$. □

Therefore, no matter which distribution $\mathcal{D}$ the random samples $X_1, \ldots, X_n$ come from, the following interval

$$\Big(X - \alpha(\delta) \cdot \frac{\sigma}{\sqrt{n}}, X + \alpha(\delta) \cdot \frac{\sigma}{\sqrt{n}}\Big). \tag{6.73}$$

is always a confidence interval for the mean μ with an approximate confidence level $1 - \delta$. This approximation becomes exact as $n \to \infty$.

Some common choices of δ are listed as follows:

- A 90% confidence interval is $(X - 1.645 \cdot \frac{\sigma}{\sqrt{n}}, X + 1.645 \cdot \frac{\sigma}{\sqrt{n}})$.
- A 95% confidence interval is $(X - 1.960 \cdot \frac{\sigma}{\sqrt{n}}, X + 1.960 \cdot \frac{\sigma}{\sqrt{n}})$.
- A 99% confidence interval is $(X - 2.576 \cdot \frac{\sigma}{\sqrt{n}}, X + 2.576 \cdot \frac{\sigma}{\sqrt{n}})$.

For Bernoulli random variables with probability p, the variance is $\sigma^2 = p(1-p)$. Consistent with our experiments in Figure 6.1, the confidence interval is narrower when p is closer to 0 or 1 than when p is closer to 0.5.

One last question is that when this confidence interval is applied in practice, how do we know the variance σ^2? A standard practice is to use the sample variance as an estimate of the variance. That is,

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - X^{(n)})^2. \tag{6.74}$$

It can be shown that $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$, so $\hat{\sigma}^2 \xrightarrow{d} \sigma^2$ as $n \to \infty$. It is not a perfect estimate, but it is very convenient to use in practice.

Next time, when you want to make sure that your good experimental results are not just due to sampling noise, you can use the confidence interval to quantify the uncertainty.

6.2 Variance is Back: Bernstein's Inequality

The central limit theorem and the confidence intervals above both point to the same lesson that variance controls the natural fluctuation scale of a sample average. However, the central limit theorem only gives an asymptotic approximation, and one may wonder whether there is a way to get a more precise upper bound for finite samples.

Hoeffding's inequality is powerful for bounded random variables, but it does not use variance. Once the support is fixed to an interval $[\ell, u]$, Hoeffding treats all distributions in essentially the same way, which can be quite wasteful.

6.2.1 Case Study: Biased Bernoulli Variables

Let us go back to Bernoulli random variables discussed in Section 6.1.4. Suppose $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ are n i.i.d. Bernoulli random variables. Let $X := \frac{1}{n} \sum_{i=1}^{n} X_i$ be the sample average. We want to derive an inequality that improves upon Hoeffding's inequality by incorporating the variance information.

Before proceeding further, let us pause for a moment to answer the question: *Where does Hoeffding's inequality lose the information about the variance?* Hoeffding's inequality is not wrong. It is a powerful tool that applies to every bounded random variable. However, when deriving lower bounds on the rate function, we invoke Hoeffding's lemma to upper bound the MGF by a generic quadratic function. This replacement makes the result distribution-free, but it also removes the fine dependence on p, and hence on the variance.

In fact, if we directly compute the rate function $I(p+\epsilon)$ for Bernoulli random variables, we obtain exactly the KL divergence between the Bernoulli distribution and the Bernoulli distribution with parameter shifted to $p+\epsilon$. This KL divergence can be much larger than the quadratic lower bound from Hoeffding's inequality when p is close to 0 or 1.

Definition 6.23 (KL divergence)**.** For any two probability mass functions p and q on a common domain, the *Kullback–Leibler divergence* (KL divergence) between p and q is defined as

$$\mathrm{KL}(p\|q) := \sum_x p(x) \log \frac{p(x)}{q(x)}, \tag{6.75}$$

where the sum is over all x in the support of p and q. If there exists x such that $p(x) > 0$ and $q(x) = 0$, then we define $\mathrm{KL}(p\|q) = +\infty$.

When p and q are real numbers, $\mathrm{KL}(p\|q)$ denotes the KL divergence between Bernoulli(p) and Bernoulli(q).

Proposition 6.24 (Rate function of Bernoulli random variables)**.** If $X \sim$ Bernoulli(p), then for every $a \in [0,1]$,

$$I_X(a) = \mathrm{KL}(a\|p) = a \log \frac{a}{p} + (1-a) \log \frac{1-a}{1-p}. \tag{6.76}$$

We leave the proof as an exercise, as it follows from basic calculus.

Now taking $a = p+\epsilon$ and $a = p-\epsilon$. we have the following Chernoff bounds.

Theorem 6.25 (Chernoff bounds for Bernoulli random variables)**.** Let $X := \frac{1}{n}\sum_{i=1}^n X_i$ be the sample average of n i.i.d. Bernoulli random variables $X_1, \ldots, X_n \sim$ Bernoulli(p) for some $p \in [0,1]$. Then for all $\epsilon \geq 0$,

$$\Pr[X \geq p+\epsilon] \leq e^{-n\,\mathrm{KL}(p+\epsilon\|p)} \quad \text{whenever} \quad p+\epsilon \leq 1, \tag{6.77}$$

$$\Pr[X \leq p-\epsilon] \leq e^{-n\,\mathrm{KL}(p-\epsilon\|p)} \quad \text{whenever} \quad p-\epsilon \geq 0. \tag{6.78}$$

To see what these bounds tell us, we can lower bound the KL divergence by more interpretable quantities. This leads to the following bounds, usually referred to as the *multiplicative Chernoff bounds*, which we state without proof.

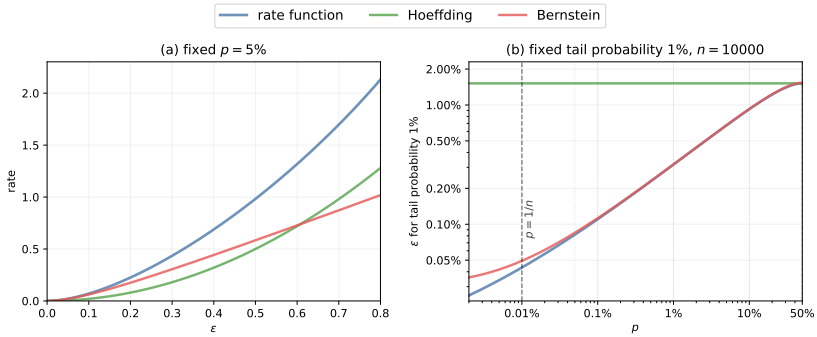


Figure 6.2: Bernstein's bound is able to track the exact rate function much more closely than Hoeffding's bound when p is small. (a) we fix $p = 5\%$ and plot the exact rate function $\mathrm{KL}(p + \epsilon \| p)$ together with the Hoeffding and Bernstein lower bounds as functions of ϵ; (b) we fix $n = 10{,}000$ and ask which ϵ makes the upper-tail probability become 1% according to each exponent.

Theorem 6.26 (Multiplicative Chernoff bounds for Bernoulli random variables)**.** Let $X := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average of n i.i.d. Bernoulli random variables $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ for some $p \in [0, 1]$. Then

$$\Pr[X \geq (1+\delta)p] \leq e^{-\delta^2 pn/(2+\delta)} \qquad \text{for all } \delta \geq 0, \tag{6.79}$$

$$\Pr[X \leq (1-\delta)p] \leq e^{-\delta^2 pn/2} \qquad \text{for all } \delta \in [0, 1], \tag{6.80}$$

$$\Pr\Big[|X - p| \geq \delta p\Big] \leq 2e^{-\delta^2 pn/3} \qquad \text{for all } \delta \in [0, 1]. \tag{6.81}$$

See Figure 6.2 for a plot of the upper-tail bounds. When p is close to 0, the rate function $I_X(p + \epsilon)$ grows much faster than the quadratic lower bound from Hoeffding's lemma as ϵ increases. If we fix the target tail probability to be at most 1% and solve for the maximum possible deviation ϵ, we can see that the exact rate function only allows for very small deviations for small p, while Hoeffding's bound remains unchanged and fails to capture this dependence on p. This illustrates that the exact rate function yields much sharper concentration when p is small, whereas the quadratic approximation in Hoeffding's lemma loses this information.

6.2.2 Bernstein's inequality

Now we introduce *Bernstein's inequality*, a general concentration inequality due to Bernstein. Unlike Hoeffding's inequality, Bernstein's inequality incorporates variance information and can therefore yield much sharper concentration when the variance is small. As shown in Figure 6.2, when applied to Bernoulli random variables, Bernstein's inequality tracks the exact rate function much more closely than Hoeffding's bound when p is small.

The key is to take variance into account in the MGF bound.

Lemma 6.27. Let X be a zero-mean random variable taking values in $[-u, u]$ for some $u \geq 0$. Then for all $t \in \mathbb{R}$ with $|t| < 3/u$,

$$\mathbb{E}[e^{tX}] \leq e^{g(t)\mathbb{E}[X^2]}, \quad \text{where} \quad g(t) := \frac{t^2/2}{1 - \frac{1}{3}u|t|}. \tag{6.82}$$

Compared with Hoeffding's lemma, this bound incorporates not only the worst-case constraint $X \in [-u, u]$ but also the variance $\mathrm{Var}(X)$. Note that the variance appears as $\mathbb{E}[X^2]$ in the bound because $\mathbb{E}[X] = 0$. We now prove the lemma.

Proof. First, we note that for all $z \in \mathbb{R}$ with $|z| < 3$, we have

$$e^z = 1 + z + \sum_{k \geq 2} \frac{z^k}{k!} \leq 1 + z + \sum_{k \geq 2} \frac{z^2 \cdot z^{k-2}}{2 \cdot 3^{k-2}} \tag{6.83}$$

$$= 1 + z + \frac{z^2}{2} \sum_{k \geq 2} \left(\frac{z}{3}\right)^{k-2} \tag{6.84}$$

$$= 1 + z + \frac{z^2/2}{1 - z/3}, \tag{6.85}$$

where the last equality follows from the geometric series sum formula.

Replacing z with tX, we obtain

$$e^{tX} \leq 1 + tX + \frac{(tX)^2/2}{1 - tX/3}. \tag{6.86}$$

Since $|X| \leq u$, we have $tX \leq |t|u$ and hence

$$e^{tX} \leq 1 + tX + \frac{(tX)^2/2}{1 - \frac{1}{3}u|t|}. \tag{6.87}$$

Taking expectation on both sides, we obtain

$$\mathbb{E}[e^{tX}] \leq 1 + \frac{t^2/2}{1 - \frac{1}{3}u|t|}\mathbb{E}[X^2] = 1 + g(t)\mathbb{E}[X^2], \tag{6.88}$$

where the tX term disappears because $\mathbb{E}[X] = 0$. Finally, we can finish the proof by noting that $1 + g(t)\mathbb{E}[X^2] \leq e^{g(t)\mathbb{E}[X^2]}$. □

Lemma 6.27 enables us to derive the following lower bound on the rate function, whose proof is left as an exercise.

Lemma 6.28. Let X be a zero-mean random variable taking values in $[-u, u]$ for some $u \geq 0$. Then for all $\epsilon \in \mathbb{R}$,

$$I_X(\epsilon) \geq \frac{\epsilon^2/2}{\mathbb{E}[X^2] + \frac{1}{3}u|\epsilon|}. \tag{6.89}$$

As in Hoeffding's inequality, we now feed this lower bound into the generic Chernoff theorem for sample averages. This yields *Bernstein's inequality*.

Theorem 6.29 (Bernstein's inequality)**.** Let $X := \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample average of n i.i.d. random variables with mean μ and variance σ^2. Assume that all these random variables take values in a bounded interval $[\mu - u, \mu + u]$ for some $u \geq 0$. Then for every $\epsilon \geq 0$,

$$\Pr[X \geq \mu + \epsilon] \leq \exp\left(-\frac{n\epsilon^2/2}{\sigma^2 + \frac{1}{3}u\epsilon}\right), \tag{6.90}$$

$$\Pr[X \leq \mu - \epsilon] \leq \exp\left(-\frac{n\epsilon^2/2}{\sigma^2 + \frac{1}{3}u\epsilon}\right). \tag{6.91}$$

Proof. Set $Y_i := X_i - \mu$. Then $\mathbb{E}[Y_i] = 0$, $\mathbb{E}[Y_i^2] = \sigma^2$, and $Y_i \in [-u, u]$. So by Lemma 6.28, $I_{Y_i}(\epsilon) \geq \frac{\epsilon^2/2}{\sigma^2 + \frac{1}{3}u\epsilon}$. Finally, apply Theorem 3.34 to the sample average of $Y_1, \ldots, Y_n$. □

Compared with Hoeffding's inequality, Bernstein's inequality has a more informative exponent because it keeps track of the variance. For variables in $[\mu - u, \mu + u]$, Hoeffding's inequality lower bounds the rate function of X by

$$I_X(\mu + \epsilon) \geq \frac{2n\epsilon^2}{((\mu + u) - (\mu - u))^2} = \frac{n\epsilon^2}{2u^2}. \tag{6.92}$$

Bernstein replaces this by

$$I_X(\mu + \epsilon) \geq \frac{n\epsilon^2/2}{\sigma^2 + \frac{1}{3}u\epsilon}. \tag{6.93}$$

This changes the qualitative picture in an important way. When ϵ is negligible compared with σ^2/u, say $\epsilon = \mathcal{O}(\frac{1}{\sqrt{n}})$, the denominator is dominated by σ^2, so the Bernstein lower bound behaves like $\approx \frac{n\epsilon^2}{2\sigma^2}$. This is better than Hoeffding because the variance can be much smaller than u^2.

When ϵ is large compared with σ^2/u, the term $u\epsilon$ dominates, and the Bernstein lower bound behaves like $\approx \frac{3}{2}n\epsilon/u$. So in the large deviation regime, the Bernstein lower bound is only linear in ϵ, whereas Hoeffding remains quadratic in ϵ. Thus, Hoeffding gives the stronger estimate.

Therefore, the two inequalities are useful in complementary ranges. Bernstein's inequality is the right tool for small deviations, especially when the variance σ^2 is tiny compared with u^2. Hoeffding's inequality is often preferable for very large deviations, where its quadratic exponent wins.

Example 6.30 (Bernstein's inequality for Bernoulli random variables)**.** Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$ be independent, and let $X := \frac{1}{n}\sum_{i=1}^n X_i$. Then $\mu = p$, $\sigma^2 = p(1-p)$, and $X_i \in [p-u, p+u]$ with $u = 1$. Bernstein's inequality therefore gives

$$\Pr\Big[|X - p| \geq \epsilon\Big] \leq 2\exp\left(-\frac{n\epsilon^2/2}{p(1-p) + \epsilon/3}\right) \tag{6.94}$$

$$\leq 2\exp\left(-\frac{n\epsilon^2/2}{p + \epsilon/3}\right). \tag{6.95}$$

If p is very small and $\epsilon \leq \frac{3}{2}p$, then the denominator is at most $\frac{3}{2}p$, so

$$\Pr\Big[|X - p| \geq \epsilon\Big] \leq 2e^{-n\epsilon^2/(3p)}. \tag{6.96}$$

Setting $\epsilon = \delta p$ for any $\delta \in [0, 1]$ recovers Equation (6.81):

$$\Pr\Big[|X - p| \geq \delta p\Big] \leq 2e^{-\delta^2 pn/3}. \tag{6.97}$$

Finally, as $n \to \infty$, *is Bernstein's inequality tight for small deviation ϵ?* To answer this, we examine the local behavior of the rate function. A careful calculation of the rate function with implicit differentiation can show that $I''(\mu) = \frac{n}{\sigma^2}$. Then as discussed in Section 3.3.4, we can approximate the rate function near μ by its second-order Taylor expansion:

$$I(\mu + \epsilon) \approx \frac{n\epsilon^2}{2\sigma^2}. \tag{6.98}$$

This matches exactly the Bernstein lower bound for small deviations. Therefore, Bernstein's inequality is as tight as Chernoff bounds for small deviations.

This is also consistent with the Gaussian approximation given by the central limit theorem. As we have seen in Theorem 5.28, the tail probability decays roughly like $e^{-n\epsilon^2/(2\sigma^2)}$. This means that Bernstein's inequality has the correct exponent in this small deviation regime.

Exercises

6.1. Let $d \geq 2$ and fix a unit vector $\boldsymbol{u} \in \mathbb{R}^d$. Let $\mathcal{U}_{\mathbb{S}}$ be the uniform distribution on the unit sphere $\mathbb{S}^{d-1}$. Prove that if $\boldsymbol{x} \sim \mathcal{U}_{\mathbb{S}}$, then as $d \to \infty$,

$$\sqrt{d}\,\langle \boldsymbol{u}, \boldsymbol{x} \rangle \xrightarrow{d} \mathcal{N}(0,1).$$

6.2. Let $\boldsymbol{A} \in \mathbb{R}^{n \times d}$ be a Gaussian random matrix, where every entry of $\boldsymbol{A}$ is sampled independently from $\mathcal{N}(0, 1/n)$.

(a) Fix a vector $\boldsymbol{x} \in \mathbb{R}^d$. Prove that as $n \to \infty$,

$$\boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x} \xrightarrow{d} \boldsymbol{x}.$$

Hint: You may use the following fact without proof: if every coordinate of a random vector converges in distribution to a constant, then the random vector converges in distribution to a constant vector.

(b) Fix another vector $\boldsymbol{y} \in \mathbb{R}^d$. Which distribution does the following random variable converge in distribution to as $n \to \infty$?

$$\sqrt{n}\,\left(\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{A}\boldsymbol{y} \rangle - \langle \boldsymbol{x}, \boldsymbol{y} \rangle\right)$$

6.3. Let $X_1, \ldots, X_n$ be n i.i.d. random samples from $\mathcal{N}(0, \sigma^2)$, and let $X^{(n)}$ be the maximum of $X_1, \ldots, X_n$. Find two sequences $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ such that

$$\frac{X^{(n)} - a_n}{b_n} \xrightarrow{d} \text{Gumbel}(0,1).$$

Here $\text{Gumbel}(0,1)$ is the standard Gumbel distribution, whose PDF and CDF are given by

$$f_G(x) = e^{-(x + e^{-x})} \qquad \text{and} \qquad F_G(x) = e^{-e^{-x}}.$$

Hint: Example 2.89 may be helpful.

Note: You may meet this Gumbel distribution once more in the *Gumbel-softmax trick*, a useful technique in machine learning for backpropagating through discrete sampling.

Chapter 7

Optimization

Many machine learning problems are formulated as optimization problems. For example, in supervised learning, we observe many input-label pairs $(\boldsymbol{x}_i, y_i)$ and want to learn a model that can map inputs to labels. This happens in a variety of settings, such as classifying images into dogs or cats, or predicting future temperature from historical weather data.

In general, we write the training dataset as $\mathcal{S} := \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$, where $\boldsymbol{x}_i \in \mathbb{R}^{d_0}$ is an input and $y_i \in \mathbb{R}$ is a label. A model for this task is often defined as a parameterized function $f_{\boldsymbol{\theta}} : \mathbb{R}^{d_0} \to \mathbb{R}$, where $\boldsymbol{\theta} \in \mathbb{R}^d$ is a vector of parameters. One simple example is the linear model

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \langle \boldsymbol{x}, \boldsymbol{\theta} \rangle. \tag{7.1}$$

In this linear example, the parameter vector has the same dimension as the input. A neural network can be viewed as a more complicated parameterized function $f_{\boldsymbol{\theta}}$, where $\boldsymbol{\theta}$ can have far more dimensions than the input. In both cases, we want to choose parameters so that the model makes accurate predictions on the training dataset $\mathcal{S}$. That is,

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) \approx y_i \qquad \text{for all } i \in [n]. \tag{7.2}$$

A standard choice for regression problems is to formulate the problem as finding parameters $\boldsymbol{\theta}$ that minimize the mean squared error between the model's predictions and the true labels:

$$\mathcal{L}(\boldsymbol{\theta}) := \frac{1}{n} \sum_{i=1}^n \ell_i(\boldsymbol{\theta}), \qquad \ell_i(\boldsymbol{\theta}) := \frac{1}{2} \big(f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) - y_i\big)^2. \tag{7.3}$$

If the model makes accurate predictions on the training dataset, then there is some *hope* that it will make accurate predictions on new inputs. You may want to take a course on machine learning theory to understand where this hope comes from, but in this chapter we will only focus on the optimization aspect of machine learning. In other words, how should we solve the following optimization problem?

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathcal{L}(\boldsymbol{\theta}). \tag{7.4}$$

Behind the recent success of deep-learning-based AI is a surprisingly simple family of algorithms: *gradient-based optimization methods*. These methods do not try to solve the optimization problem exactly. Instead, they generate a sequence of parameters and try to improve the loss gradually based on the gradient of the loss, and this simple idea turns out to be remarkably powerful.

This chapter is organized around gradient descent. We first study deterministic gradient descent through simple quadratic objectives, where the dynamics can be computed exactly. This gives intuition for the convergence rates of gradient descent and how to choose the learning rate. We then go beyond the quadratic objectives and discuss how the same ideas can be applied to smooth convex optimization. Finally, we analyze the convergence of stochastic gradient descent, where each step uses a random data point or mini-batch.

7.1 Introduction to Gradient Descent

We now study how to minimize a fixed differentiable loss function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ with gradient descent.

7.1.1 Gradient Descent

The gradient of a function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ at a point $\boldsymbol{\theta} \in \mathbb{R}^d$ is a vector in $\mathbb{R}^d$ whose i-th coordinate is the partial derivative of $\mathcal{L}$ with respect to the i-th coordinate of $\boldsymbol{\theta}$.

$$\nabla \mathcal{L}(\boldsymbol{\theta}) = \left(\frac{\partial \mathcal{L}}{\partial \theta_1}, \frac{\partial \mathcal{L}}{\partial \theta_2}, \ldots, \frac{\partial \mathcal{L}}{\partial \theta_d} \right). \tag{7.5}$$

A necessary condition for $\boldsymbol{\theta}$ to be a minimizer of $\mathcal{L}$ is that the gradient is zero:

$$\nabla \mathcal{L}(\boldsymbol{\theta}) = 0. \tag{7.6}$$

When $\mathcal{L}(\boldsymbol{\theta})$ takes a simple closed form, we may set the gradient to zero and solve for the parameter vector. However, in general, solving this gradient equation is as hard as solving the original optimization problem, since $\mathcal{L}(\boldsymbol{\theta})$ may be quite complicated, e.g., the cross-entropy loss of a language model over the entire Internet.

Gradient descent is a general optimization algorithm that iteratively updates the parameter vector $\boldsymbol{\theta}$ using the gradient of the loss. When the gradient is non-zero, taking one tiny step in the negative direction of the gradient will decrease the loss. That is, if $\eta > 0$ is small enough, then by Taylor expansion,

$$\mathcal{L}(\boldsymbol{\theta}_t - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_t)) \approx \mathcal{L}(\boldsymbol{\theta}_t) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}_t), -\eta \nabla \mathcal{L}(\boldsymbol{\theta}_t) \rangle \tag{7.7}$$

$$= \mathcal{L}(\boldsymbol{\theta}_t) - \eta \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2. \tag{7.8}$$

The following algorithm is the most basic form of gradient descent.

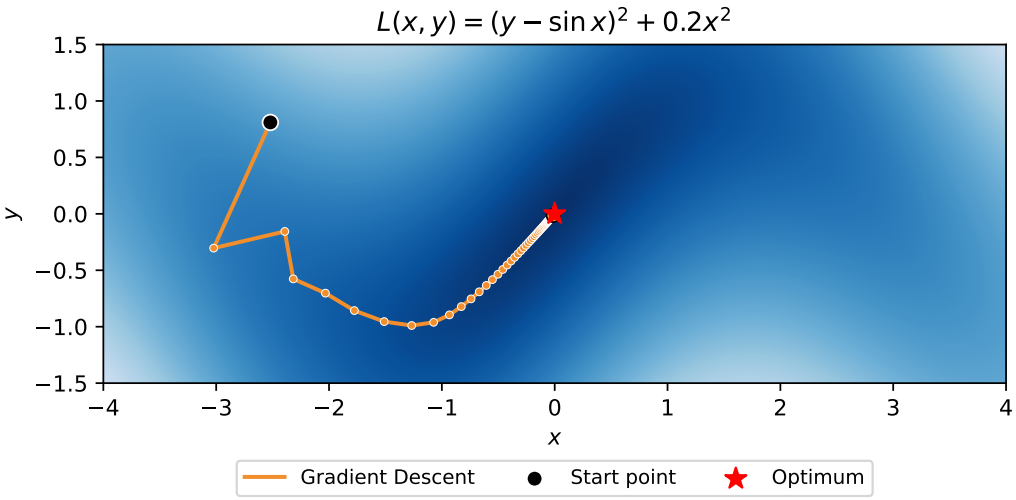


Figure 7.1: Gradient descent moves towards the minimizer through a non-convex landscape with a curved valley. The loss function is $\mathcal{L}(x, y) = (y - \sin x)^2 + 0.2x^2$.

Definition 7.1 (Gradient descent)**.** Given a differentiable loss function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$, an initial parameter vector $\boldsymbol{\theta}_0 \in \mathbb{R}^d$, and a *learning rate* $\eta > 0$ (also called a *step size*), *gradient descent* (GD) iteratively updates the parameter vector according to the following rule:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_t). \tag{7.9}$$

See Figure 7.1 for an example. Gradient descent can move towards the minimizer through iterative gradient updates.

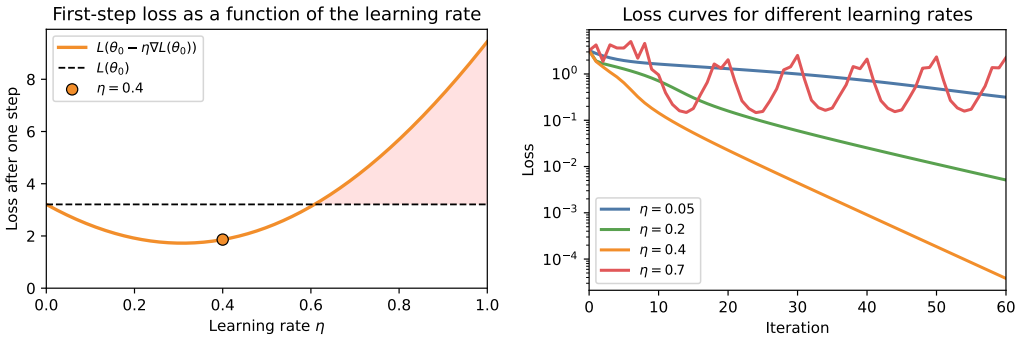
The learning rate η is a crucial hyperparameter in gradient descent. If you have trained some neural networks already, you may already know that picking the right learning rate is a bit tricky. If you have not yet trained any neural networks, you may have to try training one yourself to truly understand why we are going to spend so much time discussing how to tune the learning rate.

Although gradient descent always moves along the negative gradient direction, it is not guaranteed that the loss at the next iterate is always less than the loss at the current iterate:

$$\mathcal{L}(\boldsymbol{\theta}_t - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_t)) < \mathcal{L}(\boldsymbol{\theta}_t)\,? \tag{7.10}$$

Figure 7.2 illustrates this point by fixing the current iterate and varying only the learning rate. As the learning rate increases, the loss first decreases, but after a certain point it starts to increase again. Similarly, when we run multiple steps, a too large learning rate may prevent gradient descent from minimizing the loss.

This leads to a practical question: *how should we choose the learning rate η?* In this section, our goal is to build up some intuition about the dynamics of gradient descent. This will hopefully give us some insight into how to choose the learning rate and how to improve the design of gradient descent. Perhaps you



(a) Loss after one step. (b) Loss curves over time.

Figure 7.2: The effect of the learning rate, starting from the same point as in Figure 7.1. A moderate learning rate decreases the loss quickly, while a large learning rate can increase the loss or make the iterates unstable.

will be disappointed, but we are not going to analyze neural networks. Instead, just as physics often starts from point masses, we will start with the simplest possible system: a quadratic objective.

7.1.2 Linear Regression and Quadratic Objectives

Let us now instantiate the optimization problem from the opening in the simplest supervised learning model. We are given input-output pairs

$$(\boldsymbol{x}_1, y_1), \dots, (\boldsymbol{x}_n, y_n), \qquad \boldsymbol{x}_i \in \mathbb{R}^d, \; y_i \in \mathbb{R}, \tag{7.11}$$

and we want to fit a model that maps an input $\boldsymbol{x}$ to a prediction for y. In linear regression, the model is

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \langle \boldsymbol{x}, \boldsymbol{\theta} \rangle, \tag{7.12}$$

where the parameter vector is $\boldsymbol{\theta} \in \mathbb{R}^d$. Fitting the model means choosing $\boldsymbol{\theta}$ so that $f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) = \langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle$ is close to y_i on the training data.

A natural way to measure the prediction error of the model on one sample $(\boldsymbol{x}_i, y_i)$ is the squared loss:

$$\ell_i(\boldsymbol{\theta}) := \frac{1}{2}\big(f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) - y_i\big)^2 = \frac{1}{2}\big(\langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle - y_i\big)^2. \tag{7.13}$$

Here we add a factor $1/2$ for algebraic convenience, which cancels the factor 2 when we take derivatives. The linear regression problem asks for minimizing the mean squared error (MSE) over the training data:

$$\mathcal{L}(\boldsymbol{\theta}) := \frac{1}{n}\sum_{i=1}^{n} \ell_i(\boldsymbol{\theta}) = \frac{1}{2n}\sum_{i=1}^{n}\big(\langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle - y_i\big)^2. \tag{7.14}$$

Note that this loss function can be written more compactly using linear algebra. Let $\boldsymbol{X} \in \mathbb{R}^{n \times d}$ be the data matrix whose i-th row is $\boldsymbol{x}_i^\top$, and let $\boldsymbol{y} = (y_1, \dots, y_n)^\top$ be the vector that contains the labels. Then $\boldsymbol{X}\boldsymbol{\theta}$ is the vector that contains the predictions of the model on the training data, and $\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}$ is

the vector that contains the prediction errors. Therefore, the loss function can be written as

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2n}\|\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}\|_2^2. \tag{7.15}$$

We can further expand the square to obtain a quadratic objective:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}) &= \frac{1}{2n}(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y})^\top(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}) &(7.16)\\ &= \frac{1}{2n}(\boldsymbol{\theta}^\top\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\theta} - 2\boldsymbol{\theta}^\top\boldsymbol{X}^\top\boldsymbol{y} + \boldsymbol{y}^\top\boldsymbol{y}) &(7.17)\\ &= \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{H}\boldsymbol{\theta} - \boldsymbol{b}^\top\boldsymbol{\theta} + C_0, &(7.18)\end{aligned}$$

where

$$\boldsymbol{H} := \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{X}, \qquad \boldsymbol{b} := \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{y}, \qquad C_0 := \frac{1}{2n}\|\boldsymbol{y}\|_2^2. \tag{7.19}$$

Remark 7.2. The mean squared error penalizes large mistakes more heavily than small mistakes. It also arises naturally from the Gaussian noise model

$$y_i = \langle \boldsymbol{x}_i, \boldsymbol{\theta}\rangle + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \tag{7.20}$$

where $\boldsymbol{\theta} \in \mathbb{R}^d$ is unknown. Under this model, the likelihood of the data is

$$\prod_{i=1}^n \varphi_\sigma(r_i(\boldsymbol{\theta})), \qquad r_i(\boldsymbol{\theta}) := y_i - \langle \boldsymbol{x}_i, \boldsymbol{\theta}\rangle, \tag{7.21}$$

where φ_σ is the PDF of $\mathcal{N}(0, \sigma^2)$. As a function of $\boldsymbol{\theta}$, this quantity is called the *likelihood* of the data under the model. Maximizing this likelihood is equivalent to minimizing the mean squared error. This is because the log-likelihood is

$$\begin{aligned}\sum_{i=1}^n \log \varphi_\sigma(r_i(\boldsymbol{\theta})) &= -\frac{1}{2\sigma^2}\sum_{i=1}^n r_i(\boldsymbol{\theta})^2 - \frac{n}{2}\log(2\pi\sigma^2) &(7.22)\\ &= -\frac{n}{\sigma^2}\mathcal{L}(\boldsymbol{\theta}) - \frac{n}{2}\log(2\pi\sigma^2). &(7.23)\end{aligned}$$

Therefore, minimizing $\mathcal{L}(\boldsymbol{\theta})$ in linear regression is equivalent to performing maximum likelihood estimation (MLE) under the Gaussian noise model.

▶ Quadratic Objectives

Now we turn to study the optimization of a quadratic objective in general. Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be a quadratic function:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{H}\boldsymbol{\theta} - \boldsymbol{b}^\top\boldsymbol{\theta} + C_0, \tag{7.24}$$

where $\boldsymbol{H} \in \mathbb{R}^{d\times d}$ is symmetric, $\boldsymbol{b} \in \mathbb{R}^d$ is a vector and $C_0 \in \mathbb{R}$ is a constant.

We need to further constrain the matrix $\boldsymbol{H}$ to be positive definite, since otherwise the objective value can be made arbitrarily small.

For readers who are not familiar with linear algebra, we provide the following definitions and propositions.

Definition 7.3 (Positive definite and semidefinite matrices)**.** A symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{d\times d}$ is *positive definite*, denoted by $\boldsymbol{A} \succ \boldsymbol{0}$, if for all $\boldsymbol{v} \in \mathbb{R}^d \setminus \{\boldsymbol{0}\}$,

$$\boldsymbol{v}^\top \boldsymbol{A}\boldsymbol{v} > 0. \tag{7.25}$$

A symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{d\times d}$ is *positive semidefinite*, denoted by $\boldsymbol{A} \succeq \boldsymbol{0}$, if for all $\boldsymbol{v} \in \mathbb{R}^d \setminus \{\boldsymbol{0}\}$,

$$\boldsymbol{v}^\top \boldsymbol{A}\boldsymbol{v} \geq 0. \tag{7.26}$$

If $\boldsymbol{A}$ and $\boldsymbol{B}$ are two symmetric matrices and $\boldsymbol{A} - \boldsymbol{B}$ is positive definite, then we write $\boldsymbol{A} \succ \boldsymbol{B}$. Similarly, we write $\boldsymbol{A} \succeq \boldsymbol{B}$ if $\boldsymbol{A} - \boldsymbol{B}$ is positive semidefinite.

The following propositions are useful for analyzing symmetric matrices.

Proposition 7.4 (Properties of symmetric matrices)**.** Let $\boldsymbol{A} \in \mathbb{R}^{d\times d}$ be a symmetric matrix, i.e., $\boldsymbol{A} = \boldsymbol{A}^\top$. Then it satisfies the following properties:

1. All eigenvalues of $\boldsymbol{A}$ are real.
2. $\boldsymbol{A}$ has an orthonormal eigenbasis. That is, there exists an orthonormal basis of $\mathbb{R}^d$, $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_d\}$, such that $\boldsymbol{v}_i$ is an eigenvector of $\boldsymbol{A}$ with eigenvalue λ_i, i.e., $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$.
3. $\boldsymbol{A}$ is orthogonally diagonalizable. That is, there exist an orthogonal matrix $\boldsymbol{Q} \in \mathbb{R}^{d\times d}$ ($\boldsymbol{Q}^\top \boldsymbol{Q} = \boldsymbol{Q}\boldsymbol{Q}^\top = \boldsymbol{I}$) and a diagonal matrix $\boldsymbol{\Lambda}$ such that $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$.
4. $\boldsymbol{A} = \sum_{i=1}^d \lambda_i \boldsymbol{v}_i \boldsymbol{v}_i^\top$, where $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_d\}$ is an orthonormal eigenbasis of $\boldsymbol{A}$ and λ_i is the eigenvalue associated with $\boldsymbol{v}_i$.
5. $\boldsymbol{A} \succeq \boldsymbol{0}$ if and only if all its eigenvalues are non-negative.
6. $\boldsymbol{A} \succ \boldsymbol{0}$ if and only if all its eigenvalues are positive.

If $\boldsymbol{H}$ is not positive semidefinite, then there exists a negative eigenvalue $\lambda_i < 0$. Let $\boldsymbol{v}_i$ be an eigenvector with $\|\boldsymbol{v}_i\|_2 = 1$, associated with λ_i. Then for any $\alpha > 0$,

$$\mathcal{L}(\alpha \boldsymbol{v}_i) = \frac{1}{2}(\alpha \boldsymbol{v}_i)^\top \boldsymbol{H}(\alpha \boldsymbol{v}_i) - \boldsymbol{b}^\top(\alpha \boldsymbol{v}_i) + C_0 \tag{7.27}$$

$$= \frac{1}{2}\lambda_i \alpha^2 - \alpha \cdot \boldsymbol{b}^\top \boldsymbol{v}_i + C_0. \tag{7.28}$$

Since $\lambda_i < 0$, we can make $\mathcal{L}(\alpha \boldsymbol{v}_i)$ arbitrarily small by choosing α sufficiently large. Therefore, the objective is not bounded below and the minimum value is $-\infty$. One can show that this is not the case for linear regression, since $\boldsymbol{H} = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X}$ is positive semidefinite. The reason we further constrain $\boldsymbol{H}$ to be positive definite is just for convenience.

When $\boldsymbol{H}$ is positive definite, we can solve the equation $\nabla \mathcal{L}(\boldsymbol{\theta}) = \boldsymbol{0}$ exactly to

obtain the closed-form solution. Note that the gradient of $\mathcal{L}(\boldsymbol{\theta})$ is

$$\nabla\mathcal{L}(\boldsymbol{\theta}) = \boldsymbol{H}\boldsymbol{\theta} - \boldsymbol{b}. \tag{7.29}$$

Setting the gradient to zero gives

$$\boldsymbol{H}\boldsymbol{\theta}^* - \boldsymbol{b} = 0. \tag{7.30}$$

Therefore,

$$\boldsymbol{\theta}^* = \boldsymbol{H}^{-1}\boldsymbol{b} \tag{7.31}$$

is the minimizer of $\mathcal{L}(\boldsymbol{\theta})$. While this formula looks simple, it takes $\mathcal{O}(d^3)$ time to compute if we apply Gaussian elimination naïvely. Gradient descent provides an alternative way to compute the minimizer by iteratively updating the parameter vector, which may run faster if the convergence to the minimizer happens quickly.

▶ Exact Dynamics of Gradient Descent

Around the minimizer $\boldsymbol{\theta}^* = \boldsymbol{H}^{-1}\boldsymbol{b}$, we can write the quadratic objective as

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{H}\boldsymbol{\theta} - \boldsymbol{\theta}^\top\boldsymbol{H}\boldsymbol{H}^{-1}\boldsymbol{b} + C_0 \tag{7.32}$$

$$= \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{H}\boldsymbol{\theta} - \boldsymbol{\theta}^\top\boldsymbol{H}\boldsymbol{\theta}^* + C_0 \tag{7.33}$$

$$= \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}^*)^\top\boldsymbol{H}(\boldsymbol{\theta} - \boldsymbol{\theta}^*) + \underbrace{C_0 - \frac{1}{2}\boldsymbol{\theta}^{*\top}\boldsymbol{H}\boldsymbol{\theta}^*}_{=:\,\mathcal{L}_{\min}}. \tag{7.34}$$

Taking the derivative with respect to $\boldsymbol{\theta}$ gives

$$\nabla\mathcal{L}(\boldsymbol{\theta}) = \boldsymbol{H}(\boldsymbol{\theta} - \boldsymbol{\theta}^*). \tag{7.35}$$

So gradient descent with learning rate η updates the parameter vector by

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\boldsymbol{H}(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*). \tag{7.36}$$

Subtracting the minimizer $\boldsymbol{\theta}^*$ on both sides gives

$$\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^* = (\boldsymbol{I} - \eta\boldsymbol{H})(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*). \tag{7.37}$$

Therefore, on a quadratic objective, gradient descent acts as a linear dynamical system that repeatedly multiplies $\boldsymbol{\theta}_t - \boldsymbol{\theta}^*$ by the matrix $(\boldsymbol{I} - \eta\boldsymbol{H})$.

To analyze this dynamical system, we can write $\boldsymbol{\theta}_t - \boldsymbol{\theta}^*$ as a vector in the eigenbasis of $\boldsymbol{H}$. Let $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_d\}$ be an orthonormal eigenbasis of $\boldsymbol{H}$, where the eigenvalue associated with $\boldsymbol{v}_i$ is λ_i. Each eigenvector $\boldsymbol{v}_i$ corresponds to a direction in which the objective is curved. Along the direction $\boldsymbol{v}_i$, the objective behaves like a one-dimensional quadratic function

$$\mathcal{L}(\boldsymbol{\theta}^* + \alpha\boldsymbol{v}_i) = \frac{1}{2}\lambda_i\alpha^2 + \mathcal{L}_{\min}, \tag{7.38}$$

The larger the eigenvalue is, the more curved the objective is in that direction. We call a direction with a large eigenvalue a *high-curvature direction*, and a direction with a small eigenvalue a *low-curvature direction*.

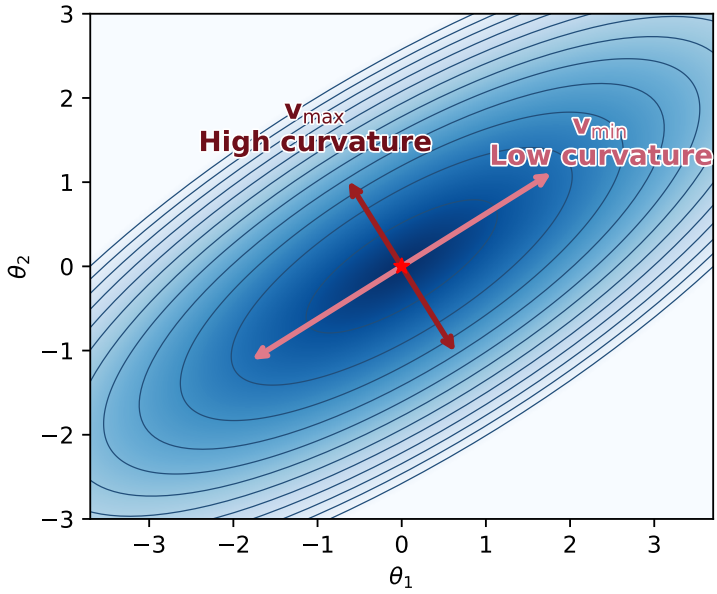


Figure 7.3: A two-dimensional quadratic objective with one high-curvature direction and one low-curvature direction.

Now we write $\boldsymbol{\theta}_t - \boldsymbol{\theta}^*$ in the eigenbasis of $\boldsymbol{H}$: let $\alpha_{t,1}, \ldots, \alpha_{t,d}$ be the coefficients of $\boldsymbol{\theta}_t - \boldsymbol{\theta}^*$ in the eigenbasis, i.e.,

$$\boldsymbol{\theta}_t - \boldsymbol{\theta}^* = \sum_{i=1}^{d} \alpha_{t,i} \boldsymbol{v}_i. \tag{7.39}$$

Remark 7.5. Here $\alpha_{t,i}$ can explicitly be written as $\alpha_{t,i} = \langle \boldsymbol{\theta}_t - \boldsymbol{\theta}^*, \boldsymbol{v}_i \rangle$, which is due to the orthogonality of the eigenbasis.

After multiplying both sides by $\boldsymbol{I} - \eta \boldsymbol{H}$, we get

$$\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^* = \sum_{i=1}^{d} \alpha_{t,i}(1 - \eta\lambda_i) \boldsymbol{v}_i. \tag{7.40}$$

Thus we can see that each coordinate in the eigenbasis evolves independently.

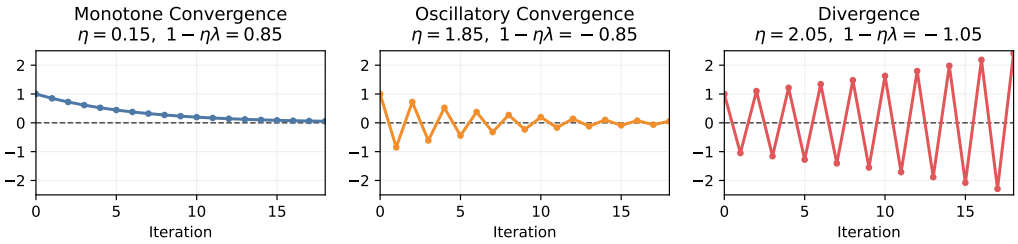
$$\alpha_{t+1,i} = \alpha_{t,i}(1 - \eta\lambda_i). \tag{7.41}$$

Unrolling this recursion gives

$$\alpha_{t,i} = \alpha_{0,i}(1 - \eta\lambda_i)^t. \tag{7.42}$$

Therefore, the parameter vector and objective value at step t are

$$\begin{cases} \boldsymbol{\theta}_t - \boldsymbol{\theta}^* = \sum_{i=1}^{d} \alpha_{0,i}(1 - \eta\lambda_i)^t \boldsymbol{v}_i, \\ \mathcal{L}(\boldsymbol{\theta}_t) = \frac{1}{2} \sum_{i=1}^{d} \lambda_i \alpha_{0,i}^2 (1 - \eta\lambda_i)^{2t} + \mathcal{L}_{\min}. \end{cases} \tag{7.43}$$



(a) $1 - \eta\lambda \in (0, 1)$. (b) $1 - \eta\lambda \in (-1, 0)$. (c) $1 - \eta\lambda < -1$.

Figure 7.4: Three possible behaviors of one coordinate under the recursion $\alpha_{t+1} = (1 - \eta\lambda)\alpha_t$. **(a)** When $\eta \in (0, \frac{1}{\lambda})$, the coordinate monotonically converges to 0. **(b)** When $\eta \in (\frac{1}{\lambda}, \frac{2}{\lambda})$, the coordinate oscillates around 0. **(c)** When $\eta \in (\frac{2}{\lambda}, +\infty)$, the coordinate diverges.

From the above equations, we can see that every coordinate is multiplied by a multiplier $1 - \eta\lambda_i$ at each step. To make the i-th coordinate converge, we need the multiplier to be less than 1 in magnitude, i.e.,

$$|1 - \eta\lambda_i| < 1, \tag{7.44}$$

which is equivalent to

$$0 < \eta < \frac{2}{\lambda_i}. \tag{7.45}$$

For all coordinates to converge simultaneously, we therefore need

$$0 < \eta < \frac{2}{\lambda_{\max}}, \tag{7.46}$$

where $\lambda_{\max}$ is the largest eigenvalue of $\boldsymbol{H}$.

If η is close to 0, all multipliers are close to 1, so progress is slow. If η is too large, then in high-curvature directions $|1 - \eta\lambda_i| > 1$, and the iterate diverges. If $1 - \eta\lambda_i$ is negative but has magnitude less than 1, that coordinate converges by oscillating across the optimum.

This gives us some intuition about what happens when we increase the learning rate η from 0:

- If η is too small, then the progress is slow along all directions.
- If η is moderate, then the progress is fast along high-curvature directions, but still slow along low-curvature directions.
- If η is too large, then some coordinates may start to oscillate or even diverge.

To choose the best learning rate, we need to balance the progress along low-curvature directions and the risk of divergence along high-curvature directions.

▶ Convergence Rate

According to Equation (7.43), the convergence rate of gradient descent on a quadratic objective is controlled by the learning rate η and the eigenvalues of

$\boldsymbol{H}$. More specifically, we have the following bounds on the convergence rate:

Theorem 7.6. Let $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \boldsymbol{H}\boldsymbol{\theta} - \boldsymbol{b}^\top\boldsymbol{\theta} + C_0$ be a quadratic objective. Assume that $\boldsymbol{H}$ is symmetric and positive definite. Let $\lambda_1, \dots, \lambda_d$ be the eigenvalues of $\boldsymbol{H}$, and $\lambda_{\max}$ be the largest eigenvalue. Define the contraction factor $\rho(\eta)$ as

$$\rho(\eta) := \max_{i\in[d]} |1 - \eta\lambda_i|. \tag{7.47}$$

Then gradient descent with learning rate $\eta > 0$ exhibits the following behaviors as $t \to \infty$:

1. If $\eta > \frac{2}{\lambda_{\max}}$ and $\alpha_{0,i} \neq 0$ for all $i \in [d]$, then
$$\mathcal{L}(\boldsymbol{\theta}_t) \to +\infty. \tag{7.48}$$
2. If $\eta < \frac{2}{\lambda_{\max}}$, then
$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} \le (\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min})\rho(\eta)^{2t}. \tag{7.49}$$
3. If $\eta < \frac{2}{\lambda_{\max}}$ and $\alpha_{0,i} \neq 0$ for all $i \in [d]$, then for some $c > 0$,
$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} \ge c \cdot \rho(\eta)^{2t}. \tag{7.50}$$

Proof. If $\eta > \frac{2}{\lambda_{\max}}$, then $|1 - \eta\lambda_i| > 1$ for some $i \in [d]$, and thus

$$\mathcal{L}(\boldsymbol{\theta}_t) \ge \frac{1}{2}\lambda_i\alpha_{0,i}^2(1-\eta\lambda_i)^{2t} + \mathcal{L}_{\min} \to +\infty. \tag{7.51}$$

If $\eta < \frac{2}{\lambda_{\max}}$, then for all $i \in [d]$, we have $|1 - \eta\lambda_i| \le \rho(\eta) \le 1$. Then

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} = \frac{1}{2}\sum_{i=1}^d \lambda_i\alpha_{0,i}^2(1-\eta\lambda_i)^{2t} \tag{7.52}$$

$$\le \frac{1}{2}\sum_{i=1}^d \lambda_i\alpha_{0,i}^2\rho(\eta)^{2t} = (\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min})\rho(\eta)^{2t}. \tag{7.53}$$

Finally, let i be the index that satisfies $|1 - \eta\lambda_i| = \rho(\eta)$, then

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} \ge \frac{1}{2}\lambda_i\alpha_{0,i}^2(1-\eta\lambda_i)^{2t} = \frac{1}{2}\lambda_i\alpha_{0,i}^2\rho(\eta)^{2t} = c\cdot\rho(\eta)^{2t}, \tag{7.54}$$

where $c = \frac{1}{2}\lambda_i\alpha_{0,i}^2$. □

Some statements in Theorem 7.6 require that $\alpha_{0,i} \neq 0$ for all $i \in [d]$. For a random initialization from a continuous distribution, this condition holds with probability 1, since the probability that $\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*$ is exactly orthogonal to any fixed eigenvector is 0. Therefore, for almost every random initialization, the objective value converges to $\mathcal{L}_{\min}$ at an exponential rate:

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} = \Theta\left(\rho(\eta)^{2t}\right). \tag{7.55}$$

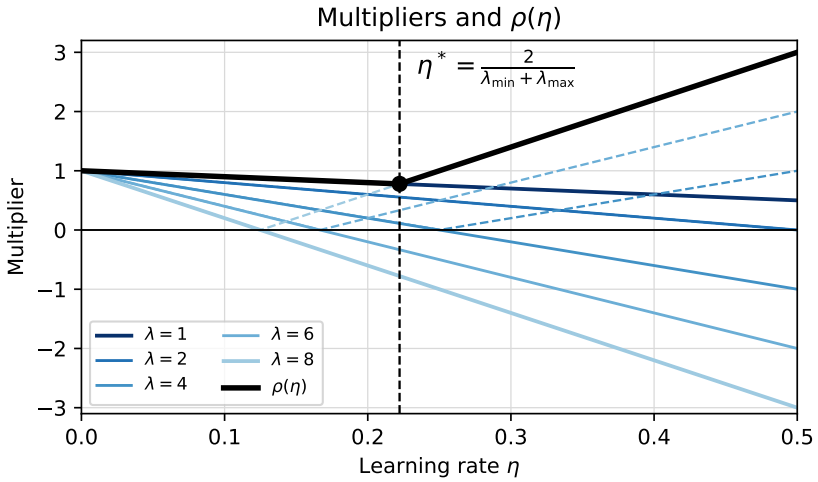


Figure 7.5: The multipliers $1 - \eta\lambda_i$ and $\rho(\eta) = \max_i |1 - \eta\lambda_i|$ as functions of the learning rate η. The intermediate eigenvalues also produce multiplier lines, but the maximum absolute value is controlled by the endpoints $\lambda_{\min}$ and $\lambda_{\max}$.

The next question is how to choose the learning rate η to minimize this convergence rate. This is simply asking for the learning rate that minimizes $\rho(\eta)$. That is,

$$\min_{\eta > 0} \rho(\eta), \qquad \text{where } \rho(\eta) := \max_{i \in [d]} |1 - \eta\lambda_i|. \tag{7.56}$$

Let $\lambda_{\min}$ and $\lambda_{\max}$ be the smallest and largest eigenvalues of $\boldsymbol{H}$, respectively. Then every $1 - \eta\lambda_i$ is bounded by

$$1 - \eta\lambda_{\max} \le 1 - \eta\lambda_i \le 1 - \eta\lambda_{\min}. \tag{7.57}$$

As we increase η from 0, both endpoints of the interval decrease from 1, and the left endpoint decreases faster than the right endpoint. If both endpoints are positive, then η is not optimal, since increasing η can further decrease $1 - \eta\lambda_{\min}$ and thus decrease $\rho(\eta)$. Therefore, the optimal η should be large enough so that the left endpoint has crossed 0. However, when η is too large, $1 - \eta\lambda_{\max}$ becomes too negative, and the absolute value of the left endpoint becomes larger. The optimal learning rate is thus attained when the two endpoints have the same absolute value:

$$1 - \eta\lambda_{\max} = -\Big(1 - \eta\lambda_{\min}\Big). \tag{7.58}$$

In other words, $1 - \eta\lambda_{\min}$ dominates $\rho(\eta)$ when η is small, and we want to make η as large as possible without letting $1 - \eta\lambda_{\max}$ dominate $\rho(\eta)$.

Solving for η gives

$$\eta^* = \frac{2}{\lambda_{\min} + \lambda_{\max}}. \tag{7.59}$$

Substituting η^* into $\rho(\eta)$ gives

$$\rho(\eta^*) = 1 - \frac{2}{\lambda_{\min} + \lambda_{\max}} \cdot \lambda_{\min} = 1 - \frac{2}{1 + \lambda_{\max}/\lambda_{\min}}. \tag{7.60}$$

Therefore, the optimal convergence rate of gradient descent on the quadratic objective $\mathcal{L}(\boldsymbol{\theta})$ is

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} = \Theta\left(\left(1 - \frac{2}{1 + \lambda_{\max}/\lambda_{\min}}\right)^{2t}\right). \tag{7.61}$$

▶ Condition Number

There is an interesting quantity that repeatedly appears in optimization theory: the *condition number*. For a symmetric positive definition matrix $\boldsymbol{H}$, the condition number is defined as follows:

Definition 7.7. Let $\boldsymbol{H} \in \mathbb{R}^{d \times d}$ be a symmetric positive definite matrix. Then the *condition number* of $\boldsymbol{H}$ is defined as

$$\kappa(\boldsymbol{H}) := \frac{\lambda_{\max}(\boldsymbol{H})}{\lambda_{\min}(\boldsymbol{H})}, \tag{7.62}$$

where $\lambda_{\min}(\boldsymbol{H})$ and $\lambda_{\max}(\boldsymbol{H})$ are the smallest and largest eigenvalues of $\boldsymbol{H}$, respectively.

Remark 7.8. In general, the condition number of a matrix $\boldsymbol{A} \in \mathbb{R}^{k \times d}$, which is not necessarily symmetric or even square, is defined as

$$\kappa(\boldsymbol{A}) = \frac{\sigma_{\max}(\boldsymbol{A})}{\sigma_{\min}(\boldsymbol{A})}, \tag{7.63}$$

where $\sigma_{\max}(\boldsymbol{A})$ and $\sigma_{\min}(\boldsymbol{A})$ are the largest and smallest singular values of $\boldsymbol{A}$, respectively. You may meet this definition more often in numerical analysis, but here we only need to consider the condition number of a symmetric positive definite matrix.

Obviously, the condition number is always greater than or equal to 1, since $\lambda_{\max}(\boldsymbol{H}) \geq \lambda_{\min}(\boldsymbol{H})$. When using the matrix $\boldsymbol{H}$ in the quadratic objective $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \boldsymbol{H}\boldsymbol{\theta} - \boldsymbol{b}^\top \boldsymbol{\theta} + C_0$, the condition number $\kappa(\boldsymbol{H})$ quantifies whether the objective is in good condition for optimization. For a matrix $\boldsymbol{H}$ with a small condition number, or a quadratic objective with such matrix $\boldsymbol{H}$, we say that the matrix or the objective is *well-conditioned*. Otherwise, we say that the matrix or the objective is *ill-conditioned*.

With this new notion, we can rewrite the optimal learning rate η^* and the optimal convergence rate of gradient descent on a quadratic objective as follows:

$$\eta^* = \frac{2}{\lambda_{\min} + \lambda_{\max}} = \frac{2}{\lambda_{\max}(1 + \kappa^{-1})}, \tag{7.64}$$

$$\rho(\eta^*) = 1 - \frac{2}{1 + \lambda_{\max}/\lambda_{\min}} = 1 - \frac{2}{1 + \kappa}, \tag{7.65}$$

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} = \Theta\left(\left(1 - \frac{2}{1 + \kappa}\right)^{2t}\right). \tag{7.66}$$

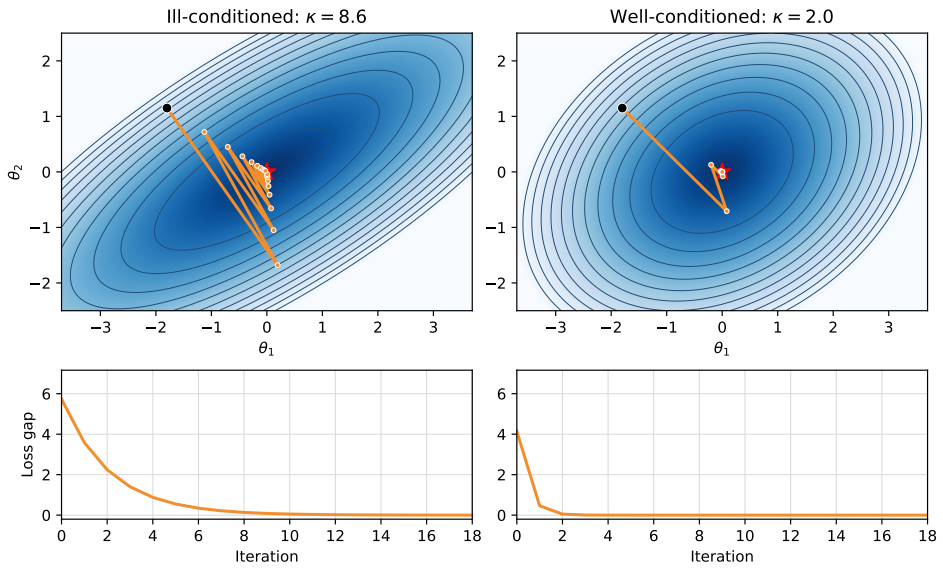


Figure 7.6: Gradient descent trajectories and the corresponding loss curves on two quadratic objectives using the optimal learning rate $\eta^* = 2/(\lambda_{\min} + \lambda_{\max})$.

For the convergence rate in the last line, we may alternatively state it as follows. Let $T(\epsilon)$ be the number of steps t to reach $\mathcal{L}(\boldsymbol{\theta}_t) \leq \mathcal{L}_{\min} + \epsilon$:

$$T(\epsilon) := \min\{t \geq 0 : \mathcal{L}(\boldsymbol{\theta}_t) \leq \mathcal{L}_{\min} + \epsilon\}. \tag{7.67}$$

Then

$$T(\epsilon) \sim \frac{\log \frac{1}{\epsilon}}{-2\log(1 - \frac{2}{1+\kappa})} \sim \frac{1}{2 \cdot \frac{2}{1+\kappa}} \log \frac{1}{\epsilon} \sim \frac{\kappa}{4} \log \frac{1}{\epsilon}, \tag{7.68}$$

where we write $\sim$ for $\epsilon \to 0$ and $\kappa \to +\infty$.

Remark 7.9. Each step of gradient descent costs $\mathcal{O}(d^2)$ time due to matrix-vector multiplication. Therefore, for well-conditioned matrices $\boldsymbol{H}$ with $\kappa \ll d$, gradient descent finds the minimizer faster than directly computing $\boldsymbol{\theta}^* = \boldsymbol{H}^{-1}b$ via Gaussian elimination.

In Figure 7.6, we visualize the gradient descent trajectories on two quadratic objectives, one is ill-conditioned and the other is well-conditioned. The following observations are worth noting:

- A small condition number means that the level sets are close to circles, while a large condition number means that the level sets are more elongated.
- Under the optimal learning rate, the convergence along the high-curvature direction is oscillatory, while the convergence along the low-curvature direction is monotonic.
- The larger the condition number is, the slower the convergence rate is.

7.1.3 Optimization of Smooth Functions

▶ Smoothness and Hessian Matrix

From the analysis of gradient descent on quadratics, a key insight is that the condition number is the key quantity that controls the convergence rate. Recall that the condition number is the ratio of the largest eigenvalue $\lambda_{\max}(\boldsymbol{H})$ to the smallest eigenvalue $\lambda_{\min}(\boldsymbol{H})$ of the matrix $\boldsymbol{H}$ in the quadratic objective. In general, there is also a quantity that generalizes $\lambda_{\max}(\boldsymbol{H})$: the smoothness of the objective function.

Definition 7.10 (Smoothness)**.** A twice differentiable function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ is β-*smooth* if for all $\boldsymbol{\theta} \in \mathbb{R}^d$,

$$\|\nabla^2 \mathcal{L}(\boldsymbol{\theta})\|_2 \leq \beta. \tag{7.69}$$

Remark 7.11. In general, a function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ is β-*smooth* if it is differentiable and $\|\nabla \mathcal{L}(\boldsymbol{\theta}) - \nabla \mathcal{L}(\boldsymbol{\theta}')\|_2 \leq \beta \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_2$ for all $\boldsymbol{\theta}, \boldsymbol{\theta}' \in \mathbb{R}^d$. This definition does not require the function to be twice differentiable, but this detail is a bit technical. Please take a more advanced course on optimization theory if you are interested.

Here we suddenly introduced two new concepts: the Hessian matrix and the operator norm $\|\cdot\|_2$. Before explaining them in more detail, let us first give you an idea why the smoothness constant β generalizes $\lambda_{\max}(\boldsymbol{H})$. In the quadratic case, we can see that $\frac{2}{\lambda_{\max}(\boldsymbol{H})}$ is a critical value for the learning rate η in gradient descent. If η is higher than this value, then the gradient descent will diverge. If η is lower than this value, then the gradient descent will converge. For optimizing smooth functions, we will see that the smoothness β plays a similar role: the loss is guaranteed to decrease if $\eta < \frac{2}{\beta}$.

To understand the above statement, we will first explain what the Hessian matrix and the operator norm are. First, the Hessian matrix $\nabla^2 \mathcal{L}(\boldsymbol{\theta})$ is a square matrix of size $d \times d$, which records the second-order partial derivatives of $\mathcal{L}(\boldsymbol{\theta})$.

Definition 7.12 (Hessian matrix)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be a function that is twice differentiable. Then the *Hessian matrix* of $\mathcal{L}(\boldsymbol{\theta})$ is defined as

$$\nabla^2 \mathcal{L}(\boldsymbol{\theta}) := \begin{bmatrix} \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_1^2} & \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_1 \partial \theta_2} & \cdots & \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_1 \partial \theta_d} \\ \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_2^2} & \cdots & \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_2 \partial \theta_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_d \partial \theta_1} & \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_d \partial \theta_2} & \cdots & \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_d^2} \end{bmatrix}. \tag{7.70}$$

That is, the Hessian matrix is a matrix whose (i, j)-th entry is the second partial derivative of $\mathcal{L}(\boldsymbol{\theta})$ with respect to the i-th and j-th variables. We denote the Hessian matrix of $\mathcal{L}(\boldsymbol{\theta})$ as $\nabla^2 \mathcal{L}(\boldsymbol{\theta})$.

It is easy to see that the Hessian matrix is symmetric, since $\frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} = \frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_j \partial \theta_i}$ for all $i, j \in [d]$. The Hessian matrix may or may not be positive semidefinite. If $\boldsymbol{\theta}$ is a local minimizer of $\mathcal{L}(\boldsymbol{\theta})$, then the Hessian matrix must be positive semidefinite. Otherwise, the Hessian matrix may not be positive semidefinite.

The symbol $\|\nabla^2 \mathcal{L}(\boldsymbol{\theta})\|_2$ does not denote the L^2-norm of any vector, but the operator norm of the Hessian matrix.

Definition 7.13 (Operator norm)**.** The *operator norm* of a matrix $\boldsymbol{A} \in \mathbb{R}^{k \times d}$ is defined as

$$\|\boldsymbol{A}\|_2 := \max_{\boldsymbol{x} \in \mathbb{R}^d \setminus \{\boldsymbol{0}\}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2}. \tag{7.71}$$

More concretely, the operator norm of a matrix $\boldsymbol{A} \in \mathbb{R}^{k \times d}$ can be characterized as follows.

Proposition 7.14. The following statements on operator norms of matrices are true:

1. For any matrix $\boldsymbol{A} \in \mathbb{R}^{k \times d}$,
$$\|\boldsymbol{A}\|_2 = \sqrt{\lambda_{\max}(\boldsymbol{A}^\top \boldsymbol{A})}. \tag{7.72}$$
2. If $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ is symmetric, then
$$\|\boldsymbol{A}\|_2 = \max\{|\lambda_{\max}(\boldsymbol{A})|, |\lambda_{\min}(\boldsymbol{A})|\}. \tag{7.73}$$
3. If $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ is symmetric positive semidefinite, then
$$\|\boldsymbol{A}\|_2 = \lambda_{\max}(\boldsymbol{A}). \tag{7.74}$$

Here $\lambda_{\max}(\,\cdot\,)$ and $\lambda_{\min}(\,\cdot\,)$ denote the largest and smallest eigenvalues of a matrix, respectively.

Proof. We can take square of the operator norm to get:

$$\|\boldsymbol{A}\|_2^2 = \max_{\boldsymbol{x} \in \mathbb{R}^d \setminus \{\boldsymbol{0}\}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_2^2}{\|\boldsymbol{x}\|_2^2} = \max_{\boldsymbol{x} \in \mathbb{R}^d \setminus \{\boldsymbol{0}\}} \frac{\boldsymbol{x}^\top \boldsymbol{A}^\top \boldsymbol{A} \boldsymbol{x}}{\|\boldsymbol{x}\|_2^2} = \max_{\boldsymbol{x} \in \mathbb{R}^d : \|\boldsymbol{x}\|_2 = 1} \boldsymbol{x}^\top \boldsymbol{A}^\top \boldsymbol{A} \boldsymbol{x}. \tag{7.75}$$

We can rewrite the right-hand side in an orthonormal eigenbasis of $\boldsymbol{A}^\top \boldsymbol{A}$ and conclude that the maximum is achieved when $\boldsymbol{x}$ is an eigenvector of $\boldsymbol{A}^\top \boldsymbol{A}$ with the largest eigenvalue. In this case,

$$\|\boldsymbol{A}\|_2^2 = \lambda_{\max}(\boldsymbol{A}^\top \boldsymbol{A}). \tag{7.76}$$

Taking the square root of both sides proves the first statement.

For the second statement, we can use the fact that $\lambda_{\max}(\boldsymbol{A}^\top \boldsymbol{A}) = \lambda_{\max}(\boldsymbol{A}^2)$ for symmetric matrices. If the eigenvalues of $\boldsymbol{A}$ are

$$\lambda_1, \lambda_2, \ldots, \lambda_d, \tag{7.77}$$

then the eigenvalues of $\boldsymbol{A}^2$ are

$$\lambda_1^2, \lambda_2^2, \ldots, \lambda_d^2. \tag{7.78}$$

The maximum is achieved at either $\lambda_{\max}(\boldsymbol{A})^2$ or $\lambda_{\min}(\boldsymbol{A})^2$. Therefore,

$$\|\boldsymbol{A}\|_2 = \sqrt{\max\{\lambda_{\max}(\boldsymbol{A})^2, \lambda_{\min}(\boldsymbol{A})^2\}} = \max\{|\lambda_{\max}(\boldsymbol{A})|, |\lambda_{\min}(\boldsymbol{A})|\}. \tag{7.79}$$

Finally, if $\boldsymbol{A}$ is symmetric positive definite, then $\lambda_{\max}(\boldsymbol{A}) \geq \lambda_{\min}(\boldsymbol{A}) > 0$. By Equation (7.79), we have

$$\|\boldsymbol{A}\|_2 = \max\{|\lambda_{\max}(\boldsymbol{A})|, |\lambda_{\min}(\boldsymbol{A})|\} = \lambda_{\max}(\boldsymbol{A}), \tag{7.80}$$

which proves the third statement. □

Remark 7.15. Another useful characterization of the operator norm is that for any matrix $\boldsymbol{A} \in \mathbb{R}^{k \times d}$, $\|\boldsymbol{A}\|_2 = \sigma_{\max}(\boldsymbol{A})$, where $\sigma_{\max}(\boldsymbol{A})$ is the largest singular value of the matrix $\boldsymbol{A}$.

In general, a function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ is β-smooth if and only if for every $\boldsymbol{\theta} \in \mathbb{R}^d$, all the eigenvalues of $\nabla^2 \mathcal{L}(\boldsymbol{\theta})$ are bounded in $[-\beta, \beta]$. We can also write this condition as

$$\beta \boldsymbol{I} \succeq \nabla^2 \mathcal{L}(\boldsymbol{\theta}) \succeq -\beta \boldsymbol{I}, \tag{7.81}$$

where $\boldsymbol{A} \succeq \boldsymbol{B}$ means that $\boldsymbol{A} - \boldsymbol{B}$ is positive semidefinite.

Example 7.16. It is time to revisit the quadratic case with these two new concepts. Let $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \boldsymbol{H} \boldsymbol{\theta} - \langle \boldsymbol{b}, \boldsymbol{\theta} \rangle + C_0$ be the quadratic objective that we studied in the previous section. Then for every $\boldsymbol{\theta} \in \mathbb{R}^d$, the Hessian matrix is

$$\nabla^2 \mathcal{L}(\boldsymbol{\theta}) = \boldsymbol{H}. \tag{7.82}$$

The Hessian matrix is symmetric, so its operator norm is

$$\|\nabla^2 \mathcal{L}(\boldsymbol{\theta})\|_2 = \max\{|\lambda_{\max}(\boldsymbol{H})|, |\lambda_{\min}(\boldsymbol{H})|\}. \tag{7.83}$$

Then $\mathcal{L}$ is β-smooth if and only if all the eigenvalues of $\boldsymbol{H}$ are bounded in $[-\beta, \beta]$. In other words,

$$\beta \boldsymbol{I} \succeq \boldsymbol{H} \succeq -\beta \boldsymbol{I}. \tag{7.84}$$

▶ Descent Lemma

To see how the smoothness constant β affects the choice of learning rate η, we introduce the following lemma, which is known as the *descent lemma* in optimization theory.

Lemma 7.17 (Descent lemma)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be a β-smooth function. Suppose we perform one step of gradient descent with learning rate η from $\boldsymbol{\theta}_t$ to $\boldsymbol{\theta}_{t+1}$, i.e.,

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_t). \tag{7.85}$$

Then

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) \leq \mathcal{L}(\boldsymbol{\theta}_t) - \eta \left(1 - \frac{\eta\beta}{2}\right) \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2. \tag{7.86}$$

In fact, this descent lemma is a simple consequence of the Taylor expansion.

Theorem 7.18. Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be a twice differentiable function. Then for all $\boldsymbol{\theta}, \boldsymbol{\theta}' \in \mathbb{R}^d$, there exists $\xi \in (0, 1)$ such that

$$\mathcal{L}(\boldsymbol{\theta}') = \mathcal{L}(\boldsymbol{\theta}) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}), \boldsymbol{\theta}' - \boldsymbol{\theta} \rangle \tag{7.87}$$

$$+ \frac{1}{2} (\boldsymbol{\theta}' - \boldsymbol{\theta})^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta} + \xi(\boldsymbol{\theta}' - \boldsymbol{\theta}))(\boldsymbol{\theta}' - \boldsymbol{\theta}). \tag{7.88}$$

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be the following function:

$$f(x) = \mathcal{L}(\boldsymbol{\theta} + x(\boldsymbol{\theta}' - \boldsymbol{\theta})). \tag{7.89}$$

Performing Taylor expansion around $x = 0$, we have that for some $\xi \in (0, 1)$,

$$f(x) = f(0) + f'(0)x + \frac{x^2}{2} f''(\xi). \tag{7.90}$$

A simple calculation shows that

$$f(0) = \mathcal{L}(\boldsymbol{\theta}_t), \tag{7.91}$$

$$f'(0) = \langle \nabla \mathcal{L}(\boldsymbol{\theta}_t - x(\boldsymbol{\theta}' - \boldsymbol{\theta})), \boldsymbol{\theta}' - \boldsymbol{\theta} \rangle \Big|_{x=0} = \langle \nabla \mathcal{L}(\boldsymbol{\theta}), \boldsymbol{\theta}' - \boldsymbol{\theta} \rangle \tag{7.92}$$

$$f''(x) = (\boldsymbol{\theta}' - \boldsymbol{\theta})^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta} + x(\boldsymbol{\theta}' - \boldsymbol{\theta}))(\boldsymbol{\theta}' - \boldsymbol{\theta}). \tag{7.93}$$

Substituting back into Equation (7.90) proves the theorem. □

Proof of Lemma 7.17. By Taylor expansion, we have that for some $\xi \in (0, 1)$,

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) = \mathcal{L}(\boldsymbol{\theta}_t) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}_t), \boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t \rangle \tag{7.94}$$

$$+ \frac{1}{2} (\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t)^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta}_t + \xi(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t))(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t). \tag{7.95}$$

Since $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_t)$, this can be rewritten as

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) = \mathcal{L}(\boldsymbol{\theta}_t) - \eta \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 \tag{7.96}$$

$$+ \frac{\eta^2}{2} (\nabla \mathcal{L}(\boldsymbol{\theta}_t))^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta}_t + \xi(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t))(\nabla \mathcal{L}(\boldsymbol{\theta}_t)). \tag{7.97}$$

Also, since $\mathcal{L}(\boldsymbol{\theta})$ is β-smooth, we have

$$\|\nabla^2 \mathcal{L}(\boldsymbol{\theta}_t - \xi \nabla \mathcal{L}(\boldsymbol{\theta}_t))\|_2 \leq \beta. \tag{7.98}$$

Therefore,

$$(\nabla \mathcal{L}(\boldsymbol{\theta}_t))^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta}_t + \xi(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t))(\nabla \mathcal{L}(\boldsymbol{\theta}_t)) \leq \beta \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2. \tag{7.99}$$

Substituting this back into the Taylor expansion, we get

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) = \mathcal{L}(\boldsymbol{\theta}_t) - \eta \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 + \frac{\eta^2}{2} \beta \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2. \tag{7.100}$$

Therefore,

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) = f(\eta) \leq \mathcal{L}(\boldsymbol{\theta}_t) - \eta \left(1 - \frac{\eta\beta}{2}\right) \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2, \tag{7.101}$$

which proves the lemma. □

An important interpretation of the descent lemma is that the loss is guaranteed to decrease if the learning rate satisfies

$$0 < \eta < \frac{2}{\beta}. \tag{7.102}$$

We have seen a similar condition in the quadratic case in Equation (7.46):

$$0 < \eta < \frac{2}{\lambda_{\max}}. \tag{7.103}$$

Therefore, the smoothness β plays a similar role as the maximum eigenvalue $\lambda_{\max}$ in the quadratic case. In fact, these two conditions are exactly the same if we take β to be its minimum possible value in the quadratic case, since this is just $\lambda_{\max}$.

Finally, a key limitation of the descent lemma is that it only guarantees a one-step decrease of the loss. To prove that the loss continues to decrease over multiple steps, we need to understand what other properties the objective function can have, and how these properties can be used to analyze the convergence of gradient descent.

7.1.4 Smooth Convex Optimization

▶ Convexity and Strong Convexity

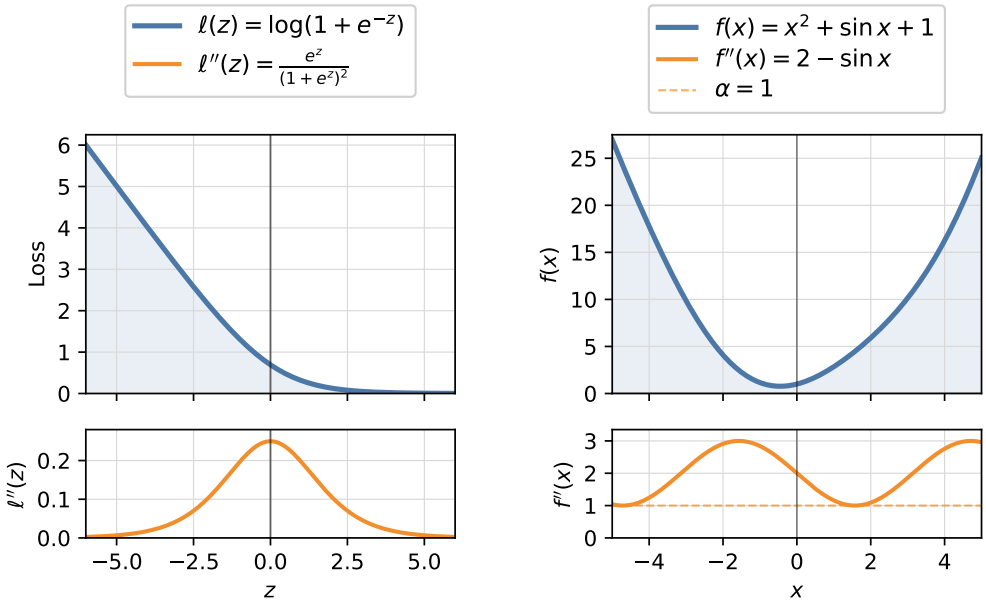
The next key property we will discuss is the convexity and strong convexity of the objective function. We have already seen the usefulness of convexity in the exercises, such as Exercise 3.2 and Exercise 3.5. Now we introduce the concepts of convexity and strong convexity through the lens of Hessian.

Definition 7.19 (Convex and strongly convex functions)**.** A twice differentiable function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ is *convex* if for all $\boldsymbol{\theta} \in \mathbb{R}^d$,

$$\nabla^2 \mathcal{L}(\boldsymbol{\theta}) \succeq \mathbf{0}. \tag{7.104}$$

It is α-*strongly convex* if for all $\boldsymbol{\theta} \in \mathbb{R}^d$,

$$\nabla^2 \mathcal{L}(\boldsymbol{\theta}) \succeq \alpha \boldsymbol{I}. \tag{7.105}$$



(a) A non-strongly convex function. (b) A strongly convex function.

Figure 7.7: The logistic loss $\ell(z) = \log(1 + e^{-z})$ is convex, but it not strongly convex because its second derivative tends to zero as $z \to \infty$. By contrast, the function $f(x) = x^2 + \sin x + 1$ is 1-strongly convex.

As usual, $\boldsymbol{A} \succeq \boldsymbol{B}$ means that $\boldsymbol{A} - \boldsymbol{B}$ is positive semidefinite. This is equivalent to saying that $\boldsymbol{v}^\top(\boldsymbol{A} - \boldsymbol{B})\boldsymbol{v} \geq 0$ for all $\boldsymbol{v} \in \mathbb{R}^d$. Another equivalent condition is that $\lambda_{\min}(\boldsymbol{A} - \boldsymbol{B}) \geq 0$.

Therefore, the convexity condition can be written as

$$\lambda_{\min}(\nabla^2 \mathcal{L}(\boldsymbol{\theta})) \geq 0. \tag{7.106}$$

Similarly, the strong convexity condition can be written as

$$\lambda_{\min}(\nabla^2 \mathcal{L}(\boldsymbol{\theta})) \geq \alpha. \tag{7.107}$$

From this perspective, the parameter α in strong convexity plays a role similar to $\lambda_{\min}(\boldsymbol{H})$ for quadratic objectives. The difference is that α gives a uniform lower bound for the eigenvalues of all Hessian matrices $\nabla^2 \mathcal{L}(\boldsymbol{\theta})$, instead of the smallest eigenvalue of a single matrix $\boldsymbol{H}$ in the quadratic case. In fact, for a quadratic objective, the function is α-strongly convex if and only if $\alpha \leq \lambda_{\min}(\boldsymbol{H})$.

In the one-dimensional case, this just means that the second derivative is non-negative or is greater than some constant α. See Figure 7.7 for some examples.

Now we state and prove some basic consequences of convexity and strong convexity.

Proposition 7.20 (Basic consequences of convexity)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be twice differentiable and convex. Then:

1. For all $\boldsymbol{\theta}, \boldsymbol{u} \in \mathbb{R}^d$,
$$\mathcal{L}(\boldsymbol{u}) \geq \mathcal{L}(\boldsymbol{\theta}) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}), \boldsymbol{u} - \boldsymbol{\theta} \rangle. \tag{7.108}$$
2. For any $\boldsymbol{\theta}^* \in \mathbb{R}^d$, $\nabla \mathcal{L}(\boldsymbol{\theta}^*) = \mathbf{0}$ if and only if $\boldsymbol{\theta}^*$ is a minimizer of $\mathcal{L}$.

Proof. For the first claim, we use the Taylor expansion around $\boldsymbol{\theta}$. Then for some $\xi \in (0, 1)$,
$$\mathcal{L}(\boldsymbol{u}) = \mathcal{L}(\boldsymbol{\theta}) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}), \boldsymbol{u} - \boldsymbol{\theta} \rangle + \frac{1}{2} (\boldsymbol{u} - \boldsymbol{\theta})^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta} + \xi(\boldsymbol{u} - \boldsymbol{\theta}))(\boldsymbol{u} - \boldsymbol{\theta}). \tag{7.109}$$
Since $\mathcal{L}$ is convex, $\nabla^2 \mathcal{L} \succeq \mathbf{0}$, so the last term is non-negative. This proves Equation (7.108).

For the second claim, we apply Equation (7.108) with $\boldsymbol{\theta} = \boldsymbol{\theta}^*$, then for every $\boldsymbol{u} \in \mathbb{R}^d$,
$$\mathcal{L}(\boldsymbol{u}) \geq \mathcal{L}(\boldsymbol{\theta}^*) + \langle \mathbf{0}, \boldsymbol{u} - \boldsymbol{\theta}^* \rangle = \mathcal{L}(\boldsymbol{\theta}^*). \tag{7.110}$$
Therefore, $\boldsymbol{\theta}^*$ is a minimizer of $\mathcal{L}$.

Conversely, if $\boldsymbol{\theta}^*$ is a minimizer of $\mathcal{L}$, then as long as $\mathcal{L}$ is differentiable, it can be shown that $\nabla \mathcal{L}(\boldsymbol{\theta}^*) = \mathbf{0}$. For completeness, we provide a proof here. Assume to the contrary that $\nabla \mathcal{L}(\boldsymbol{\theta}^*) \neq \mathbf{0}$. Consider the function $\phi(t) := \mathcal{L}(\boldsymbol{\theta}^* + t \nabla \mathcal{L}(\boldsymbol{\theta}^*))$. Since $\mathcal{L}$ is differentiable, $\phi(t)$ is also differentiable. The derivative of $\phi(t)$ at 0 is
$$\phi'(0) = \lim_{t \to 0} \frac{\phi(t) - \phi(0)}{t}. \tag{7.111}$$
If $\phi'(0) > 0$, then for some $t > 0$, we have $\phi(t) > \phi(0)$. If $\phi'(0) < 0$, then for some $t < 0$, we have $\phi(t) > \phi(0)$. In both cases, $\mathcal{L}(\boldsymbol{\theta}^* + t \nabla \mathcal{L}(\boldsymbol{\theta}^*)) > \mathcal{L}(\boldsymbol{\theta}^*)$ for some $t \neq 0$, which contradicts the assumption that $\boldsymbol{\theta}^*$ is a minimizer. Therefore, we must have $\phi'(0) = 0$. However, by the chain rule, we have
$$\phi'(0) = \langle \nabla \mathcal{L}(\boldsymbol{\theta}^*), \nabla \mathcal{L}(\boldsymbol{\theta}^*) \rangle = \|\nabla \mathcal{L}(\boldsymbol{\theta}^*)\|_2^2. \tag{7.112}$$
Therefore, we must have $\|\nabla \mathcal{L}(\boldsymbol{\theta}^*)\|_2^2 = 0$, which contradicts to $\nabla \mathcal{L}(\boldsymbol{\theta}^*) \neq \mathbf{0}$. □

Proposition 7.21 (Basic consequences of strong convexity)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be twice differentiable and α-strongly convex. Then:

1. For all $\boldsymbol{\theta}, \boldsymbol{u} \in \mathbb{R}^d$,
$$\mathcal{L}(\boldsymbol{u}) \geq \mathcal{L}(\boldsymbol{\theta}) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}), \boldsymbol{u} - \boldsymbol{\theta} \rangle + \frac{\alpha}{2} \|\boldsymbol{u} - \boldsymbol{\theta}\|_2^2. \tag{7.113}$$
2. The function $\mathcal{L}$ has a unique minimizer $\boldsymbol{\theta}^*$.
3. At the unique minimizer $\boldsymbol{\theta}^*$, it holds that for all $\boldsymbol{u} \in \mathbb{R}^d$,
$$\mathcal{L}(\boldsymbol{u}) - \mathcal{L}(\boldsymbol{\theta}^*) \geq \frac{\alpha}{2} \|\boldsymbol{u} - \boldsymbol{\theta}^*\|_2^2. \tag{7.114}$$

Proof. For the first claim, we again use the Taylor expansion around $\boldsymbol{\theta}$. Then for some $\xi \in (0,1)$,

$$\mathcal{L}(\boldsymbol{u}) = \mathcal{L}(\boldsymbol{\theta}) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}), \boldsymbol{u} - \boldsymbol{\theta} \rangle + \frac{1}{2}(\boldsymbol{u} - \boldsymbol{\theta})^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta} + \xi(\boldsymbol{u} - \boldsymbol{\theta}))(\boldsymbol{u} - \boldsymbol{\theta}). \quad (7.115)$$

Since $\mathcal{L}$ is convex, $\nabla^2 \mathcal{L} \succeq \alpha \boldsymbol{I}$, so the last term can be lower bounded by

$$\frac{1}{2}(\boldsymbol{u} - \boldsymbol{\theta})^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta} + \xi(\boldsymbol{u} - \boldsymbol{\theta}))(\boldsymbol{u} - \boldsymbol{\theta}) \geq \frac{\alpha}{2}\|\boldsymbol{u} - \boldsymbol{\theta}\|_2^2. \quad (7.116)$$

Combining the two inequalities gives Equation (7.113).

For the second claim, we first prove existence. Fix any point $\boldsymbol{\theta}_0 \in \mathbb{R}^d$. By Equation (7.113),

$$\mathcal{L}(\boldsymbol{u}) \geq \mathcal{L}(\boldsymbol{\theta}_0) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}_0), \boldsymbol{u} - \boldsymbol{\theta}_0 \rangle + \frac{\alpha}{2}\|\boldsymbol{u} - \boldsymbol{\theta}_0\|_2^2. \quad (7.117)$$

The right-hand side goes to $+\infty$ as $\|\boldsymbol{u}\|_2 \to \infty$, This implies that $\mathcal{L}(\boldsymbol{u}) \to +\infty$ as well. Therefore, it is possible to find a radius $R > 0$ such that for all $\boldsymbol{u} \in \mathbb{R}^d$ with $\|\boldsymbol{u} - \boldsymbol{\theta}_0\|_2 > R$,

$$\mathcal{L}(\boldsymbol{u}) > \mathcal{L}(\boldsymbol{\theta}_0). \quad (7.118)$$

Since $\mathcal{L}$ is continuous, it attains its minimum over the closed ball $B_R(\boldsymbol{\theta}_0) := \{\boldsymbol{u} \in \mathbb{R}^d : \|\boldsymbol{u} - \boldsymbol{\theta}_0\|_2 \leq R\}$ at some point $\boldsymbol{\theta}^* \in B_R(\boldsymbol{\theta}_0)$. Moreover, every point outside $B_R(\boldsymbol{\theta}_0)$ has value strictly larger than $\mathcal{L}(\boldsymbol{\theta}_0)$, so this point $\boldsymbol{\theta}^*$ is a minimizer of $\mathcal{L}$ over the entire space $\mathbb{R}^d$.

It remains to prove uniqueness. Suppose there is another minimizer $\boldsymbol{\theta}_1 \in \mathbb{R}^d$ with $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_0$. Then by Equation (7.113),

$$\mathcal{L}(\boldsymbol{\theta}_1) \geq \mathcal{L}(\boldsymbol{\theta}_0) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}_0), \boldsymbol{\theta}_1 - \boldsymbol{\theta}_0 \rangle + \frac{\alpha}{2}\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_0\|_2^2 \quad (7.119)$$

$$= \mathcal{L}(\boldsymbol{\theta}_0) + \frac{\alpha}{2}\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_0\|_2^2. \quad (7.120)$$

However, $\boldsymbol{\theta}_0$ and $\boldsymbol{\theta}_1$ have the same minimum value, so it is impossible to have $\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_0\|_2 > 0$. Thus the minimizer is unique.

The last claim is essentially the same as Equation (7.120), just we do not require $\boldsymbol{\theta}_1$ to be the minimizer. □

▶ Logistic regression

A basic example of a convex loss in machine learning is logistic regression. In many scenarios, we aim to classify data into one of several categories, such as distinguishing images of dogs from cats. Linear regression is not suitable here, because the labels represent discrete categories rather than continuous numerical values.

In logistic regression, we focus on the binary classification setting, where each input is labeled with either -1 or $+1$:

$$(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n), \qquad \boldsymbol{x}_i \in \mathbb{R}^d, \quad y_i \in \{-1, +1\}. \quad (7.121)$$

Similar to linear regression, we use a linear model to predict the label:

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \langle \boldsymbol{x}, \boldsymbol{\theta} \rangle. \tag{7.122}$$

If $f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) > 0$, the model predicts $+1$. Conversely, if $f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) < 0$, it predicts -1. This model is also called a *linear classifier*. So for classification problems, we often use the margin $\gamma_i(\boldsymbol{\theta})$ as a measure of how correct the prediction is for the i-th example:

$$\gamma_i(\boldsymbol{\theta}) := y_i \langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle \tag{7.123}$$

If $\gamma_i(\boldsymbol{\theta})$ is large and positive, then the i-th input is classified correctly with high confidence. If $\gamma_i(\boldsymbol{\theta})$ is negative, then the i-th input is classified incorrectly.

We therefore want a loss that is small when $\gamma_i(\boldsymbol{\theta})$ is large and positive, and large when $\gamma_i(\boldsymbol{\theta})$ is negative. Of course, one can choose the $0-1$ loss:

$$\ell_i(\boldsymbol{\theta}) := \mathbb{1}_{[\gamma_i(\boldsymbol{\theta}) < 0]}. \tag{7.124}$$

But this loss is not even continuous, which makes it difficult to optimize. A surrogate loss that is continuous and smooth is the logistic loss:

$$\ell(z) := \log\bigl(1 + e^{-z}\bigr), \tag{7.125}$$

$$\ell_i(\boldsymbol{\theta}) := \ell(y_i \langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle) = \log\bigl(1 + e^{-y_i \langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle}\bigr). \tag{7.126}$$

See Figure 7.7 for a visuliation of the logistic loss. This loss is convex and is monotone decreasing as a function of the margin $\gamma_i(\boldsymbol{\theta})$. Also it is an upper bound for the $0-1$ loss:

$$\log\bigl(1 + e^{-\gamma_i(\boldsymbol{\theta})}\bigr) \geq \log 2 \cdot \mathbb{1}_{[\gamma_i(\boldsymbol{\theta}) < 0]}. \tag{7.127}$$

The logistic regression problem asks for minimizing the average logistic loss over the training data:

$$\mathcal{L}(\boldsymbol{\theta}) := \frac{1}{n} \sum_{i=1}^{n} \ell_i(\boldsymbol{\theta}) \tag{7.128}$$

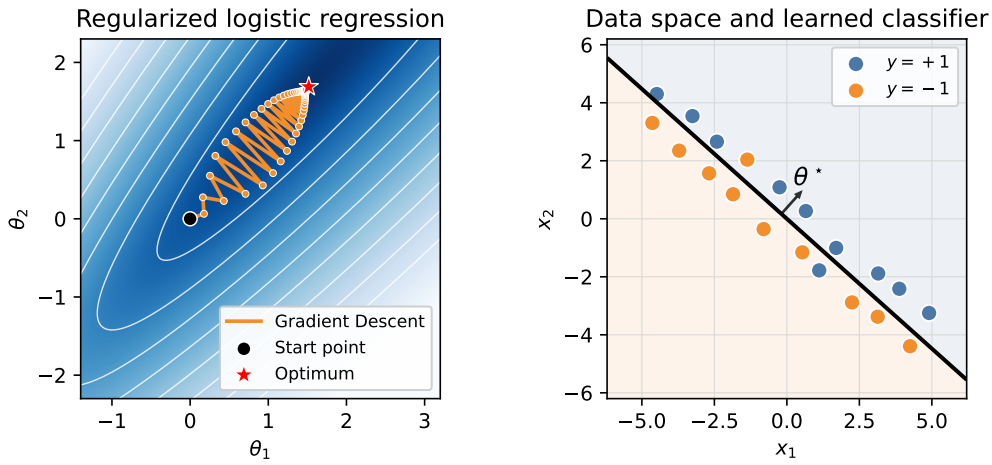
In practice, we often add a regularization term to the loss. Recall that a linear classifier classifies an input $\boldsymbol{x}$ according to the sign of $\langle \boldsymbol{\theta}, \boldsymbol{x} \rangle$. Multiplying $\boldsymbol{\theta}$ by a large constant does not change the classification result, but it can make the model much more confident on the training data and further decrease the logistic loss. However, some labels may be wrong due to errors in data annotation, and the model should not become too confident about fitting every training example. A standard way to prevent this is to penalize the norm of the parameter vector:

$$\mathcal{L}(\boldsymbol{\theta}) := \frac{1}{n} \sum_{i=1}^{n} \ell_i(\boldsymbol{\theta}) + \frac{\lambda}{2} \|\boldsymbol{\theta}\|_2^2, \tag{7.129}$$

where $\lambda \geq 0$ is the regularization strength. Roughly speaking, among classifiers that fit the training data similarly well, this regularization term prefers the one with smaller weights.

Now we check convexity using the Hessian. A direct differentiation gives

$$\nabla \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \ell'(\gamma_i(\boldsymbol{\theta})) y_i \boldsymbol{x}_i + \lambda \boldsymbol{\theta}. \tag{7.130}$$



(a) Gradient descent in parameter space. (b) The learned classifier in the data space.

Figure 7.8: A visualization of logistic regression. **(a)** Gradient descent approaches the optimum in the parameter space. **(b)** The learned linear classifier separates the training data into two categories.

Differentiating again gives

$$\nabla^2 \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \ell''(\gamma_i(\boldsymbol{\theta})) \boldsymbol{x}_i \boldsymbol{x}_i^\top + \lambda \boldsymbol{I}. \tag{7.131}$$

Since the logistic loss is convex with respect to the margin $\gamma_i(\boldsymbol{\theta})$, we have $\ell''(\gamma_i(\boldsymbol{\theta}))\boldsymbol{x}_i\boldsymbol{x}_i^\top \succeq \boldsymbol{0}$. Summing over all examples, we get

$$\nabla^2 \mathcal{L}(\boldsymbol{\theta}) \succeq \lambda \boldsymbol{I}. \tag{7.132}$$

Therefore, $\mathcal{L}(\boldsymbol{\theta})$ is λ-strongly convex. Of course, if $\lambda = 0$, then $\mathcal{L}(\boldsymbol{\theta})$ may only be convex, but not strongly convex.

For later use, we also note the smoothness constant for logistic regression. Since $\ell''(z) \leq 1/4$,

$$\nabla^2 \mathcal{L}(\boldsymbol{\theta}) \preceq \frac{1}{4n} \boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}. \tag{7.133}$$

Therefore, the ojective $\mathcal{L}(\boldsymbol{\theta})$ is β-smooth with

$$\beta = \frac{1}{4n} \lambda_{\max}(\boldsymbol{X}^\top \boldsymbol{X}) + \lambda. \tag{7.134}$$

Remark 7.22. Some empirical machine learning researchers have done experiments for so long that they have lost the ability to think about anything mathematically. They may crash into a theory paper and say: "Hey, I am using softmax plus cross-entropy loss for multi-class classification. Why study logistic loss? You theory guys always study unrealistic things." This comment is ridiculous but please do not get tired of explaining to these people that logistic loss is exactly softmax plus cross-entropy loss in the binary case:

$$\log(1 + e^{-y_i \langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle}) = -\log[\mathrm{softmax}(\langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle, -\langle \boldsymbol{x}_i, \boldsymbol{\theta} \rangle)]_{y_i}, \tag{7.135}$$

where softmax is a function that maps a vector $\boldsymbol{v} = (v_1, \ldots, v_k)$ to the probability vector $\frac{1}{\sum_{i=1}^{k} e^{v_i}}(e^{v_1}, \ldots, e^{v_k})$. We study logistic loss for simplicity, but many conclusions in the binary case can be easily generalized to the multi-class case.

▶ Convergence Rate

Now we consider the optimization of any objective function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ that is twice differentiable, α-strongly convex and β-smooth. This means that for all $\boldsymbol{\theta} \in \mathbb{R}^d$,

$$\beta \boldsymbol{I} \succeq \nabla^2 \mathcal{L}(\boldsymbol{\theta}) \succeq \alpha \boldsymbol{I}. \tag{7.136}$$

Let $\boldsymbol{\theta}^*$ be the unique minimizer of $\mathcal{L}$. We study gradient descent with learning rate $\eta > 0$. The descent lemma (Lemma 7.17) explains how much one step decreases the loss:

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) \leq \mathcal{L}(\boldsymbol{\theta}_t) - \eta \left(1 - \frac{\beta\eta}{2}\right) \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2. \tag{7.137}$$

However, this does not tell us how the loss decreases over multiple steps. For this, we need to know that the gradient is large whenever the loss is still far from optimal. One formulation of this property is the *Polyak–Łojasiewicz condition*.

Definition 7.23 (Polyak–Łojasiewicz condition)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be differentiable and have a minimizer $\boldsymbol{\theta}^*$. We say that $\mathcal{L}$ satisfies the *Polyak–Łojasiewicz condition*, or *PL condition*, with parameter $\mu > 0$ if

$$\frac{1}{2}\|\nabla \mathcal{L}(\boldsymbol{\theta})\|_2^2 \geq \mu(\mathcal{L}(\boldsymbol{\theta}) - \mathcal{L}(\boldsymbol{\theta}^*)) \qquad \text{for all } \boldsymbol{\theta} \in \mathbb{R}^d. \tag{7.138}$$

This PL condition states that the gradient is large whenever there is a large gap between the current loss and the optimal loss. While this condition can hold even for some non-convex functions, here we will only need it as a consequence of strong convexity.

Lemma 7.24 (Strong convexity implies PL)**.** If $\mathcal{L}(\boldsymbol{\theta})$ is twice differentiable and α-strongly convex, then $\mathcal{L}(\boldsymbol{\theta})$ satisfies the PL condition with parameter α.

Proof. Apply Equation (7.113) with $\boldsymbol{u} = \boldsymbol{\theta}^*$:

$$\mathcal{L}(\boldsymbol{\theta}^*) \geq \mathcal{L}(\boldsymbol{\theta}) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}), \boldsymbol{\theta}^* - \boldsymbol{\theta} \rangle + \frac{\alpha}{2}\|\boldsymbol{\theta}^* - \boldsymbol{\theta}\|_2^2. \tag{7.139}$$

The right-hand side is at least the minimum over all displacement vectors $\boldsymbol{v}$:

$$\mathcal{L}(\boldsymbol{\theta}^*) \geq \mathcal{L}(\boldsymbol{\theta}) + \inf_{\boldsymbol{v} \in \mathbb{R}^d} \left\{ \langle \nabla \mathcal{L}(\boldsymbol{\theta}), \boldsymbol{v} \rangle + \frac{\alpha}{2}\|\boldsymbol{v}\|_2^2 \right\} \tag{7.140}$$

$$= \mathcal{L}(\boldsymbol{\theta}) - \frac{1}{2\alpha}\|\nabla \mathcal{L}(\boldsymbol{\theta})\|_2^2. \tag{7.141}$$

Rearranging gives Equation (7.138). □

Now we are ready to bound the convergence rate of gradient descent.

Theorem 7.25. Let $\mathcal{L}(\boldsymbol{\theta})$ be a twice differentiable, α-strongly convex and β-smooth objective function. Let $\mathcal{L}_{\min}$ be the minimum value of $\mathcal{L}(\boldsymbol{\theta})$. For gradient descent with learning rate $\eta \in (0, 2/\beta)$, we have

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} \leq \rho(\eta)^t (\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}), \tag{7.142}$$

where the contraction factor $\rho(\eta)$ is defined by $\rho(\eta) := 1 - 2\alpha\eta + \alpha\beta\eta^2$.

Proof. By the descent lemma from Equation (7.137),

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) \leq \mathcal{L}(\boldsymbol{\theta}_t) - \eta \left(1 - \frac{\beta\eta}{2}\right) \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2. \tag{7.143}$$

By the PL condition implied by strong convexity in Lemma 7.24,

$$\|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 \geq 2\alpha(\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min}). \tag{7.144}$$

Putting these two inequalities together, we get

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) \leq \mathcal{L}(\boldsymbol{\theta}_t) - \eta \left(1 - \frac{\beta\eta}{2}\right) \cdot 2\alpha(\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min}). \tag{7.145}$$

Subtracting $\mathcal{L}_{\min}$ from both sides, we get

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) - \mathcal{L}_{\min} \leq \mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} - \eta \left(1 - \frac{\beta\eta}{2}\right) \cdot 2\alpha(\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min}). \tag{7.146}$$

Rearranging gives

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) - \mathcal{L}_{\min} \leq \left(1 - 2\alpha\eta + \alpha\beta\eta^2\right) (\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min}). \tag{7.147}$$

Iterating this inequality proves the theorem. □

From Theorem 7.25, we can see that the convergence rate of gradient descent is determined by the following quantity:

$$\rho(\eta) = 1 - 2\alpha\eta + \alpha\beta\eta^2. \tag{7.148}$$

Minimizing this quantity over $\eta > 0$ gives

$$\eta^* = \frac{1}{\beta}, \qquad \rho(\eta^*) = 1 - \frac{\alpha}{\beta}. \tag{7.149}$$

Recall that α plays a similar role as $\lambda_{\min}(\boldsymbol{H})$ for quadratic objectives, and β plays a similar role as $\lambda_{\max}(\boldsymbol{H})$ for quadratic objectives. So $\frac{\alpha}{\beta}$ should be a similar quantity as the condition number $\kappa(\boldsymbol{H})$ of quadratic objectives. Once again, we meet the condition number!

▶ Condition Number

For the optimization of smooth strongly convex functions, the condition number is a key quantity that controls the convergence rate. We define the condition number as follows.

Definition 7.26 (Condition number for smooth strongly convex functions)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be a twice differentiable, α-strongly convex and β-smooth objective function, where α and β are the best possible constants for the strong convexity and smoothness of $\mathcal{L}$. The *condition number* of $\mathcal{L}$ is defined as

$$\kappa(\mathcal{L}) := \frac{\beta}{\alpha}. \tag{7.150}$$

This is the same idea as the condition number in the quadratic case in Definition 7.7. In fact, if $\mathcal{L}$ is a quadratic objective with coefficient matrix $\boldsymbol{H}$, then $\kappa(\mathcal{L}) = \kappa(\boldsymbol{H})$.

Based on the definition of the condition number, we can write the convergence rate of gradient descent under the learning rate $\eta^* = \frac{1}{\beta}$ as:

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} \le \left(1 - \frac{1}{\kappa}\right)^t (\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}) = \mathcal{O}\left(\left(1 - \frac{1}{\kappa}\right)^t\right). \tag{7.151}$$

From here, we can see a similar asymptotic behavior as the quadratic case in Equation (7.68): the larger the condition number is, the slower the convergence rate is.

▶ How to Choose the Learning Rate

Careful readers may have already noticed that the convergence rate in Equation (7.151) is slower than the quadratic case in Equation (7.66). The recommended learning rate $\eta^* = \frac{1}{\beta}$ is also not optimal for quadratic objectives.

In fact, this is because the convergence analysis above is not tight enough. A more refined analysis can show the following bound:

Theorem 7.27. Under the same setting as Theorem 7.25, we have

$$\|\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^*\|_2 \le \rho(\eta)\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|_2, \tag{7.152}$$

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} \le \kappa \cdot \rho(\eta)^{2t}(\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}), \tag{7.153}$$

where $\rho(\eta) := \max\{|1 - \eta\alpha|, |1 - \eta\beta|\}$.

Proof. Since $\nabla\mathcal{L}(\boldsymbol{\theta}^*) = \boldsymbol{0}$, we can write

$$\nabla\mathcal{L}(\boldsymbol{\theta}_t) = \nabla\mathcal{L}(\boldsymbol{\theta}_t) - \nabla\mathcal{L}(\boldsymbol{\theta}^*) = \int_0^1 \nabla^2\mathcal{L}(\boldsymbol{\theta}^* + s(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*))(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*)\, ds, \tag{7.154}$$

where we use the fact that the derivative of $\nabla\mathcal{L}(\boldsymbol{\theta}^* + s(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*))$ with respect to s is $\nabla^2\mathcal{L}(\boldsymbol{\theta}^* + s(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*))(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*)$. Therefore, we can write $\nabla\mathcal{L}(\boldsymbol{\theta}_t)$ as

$$\nabla\mathcal{L}(\boldsymbol{\theta}_t) = \boldsymbol{A}_t(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*), \quad \text{where} \quad \boldsymbol{A}_t := \int_0^1 \nabla^2\mathcal{L}(\boldsymbol{\theta}^* + s(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*))\, ds. \tag{7.155}$$

Each integrand is a symmetric positive definite matrix, and for every $\boldsymbol{v} \in \mathbb{R}^d$,

$$\beta\|\boldsymbol{v}\|_2^2 \ge \boldsymbol{v}^\top\nabla^2\mathcal{L}(\boldsymbol{\theta}^* + s(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*))\boldsymbol{v} \ge \alpha\|\boldsymbol{v}\|_2^2. \tag{7.156}$$

Integrating over s gives

$$\beta\|\boldsymbol{v}\|_2^2 \geq \boldsymbol{A}_t \geq \alpha\|\boldsymbol{v}\|_2^2. \tag{7.157}$$

So all the eigenvalues of $\boldsymbol{A}_t$ lie between α and β.

Now we can write gradient descent as

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \boldsymbol{A}_t(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*). \tag{7.158}$$

Subtracting $\boldsymbol{\theta}^*$ from both sides gives

$$\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^* = (\boldsymbol{I} - \eta \boldsymbol{A}_t)(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*). \tag{7.159}$$

The eigenvalues of $\boldsymbol{I} - \eta \boldsymbol{A}_t$ are all between $1 - \eta\alpha$ and $1 - \eta\beta$. So we have

$$\|\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^*\|_2 \leq \rho(\eta)\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|_2, \tag{7.160}$$

where $\rho(\eta) := \max\{|1 - \eta\alpha|, |1 - \eta\beta|\}$. Iterating this inequality gives the first bound.

For the second bound, note that smoothness and strong convexity imply

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}(\boldsymbol{\theta}^*) \leq \frac{\beta}{2}\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|_2^2 \tag{7.161}$$

and

$$\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}(\boldsymbol{\theta}^*) \geq \frac{\alpha}{2}\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|_2^2. \tag{7.162}$$

Combining these inequalities gives

$$\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} \leq \frac{\beta}{2}\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|_2^2 \tag{7.163}$$

$$\leq \frac{\beta}{2}\rho(\eta)^2 t\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|_2^2 \leq \kappa \cdot \rho(\eta)^t(\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}), \tag{7.164}$$

which proves the second bound. □

From the refined analysis above, we can exactly recover the optimal learning rate and the optimal convergence rate in the quadratic case:

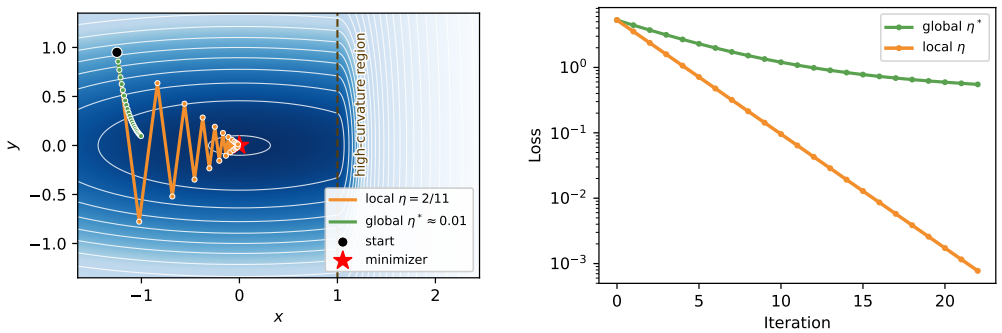
$$\eta^* = \frac{2}{\alpha + \beta}, \tag{7.165}$$

$$\rho(\eta^*) = \frac{\beta - \alpha}{\beta + \alpha} = 1 - \frac{2}{1 + \kappa}. \tag{7.166}$$

This sounds like a good result. So should we always try to use a learning rate that is as close to $\eta^* = \frac{2}{\alpha+\beta}$ as possible? While the answer is yes for quadratic objectives, it is not necessarily true for general strongly convex objectives. The main reason is that α and β are global constants and may not reflect the local loss landscape that gradient descent explores.

Here is a two-dimensional example that illustrates this point. Consider the objective

$$\mathcal{L}(x, y) = \begin{cases} \frac{1}{2}x^2 + 5y^2, & x \leq 1, \\ \frac{1}{2}x^2 + 5y^2 + 100(x-1)^2, & x > 1. \end{cases} \tag{7.167}$$



(a) Gradient descent in parameter space.

(b) Loss along the iterates.

Figure 7.9: Choosing a learning rate based on the global constants leads to a stable but overly conservative trajectory.

On the left half-plane $x \leq 1$, the objective is just the quadratic $\frac{1}{2}x^2 + 5y^2$, whose Hessian is $\mathrm{diag}(1, 10)$. On the right half-plane $x > 1$, the extra term $100(x-1)^2$ changes the Hessian to $\mathrm{diag}(201, 10)$ and creates a high-curvature wall. The objective is thus α-strongly convex and β-smooth with

$$\alpha = 1, \qquad \beta = 201. \tag{7.168}$$

Now initialize gradient descent at $\boldsymbol{\theta}_0 = (-1.25, 0.95)$. If we set the learning rate according to the global constants, then

$$\eta^* = \frac{2}{\alpha + \beta} = \frac{1}{101} \approx 0.01. \tag{7.169}$$

As shown in Figure 7.9, this gives a stable but very conservative trajectory. The iterates move towards the origin while staying in the local quadratic region $x \leq 1$. While the iterates never encounter the high-curvature region, the step size is still chosen for the stability in that region, which can be quite wasteful.

Instead, we may choose a learning rate based on the local quadratic region:

$$\eta = \frac{2}{1 + 10} = \frac{2}{11} \approx 0.18. \tag{7.170}$$

The iterates remain in the left half-plane and converge much faster. The y-coordinate oscillates but the amplitude decreases steadily.

Therefore, the takeaway is that $\eta^* = 2/(\alpha + \beta)$ is the optimal learning rate for quadratic objectives, but it can be overly conservative for general objectives. In practice, the learning rate should be chosen based on the local landscape explored by the algorithm rather than the global constants.

7.2 Stochastic Gradient Descent

So far, we have assumed that it is possible to compute the gradient $\nabla \mathcal{L}(\boldsymbol{\theta})$ exactly, and then we can run gradient descent to minimize the objective function

$\mathcal{L}(\boldsymbol{\theta})$. However, in many scenarios of interest, including linear regression and logistic regression, $\mathcal{L}(\boldsymbol{\theta})$ is the average of many individual loss functions $\ell_i(\boldsymbol{\theta})$, where each $\ell_i(\boldsymbol{\theta})$ is the loss on the i-th data point:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \ell_i(\boldsymbol{\theta}). \tag{7.171}$$

In this case, evaluating the gradient $\nabla\mathcal{L}(\boldsymbol{\theta})$ requires computing the gradient of each $\ell_i(\boldsymbol{\theta})$ and averaging them, which costs $\mathcal{O}(n)$ gradient evaluations. When the dataset contains millions or even billions of data points, computing the full gradient can become prohibitively expensive.

A compute-efficient variant of gradient descent is *stochastic gradient descent* (SGD), which replaces the full gradient with a random unbiased estimate. One of the most popular variants of SGD is *mini-batch SGD*, which works as follows. At each step t, we randomly sample B data points to form a *mini-batch* $(i_{t,b})_{b=1}^{B}$, where B is called the *batch size*. We then compute the gradient of the loss on the mini-batch:

$$\boldsymbol{g}_t = \frac{1}{B}\sum_{b=1}^{B} \nabla\ell_{i_{t,b}}(\boldsymbol{\theta}_t). \tag{7.172}$$

Then we update the parameter vector by

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \boldsymbol{g}_t. \tag{7.173}$$

To distinguish gradient descent from mini-batch SGD, we call the former *full-batch GD* and reserve the term *gradient descent* for any algorithm whose core idea is to follow the gradient. Mini-batch SGD requires only $\mathcal{O}(B)$ gradient evaluations per step, much less than the $\mathcal{O}(n)$ required by full-batch GD. However, it introduces noise into the updates, which may slow down the convergence compared to full-batch GD. Still, in many practical scenarios, mini-batch SGD achieves better overall performance. Can we analyze its convergence theoretically?

7.2.1 SGD as a Stochastic Process

First, *what kind of mathematical object is SGD?* If there is no randomness, then $(\boldsymbol{\theta}_t)_{t\geq 0}$ in SGD is simply a sequence of vectors. With randomness, $(\boldsymbol{\theta}_t)_{t\geq 0}$ is a sequence of random vectors. Such a sequence is called a *stochastic process*.

Definition 7.28 (Stochastic process)**.** A *stochastic process* is a collection of random variables, or other random objects such as random vectors, $(X_t)_{t\in\mathcal{T}}$ defined on a common probability space and indexed by a set $\mathcal{T}$, called the *index set*.

- *Discrete-time stochastic process*: the index set $\mathcal{T}$ is countable, e.g., $\mathcal{T} = \mathbb{N}$.
- *Continuous-time stochastic process*: the index set $\mathcal{T}$ is a continuous interval, e.g., $\mathcal{T} = [0, \infty)$.

In fact, we have met discrete-time stochastic processes several times. For example, the random walk in Example 2.95 is a discrete-time stochastic process, where each X_t is the position of the drunk man after t steps. The martingale in Definition 3.50 is another example, a useful concept that helps us prove the McDiarmid inequality (Theorem 3.60) in Chapter 3. Now the iterates $(\boldsymbol{\theta}_t)_{t\geq 0}$ in SGD are another example of a stochastic process, where each $\boldsymbol{\theta}_t$ is a random vector. We have not seen any continuous-time stochastic processes yet, but we will see some examples in Chapter 8.

A special type of stochastic process is a *Markov chain*, which is also the type of stochastic process that SGD belongs to.

Definition 7.29 (Markov chain)**.** A *Markov chain* is a discrete-time stochastic process $(X_t)_{t\geq 0}$ such that for any sets $A_0, A_1, \ldots, A_{t+1}$ with

$$\Pr[X_0 \in A_0, X_1 \in A_1, \ldots, X_t \in A_t] > 0,$$

we have

$$\begin{aligned} &\Pr[X_{t+1} \in A_{t+1} \mid X_0 \in A_0, X_1 \in A_1, \ldots, X_t \in A_t] \\ &\quad = \Pr[X_{t+1} \in A_{t+1} \mid X_t \in A_t]. \end{aligned} \tag{7.174}$$

That is, given the current state X_t, the next state X_{t+1} is independent of the previous history.

A Markov chain can be viewed as an iterative algorithm with state X_t. At each step, the next state X_{t+1} is sampled according to a stochastic rule that depends only on the current state X_t. If the state space is finite, this transition rule can be summarized by the *transition probabilities*

$$p_t(x, y) := \Pr[X_{t+1} = y \mid X_t = x]. \tag{7.175}$$

More generally, for a state space such as $\mathbb{R}^d$, it is better to describe the transition rule on sets:

$$K_t(x, B) := \Pr[X_{t+1} \in B \mid X_t = x]. \tag{7.176}$$

Here $K_t(x, B)$ is the probability that the next state lies in B, given that the current state is x. Such a function K_t is called the *Markov transition kernel* of the Markov chain. In the finite-state case, this kernel can be connected to the transition probabilities by

$$K_t(x, B) = \sum_{y \in B} p_t(x, y). \tag{7.177}$$

If the transition kernel does not depend on t, that is, if there exists a kernel K such that $K_t = K$ for all t, then the chain is called a *time-homogeneous Markov chain*. Otherwise, it is called a *time-inhomogeneous Markov chain*.

SGD is a time-homogeneous Markov chain in parameter space. Now we define the most general form of SGD.

Definition 7.30 (Stochastic gradient descent)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be a differentiable loss function. Suppose we have access to a *stochastic gradient oracle* such that when queried at $\boldsymbol{\theta} \in \mathbb{R}^d$, the oracle returns a random vector $\boldsymbol{g}$ with distribution $\mathcal{G}_{\boldsymbol{\theta}}$. This random vector is assumed to provide an unbiased estimate of the gradient, in the sense that

$$\mathbb{E}_{\boldsymbol{g}\sim\mathcal{G}_{\boldsymbol{\theta}}}[\boldsymbol{g}] = \nabla\mathcal{L}(\boldsymbol{\theta}). \tag{7.178}$$

Given an initial parameter vector $\boldsymbol{\theta}_0 \in \mathbb{R}^d$, and a learning rate $\eta > 0$, *stochastic gradient descent* (SGD) proceeds step by step: at step t, it queries the oracle at the current point $\boldsymbol{\theta}_t$ and updates

$$\boldsymbol{g}_t \sim \mathcal{G}_{\boldsymbol{\theta}_t}, \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\boldsymbol{g}_t. \tag{7.179}$$

Once $\boldsymbol{\theta}_t$ is known, the distribution of the next iterate is determined by the oracle $\mathcal{G}_{\boldsymbol{\theta}_t}$ and the learning rate η. SGD can be viewed as a time-homogeneous Markov chain with the Markov transition kernel

$$K_t(\boldsymbol{\theta}, B) := \Pr_{\boldsymbol{g}\sim\mathcal{G}_{\boldsymbol{\theta}}}[\boldsymbol{\theta} - \eta\boldsymbol{g} \in B], \tag{7.180}$$

which does not change with time t.

Example 7.31 (Single-sample SGD)**.** In single-sample SGD, the stochastic gradient oracle is provided by sampling one data point from the dataset and computing the gradient of the loss on this single sample: when queried at $\boldsymbol{\theta}$, the oracle returns

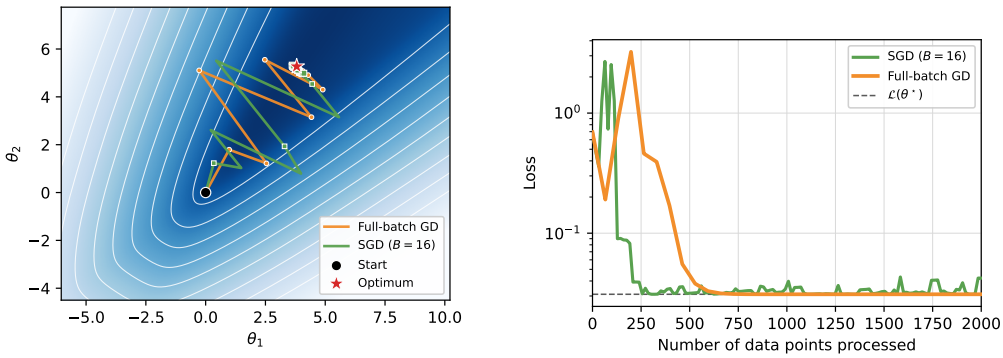
$$\boldsymbol{g} = \nabla\ell_i(\boldsymbol{\theta}), \tag{7.181}$$

where i is sampled uniformly at random from $[n]$. The mean and covariance of the gradient are given by

$$\mathbb{E}[\boldsymbol{g}] = \nabla\mathcal{L}(\boldsymbol{\theta}), \qquad \mathrm{Cov}(\boldsymbol{g}) = \boldsymbol{\Sigma}(\boldsymbol{\theta}), \tag{7.182}$$

where $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ is the *gradient noise covariance matrix*:

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n}\left(\nabla\ell_i(\boldsymbol{\theta}) - \nabla\mathcal{L}(\boldsymbol{\theta})\right)\left(\nabla\ell_i(\boldsymbol{\theta}) - \nabla\mathcal{L}(\boldsymbol{\theta})\right)^{\top}. \tag{7.183}$$



(a) Trajectories in parameter space.

(b) Loss curves.

Figure 7.10: Full-batch GD and mini-batch SGD for logistic regression. Full-batch GD follows a deterministic path, while mini-batch SGD follows a stochastic path due to gradient noise. Comparing the loss curves of SGD and full-batch GD under the same total number of data points processed, SGD reduces the loss faster than full-batch GD in the short term, but full-batch GD achieves a more steady loss reduction in the long term.

Example 7.32 (Mini-batch SGD)**.** Mini-batch SGD samples B data points from the dataset to reduce the variance of the gradient estimate in comparison to single-sample SGD. When queried at $\boldsymbol{\theta}$, the stochastic gradient oracle returns

$$\boldsymbol{g} = \frac{1}{B} \sum_{b=1}^{B} \nabla \ell_{i_b}(\boldsymbol{\theta}), \tag{7.184}$$

where $i_1, \ldots, i_B$ are i.i.d. samples from the uniform distribution on $[n]$. One can check that the mean and covariance of the gradient are

$$\mathbb{E}[\boldsymbol{g}] = \nabla \mathcal{L}(\boldsymbol{\theta}), \qquad \mathrm{Cov}(\boldsymbol{g}) = \frac{1}{B} \boldsymbol{\Sigma}(\boldsymbol{\theta}), \tag{7.185}$$

where $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ is the gradient noise covariance matrix. The factor $\frac{1}{B}$ here shows that the variance of the gradient estimate can be reduced by increasing the batch size B.

Remark 7.33. A subtle issue here is that in practice, mini-batches may be sampled without replacement. In this case, the next gradient estimate depends on previous ones, so SGD is no longer strictly a Markov chain. The version presented here, with independent sampling, is the most common one used in theoretical analysis.

Example 7.34 (SGD with Gaussian Noise)**.** When B is large enough, the central limit theorem implies that the gradient estimate is approximately Gaussian:

$$\mathcal{G}_{\boldsymbol{\theta}} \text{ can be modeled by } \mathcal{N}\left(\nabla\mathcal{L}(\boldsymbol{\theta}), \frac{1}{B}\boldsymbol{\Sigma}(\boldsymbol{\theta})\right). \tag{7.186}$$

Motivated by this approximation, some papers may study SGD with Gaussian noise, where the stochastic gradient oracle returns

$$\boldsymbol{g} = \nabla\mathcal{L}(\boldsymbol{\theta}) + \boldsymbol{\xi} \tag{7.187}$$

for some zero-mean Gaussian noise $\boldsymbol{\xi}$. This variant of SGD is just for theoretical analysis, not for practical use.

7.2.2 SGD for Smooth Convex Optimization

▶ Setup

For full-batch GD, the smooth strongly convex case is already well understood. In Theorem 7.25, we have already shown how to analyze the convergence of full-batch GD for smooth strongly convex objectives. The proof used two ideas: the descent lemma Lemma 7.17, which gives a one-step decrease of the loss based on the gradient norm, and the PL condition Definition 7.23, which says that the gradient norm cannot be too small when the loss is still far from the minimum.

Now we ask how much of this analysis survives for SGD. We keep the same objective function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$, which is assumed to be twice differentiable, α-strongly convex and β-smooth. At step t, SGD queries the stochastic gradient oracle at $\boldsymbol{\theta}_t$ and updates

$$\boldsymbol{g}_t \sim \mathcal{G}_{\boldsymbol{\theta}_t}, \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\boldsymbol{g}_t. \tag{7.188}$$

For the analysis here, we assume that the oracle is unbiased and has bounded noise: for all $\boldsymbol{\theta} \in \mathbb{R}^d$,

$$\mathbb{E}_{\boldsymbol{g}\sim\mathcal{G}_{\boldsymbol{\theta}}}[\boldsymbol{g}] = \nabla\mathcal{L}(\boldsymbol{\theta}), \qquad \mathbb{E}_{\boldsymbol{g}\sim\mathcal{G}_{\boldsymbol{\theta}}}\left[\|\boldsymbol{g} - \nabla\mathcal{L}(\boldsymbol{\theta})\|_2^2\right] \le \sigma^2. \tag{7.189}$$

After conditioning on the current iterate in SGD, these conditions become

$$\mathbb{E}[\boldsymbol{g}_t \mid \boldsymbol{\theta}_t] = \nabla\mathcal{L}(\boldsymbol{\theta}_t), \qquad \mathbb{E}\Big[\|\boldsymbol{g}_t - \nabla\mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 \mid \boldsymbol{\theta}_t\Big] \le \sigma^2. \tag{7.190}$$

Here, the parameter σ^2 measures the size of the noise introduced by the oracle. For mini-batch SGD, increasing the batch size B reduces the noise level σ^2 inversely with B. This bounded noise condition is common in optimization theory. While in some cases the gradient noise may not have a uniform upper bound over all $\boldsymbol{\theta} \in \mathbb{R}^d$, assuming a bound simplifies the analysis and makes it easier to understand how the noise level affects the convergence rate.

▶ Descent Lemma

We start with the analogue of the full-batch descent lemma Lemma 7.17. An immediate difference from full-batch GD is that the loss at each step is no longer deterministic. Even if $\boldsymbol{\theta}_t$ is fixed, the oracle output $\boldsymbol{g}_t$ can be random, and therefore the loss at the next iterate

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) \tag{7.191}$$

is a random variable. A natural way to analyze this is to take conditional expectation given $\boldsymbol{\theta}_t$.

Lemma 7.35 (Descent lemma for SGD)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be a β-smooth function, and assume that the stochastic gradient oracle satisfies the conditions in Equation (7.189). Suppose we perform one step of SGD with learning rate η from $\boldsymbol{\theta}_t$ to $\boldsymbol{\theta}_{t+1}$, i.e.,

$$\boldsymbol{g}_t \sim \mathcal{G}_{\boldsymbol{\theta}_t}, \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \boldsymbol{g}_t. \tag{7.192}$$

Then

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_{t+1}) \mid \boldsymbol{\theta}_t] \le \mathcal{L}(\boldsymbol{\theta}_t) - \eta \left(1 - \frac{\beta\eta}{2}\right) \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 + \frac{\beta\eta^2}{2}\sigma^2. \tag{7.193}$$

Proof. By Taylor expansion, we have that for some $\xi_t \in (0, 1)$,

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) = \mathcal{L}(\boldsymbol{\theta}_t) + \langle \nabla \mathcal{L}(\boldsymbol{\theta}_t), \boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t \rangle \tag{7.194}$$

$$+ \frac{1}{2}(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t)^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta}_t + \xi_t(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t))(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t). \tag{7.195}$$

Since $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \boldsymbol{g}_t$, this can be rewritten as

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) = \mathcal{L}(\boldsymbol{\theta}_t) - \eta \langle \nabla \mathcal{L}(\boldsymbol{\theta}_t), \boldsymbol{g}_t \rangle + \frac{\eta^2}{2} \boldsymbol{g}_t^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta}_t + \xi_t(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t))\boldsymbol{g}_t. \tag{7.196}$$

Also, since $\mathcal{L}$ is β-smooth, we have

$$\boldsymbol{g}_t^\top \nabla^2 \mathcal{L}(\boldsymbol{\theta}_t + \xi_t(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t))\boldsymbol{g}_t \le \beta \|\boldsymbol{g}_t\|_2^2. \tag{7.197}$$

Therefore,

$$\mathcal{L}(\boldsymbol{\theta}_{t+1}) \le \mathcal{L}(\boldsymbol{\theta}_t) - \eta \langle \nabla \mathcal{L}(\boldsymbol{\theta}_t), \boldsymbol{g}_t \rangle + \frac{\beta\eta^2}{2} \|\boldsymbol{g}_t\|_2^2. \tag{7.198}$$

Taking conditional expectation given $\boldsymbol{\theta}_t$, the linear term is

$$\mathbb{E}[\langle \nabla \mathcal{L}(\boldsymbol{\theta}_t), \boldsymbol{g}_t \rangle \mid \boldsymbol{\theta}_t] = \langle \nabla \mathcal{L}(\boldsymbol{\theta}_t), \mathbb{E}[\boldsymbol{g}_t \mid \boldsymbol{\theta}_t] \rangle = \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2. \tag{7.199}$$

For the quadratic term, write

$$\boldsymbol{g}_t = \nabla \mathcal{L}(\boldsymbol{\theta}_t) + \boldsymbol{\xi}_t, \qquad \mathbb{E}[\boldsymbol{\xi}_t \mid \boldsymbol{\theta}_t] = \mathbf{0}. \tag{7.200}$$

Then the cross term vanishes after conditioning:

$$\mathbb{E}\Big[\|\boldsymbol{g}_t\|_2^2 \mid \boldsymbol{\theta}_t\Big] = \mathbb{E}\Big[\|\nabla \mathcal{L}(\boldsymbol{\theta}_t) + \boldsymbol{\xi}_t\|_2^2 \mid \boldsymbol{\theta}_t\Big] \tag{7.201}$$

$$= \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 + 2\langle \nabla \mathcal{L}(\boldsymbol{\theta}_t), \mathbb{E}[\boldsymbol{\xi}_t \mid \boldsymbol{\theta}_t] \rangle + \mathbb{E}\Big[\|\boldsymbol{\xi}_t\|_2^2 \mid \boldsymbol{\theta}_t\Big] \tag{7.202}$$

$$= \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 + \mathbb{E}\Big[\|\boldsymbol{\xi}_t\|_2^2 \mid \boldsymbol{\theta}_t\Big] \le \|\nabla \mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 + \sigma^2. \tag{7.203}$$

Substituting these two estimates proves Equation (7.193). □

The above descent lemma for SGD gives almost the same statement as the full-batch descent lemma (Lemma 7.17), except for the additional positive term that appears at the end:

$$\frac{\beta\eta^2}{2}\sigma^2. \tag{7.204}$$

This term is induced by the gradient noise. From this term, we can see which factors make SGD worse than full-batch GD: SGD can be much worse than full-batch GD when the smoothness constant β is large, the learning rate η is large, or the gradient noise level σ^2 is large.

▶ Convergence Rate

As in the full-batch case, a one-step descent lemma is not enough by itself. To obtain a convergence rate, we use strong convexity to provide a lower bound on the gradient norm, and proceed similarly to Theorem 7.25 for full-batch GD.

Theorem 7.36. Let $\mathcal{L}(\boldsymbol{\theta})$ be a twice differentiable, α-strongly convex and β-smooth objective function, and assume that the stochastic gradient oracle satisfies the conditions in Equation (7.189). For SGD with learning rate $\eta \in (0, 2/\beta)$, we have

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_t)] - \mathcal{L}_{\min} \leq \rho(\eta)^t(\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}) + \frac{\beta\eta\sigma^2}{4\alpha(1-\beta\eta/2)}, \tag{7.205}$$

where

$$\rho(\eta) := 1 - 2\alpha\eta + \alpha\beta\eta^2. \tag{7.206}$$

Proof. By the descent lemma from Equation (7.193),

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_{t+1}) \mid \boldsymbol{\theta}_t] \leq \mathcal{L}(\boldsymbol{\theta}_t) - \eta\left(1 - \frac{\beta\eta}{2}\right)\|\nabla\mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 + \frac{\beta\eta^2}{2}\sigma^2. \tag{7.207}$$

By the PL condition implied by strong convexity in Lemma 7.24,

$$\|\nabla\mathcal{L}(\boldsymbol{\theta}_t)\|_2^2 \geq 2\alpha(\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min}). \tag{7.208}$$

Putting these two inequalities together, we get

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_{t+1}) \mid \boldsymbol{\theta}_t] \leq \mathcal{L}(\boldsymbol{\theta}_t) - \eta\left(1 - \frac{\beta\eta}{2}\right)\cdot 2\alpha(\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min}) + \frac{\beta\eta^2}{2}\sigma^2. \tag{7.209}$$

Subtracting $\mathcal{L}_{\min}$ from both sides, we get

$$\begin{aligned}\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_{t+1}) - \mathcal{L}_{\min} \mid \boldsymbol{\theta}_t] \leq \mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min} \\ - \eta\left(1 - \frac{\beta\eta}{2}\right)\cdot 2\alpha(\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min}) + \frac{\beta\eta^2}{2}\sigma^2.\end{aligned} \tag{7.210}$$

Rearranging gives

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_{t+1}) - \mathcal{L}_{\min} \mid \boldsymbol{\theta}_t] \leq \underbrace{\left(1 - 2\alpha\eta + \alpha\beta\eta^2\right)}_{=:\rho(\eta)}(\mathcal{L}(\boldsymbol{\theta}_t) - \mathcal{L}_{\min}) + \frac{\beta\eta^2}{2}\sigma^2. \quad (7.211)$$

Taking expectation over $\boldsymbol{\theta}_t$, we obtain

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_{t+1})] - \mathcal{L}_{\min} \leq \rho(\eta)(\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_t)] - \mathcal{L}_{\min}) + \frac{\beta\eta^2}{2}\sigma^2. \quad (7.212)$$

Iterating this recursion,

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_t)] - \mathcal{L}_{\min} \leq \rho(\eta)^t(\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}) + \frac{\beta\eta^2}{2}\sigma^2 \sum_{s=0}^{t-1} \rho(\eta)^s \quad (7.213)$$

$$\leq \rho(\eta)^t(\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}) + \frac{\beta\eta^2\sigma^2}{2(1-\rho(\eta))}. \quad (7.214)$$

Since

$$1 - \rho(\eta) = 2\alpha\eta\left(1 - \frac{\beta\eta}{2}\right), \quad (7.215)$$

the last term can be simplified as

$$\frac{\beta\eta^2\sigma^2}{2(1-\rho(\eta))} = \frac{\beta\eta^2\sigma^2}{2(2\alpha\eta(1-\frac{\beta\eta}{2}))} = \frac{\beta\eta\sigma^2}{4\alpha(1-\beta\eta/2)}, \quad (7.216)$$

which proves the theorem. □

Remark 7.37. For anyone who wonders if it is possible to instead prove that $\mathcal{L}(\boldsymbol{\theta}_t)$ is small with high probability, use Azuma's inequality!

▶ How to Choose the Learning Rate

Let us take a closer look at the convergence rate of SGD in Theorem 7.36:

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_t)] - \mathcal{L}_{\min} \leq \underbrace{\rho(\eta)^t(\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min})}_{\text{contraction term}} + \underbrace{\frac{\beta\eta\sigma^2}{4\alpha(1-\beta\eta/2)}}_{\text{noise term}}. \quad (7.217)$$

The first term is the same geometric contraction term as in full-batch GD. But the second term is new, as it comes from gradient noise.

To choose the learning rate, a natural first idea is to make the contraction factor $\rho(\eta)$ as small as possible, following the same strategy as in full-batch GD. This leads to the same optimal learning rate as in full-batch GD: $\eta^* = 1/\beta$. Then Equation (7.217) can be rewritten as

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_t)] - \mathcal{L}_{\min} \leq \left(1 - \frac{1}{\kappa}\right)^t (\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}) + \frac{\sigma^2}{2\alpha}. \quad (7.218)$$

where $\kappa := \beta/\alpha$ is the condition number. The contraction term decays exponentially, but the noise term remains constant! This shows that we should not only minimize $\rho(\eta)$, but also take the noise term into account.

Let us now look more carefully at how to choose the learning rate. The contraction term and the noise term are actually competing with each other. A larger η makes the contraction term decay faster, but it also increases the noise term. A smaller η reduces the noise term, but it slows down the contraction. Therefore, the learning rate should be chosen by balancing these two effects.

Now fix the total number of steps T to be a large number. If we want the final expected loss gap to go to zero as $T \to \infty$, the learning rate cannot stay fixed. Otherwise, the noise term dominates and the expected loss gap does not go to zero.

For small η, the contraction term can be approximated as

$$\rho(\eta)^T = \exp\left(-2\alpha\eta T + \mathcal{O}(\eta^2 T)\right). \tag{7.219}$$

Thus, the convergence bound has the form

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_T)] - \mathcal{L}_{\min} = \mathcal{O}\Big(\exp(-2\alpha\eta T + \mathcal{O}(\eta^2 T)) + \eta\Big). \tag{7.220}$$

Balancing the two terms gives

$$\eta^* = \Theta\left(\frac{\log T}{T}\right), \qquad \mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_T)] - \mathcal{L}_{\min} = \mathcal{O}\left(\frac{\log T}{T}\right). \tag{7.221}$$

Note that this is a very different behavior from full-batch GD, since now the best learning rate is no longer a constant, but depends on the training horizon T.

In fact, one can do better by letting the learning rate decay over time. For example, choose

$$\eta_t = \frac{2}{\alpha t + \beta}, \tag{7.222}$$

With this schedule, one can show that the convergence rate is improved to

$$\mathbb{E}[\mathcal{L}(\boldsymbol{\theta}_T)] - \mathcal{L}_{\min} = \mathcal{O}\left(\frac{1}{T}\right), \tag{7.223}$$

which removes the $\log T$ factor in the case of constant learning rate.

Overall, these results show that in SGD, the learning rate should balance progress and noise over the course of training, and its best choice can depend on the training horizon.

Exercises

7.1. In Section 7.1.2, we have shown that the convergence rate of gradient descent on quadratic objectives is controlled by the following quantity:

$$\rho(\eta) := \max_{i\in[d]} |1 - \eta\lambda_i|.$$

Under the optimal choice of η, which is $\eta = \frac{2}{\lambda_{\min}+\lambda_{\max}}$, we have

$$\rho^* = 1 - \frac{2}{1+\kappa}.$$

(a) Suppose you really hate oscillatory behavior and really want to make the convergence entirely stable, which means that you want to ensure $1 - \eta\lambda_i \in [0, 1)$ for all $i \in [d]$. Show that under the optimal choice of η in this regime, we have $\rho(\eta) = 1 - \frac{1}{\kappa}$.

Note: This means if you insist on having a stable convergence, the number of steps to reach $\mathcal{L}(\boldsymbol{\theta}_t) \le \mathcal{L}_{\min} + \epsilon$ will be increased from $T(\epsilon) \sim \frac{\kappa}{4}\log\frac{1}{\epsilon}$ (Equation (7.68)) to $T(\epsilon) \sim \frac{\kappa}{2}\log\frac{1}{\epsilon}$.

(b) In practice, we cannot compute the optimal learning rate exactly, since computing the maximum and minimum eigenvalues can be even slower than solving the optimization problem itself. A common practice is to use grid search: start with a base learning rate η_0, and then try gradient descent with learning rate $\eta_k = \alpha^k\eta_0$ for many $k \in \mathbb{Z}$, where $\alpha > 1$ is a constant. Suppose that we are unlucky with the choice of η_0, but we search very thoroughly over k. Then the convergence rate is controlled by

$$\rho^*_\alpha := \sup_{\eta_0>0} \inf_{k\in\mathbb{Z}} \rho(\alpha^k\eta_0).$$

Express ρ^*_α in terms of α and κ. You can assume that κ is so large that $\alpha > 1 + \frac{1}{\kappa}$.

Note: Typical choices of α in deep learning are $\sqrt{2}, 2, \sqrt{10}, 10$, depending on your compute budget.

Note: For those who get used to measuring time complexity with big-$\mathcal{O}$ notation and get annoyed by this very precise calculation with constant factors: constant factors can be very important in practice. For example, reducing a 6-month training time by a factor of 2 is a huge deal!

7.2. Let $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{H}\boldsymbol{\theta}$ be a quadratic objective, where $\boldsymbol{H} \in \mathbb{R}^{d\times d}$ is symmetric. Assume that $\boldsymbol{H}$ has at least one negative eigenvalue. Show that if we run gradient descent with any learning rate $0 < \eta < \frac{2}{\lambda_{\max}(\boldsymbol{H})}$ and random initialization $\boldsymbol{\theta}_0 \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, then for any $M > 0$, with probability 1 we have $\mathcal{L}(\boldsymbol{\theta}_t) < -M$ for some $t \ge 0$.

Note: In general, we say that $\boldsymbol{\theta}$ is a saddle point if $\nabla\mathcal{L}(\boldsymbol{\theta}) = \boldsymbol{0}$ but $\boldsymbol{\theta}$ is not a local minimizer of $\mathcal{L}$. It can be shown that if $\nabla^2\mathcal{L}(\boldsymbol{\theta})$ has at least one negative eigenvalue at a saddle point $\boldsymbol{\theta}$, then it happens with probability 1

that gradient descent with random initialization will not converge to this saddle point. Check the *stable manifold theorem* if you are interested.

Chapter 8

Continuous Approximations

In the previous chapter (Chapter 7), we analyzed full-batch GD and stochastic gradient descent directly as discrete-time algorithms, and we have seen that these algorithms can exhibit very rich behaviors despite the simplicity of their update rules. However, their discrete nature makes them difficult to analyze, and convergence results are usually established through a series of technical lemmas.

If the goal is to just obtain a rough understanding of the behavior of these algorithms, an alternative viewpoint is to model them as continuous-time processes. More concretely, we replace the discrete sequence

$$\boldsymbol{\theta}_0, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots \tag{8.1}$$

by a continuous trajectory

$$\boldsymbol{\theta}(t), \quad t \in [0, \infty). \tag{8.2}$$

Under this view, the discrete sequence is some form of discretization of a continuous trajectory, obtained by taking snapshots of the trajectory at discrete time points. This modeling is of course not exact, but sometimes it can help us understand discrete sequences more intuitively.

In this chapter, we discuss two continuous-time approximations that frequently arise in the study of artificial intelligence. We first introduce *ordinary differential equations* (ODEs), then use them to provide a continuous approximation of full-batch gradient descent, leading to the notion of gradient flow. We then turn to SGD, whose stochastic updates make it difficult to describe by ODEs. To address this, we introduce *stochastic differential equations* (SDEs), which is essentially a stochastic version of ODEs. We next present an alternative viewpoint of SDEs based on *partial differential equations* (PDEs), which describes the evolution of probability densities over time rather than individual trajectories. Finally, we illustrate how SDEs can be used to develop a class of generative models that outperform GANs, known as *diffusion models*.

8.1 Ordinary Differential Equations

An ordinary differential equation (ODE) is a mathematical equation that describes the evolution of a continuous trajectory over time. In this section, we focus on first-order ODEs, where the time derivative is a function of the current state only:

$$\dot{\boldsymbol{x}}(t) = F(\boldsymbol{x}(t)). \tag{8.3}$$

Here $\boldsymbol{x}(t)$ is the state at time t, and the function F tells us how the state should change when the current state is $\boldsymbol{x}(t)$. In this formula and throughout this section, putting a dot above a variable denotes its derivative with respect to time, namely the *time derivative*:

$$\dot{\boldsymbol{x}}(t) := \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t}(t). \tag{8.4}$$

8.1.1 Gradient Flow Approximation

Recall that full-batch GD updates the parameter by

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_k). \tag{8.5}$$

Subtracting $\boldsymbol{\theta}_k$ from both sides and dividing by η gives

$$\frac{\boldsymbol{\theta}_{k+1} - \boldsymbol{\theta}_k}{\eta} = -\nabla \mathcal{L}(\boldsymbol{\theta}_k). \tag{8.6}$$

The left-hand side now looks like a finite-difference approximation of a time derivative. To make this idea precise, we introduce a continuous trajectory $\boldsymbol{\theta}(t)$ and view the discrete iterates as samples of this trajectory. The denominator in the finite difference is η, so we use η as the time gap between two consecutive samples. Thus we set

$$t_k = k\eta, \tag{8.7}$$

and regard $\boldsymbol{\theta}_k$ as an approximation of $\boldsymbol{\theta}(t_k)$. Then

$$\frac{\boldsymbol{\theta}_{k+1} - \boldsymbol{\theta}_k}{\eta} \approx \frac{\boldsymbol{\theta}(t_{k+1}) - \boldsymbol{\theta}(t_k)}{t_{k+1} - t_k} \approx \dot{\boldsymbol{\theta}}(t_k), \tag{8.8}$$

Therefore, it is natural to constrain $\boldsymbol{\theta}(t)$ to satisfy

$$\dot{\boldsymbol{\theta}}(t) = -\nabla \mathcal{L}(\boldsymbol{\theta}(t)). \tag{8.9}$$

More generally, we have the following definition.

Definition 8.1 (Gradient flow)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be differentiable. A differentiable trajectory $\boldsymbol{\theta}(t)$ with $t \in [0, \infty)$ is called a *gradient flow* (GF) of $\mathcal{L}$ if it satisfies the following ODE:

$$\dot{\boldsymbol{\theta}}(t) = -\nabla \mathcal{L}(\boldsymbol{\theta}(t)). \tag{8.10}$$

Gradient flow is a useful tool for understanding full-batch GD for two reasons. First, when the learning rate is very small, full-batch GD and GF should have

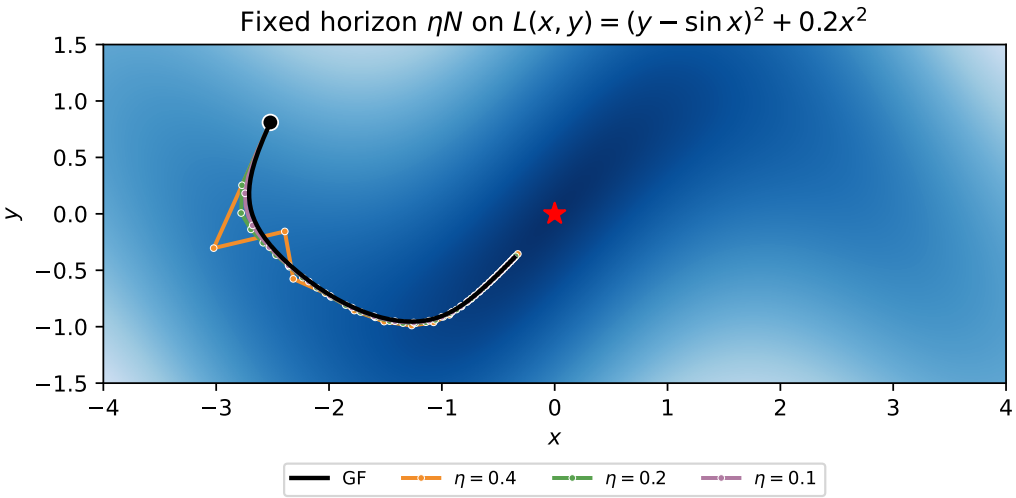


Figure 8.1: GD trajectories on $L(x, y) = (y - \sin x)^2 + 0.2x^2$ with the same continuous-time horizon ηN. As η becomes smaller, the discrete GD trajectory tracks the GF trajectory in the valley more closely.

almost the same trajectory. If $\boldsymbol{\theta}(t)$ is the GF with the same initialization as GD, i.e., $\boldsymbol{\theta}(0) = \boldsymbol{\theta}_0$, then we expect

$$\boldsymbol{\theta}_k \approx \boldsymbol{\theta}(k\eta) \quad \text{for all } k \geq 0. \tag{8.11}$$

Second, even if small learning rates may not be our main interest, GF can still help us understand full-batch GD. A reasonable GD analysis should not break when the learning rate is made smaller, as it did in the previous chapter. For example, the descent lemma gives a one-step decrease whenever $0 < \eta < 2/\beta$ in Lemma 7.17, and the convergence analysis for the strongly convex case holds for all $0 < \eta < 2/\beta$ in Theorem 7.25. Therefore, when we do not have a good understanding of full-batch GD with finite learning rates, it is useful to first analyze its continuous-time approximation, GF, expect that many key ingredients of the analysis will carry over to the discrete-time case.

8.1.2 Gradient Flow for Smooth Convex Optimization

Recall that the descent lemma for discrete GD in Lemma 7.17 shows that, when the learning rate is chosen properly, one GD step decreases the loss. To analyze the convergence of GF, we want the same kind of guarantee. Since a GF trajectory changes continuously, the one-step loss decrease should be replaced by the time derivative of the loss along the trajectory.

Lemma 8.2 (Descent lemma for gradient flow)**.** Let $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be differentiable, and $\boldsymbol{\theta}(t)$ be a GF of $\mathcal{L}$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{L}(\boldsymbol{\theta}(t)) = -\|\nabla\mathcal{L}(\boldsymbol{\theta}(t))\|_2^2 \leq 0. \tag{8.12}$$

Proof. By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{L}(\boldsymbol{\theta}(t)) = \langle \nabla \mathcal{L}(\boldsymbol{\theta}(t)), \dot{\boldsymbol{\theta}}(t) \rangle. \tag{8.13}$$

Since $\boldsymbol{\theta}(t)$ is a GF, we have $\dot{\boldsymbol{\theta}}(t) = -\nabla \mathcal{L}(\boldsymbol{\theta}(t))$. Substituting this into the previous equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{L}(\boldsymbol{\theta}(t)) = -\|\nabla \mathcal{L}(\boldsymbol{\theta}(t))\|_2^2, \tag{8.14}$$

which proves the lemma. □

Compared with the discrete descent lemma, the proof is much cleaner. In the discrete case, we needed a Taylor expansion to control the error from one finite step. For GF, the step has become infinitesimal, so we just need to apply the chain rule. The result is also an equality rather than just an inequality, which tells us exactly how fast the loss decreases along the trajectory.

Now we further assume that $\mathcal{L}$ is strongly convex and derive a convergence rate for GF. By Lemma 7.24, strong convexity gives the PL condition, which relates the gradient norm to the loss gap. Combining it with the GF descent lemma gives the following theorem.

Theorem 8.3. Assume $\mathcal{L}(\boldsymbol{\theta})$ be a twice differentiable and α-strongly convex objective function. Let $\mathcal{L}_{\min}$ be the minimum value of $\mathcal{L}(\boldsymbol{\theta})$. If $\boldsymbol{\theta}(t)$ is a GF of $\mathcal{L}$, then

$$\mathcal{L}(\boldsymbol{\theta}(t)) - \mathcal{L}_{\min} \leq e^{-2\alpha t}(\mathcal{L}(\boldsymbol{\theta}(0)) - \mathcal{L}_{\min}). \tag{8.15}$$

Proof. Let $\Delta(t) = \mathcal{L}(\boldsymbol{\theta}(t)) - \mathcal{L}_{\min}$. By Lemma 8.2,

$$\dot{\Delta}(t) = -\|\nabla \mathcal{L}(\boldsymbol{\theta}(t))\|_2^2. \tag{8.16}$$

By Lemma 7.24, strong convexity implies the PL condition:

$$\|\nabla \mathcal{L}(\boldsymbol{\theta}(t))\|_2^2 \geq 2\alpha\Delta(t). \tag{8.17}$$

Therefore,

$$\dot{\Delta}(t) \leq -2\alpha\Delta(t). \tag{8.18}$$

Dividing both sides by $\Delta(t)$ gives

$$\frac{\dot{\Delta}(t)}{\Delta(t)} \leq -2\alpha. \tag{8.19}$$

Since $\frac{\dot{\Delta}(t)}{\Delta(t)} = \frac{\mathrm{d}}{\mathrm{d}t}\log\Delta(t)$, integrating from 0 to t gives

$$\log\Delta(t) - \log\Delta(0) \leq -2\alpha t. \tag{8.20}$$

Exponentiating both sides proves the theorem. □

This is the continuous-time analogue of the discrete-time bound for full-batch GD in Theorem 7.25:

$$\mathcal{L}(\boldsymbol{\theta}_k) - \mathcal{L}_{\min} \leq \rho(\eta)^k(\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}), \quad \text{where} \quad \rho(\eta) = 1 - 2\alpha\eta + \alpha\beta\eta^2. \tag{8.21}$$

Let us see how this discrete-time convergence rate becomes the exponential decay rate in the GF bound when the learning rate goes to zero. Fix a time horizon T. As one GD step corresponds to a time interval of length η, reaching time T requires T/η steps. The discrete-time bound then becomes

$$\mathcal{L}(\boldsymbol{\theta}_K) - \mathcal{L}_{\min} \leq \rho(\eta)^{T/\eta}(\mathcal{L}(\boldsymbol{\theta}_0) - \mathcal{L}_{\min}). \tag{8.22}$$

As $\eta \to 0$, the contraction factor in the GD bound converges to

$$\rho(\eta)^{T/\eta} = \left(1 - 2\alpha\eta + \alpha\beta\eta^2\right)^{T/\eta} \tag{8.23}$$

$$= \exp\left(T \cdot \frac{1}{\eta}\log(1 - 2\alpha\eta + \alpha\beta\eta^2)\right) \tag{8.24}$$

$$= \exp\left(-2\alpha T + \mathcal{O}(\eta)\right) \to e^{-2\alpha T}, \tag{8.25}$$

which is exactly the decay factor in the GF bound at the same time T. In this sense, the GF convergence rate is the continuous-time limit of the GD convergence rate.

The continuous approximation simplifies the convergence proof and makes the main ideas easier to see. The first insight is that we need a descent lemma, which controls how the loss decreases. The second insight is that the descent lemma alone is not enough. We also need a lower bound on the gradient norm in terms of the loss gap, which turns the descent lemma into a convergence rate. The corresponding GD proof is more complicated because it also needs to track the effect of the learning rate, but this is necessary when we really want to understand learning rates theoretically.

8.1.3 Gradient Descent as Euler Method

In Section 8.1.1, our derivation of the gradient flow approximation involves several heuristic approximation steps, whose validity is not entirely clear. Now we ask the following question: *when the learning rate is very small, how close is the gradient flow approximation to the full-batch GD trajectory?*

To answer this, we take a higher-level view and study a more general problem. Consider the following ODE:

$$\dot{\boldsymbol{x}}(t) = F(\boldsymbol{x}(t)), \qquad \boldsymbol{x}(0) = \boldsymbol{x}_0. \tag{8.26}$$

A standard way to discretize this ODE is called the *Euler method*, which is the method introduced by Leonhard Euler in his book *Institutiones Calculi Integralis* (Foundations of Integral Calculus) [Eul70]. The idea is very simple: instead of moving $\boldsymbol{x}$ continuously, we move it in discrete steps of size η.

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \eta F(\boldsymbol{x}_k). \tag{8.27}$$

Here $\eta > 0$ is called the *step size*.

It is easy to see that gradient descent is a special case of the Euler method. When the ODE is the gradient flow of a loss function $\mathcal{L}$, we have $F(\boldsymbol{x}) = -\nabla\mathcal{L}(\boldsymbol{x})$, so the Euler method becomes $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \eta\nabla\mathcal{L}(\boldsymbol{x}_k)$, which is full-batch GD.

We now state a general theorem that establishes an error bound that quantifies how well the Euler method approximates the ODE solution.

Theorem 8.4 (Error bound for Euler method)**.** Let $F : \mathbb{R}^d \to \mathbb{R}^d$ be an L-Lipschitz function with $L > 0$. That is,

$$\|F(\boldsymbol{x}) - F(\boldsymbol{y})\|_2 \leq L\|\boldsymbol{x} - \boldsymbol{y}\|_2 \quad \text{for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d. \tag{8.28}$$

Let $\eta > 0$ be the step size and N be a positive integer, and let $T = \eta N$. Let $\boldsymbol{x}(t)$ be the solution to the ODE

$$\dot{\boldsymbol{x}}(t) = F(\boldsymbol{x}(t)), \qquad \boldsymbol{x}(0) = \boldsymbol{x}_0. \tag{8.29}$$

Assume that for some $M \geq 0$, $\|\ddot{\boldsymbol{x}}(t)\|_2 \leq M$ for all $t \in [0, T]$. Then the Euler method with step size $\eta > 0$,

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \eta F(\boldsymbol{x}_k) \quad \text{for } 0 \leq k < N, \tag{8.30}$$

tracks the ODE solution $\boldsymbol{x}(t)$ with the following error bound:

$$\max_{0 \leq k \leq N} \|\boldsymbol{x}_k - \boldsymbol{x}(t_k)\|_2 \leq \frac{M}{2L}\left(e^{LT} - 1\right)\eta, \tag{8.31}$$

where $t_k = k\eta$ for all $0 \leq k \leq N$.

Proof. By the integral form of Taylor expansion,

$$\boldsymbol{x}(t_{k+1}) = \boldsymbol{x}(t_k) + \eta\dot{\boldsymbol{x}}(t_k) + \int_{t_k}^{t_{k+1}} \int_{t_k}^{s} \ddot{\boldsymbol{x}}(r)\, \mathrm{d}r\, \mathrm{d}s \tag{8.32}$$

$$= \boldsymbol{x}(t_k) + \eta F(\boldsymbol{x}(t_k)) + \frac{\eta^2}{2}\boldsymbol{r}_k, \tag{8.33}$$

where $\boldsymbol{r}_k$ is a vector with $\|\boldsymbol{r}_k\|_2 \leq M$.

Let $\boldsymbol{\delta}_k := \boldsymbol{x}_k - \boldsymbol{x}(t_k)$. According to the Euler method,

$$\boldsymbol{\delta}_{k+1} = \boldsymbol{x}_{k+1} - \boldsymbol{x}(t_{k+1}) \tag{8.34}$$

$$= \boldsymbol{x}_k + \eta F(\boldsymbol{x}_k) - \boldsymbol{x}(t_k) - \eta F(\boldsymbol{x}(t_k)) - \frac{\eta^2}{2}\boldsymbol{r}_k. \tag{8.35}$$

Therefore,

$$\|\boldsymbol{\delta}_{k+1}\|_2 \leq \|\boldsymbol{x}_k - \boldsymbol{x}(t_k)\|_2 + \eta\|F(\boldsymbol{x}_k) - F(\boldsymbol{x}(t_k))\|_2 + \frac{\eta^2}{2}\|\boldsymbol{r}_k\|_2 \tag{8.36}$$

$$\leq \|\boldsymbol{\delta}_k\|_2 + \eta L\|\boldsymbol{\delta}_k\|_2 + \frac{\eta^2}{2}M \tag{8.37}$$

$$\leq (1 + L\eta)\|\boldsymbol{\delta}_k\|_2 + \frac{M}{2}\eta^2. \tag{8.38}$$

Since $\boldsymbol{\delta}_0 = \mathbf{0}$, iterating the recursion gives

$$\|\boldsymbol{\delta}_k\|_2 \leq \frac{M}{2}\eta^2 \sum_{j=0}^{k-1}(1 + L\eta)^j = \frac{M}{2}\eta^2 \cdot \frac{(1 + L\eta)^k - 1}{L\eta} \tag{8.39}$$

$$= \frac{M}{2L}\Big((1 + L\eta)^k - 1\Big)\eta. \tag{8.40}$$

For all $0 \le k \le N$, we have $k\eta \le T$. Finally, noting that $(1+L\eta)^k \le e^{Lk\eta} \le e^{LT}$, we prove the bound. □

The theorem makes precise how the Euler approximation behaves as the step size decreases. Fix the time horizon $T = \eta N$. If we make η smaller, which means we take more steps $N = T/\eta$ over the same time interval, then the error bound $\frac{M}{2L}\left(e^{LT} - 1\right)\eta$ implies that the approximation error goes to zero linearly in η. This means a smaller step size makes the discrete Euler trajectory closer to the ODE solution.

The other quantities in the bound also tell us what makes the approximation easier or harder. The constant M appears in the numerator, so a larger M gives a worse bound. This is intuitive because M controls how much the ODE solution bends over time. If the ODE solution bends a lot, then approximating it by the piecewise linear Euler trajectory inevitably loses accuracy.

The Lipschitz constant L appears in the factor $e^{LT} - 1$, so a larger L gives a worse bound. This is also intuitive because a larger L means that F changes more rapidly near the current point $\boldsymbol{x}_k$. Starting from the same point, Euler follows the fixed vector $F(\boldsymbol{x}_k)$ during the whole step, while the ODE follows a vector field that changes continuously, so the two trajectories can deviate more when L is larger. The time horizon T appears in the same factor $e^{LT} - 1$. A larger T also gives a worse bound, which is intuitive because local approximation errors have more time to accumulate.

Now return to gradient flow. In this case, $F(\boldsymbol{\theta}) = -\nabla\mathcal{L}(\boldsymbol{\theta})$. If $\mathcal{L}$ is β-smooth, then $\nabla\mathcal{L}$ is β-Lipschitz, so F is also β-Lipschitz. Thus the constant L in the error bound can be taken as β.

The remaining quantity M is an upper bound on the acceleration of the GF trajectory. Since $\dot{\boldsymbol{\theta}}(t) = -\nabla\mathcal{L}(\boldsymbol{\theta}(t))$, differentiating in time gives

$$\ddot{\boldsymbol{\theta}}(t) = -\nabla^2\mathcal{L}(\boldsymbol{\theta}(t))\dot{\boldsymbol{\theta}}(t) = \nabla^2\mathcal{L}(\boldsymbol{\theta}(t))\nabla\mathcal{L}(\boldsymbol{\theta}(t)). \tag{8.41}$$

Therefore, we can take M as an upper bound on $\|\nabla^2\mathcal{L}(\boldsymbol{\theta}(t))\nabla\mathcal{L}(\boldsymbol{\theta}(t))\|_2$ over the time interval $[0, T]$. In particular, if $\mathcal{L}$ is β-smooth and $\|\nabla\mathcal{L}(\boldsymbol{\theta}(t))\|_2 \le G$, then one simple choice is

$$M := \beta G. \tag{8.42}$$

With these choices, the error bound becomes the following:

$$\max_{0\le k\le N} \|\boldsymbol{\theta}_k - \boldsymbol{\theta}(t_k)\|_2 \le \frac{G}{2}\left(e^{\beta T} - 1\right)\eta. \tag{8.43}$$

Therefore, full-batch GD tracks the GF closely when the learning rate η is small, the gradient norm at each step is small, the smoothness constant β is small, and the time horizon T is not too long.

8.2 Stochastic Differential Equations

Although gradient flow provides a useful continuous approximation of full-batch GD, this ODE-based viewpoint is not sufficient for understanding SGD. The reason is that SGD is inherently a stochastic process, whereas gradient flow and other ODE models are fully deterministic. Any deterministic approximation of SGD necessarily discards gradient noise, but this noise plays a crucial role in shaping the behavior of SGD.

Therefore, any meaningful continuous-time approximation of SGD must itself be stochastic. This motivates the introduction of *stochastic differential equations* (SDEs), which can be viewed as stochastic analogues of ODEs. An ODE describes the evolution of a deterministic trajectory over time, whereas an SDE describes the evolution of a continuous-time stochastic process (Definition 7.28). We now introduce SDEs in detail.

8.2.1 Stochastic Difference Equations

First, let us consider a discrete-time stochastic process $(X_k)_{k\geq 0}$ that evolves according to the following recursion:

$$X_{k+1} = X_k + \eta\mu(X_k) + \varepsilon\xi_k, \quad \xi_k \sim \mathcal{Z}(X_k). \tag{8.44}$$

Here, $\mu : \mathbb{R} \to \mathbb{R}$ is a function that determines the change of X_k in expectation, $\mathcal{Z}(x)$ is a zero-mean probability distribution on $\mathbb{R}$ that determines the random fluctuation of X_k. $\mu(X_k)$ is commonly referred to as the *drift* term of the process, and ξ_k is the *noise* term, or the *diffusion* term of the process. The drift and noise terms are scaled by η and ε, respectively, which are both positive constants. We refer to Equation (8.44) as a *stochastic difference equation*.

SGD can be seen as a multidimensional variant of the stochastic difference equation. Following the stochastic gradient oracle notation from the previous chapter, one step of SGD is given by

$$\boldsymbol{g}_k \sim \mathcal{G}_{\boldsymbol{\theta}_k}, \tag{8.45}$$

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta\boldsymbol{g}_k \tag{8.46}$$

$$= \boldsymbol{\theta}_k - \eta\nabla\mathcal{L}(\boldsymbol{\theta}_k) - \eta\boldsymbol{\xi}_k, \tag{8.47}$$

where $\boldsymbol{\xi}_k := \boldsymbol{g}_k - \nabla\mathcal{L}(\boldsymbol{\theta}_k)$. Therefore, we can regard $-\nabla\mathcal{L}(\boldsymbol{\theta}_k)$ as the drift term and $-\boldsymbol{\xi}_k$ as the noise term. Under this view, we have noise scale $\varepsilon = \eta$.

Now we ask the following question: *if we take $\eta \to 0$, does the stochastic difference equation approach any continuous-time stochastic process?* For this to be the case, we have to scale the noise level ε appropriately when $\eta \to 0$.

We now give a heuristic argument to determine the correct scaling of ε. For simplicity, we ignore the state dependence of the drift and noise terms, and assume that $\mu(x) = \mu$ is a constant and $\mathcal{Z}(x) = \mathcal{Z}$ is a fixed distribution with constant variance σ^2. Now we consider a time interval $[0, T]$ and associate each step with a time interval of length η, leading to $N = T/\eta$ steps. After N steps,

the recursion gives

$$X_N = X_0 + N\eta\mu + \varepsilon \sum_{k=0}^{N-1} \xi_k. \tag{8.48}$$

Therefore, the mean and variance of X_N are given by

$$\mathbb{E}[X_N] = X_0 + N\eta\mu = X_0 + T\mu, \tag{8.49}$$

$$\mathrm{Var}(X_N) = \varepsilon^2 \sum_{k=0}^{N-1} \mathrm{Var}(\xi_k) = \varepsilon^2 N\sigma^2 = T\sigma^2 \frac{\varepsilon^2}{\eta}. \tag{8.50}$$

If we set $\varepsilon = \eta$ as in SGD, then we have $\mathrm{Var}(X_N) = T\sigma^2\eta$, which goes to zero as $\eta \to 0$. This means that the noise disappears if we scale ε linearly with η. In general, the same phenomenon happens if $\varepsilon^2 = o(\eta)$. Conversely, if $\varepsilon^2 = \omega(\eta)$, then $\mathrm{Var}(X_N)$ blows up and the marginal distribution of X_N does not converge to anything.

The only non-trivial scaling is therefore

$$\varepsilon^2 = \Theta(\eta). \tag{8.51}$$

In particular, if we set $\varepsilon^2 = \eta$, i.e., $\varepsilon = \sqrt{\eta}$, then

$$\mathrm{Var}(X_N) = T\sigma^2, \tag{8.52}$$

so the noise remains visible when $\eta \to 0$. In this case, the stochastic difference equation becomes

$$X_{k+1} = X_k + \underbrace{\eta\mu(X_k)}_{\text{drift}} + \underbrace{\sqrt{\eta}\xi_k}_{\text{noise}}, \qquad \xi_k \sim \mathcal{Z}(X_k). \tag{8.53}$$

Overall, this heuristic argument suggests that to make the stochastic difference equation converge to a continuous-time stochastic process, a necessary condition is that the noise scale ε scales as $\sqrt{\eta}$. What remains is to identify the continuous-time stochastic process that approximates Equation (8.53).

8.2.2 Brownian Motion

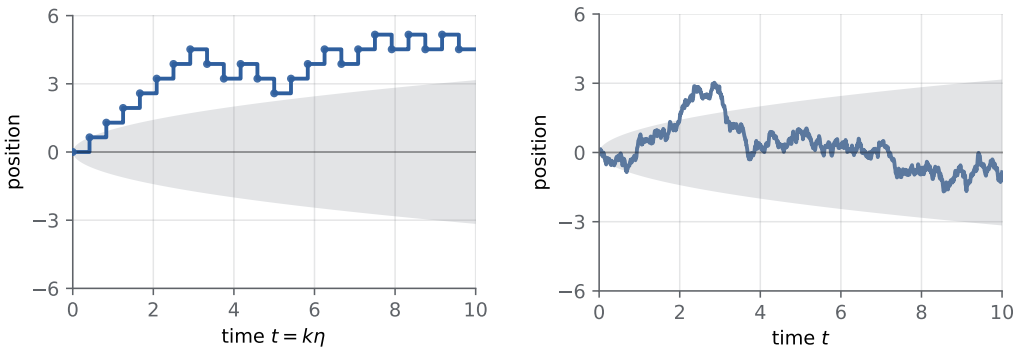
We now study the simplest case of Equation (8.53). Set

$$\mu(x) = 0, \qquad \mathcal{Z}(x) = \mathcal{N}(0,1) \quad \text{for all } x. \tag{8.54}$$

Then the stochastic difference equation becomes

$$X_{k+1} = X_k + \sqrt{\eta}\xi_k, \qquad \xi_k \sim \mathcal{N}(0,1), \qquad X_0 = 0. \tag{8.55}$$

Now associate each step with a time interval of length η, and the first N steps are now associated with the time interval $[0, N\eta]$. Since there are infinite number of steps, this essentially discretizes the time interval $[0, +\infty)$ into infinitely many small time intervals of length η. Imagine that we make η smaller and smaller, then the length of each time interval becomes infinitesimal. In the limit, the whole process becomes a continuous-time stochastic process, which we denote as $\{W(t)\}_{t\geq 0}$.



(a) Discrete random walk.

(b) Brownian motion.

Figure 8.2: A discrete one-dimensional random walk ($X_{k+1} = X_k \pm \sqrt{\eta}$), which is a discrete-time stochastic process, and Brownian motion, which is a continuous-time stochastic process. The shaded region shows the band for one standard deviation $\pm\sqrt{t}$.

It takes some effort to formally define how this limit is taken, but if we do not insist on full rigor, then for every time $t \geq 0$, we can find a state X_k associated with $W(t)$, where $k = t/\eta$. We can directly compute the state X_k by

$$X_k = \sqrt{\eta} \sum_{j=0}^{k-1} \xi_j. \tag{8.56}$$

The sum above is a sum of Gaussian random variables, so the marginal distribution of X_k is

$$W(t) = X_k \sim \mathcal{N}(0, t). \tag{8.57}$$

Similarly, if we fix two time points $0 \leq s < t$, then

$$W(t) - W(s) \sim \mathcal{N}(0, t - s). \tag{8.58}$$

This shows some basic properties of the stochastic process $\{W(t)\}_{t \geq 0}$, although we still have not discussed how this process is formally constructed, and we will not do so here. In fact, the process $\{W(t)\}_{t \geq 0}$ is called *standard Brownian motion*, or equivalently a *Wiener process*. The name *Brownian motion* honors the botanist Robert Brown (1773-1858), who observed that tiny particles such as pollen grains exhibit a continuous jittery motion in water. The name *Wiener process* honors the mathematician Norbert Wiener (1894-1964), who gave a rigorous mathematical construction of this continuous-time stochastic process. We will use the names *Brownian motion* and *Wiener process* interchangeably.

More formally, a Wiener process is defined as a continuous-time stochastic process that satisfies the following conditions.

Definition 8.5 (Wiener process)**.** A continuous-time stochastic process $\{W(t)\}_{t\geq 0}$ is a *Wiener process* (or *standard Brownian motion*) if it satisfies:

1. $W(0) = 0$;
2. The increments are independent: for every $0 \leq t_0 < t_1 < \cdots < t_n$, the following random variables are independent:
$$W(t_1) - W(t_0),\ W(t_2) - W(t_1),\ \ldots,\ W(t_n) - W(t_{n-1}). \tag{8.59}$$
3. The increments are Gaussian: for any $0 \leq s < t$,
$$W(t) - W(s) \sim \mathcal{N}(0, t-s). \tag{8.60}$$
4. When viewed as a function of time, $t \mapsto W(t)$ is continuous.

The four conditions describe how a Wiener process behaves over time. The first condition fixes the starting point of the process at the origin. The second condition says that changes over non-overlapping time intervals are independent. That is, knowing the change over one interval gives no information about the change over another. For example, if $0 < s < t$, then the change $W(s) - W(0)$ during $[0, s]$ is independent of the later change $W(t) - W(s)$ during $[s, t]$. Note that this does not mean that $W(s)$ and $W(t)$ are independent, because $W(t)$ contains the earlier value $W(s)$.

The third condition specifies the distribution of each change. Over an interval $[s, t]$, the increment has mean 0 and variance $t - s$:
$$W(t) - W(s) \sim \mathcal{N}(0, t-s). \tag{8.61}$$
Its distribution depends only on the interval length $t - s$, not on the starting time s or the ending time t. In particular, its standard deviation, and hence its typical size, is $\sqrt{t-s}$.

The fourth condition restricts $W(t)$ to be continuous in t. That is, small changes in time produce small changes in the value of the process. It is important to note that this does not mean that $W(t)$ is differentiable with respect to t. For $W(t)$ to be differentiable at time t, the ratio
$$\frac{W(t+h) - W(t)}{h} \tag{8.62}$$
must approach a finite value as h approaches zero. However, the change in W over the time interval $[t, t+h]$, namely $W(t+h) - W(t)$, typically has size $\sqrt{h}$. Therefore, the ratio typically has size
$$\frac{\sqrt{h}}{h} = \frac{1}{\sqrt{h}}, \tag{8.63}$$
which increases as h decreases. Thus, the change in W over $[t, t+h]$ does not become small quickly enough relative to the interval length h for a derivative to

exist. In fact, the Wiener process is notorious for being continuous but nowhere differentiable with probability one.

The second condition also implies that the Wiener process belongs to a broader class of stochastic processes, called *continuous-time Markov processes*. For a Markov process, once the present state is known, knowing the earlier states gives no additional information about the future.

Definition 8.6 (Continuous-time Markov process)**.** A continuous-time stochastic process $\{X(t)\}_{t\geq 0}$ is a *Markov process* if, for every choice of times

$$0 \leq t_0 < t_1 < \cdots < t_m, \tag{8.64}$$

the conditional distribution of $X(t_m)$ given all the earlier states depends only on the most recent state $X(t_{m-1})$. That is,

$$\Pr[X(t_m) \in A \mid X(t_0), \ldots, X(t_{m-1})] = \Pr[X(t_m) \in A \mid X(t_{m-1})] \tag{8.65}$$

for every set A for which these probabilities are defined.

To see why a Wiener process must be Markov, consider two times $s < t$. We can write

$$W(t) = W(t) - W(0) = \big(W(s) - W(0)\big) + \big(W(t) - W(s)\big), \tag{8.66}$$

where the increment $W(t)-W(s)$ is independent of all earlier increments, including $W(s) - W(0)$. Therefore, conditioned on $W(s)$, the distribution of $W(t)$ is $\mathcal{N}(W(s) - W(0), t - s) = \mathcal{N}(W(s), t - s)$. This conditional distribution depends only on $W(s)$, not on any earlier values, so the Wiener process is a continuous-time Markov process.

8.2.3 Stochastic Differential Equations

In the stochastic difference equation from Equation (8.53), each discrete update combines a deterministic drift with a random fluctuation, and both terms may depend on the current state. To construct a continuous-time approximation with the same structure, we introduce *stochastic differential equations* (SDEs), which may be viewed as stochastic versions of ODEs.

▶ One-dimensional SDEs

A one-dimensional SDE has the form

$$\mathrm{d}X(t) = \underbrace{\mu(X(t))\ \mathrm{d}t}_{\text{drift}} + \underbrace{\sigma(X(t))\ \mathrm{d}W(t)}_{\text{diffusion}}. \tag{8.67}$$

This SDE describes a continuous-time stochastic process $\{X(t)\}_{t\geq 0}$. An initial value such as $X(0) = x_0$ must also be specified. Under appropriate conditions on μ and σ, the SDE and the initial value together determine the process $X(t)$.

To make sense of the equation, first consider the case in which the diffusion coefficient is zero everywhere. Only the drift term remains, and the equation

becomes

$$\mathrm{d}X(t) = \mu(X(t))\ \mathrm{d}t. \tag{8.68}$$

The equation above is simply an ODE written in differential form, which may look less familiar. Dividing both sides by $\mathrm{d}t$ gives the usual derivative form

$$\frac{\mathrm{d}X(t)}{\mathrm{d}t} = \mu(X(t)), \tag{8.69}$$

or equivalently

$$\dot{X}(t) = \mu(X(t)). \tag{8.70}$$

The function $\mu : \mathbb{R} \to \mathbb{R}$ is called the *drift coefficient* and determines the deterministic part of the motion of $X(t)$.

The full SDE adds the term $\sigma(X(t))\ \mathrm{d}W(t)$ to this ODE, where $W(t)$ is a Wiener process. This term introduces random motion, and the *diffusion coefficient* $\sigma : \mathbb{R} \to \mathbb{R}$ determines its scale. To understand how the equation describes motion overall, consider a short time interval $[t, t + \Delta t]$. Over this interval, $\mathrm{d}t$ can be understood as the elapsed time, while $\mathrm{d}W(t)$ can be understood as the change in the Brownian motion. Thus, over this interval, the two differential symbols can be understood as the following finite differences:

$$\mathrm{d}t \text{ “=” } (t + \Delta t) - t = \Delta t, \qquad \mathrm{d}W(t) \text{ “=” } W(t + \Delta t) - W(t). \tag{8.71}$$

The SDE therefore gives the following approximation:

$$X(t + \Delta t) - X(t) \approx \mu(X(t))\Delta t + \sigma(X(t))\big(W(t + \Delta t) - W(t)\big). \tag{8.72}$$

Once the current value $X(t)$ is fixed, the only random quantity on the right-hand side is the Brownian increment. Since

$$W(t + \Delta t) - W(t) \sim \mathcal{N}(0, \Delta t), \tag{8.73}$$

the change $X(t + \Delta t) - X(t)$ has approximately the conditional distribution

$$\mathcal{N}\left(\mu(X(t))\Delta t, \sigma(X(t))^2 \Delta t\right). \tag{8.74}$$

Thus, viewed as a stochastic difference equation, Equation (8.72) shows that each update consists of a deterministic drift $\mu(X(t))\Delta t$ and a random fluctuation $\sigma(X(t))\big(W(t+\Delta t) - W(t)\big)$. Heuristically, the SDE is the limit of these updates as the time step Δt approaches zero.

▶ Integral Form of SDEs and Itô Integrals

The heuristic argument above gives some intuition for what an SDE is, but it is not yet a formal definition of an SDE. Recall that when the diffusion coefficient is zero, the SDE reduces to an ODE, as shown in Equation (8.69). Any function $X(t)$ whose function values and derivatives satisfy the ODE is said to be a solution to the ODE. For an SDE, one tempting way to define a solution for the SDE is to require that the function values and derivatives satisfy the following function equation:

$$\frac{\mathrm{d}X(t)}{\mathrm{d}t} = \mu(X(t)) + \sigma(X(t))\frac{\mathrm{d}W(t)}{\mathrm{d}t}. \tag{8.75}$$

However, this equation does not make sense at all! The Wiener process $W(t)$ is nowhere differentiable, so the derivative $\frac{\mathrm{d}W(t)}{\mathrm{d}t}$ does not exist.

For this reason, an SDE solution is defined as any function $X(t)$ that satisfies the following integral form of the SDE:

$$X(t) = X(0) + \int_0^t \mu(X(\tau))\ \mathrm{d}\tau + \int_0^t \sigma(X(\tau))\ \mathrm{d}W(\tau). \tag{8.76}$$

The first integral is an ordinary integral that accumulates the drift term. The second integral accumulates the random changes produced by Brownian motion. It is called an *Itô integral*.

To define the Itô integral, divide $[0, t]$ into n equal subintervals of length $\Delta t = t/n$, so that the subintervals are $[t_0, t_1], [t_1, t_2], \ldots, [t_{n-1}, t_n]$ with $t_k = k\Delta t$. On each subinterval $[t_k, t_{k+1}]$, multiply the Brownian increment by the diffusion coefficient evaluated at the left endpoint of the subinterval t_k. Adding these terms gives the random sum

$$\sum_{k=0}^{n-1} \sigma(X(t_k))(W(t_{k+1}) - W(t_k)). \tag{8.77}$$

As the partition becomes finer, the lengths of all its subintervals approach zero. Under appropriate conditions, the random sum above approaches a limit, which is defined to be the Itô integral

$$\int_0^t \sigma(X(\tau))\ \mathrm{d}W(\tau) = \lim_{n\to\infty} \sum_{k=0}^{n-1} \sigma(X(t_k))(W(t_{k+1}) - W(t_k)). \tag{8.78}$$

Technically, the limit is defined in the sense of convergence in probability (Definition 6.14), but you should think of it as an ordinary limit and forget about the technical details here. Otherwise, you may fail to see the forest for the trees. Throughout the section, we will be deliberately loose about the precise mode of convergence.

Example 8.7. If the integrand is the constant 1, then

$$\int_0^t \mathrm{d}W(\tau) = \lim_{n\to\infty} \sum_{k=0}^{n-1} (W(t_{k+1}) - W(t_k)) = W(t) - W(0) = W(t). \tag{8.79}$$

More generally, for any constant $c \in \mathbb{R}$,

$$\int_0^t c\ \mathrm{d}W(\tau) = cW(t). \tag{8.80}$$

The Itô integral also satisfies many familiar rules for ordinary integrals. For example, it is linear in the integrand:

$$\int_0^t (af(\tau) + bg(\tau))\ \mathrm{d}W(\tau) = a\int_0^t f(\tau)\ \mathrm{d}W(\tau) + b\int_0^t g(\tau)\ \mathrm{d}W(\tau). \tag{8.81}$$

Remark 8.8 (Itô and Stratonovich conventions)**.** We need to be careful about the choice of where to evaluate the coefficient in the stochastic integral. The

Itô convention uses the left endpoint t_k, so the coefficient is determined before the Brownian increment over $[t_k, t_{k+1}]$ occurs. Another common choice is the *Stratonovich convention*, which uses the midpoint of each subinterval. That is,

$$\int_0^t \sigma(X(\tau)) \circ \mathrm{d}W(\tau) = \lim_{n\to\infty} \sum_{k=0}^{n-1} \frac{\sigma(X(t_k))+\sigma(X(t_{k+1}))}{2} (W(t_{k+1}) - W(t_k)), \quad (8.82)$$

where $\circ\, \mathrm{d}W(\tau)$ is the notation for the Stratonovich integral. An SDE based on the Stratonovich convention is called a *Stratonovich SDE*:

$$\mathrm{d}X(t) = \mu(X(t))\ \mathrm{d}t + \sigma(X(t)) \circ \mathrm{d}W(t). \quad (8.83)$$

The two conventions generally give different integrals. If σ is differentiable, the Stratonovich SDE above is equivalent to the Itô SDE

$$\mathrm{d}X(t) = \left[\mu(X(t)) + \frac{1}{2}\sigma(X(t))\sigma'(X(t))\right] \mathrm{d}t + \sigma(X(t))\ \mathrm{d}W(t). \quad (8.84)$$

Unless stated otherwise, all stochastic integrals and SDEs in this chapter use the Itô convention.

▶ Euler–Maruyama Method

The integral form above tells us precisely what an SDE means. We now return to the stochastic difference equation that motivated this discussion and ask whether it can be approximated by an SDE when the step size is small.

Recall from Equation (8.53) that retaining non-trivial random fluctuations as $\eta \to 0$ requires the noise to be scaled by $\sqrt{\eta}$. The resulting stochastic difference equation is

$$X_{k+1} = X_k + \eta\mu(X_k) + \sqrt{\eta}\,\xi_k, \qquad \xi_k \sim \mathcal{Z}(X_k). \quad (8.85)$$

We denote by $\sigma^2(x)$ the variance of the noise distribution at state x, and compare the above recursion with the SDE in Equation (8.67):

$$\mathrm{d}X(t) = \mu(X(t))\ \mathrm{d}t + \sigma(X(t))\ \mathrm{d}W(t). \quad (8.86)$$

Fix a time horizon $T > 0$ and take $N = T/\eta$ updates over $[0, T]$. The k-th update then corresponds to time $t_k = k\eta$, so X_k corresponds to $X(t_k)$. Conditioned on X_k, one step of Equation (8.85) has mean $\eta\mu(X_k)$ and variance $\eta\sigma^2(X_k)$, which are the same as the local mean and variance of the SDE over the interval $[t_k, t_{k+1}]$, as discussed in Equation (8.72). The stochastic difference equation can therefore be seen as an Euler method for this SDE, analogous to the Euler method for ODEs in Equation (8.27).

A special case is that the noise distribution is Gaussian, i.e.,

$$\mathcal{Z}(x) = \mathcal{N}(0, \sigma(x)^2), \quad (8.87)$$

for which $\sqrt{\eta}\,\xi_k$ has the same distribution as $\sigma(X_k)(W(t_{k+1}) - W(t_k))$. In this Gaussian case, Equation (8.85) is called the *Euler–Maruyama method* for the SDE in Equation (8.67).

Following a similar argument as in the proof of Theorem 8.4, one can show that under some regularity conditions on the drift and noise terms, Equation (8.85)

and the SDE in Equation (8.67) approximate each other in the following sense:

$$X_N \xrightarrow{d} X(T) \quad \text{as } \eta \to 0. \tag{8.88}$$

We will not provide the proof here, but readers interested in the precise assumptions, convergence rates and proof can see Chapter 14 of [KP92].

▶ Connection to the Central Limit Theorem

While we skip the proof of Equation (8.88), we can explain why the stochastic difference equation converges to the SDE in the special case where the drift and diffusion coefficients are constant. That is,

$$\mathrm{d}X(t) = \mu \, \mathrm{d}t + \sigma \, \mathrm{d}W(t), \qquad X(0) = x. \tag{8.89}$$

Its integral form is

$$X(t) = x + \int_0^t \mu \, \mathrm{d}\tau + \int_0^t \sigma \, \mathrm{d}W(\tau). \tag{8.90}$$

It is easy to see that $\int_0^t \mu \, \mathrm{d}\tau = \mu t$ and $\int_0^t \sigma \, \mathrm{d}W(\tau) = \sigma W(t)$. Therefore,

$$X(t) = x + \mu t + \sigma W(t). \tag{8.91}$$

Since $W(t) \sim \mathcal{N}(0, t)$, we can directly write down the marginal distribution of $X(t)$:

$$X(t) \sim \mathcal{N}(x + \mu t, \sigma^2 t). \tag{8.92}$$

Consider the corresponding stochastic difference equation. Let ξ_k for $k \geq 0$ be independent with common distribution $\mathcal{Z}$. Suppose that $\mathcal{Z}$ has mean 0 and variance σ^2. It need not be Gaussian. Fix a time interval $[0, T]$, set $\eta = T/N$, and consider the constant-coefficient stochastic difference equation

$$X_{k+1} = X_k + \eta\mu + \sqrt{\eta}\,\xi_k, \qquad X_0 = x. \tag{8.93}$$

After N updates, adding the changes in Equation (8.93) gives

$$X_N = x + \mu T + \sqrt{\eta} \sum_{k=0}^{N-1} \xi_k \tag{8.94}$$

$$= x + \mu T + \sqrt{T} \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \xi_k \right). \tag{8.95}$$

Do you feel any familiar pattern here? If not, you have not truly mastered the central limit theorem yet! By the central limit theorem in Theorem 6.16,

$$\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \xi_k \xrightarrow{d} \mathcal{N}(0, \sigma^2). \tag{8.96}$$

Consequently,

$$X_N \xrightarrow{d} \mathcal{N}(x + \mu T, \sigma^2 T), \tag{8.97}$$

which is exactly the distribution of $X(T)$ obtained from Equation (8.91).

8.2.4 Itô's Lemma

One of the most useful tools in ordinary calculus is the *chain rule*. If $X(t)$ satisfies an ODE, it tells us how to determine the evolution of $f(X(t))$ from the evolution of $X(t)$. In differential notation,

$$\mathrm{d}f(X(t)) = f'(X(t))\ \mathrm{d}X(t). \tag{8.98}$$

We would like to use the same idea when $X(t)$ satisfies an SDE. The calculation is more complicated because two types of small changes $\mathrm{d}t$ and $\mathrm{d}W(t)$ are involved. More specifically, consider a small time interval $[t, t+\Delta t]$ and let $\Delta W := W(t+\Delta t) - W(t)$ be the corresponding Brownian increment. Then the Taylor expansion of $f(X(t+\Delta t))$ gives

$$\begin{aligned} f(X(t+\Delta t)) - f(X(t)) = {} & f'(X(t))\Big(\mu(X(t))\Delta t + \sigma(X(t))\Delta W\Big) \\ & + \frac{1}{2} f''(X(t))\Big(\mu(X(t))\Delta t + \sigma(X(t))\Delta W\Big)^2 \\ & + \text{higher-order terms.} \end{aligned} \tag{8.99}$$

While in the ODE case, the second-order terms are discarded, these terms cannot be discarded in the SDE case. This is because Brownian motion has a different scaling behavior than deterministic functions. Over an interval of length Δt, a Brownian increment ΔW typically has size $\sqrt{\Delta t}$. Its square therefore has size Δt and cannot be discarded. The following proposition summarizes the rules for handling the second-order terms, which are usually called the *Itô rules*.

Proposition 8.9 (Itô rules)**.** For one-dimensional Brownian motion,

$$(\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\ \mathrm{d}W(t) = 0, \qquad (\mathrm{d}W(t))^2 = \mathrm{d}t. \tag{8.100}$$

More formally, we divide an interval $[0, T]$ into n equal subintervals of length $\Delta t = T/n$, so that the subintervals are $[t_0, t_1], [t_1, t_2], \ldots, [t_{n-1}, t_n]$ with $t_k = k\Delta t$. If $\{W(t)\}_{t\geq 0}$ is a one-dimensional Wiener process, then

$$\int_0^T (\mathrm{d}t)^2 := \lim_{n\to\infty} \sum_{k=0}^{n-1} (\Delta t)^2 = 0, \tag{8.101}$$

$$\int_0^T \mathrm{d}t\ \mathrm{d}W(t) := \lim_{n\to\infty} \sum_{k=0}^{n-1} \Delta t \Delta W_k = 0, \tag{8.102}$$

$$\int_0^T (\mathrm{d}W(t))^2 := \lim_{n\to\infty} \sum_{k=0}^{n-1} (\Delta W_k)^2 = T = \int_0^T \mathrm{d}t, \tag{8.103}$$

where we denote the Brownian increment over the k-th subinterval by $\Delta W_k := W(t_{k+1}) - W(t_k)$, and the limit is defined in the sense of convergence in probability (Definition 6.14).

Proof. For proving $(\mathrm{d}t)^2 = 0$, we just need to note that

$$\sum_{k=0}^{n-1} (\Delta t)^2 = n \cdot (\Delta t)^2 = \Delta t \cdot T \to 0. \tag{8.104}$$

For proving $\mathrm{d}t\ \mathrm{d}W(t) = 0$, we just need to note that the Brownian increments telescope:

$$\sum_{k=0}^{n-1} \Delta t\, \Delta W_k = \Delta t\big(W(T) - W(0)\big) = \Delta t \cdot W(T) \to 0, \tag{8.105}$$

where the limit holds because $W(T)$ is finite with probability 1 and $\Delta t \to 0$.

For proving $(\mathrm{d}W(t))^2 = \mathrm{d}t$, let

$$Q_n = \sum_{k=0}^{n-1} (\Delta W_k)^2. \tag{8.106}$$

The Brownian increments are independent, and each has distribution $\mathcal{N}(0, \Delta t)$. Therefore, the mean and variance of $(\Delta W_k)^2$ are

$$\mathbb{E}[(\Delta W_k)^2] = \Delta t, \tag{8.107}$$

$$\mathrm{Var}\big((\Delta W_k)^2\big) = \mathbb{E}[(\Delta W_k)^4] - (\mathbb{E}[(\Delta W_k)^2])^2 \tag{8.108}$$

$$= 3(\Delta t)^2 - (\Delta t)^2 = 2(\Delta t)^2. \tag{8.109}$$

Now we rewrite Q_n as

$$Q_n = \frac{1}{n} \sum_{k=0}^{n-1} (\sqrt{n}\Delta W_k)^2. \tag{8.110}$$

The terms $Z_k := \big(\sqrt{n}\Delta W_k\big)^2$ are independent random variables with common mean T and variance $2T^2$. By the weak law of large numbers in Theorem 6.13,

$$\frac{1}{n} \sum_{k=0}^{n-1} Z_k \xrightarrow{\mathrm{P}} \mathbb{E}[Z_1] = T. \tag{8.111}$$

Therefore, $Q_n \xrightarrow{\mathrm{P}} T$. □

Remark 8.10 (Quadratic variation)**.** For a stochastic process $\{X(t)\}_{t\geq 0}$, its *quadratic variation* over $[0, t]$ is the limit of sums of squared increments along partitions of $[0, t]$ whose subintervals become arbitrarily short. That is,

$$[X]_t = \lim_{\|P\|\to 0} \sum_{k=0}^{n-1} (X_{t_{k+1}} - X_{t_k})^2, \tag{8.112}$$

where P ranges over all partitions of $[0, t]$ into subintervals with endpoints $0 = t_0 < t_1 < \cdots < t_n = t$, and the norm of a partition is defined as the length of the longest subinterval $\|P\| = \max_{0\leq k<n}\{t_{k+1} - t_k\}$. In this context, the quadratic variation of a Wiener process is $[W]_t = t$.

Itô's rules suggest that when we perform the Taylor expansion as in Equation (8.99), the second-order terms cannot be discarded and should retain the

term associated with $(\mathrm{d}W(t))^2$. Consequently, the ordinary chain rule (Equation (8.98)) does not apply directly to the solution of an SDE. The correct version of the chain rule for SDEs is called *Itô's lemma*.

Lemma 8.11 (Itô's lemma)**.** Let $X(t)$ satisfy the one-dimensional SDE

$$\mathrm{d}X(t) = \mu(X(t))\ \mathrm{d}t + \sigma(X(t))\ \mathrm{d}W(t). \tag{8.113}$$

If $f : \mathbb{R} \to \mathbb{R}$ is twice differentiable, then

$$\begin{aligned}\mathrm{d}f(X(t)) = &\left[f'(X(t))\mu(X(t)) + \frac{1}{2}f''(X(t))\sigma(X(t))^2\right]\mathrm{d}t \\ &+ f'(X(t))\sigma(X(t))\ \mathrm{d}W(t).\end{aligned} \tag{8.114}$$

"Proof". A rigorous proof requires a careful treatment of the limits defining the stochastic integrals. Here we give a proof with physics-level rigor.

Recall the Taylor expansion in Equation (8.99). Replacing Δt and ΔW with $\mathrm{d}t$ and $\mathrm{d}W(t)$ gives

$$\begin{aligned}\mathrm{d}f(X(t)) = &f'(X(t))\Big(\mu(X(t))\ \mathrm{d}t + \sigma(X(t))\ \mathrm{d}W(t)\Big) \\ &+ \frac{1}{2}f''(X(t))\Big(\mu(X(t))\ \mathrm{d}t + \sigma(X(t))\ \mathrm{d}W(t)\Big)^2.\end{aligned} \tag{8.115}$$

Terms of degree three and above are omitted because repeated use of the Itô rules reduces them to zero. Expanding the square gives

$$\mathrm{d}f(X(t)) = f'(X(t))\mu(X(t))\ \mathrm{d}t + f'(X(t))\sigma(X(t))\ \mathrm{d}W(t) \tag{8.116}$$

$$+ \frac{1}{2}f''(X(t))\mu(X(t))^2\underbrace{(\mathrm{d}t)^2}_{=0} \tag{8.117}$$

$$+ f''(X(t))\mu(X(t))\sigma(X(t))\underbrace{\mathrm{d}t\ \mathrm{d}W(t)}_{=0} \tag{8.118}$$

$$+ \frac{1}{2}f''(X(t))\sigma(X(t))^2\underbrace{(\mathrm{d}W(t))^2}_{=\mathrm{d}t} \tag{8.119}$$

$$= f'(X(t))\mu(X(t))\ \mathrm{d}t + f'(X(t))\sigma(X(t))\ \mathrm{d}W(t) \tag{8.120}$$

$$+ \frac{1}{2}f''(X(t))\sigma(X(t))^2\,\mathrm{d}t, \tag{8.121}$$

where the last equality follows from Proposition 8.9. Rearranging the terms gives the desired formula. □

8.2.5 Ornstein–Uhlenbeck Process

▶ Definition

As an application of Itô's lemma, let us solve a classical SDE: the Ornstein–Uhlenbeck process, commonly abbreviated as the OU process.

$$\mathrm{d}X(t) = -\gamma X(t)\ \mathrm{d}t + \sigma\ \mathrm{d}W(t), \qquad X(0) = x_0. \tag{8.122}$$

Here $\gamma > 0$ and $\sigma > 0$. The drift $-\gamma X(t)\,\mathrm{d}t$ pushes $X(t)$ towards the origin, while the diffusion term $\sigma\,\mathrm{d}W(t)$ continually perturbs the process.

In physics, $X(t)$ can be interpreted as the velocity of a particle moving through a fluid. Friction tends to slow the particle down, while random collisions with the surrounding molecules continually change its velocity.

The same process also arises as a local approximation of SGD near a minimizer. Suppose the loss near a local minimizer θ^* is approximated by the quadratic

$$\mathcal{L}(\theta) = \frac{\gamma}{2}(\theta - \theta^*)^2, \qquad \nabla\mathcal{L}(\theta) = \gamma(\theta - \theta^*). \tag{8.123}$$

With gradient noise, one SGD step takes the form

$$\theta_{k+1} = \theta_k - \eta(\gamma(\theta_k - \theta^*) + \xi_k). \tag{8.124}$$

Now let $X_k = \theta_k - \theta^*$ be the centered coordinate. Subtracting θ^* from both sides of the SGD update gives

$$X_{k+1} = (1 - \eta\gamma)X_k - \eta\xi_k. \tag{8.125}$$

Suppose the gradient noise ξ_k satisfies $\mathbb{E}[\xi_k] = 0$ and $\mathrm{Var}(\xi_k) = \sigma_0^2$. According to the SDE approximation, this recursion corresponds to the following OU process:

$$\mathrm{d}X(t) = -\gamma X(t)\ \mathrm{d}t + \sqrt{\eta}\sigma_0\ \mathrm{d}W(t). \tag{8.126}$$

Finally, the OU process also serves as the forward noising process in diffusion models, as we will see in Section 8.3.3.

▶ Solution

Now we derive a closed-form solution for the OU process. If there is no noise, then the OU process is a deterministic ODE:

$$\mathrm{d}X(t) = -\gamma X(t)\ \mathrm{d}t. \tag{8.127}$$

Let $Y(t) = e^{\gamma t}X(t)$. Then

$$\begin{aligned}
\mathrm{d}Y(t) &= \gamma e^{\gamma t}X(t)\ \mathrm{d}t + e^{\gamma t}\,\mathrm{d}X(t) && (8.128)\\
&= (\gamma X(t) - \gamma X(t))\,\mathrm{d}t && (8.129)\\
&= 0. && (8.130)
\end{aligned}$$

Therefore, $Y(t) = Y(0) = e^{\gamma\cdot 0}x_0 = x_0$, and thus $X(t) = e^{-\gamma t}x_0$.

But now we have noise. All we need to do is to apply Itô's lemma to $Y(t)$ instead.

$$\begin{aligned}
\mathrm{d}Y(t) &= \gamma e^{\gamma t}X(t)\ \mathrm{d}t + e^{\gamma t}\ \mathrm{d}X(t) && (8.131)\\
&= \gamma e^{\gamma t}X(t)\ \mathrm{d}t + e^{\gamma t}\left(-\gamma X(t)\ \mathrm{d}t + \sigma\ \mathrm{d}W(t)\right) && (8.132)\\
&= \sigma e^{\gamma t}\ \mathrm{d}W(t). && (8.133)
\end{aligned}$$

Integrating from 0 to t and using $Y(0) = X(0) = x_0$ gives

$$Y(t) = x_0 + \sigma\int_0^t e^{\gamma\tau}\ \mathrm{d}W(\tau). \tag{8.134}$$

To compute the stochastic integral here, let $\Delta t = t/n$ and $t_k = k\Delta t$. By the definition of the Itô integral,

$$\int_0^t e^{\gamma\tau}\, \mathrm{d}W(\tau) = \lim_{n\to\infty} \sum_{k=0}^{n-1} e^{\gamma t_k}(W(t_{k+1}) - W(t_k)). \tag{8.135}$$

For each k, the increment $W(t_{k+1}) - W(t_k)$ is distributed as $\mathcal{N}(0, \Delta t)$. By Gaussian linearity, the finite sum is also distributed as Gaussian. Its mean is always 0, and its variance is

$$\sum_{k=0}^{n-1} e^{2\gamma t_k}\Delta t = \Delta t \sum_{k=0}^{n-1} e^{2\gamma k\Delta t} = \Delta t \frac{e^{2\gamma t} - 1}{e^{2\gamma\Delta t} - 1} \to \frac{e^{2\gamma t} - 1}{2\gamma}, \tag{8.136}$$

as $n \to \infty$. It follows that

$$\int_0^t e^{\gamma\tau}\, \mathrm{d}W(\tau) \sim \mathcal{N}\left(0, \frac{e^{2\gamma t} - 1}{2\gamma}\right), \tag{8.137}$$

and therefore

$$Y(t) \sim \mathcal{N}\left(x_0,\, \tfrac{\sigma^2}{2\gamma}(e^{2\gamma t} - 1)\right). \tag{8.138}$$

Finally, using $X(t) = e^{-\gamma t}Y(t)$ gives the marginal distribution of $X(t)$:

$$X(t) \sim \mathcal{N}\left(e^{-\gamma t}x_0,\, \tfrac{\sigma^2}{2\gamma}(1 - e^{-2\gamma t})\right). \tag{8.139}$$

▶ Stationary Distribution

An important consequence of the above analysis is that the marginal distribution approaches a fixed limit as $t \to \infty$. Its mean converges to 0, and its variance converges to $\frac{\sigma^2}{2\gamma}$. Since $X(t)$ is Gaussian, the MGF convergence criterion in Theorem 6.11 gives

$$X(t) \xrightarrow{d} \mathcal{N}\Big(0,\, \tfrac{\sigma^2}{2\gamma}\Big). \tag{8.140}$$

This limiting Gaussian is the *stationary distribution* of the OU process. Two important properties of this stationary distribution are: (1) If $X(0)$ is sampled from this distribution, then $X(t)$ has the same distribution for all $t \geq 0$; (2) No matter where $X(0)$ starts, the marginal distribution of $X(t)$ approaches this stationary distribution.

Viewing the OU process as a continuous-time approximation of SGD, we see that its stationary behavior matches our earlier analysis in Equation (7.217). There, we obtained a bound on the expected loss of SGD with a fixed learning rate. The contraction term in the bound decays to 0 as $t \to 0$, but the noise term does not. We therefore found that the expected loss of SGD with a fixed learning rate does not converge to the minimum value.

The stationary distribution of the OU process makes this behavior explicit. Recall that $X(t) = \theta(t) - \theta^*$ and $\sigma = \sqrt{\eta}\sigma_0$, so we have

$$\theta(t) \xrightarrow{d} \mathcal{N}\left(\theta^*, \tfrac{\eta\sigma_0^2}{2\gamma}\right). \tag{8.141}$$

Since the expected loss is quadratic,

$$\mathbb{E}[\mathcal{L}(\theta(t)) - \mathcal{L}(\theta^*)] = \frac{\gamma}{2}\mathbb{E}[X(t)^2] \to \frac{\gamma}{2} \cdot \frac{\eta\sigma_0^2}{2\gamma} = \frac{\eta\sigma_0^2}{4}. \tag{8.142}$$

Therefore, the expected loss does not converge to the minimum value, which agrees with our earlier analysis of SGD. More importantly, our earlier analysis of SGD only provides an upper bound on the expected loss, but this SDE-based analysis precisely characterizes the long-term behavior of the iterates: instead of converging to the minimizer, they fluctuate around it approximately following the Gaussian distribution $\mathcal{N}\left(\theta^*, \frac{\eta\sigma_0^2}{2\gamma}\right)$.

8.2.6 Multidimensional SDEs

The one-dimensional SDE extends naturally to a process whose state has several coordinates.

▶ Definition

A multidimensional SDE has the form

$$\mathrm{d}\boldsymbol{x}(t) = \boldsymbol{\mu}(\boldsymbol{x}(t))\ \mathrm{d}t + \boldsymbol{\sigma}(\boldsymbol{x}(t))\ \mathrm{d}\boldsymbol{w}(t). \tag{8.143}$$

where the state $\boldsymbol{x}(t)$ lies in $\mathbb{R}^d$. The drift $\boldsymbol{\mu} : \mathbb{R}^d \to \mathbb{R}^d$ is now a vector. The process

$$\boldsymbol{w}(t) = \begin{bmatrix} W_1(t) \\ \vdots \\ W_d(t) \end{bmatrix} \tag{8.144}$$

is a *d-dimensional Wiener process* (or *standard d-dimensional Brownian motion*), which consists of d independent one-dimensional Wiener processes. Consequently, for any $0 \le s < t$, the Brownian increment over $[s, t]$ has distribution

$$\boldsymbol{w}(t) - \boldsymbol{w}(s) \sim \mathcal{N}(\mathbf{0}, (t-s)\boldsymbol{I}). \tag{8.145}$$

This is similar to the one-dimensional case, but now the distribution of a Brownian increment is the d-dimensional Gaussian $\mathcal{N}(\mathbf{0}, (t-s)\boldsymbol{I})$ rather than the one-dimensional Gaussian $\mathcal{N}(0, t-s)$.

The diffusion coefficient $\boldsymbol{\sigma} : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ is a matrix because it transforms the vector increment of $\boldsymbol{w}(t)$ into the vector-valued random change in $\boldsymbol{x}(t)$. It can scale the noise differently in different directions and can also mix its components.

As in the one-dimensional case in Equation (8.76), a solution is defined through the integral form

$$\boldsymbol{x}(t) = \boldsymbol{x}(0) + \int_0^t \boldsymbol{\mu}(\boldsymbol{x}(\tau))\ \mathrm{d}\tau + \int_0^t \boldsymbol{\sigma}(\boldsymbol{x}(\tau))\ \mathrm{d}\boldsymbol{w}(\tau), \tag{8.146}$$

where the first and second integrals accumulate the drift and the Brownian increments, respectively.

▶ Diffusion Matrix

As in the one-dimensional case in Equation (8.72), over a short time interval $[t, t+\Delta t]$ the multidimensional SDE gives the approximation

$$\boldsymbol{x}(t+\Delta t) - \boldsymbol{x}(t) \approx \Delta t\, \boldsymbol{\mu}(\boldsymbol{x}(t)) + \boldsymbol{\sigma}(\boldsymbol{x}(t))\big(\boldsymbol{w}(t+\Delta t) - \boldsymbol{w}(t)\big). \tag{8.147}$$

Conditioned on the current state $\boldsymbol{x}(t)$, the Gaussian linearity (Theorem 5.39) implies that the change $\boldsymbol{x}(t+\Delta t) - \boldsymbol{x}(t)$ is approximately Gaussian with the following mean and covariance:

$$\mathcal{N}\Big(\Delta t\, \boldsymbol{\mu}(\boldsymbol{x}(t)), \Delta t\, \boldsymbol{\sigma}(\boldsymbol{x}(t))\boldsymbol{\sigma}(\boldsymbol{x}(t))^\top\Big). \tag{8.148}$$

Therefore, the multidimensional SDE in Equation (8.143) essentially says that at each time step, the state $\boldsymbol{x}(t)$ changes by a Gaussian random vector with mean $\Delta t\, \boldsymbol{\mu}(\boldsymbol{x}(t))$ and covariance $\Delta t\, \boldsymbol{\sigma}(\boldsymbol{x}(t))\boldsymbol{\sigma}(\boldsymbol{x}(t))^\top$. This motivates the following definition of the *diffusion matrix* $\boldsymbol{C}(\boldsymbol{x})$.

$$\boldsymbol{C}(\boldsymbol{x}) := \boldsymbol{\sigma}(\boldsymbol{x})\boldsymbol{\sigma}(\boldsymbol{x})^\top. \tag{8.149}$$

With this notation, the approximate conditional distribution of the change can be written more compactly as

$$\mathcal{N}\Big(\Delta t\, \boldsymbol{\mu}(\boldsymbol{x}(t)), \Delta t\, \boldsymbol{C}(\boldsymbol{x}(t))\Big). \tag{8.150}$$

Note that the conditional distribution above depends on the diffusion coefficient $\boldsymbol{\sigma}$ only through the diffusion matrix $\boldsymbol{C}$. Consequently, two SDEs with the same drift vector and diffusion matrix essentially describe the same process, even if their diffusion coefficients are different. More precisely, consider

$$\mathrm{d}\boldsymbol{x}_1(t) = \boldsymbol{\mu}(\boldsymbol{x}_1(t))\ \mathrm{d}t + \boldsymbol{\sigma}_1(\boldsymbol{x}_1(t))\ \mathrm{d}\boldsymbol{w}(t), \tag{8.151}$$

$$\mathrm{d}\boldsymbol{x}_2(t) = \boldsymbol{\mu}(\boldsymbol{x}_2(t))\ \mathrm{d}t + \boldsymbol{\sigma}_2(\boldsymbol{x}_2(t))\ \mathrm{d}\boldsymbol{w}(t), \tag{8.152}$$

where the diffusion coefficients satisfy

$$\boldsymbol{\sigma}_1(\boldsymbol{x})\boldsymbol{\sigma}_1(\boldsymbol{x})^\top = \boldsymbol{\sigma}_2(\boldsymbol{x})\boldsymbol{\sigma}_2(\boldsymbol{x})^\top \tag{8.153}$$

for every $\boldsymbol{x} \in \mathbb{R}^d$. If $\boldsymbol{x}_1(0)$ and $\boldsymbol{x}_2(0)$ have the same distribution, then, for every fixed $t \geq 0$, $\boldsymbol{x}_1(t)$ and $\boldsymbol{x}_2(t)$ have the same marginal distribution. More generally, the two processes have the same distribution in the sense that for any set of time points $t_1, \ldots, t_k$, the random vectors $(\boldsymbol{x}_1(t_1), \ldots, \boldsymbol{x}_1(t_k))$ and $(\boldsymbol{x}_2(t_1), \ldots, \boldsymbol{x}_2(t_k))$ have the same distribution.

Accordingly, an Euler method for a multidimensional SDE only needs to match the drift vector and diffusion matrix. Consider the following stochastic difference equation with step size η:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \eta\boldsymbol{\mu}(\boldsymbol{x}_k) + \sqrt{\eta}\,\boldsymbol{\xi}_k, \qquad \boldsymbol{\xi}_k \sim \mathcal{Z}(\boldsymbol{x}_k). \tag{8.154}$$

If the noise term $\boldsymbol{\xi}_k$ satisfies $\mathbb{E}[\boldsymbol{\xi}_k] = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\xi}_k) = \boldsymbol{C}(\boldsymbol{x}_k)$, then this stochastic difference equation can be approximated by the SDE in Equation (8.143):

$$\mathrm{d}\boldsymbol{x}(t) = \boldsymbol{\mu}(\boldsymbol{x}(t))\ \mathrm{d}t + \boldsymbol{\sigma}(\boldsymbol{x}(t))\ \mathrm{d}\boldsymbol{w}(t), \tag{8.155}$$

The approximation holds in the same sense as in the one-dimensional case in Equation (8.88). That is, $\boldsymbol{x}_N \xrightarrow{d} \boldsymbol{x}(T)$ as $\eta \to 0$.

▶ Itô's Lemma

We have extended an SDE from a one-dimensional state to a multidimensional state. We now make the same extension for Itô's lemma in Lemma 8.11.

For a function $f : \mathbb{R}^d \to \mathbb{R}$, let ∇f denote its gradient and $\nabla^2 f$ denote its Hessian matrix. For any two matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{d\times d}$, define their *Frobenius inner product* by

$$\langle \boldsymbol{A}, \boldsymbol{B} \rangle := \sum_{i=1}^{d}\sum_{j=1}^{d} A_{ij}B_{ij}. \tag{8.156}$$

For the Hessian $\nabla^2 f$ and the diffusion matrix $\boldsymbol{C}$, the Frobenius inner product is

$$\langle \nabla^2 f, \boldsymbol{C} \rangle = \sum_{i=1}^{d}\sum_{j=1}^{d} \frac{\partial^2 f}{\partial x_i \partial x_j} C_{ij}. \tag{8.157}$$

The multidimensional Itô's lemma can now be stated as follows.

Lemma 8.12 (Multidimensional Itô's lemma)**.** Let $\boldsymbol{x}(t)$ satisfy the multidimensional SDE

$$\mathrm{d}\boldsymbol{x}(t) = \boldsymbol{\mu}(\boldsymbol{x}(t))\ \mathrm{d}t + \boldsymbol{\sigma}(\boldsymbol{x}(t))\ \mathrm{d}\boldsymbol{w}(t). \tag{8.158}$$

If $f : \mathbb{R}^d \to \mathbb{R}$ is twice differentiable, then

$$\begin{aligned}\mathrm{d}f(\boldsymbol{x}(t)) = &\left[\langle \nabla f(\boldsymbol{x}(t)), \boldsymbol{\mu}(\boldsymbol{x}(t))\rangle + \frac{1}{2}\langle \nabla^2 f(\boldsymbol{x}(t)), \boldsymbol{C}(\boldsymbol{x}(t))\rangle\right]\mathrm{d}t \\ &+ \nabla f(\boldsymbol{x}(t))^\top \boldsymbol{\sigma}(\boldsymbol{x}(t))\ \mathrm{d}\boldsymbol{w}(t).\end{aligned} \tag{8.159}$$

Equivalently, in coordinates,

$$\begin{aligned}\mathrm{d}f(\boldsymbol{x}(t)) = &\left[\sum_{i=1}^{d}\frac{\partial f}{\partial x_i}(\boldsymbol{x}(t))\mu_i(\boldsymbol{x}(t)) + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d}\frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{x}(t))C_{ij}(\boldsymbol{x}(t))\right]\mathrm{d}t \\ &+ \nabla f(\boldsymbol{x}(t))^\top \boldsymbol{\sigma}(\boldsymbol{x}(t))\ \mathrm{d}\boldsymbol{w}(t).\end{aligned}$$

"Proof". As in the proof of Lemma 8.11, we give a heuristic derivation using the Itô differential rules, with physics-level rigor.

We first extend the Itô rules in Proposition 8.9 to the components of the d-dimensional Wiener process. Each $W_i(t)$ is a one-dimensional Wiener process, so we have

$$\big(\mathrm{d}W_i(t)\big)^2 = \mathrm{d}t, \qquad \mathrm{d}t\ \mathrm{d}W_i(t) = 0. \tag{8.160}$$

When $i \neq j$, the Brownian increments of $W_i(t)$ and $W_j(t)$ are independent. By Gaussian linearity, $\big(W_i(t) + W_j(t)\big)/\sqrt{2}$ and $\big(W_i(t) - W_j(t)\big)/\sqrt{2}$ are both one-dimensional Wiener processes. The one-dimensional Itô rules then imply that

$$\big(\mathrm{d}W_i(t) + \mathrm{d}W_j(t)\big)^2 = 2\ \mathrm{d}t, \tag{8.161}$$

$$\big(\mathrm{d}W_i(t) - \mathrm{d}W_j(t)\big)^2 = 2\ \mathrm{d}t. \tag{8.162}$$

Subtracting these equations gives

$$\begin{aligned} \mathrm{d}W_i(t)\ \mathrm{d}W_j(t) &= \frac{1}{4}\Big(\big(\mathrm{d}W_i(t)+\mathrm{d}W_j(t)\big)^2-\big(\mathrm{d}W_i(t)-\mathrm{d}W_j(t)\big)^2\Big) \\ &= \frac{1}{4}(2\ \mathrm{d}t-2\ \mathrm{d}t)=0. \end{aligned} \tag{8.163}$$

The multidimensional Itô rules can then be summarized as

$$\mathrm{d}W_i(t)\ \mathrm{d}W_j(t) = \mathbb{1}_{[i=j]}\ \mathrm{d}t, \tag{8.164}$$

$$\mathrm{d}t\ \mathrm{d}W_i(t) = 0, \tag{8.165}$$

$$(\mathrm{d}t)^2 = 0. \tag{8.166}$$

Equivalently, in matrix form,

$$\mathrm{d}\boldsymbol{w}(t)\ \mathrm{d}\boldsymbol{w}(t)^\top = \boldsymbol{I}\ \mathrm{d}t. \tag{8.167}$$

With these rules in hand, the second-order Taylor expansion of $f(\boldsymbol{x}(t))$ gives

$$\mathrm{d}f(\boldsymbol{x}(t)) = \langle \nabla f(\boldsymbol{x}(t)), \mathrm{d}\boldsymbol{x}(t)\rangle + \frac{1}{2}\langle \nabla^2 f(\boldsymbol{x}(t)), \mathrm{d}\boldsymbol{x}(t)\,\mathrm{d}\boldsymbol{x}(t)^\top\rangle. \tag{8.168}$$

Terms of degree three and above are omitted because repeated use of the Itô rules reduces them to zero. Substituting the SDE into the first-order term gives

$$\langle \nabla f(\boldsymbol{x}(t)), \mathrm{d}\boldsymbol{x}(t)\rangle = \langle \nabla f(\boldsymbol{x}(t)), \boldsymbol{\mu}(\boldsymbol{x}(t))\rangle\,\mathrm{d}t + \nabla f(\boldsymbol{x}(t))^\top \boldsymbol{\sigma}(\boldsymbol{x}(t))\,\mathrm{d}\boldsymbol{w}(t). \tag{8.169}$$

To calculate the second-order term, we expand the product $\mathrm{d}\boldsymbol{x}(t)\ \mathrm{d}\boldsymbol{x}(t)^\top$ and use the Itô rules:

$$\begin{aligned} \mathrm{d}\boldsymbol{x}(t)\ \mathrm{d}\boldsymbol{x}(t)^\top &= \boldsymbol{\mu}(\boldsymbol{x}(t))\underbrace{(\mathrm{d}t)^2}_{=0}\boldsymbol{\mu}(\boldsymbol{x}(t))^\top \\ &\quad + \boldsymbol{\mu}(\boldsymbol{x}(t))\underbrace{\mathrm{d}t\ \mathrm{d}\boldsymbol{w}(t)^\top}_{=\mathbf{0}}\boldsymbol{\sigma}(\boldsymbol{x}(t))^\top \\ &\quad + \boldsymbol{\sigma}(\boldsymbol{x}(t))\underbrace{\mathrm{d}\boldsymbol{w}(t)\ \mathrm{d}t}_{=\mathbf{0}}\boldsymbol{\mu}(\boldsymbol{x}(t))^\top \\ &\quad + \boldsymbol{\sigma}(\boldsymbol{x}(t))\underbrace{\mathrm{d}\boldsymbol{w}(t)\ \mathrm{d}\boldsymbol{w}(t)^\top}_{=\boldsymbol{I}\,\mathrm{d}t}\boldsymbol{\sigma}(\boldsymbol{x}(t))^\top \\ &= \boldsymbol{\sigma}(\boldsymbol{x}(t))\boldsymbol{\sigma}(\boldsymbol{x}(t))^\top\ \mathrm{d}t \\ &= \boldsymbol{C}(\boldsymbol{x}(t))\ \mathrm{d}t. \end{aligned} \tag{8.170}$$

The second-order term is then

$$\frac{1}{2}\langle \nabla^2 f(\boldsymbol{x}(t)), \mathrm{d}\boldsymbol{x}(t)\ \mathrm{d}\boldsymbol{x}(t)^\top\rangle = \frac{1}{2}\langle \nabla^2 f(\boldsymbol{x}(t)), \boldsymbol{C}(\boldsymbol{x}(t))\rangle\ \mathrm{d}t. \tag{8.171}$$

Combining the first-order and second-order terms proves the lemma. □

8.2.7 SDE Approximation of SGD and the Linear Scaling Rule

We now apply SDEs to approximate SGD in continuous time. This approximation separates the motion produced by the loss gradient from the random fluctuations caused by mini-batch sampling. It also helps answer a practical question: *How should the learning rate change when the batch size changes?*

▶ SDE Approximation of SGD

Consider mini-batch SGD with batch size B. Following Definition 7.30 and example 7.32, its update is

$$\boldsymbol{g}_k \sim \mathcal{G}_{\boldsymbol{\theta}_k}, \qquad \boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta \boldsymbol{g}_k. \tag{8.172}$$

Here $\eta > 0$ is the learning rate, and $\boldsymbol{g}_k$ is the mini-batch gradient at $\boldsymbol{\theta}_k$.

Now we write the mini-batch gradient as the full gradient plus a noise term:

$$\boldsymbol{\xi}_k := \boldsymbol{g}_k - \nabla \mathcal{L}(\boldsymbol{\theta}_k), \qquad \boldsymbol{g}_k = \nabla \mathcal{L}(\boldsymbol{\theta}_k) + \boldsymbol{\xi}_k. \tag{8.173}$$

The vector $\boldsymbol{\xi}_k$ is the mini-batch gradient noise. According to Example 7.32, this noise satisfies

$$\mathbb{E}[\boldsymbol{\xi}_k] = \mathbf{0}, \qquad \mathrm{Cov}(\boldsymbol{\xi}_k) = \frac{1}{B}\boldsymbol{\Sigma}(\boldsymbol{\theta}_k), \tag{8.174}$$

where $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ is the gradient noise covariance matrix, defined by

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}) := \frac{1}{n}\sum_{i=1}^{n}\Big(\nabla \ell_i(\boldsymbol{\theta}) - \nabla \mathcal{L}(\boldsymbol{\theta})\Big)\Big(\nabla \ell_i(\boldsymbol{\theta}) - \nabla \mathcal{L}(\boldsymbol{\theta})\Big)^{\top}. \tag{8.175}$$

The factor $1/B$ appears because the mini-batch gradient averages B independent single-sample gradients.

As in the gradient flow approximation in Section 8.1.1, associate one SGD update with a time interval of length η. Substituting the decomposition of $\boldsymbol{g}_k$ into Equation (8.172) and extracting a factor of $\sqrt{\eta}$ from the noise term gives

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \eta\Big(-\nabla \mathcal{L}(\boldsymbol{\theta}_k)\Big) + \sqrt{\eta}\Big(-\sqrt{\eta}\,\boldsymbol{\xi}_k\Big). \tag{8.176}$$

This is exactly the form of the multidimensional stochastic difference equation in Equation (8.154). Its drift vector is $-\nabla \mathcal{L}(\boldsymbol{\theta}_k)$, and the noise term that follows $\sqrt{\eta}$ is $-\sqrt{\eta}\,\boldsymbol{\xi}_k$, which has covariance

$$\mathrm{Cov}\Big(-\sqrt{\eta}\,\boldsymbol{\xi}_k\Big) = (\sqrt{\eta})^2 \mathrm{Cov}(\boldsymbol{\xi}_k) = \frac{\eta}{B}\boldsymbol{\Sigma}(\boldsymbol{\theta}_k). \tag{8.177}$$

The corresponding SDE is therefore

$$\mathrm{d}\boldsymbol{\theta}(t) = -\nabla \mathcal{L}(\boldsymbol{\theta}(t))\ \mathrm{d}t + \sqrt{\frac{\eta}{B}}\,\boldsymbol{\Sigma}(\boldsymbol{\theta}(t))^{1/2}\ \mathrm{d}\boldsymbol{w}(t). \tag{8.178}$$

The drift vector of this SDE is $-\nabla \mathcal{L}(\boldsymbol{\theta})$, the same as in gradient flow. The interesting part is that the SDE also contains a diffusion term, which does not appear in gradient flow.

The diffusion coefficient here is $\sqrt{\frac{\eta}{B}}\,\boldsymbol{\Sigma}(\boldsymbol{\theta})^{1/2}$, which contains a new notation $\boldsymbol{\Sigma}(\boldsymbol{\theta})^{1/2}$ that needs some explanation. In general, for any symmetric positive semidefinite matrix $\boldsymbol{A}$, we use $\boldsymbol{A}^{1/2}$ to denote the unique symmetric positive semidefinite matrix satisfying $(\boldsymbol{A}^{1/2})^2 = \boldsymbol{A}$. More explicitly, suppose that $\boldsymbol{A}$ has the eigendecomposition

$$\boldsymbol{A} = \boldsymbol{Q}\,\mathrm{diag}(\lambda_1, \ldots, \lambda_d)\,\boldsymbol{Q}^{\top}, \tag{8.179}$$

where $\boldsymbol{Q}$ is orthogonal and $\lambda_1, \ldots, \lambda_d$ are non-negative. Then its square root is

$$\boldsymbol{A}^{1/2} = \boldsymbol{Q}\,\mathrm{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_d})\,\boldsymbol{Q}^{\top}. \tag{8.180}$$

Since $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ is a covariance matrix, it is symmetric positive semidefinite, so $\boldsymbol{\Sigma}(\boldsymbol{\theta})^{1/2}$ is well-defined.

We can now verify that the diffusion coefficient here gives the desired diffusion matrix. Since $\boldsymbol{\Sigma}(\boldsymbol{\theta})^{1/2}$ is symmetric,

$$\begin{aligned}&\Big(\sqrt{\frac{\eta}{B}}\,\boldsymbol{\Sigma}(\boldsymbol{\theta})^{1/2}\Big)\Big(\sqrt{\frac{\eta}{B}}\,\boldsymbol{\Sigma}(\boldsymbol{\theta})^{1/2}\Big)^{\top}\\ &\qquad=\frac{\eta}{B}\boldsymbol{\Sigma}(\boldsymbol{\theta})^{1/2}(\boldsymbol{\Sigma}(\boldsymbol{\theta})^{1/2})^{\top}=\frac{\eta}{B}\boldsymbol{\Sigma}(\boldsymbol{\theta}).\end{aligned} \tag{8.181}$$

At $\boldsymbol{\theta}=\boldsymbol{\theta}_k$, this is exactly the covariance of $-\sqrt{\eta}\,\boldsymbol{\xi}_k$ computed above. Thus, the SDE has the same drift vector and diffusion matrix as SGD.

▶ Linear Scaling Rule

Imagine that you are running mini-batch SGD on a collection of GPUs with batch size B. At each step, the B data points are divided across the GPUs. The GPUs compute their gradients in parallel, and SGD averages the gradients before updating the parameter vector once. If you plan to process D data points in total, the number of sequential SGD steps is

$$T=\frac{D}{B}. \tag{8.182}$$

Now suppose the number of available GPUs increases by a factor of κ. If the amount of work assigned to each GPU remains unchanged, the GPUs can process κB data points in parallel. You therefore want to increase the batch size from B to κB. Ideally, each step still takes roughly the same amount of time. Processing the same D data points now requires only

$$T'=\frac{D}{\kappa B}=\frac{T}{\kappa} \tag{8.183}$$

sequential steps. This gives a potential speedup of a factor of κ.

However, after processing the same D data points, the faster run need not achieve the same loss as the original run. If the faster run followed the same training trajectory using fewer sequential steps, then the additional GPUs would accelerate the optimization without changing its outcome. However, increasing the batch size generally changes the training trajectory. The new trajectory can be worse because the learning rate and other hyperparameters may be carefully tuned for the original batch size B. A practical question then arises: *Can we adjust the learning rate to approximately preserve the original trajectory?*

The SDE approximation gives a direct answer. In Equation (8.178), the learning rate and the batch size enter the diffusion coefficient through the factor $\sqrt{\frac{\eta}{B}}$. To keep the SDE unchanged, we should require

$$\frac{\eta'}{\kappa B}=\frac{\eta}{B}. \tag{8.184}$$

Solving this equation for η' leads to the famous *linear scaling rule* (LSR):

$$\eta'=\kappa\eta. \tag{8.185}$$

When the batch size is multiplied by κ, LSR prescribes multiplying the learning rate by κ as well. Thus both the drift and the diffusion coefficient of the SDE remain unchanged. The SDE therefore predicts the same distribution over training trajectories. The new run also covers the same continuous-time horizon because

$$T'\eta' = \frac{T}{\kappa} \cdot \kappa\eta = T\eta. \tag{8.186}$$

In deep learning, LSR is a crucial tool for ensuring the performance of SGD on large batch sizes [Kri14]. In particular, in the famous paper titled "Accurate, Large Minibatch SGD: Training ImageNet in 1 Hour" [GDG+17], Goyal et al. at Facebook, with Kaiming He as the last author, used the linear scaling rule to train ResNet-50 on ImageNet in one hour. Their training used a mini-batch size of 8192 distributed over 256 GPUs while matching the accuracy achieved with small mini-batches.

Beyond distributed training, the SDE viewpoint also helps explain how gradient noise affects the behavior and generalization of SGD [JKA+17, SED20]. Readers interested in the mathematical foundations and practical validity of the SDE approximation may refer to [LTE19, LMA21].

8.3 PDE Viewpoint of SDEs

An SDE describes how a random state changes locally. Given its current value $X(t)$, the drift and diffusion terms specify how it evolves over the next short time interval. Sometimes, however, we want a more macroscopic description. Rather than asking how one current state changes, we ask how the entire marginal distribution of $X(t)$ evolves over time.

Answering this question requires a new mathematical tool. If $p_t(x)$ denotes the probability density function of $X(t)$, then $p_t(x)$ depends on both time t and position x, and its evolution is described by a *partial differential equation* (PDE). This viewpoint will also lead us to diffusion models later in the section.

8.3.1 From SDE to PDE

We begin by studying the one-dimensional case. Consider the following SDE:

$$\mathrm{d}X(t) = \mu(X(t))\ \mathrm{d}t + \sigma(X(t))\ \mathrm{d}W(t). \tag{8.187}$$

Let $p_t(x)$ be the probability density function (PDF) of $X(t)$. The following *Fokker–Planck equation* describes how $p_t(x)$ evolves over time through a partial differential equation (PDE).

Theorem 8.13 (One-dimensional Fokker–Planck equation)**.** If $\{X(t)\}_{t\geq 0}$ evolves according to the SDE (Equation (8.187)), then the PDF $p_t(x)$ of $X(t)$ satisfies:

$$\frac{\partial p_t}{\partial t}(x) = -\frac{\partial}{\partial x}(\mu(x)p_t(x)) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(\sigma(x)^2 p_t(x)). \tag{8.188}$$

"Proof". Here we give a derivation with physics-level rigor. Recall how we used test functions in Section 5.4. Instead of working with a distribution directly, we probe it through the expectations of test functions. We use the same idea here over time.

Choose a twice differentiable test function $\varphi : \mathbb{R} \to \mathbb{R}$ that is zero outside a bounded interval. Let $\mathcal{A}^*$ be the following differential operator:

$$(\mathcal{A}^* p)(x) = -\frac{\partial}{\partial x}(\mu(x)p(x)) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(\sigma(x)^2 p(x)). \tag{8.189}$$

Our goal is to show that

$$\int_{\mathbb{R}} \varphi(x)\frac{\partial p_t}{\partial t}(x)\,\mathrm{d}x = \int_{\mathbb{R}} \varphi(x)(\mathcal{A}^* p_t)(x)\,\mathrm{d}x. \tag{8.190}$$

Once this is shown, we can conclude that $\frac{\partial p_t}{\partial t} = \mathcal{A}^* p_t$ since the choice of test function is arbitrary.

To show this, a key idea is to apply Itô's lemma to $\varphi(X(t))$.

$$\mathrm{d}\varphi(X(t)) = (\mathcal{A}\varphi)(X(t))\ \mathrm{d}t + \sigma(X(t))\varphi'(X(t))\ \mathrm{d}W(t), \tag{8.191}$$

where $\mathcal{A}$ is the differential operator

$$(\mathcal{A}\varphi)(x) = \mu(x)\frac{\partial}{\partial x}\varphi(x) + \frac{1}{2}\sigma(x)^2\frac{\partial^2}{\partial x^2}\varphi(x). \tag{8.192}$$

Taking expectations gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\varphi(X(t))] = \mathbb{E}[(\mathcal{A}\varphi)(X(t))]. \tag{8.193}$$

Expanding the expectations into integrals gives

$$\int_{\mathbb{R}} \varphi(x)\frac{\partial p_t}{\partial t}(x)\ \mathrm{d}x = \int_{\mathbb{R}} (\mathcal{A}\varphi)(x)p_t(x)\ \mathrm{d}x. \tag{8.194}$$

It remains to show that the right-hand side of Equation (8.190) is equal to the right-hand side of Equation (8.194). But this is straightforward to show by integrating by parts:

$$\int_{\mathbb{R}} \mu(x)p_t(x)\frac{\partial}{\partial x}\varphi(x)\ \mathrm{d}x = -\int_{\mathbb{R}} \frac{\partial}{\partial x}(\mu(x)p_t(x))\varphi(x)\ \mathrm{d}x, \tag{8.195}$$

$$\frac{1}{2}\int_{\mathbb{R}} \sigma(x)^2 p_t(x)\frac{\partial^2}{\partial x^2}\varphi(x)\ \mathrm{d}x = \frac{1}{2}\int_{\mathbb{R}} \frac{\partial^2}{\partial x^2}(\sigma(x)^2 p_t(x))\varphi(x)\ \mathrm{d}x. \tag{8.196}$$

Adding these two equations together gives

$$\int_{\mathbb{R}} (\mathcal{A}\varphi)(x)p_t(x)\ \mathrm{d}x = \int_{\mathbb{R}} \varphi(x)(\mathcal{A}^* p_t)(x)\,\mathrm{d}x, \tag{8.197}$$

which completes the proof. □

Remark 8.14 (Generator)**.** The differential operator $\mathcal{A}$ introduced above is called the *generator* of the SDE. It summarizes how the expected value of a test function changes over time.

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\varphi(X(t))] = \mathbb{E}[(\mathcal{A}\varphi)(X(t))]. \tag{8.198}$$

Remark 8.15 (Adjoint operator)**.** The relationship between $\mathcal{A}$ and $\mathcal{A}^*$ can be written using an inner product between functions. For two functions f and g, let

$$\langle f, g\rangle := \int_{\mathbb{R}} f(x)g(x)\,\mathrm{d}x. \tag{8.199}$$

Then the integration by parts calculation in the proof essentially shows that

$$\langle \mathcal{A}\varphi, p\rangle = \langle \varphi, \mathcal{A}^* p\rangle. \tag{8.200}$$

In general, for two operators $\mathcal{A}$ and $\mathcal{A}^*$ that satisfy Equation (8.200) for all pairs of functions φ and p, we say that $\boldsymbol{A}$ and $\boldsymbol{A}^*$ are *adjoint* to each other. This is analogous to the transpose of a matrix, which satisfies $\langle \boldsymbol{A}\boldsymbol{u}, \boldsymbol{v}\rangle = \langle \boldsymbol{u}, \boldsymbol{A}^\top \boldsymbol{v}\rangle$. Readers interested in the general theory of adjoint operators can take a course on functional analysis.

We now consider the multidimensional SDE

$$\mathrm{d}\boldsymbol{x}(t) = \boldsymbol{\mu}(\boldsymbol{x}(t))\,\mathrm{d}t + \boldsymbol{\sigma}(\boldsymbol{x}(t))\,\mathrm{d}\boldsymbol{w}(t). \tag{8.201}$$

Let $p_t(\boldsymbol{x})$ be the PDF of $\boldsymbol{x}(t)$. Recall that the diffusion matrix is

$$\boldsymbol{C}(\boldsymbol{x}) := \boldsymbol{\sigma}(\boldsymbol{x})\boldsymbol{\sigma}(\boldsymbol{x})^\top. \tag{8.202}$$

The same argument as in the one-dimensional case gives the following result, which we state without proof.

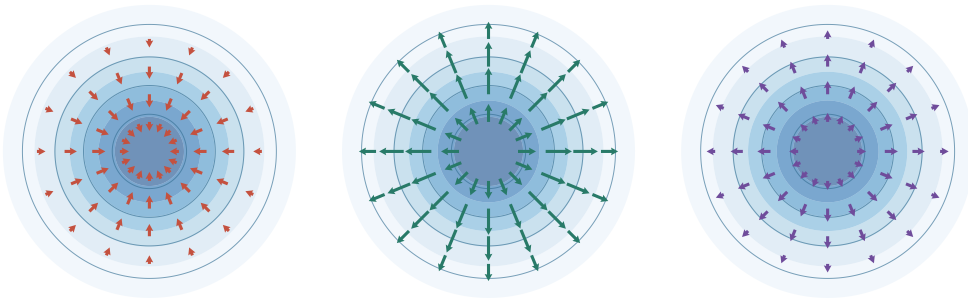
Theorem 8.16 (Multidimensional Fokker–Planck equation)**.** If $\{\boldsymbol{x}(t)\}_{t\geq 0}$ evolves according to the SDE (Equation (8.201)), then the PDF $p_t(\boldsymbol{x})$ of $\boldsymbol{x}(t)$ satisfies

$$\frac{\partial p_t}{\partial t}(\boldsymbol{x}) = -\sum_{i=1}^{d} \frac{\partial}{\partial x_i}(\mu_i(\boldsymbol{x})p_t(\boldsymbol{x})) + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d} \frac{\partial^2}{\partial x_i \partial x_j}(C_{ij}(\boldsymbol{x})p_t(\boldsymbol{x})). \tag{8.203}$$

The Fokker–Planck equation can be read as a description of how probability flows through space. Define the *probability current* $\boldsymbol{J}_t(\boldsymbol{x}) = (J_{t,1}(\boldsymbol{x}), \ldots, J_{t,d}(\boldsymbol{x}))$, where

$$J_{t,i}(\boldsymbol{x}) := \mu_i(\boldsymbol{x})p_t(\boldsymbol{x}) - \frac{1}{2}\sum_{j=1}^{d} \frac{\partial}{\partial x_j}(C_{ij}(\boldsymbol{x})p_t(\boldsymbol{x})). \tag{8.204}$$

Intuitively, $\boldsymbol{J}_t(\boldsymbol{x})$ describes how probability is flowing near $\boldsymbol{x}$ at time t. The direction of $\boldsymbol{J}_t(\boldsymbol{x})$ indicates where probability is moving, and its magnitude indicates how strongly it is flowing. The first term in Equation (8.204) represents the



(a) $\mathrm{d}\boldsymbol{x}(t) = -\frac{1}{2}\boldsymbol{x}(t)\,\mathrm{d}t$. (b) $\mathrm{d}\boldsymbol{x}(t) = \mathrm{d}\boldsymbol{w}(t)$. (c) $\mathrm{d}\boldsymbol{x}(t) = -\frac{1}{2}\boldsymbol{x}(t)\,\mathrm{d}t + \mathrm{d}\boldsymbol{w}(t)$.

Figure 8.3: Probability currents of an OU process when $p_t = \mathcal{N}(\mathbf{0}, \frac{1}{2}\boldsymbol{I})$. **(a)** The drift term carries probability towards the origin. **(b)** The diffusion term spreads probability outward. **(c)** For the full OU process, the diffusion happens to be stronger than the drift, so the total current points outward.

probability mass near $\boldsymbol{x}$ being carried along by the drift $\boldsymbol{\mu}(\boldsymbol{x})$. The second term represents spreading caused by random fluctuations, while the diffusion matrix $\boldsymbol{C}(\boldsymbol{x})$ shapes and distorts this spreading by controlling its strength in different directions.

With this notation, the Fokker–Planck equation becomes

$$\frac{\partial p_t}{\partial t} + \nabla \cdot \boldsymbol{J}_t = 0, \tag{8.205}$$

where $\nabla \cdot \boldsymbol{J}_t := \sum_{i=1}^{d} \frac{\partial J_{t,i}}{\partial x_i}$ denotes the divergence of the probability current. Physically, the divergence measures the net outward flow from a small neighborhood of a point. A positive value means that more probability flows out of the neighborhood than into it, while a negative value means that more probability flows in than out.

For any region Ω, Equation (8.205) gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega} p_t(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x} = -\int_{\partial\Omega} \boldsymbol{J}_t(\boldsymbol{x}) \cdot \mathrm{d}\boldsymbol{S}, \tag{8.206}$$

where $\mathrm{d}\boldsymbol{S}$ is the outward-oriented vector surface element on $\partial\Omega$. The left-hand side is the rate of change of the probability inside Ω, and the right-hand side is the negative of the total outward probability flow across its boundary. This identity is intuitive because, when we sum the changes in probability over all points inside Ω, flows between points inside Ω cancel, leaving only the flow across its boundary.

Example 8.17 (Constant diffusion matrix)**.** Suppose the diffusion matrix $\boldsymbol{C}(\boldsymbol{x}) = \boldsymbol{C}$ is constant. Then the Fokker–Planck equation becomes

$$\frac{\partial p_t}{\partial t}(\boldsymbol{x}) = -\nabla \cdot (\boldsymbol{\mu}(\boldsymbol{x}) p_t(\boldsymbol{x})) + \frac{1}{2}\langle \boldsymbol{C}, \nabla^2 p_t(\boldsymbol{x})\rangle. \tag{8.207}$$

The corresponding probability current is

$$\boldsymbol{J}_t(\boldsymbol{x}) = \boldsymbol{\mu}(\boldsymbol{x}) p_t(\boldsymbol{x}) - \frac{1}{2}\boldsymbol{C}\nabla p_t(\boldsymbol{x}). \tag{8.208}$$

The first term carries probability along the drift, while the second drives it downhill on the density landscape in the direction of $-\nabla p_t$, with $\boldsymbol{C}$ stretching the resulting flow.

Example 8.18 (Brownian motion)**.** For standard Brownian motion, $\boldsymbol{\mu} = \boldsymbol{0}$ and $\boldsymbol{C} = \boldsymbol{I}$. Its Fokker–Planck equation is therefore the heat equation

$$\frac{\partial p_t}{\partial t}(\boldsymbol{x}) = \frac{1}{2}\Delta p_t(\boldsymbol{x}), \tag{8.209}$$

where $\Delta p := \sum_{i=1}^{d} \frac{\partial^2 p}{\partial x_i^2}$ is the Laplacian of p. The corresponding probability current is

$$\boldsymbol{J}_t(\boldsymbol{x}) = -\frac{1}{2}\nabla p_t(\boldsymbol{x}). \tag{8.210}$$

Thus the current points from regions of higher density towards regions of lower density.

Recall from Equation (8.145) that a Brownian increment over elapsed time t has distribution $\mathcal{N}(\boldsymbol{0}, t\boldsymbol{I})$. So if the Brownian motion starts from $\boldsymbol{x}_0$, its value after time t has distribution $\mathcal{N}(\boldsymbol{x}_0, t\boldsymbol{I})$, whose PDF is

$$p_t(\boldsymbol{x}) = \frac{1}{(2\pi t)^{d/2}} \exp\left(-\frac{\|\boldsymbol{x} - \boldsymbol{x}_0\|_2^2}{2t}\right). \tag{8.211}$$

One can verify that this PDF solves Equation (8.209).

8.3.2 Langevin Dynamics and Gibbs Distributions

In Example 7.34, we introduced an idealized model of SGD where the gradient noise is Gaussian. We now study a particularly useful scaling of this noise:

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta\nabla\mathcal{L}(\boldsymbol{\theta}_k) + \sqrt{2\tau\eta}\,\boldsymbol{\xi}_k, \qquad \boldsymbol{\xi}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}). \tag{8.212}$$

Here $\tau > 0$ is called the *temperature*. If one iteration is viewed as a time step of length η, then this update is an Euler–Maruyama step for the SDE

$$\mathrm{d}\boldsymbol{\theta}(t) = -\nabla\mathcal{L}(\boldsymbol{\theta}(t))\ \mathrm{d}t + \sqrt{2\tau}\ \mathrm{d}\boldsymbol{w}(t). \tag{8.213}$$

This SDE is called *overdamped Langevin dynamics*. If p_t is the PDF of $\boldsymbol{\theta}(t)$, then its Fokker–Planck equation and probability current are

$$\frac{\partial p_t}{\partial t} = -\nabla \cdot \boldsymbol{J}_t, \qquad \boldsymbol{J}_t = -p_t\nabla\mathcal{L} - \tau\nabla p_t. \tag{8.214}$$

The first term moves probability towards lower loss, while the second spreads it out. As time evolves, we might expect these two effects to balance and the PDF to settle into a steady state. Indeed, we already encountered a steady state of this kind for the Ornstein–Uhlenbeck process in Section 8.2.5, where we called it a stationary distribution. We now define this term for general SDEs.

Definition 8.19 (Stationary distribution)**.** A probability distribution with PDF p_∞ is *stationary* for an SDE if starting the process with PDF $p_0 = p_\infty$ implies $p_t = p_\infty$ for every $t \geq 0$. In terms of the Fokker–Planck equation, this means that p_∞ satisfies

$$0 = -\sum_{i=1}^{d} \frac{\partial}{\partial x_i}(\mu_i(\boldsymbol{x})p_\infty(\boldsymbol{x})) + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d} \frac{\partial^2}{\partial x_i \partial x_j}(C_{ij}(\boldsymbol{x})p_\infty(\boldsymbol{x})). \quad (8.215)$$

For Langevin dynamics, the stationary distribution has a simple form.

Theorem 8.20 (Stationary distribution of Langevin dynamics)**.** For Langevin dynamics in Equation (8.213), suppose

$$Z_\tau := \int_{\mathbb{R}^d} e^{-\mathcal{L}(\boldsymbol{\theta})/\tau} \, \mathrm{d}\boldsymbol{\theta} < \infty. \quad (8.216)$$

Then the PDF

$$p_\infty(\boldsymbol{\theta}) = \frac{1}{Z_\tau} e^{-\mathcal{L}(\boldsymbol{\theta})/\tau} \quad (8.217)$$

is the PDF of the unique stationary distribution.

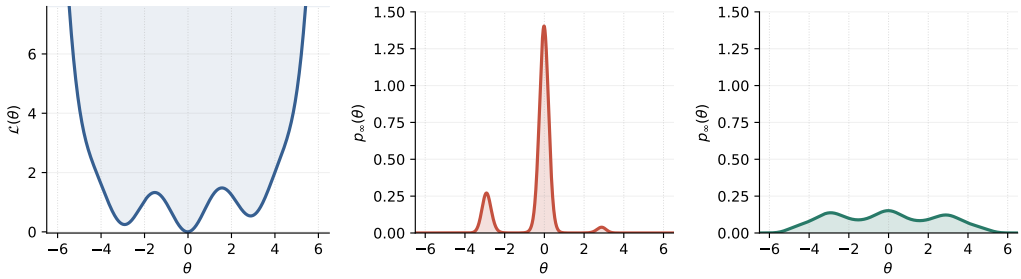
The distribution specified by Equation (8.217) is called the *Gibbs distribution* associated with $\mathcal{L}$. The parameter τ is called the *temperature*. The value of its PDF at $\boldsymbol{\theta}$ is proportional to $e^{-\mathcal{L}(\boldsymbol{\theta})/\tau}$, and Z_τ is chosen so that the PDF integrates to 1. Because Z_τ depends on the temperature τ, it is a function of τ and is called the *partition function* associated with $\mathcal{L}$. These names come from statistical mechanics, where the Gibbs distribution describes the probability of a system being in each possible state at thermal equilibrium.

Since the PDF is proportional to $e^{-\mathcal{L}(\boldsymbol{\theta})/\tau}$, it decreases exponentially as $\mathcal{L}(\boldsymbol{\theta})$ increases. The factor $1/\tau$ determines how quickly this decrease occurs. At low temperature, $1/\tau$ is large, so even a small increase in loss causes a large decrease in the PDF, and probability concentrates sharply near the global minimizer of $\mathcal{L}$. At high temperature, $1/\tau$ is small, so differences in loss have a weaker effect and probability spreads across several basins of the loss landscape. This contrast is illustrated in Figure 8.4.

Now we prove the theorem.

Proof. Let p_∞ be the PDF of any stationary distribution. Setting the left-hand side of Equation (8.214) to zero gives

$$0 = -\nabla \cdot \boldsymbol{J}_\infty, \qquad \boldsymbol{J}_\infty = -p_\infty \nabla \mathcal{L} - \tau \nabla p_\infty. \quad (8.218)$$



(a) Loss. (b) Low temperature. (c) High temperature.

Figure 8.4: A visualization of the Gibbs distribution over a non-convex loss landscape. At low temperature ($\tau = 0.15$), the Gibbs distribution concentrates sharply near the global minimizer. At high temperature ($\tau = 2.5$), it spreads more broadly across several basins of the loss landscape.

A key observation here is that the probability current can be rewritten as

$$\boldsymbol{J}_\infty(\boldsymbol{\theta}) = -\tau e^{-\mathcal{L}(\boldsymbol{\theta})/\tau} \nabla q(\boldsymbol{\theta}), \qquad q(\boldsymbol{\theta}) := e^{\mathcal{L}(\boldsymbol{\theta})/\tau} p_\infty(\boldsymbol{\theta}). \tag{8.219}$$

In coordinates, the stationary condition is

$$\sum_{i=1}^{d} \frac{\partial J_{\infty,i}}{\partial \theta_i} = 0. \tag{8.220}$$

Multiplying by q, integrating over $\mathbb{R}^d$, and applying integration by parts in each coordinate gives

$$0 = \int_{\mathbb{R}^d} \sum_{i=1}^{d} q \frac{\partial J_{\infty,i}}{\partial \theta_i} \, \mathrm{d}\boldsymbol{\theta} = \sum_{i=1}^{d} \int_{\mathbb{R}^d} q \frac{\partial J_{\infty,i}}{\partial \theta_i} \, \mathrm{d}\boldsymbol{\theta} \tag{8.221}$$

$$= -\sum_{i=1}^{d} \int_{\mathbb{R}^d} \frac{\partial q}{\partial \theta_i} J_{\infty,i} \, \mathrm{d}\boldsymbol{\theta}, \tag{8.222}$$

Substituting the expression for $\boldsymbol{J}_\infty$ above gives

$$0 = \tau \sum_{i=1}^{d} \int_{\mathbb{R}^d} e^{-\mathcal{L}(\boldsymbol{\theta})/\tau} \left(\frac{\partial q}{\partial \theta_i}(\boldsymbol{\theta}) \right)^2 \mathrm{d}\boldsymbol{\theta} = \tau \int_{\mathbb{R}^d} e^{-\mathcal{L}(\boldsymbol{\theta})/\tau} \|\nabla q(\boldsymbol{\theta})\|_2^2 \, \mathrm{d}\boldsymbol{\theta}. \tag{8.223}$$

The weight $e^{-\mathcal{L}(\boldsymbol{\theta})/\tau}$ is strictly positive, and the squared norm $\|\nabla q(\boldsymbol{\theta})\|_2^2$ is non-negative. The integral can therefore vanish only if $\nabla q(\boldsymbol{\theta}) = \mathbf{0}$, so $q(\boldsymbol{\theta})$ is a constant function. Therefore, $p_\infty(\boldsymbol{\theta})$ must have the form

$$p_\infty(\boldsymbol{\theta}) = c e^{-\mathcal{L}(\boldsymbol{\theta})/\tau} \tag{8.224}$$

for some constant $c \in \mathbb{R}$. The condition $\int_{\mathbb{R}^d} p_\infty(\boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\theta} = 1$ forces $c = 1/Z_\tau$. It can therefore be verified that the PDF in Equation (8.217) is the unique solution to the stationary condition. □

When $\mathcal{L}(\boldsymbol{\theta})$ is quadratic, Langevin dynamics is simply the Ornstein–Uhlenbeck process studied in Section 8.2.5. There, we obtained its stationary distribution and showed that the marginal distribution converges to it from any initial state.

Example 8.21 (One-dimensional Ornstein–Uhlenbeck process)**.** Consider one-dimensional Langevin dynamics with

$$\mathcal{L}(x) = \frac{\gamma x^2}{2}, \qquad \tau = \frac{\sigma^2}{2}, \tag{8.225}$$

where $\gamma > 0$ and $\sigma > 0$. Since $\mathcal{L}'(x) = \gamma x$, the Langevin SDE becomes

$$\mathrm{d}X(t) = -\gamma X(t)\ \mathrm{d}t + \sigma\ \mathrm{d}W(t). \tag{8.226}$$

This is exactly the Ornstein–Uhlenbeck process defined in Equation (8.122). The corresponding Fokker–Planck equation is

$$\frac{\partial p_t}{\partial t}(x) = \gamma \frac{\partial}{\partial x}(x p_t(x)) + \frac{\sigma^2}{2}\frac{\partial^2 p_t}{\partial x^2}(x). \tag{8.227}$$

The Gibbs distribution from Theorem 8.20 has PDF

$$p_\infty(x) = \sqrt{\tfrac{\gamma}{\pi\sigma^2}}\, e^{-\gamma x^2/\sigma^2}. \tag{8.228}$$

This is the PDF of $\mathcal{N}(0, \frac{\sigma^2}{2\gamma})$, the same stationary distribution obtained from the explicit solution in Equation (8.139).

Example 8.22 (Multidimensional Ornstein–Uhlenbeck process)**.** The same example extends to $\mathbb{R}^d$. Consider Langevin dynamics with

$$\mathcal{L}(\boldsymbol{x}) = \frac{\gamma\|\boldsymbol{x}\|_2^2}{2}, \qquad \tau = \frac{\sigma^2}{2}. \tag{8.229}$$

The resulting SDE is

$$\mathrm{d}\boldsymbol{x}(t) = -\gamma \boldsymbol{x}(t)\ \mathrm{d}t + \sigma\ \mathrm{d}\boldsymbol{w}(t). \tag{8.230}$$

The corresponding Fokker–Planck equation is

$$\frac{\partial p_t}{\partial t}(\boldsymbol{x}) = \gamma \nabla \cdot (\boldsymbol{x} p_t(\boldsymbol{x})) + \frac{\sigma^2}{2}\Delta p_t(\boldsymbol{x}). \tag{8.231}$$

The Gibbs distribution from Theorem 8.20 has PDF

$$p_\infty(\boldsymbol{x}) = \left(\tfrac{\gamma}{\pi\sigma^2}\right)^{d/2} e^{-\gamma\|\boldsymbol{x}\|_2^2/\sigma^2}. \tag{8.232}$$

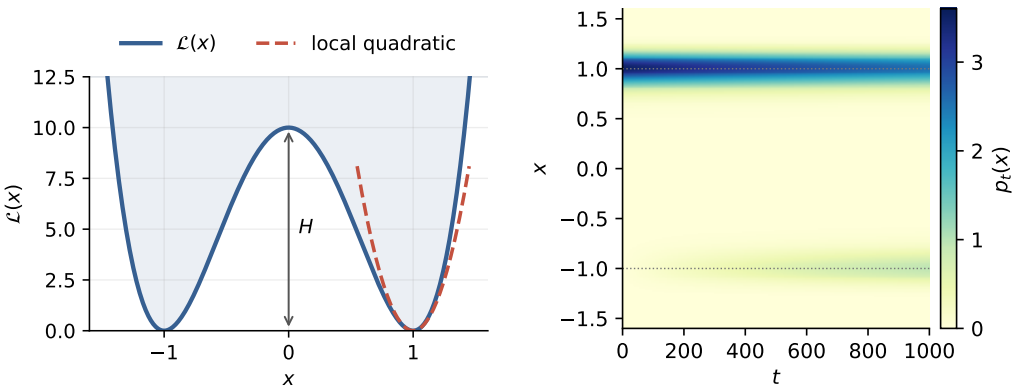
We can verify this conclusion through direct calculation. Suppose $\boldsymbol{x}(0) = \boldsymbol{x}_0$. Repeating the calculation from Section 8.2.5 coordinatewise, we obtain

$$\boldsymbol{x}(t) \sim \mathcal{N}\left(e^{-\gamma t}\boldsymbol{x}_0, \frac{\sigma^2}{2\gamma}(1 - e^{-2\gamma t})\boldsymbol{I}\right). \tag{8.233}$$

As $t \to \infty$, the mean tends to $\mathbf{0}$ and the covariance tends to $\sigma^2/(2\gamma)\boldsymbol{I}$. Therefore,

$$\boldsymbol{x}(t) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}, \frac{\sigma^2}{2\gamma}\boldsymbol{I}\right), \tag{8.234}$$

which is exactly the stationary distribution obtained above.



(a) Double-well loss.

(b) Evolution of the PDF.

Figure 8.5: Slow convergence for Langevin dynamics on the double-well loss $\mathcal{L}(x) = H(x^2-1)^2$, with $H = 10$ and $\tau = 1$. Starting from $X(0) = 1$, the marginal distribution p_t of $X(t)$ quickly settles near the minimizer $x = 1$ but transfers probability to the other minimizer $x = -1$ much more slowly.

For these quadratic losses, convergence to the stationary distribution is exponentially fast. For a general loss, however, convergence to the Gibbs distribution can be much slower, as the following example illustrates.

Example 8.23 (A double-well loss)**.** Consider one-dimensional Langevin dynamics with

$$\mathcal{L}(x) = H(x^2-1)^2, \qquad H > 0. \tag{8.235}$$

This loss has two local minimizers at $x = -1$ and $x = 1$, separated by a barrier of height H at $x = 0$. The corresponding Gibbs distribution is

$$p_\infty(x) = \frac{1}{Z_\tau} e^{-H(x^2-1)^2/\tau}, \tag{8.236}$$

which places probability equally near both minimizers.

Suppose that the process starts from $X(0) = 1$. Writing $x = 1 + y$ and expanding the loss around this minimizer gives

$$\mathcal{L}(1+y) = 4Hy^2 + \mathcal{O}(y^3). \tag{8.237}$$

Consequently, $X(t)$ behaves approximately like an Ornstein–Uhlenbeck process near $x = 1$ and quickly settles there. Approaching the Gibbs distribution, however, also requires probability to reach the other well $x = -1$. To do so, the process must cross the barrier at $x = 0$. When H is large relative to τ, such crossings are rare, so the process can remain trapped near one minimizer for a very long time.

8.3.3 Diffusion Models

We now return to the image generation problem discussed in Section 5.4.3, where real images are viewed as samples from an unknown distribution $\mathcal{D}_{\text{data}}$, and the goal is to generate new images from approximately the same distribution. We have introduced GANs, where a generator and a discriminator are trained together through a zero-sum game. Although GANs can generate impressive images, training two neural networks against each other can be very tricky.

For several years, much of the progress in generative modeling came from designing better objectives and better regularization for GANs, so it was natural to expect the next major advance to be a still better GAN. However, what happened instead was the rise of a different class of generative models, called *diffusion models*. This new class of models began to attract widespread attention around 2020 and soon became a dominant approach to generative modeling.

Rather than further modifying the game between a generator and a discriminator, diffusion models abandon this game altogether. At the heart of their approach is a gradual transformation between data and noise. Song et al. [SSDK+21] provided an influential unifying perspective, showing that the gradual addition and removal of noise underlying several existing diffusion models can be formulated as a forward SDE and its corresponding reverse-time SDE. We adopt this SDE-based viewpoint throughout this section.

At a high level, diffusion models involve three key ideas:

1. **Forward diffusion.** Start with $\boldsymbol{x}(0) \sim \mathcal{D}_{\text{data}}$ and gradually add noise using an Ornstein–Uhlenbeck process. At a large time T, the marginal distribution p_T is close to that of a standard Gaussian $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$, from which we can easily sample.
2. **Reverse diffusion.** Transform p_T back into $\mathcal{D}_{\text{data}}$ using a reverse-time SDE. Its drift depends on $\nabla \log p_t(\boldsymbol{x})$, known as the *score function*.
3. **Learning and sampling.** Train a neural network to learn the score from training data. To generate a new sample, we start with Gaussian noise and simulate the reverse SDE using the learned score.

▶ Forward Diffusion

The first step is to turn data into noise in a controlled way. The multidimensional Ornstein–Uhlenbeck process studied in Example 8.22 provides exactly such a mechanism: its drift gradually shrinks the initial signal, while its Brownian term injects Gaussian noise. Setting $\gamma = 1/2$ and $\sigma = 1$ in Equation (8.230), we define the forward diffusion by

$$\mathrm{d}\boldsymbol{x}(t) = -\frac{1}{2}\boldsymbol{x}(t)\ \mathrm{d}t + \mathrm{d}\boldsymbol{w}(t). \tag{8.238}$$

Conditioned on the initial value $\boldsymbol{x}(0) = \boldsymbol{x}_0$, Equation (8.233) gives

$$\boldsymbol{x}(t) \sim \mathcal{N}\left(e^{-t/2}\boldsymbol{x}_0, (1 - e^{-t})\boldsymbol{I}\right). \tag{8.239}$$

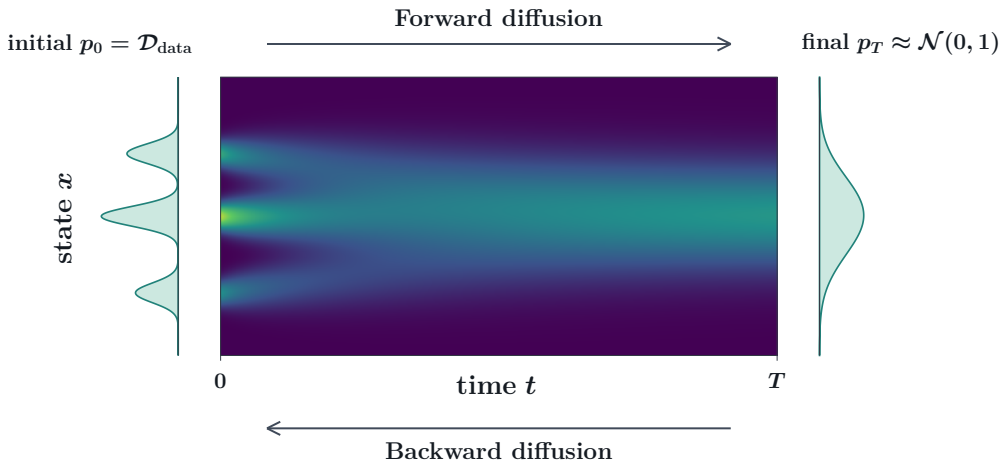


Figure 8.6: An illustration of the main idea of diffusion models. Starting from the data distribution $p_0 = \mathcal{D}_{\text{data}}$, the forward diffusion gradually removes its structure and produces a distribution p_T that is close to $\mathcal{N}(0,1)$. A diffusion model learns the score needed to run the reverse diffusion, which turns noise back into data. The heat map shows the intermediate PDFs p_t at each time t.

Thus, as t increases, the contribution of the clean data point $\boldsymbol{x}_0$ decays while the noise variance increases. When the clean point is itself drawn from the data distribution, we have the following representation of the marginal distribution.

Theorem 8.24 (Forward diffusion)**.** Let $\boldsymbol{x}(0) \sim \mathcal{D}_{\text{data}}$, let $\boldsymbol{x}(t)$ follow Equation (8.238), and let $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ be independent of $\boldsymbol{x}(0)$. Then, for every $t > 0$,

$$\boldsymbol{x}(t) \sim e^{-t/2}\boldsymbol{x}(0) + \sqrt{1 - e^{-t}}\,\boldsymbol{z}. \tag{8.240}$$

Moreover,

$$\boldsymbol{x}(t) \xrightarrow{d} \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}) \qquad \text{as } t \to \infty. \tag{8.241}$$

Proof. Fix $\boldsymbol{x}(0) = \boldsymbol{x}_0$. Since $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, the random vector

$$e^{-t/2}\boldsymbol{x}_0 + \sqrt{1 - e^{-t}}\,\boldsymbol{z} \tag{8.242}$$

has distribution

$$\mathcal{N}\left(e^{-t/2}\boldsymbol{x}_0, (1 - e^{-t})\boldsymbol{I}\right), \tag{8.243}$$

which is exactly the conditional distribution of $\boldsymbol{x}(t)$ in Equation (8.239). Sampling $\boldsymbol{x}_0 \sim \mathcal{D}_{\text{data}}$ therefore gives Equation (8.240).

Finally, as $t \to \infty$, $e^{-t/2} \to 0$ and $\sqrt{1 - e^{-t}} \to 1$. Hence the right-hand side of Equation (8.240) converges in distribution to $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$. □

The representation above shows that the marginal distribution p_t of $\boldsymbol{x}(t)$ is a Gaussian mixture obtained by placing a Gaussian with covariance $(1 - e^{-t})\boldsymbol{I}$ around each scaled data point $e^{-t/2}\boldsymbol{x}_0$. As t increases, these centers contract towards the origin and the covariance approaches $\boldsymbol{I}$. Consequently, the mixture loses the structure of p_0 and approaches the standard Gaussian $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$.

▶ Reverse Diffusion

As shown in Theorem 8.24, forward diffusion transforms the data distribution into the standard Gaussian. Now suppose that we could reverse this forward diffusion process, that is, given a standard Gaussian sample, we could turn it into a sample from the data distribution. Then this would perfectly solve the generation problem because sampling from a standard Gaussian is easy, and running the reverse diffusion would convert the Gaussian into a new data point. *But can we reverse the forward diffusion?*

For an ODE, reversing time is easy. If

$$\frac{\mathrm{d}\boldsymbol{x}(t)}{\mathrm{d}t} = F(\boldsymbol{x}(t)), \tag{8.244}$$

then the reversed trajectory $\boldsymbol{y}(s) := \boldsymbol{x}(T - s)$ satisfies

$$\frac{\mathrm{d}\boldsymbol{y}(s)}{\mathrm{d}s} = -F(\boldsymbol{y}(s)). \tag{8.245}$$

Therefore, if we want to reverse an ODE, we should simply negate the velocity field $F(\boldsymbol{x}(t))$.

It is therefore tempting to reverse the forward OU process in Equation (8.238) by negating its drift. Since its forward drift is $-\frac{1}{2}\boldsymbol{x}$, this would suggest the SDE

$$\mathrm{d}\boldsymbol{y}(s) = \frac{1}{2}\boldsymbol{y}(s)\ \mathrm{d}s + \mathrm{d}\bar{\boldsymbol{w}}(s), \tag{8.246}$$

where $\boldsymbol{y}(0)$ has PDF p_T, s is the time elapsed in the reverse direction, and $\bar{\boldsymbol{w}}(s)$ is a standard Brownian motion. Is the distribution of $\boldsymbol{y}(T)$ the same as $\boldsymbol{x}(0)$?



This is not the right way to reverse the forward diffusion!

To understand why this fails, we compare the density evolution of the above SDE with the density evolution required of a true reverse process. If p_t is the forward density, then the reverse process should visit the same marginal distributions in the opposite order. Its PDF at reverse time s must be

$$q_s(\boldsymbol{x}) := p_{T-s}(\boldsymbol{x}). \tag{8.247}$$

In particular, it starts from $q_0 = p_T$ and ends at $q_T = p_0$, the PDF of $\mathcal{D}_{\text{data}}$.

The Fokker–Planck equation for the forward OU process is

$$\frac{\partial p_t}{\partial t}(\boldsymbol{x}) = \frac{1}{2}\nabla \cdot (\boldsymbol{x} p_t(\boldsymbol{x})) + \frac{1}{2}\Delta p_t(\boldsymbol{x}). \tag{8.248}$$

Differentiating $q_s = p_{T-s}$ with respect to s gives the evolution required of the reverse density.

$$\frac{\partial q_s}{\partial s}(\boldsymbol{x}) = -\frac{1}{2}\nabla \cdot (\boldsymbol{x} q_s(\boldsymbol{x})) - \frac{1}{2}\Delta q_s(\boldsymbol{x}). \tag{8.249}$$

However, the Fokker–Planck equation for the SDE in Equation (8.246) is

$$\frac{\partial q_s}{\partial s}(\boldsymbol{x}) = -\frac{1}{2}\nabla \cdot (\boldsymbol{x} q_s(\boldsymbol{x})) + \frac{1}{2}\Delta q_s(\boldsymbol{x}). \tag{8.250}$$

But these two equations have opposite signs for the diffusion terms!

Is it because that we did not negate the diffusion term in Equation (8.246)? In fact, negating the diffusion term does not essentially change the stochastic process, as discussed in Section 8.2.6. To really fix the problem, we need a significant correction to the drift term.

Theorem 8.25 (Reverse diffusion)**.** Let $\boldsymbol{y}(0) \sim p_T$, and let $\boldsymbol{y}(s)$ follow the SDE

$$\mathrm{d}\boldsymbol{y}(s) = \left(\frac{1}{2}\boldsymbol{y}(s) + \nabla \log q_s(\boldsymbol{y}(s))\right)\mathrm{d}s + \mathrm{d}\bar{\boldsymbol{w}}(s), \tag{8.251}$$

where $\bar{\boldsymbol{w}}(s)$ is a standard Brownian motion and q_s is the PDF of $\boldsymbol{y}(s)$. Then $\boldsymbol{y}(s) \sim p_{T-s}$ at every reverse time s. In particular, $\boldsymbol{y}(T) \sim \mathcal{D}_{\text{data}}$.

Before proving the theorem, we note that the new term $\nabla \log q_s$ appearing in the theorem is called the *score function*, a concept from statistics. In general, the score function of a PDF p is defined as

$$s(\boldsymbol{x}) := \nabla \log p(\boldsymbol{x}) = \frac{\nabla p(\boldsymbol{x})}{p(\boldsymbol{x})}. \tag{8.252}$$

A score function points in the direction of increasing density, and thus pushes a noisy point towards high-density regions in Equation (8.251).

Example 8.26 (Score of a standard Gaussian)**.** If $q(\boldsymbol{x})$ is the density of $\mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, then $\nabla \log q(\boldsymbol{x}) = -\boldsymbol{x}$.

Proof of Theorem 8.25. The PDF q_s of $\boldsymbol{y}(s)$ satisfies the Fokker–Planck equation

$$\frac{\partial q_s}{\partial s}(\boldsymbol{x}) = -\nabla \cdot \left(\left(\frac{1}{2}\boldsymbol{x} + \nabla \log q_s(\boldsymbol{x})\right) q_s(\boldsymbol{x})\right) + \frac{1}{2}\Delta q_s(\boldsymbol{x}). \tag{8.253}$$

Note that $\nabla \log q_s(\boldsymbol{x})\, q_s(\boldsymbol{x}) = \nabla q_s(\boldsymbol{x})$. Substituting this into the Fokker–Planck equation gives

$$\frac{\partial q_s}{\partial s}(\boldsymbol{x}) = -\nabla \cdot (\boldsymbol{x} q_s(\boldsymbol{x})) - \nabla \cdot \nabla q_s(\boldsymbol{x}) + \frac{1}{2}\Delta q_s(\boldsymbol{x}) \tag{8.254}$$

$$= -\frac{1}{2}\nabla \cdot (\boldsymbol{x} q_s(\boldsymbol{x})) - \frac{1}{2}\Delta q_s(\boldsymbol{x}), \tag{8.255}$$

where the last line uses the fact that $\nabla \cdot \nabla q_s(\boldsymbol{x}) = \Delta q_s(\boldsymbol{x})$. This exactly matches Equation (8.249), and thus $q_s = p_{T-s}$. □

Remark 8.27 (Initial condition)**.** The theorem starts from the exact distribution p_T. In practice, T is chosen large enough that p_T is close to $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$, and generation starts from a standard Gaussian sample instead.

Remark 8.28 (Time-dependent reverse dynamics)**.** Note that the reverse diffusion process in Theorem 8.25 is time-dependent, as the score function $\nabla \log q_s$ depends on the reverse time s. Nevertheless, Itô's lemma and the Fokker–Planck equation also apply in this time-dependent setting. More generally, we may consider an SDE of the form

$$\mathrm{d}\boldsymbol{x}(t) = \boldsymbol{\mu}_t(\boldsymbol{x}(t))\ \mathrm{d}t + \boldsymbol{\sigma}_t(\boldsymbol{x}(t))\ \mathrm{d}\boldsymbol{w}(t). \tag{8.256}$$

The results in Lemma 8.12 and theorem 8.16 continue to hold with $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ replaced by $\boldsymbol{\mu}_t$ and $\boldsymbol{\sigma}_t$. We used this time-dependent extension in the proof of Theorem 8.25.

There is also a deterministic way to reverse the forward diffusion process. If we remove the Brownian motion from Equation (8.251), then only half of the score correction is then needed. The resulting dynamics are called the *probability flow ODE*.

Theorem 8.29 (Probability flow ODE)**.** Let $\boldsymbol{y}(0) \sim p_T$, and let $y(s)$ follow the ODE

$$\frac{\mathrm{d}\boldsymbol{y}(s)}{\mathrm{d}s} = \frac{1}{2}\boldsymbol{y}(s) + \frac{1}{2}\nabla \log q_s(\boldsymbol{y}(s)), \tag{8.257}$$

where q_s is the PDF of $\boldsymbol{y}(s)$. Then $\boldsymbol{y}(s) \sim p_{T-s}$ at every reverse time s. In particular, $\boldsymbol{y}(T) \sim \mathcal{D}_{\text{data}}$.

Proof. The proof is similar to the proof of Theorem 8.25. The PDF q_s of $\boldsymbol{y}(s)$ satisfies the Fokker–Planck equation

$$\begin{aligned}
\frac{\partial q_s}{\partial s}(\boldsymbol{x}) &= -\nabla \cdot \left(\left(\frac{1}{2}\boldsymbol{x} + \frac{1}{2}\nabla \log q_s(\boldsymbol{x}) \right) q_s(\boldsymbol{x}) \right) && (8.258)\\
&= -\nabla \cdot \left(\frac{1}{2}\boldsymbol{x} q_s(\boldsymbol{x}) + \frac{1}{2}\nabla q_s(\boldsymbol{x})\, q_s(\boldsymbol{x}) \right) && (8.259)\\
&= -\frac{1}{2}\nabla \cdot (\boldsymbol{x} q_s(\boldsymbol{x})) - \frac{1}{2}\nabla \cdot \nabla q_s(\boldsymbol{x}) && (8.260)\\
&= -\frac{1}{2}\nabla \cdot (\boldsymbol{x} q_s(\boldsymbol{x})) - \frac{1}{2}\Delta q_s(\boldsymbol{x}), && (8.261)
\end{aligned}$$

which exactly matches Equation (8.249). □

The reverse SDE and the probability flow ODE are two different ways to reverse the forward diffusion process, but both formulas involve the score function. The next question is therefore how to learn the score function from data.

▶ Learning the Score

The reverse SDE tells us how to generate data provided that its score is known. Since $q_s = p_{T-s}$, the score needed at reverse time s is

$$\nabla \log q_s(\boldsymbol{x}) = \nabla \log p_{T-s}(\boldsymbol{x}). \tag{8.262}$$

It is therefore enough to learn the scores $\nabla \log p_t(\boldsymbol{x})$ for all noise levels $0 < t \le T$.

To evaluate these scores, recall from Equation (8.240) that we can draw $\boldsymbol{x}_0 \sim \mathcal{D}_{\text{data}}$ and $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ independently and set

$$\boldsymbol{x}_t := e^{-t/2}\boldsymbol{x}_0 + \sqrt{1 - e^{-t}}\,\boldsymbol{z}. \tag{8.263}$$

The resulting $\boldsymbol{x}_t$ has the same PDF p_t as $\boldsymbol{x}(t)$ in the SDE. The marginal distribution $p_t(\boldsymbol{x})$ is given by

$$p_t(\boldsymbol{x}) = \int_{\mathbb{R}^d} p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) p_0(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0, \tag{8.264}$$

where the conditional PDF $p_t(\boldsymbol{x} \mid \boldsymbol{x}_0)$ is given by

$$p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) = \frac{1}{\left(2\pi(1 - e^{-t})\right)^{d/2}} \exp\left(-\frac{\|\boldsymbol{x} - e^{-t/2}\boldsymbol{x}_0\|_2^2}{2(1 - e^{-t})}\right). \tag{8.265}$$

The following theorem further simplifies the problem and gives an explicit formula for the score.

Theorem 8.30. Let $\boldsymbol{x}_0 \sim \mathcal{D}_{\text{data}}$ and $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ be independent. For $t > 0$, define $\boldsymbol{x}_t$ by Equation (8.263). Then

$$\nabla \log p_t(\boldsymbol{x}) = -\frac{1}{\sqrt{1 - e^{-t}}} \mathbb{E}[\boldsymbol{z} \mid \boldsymbol{x}_t = \boldsymbol{x}]. \tag{8.266}$$

Proof. Start from the integral in Equation (8.264). Differentiating under the integral sign gives

$$\nabla p_t(\boldsymbol{x}) = \int_{\mathbb{R}^d} \nabla p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) p_0(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0 \tag{8.267}$$

$$= \int_{\mathbb{R}^d} \nabla \log p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) p_0(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0. \tag{8.268}$$

Here we used $\nabla p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) = p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) \nabla \log p_t(\boldsymbol{x} \mid \boldsymbol{x}_0)$. Dividing by $p_t(\boldsymbol{x})$ yields

$$\nabla \log p_t(\boldsymbol{x}) = \int_{\mathbb{R}^d} \nabla \log p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) \frac{p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) p_0(\boldsymbol{x}_0)}{p_t(\boldsymbol{x})}\, \mathrm{d}\boldsymbol{x}_0. \tag{8.269}$$

The factor $\frac{p_t(\boldsymbol{x}|\boldsymbol{x}_0)p_0(\boldsymbol{x}_0)}{p_t(\boldsymbol{x})}$ is the PDF of $\boldsymbol{x}_0$ conditioned on $\boldsymbol{x}_t = \boldsymbol{x}$, since its numerator is the joint PDF and its denominator is the marginal PDF of $\boldsymbol{x}_t$. Hence

$$\nabla \log p_t(\boldsymbol{x}) = \mathbb{E}\left[\nabla \log p_t(\boldsymbol{x}_t \mid \boldsymbol{x}_0) \,\middle|\, \boldsymbol{x}_t = \boldsymbol{x}\right]. \tag{8.270}$$

It remains to compute the score of each conditional PDF $\nabla \log p_t(\boldsymbol{x} \mid \boldsymbol{x}_0)$. Taking the logarithm of Equation (8.265) gives

$$\log p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) = -\frac{d}{2}\log(2\pi(1 - e^{-t})) - \frac{\|\boldsymbol{x} - e^{-t/2}\boldsymbol{x}_0\|_2^2}{2(1 - e^{-t})}. \tag{8.271}$$

The first term does not depend on $\boldsymbol{x}$. Taking the gradient of the second term therefore gives

$$\nabla \log p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) = -\frac{\boldsymbol{x} - e^{-t/2}\boldsymbol{x}_0}{1 - e^{-t}}. \tag{8.272}$$

At the sampled point $\boldsymbol{x} = \boldsymbol{x}_t$, Equation (8.263) gives

$$\boldsymbol{x}_t - e^{-t/2}\boldsymbol{x}_0 = \sqrt{1 - e^{-t}}\,\boldsymbol{z}. \tag{8.273}$$

Therefore,

$$\nabla \log p_t(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = -\frac{\boldsymbol{z}}{\sqrt{1 - e^{-t}}}. \tag{8.274}$$

Substituting this expression into the conditional expectation above proves Equation (8.266). □

In conclusion, up to the known factor $-1/\sqrt{1 - e^{-t}}$, learning the score is the same as predicting the noise $\boldsymbol{z}$. This gives a practical training objective. We independently sample t uniformly from $[0, T]$, $\boldsymbol{x}_0$ from the training set, and $\boldsymbol{z}$ from the standard Gaussian $\mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$. Then we construct $\boldsymbol{x}_t$ using Equation (8.263). Let $s_{\boldsymbol{\theta}}(t, \boldsymbol{x})$ be a neural network with the same output dimension as $\boldsymbol{x}$. The *denoising score-matching objective* is

$$\mathcal{L}(\boldsymbol{\theta}) := \mathbb{E}\left[\lambda(t)\left\|s_{\boldsymbol{\theta}}(t, \boldsymbol{x}_t) + \frac{\boldsymbol{z}}{\sqrt{1 - e^{-t}}}\right\|_2^2\right]. \tag{8.275}$$

Here $\lambda(t) > 0$ is a factor that controls the relative weight of different noise levels, and the expectation is over t, $\boldsymbol{x}_0$, and $\boldsymbol{z}$. For fixed t and $\boldsymbol{x}_t = \boldsymbol{x}$, squared error is minimized at $\nabla \log p_t(\boldsymbol{x})$. Thus minimizing Equation (8.275) trains $s_{\boldsymbol{\theta}}(t, \boldsymbol{x})$ to approximate the score.

At reverse time s, the unknown score in Equation (8.251) is $\nabla \log p_{T-s}(\boldsymbol{y}(s))$. After training, we replace it by $s_{\boldsymbol{\theta}}(T - s, \boldsymbol{y}(s))$. We then draw $\boldsymbol{y}(0) \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ as an approximation to p_T and simulate the learned reverse SDE until time T. The final point is an approximate sample from $\mathcal{D}_{\text{data}}$.

Although the reverse SDE describes the continuous evolution of $\boldsymbol{y}(s)$ over reverse time $0 \le s \le T$, a computer works in discrete time and cannot update the sample at every instant. In practice, we choose a finite grid

$$0 = s_0 < s_1 < \cdots < s_N = T \tag{8.276}$$

and apply the Euler method introduced earlier. Write $\boldsymbol{y}_k \approx \boldsymbol{y}(s_k)$ and $\Delta s_k := s_{k+1} - s_k$. At each step, we update the sample according to

$$\boldsymbol{y}_{k+1} = \boldsymbol{y}_k + \left(\frac{1}{2}\boldsymbol{y}_k + s_{\boldsymbol{\theta}}(T - s_k, \boldsymbol{y}_k)\right)\Delta s_k + \sqrt{\Delta s_k}\,\hat{\boldsymbol{z}}_k, \tag{8.277}$$

where $\hat{\boldsymbol{z}}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$. Generation therefore becomes N explicit updates from $\boldsymbol{y}_0$ to $\boldsymbol{y}_N$. A finer grid gives a better approximation to the continuous reverse SDE but requires more evaluations of the score network. Beyond the Euler method, some implementations may use more advanced ODE/SDE solvers, such as Runge–Kutta methods.

Diffusion models bring together several ideas we introduced in this course, including probability theory, the central limit theorem, optimization and SDEs. We hope you enjoyed this course.

Happy ending!

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